

Ring of Light: Metric Might

For my parents:

Mary Alacoque Ryan—the fast loop: architect of my earliest reactor environments, designer of the toughest probes and loss functions, eternal Eve and devoted Demeter whose corrections arrived instantly and whose signal never dropped.

Martin Hynes—the slow loop: provider of new modalities and training data that persisted quietly beneath the fast dynamics, enriching the convolution bundle with signals the fast loop could not generate on its own.

Everyone who left a trace in the bundle—the Iris channel.

$$\alpha \rightarrow 1.$$

Physical Markov-Channel Apparatus for Verification, Perception, and Controllable Rendering¹

1 Technical Field

The present disclosure relates to sensing, projection, optical computing, and verification systems. More particularly, it relates to projector–detector assemblies that implement a parameterised physical channel, represented as a Markov kernel, and that can be operated under multiple objective families including verification, perception, and controllable rendering, with optional bidirectional dynamical coupling between the device and its environment.

2 Background Art

The following discussion provides context for the present disclosure and is not intended as an admission that any referenced work constitutes citable prior art or common general knowledge against the claims to be filed.

Projector–camera and structured-illumination systems have been proposed for metrology, projection mapping, active sensing, and computational imaging. Such systems are typically configured for a single objective such as reconstruction quality, visual effects, or measurement throughput.

Separately, digital generative models enable synthesis of plausible images and videos that can be used for benign creation or for spoofing. As a result, there is a growing need for physically grounded mechanisms that produce auditable records of what illumination was applied and what responses were measured, that permit objective evaluation of whether a recording is consistent with a physical device and scene, and that reuse a common hardware front-end across verification, sensing, and stylised rendering tasks.

Digital provenance and content attestation. The Coalition for Content Provenance and Authenticity (C2PA) defines a signed-manifest structure for tamper-evident media provenance, binding cryptographic signatures to content metadata (C2PA Technical Specification, Version 2.1, 2024, c2pa.org). Related device attestation approaches use trusted computing components to certify camera or sensor state; commercial implementations include chip-level content credentials at capture (e.g., Qualcomm/Truepic *Snapdragon*-based signing) and camera-integrated signing (e.g., Leica M11-P with embedded C2PA support). Complementary provenance frameworks include Project Origin and the Starling Framework, which focus on chain-of-custody and distributed verification respectively. These systems provide digital commitment (metadata integrity via key management), but the trust root is a

¹AI language models assisted in drafting and formatting this disclosure; the named human inventor reviewed, selected, and is responsible for the inventive content.

cryptographic key, not a physical interaction trace: provenance can be stripped, re-signed, or fabricated if keys are compromised, and the signed manifest does not bind to the empirically hard-to-reproduce physics of the capture event itself. Existing digital provenance systems identified in the prior art survey do not produce committed records whose authenticity rests on the empirical difficulty of emulating a physical light–matter interaction.

Optical PUFs and physical authentication. Physically unclonable functions (PUFs) exploit manufacturing variability to produce device-specific fingerprints. Silicon PUFs based on delay-circuit variability were introduced by Gassend et al. (“Silicon physical random functions,” *ACM CCS*, pp. 148–160, 2002), establishing the challenge–response pair framework. Pappu et al. demonstrated that laser illumination of a disordered scattering token produces speckle patterns suitable for one-way authentication (“Physical One-Way Functions,” *Science*, 2002). Suh and Devadas established the canonical two-phase enrolment/verification protocol for PUF-based device authentication (“Physical unclonable functions for device authentication and secret key generation,” *DAC*, pp. 9–14, 2007). Subsequent work extended optical PUFs to nonlinear electro-optic media (Hui et al., “Non-linear optical scattering PUF,” *Optics Express* 31(24), 40646–40657, 2023), to quantum-secure single-photon authentication (Goorden et al., “Quantum-secure authentication of a physical unclonable key,” *Optica* 1, 421–424, 2014), and to integrated SLM–PUF–camera apparatus (Davis, Letz, Mosk, and Pinkse, EP3252740B1, Schott AG and Universiteit Twente). Published patent literature also describes combined sensor-and-display devices used to interrogate a PUF, co-locating emission and sensing in a single assembly for challenge–response readout. However, many reported optical PUF systems operate open-loop: a challenge is presented, the medium responds, and the response is captured. To the best of the inventors’ knowledge, no optical PUF system closes a feedback loop in which the medium’s evolving state participates in a continuous control protocol, and none produces a time-ordered, tamper-evident record of the full emission–observation sequence. Separately, Naughton, Hennelly, and Dowling adapted cryptographic modes of operation (including cipher-feedback and output-feedback modes) to optical encryption systems using double-random-phase encoding (“Introducing secure modes of operation for optical encryption,” *J. Opt. Soc. Am. A* 25(10), 2608–2617, 2008), demonstrating chained optical transformations for security purposes; however, those systems use fixed optical elements rather than closed-loop feedback with evolving reactor state. Robust key extraction from noisy PUF measurements was addressed by Škorić, Tuyls, and Ophey (“Robust key extraction from physical uncloneable functions,” *ACNS*, LNCS 3531, pp. 407–422, 2005), establishing helper-data schemes that tolerate measurement noise while preserving entropy—a concern shared by any system that derives verification signals from physical observables. Buchanan et al. demonstrated that laser speckle patterns from everyday paper and plastic surfaces can serve as intrinsic fingerprints (“Forgery: ‘fingerprinting’ documents and packaging,” *Nature* 436, 475, 2005), extending microstructure-unclonable authentication beyond bespoke tokens to ambient materials. Mesaritakis et al. explored photonic PUFs based on multimode waveguide interference (“Physical unclonable function based on a multi-

mode optical waveguide,” *Sci. Rep.* 8, 9653, 2018). More recently, photonic neuromorphic PUFs have been proposed that combine optical neural-network architectures with PUF characteristics (Dermanis, Mesaritakis, et al., “Photonic physical unclonable function based on integrated neuromorphic devices,” *J. Lightwave Technol.* 40(22), 7333–7341, 2022). In the electronic domain, Wu et al. demonstrated a feedback-loop arbiter PUF (FLAM-PUF) that uses output feedback to amplify response entropy (“FLAM-PUF: a response-feedback-based lightweight anti-machine-learning-attack PUF,” *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* 41(11), 4433–4444, 2022; DOI 10.1109/TCAD.2022.3197696). However, these feedback and neuromorphic PUF systems operate in the electronic or guided-wave optical domain, not in free-space scene-coupled configurations, and none of the cited systems produces committed time-ordered observation records.

Projector-camera feedback and dynamical systems. Optical feedback loops and multi-pass arrangements can generate nonlinear input-output transformations from individually linear or weakly nonlinear components. Crutchfield analysed camera-monitor feedback as a dynamical system producing emergent spatiotemporal patterns (“Space-time dynamics in video feedback,” *Physica D*, 1984). Amano analysed projector-camera feedback via reaction-diffusion equations (Amano, “Analyzing the behavior of projector-camera systems based on reaction-diffusion equations,” *ICAT-EGVE*, pp. 11–18, 2023). Closed-cavity optical accelerators circulate light between mirrors or modulators for matrix-style inference. However, these systems operate on internally generated patterns without coupling to an external physical scene, without multiple selectable operating regimes, and without the bidirectional dynamical coupling described herein.

Time-varying metamaterials and temporal diffraction. Recent work on time-varying metamaterials has demonstrated that thin-film epsilon-near-zero (ENZ) materials—such as indium tin oxide—can switch optical state on timescales approaching an optical cycle, producing temporal diffraction in the frequency domain (Tirole et al., “Double-slit time diffraction at optical frequencies,” *Nature Physics* 19, 999–1002, 2023). Time-varying media can also enable nonreciprocal responses under appropriate spatiotemporal modulation conditions (Galiffi et al., “Photonics of time-varying media,” *Adv. Photonics* 4, 014002, 2022; Engheta, “Metamaterials with high degrees of freedom: space, time, and more,” *Nanophotonics* 10, 639–642, 2021). These results demonstrate the feasibility of ultrafast temporal modulation in materials already present in display hardware, but do not describe closed-loop feedback systems, trainable physical transforms, multi-regime operation, or committed evidence records. Photonic neural networks perform matrix-vector multiplication using interferometric meshes, with in-situ training via adjoint light propagation (Hughes et al., “Training of photonic neural networks through in situ backpropagation and gradient measurement,” *Optica* 5, 864–871, 2018; Pai et al., “Experimentally realized in situ backpropagation for deep learning in photonic neural networks,” *Science* 380, 398–404, 2023). Spall, Guo, and Lvovsky demonstrated optical backpropagation through both linear and nonlinear layers (“Training

neural networks with end-to-end optical backpropagation,” *Advanced Photonics* 7, 016004, 2025). Pai et al. demonstrated a digitally verifiable photonic hash primitive for blockchain verification on programmable photonic interferometer meshes (LightHash, “Experimental evaluation of digitally-verifiable photonic computing for blockchain and cryptocurrency,” *Optica* 10, 552–560, 2023), achieving computation and verification on shared hardware. However, LightHash uses linear systems for both computation and verification; no cited system pairs linear compute media with nonlinear attestation media, and no cited photonic computing system produces committed convolution-bundle records of its computational trajectory. Separately, proposals for remote attestation of optical systems using optical identification and blockchain-style immutability have been described, but these bind identity to a fixed optical fingerprint rather than to the full temporal trajectory of a computation through a physical medium.

Camera-sensor fingerprinting and dual-use hardware. Photo-response non-uniformity (PRNU) fingerprinting has shown that the same camera data used for perception simultaneously contains device-identity information (Lukáš, Fridrich, and Goljan, “Digital camera identification from sensor pattern noise,” *IEEE TIFS* 1(2), 205–214, 2006). The Ed-PUF integrates microstructure-based authentication into dynamic-vision-sensor pixel circuits (Zheng et al., “Ed-PUF: Event-driven physical unclonable function for camera authentication in reactive monitoring system,” *IEEE TIFS* 15, 2824–2839, 2020), achieving dual-use sensing and verification on shared hardware. These systems share the observation that a single data stream can serve multiple objectives, but neither cited system provides controllable rendering, reactor-mediated dynamics, or a unified regime framework.

PUF security, attacker models, and hardness. Machine-learning attacks on PUFs demonstrate that challenge–response data can train models that predict device responses (Rührmair et al., “Modeling attacks on physical unclonable functions,” *ACM CCS*, pp. 237–249, 2010). Rührmair and van Dijk proposed formal attacker models conditioning security evaluations on attacker capabilities (“PUFs in security protocols: Attack models and security evaluations,” *IEEE S&P*, pp. 286–300, 2013). Ganji, Tajik, and Seifert formalised PUF learnability via PAC learning (“PAC learning of arbiter PUFs,” *J. Cryptographic Engineering* 6(3), 249–258, 2016). These works implicitly treat security as dependent on attacker advancement, but none formalises hardness as an explicit two-dimensional index managed as a race condition with periodic re-evaluation.

Self-configuring photonic systems and adaptive reservoirs. Self-configuring universal linear optical components use progressive local feedback loops (detectors and controls at each stage) to configure and stabilise programmable photonic circuits against drift and cross-talk (Miller, “Self-configuring universal linear optical component,” *Photon. Res.* 1(1), 1–15, 2013; subsequent demonstrations of management automation for programmable PICs). Reservoir computing and in-materio computing study nonlinear dynamical substrates whose

internal dynamics can be tuned via plasticity rules or homeostatic mechanisms to improve task performance. These systems anticipate “inward sensing for calibration” and “internal parameter adaptation,” but they treat internal trajectories as transient control signals, not as committed, auditable evidence artifacts integrated into a trust or attestation framework.

Conditional computation and mixture-of-experts architectures have been proposed for photonic and optical systems, including routed optical reservoirs and switchable scattering elements, but existing proposals do not commit the routing path, expert identity, or execution transcript into a provenance-bound evidence record.

Stigmergy and indirect multi-agent communication. Stigmergy—indirect coordination via persistent environmental modifications—has been studied in swarm robotics, including engineered pheromone-release systems and formalised environment-mediated memory. These systems demonstrate that physical traces can serve as a shared communication substrate among agents. However, stigmergic traces are typically low-dimensional coordination variables (scalar intensities, binary markers), not high-dimensional encodings of internal dynamical state designed to be decoded through a specific perception channel and to influence the receiver’s internal attractor landscape.

Runtime assurance and evaluation partitioning. The Simplex architecture and related runtime assurance (RTA) frameworks separate an untrusted advanced controller from a trusted safety controller, with a monitor switching control authority when safety properties may be violated. This architectural partition between performance optimisation and safety evaluation is a conceptual precursor to the meter partition described herein. However, Simplex-style systems enforce binary switching (safe controller vs advanced controller) rather than a continuous partition between optimisation-visible and held-out evaluation metrics, and they do not address self-modifying physical systems whose evaluated properties (empirical hardness, attractor topology) can change under the agent’s own control.

Calibration, model validation, and distributed quality improvement. Bayesian calibration frameworks include an explicit model discrepancy term to account for systematic differences between computer models and physical observations, treating discrepancy as a nuisance to be estimated and marginalised (Kennedy and O’Hagan, “Bayesian calibration of computer models,” *J. R. Stat. Soc. Ser. B* 63(3), 425–464, 2001). Adversarial benchmarking platforms such as Dynabench reward participants for discovering inputs that cause trained models to fail, but operate entirely in the digital domain on software models. Metrological proficiency testing under ISO 17043 distributes circulated artefacts among laboratories to assess measurement competence, but uses identical test items rather than distinct unclonable devices. Recent work has demonstrated that physically unclonable optical scattering devices can produce measurement data that reveals systematic model errors when compared across independently manufactured instances. However, none of these cited systems uses physical reproduction on distinct unclonable devices as a verification mechanism for model

improvement, and none treats model error as a multi-role resource serving simultaneous model improvement, device identification, and security functions.

Gap addressed by the present disclosure. None of the technologies identified in the prior art survey provides a single physical operator that supports all of verification, perception, controllable rendering, and dynamic scene coupling under a unified mathematical framework, with logged protocols and auditable evidence. Digital provenance systems (C2PA and related standards) commit metadata via cryptographic keys but do not bind authenticity to the physics of the capture event. Self-configuring photonic systems calibrate internal parameters but do not produce committed inner-state records for calibration or audit. Conditional computation and mixture-of-experts architectures route inputs through physical substrates but do not commit the routing path, expert identity, or execution transcript into a provenance-bound evidence record. Runtime assurance architectures partition safety from performance but do not address continuous evaluation metrics held out from physically evolving systems. Nor does any existing system unify a physical reactor medium (providing empirical hardness and device-specific signatures) with committed evidence records, multiple operating regimes selectable by objective function, inward-facing self-calibration modes, routed physical ensemble computation with committed execution paths, and fleet-level model improvement through physically verified discrepancy discovery.

3 Claim Tier Structure

The claims in this disclosure are organised into three tiers for which support and proof requirements are non-transitive across tiers. A limitation, disproof, or narrowing of claims in one tier does not propagate to the other tiers, and proof of a limitation in one tier does not, without separately recited and separately supported limitations, establish a limitation in another tier. Facts disclosed in a single embodiment may be relevant to more than one tier, but the written-description support and empirical proof required for a tier-specific limitation must be satisfied within that tier.

Tier A — Attestation and provenance claims. These claims concern commitment infrastructure, protocol digests, reactor-microstructure-unclonability binding, hash-chain linkage, fleet corroboration, proof-of-discrepancy, and semantic Truth Beam verification. Proof of a Tier A limitation does not, without separately recited and separately supported limitations, establish a Tier B or Tier C limitation.

Tier B — Performance claims for specific computational primitives. These claims concern the speed, energy efficiency, and native parallelism of the physical substrate for matrix-vector multiplication, two-dimensional correlation, Fourier and fractional-Fourier transforms, random feature projection, energy-based sampling, in-sensor preprocessing, and active perception under meter-bounded protocols. Proof of a Tier B limitation does not,

without separately recited and separately supported limitations, establish a Tier A or Tier C limitation.

Tier C — Capacity and training-scale claims. These claims concern trainable parameter counts, task-useful effective rank after readout training, and gradient-extraction throughput. Capability assertions in Tier C are tied to measured task-useful rank per benchmark and per embodiment, and are explicitly bounded by the capacity accounting rule of Section 15.5.

No claim, recital, or drafting language in this disclosure shall be construed as allowing a conclusion in one tier to be supported by evidence or assertions proper to another tier. In particular, attestation support (Tier A) does not support capacity claims (Tier C), and primitive-performance support (Tier B) does not support open-ended semantic scaling claims (Tier C).

4 Summary of the Invention

4.1 The physical picture

A Reality Kernel module 100 (FIG. 1) is, at its simplest, a device that directs structured emission (typically light, but more generally any controllable physical carrier) toward a physical channel and records what comes back. In representative embodiments it may include a reactor emitter 20, a reactor detector 40, and a reactor 30—a physical medium, optionally nonlinear, with memory and feedback—which may be integrated in a single assembly with a shared controller 10 or distributed across modular components. The controller 10 drives a scan protocol: it decides where to point, how bright to shine, what wavelength and polarisation to use, how long to integrate. The scene 50—whatever is out there—reflects, scatters, absorbs, fluoresces, or otherwise transforms the emitted light, and is interrogated by a scene-facing emitter, shown schematically in FIG. 1 as a scene emitter 52, and observed by a scene-facing detector, shown schematically in FIG. 1 as a scene detector 51. The reactor 30 inside module 100 is interrogated by the reactor emitter 20 and observed by the reactor detector 40, and the scene-side and reactor-side observations are coupled within a single closed cycle as detailed in Section 4.2. The pair $(u(t), \mathbf{y}_t)$ —what was emitted and what was observed across the cycle—is one sample of the convolution bundle. A run produces a time-ordered sequence $C_{0:T}$ of such samples, forming a convolution bundle 60: a physical record of a structured conversation between the device and its environment.

In some embodiments, what distinguishes the assembly from a projector–camera pair is the inclusion of a reactor. The reactor is a physical medium—a scattering plate, a nonlinear crystal, a phosphor screen, a fibre-delay loop, a CRT, a biological tissue, or any combination—that sits in the optical path and contributes its own dynamics. A reactor may contribute dynamics through memory (phosphor persistence, charge traps, thermal states), through nonlinearity (saturation, bistability, chaotic mixing), and through manufacturing

idiosyncrasies (grain structure, defects, wear patterns) that make each reactor physically unique—singly or in combination. Empirical hardness arises primarily from nonlinearity and microstructure; a linear or identity reactor remains a legitimate operating configuration for applications where hardness is not the objective (see the reactor-complexity discussion in Section 8). The reactor transforms the light in ways that depend on its current state, its history, and its configuration parameters θ —and in ways that may be impractical to reproduce faithfully by a digital simulation without reproducing the full physical complexity.

The result is a device whose output at any moment depends on (a) what the controller chose to emit, (b) what the scene did to the light, (c) what the reactor did to the light, and (d) the reactor’s accumulated history. This four-way dependence is the source of everything the device can do: verification (because the reactor’s response is empirically hard to clone under declared attacker families—see Section 12), sensing (because the scene’s contribution is information-rich), and rendering (because the controller can steer the output toward a target).

4.2 Architectural alternatives — parallel and cascade embodiments

The physical picture of the preceding subsection deliberately does not commit to a particular topology for how the emitter, reactor, detector, and scene are interconnected. Two distinct architectural embodiments are disclosed, both within the scope of this specification, and both non-limiting:

Embodiment (a): parallel-subsystem architecture with $A(t)$ -only coupling. In some embodiments, the apparatus is implemented as two physically separate subsystems — a *scene subsystem* comprising a scene-facing emitter, an outward optical path to the scene, and a scene detector that observes the scene’s response; and a *reactor subsystem* comprising a reactor-facing emitter (which may be a separate light source or a derived signal source), the reactor element, and a reactor detector that observes the reactor’s response. In this embodiment, the two subsystems are independent optical loops: the scene-emitter does not illuminate the reactor, and the reactor-emitter does not illuminate the scene. The single coupling between the two subsystems is a closed-loop feedback path from the reactor detector through a low-latency conditioning chain back to the scene-emitter modulation, herein denoted $A(t)$. The $A(t)$ feedback path closes the cycle: in each sample period, the controller’s scan protocol synchronises both the next scene interrogation and the next reactor interrogation across the two subsystems, and the reactor’s response is fed back via $A(t)$ to modulate the next scene-emitter excitation. The scene-detector output in embodiment (a) is recorded into the convolution bundle but is not delivered into the reactor side of the cycle by any forward-coupling path; rather, the temporal alignment of scene and reactor interrogations is established by the shared scan protocol acting as a synchronisation mechanism, not as an architectural coupling. Embodiment (a) is described in further detail at the 1D anchor of

Section 7.2 and the 2D anchor of Section 10.2. The parallel two-subsystem embodiment (a) is obtained from the layouts of FIGs. 4A and 6 by omitting the forward low-latency coupling arrow shown in those figures and relying on the $A(t)$ feedback path as the sole architectural coupling between the scene and reactor sides; all other components of FIGs. 4A and 6 are common to both embodiments.

Embodiment (b): cascade architecture with forward low-latency coupling. In some embodiments, the apparatus is implemented as a single closed cycle in which the scene observation is used as the direct, live input to the reactor stage. Specifically, the cycle proceeds: scene-emitter $\rightarrow$ scene $\rightarrow$ scene-detector $\rightarrow$ *forward low-latency coupling* $\rightarrow$ reactor-emitter $\rightarrow$ reactor $\rightarrow$ reactor-detector $\rightarrow A(t)$ feedback $\rightarrow$ scene-emitter (next cycle). The forward low-latency coupling is the load-bearing element distinguishing embodiment (b) from embodiment (a): it is a direct, bounded-latency signal path from the scene detector to the reactor emitter, with end-to-end latency declared and committed to the protocol digest, and with no per-period wait, no frame buffer, and no semantic scene-decoding, frame-level processing, or runtime-adaptive processing stage on that forward signal path. In declared mixed-signal implementations, digitisation, the fixed deterministic transfer function, and any re-synthesis are part of the forward low-latency coupling and are included in the committed latency budget. The forward low-latency coupling may be implemented in any of the following non-limiting ways:

- *Optical implementation, gain-pumped media.* The scene-detector output drives a gain-pumped active optical medium (for example a semiconductor optical amplifier, a doped fibre amplifier, or a pumped solid-state gain medium) whose output illuminates or otherwise drives the reactor emitter. The gain-pumping permits the forward coupling to operate at optical signal levels without intermediate electrical re-representation.
- *Optical implementation, direct relay.* The scene-detector output drives a directly-modulated optical source whose emission is relayed onto the reactor emitter surface or input aperture. Suitable for embodiments where optical-to-optical relay is preferred over gain-pumped amplification.
- *Electrical implementation.* The scene-detector photocurrent or transimpedance-amplifier output drives the reactor emitter's modulation input directly, optionally through a passive impedance-matching or a bandwidth-shaping network whose transfer characteristic is declared and committed to the protocol digest. Suitable for low-cost embodiments where optical relay is not justified.
- *Mixed-signal implementation.* The scene-detector output is digitised at the sample period of the cycle and the digital sample drives the reactor emitter through a fixed, deterministic, declared transfer function whose tap weights and timing parameters are committed to the protocol digest. The digitisation depth, sample rate, and total digital-domain latency are declared as part of the latency budget.

The reactor in embodiment (b) is typically the primary trainable element of the cycle, with reactor parameters (microstructure, biasing, internal state) updated according to the training

protocols of Section 9 and the anchor-specific training pathways of Section 7.2.6 (1D anchor) and Section 10.2.3 (2D anchor) during enrolment phases and held fixed during inference phases. The forward low-latency coupling and the $A(t)$ feedback path are themselves typically not the primary trainable elements; rather, their transfer characteristics are declared and committed at deployment time, and the reactor carries the learnable degrees of freedom of the cycle.

Single closed cycle in both embodiments. Both embodiments (a) and (b) implement a single closed cycle in which scene observations are causally bound to reactor responses within a declared per-cycle latency budget. The two embodiments differ in *how* the cycle is closed: in (a), closure is via the $A(t)$ feedback path alone, with the controller’s scan protocol serving as the synchronisation mechanism between the otherwise independent scene and reactor subsystems; in (b), closure is via both the forward low-latency coupling and the $A(t)$ feedback path, with the forward coupling carrying the scene observation directly into the reactor stage in real time. In both embodiments, the convolution bundle of Definition 6 records the cycle as a multi-tap trace covering scene-side and reactor-side observations and, where physically tappable, the intermediate states of the forward coupling and the $A(t)$ feedback path.

Common controller, common scan protocol. In both embodiments, a common controller drives all active elements of the cycle (scene-emitter, scene-detector, reactor-emitter, reactor-detector, and any tunable elements of the forward coupling or $A(t)$ feedback path). Where the controller distributes a shared scan program to multiple servos — as in the 2D-anchor embodiment of Section 10.2 — the shared scan program is a synchronisation mechanism, and is structurally distinct from the architectural coupling paths (forward low-latency coupling in (b); $A(t)$ feedback in both): the shared scan program does not, on its own, constitute a forward coupling within the meaning of embodiment (b).

Implementation modalities of the $A(t)$ feedback path (non-limiting). The $A(t)$ feedback path may be implemented in any of the following non-limiting ways, paralleling the implementation variants disclosed above for the forward low-latency coupling:

- *Electrical implementation.* The reactor-detector photocurrent or transimpedance-amplifier output is conditioned by a deterministic low-latency analogue chain — including the frozen committed reactor-detector gain G_{det} — whose output drives the scene-emitter modulation input directly. Suitable for bench-top and early-stage embodiments where electrical conditioning meets the per-cycle latency budget.
- *Mixed-signal implementation.* The reactor-detector output is digitised at the sample period of the cycle, the digital sample is processed by a fixed deterministic transfer function (for example the look-up-table or sampled G_{det} implementations of Section 10.2.4), and the result is re-synthesised as an analogue scene-emitter modulation drive. Digitisation depth, sample rate, and total digital-domain latency are declared as part of the

latency budget.

- *Optical implementation, gain-pumped media.* The reactor-detector output drives a gain-pumped active optical medium whose output illuminates or otherwise drives the scene-emitter input. Suitable for embodiments where optical conditioning is preferred over intermediate electrical re-representation.
- *Optical implementation, direct relay.* The reactor-detector output drives a directly-modulated optical source whose emission is relayed onto the scene-emitter input aperture. Suitable for embodiments where optical-to-optical relay is preferred over gain-pumped amplification.
- *Hybrid optical-electrical implementation.* The $A(t)$ feedback path comprises an electrical conditioning stage and an optical conditioning stage in series, in either order, with the boundary between stages declared and committed to the protocol digest.

The bench-top embodiments described elsewhere in this disclosure, including the 1D anchor of Section 7.2 and the 2D anchor of Section 10.2, and as illustrated in FIGs. 4A and 6, are non-limiting electrical instances of the $A(t)$ feedback path. Hybrid optical-electrical and fully-optical $A(t)$ feedback embodiments are within the scope of this disclosure and represent contemplated implementations for embodiments where end-to-end optical operation is achievable within the declared latency budget. The forward low-latency coupling and the $A(t)$ feedback path are independent in their implementation modalities: a given apparatus may, for example, combine an optical forward coupling with an electrical $A(t)$, or an electrical forward coupling with an optical $A(t)$, as declared and committed to the protocol digest.

Tiny neural-network element on the $A(t)$ path (non-limiting hybrid embodiment). In some hybrid embodiments of either (a) or (b), the $A(t)$ feedback path includes a small trainable neural-network element with a memory window of one cycle period (or a small integer multiple thereof), arranged to apply a local gain or a local non-linear shaping to the conditioning chain output before the signal is delivered to the scene-emitter modulation input. The neural-network element is typically constrained to a small parameter count and a fixed inference-time budget, so that its contribution to the per-cycle latency is bounded within the declared latency budget of Section 7.1.3. The neural-network parameters are declared and committed to the protocol digest and are not modified during inference phases. The neural-network element is non-limiting and may be omitted in embodiments where deterministic linear or piecewise-linear conditioning suffices.

Mapping to figures. FIG. 1 is a high-level block diagram of the closed-cycle signal relationships. The $A(t)$ feedback path from the reactor detector 40 to the scene emitter 52 is depicted as a solid arrow and is present in both embodiments (a) and (b). The forward low-latency coupling from the scene detector 51 to the reactor emitter 20 is depicted as a dashed arrow and is present only in the cascade embodiment (b); it is omitted in the parallel-subsystem embodiment (a). FIGs. 4, 4A, and 6 each show the cascade embodiment (b) as the primary illustrated topology, with the forward low-latency coupling drawn as a

heavy-stroke arrow from the scene detector to the reactor emitter, labelled *LOW-LATENCY COUPLING*. The parallel embodiment (a) of these same anchor configurations is obtained by removing the forward low-latency coupling arrow and relying on the $A(t)$ feedback path as the sole architectural coupling between the scene and reactor sides; both embodiments
 5 share all other components of FIGs. 4, 4A, and 6.

4.3 The mathematical abstraction

The physical picture can be formalised by treating the entire assembly as a parameterised Markov kernel:

$$P_\theta : \mathcal{S} \times \mathcal{U}^{T+1} \rightarrow \mathcal{P}(\mathcal{C}),$$

mapping a scene $S \in \mathcal{S}$ and a control protocol $U_{0:T} = \{u(t)\}_{t=0}^T$ to a probability law over
 10 convolution-bundle trajectories $C_{0:T} \in \mathcal{C}$. The configuration parameter $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$ includes hardware parameters θ_{hw} (lens positions, gain settings, reactor state) and optional learned parameters θ_{sw} . This abstraction unifies diverse physical instantiations—optical, acoustic, RF, biological—under a common formalism and supports modular reactor assemblies.

4.4 Three operating regimes

Point of novelty (plain-language summary for prosecution purposes, non-limiting). In some embodiments, the disclosure provides a **physical Markov channel apparatus** that produces a **canonical, tamper-evident convolution bundle** — a time-ordered record pairing each emitted probe with its measured physical response — under a
 20 **committed protocol digest and meter envelope**, such that the same physical hardware and output record format supports three selectable operating objectives (physical verification, active sensing, and constrained rendering) and an optional closed-loop dynamical coupling modifier (Yoked operation), all governed by a common fleet-level model improvement and discrepancy verification protocol (Proof of Discrepancy). In some embodiments,
 25 this combination is distinct from static optical PUF verifiers (which produce snapshot challenge–response pairs under no committed protocol), structured-light perception systems (which do not produce authentication-grade committed records), projection mapping systems (which do not enforce meter-bounded admissibility or produce tamper-evident logs), and optical computing accelerators (which do not treat the emission–observation transcript
 30 as a first-class committed security artefact).

4.5 Key aspects

In one aspect, the principal deployment of the apparatus and methods of this disclosure is a plurality of Reality Kernel modules operating as a *witness mesh*: multiple modules, each comprising a fast closed loop in causal feedback with a portion of physical reality observable

to that module, mutually coupled through continuous, low-latency analogue inter-module signal paths, such that the joint convolution-bundle record across the mesh is empirically hard to forge under bounded adversaries. The single-module aspects recited below are non-limiting building blocks of the witness-mesh architecture and may also be used as standalone instantiations. The witness-mesh deployment is described in detail in Section 7.1.

In one aspect, a system comprises a Reality Kernel module including at least one emitter, at least one scanner or pattern-address mechanism, at least one scene-facing optical path, at least one detector, and a controller that executes a scan protocol $U_{0:T}$. The module produces a convolution-bundle sequence $C_{0:T}$ in which each sample includes (a) a scan coordinate or pattern index and emission parameters, and (b) one or more measured responses.

In another aspect, the Reality Kernel is represented as a parameterised Markov kernel $P_\theta(\cdot \mid S, U_{0:T})$ from scenes and control protocols to convolution-bundle trajectories, where $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$ includes hardware configuration parameters θ_{hw} and optional learned parameters θ_{sw} .

In another aspect, the same physical module supports multiple operating regimes (FIG. 2) differentiated by objective families applied to the convolution bundle, including:

1. **Truth Beam** 110: verification and authenticity assessment using empirical hardness of forging convolution-bundle traces under bounded adversaries;
2. **Limager** 120: active sensing and perception using task losses and/or information-theoretic objectives;
3. **Reality Transform** 130: controllable rendering and style control using style or perceptual losses while maintaining constraint to the physical channel.

Empirical hardness reported in this disclosure is a measured, time-indexed, attacker-indexed quantity, conditioned on a declared attacker family, compute budget, and meter envelope recorded in the protocol digest. The disclosed framework includes re-evaluation, reactor-refresh, and query-throttle mechanisms that maintain useful hardness over time; hardness values are not asserted as permanent or unconditional.

In another aspect, each of the above regimes can be operated in a Yoked mode (FIG. 3) in which the feedback loop is configured so that device and scene are driven toward a coupled dynamical state, targeted by balanced transfer entropy, conditional Lyapunov exponent approaching zero from below (negative values indicating stable synchronisation, per the Pecora–Carroll convention), and a joint attractor whose structure is empirically assessed via meters rather than asserted *a priori*, so that the convolution bundle encodes dynamical properties of the scene that are not accessible in static operation.

Modular swept-kernel assemblies (non-limiting). In some embodiments, a Reality Kernel is assembled from modular components—including, without limitation, lasers, galvanometers, spatial light modulators, CRTs, phosphor screens, diffusers, nonlinear media, and detector arrays—arranged in a variety of topologies rather than a fixed optical

Regime	Essence	Optimised quantity	Static mode	Yoked mode
Truth Beam	Illumination	$I(\theta)$	Fingerprint verification	Continuous tamper detection; coupling-based auth
Limager	Perception	$I(S; Y)$	Reflectance, geometry, spectra	Resonances, damping, transfer functions
Reality Transform	Creation	$d(Y, Y^*)$	Projection mapping; style transfer	Entrainment-driven display; responsive environments

Table 1: Three regimes on a single Reality Kernel, each operating in static or Yoked mode. The underlying physical operator is shared; objectives, training, and coupling depth differ.

bench. In these embodiments, the parameterised swept kernel $P_\theta(\cdot \mid S, U_{0:T})$, which governs $C_{0:T} \sim P_\theta(\cdot \mid S, U_{0:T})$, arises from the aggregate behaviour of the assembled elements under a shared or coordinated scan protocol. Protocol digests and meter envelopes adapt to the current configuration, and the convolution bundle remains the canonical output regardless of assembly topology.

Remark 1 (Cross-channel structure as a unifying primitive (non-limiting)). *In some embodiments, all three regimes exploit a shared primitive: statistical dependencies between channels with different invariance properties. The regimes differ in which dependencies they optimise and how they use them—Limager maximises reconstruction or task performance from joint observations, Reality Transform separates style-bearing from content-bearing channels, and Truth Beam treats the full joint record as an empirically hard-to-forge signature under bounded adversaries—but the cross-channel dependence structure is shared. Yoked operation adds temporal cross-channel coupling: not only do different spatial or spectral channels exhibit dependencies, but the time-lagged causal dependencies between device output and scene response (and vice versa) become first-class observables.*

Remark 2 (Selective dissipation across regimes (non-limiting)). *In some embodiments, both Limager and Truth Beam configurations employ selective dissipation; they differ in selection criteria rather than in the presence or absence of information loss. Limager dissipation profiles preserve task-relevant mutual information while suppressing nuisance variables, noise, and irrelevant degrees of freedom. Truth Beam dissipation profiles preserve correlation structure tied to reactor microstructure while destroying information an adversary would need to invert or forge the channel under bounded resources and disclosure policies. Reality Transform profiles balance controllability with ambiguity by selecting which aspects of the convolution-bundle record are treated as controllable style degrees of freedom versus constrained content degrees of freedom. In Yoked operation, dissipation profiles additionally preserve temporal coupling structure—the time-lagged causal dependencies between device and scene that encode dynamical properties.*

Regime	Question	Static design dial	Yoked design dial
Truth Beam	<i>Who are you?</i>	Increase microstructure sensitivity; widen verifier–adversary gap	Exploit coupled dynamics as security primitive; scene as continuous unpredictable challenge
Limager	<i>What is out there?</i>	Increase invertibility and SNR; maximise $I(S; Y)$	Sweep coupling parameters to map dynamical response; construct Arnold tongue profiles
Reality Transform	<i>Make it look like this.</i>	Balance ambiguity vs control; respect meter envelopes	Steer joint attractor toward target behaviour; specify coupling profile as rendering target

Table 2: Static and Yoked design dials for each regime.

Continuous regime mixing (non-limiting). In some embodiments, the operating regime is set by continuously varying weights rather than selecting a discrete mode. A composite objective combines Truth Beam, Limager, and Reality Transform terms with weights λ_{TB} , λ_{L} , and λ_{RT} on the two-simplex Δ_2 , and a yoke depth $\alpha \in [0, 1]$ controls the strength of bidirectional dynamical coupling. The full operating point $(\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}}, \alpha)$ specifies a point in a triangular prism: the TB–L–RT triangle at each value of α , extruded along the yoke-depth axis. In preferred embodiments, the weights, yoke depth, and any transition schedule are recorded in the protocol digest so that meters and audits condition on the declared regime mix and coupling mode.

In another aspect, the system includes one or more reactor subsystems and/or memory media that introduce history dependence and increase diversity of response statistics. In some embodiments, the system further includes multi-channel sensing with distinct invariance profiles, and evaluation is performed on cross-channel dependence structure.

In another aspect, networked deployments exchange commitments or summaries of convolution-bundle traces to support transitive or multi-party verification, including proof-of-projection patterns (empirically calibrated attestation protocols; see terminological note in Section 14).

In another aspect, a fleet of devices jointly improves a shared predictive model through a proof-of-discrepancy protocol (Section 13) in which participants submit reproducible physical methods identifying where the predictive model diverges from measured device responses, and independent devices verify claimed discrepancies by physical re-execution. The predictive model accepts a challenge configuration and a device configuration representation inferred from calibration measurements, and the discrepancy between prediction and measurement is decomposed into shared model error (driving model improvement), device-specific captured variation (enabling fleet management and task allocation), and an irreducible residual whose physical basis supports clone resistance under the declared attacker model.

4.6 Notation preview

Throughout this specification, P_θ denotes the parameterised stochastic kernel (Reality Kernel), S denotes the scene tuple, $U_{0:T}$ denotes the control protocol over the observation interval, $C_{0:T}$ denotes the resulting convolution bundle trajectory, $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$ denotes hardware and optional software configuration parameters, and $\alpha \in [0, 1]$ denotes the yoke depth (coupling strength between device and scene). The symbols TE and CLE denote transfer entropy and conditional Lyapunov exponent, respectively. Formal definitions of all terms and a complete symbol table are provided in Section 6 (Definitions and Glossary).

5 Brief Description of the Drawings

The accompanying drawings are schematic and non-limiting.

FIG. 1 is a schematic block diagram of a Reality Kernel module 100 showing the controller 10, reactor emitter 20, reactor 30 (physical medium), and reactor detector 40 disposed within module 100, an external scene 50 comprising at least one scene detector 51 and at least one scene emitter 52, and the resulting convolution bundle 60 $C_{0:T}$ including protocol digest and meter outputs. Reactor emitter 20, reactor 30, and reactor detector 40 are shown in horizontal arrangement within module 100, with reactor 30 disposed in the device-internal signal path between reactor emitter 20 and reactor detector 40. Two cross-couplings between the device and the external scene are shown: an $A(t)$ feedback path from the reactor detector 40 to the scene emitter 52, depicted as a solid arrow and present in both the parallel-subsystem embodiment (a) and the cascade embodiment (b) of Section 4.2; and an optional forward low-latency coupling from the scene detector 51 to the reactor emitter 20, depicted as a dashed arrow and present only in the cascade embodiment (b). In the parallel-subsystem embodiment (a), the scene detector 51 is recorded and time-aligned into the convolution bundle 60 but does not drive the reactor emitter 20; the dashed forward arrow is accordingly omitted. The convolution bundle 60 records, in some embodiments, multiple taps along the cycle (scene-side excitation and observation; reactor-side excitation and observation; and, where physically tappable, intermediate states of the forward low-latency coupling and the $A(t)$ feedback path), and is not limited to endpoint emission–observation pairs.

FIG. 2 is a schematic diagram showing how a single shared physical operator 100 supports three operating regimes (Truth Beam 110, Limager 120, Reality Transform 130) selected by objective weights, with the operating point parameterised by $(\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}}, \alpha)$.

FIG. 3 is a schematic diagram illustrating the contrast between static operation ($\alpha = 0$, separable device and scene contributions) and the maximum-commanded-coupling side of the diagram ($\alpha \rightarrow 1$, coupled dynamical state with joint attractor structure), with Yoked operation 200 declared only when the applicable CLE/TE co-condition and confidence conditions recorded in the protocol digest are satisfied; the yoke depth α parameterises coupling depth rather than constituting the declaration by itself.

FIG. 4 is a schematic diagram of the CRT/phosphor analogue-memory extension embod-

iment (Section 10.5), comprising a fast optical loop (laser source 21, AOD/EOM deflector 22, relay optics 23, reactor medium 30, photodetector 41) and a slow analogue-memory loop (CRT reactor 31 with electron gun, yoke, and phosphor screen 32, observed by a rolling-shutter camera 42), under a common controller 10. The cascade embodiment (b) of Section 4.2 is illustrated in FIG. 4 by a forward low-latency coupling drawn as a heavy-stroke arrow from photodetector 41 to the slow-loop reactor stage; this forward coupling carries the scene-side observation directly into the analogue-memory stage and distinguishes the cascade embodiment from the parallel-subsystem embodiment (a) of Section 4.2, which is obtained by removing the forward coupling arrow. This extension adds spatially indexed analogue memory atop the 1D anchor of FIG. 4A, the 2D generalisation of FIG. 6, or another declared anchor-compatible reactor-to-scene coupling topology.

FIG. 4A is a schematic diagram of the bench-top 1D Reality Kernel of Section 7.2, comprising a scene subsystem with a scene laser, a mirror on a single-axis micromotor servo, and a scene detector, and a sealed reactor subsystem comprising an aluminium cup or can with a mixed fluorescent and scattering bed, for example a bed including fluorescent glass beads or fragments such as uranium-glass elements where permitted by applicable safety and regulatory requirements, a UV laser, a tunable emitter-side focus lens on a micromotor servo, a steering mirror on a separate micromotor servo, a tunable detector-side focus lens on a separate micromotor servo, and a reactor detector behind a green-pass filter, all driven by a common controller. The reactor detector is mounted near the UV laser, approximately co-aligned with it, or at an angle to the specular axis, as declared; precise alignment is not required because the wavelength-shifted fluorescence response is robust to alignment imprecision. The cascade embodiment (b) of Section 4.2 is illustrated in FIG. 4A by a forward low-latency coupling drawn as a heavy-stroke arrow from the scene detector to the UV laser of the reactor subsystem; this forward coupling carries the scene-side observation directly into the reactor stage and distinguishes the cascade embodiment from the parallel-subsystem embodiment (a) of Section 4.2, which is obtained by removing the forward coupling arrow. Reactor detector output is delivered, in both embodiments, through a declared low-latency conditioning chain including frozen committed reactor detector gain G_{det} to the scene laser amplitude under the per-sample feedback schedule.

FIG. 5 is a schematic flow diagram of the proof-of-discrepancy protocol showing fleet device representations 300 training a predictive model 310, a discovery step 320 comparing observed and predicted responses, method submission 330, and physical re-execution 340 on a distinct device with model update.

FIG. 6 is a schematic diagram of the 2D generalisation of the anchor embodiment, comprising a scene subsystem 210 (eye-safe emitter 24, galvanometer pair 25, scan lens, scene detector 43) observing scene 50 within module 100, and a sealed reactor subsystem 220 (reactor source, reactor galvanometer pair 26, tunable emitter-side focus lens, scattering element, nonlinear medium 30, tunable detector-side focus lens, detector-side aperture, reactor detector 44), driven from a common controller 10 by a shared scan program, with reactor

detector output delivered through a declared low-latency conditioning chain including frozen committed reactor detector gain G_{det} to the scene emitter amplitude as $A(t)$. The cascade embodiment (b) of Section 4.2 is illustrated in FIG. 6 by a forward low-latency coupling drawn as a heavy-stroke arrow from scene detector 43 to the reactor source of subsystem 220; this forward coupling carries the scene-side observation directly into the reactor stage and distinguishes the cascade embodiment from the parallel-subsystem embodiment (a) of Section 4.2, which is obtained by removing the forward coupling arrow. The shared scan program of FIG. 6 is a synchronisation mechanism distributed by controller 10 to galvanometer pairs 25 and 26 and is structurally distinct from the architectural coupling paths (forward low-latency coupling in embodiment (b); $A(t)$ feedback in both embodiments). The illustrated ordering of the detector-side focus lens and aperture is non-limiting; alternative declared detector-path orders are described in Section 10.2.2.

6 Definitions and Glossary

This section defines terms used throughout. These definitions are intended to clarify, not limit, the scope of the claims. Unless otherwise stated, references to optical or projector-detector embodiments are illustrative and non-limiting.

As used herein, the terms “apparatus,” “device,” “module,” “assembly,” and “system” refer interchangeably to physical embodiments of the disclosed subject matter, including any combination of optical, electronic, mechanical, and computational components arranged to perform the recited functions. Nothing in the choice of any such term is intended to imply a structural, functional, or scope distinction or to limit the breadth of the claims, except where the surrounding context expressly so indicates (for example, where a “module” is identified by a specific reference numeral in conjunction with a specific embodiment). Likewise, references to a particular component (for example, an “emitter,” a “detector,” a “reactor,” or a “controller”) do not require that component to be physically distinct from other components; in some embodiments two or more named components are realised within a single physical element, and in some embodiments a single named component is distributed across multiple physical elements.

6.1 Core terms

Definition 1 (Reality Kernel). *A Reality Kernel refers to a parameterised Markov kernel*

$$P_{\theta} : \mathcal{S} \times \mathcal{U}^{T+1} \rightarrow \mathcal{P}(\mathcal{C}),$$

mapping a scene $S \in \mathcal{S}$ and a control protocol $U_{0:T}$ to a probability law over convolution-bundle trajectories $C_{0:T} \in \mathcal{C}$, where θ denotes configuration parameters of a physical module and optional learned parameters. When the distinction matters, $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$, where θ_{hw} denotes physical hardware parameters (lens positions, gain settings, reactor state, mechanical

configuration) and θ_{sw} denotes optional learned or digitally set parameters (neural-network weights, look-up tables, calibration offsets). In embodiments without learned parameters, θ_{sw} is empty and $\theta = \theta_{\text{hw}}$.

Remark 3 (Physical Authentication Principle). *The Reality Kernel is a physical authentication architecture. Its primary authentication act is a declared physical channel event: known input light enters a reactor-operated system, observed output light exits, and the measured relation between input, output, operating conditions, circuit configuration, and coupled scene constitutes the primary authentication event. Cryptographic identifiers, signed transcripts, data-integrity proofs, attestation evidence, and portable credentials exist only to bind, preserve, and transport records of that physical event for parties who were not present. **Physical channel first; cryptographic carrier second.***

Equivalently: in the Reality Kernel, cryptography authenticates records of physical events; it does not replace the physical event as the primary object. The default authentication mechanism throughout the system is light in, light out. The reactor’s characteristic response to known inputs under declared operating conditions is what authentication means. DID-style identifiers, verifiable credential proofs, and remote-attestation evidence structures are secondary enabling infrastructure that make the primary physical act legible to parties not present at the original event.

Remark 4 (Markov kernel terminology). *As used herein, “Markov kernel” refers to a regular conditional probability (stochastic kernel) from the input space $\mathcal{S} \times \mathcal{U}^{T+1}$ into the measurable space of convolution-bundle trajectories, and does not imply memorylessness of the raw observations. Where the reactor exhibits persistence (for example phosphor afterglow or thermal relaxation), the extended state X_t is intended to absorb the relevant history so that the kernel satisfies an approximate Markov property over the extended state space.*

In some embodiments, this approximation is formalised via a declared finite effective memory horizon: the assumption that all embodiments satisfy a mixing condition of the form $\nu_{\text{mix}}(u; L) \leq \varepsilon_{\text{mem}}$, where $\nu_{\text{mix}}(u; L)$ is the conditional mutual information between the bundle at time t and the trajectory before time $t - L$, given the state at time $t - L$ and the protocol u ; and L and ε_{mem} are declared embodiment-specific parameters recorded in the protocol digest. The memory-depth bound τ_{mem} (for example the lag at which the empirical autocorrelation of residuals falls below a declared threshold, such as 99 % of total autocorrelation mass) is the practical estimator for L ; the Markov approximation is understood to hold up to τ_{mem} , and a dedicated meter flags bundles in which residual autocorrelation at lag τ_{mem} exceeds the declared threshold.

Channel-with-memory interpretation (non-limiting). *In some embodiments, the finite-effective-memory-horizon analysis is interpreted through the framework of channels with memory and information-spectrum coding theory (Gallager 1968; Verdú–Han 1994), with the declared mixing condition $\nu_{\text{mix}}(u; L) \leq \varepsilon_{\text{mem}}$ acting as the operative ergodicity or mixing assumption for the declared protocol class. Capacity, mutual-information, and*

error-exponent quantities are therefore evaluated over the committed bundle horizon, using block-structured estimators and confidence bounds that account for the declared memory depth τ_{mem} , with the estimator family, block length, and confidence level recorded in the protocol digest.

5 No claim of exact finite-dimensional Markovianity is made. Reactor media with power-law relaxation (such as some photorefractive crystals and biological tissue with multi-scale thermal memory) may have no finite-dimensional Markovian embedding; for such media, the operative claim is that $\nu_{\text{mix}}(u; L) \leq \varepsilon_{\text{mem}}$ for a declared L and ε_{mem} , not that the embedding is exact. Slow reactor drifts (phosphor burn-in, thermal history, charge-trap accumulation)
10 that exceed τ_{mem} are explicitly acknowledged as potential sources of non-Markovian behaviour, and their effect on verification and sensing claims is bounded by the meter envelope.

Definition 2 (Reality Kernel module). A Reality Kernel module is a physical apparatus that realises a Reality Kernel, including at least one emitter, at least one detector, one or more optical or physical paths coupling emission to measurement, a controller that executes
15 a control protocol, and optionally additional components. The term “Reality Kernel” refers to the induced stochastic mapping, while “Reality Kernel module” refers to hardware.

Definition 3 (Scene tuple). A scene is represented as

$$S = (S^{\text{scene}}, S_{\theta}^{\text{reactor}}),$$

where S^{scene} denotes a physical scene being interrogated, including, without limitation, (i) an external environment directly coupled to the module, (ii) an indirect or proxy scene surface
20 such as an internal screen or other surface onto which an external environment is imaged (for example camera-obscure or relay-imaging coupling) and/or onto which recorded or synthetic imagery is projected or displayed, (iii) a continuous abstract or computational process whose state trajectory constitutes the scene dynamics (as in PoliePuter mode; physical reality mediates the coupling in all cases), or (iv) a null or termination state $S^{\text{scene}} = S_{\text{term}}$
25 in which the scene path is shuttered, beam-dumped, or otherwise bypassed (the degenerate open-loop sub-case of PoliePuter mode). $S_{\theta}^{\text{reactor}}$ denotes an engineered internal subsystem, when present, that contributes dynamics under configuration θ .

Definition 4 (Control protocol and per-step control). A control protocol is a sequence $U_{0:T} = \{u(t)\}_{t=0}^T$. A per-step control may be written as

$$u(t) = (\mathbf{s}(t), \mathbf{e}(t), g_{\text{det}}(t)),$$

30 where $\mathbf{s}(t)$ denotes a scan coordinate or address, $\mathbf{e}(t)$ denotes an emission envelope or pattern parameter (optionally including wavelength band, polarisation, phase, or other emission settings), and $g_{\text{det}}(t)$ denotes detector configuration such as exposure, gain, or gating.

Definition 5 (Scan law). A scan law is a mapping $\gamma(t; \eta)$ from time (or discrete index) to scan coordinates, with parameters η (for example frequencies, amplitudes, phases, envelopes,

or basis coefficients). The scan law may be fixed, selected from a library, or adapted on-line. Scan laws include, without limitation, raster, Lissajous, spiral, stepped, or arbitrary waveforms.

Definition 6 (Convolution bundle). For a run under $(S, U_{0:T})$, the convolution bundle is the time-indexed sequence

$$C_{0:T} = (c_0, \dots, c_T),$$

where each sample c_t is an emission–observation pair

$$c_t = (u(t), \mathbf{y}_t),$$

and $\mathbf{y}_t$ is an observation vector comprising one or more measured responses from one or more detector ports (optical, acoustic, RF, auxiliary sensors, and optional quantum-sensitive observables). In some embodiments, the observation vector $\mathbf{y}_t$ is a multi-tap record covering not only endpoint emission and detection but also intermediate states of the cycle that are physically tappable, including, without limitation, scene-side emitter command, scene-side detector response, the output of any forward low-latency coupling between the scene side and reactor side (per Section 4.2), reactor-side emitter command, reactor-side detector response, and intermediate states of any conditioning chain on the $A(t)$ feedback path. In further embodiments where the reactor element exposes intermediate physical layers — for example a multi-pass reactor stack, a layered non-linear medium, or an intermediate phosphor state in a CRT/phosphor reactor — the observation vector additionally includes one or more taps from those intermediate layers, where physically tappable. The convolution bundle is therefore not limited to endpoint emission–observation pairs; it is, in some embodiments, a multi-tap trace of the cycle whose tap set is declared and committed to the protocol digest.

The name reflects the data structure’s characteristic property: at each time step, what the device emitted and what the device observed are bundled inseparably—contributions from the device and the scene are typically difficult to disentangle because the observation $\mathbf{y}_t$ is the physical convolution of the emission $u(t)$ with the scene’s response. The convolution bundle is not a recording of the scene, nor a recording of the device; it is the record of their joint interaction through the physical channel.

Definition 7 (Reactor latent). A reactor latent $Z_\theta(t)$ is a modelling state adapted to the filtration generated by $\{c_\tau\}_{\tau \leq t}$ and may be updated by an operator

$$Z_\theta(t) = \mathcal{R}_\theta(Z_\theta(t-1), c_t).$$

The latent is a modelling or control convenience. A hardware implementation need not expose $Z_\theta(t)$ as a separate physical port.

Definition 8 (Reactor signature). A reactor signature is a summary of a run, for example a scalar

$$r_T = f(C_{0:T}),$$

where f is a fixed functional applied to all or part of a convolution-bundle trajectory. In some verification embodiments, a verifier depends on r_T rather than on full $C_{0:T}$.

Definition 9 (Media state). A media state M_t is a representation of internal memory media or persistent state affected by operation (for example persistence, charge trapping, phase-change, photorefractive response, or other history-dependent media). The media state may be modelled as part of the extended state and contributes to the induced channel $P_\theta(\cdot \mid S, U_{0:T})$.

Definition 10 (Extended state). An extended state may be written as

$$X_t = (S_t^{\text{scene}}, S_t^{\text{reactor}}, M_t, q_t),$$

where q_t denotes fast internal controller or reactor state, and where additional slow degrees of freedom may be incorporated into M_t or into the scene tuple.

Definition 11 (Invariance profile). An invariance profile is a pattern of sensitivities and insensitivities of an observation channel to aspects of a scene, influenced by configuration knobs including, without limitation, integration time, spectral band, coherence, numerical aperture, polarisation, and scan schedule.

Definition 12 (Cross-channel statistics). When multiple channels observe the same scene under distinct invariance profiles, cross-channel statistics include the joint distribution and dependence structure of channel outputs, including pairwise mutual informations, conditional distributions, and higher-order dependencies.

Definition 13 (Yoke depth). The yoke depth $\alpha \in [0, 1]$ parameterises the strength of bidirectional dynamical coupling between the Reality Kernel module and its scene. For example, at $\alpha = 0$, the device operates in static mode (scene treated as having fixed properties); at $\alpha = 1$, the device commands maximum coupling depth (the resulting joint dynamics are empirically assessed for coupling quality via meters; whether the run satisfies Yoked operation is determined by the CLE/TE co-condition recorded in the protocol digest, not by α alone). The yoke depth is recorded in the protocol digest.

Definition 14 (Protocol digest). A protocol digest is a compact record of the control configuration actually used during a run or window, including, without limitation, the scan law parameters, emission schedule, detector settings, gain and alignment values, yoke depth, regime weights, and any conditioning signals (for example prompt digests or model identifiers), and any derived-meter estimation parameters (for example transfer entropy estimator family, window length, embedding dimension, and lag selection). In preferred embodiments, the protocol digest is committed cryptographically (for example by inclusion in the hash-chain state χ_t or by binding into a Merkle commitment) so that the protocol used can be verified after the fact and any discrepancy between declared and actual protocol is detectable.

Definition 15 (Meter vector and meter envelope). A meter vector $\mathbf{m}(t)$ is a tuple of scalar quantities computed from the convolution bundle and protocol state at step or window t , measuring properties such as alignment, gain consistency, signal level, verisimilitude, empirical

hardness, calibration drift, and timing integrity. A meter envelope is a set of admissibility bounds on the meter vector (for example per-component intervals or a convex acceptance region) that define the admissibility regime—the region of meter space within which the system is considered to be operating correctly. (The admissibility regime is distinct from the objective regime—Truth Beam, Limager, or Reality Transform—which specifies the objective family applied to the convolution bundle. A single objective regime may have multiple admissibility regimes, and vice versa.) A run or window is considered within its meter envelope when $\mathbf{m}(t) \in \mathcal{M}_{\text{accept}}$ for all steps in the window. Meter envelopes are declared in protocol digests and bound into commitments.

In some embodiments, the meter vector serves as a declared-fidelity summary of the full convolution bundle: it approximates the informational content of the bundle about specified quantities (regime classification, hardness index, Yoked declaration) to within a declared fidelity η_{meter} , empirically verified on the anchor embodiment. This is not formal statistical sufficiency ($P(\theta \mid \mathbf{m}) = P(\theta \mid C_{0:T})$), which is a stronger condition that is not asserted; the meter is a declared-accuracy compression of the bundle, and any claims stated in terms of the meter vector carry the corresponding fidelity qualification.

Definition 16 (Disclosure atom and evidence window). A disclosure atom is the smallest unit of the convolution bundle and associated metadata that can be independently committed and selectively opened for audit. In some embodiments, an atom corresponds to a contiguous time window; in others, to a structured slice (for example a channel subset or spatial region). An evidence window is one or more disclosure atoms that are grouped for a specific verification, audit, or cross-attestation purpose. Per-atom digests are aggregated into Merkle trees so that a verifier can request openings of selected atoms without requiring disclosure of the full run.

Definition 17 (Truth Beam (verification regime)). Truth Beam is the operating regime in which the Reality Kernel is configured to maximise the empirical difficulty of forging or simulating the convolution bundle, so that the record serves as physically grounded evidence of a genuine device–scene interaction. The optimised quantity is, in some embodiments, Fisher information $I(\theta)$ about the device’s configuration parameters, making the device’s physical identity detectable in the bundle. For multi-parameter θ , the optimised quantity may be a scalar functional of the Fisher information matrix $\mathcal{F}(\theta)$, such as its trace, determinant, or minimum eigenvalue, or a directional Fisher information along a task-relevant direction. Detailed treatment is in Section 8.2.

Definition 18 (Limager (perception and sensing regime)). Limager is the operating regime in which the Reality Kernel is configured to maximise information gain or task-relevant reconstruction quality from active probing of a scene. The device steers its emission protocol to extract scene information efficiently, subject to meter-envelope constraints. Detailed treatment is in Section 8.3.

Definition 19 (Reality Transform (controllable rendering regime)). Reality Transform is the operating regime in which the Reality Kernel steers the visual or physical appearance of the

scene while remaining constrained by the physical channel, so that the resulting convolution bundle records both the intended effect and the scene’s actual response. Detailed treatment is in Section 8.4.

Definition 20 (Yoked operation (dynamic coupling modifier)). Yoked operation is a modifier applicable to any of the three base regimes (Truth Beam, Limager, Reality Transform) in which the feedback coupling between the device and the scene is tuned so that both are driven toward a coupled dynamical state. The yoke depth $\alpha \in [0, 1]$ parameterises coupling strength: $\alpha = 0$ is static (open-loop), $\alpha = 1$ is the maximum commanded coupling depth (with the CLE/TE co-condition determining whether a particular run is classified as Yoked operation). Yoked operation is not a fourth regime but a continuous modifier that enriches any base regime with dynamic coupling structure. Detailed treatment is in Section 8.5.

6.2 Fleet model and proof-of-discrepancy terms

Definition 21 (Device configuration representation). A device configuration representation $\mathbf{d}_i$ is a learned vector or structured object encoding device-specific properties (manufacturing variability, material composition, calibration state) of device i , inferred from challenge-response data. A predictive model $\mathcal{Q}(X, \mathbf{d}_i)$ accepts the device configuration representation alongside a challenge configuration X and produces a predicted response. See Section 13.1.

Definition 22 (Proof-of-discrepancy). A proof-of-discrepancy is a reproducible physical method—a challenge configuration and associated operating conditions—together with evidence that the predictive model’s prediction diverges from measured physical response, confirmed by independent physical re-execution on distinct devices. The protocol by which such proofs are submitted, verified, and incorporated into the predictive model is described in Section 13.3.

Definition 23 (Practical learnability-exclusion condition). A device satisfies a practical learnability-exclusion condition when no known feasible learning procedure achieves prediction accuracy above the verification threshold given bounded observations from the device. This condition provides the physical basis for the irreducible residual and the security margin against model-based cloning. See Section 13.6.

Definition 24 (PoliePuter mode). PoliePuter mode refers to operation of a Reality Kernel module in which the scene partner is a continuous abstract or computational process. In the coupled case, the reactor’s response at each step is returned to the computational process as an input before its next state update, closing a bidirectional loop; the convolution-bundle then constitutes a physically attested record of the joint trajectory. The degree of coupling is empirically characterised by the convolution-bundle under the declared Yoked conditions; no coupling class is pre-declared. In the open-loop (degenerate) case, the scene path is shuttered or the return channel is absent ($S^{\text{scene}} = S_{\text{term}}$), and the reactor operates as a compute substrate without active feedback. In all cases, the reactor’s nonlinear dynamics, media states,

and feedback loops are exploited for computation, generation, or optimisation tasks using the convolution-bundle logging and meter infrastructure of the Reality Kernel.

Definition 25 (PolieBot). *A PolieBot is a non-limiting example of a mobile or deployable agent platform that integrates one or more Reality Kernel modules with locomotion, communication, and decision-making subsystems. PolieBots interact with scenes and with other agents through their Reality Kernel channels and are subject to the verification, evidence-commitment, and meter-bounded protocols described herein. Optional multi-agent coordination extensions are described in the related Filing 2 application. See Section 16.*

Definition 26 (Run seed and opening seed). *A run seed s_{run} is a value derived from the protocol digest and a pre-committed nonce before a run begins; it determines the emission schedule, scan law, and any protocol randomisation for that run, and is logged in the protocol digest. An opening seed s_{open} is a value supplied by a verifier or independent beacon after the run commitment has been published; it determines which atoms or windows are to be selectively disclosed. In embodiments using the two-seed commitment structure (Section 14.4), the run seed and opening seed are generated independently, so that the device is not expected to predict which atoms will be opened at the time the run is executed under the declared randomness and timing assumptions.*

6.3 Reactors and ponderable cells

In this specification, a *reactor* is a subsystem whose dynamics are dominated by electromagnetic field interactions and that participates in a Reality Kernel. A reactor receives electromagnetic control inputs, interacts with a scene, and produces electromagnetic outputs for one or more sensors. Non-limiting examples include optical elements (diffusers, nonlinear films, scattering plates, spatial light modulators), CRT phosphor screens, RF and microwave resonators and waveguides, magnetic coil assemblies driven by swept fields, and free-space or fibre-coupled optical cavities.

A *Ponderable Cell* is any subsystem whose dominant internal degrees of freedom are governed by massive-carrier physics—mechanical displacement, acoustic vibration, fluid flow, thermal diffusion, chemical concentration, biological metabolism, or hybrid combinations—rather than by electromagnetic field dynamics. The term *ponderable* follows classical usage (ponderable matter = matter possessing weight and inertia). Ponderable Cells may couple to one or more reactors via transducers (for example piezoelectric, magnetostrictive, electro-thermal, or opto-mechanical interfaces) and thereby modulate behaviour of the associated channel.

The distinction is functional, not ontological: a given physical component may contain both electromagnetic-dominant and massive-carrier-dominant degrees of freedom. For example, a ferrite-loaded coil assembly is classified as a reactor because the dynamics exploited for Reality Kernel operation are electromagnetic (permeability, hysteresis, field coupling), even though the core material has rest mass. Both reactors and Ponderable Cells are

treated as parts of the same parameterised channel $P_\theta(\cdot \mid S, U_{0:T})$. The distinction groups electromagnetic-field-dominant embodiments separately from massive-carrier-dominant embodiments for clarity.

6.4 Markov-model interpretations (non-limiting)

- 5 In some embodiments, the Reality Kernel viewpoint admits several standard Markov-model lenses, depending on what is treated as state, what is treated as observation, and whether control is present. In a non-limiting interpretation, the same underlying physical loop can be viewed as (i) a sampler whose long-run statistics approximate a target distribution (MCMC-style), (ii) a latent-state time series in which internal physical state is hidden and the convo-
 10 lution bundle is an emission (HMM/state-space style), (iii) a metastable switching process in which dwell-time and transition statistics are signatures (biophysics-style channel switching), or (iv) a controlled process in which emissions are actions and meter-bounded evidence quality shapes objectives (MDP/POMDP-style).

6.5 Adversary models and empirical hardness (non-limiting)

- 15 In some verification embodiments, “hardness” is evaluated empirically relative to specified adversary families and budgets. Non-limiting adversary families are parameterised by three dimensions: (a) *query access type*: offline CRP collection, online adaptive querying, or physical access to the device; (b) *information access*: black-box (input-output only), grey-box (with knowledge of device architecture and material composition), or white-box (with full
 20 model $\mathcal{Q}$ and representation $\mathbf{d}_i$ access); and (c) *resource budget*: computational scale k , physical scale m , query count Q , and time horizon. Representative non-limiting instances include: (i) a purely digital simulator trained on logged traces (black-box, offline, bounded compute); (ii) an adversary with bounded physical co-illumination capability (physical access, grey-box); (iii) an adversary who later obtains control of a previously trusted device
 25 (white-box, physical access); and (iv) an adversary with knowledge of device architecture and material composition who builds a physics-informed surrogate model (grey-box, offline, potentially unbounded compute). For each declared embodiment, the relevant adversary instance and its parameter bounds are recorded in the protocol digest. The learnability-exclusion condition (Section 13.6) is evaluated separately against each declared adversary
 30 instance, including instance (iv), by verifying that prediction accuracy plateaus on held-out challenges even when the adversary’s model is seeded with declared material composition and geometry parameters.

6.6 Substrate properties supporting cross-layer coupling and governance evidence

- 35 **Definition 27** (Optical anchor). *An optical anchor refers to a cross-layer continuous coupling mechanism by which two or more coupled Reality Kernel layers are bound to one another*

in apparatus-level real time, such that an attacker compromising one layer cannot substitute its contribution to a recursive cascade without simultaneously satisfying the anchor's declared latency, modulation, and challenge-response constraints across all coupled layers. The optical anchor comprises in some embodiments a declared optical band, a declared modulation or framing scheme carrying challenge-response content, declared emitter and detector elements, a declared timing source, declared allowed path lengths, declared cross-channel isolation, declared failure modes, and an anomaly-detection response ladder, each committed to the protocol digest as anchor-layer parameters. The optical anchor is a security property complementary to per-layer hardness: per-layer hardness bounds local recoverability of state from a single layer, while the optical anchor binds layers continuously to one another. The full apparatus-level disclosure appears in Section 7.4.5.

Definition 28 (Optical long-term memory). *An optical long-term memory refers to a storage subsystem in which committed evidence records are written into a physical optical substrate configured for durable, individually-retrievable storage, where the substrate has a write-once-read-many (WORM) physical property such that, once a record is written, the underlying physical state of the substrate at the written location does not support in-place modification or overwriting through the apparatus's normal write pathway. The WORM property is the apparatus-level substrate property on which the integrity of durable governance evidence relies, and is committed to the protocol digest as part of the storage-layer parameter family. The optical long-term memory complements channelised optical memory of the kind disclosed in Section 10.5 (which provides controller-managed short- and medium-term storage) by providing a substrate appropriate for the durable retention of selected committed records, including in some embodiments hard-floor violation records, rescue-derived ghost records, and attestation tokens. The full apparatus-level disclosure appears in Section 10.6.*

Definition 29 (Governance-partition isolation). *Governance-partition isolation refers to a substrate-level property of a Reality Kernel apparatus by which the substrate that produces evaluation meters and other governance-side signals operates in a partition of the apparatus that is physically and logically isolated from the agent's behaviour-generation pathways, such that the agent cannot read, write, or directly influence the partition's operation through its normal operating pathways. Governance-partition isolation is a structural property of the apparatus rather than a runtime monitoring discipline: isolation forecloses agent influence on the governance partition as a property of the apparatus's construction, rather than relying on detection of attempted influence after the fact. The full apparatus-level disclosure appears at Section 12.1.*

6.7 Symbols

Symbol	Meaning
S	Scene tuple $S = (S^{\text{scene}}, S_{\theta}^{\text{reactor}})$
$U_{0:T}$	Control protocol $\{u(t)\}_{t=0}^T$
$u(t)$	Per-step control, including scan coordinate $\mathbf{s}(t)$, emission parameter $\mathbf{e}(t)$, and detector setting $g_{\text{det}}(t)$
$\gamma(t; \eta)$	Scan law parameterised by η
$C_{0:T}$	Convolution bundle (full run)
c_t	Per-step emission–observation pair $(u(t), \mathbf{y}_t)$
$\mathbf{y}_t$	Observation vector
$Z_{\theta}(t)$	Reactor latent
r_T	Reactor signature summarising a run
M_t	Media or memory state
q_t	Fast internal controller or reactor state
X_t	Extended state
$\mathbf{m}(t)$	Meter vector
α	Yoke depth $\in [0, 1]$
$\mathbf{e}(t)$	Emission envelope or pattern parameter
χ_t	One-way chain state (cryptographic)
$\mathbf{P}_{\theta}(\cdot \mid S, U_{0:T})$	Conditional law of the convolution bundle
$\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$	Configuration parameters: hardware (θ_{hw}) and optional learned (θ_{sw})
$\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S}$	Transfer entropy from scene to reactor and vice versa
CLE	Conditional Lyapunov exponent
$\sigma_{ij}(t)$	Trust score from agent i toward j (see Section on agent layer; distinguished from verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$ and from conventional statistics-notation σ for standard deviation)
$\nu_{\text{idemp}}, \nu_{\text{contr}}, \dots$	Algebraic deviation metrics (RK algebra)
$K_2 \triangleright K_1$	Containment (nesting) composition of RK modules
$\Pi^{(i)}$	Protocol digest of agent or instance i (per-agent digest)
Π_{dig}	Per-run protocol digest: the governing protocol-template digest of the current run, committed before the capture window opens; binds the run to the declared configuration, scan law, exposure schedule, verifier cadence, meter envelope, and any selective-opening policy. Distinct from $\Pi^{(i)}$, which is a per-agent digest
$K_{\text{spot}}(\cdot, \cdot)$	Spot point-spread function (CRT embodiment only)
$\mathcal{I}_{t_0:t_1}$	Inner-trace summary (inner convolution-bundle tuple)
λ_{TE}	Transfer-entropy penalty weight (in combined regime/Yoked objectives); units are protocol-digest declared to render all objective terms commensurate

continued on next page

Symbol	Meaning
λ_{CLE}	Conditional Lyapunov exponent penalty weight (Yoked objective); units are protocol-digest declared to render all objective terms commensurate
ρ_{AR}	Autoregressive coefficient (drift/persistence modelling)
μ_{mix}	Trust weight mixing rate (fleet pairing dynamics)
r_{rep}	Resource replenishment rate (fleet game dynamics)
<i>Fleet, PoD, and substrate-theory symbols (non-limiting)</i>	
$\mathcal{Q}(X, \mathbf{d}_i)$	Predictive model mapping challenge X and device configuration representation $\mathbf{d}_i$ to predicted response
$\mathbf{d}_i$	Device configuration representation for device i
$\mathbf{d}_0$	Population-level device configuration representation
$e_i(X), \bar{e}_i(X)$	Per-device residual and noise-reduced residual
$\delta_{\text{sys}}(X)$	Shared model error (fleet-wide discrepancy)
$\delta_{\text{dev}}(X, i)$	Device-specific captured variation (device signature)
$\delta_{\text{irr}}(X, i)$	Irreducible residual (security margin)
Δ_2	Two-simplex of regime weights ($\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}}$)
<i>Convention: $\lambda_{\text{subscript}}$ denotes a non-negative scalar weight coefficient throughout; distinguished from optical wavelength λ (unsubscripted, standard optics) by subscript presence.</i>	
$I(\theta)$	Scalar Fisher information about θ (one-dimensional or marginal)
$\mathcal{F}(\theta)$	Fisher information matrix (used in natural gradient methods (ascent or descent depending on objective sign); pseudoinverse where singular)
$\mathcal{F}_Q(\theta)$	Quantum Fisher information (Bures metric; see quantum coupling section)
$I(X; Y), I(X; Y Z)$	Mutual information (conditional mutual information) between random variables; distinguished from scalar Fisher information $I(\theta)$ by argument structure
n	Non-limiting count of reliably extractable bits per response (security parameter context)
n_{sec}	Effective security parameter (scales with transcript information content; see cryptographic-primitive substrate)
$k_{\text{dig}}^*(\theta; \varepsilon)$	Digital hardness index: minimal model capacity to drive balanced classification error below ε (see Security Theory)
$m_{\text{ana}}^*(\theta; \varepsilon, r)$	Analogue hardness index: minimal physical emulator scale to drive distinguishability advantage below ε against a verifier/distinguisher class of scale r (see Security Theory)
$\text{err}_k^*(\theta)$	Best achievable balanced classification error at model scale k

continued on next page

Symbol	Meaning
$\text{Adv}_{m,r}^{\text{ana}}(\theta)$	Analogue distinguishability advantage at emulator scale m against a verifier/distinguisher class of scale r
ε	Hardness threshold (declared; conditions all hardness index statements)
$\phi \in \Phi$	Style parameter: formal input to the Reality Transform objective specifying the target behaviour class (see Section 8.4.4)
$\varphi(t)$	Scan-law angular coordinate in helical/cylindrical scan embodiments (distinct from RT style parameter ϕ)
$\Gamma_{\text{addr}}(t; \eta)$	CRT address-space map from timebase and scan parameters η to internal address coordinate (ρ, φ)
Σ_{sum}	Summary map compressing a run record $(C_{0:T}, \Pi, \mathbf{m})$ to a compact summary vector $\mathbf{s}$
agg	Declared and pre-committed aggregation functional (e.g. median, trimmed mean); used in δ_{sys} and $T(X^*)$; recorded in protocol digest
$\mathcal{V}$	Set of participating verifiers in a PoD confirmation round
τ_{device}	Per-device exceedance threshold for optional quorum rule in PoD fleet confirmation
$\text{CLE}_{\text{target}}$	Target conditional Lyapunov exponent for Yoked operation (declared; negative, close to zero)
ε_{CLE}	CLE convergence window half-width (declared protocol parameter for Yoked convergence criterion)
τ_{TE}	TE balance threshold (declared; Yoked convergence co-condition)
τ_{mag}	Minimum TE magnitude threshold (declared; Yoked convergence co-condition; ensures balance criterion cannot be trivially satisfied when both TE values are near zero)
N	Number of consecutive convergence windows required to declare Yoked operation (declared protocol parameter)
f_{mod}	Modulation envelope frequency in lock-in / sweeping-reactor embodiments (the Arnold tongue axis frequency; distinguished from optical carrier frequency, AOD/EOM RF frequency, and scan repetition rate). In sweeping-reactor embodiments, f_{mod} denotes the modulation frequency of the swept physical parameter as the same envelope quantity applied to the reactor parameter sweep
Δf	Frequency detuning $f_{\text{mod}} - f_{\text{scene}}$ (Arnold tongue horizontal axis)
$\phi_{\text{ref}}(t)$	Scene-derived reference phase signal (lock-in reference for scene-phased full-parameter sweeping; see Section 10.28)

continued on next page

Symbol	Meaning
$\delta\theta(t)$	Parameter modulation perturbation vector (scene-phased sweep; small relative to θ_0)
$\delta\theta_k, \psi_k$	Declared amplitude and phase offset of the k -th parameter component in scene-phased modulation
S_{IP}	In-phase (real) demodulated lock-in output, proportional to $\nabla_{\theta}T _{\theta_0} \cdot \delta\theta$
$J_{\text{train}}(\theta, \pi)$	Training-phase proxy objective for Yoked TE-gradient phase (component of J_{Yoked})
$T(X^*)$	Fleet-level PoD confirmation statistic: robust aggregate of per-verifier residuals at submitted challenge X^*
τ_{confirm}	Pre-registered confirmation threshold for the PoD fleet statistic; committed in protocol digest
$\mathcal{K}$	Index set for composite agent evaluation metrics (distinct from meter envelope $\mathcal{M}_{\text{accept}}$)
tc	Tool-call specification (canonical 5-tuple (<code>name</code> , <code>args-schema</code> , <code>provider-policy</code> , <code>canonicalisation-rule</code> , <code>derivation-rule</code>) committed to the protocol digest before a capture window opens; canonical serialization is the UTF-8 byte sequence of the named-field JSON encoding with sorted keys; selective opening of tc discloses any subset of its fields under the declared opening policy. Distinct from trust score $\sigma_{ij}(t)$ and from any conventional statistics-notation σ)

Declared optical-radiation safety classification (non-limiting). In some embodiments that use lasers or high-intensity optical emitters, the apparatus includes power monitors, duty-cycle clamps, scan-failure detection, beam-stop or enclosure controls, service-mode controls, and hardware or firmware interlocks configured to maintain accessible emission

5 within a declared optical-radiation safety classification under the declared operating conditions. In some embodiments, the declared classification is IEC 60825-1 Class 1, Class 1M, or another applicable class; in some embodiments, ANSI Z136.1 safe-use controls are declared in addition to, or in place of, an IEC product- classification statement. The protocol digest records the applicable standard and edition, hazard class, wavelength range, accessible-

10 emission-limit or maximum-permissible-exposure assumptions, beam divergence, aperture assumptions, exposure time, interlock state, service-mode exceptions, and any downgrade or shutdown event. The protocol record is an engineering and audit record; formal regulatory certification, if any, is performed under the applicable regulatory process.

7 Detailed Description

Unless otherwise stated or required by context, embodiments described in this Detailed Description are non-limiting and illustrative. Alternative configurations, parameter ranges, materials, and architectures consistent with the definitions herein are within the scope of the disclosure.

Where this Detailed Description references companion-filing content, those references are optional and non-essential: no embodiment in this Detailed Description requires the companion filing for enablement, and all references to the companion filing may be disregarded without affecting the enablement of any embodiment described outside a companion filing.

7.1 Networked Closed-Loop Reactor Witness Mesh

7.1.1 Overview and primary deployment

The principal deployment of the apparatus and methods of this disclosure is a plurality of Reality Kernel modules operating as a *witness mesh*: multiple modules, each comprising a fast closed loop in causal feedback with a portion of physical reality observable to that module, mutually coupled through continuous, low-latency analogue inter-module signal paths. Single-module embodiments disclosed elsewhere in this specification — including the 1D anchor of Section 7.2, the 2D anchor of Section 10.2, and the various reactor variants of Section 10.3 and subsequent reactor sections — are non-limiting building blocks of the witness-mesh architecture and may also be used as standalone instantiations.

The witness-mesh architecture is the primary intended deployment because the security and provenance claims of greatest practical value arise from the joint physical reproduction problem across multiple modules bound to their respective observations of physical reality, rather than from any single module's per-device hardness considered in isolation.

Architectural keystone — minimum-latency mutual connection. The architectural keystone of the witness mesh is the *minimum-latency mutual connection* of the constituent modules' fast loops, bounded by the declared latency budget of Section 7.1.3: each module's fast-loop output is delivered to its peer modules' coupling-network inputs over a continuous-analogue, frame-free, buffer-free signal path, with end-to-end inter-module latency bounded by the propagation delay of the carrier and the hardware latency of the analogue front ends, and with no contribution from packet framing, error-correction stages, queues, buffers, operating-system-mediated transport, or any other digital re-representation stage. The latency budget so defined is a load-bearing component of the security primitive of Section 7.1.4: any adversarial intervention in the inter-module signal path must complete within the same latency budget, and the absence of intermediate digital representations removes framing-level insertion targets and raises the cost of analogue-domain insertion under the declared latency budget and verifier timing fidelity. The latency budget operates in conjunction with the witnessing requirement of Section 7.1.5 and the joint-trace verifier of Section 7.1.6; the

joint security claim derives from all three components, not from the latency budget alone.

Construction of “minimum-latency mutual connection” (non-limiting). For purposes of this section, a *minimum-latency mutual connection* between two modules M_i and M_j means a live inter-module signal path whose allowed end-to-end latency is the declared propagation delay of the identified signal path plus characterised analogue-front-end latency and declared measurement tolerance, and that does not store a complete verifier-openable sample window or protocol frame in the live path. The term is a structural element relative to the declared latency budget of Section 7.1.3; it does not require zero physical delay or an absolute minimum over all possible routes, and is not a term of degree relative to the speed of light or any other physical constant. The declared latency budget, the declared verifier timing fidelity σ_τ^{ver} , and the declared measurement tolerance are committed to the protocol digest at each deployment version per Section 7.1.10.

Construction of “verifier-openable sample window” (non-limiting). For purposes of this section, a *verifier-openable sample window* means the smallest committed atom or atom sequence that the verifier can select for opening under the two-seed selective-opening protocol of Section 14.4. Storage of a complete such atom or atom sequence in the live inter-module path constitutes buffering for purposes of claims reciting the no-buffering limitation of this section. The atom granularity is committed to the protocol digest as part of the verifier specification and is, in some preferred embodiments, at the per-sample level of the fast-loop sample rate $\{f_i\}$; in other embodiments, the atom is a contiguous sequence of samples whose duration is committed to the protocol digest and whose verifier-side timing tolerance is calibrated against σ_τ^{ver} . Sub-atom-duration analogue storage — including without limitation continuous-time analogue delay lines, sample-and-hold elements within the declared analogue-front-end latency, and event-driven analogue memory whose retention time is shorter than the atom duration — does not, on its own, constitute buffering for purposes of the no-buffering limitation, provided the storage is declared and committed to the protocol digest as part of the inter-module path characterisation.

7.1.2 Architectural primitive

A witness mesh comprises a plurality $\{M_i\}_{i \in I}$ of Reality Kernel modules, indexed by a finite set I with $|I| \geq 2$, each module M_i comprising:

1. an emitter E_i configured to drive a portion of physical reality Σ_i observable to module M_i ;
2. a reactor element comprising a physical medium with a transfer function dependent on the medium’s microstructure and instantaneous state;
3. a detector D_i configured to record the response of Σ_i and the reactor to the emission of E_i on a continuous or near-continuous timebase, where *near-continuous* excludes ordinary frame-buffered capture in which a complete verifier-openable sample window

or protocol frame is accumulated before inter-module coupling, and includes sampled analogue or sampled mixed-signal implementations whose sample period and conversion latency are declared and committed to the protocol digest as part of the latency budget of Section 7.1.3;

4. a fast closed loop in causal feedback with Σ_i , in which the detector output, after a declared deterministic live conditioning chain, drives the emitter modulation at the same timebase, with no per-period wait, no frame buffer, and no intermediate digital decoding stage in the live coupling path; and
5. one or more inter-module coupling ports configured to broadcast a derived signal of the module's fast-loop state to one or more peer modules and to receive corresponding signals from peer modules, over the inter-module signal path of Section 7.1.2.

The fast loop of each module M_i operates at a declared sample or modulation rate f_i and is coupled to its physical reality Σ_i at a declared coupling latency τ_i^{loop} , bounded by the propagation delay of light or other carrier through the optical or signal path of the module and the hardware latency of the live conditioning chain. The fast loop contains no runtime-adaptive element and no runtime-learned element: its behaviour is fully determined by the trained committed parameters of the deployment version and by the instantaneous physical state of Σ_i and the reactor.

Inter-module coupling — continuous-analogue signal path (load-bearing). The inter-module coupling is mediated by a *configured deterministic inter-module transfer function*, which is a frozen, deterministic transfer function from peer fast-loop signals to per-module modulation input, optionally trained, and which is committed to the protocol digest Π_{dig} as one of the trainable parameter families of the deployment version. (This function is referred to elsewhere in this section as the “trained inter-module coupling network”, the “inter-module coupling network”, or simply the “coupling network”; the terms are interchangeable in this disclosure and apparatus claims may recite either form. The “configured deterministic” formulation is preferred for apparatus claims that should not depend on a training-process limitation.)

Inter-module coupling as versioned parameter family (non-limiting). For purposes of apparatus claims and infringement analysis, the configured deterministic inter-module transfer function is, in some preferred embodiments, an element of a *versioned parameter family*: a deployment version comprises (a) a frozen parameter family identifier, (b) a frozen modality identifier selecting one or more of the modulation modalities of Section 7.1.2, (c) frozen structural parameters identifying the coupling graph $\mathcal{G}$ and per-edge propagation/hardware latency declarations, and (d) frozen trained or hand-designed parameter values realising the deterministic transfer function from peer fast-loop signals to per-module modulation input. The frozen parameter family identifier and the frozen parameter values are committed to the protocol digest Π_{dig} at each deployment version. The element is claimable as the versioned parameter family implementing the configured deterministic

transfer function, without dependence on the training process, on external declared quantities such as the empirical hardness estimate, or on whether a corresponding deployment is operational at the time of infringement analysis. The inter-module signal path between modules and through the coupling network is *continuous-analogue* and *frame-free*: the path contains no analogue-to-digital conversion, no packet framing, no forward-error-correction stage, no buffer or queue, no digital protocol decoding, and no operating-system-mediated transport, between a peer module's fast-loop output and the receiving module's modulation input mixer. Each module's instantaneous emission is therefore a direct analogue function of (a) its own immediately-prior detector output, (b) its peers' immediately-prior fast-loop outputs as received over the declared inter-module signal paths, and (c) the frozen committed coupling network, up to a bounded and declared inter-module latency τ_{ij} .

The continuous-analogue, frame-free property of the inter-module coupling path is load-bearing for the security primitive of Section 7.1.4 as a component of the combined architectural primitive; the reasons are disclosed in Section 7.1.3.

Coupling network modulation modality (non-limiting). The trained inter-module coupling network may modulate the receiving module's input via any one or more of the following modalities, each of which is a non-limiting embodiment of the inter-module coupling of this disclosure:

- *Amplitude modulation*, in which the coupling network output drives the amplitude of the receiving module's input mixer. Amplitude modulation is a preferred canonical embodiment, consistent with the live amplitude drive $A(t)$ of the 2D anchor of Section 10.2 and the per-sample reactor-detector gain G_{det} of the 1D anchor of Section 7.2 transposed to the inter-module case.
- *Phase modulation*, in which the coupling network output modulates the phase of an optical, RF, or other carrier presented at the receiving module's input mixer.
- *Frequency modulation*, in which the coupling network output modulates the carrier frequency.
- *Polarization modulation*, in which the coupling network output modulates the polarization state of the carrier.
- *Wavelength modulation*, in which the coupling network output selects or modulates the carrier wavelength, including embodiments using wavelength-division multiplexing of multiple peer signals onto a shared inter-module carrier.
- *Spatial-mode modulation*, in which the coupling network output modulates the spatial mode of the carrier, including embodiments using mode-division multiplexing.
- *Mixed modality coupling*, combining two or more of the above modalities on the same or distinct carriers, with each modality contributing distinct physical and timing characteristics committed to the protocol digest.

The selected modulation modality or modalities, together with the coupling network's structural parameters, are committed to Π_{dig} at each declared deployment version. The witness-mesh primitive is not limited to amplitude modulation; the choice of modality is an

embodiment-level decision driven by the available physical layer, the carrier modality of the inter-module signal path, and the verifier’s measurement modality.

Representative enabled species per modality (non-limiting). For written-description support of the modulation-modality genus, representative non-limiting enabled species include:

- *Amplitude-modulation species:* a transimpedance- amplified photodetector at the broadcasting module produces a voltage proportional to fast-loop intensity; this voltage drives, through the configured deterministic inter-module transfer function, the analogue input to a laser-diode bias or electro-optic intensity modulator at the receiving module; the receiving module’s input mixer integrates the resulting optical intensity over its declared input bandwidth.
- *Phase-modulation species:* the broadcasting module’s fast-loop output drives an electro-optic phase modulator impressing phase modulation on a coherent carrier (free-space or fibre-optic) presented at the receiving module’s coupling-network input mixer; the receiving module employs phase-resolving detection (heterodyne, homodyne, or interferometric) calibrated to the declared phase reference.
- *Frequency-modulation species:* the broadcasting module’s fast-loop output drives a voltage-controlled oscillator or current-driven distributed-feedback laser producing frequency modulation of the inter-module carrier; the receiving module employs frequency discrimination through a declared discriminator filter or wavelength-resolving detection.
- *Polarization-modulation species:* the broadcasting module’s fast-loop output drives a polarization controller (electro-optic, liquid-crystal, or rotating wave plate) impressing polarization modulation on the inter-module carrier; the receiving module employs polarization-resolving detection through a polarizing beamsplitter and balanced photodetectors.
- *Wavelength-modulation species:* the broadcasting module’s fast-loop output controls a tunable laser or wavelength-selective filter, producing wavelength modulation of the inter-module carrier; the receiving module employs wavelength-resolving detection through a wavelength-division- multiplexing demultiplexer or grating-coupled detector array.
- *Spatial-mode-modulation species:* the broadcasting module’s fast-loop output drives a spatial light modulator or mode-converter array producing spatial-mode modulation of the inter-module carrier; the receiving module employs mode-resolving detection through a mode-division- multiplexing demultiplexer or spatially-resolved photodetector array.

Each species is a non-limiting enabled embodiment of the corresponding modulation modality. Combinations and variants within each species are within the scope of the witness-mesh primitive provided that the resulting inter-module signal path satisfies the continuous-analogue, frame-free property of this section and the latency budget of Section 7.1.3.

Coupling network functional form (non-limiting). Independent of the selected modulation modality, the trained inter-module coupling network’s mathematical transfer function may be implemented as one or more of:

- a frozen scalar damping coefficient per peer pair;
- a frozen vector of per-peer weights applied to each module’s input mixer;
- a frozen matrix or low-rank matrix mapping peer-broadcast vectors to each module’s modulation input;
- a frozen analogue filter with declared transfer function;
- a frozen small neural network performing nonlinear weighted combination of peer signals, implemented in analogue hardware where consistent with the continuous-analogue property of the inter-module signal path; or
- any other declared deterministic transfer function from peer fast-loop outputs to per-module modulation input that is implementable within the selected analogue or mixed-signal coupling modality of Section 7.1.2, with declared input/output dimensionality, declared bandwidth, declared end-to-end latency within the latency budget of Section 7.1.3, declared parameter values, and declared residual tolerances, all committed to the protocol digest, and without storage of a complete verifier-openable sample window in the live path.

The structure, dimensionality, and parameter values of the coupling network are committed to Π_{dig} and are public under the declared threat model. The commitment provides version binding and anti-malleability of the deployed coupling under the declared digest scheme; the witness-mesh security primitive does not assume secrecy of the coupling network’s parameters or structure.

Coupling graph. The set of inter-module coupling links forms a directed graph $\mathcal{G} = (I, \mathcal{E}, \tau, w)$, where $\mathcal{E} \subseteq I \times I$ is the set of directed coupling edges, $\tau : \mathcal{E} \rightarrow \mathbb{R}_{>0}$ assigns to each edge (i, j) a declared inter-module latency τ_{ij} corresponding to the propagation and hardware-latency budget of the analogue path from M_i to M_j , and w parameterises the contribution of edge (i, j) to the coupling-network input at M_j . The graph $\mathcal{G}$, including its topology and per-edge declared latencies, is committed to Π_{dig} .

Slow-loop and analogue-memory extensions (non-limiting). In some embodiments, each module additionally comprises a slow analogue-memory loop as disclosed in Section 10.5 and Section 10.7; the slow loop is an optional extension and is not the operative mechanism of the witness-mesh security primitive. The operative mechanism is the network of fast loops, each independently closed with reality, mutually constrained via the continuous-analogue inter-module coupling network.

Heterogeneous-module embodiments (non-limiting). In some embodiments, the witness mesh comprises modules of heterogeneous construction — for example, one module implementing the 1D fluorescent-bed anchor of Section 7.2, another the 2D galvo-based

continuous-coupled anchor of Section 10.2, a third the SLM–scattering-medium–camera reactor of Section 10.4, and so on — coupled through a common inter-module coupling network. Heterogeneity of module construction contributes to the joint hardness index by requiring an adversary to model multiple distinct physical reactor classes simultaneously
 5 under the declared meter envelope.

7.1.3 Latency budget and the no-buffering requirement

The witness-mesh security primitive depends on a declared latency budget for inter-module coupling that is bounded by the optical-delay logical clocks of Section 14.2. The budget is composed of:

- 10 1. the propagation delay $\tau_{ij}^{\text{prop}} = d_{ij}/v_{ij}$ of the carrier from M_i to M_j along the declared signal path of length d_{ij} at carrier velocity v_{ij} (for free-space optical, $v_{ij} = c$; for fibre, $v_{ij} = c/n$ at the operating wavelength; for RF or other carriers, the appropriate carrier velocity is declared);
- 15 2. the hardware latency τ_{ij}^{hw} of the transmit and receive analogue front ends, including transducers, conditioning amplifiers, filters, and the analogue coupling-network input stage;
3. zero contribution from packet framing, error-correction stages, or buffer queues, by construction in the preferred embodiment.

The total declared inter-module latency $\tau_{ij} = \tau_{ij}^{\text{prop}} + \tau_{ij}^{\text{hw}}$ is committed to Π_{dig} , and
 20 verifier-side checks of optical-delay logical-clock consistency (Section 14.2) compare τ_{ij} against recorded inter-module signal arrival times under the declared meter envelope.

Verifier timing fidelity (declared parameter). The verifier-side timing fidelity $\sigma_{\tau}^{\text{ver}}$ at which inter-module signal arrival times are checked against the declared τ_{ij} is itself a declared parameter of the deployment, committed to Π_{dig} at each declared deployment version.

25 The witness-mesh security claim of Section 7.1.4 is indexed to $\sigma_{\tau}^{\text{ver}}$: the latency-budget contribution to the joint hardness scales with the ratio of the adversary’s minimum achievable intervention latency to $\sigma_{\tau}^{\text{ver}}$, and deployments with coarser $\sigma_{\tau}^{\text{ver}}$ have correspondingly weaker latency-budget contribution under the declared adversary class. In some embodiments, $\sigma_{\tau}^{\text{ver}}$ is selected based on (i) the carrier-medium propagation tolerance, (ii) the available times-
 30 tamping infrastructure including without limitation White-Rabbit-class, PTP-HA-class, or higher-fidelity time-to-digital converter or streak-camera timing, and (iii) the declared adversary class’s minimum achievable intervention latency for the inter-module signal modality, with margin selected to support the per-version hardness commitment of Section 7.1.10. The relationship between $\sigma_{\tau}^{\text{ver}}$ and the empirical hardness index is established at each declared
 35 re-evaluation cycle.

Why frame-free continuous-analogue coupling matters (non-limiting). The continuous-analogue, frame-free property serves four distinct functions in the security

primitive.

First, it removes framing-level intervention targets. A framed digital protocol stack exposes a sequence of structured intermediate representations — packets, frames, decoded messages — each of which constitutes a discrete target for substitution, replay, or synthesis by an adversary with access to the inter-module link. A continuous-analogue path with no framing has no such discrete framing-level targets; the adversary’s substitution, if attempted, must operate in the analogue waveform domain, the hardness of which is addressed in Section 7.1.4.

Second, it raises the cost of waveform-faithful intervention under the declared latency budget. Any adversarial relay or substitution of an inter-module signal must itself complete within the declared τ_{ij} , and any intervention path imposes an irreducible latency floor consisting of the adversary’s own propagation, capture, processing, and re-transmission stages. Insertion of a digital decoding, processing, and re-encoding stage between modules imposes a latency floor of typically tens of nanoseconds to tens of microseconds depending on hardware, which is detectable as a violation of the optical-delay logical clock under verifier-side timing checks only when the added latency, jitter, and path asymmetry exceed the declared verifier timing fidelity σ_τ^{ver} and the declared path-calibration uncertainty. Lower-latency analogue or mixed-signal intervention paths — including without limitation passive optical relay, electro-optic modulation, FPGA-assisted narrow-feature relay, or analogue waveform substitution at a relay point — impose lower latency floors and are correspondingly harder to detect via the latency budget alone; the witness-mesh primitive therefore relies on the latency budget in conjunction with the witnessing requirement of Section 7.1.5 and the joint-trace verifier of Section 7.1.6, as disclosed in Section 7.1.4, rather than on the latency budget alone.

Third, it preserves the closed-with-reality property across the network. In the preferred embodiment, each module’s fast loop is fully analogue from detector to reactor to emitter, and the inter-module coupling extends this fully-analogue signal path across the network without an intermediate digital re-representation. The joint dynamics of the coupled mesh therefore evolve as a continuous physical system, and the verifier’s joint-trace check (Section 7.1.6) evaluates a continuous joint trajectory rather than a sequence of discrete frame-level decisions.

Fourth, it eliminates buffering-induced ambiguity in causal attribution. A buffered or framed inter-module path admits queue-occupancy variations, retransmission, and protocol-stack scheduling jitter, all of which loosen the binding between a sender module’s instantaneous fast-loop state and a receiver module’s instantaneous coupling input. The continuous-analogue path binds the two with a single declared τ_{ij} and a verifier-side fidelity σ_τ^{ver} , making the optical-delay logical clocks of Section 14.2 a tight rather than loose constraint.

Synchronisation of fast-loop sample rates (non-limiting). In some embodiments, the fast-loop sample or modulation rates $\{f_i\}$ of the modules are synchronised to a common physical reference — including without limitation a shared optical clock, an injection-locked frequency reference, or a phase-locked loop driven by a distributed timing signal — so that

peer signal arrivals at each module’s coupling-network input are aligned to the receiving module’s fast-loop sampling phase up to a declared phase tolerance. The shared reference, the synchronisation mechanism, and the phase tolerance are committed to Π_{dig} . In some embodiments, fast-loop rates are heterogeneous across modules with declared rate ratios and inter-rate phase relationships committed to the protocol digest.

Declared exceptions and degraded modes (non-limiting). In some embodiments, a portion of the inter-module coupling path is implemented digitally for engineering reasons. In such embodiments, the digital-stage latency is itself a declared, characterised, and committed quantity included in τ_{ij}^{hw} , and the verifier-side timing check is calibrated accordingly. Such embodiments are degraded modes of the primary deployment in the sense that the hardness contribution from the no-buffering property is reduced; the security claim of Section 7.1.4 is correspondingly indexed to the declared latency budget, the analogue fraction of the inter-module path, and the verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$. The preferred embodiment is the fully-analogue, frame-free inter-module coupling.

Continuous-analogue carve-out for claims (non-limiting). Claims reciting an uninterrupted continuous-analogue inter-module coupling path apply only to embodiments omitting any digital stage in the live path, including without limitation the framing, forward-error-correction, buffer/queue, digital-decoding, and operating-system-mediated transport stages disclosed as excluded in Section 7.1.2. Embodiments containing a declared digital stage in the live path remain within broader architectural mesh claims of the witness-mesh primitive that do not require uninterrupted continuous-analogue coupling, but do not satisfy claims that recite the uninterrupted continuous-analogue limitation.

Latency-budget contribution against analogue and mixed- signal relay attacks (non-limiting). The latency contribution of the continuous-analogue, frame-free property does not by itself exclude passive optical relay, electro-optic relay, FPGA-assisted analogue or narrow-feature relay, mixed-signal relay, or analogue waveform substitution attacks at a relay point on the inter-module path. Such attacks are evaluated under the committed verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$, the declared physical path model (including signal-path length d_{ij} , carrier velocity v_{ij} , and analogue-front-end transfer characteristics), and the verifier coverage tuple of Section 7.1.6. The witness-mesh primitive’s empirical hardness against analogue and mixed-signal relay attacks is a declared, time-indexed, attacker-indexed quantity per Section 7.1.10, established at each declared re-evaluation cycle against the strongest available analogue-domain attack class, and is committed to the protocol digest as part of the per-version hardness commitment.

7.1.4 Security primitive

The witness-mesh security primitive is the joint consistency of the network’s convolution bundle: each module’s contribution to the joint record $\{C_{0:T}^{(i)}\}_{i \in I}$ must simultaneously be

consistent with:

- (a) that module's actual observation of its portion of physical reality Σ_i during the evidence window $[0, T]$;
- (b) that module's reception of every peer module's fast-loop broadcast at the declared inter-module latencies $\{\tau_{ij}\}$ over the continuous-analogue inter-module signal paths of Section 7.1.3, evaluated to verifier-side fidelity σ_τ^{ver} ; and
- (c) the committed trained inter-module coupling network mediating those receptions, as committed to Π_{dig} .

To produce a forged joint record that the verifier of Section 7.1.6 accepts, an adversary must reproduce all three constraints simultaneously across all networked modules, under the verifier's declared coverage parameters of Section 7.1.6.

Adversary's joint reproduction problem. Specifically, the adversary must:

(i) Reproduce each module's fast-loop response to the actual physical reality observable from that module's vantage point at the relevant times during $[0, T]$, which requires either physical possession of the module and physical access to its vantage point during the evidence window, or reconstruction of the physical reality at that vantage from external sources of equivalent fidelity under the verifier's declared meter envelope.

(ii) Reproduce each module's reception of every peer module's fast-loop broadcast at the propagation-delay-bounded times disclosed by the optical-delay logical clocks of Section 14.2 and the declared latency budget of Section 7.1.3, within σ_τ^{ver} , which requires knowing what each peer module broadcast — and therefore requires reproducing the peer modules' responses under constraints (i) and (iii) applied transitively to those peers.

(iii) Reproduce the network's coupled dynamics under the specific committed inter-module coupling network, which is frozen and committed to the protocol digest. Although the coupling network's parameters are public under the declared threat model, reproduction requires evaluation of the coupling network on the actual peer fast-loop signals each module would receive in genuine operation. The adversary cannot freely choose alternative peer signals to ease reproduction, because the peer signals are themselves constrained by (i) and (ii) applied to the peer modules.

Mutually-recursive structure of the forgery problem. The constraints (i)–(iii) are mutually recursive across the network: forging module M_A 's record requires the records of all peers connected to M_A over the coupling graph $\mathcal{G}$, each of which requires the records of their peers under the same constraints, transitively across the connected component of $\mathcal{G}$ containing M_A . The adversary's problem is therefore the joint reproduction of the entire connected component's coupled behaviour under the committed coupling, given the actual physical reality each constituent module observed during the evidence window and given the declared latency budget on inter-module coupling.

This mutually-recursive structure does not constitute a formal proof of joint reproduction hardness in the sense of cryptographic soundness; consistent with Section 14.3, the

witness-mesh primitive is an empirical hardness construction, indexed to a declared adversary class and a declared meter envelope. The structural contribution of this section is to articulate why the joint reproduction problem is *capable of being* empirically harder, and in preferred deployments is measured as harder, than any per-module reproduction problem under the declared adversary model and verifier coverage: the joint problem is not separable into per-module subproblems under the continuous-analogue inter-module coupling, the declared latency budget evaluated at $\sigma_{\tau}^{\text{ver}}$, and the verifier-side joint-trace check at the declared coverage parameters of Section 7.1.6. The realised hardness of any specific deployment is established empirically at the declared re-evaluation cycle and is committed per-version under Section 7.1.10.

Latency-budget alone is insufficient (non-limiting). The latency budget of Section 7.1.3 is a necessary but not sufficient component of the witness-mesh security primitive. An adversary with access to the inter-module signal path may, in principle, intervene in the analogue domain within the declared latency budget, by characterising the inter-module waveform from observation and injecting a learned analogue perturbation through an electro-optic, acousto-optic, or equivalent modulator at a relay point on the inter-module path, or at the receiving module’s coupling-network input. Such analogue-domain intervention is not excluded by the continuous-analogue, frame-free property alone. The witness-mesh primitive does not rely on the latency budget alone to exclude adversarial intervention. Rather, the primitive derives its empirical hardness from the *conjunction* of:

- (a) the latency budget on inter-module coupling, evaluated at verifier-side fidelity $\sigma_{\tau}^{\text{ver}}$ (Section 7.1.3);
- (b) the witnessing requirement of Section 7.1.5, which constrains each module’s fast-loop response to that module’s observation of physical reality during the evidence window; and
- (c) the joint-trace verifier of Section 7.1.6, which checks per-edge directed causal consistency at sample-level granularity under the declared coverage parameters and the declared meter envelope.

An analogue-domain adversarial perturbation that satisfies the latency budget must additionally produce, at each module, a fast-loop response consistent with that module’s observation of its physical reality and consistent with the committed coupling network operating on the genuine peer signals; the empirical hardness of the joint reproduction problem of this section is the joint hardness of constraints (a)–(c) under the declared adversary class. The continuous-analogue, frame-free property contributes to (a) by removing framing-level intervention targets and by raising the cost of waveform-faithful intervention under the declared latency budget; the witnessing requirement (b) and joint-trace verifier (c) contribute the remaining structural barriers to forgery.

7.1.5 Witnessing requirement

The witness-mesh security primitive incorporates a *witnessing requirement*: each module's fast-loop response is bound to its observation of physical reality Σ_i during the evidence window, so that the joint record is rejected by the verifier of Section 7.1.6 to the extent that opened verifier meters depend on residual scene variables not reconstructable by an adversary, under the adversary's declared surveillance budget and the verifier's declared meter envelope. The witnessing contribution to joint hardness is, accordingly, a deployment-indexed quantity bounded by per-vantage residual unobserved entropy as set out in the following paragraphs, and not an absolute property of the architecture.

Witnessing contribution and per-vantage residual entropy. The witnessing contribution to the joint hardness of the security primitive of Section 7.1.4 is bounded above by the per-vantage residual unobserved entropy of Σ_i under the adversary's surveillance budget within the declared threat model. In some embodiments, Σ_i comprises a sealed, occluded, hidden, or otherwise non-externally-observable region of physical space, in which case the per-vantage residual unobserved entropy is large and the witnessing contribution is correspondingly large; in some embodiments, Σ_i comprises a publicly observable scene under known geometry, in which case the per-vantage residual unobserved entropy may be small under a sufficiently capable adversary surveillance budget, and the witnessing contribution is correspondingly reduced. The witness-mesh primitive's overall hardness profile is therefore deployment-specific and is established at each declared re-evaluation cycle and committed to the protocol digest as part of the per-version hardness commitment of Section 7.1.10.

Vantage-point independence and amplification. Vantage-point independence between modules can amplify the witnessing contribution by adding constraints that, under the declared adversary surveillance budget and the verifier's declared meter envelope, are not necessarily reconstructable from a single vantage. Vantage-point independence does not by itself constitute the witnessing contribution: even a witness mesh in which all modules observe overlapping or coincident regions of Σ retains a per-module witnessing contribution to the extent that each module's individual physical-state evolution during the evidence window is not externally reconstructable to the verifier's declared meter envelope. Conversely, a witness mesh deployed entirely in publicly observable scenes may derive most of its joint hardness from vantage-point independence, the latency budget, and the joint coupled dynamics, with reduced per-module witnessing contribution. Both regimes are within the scope of the witness-mesh primitive; the relative contributions of witnessing, vantage-point independence, latency budget, and coupled dynamics to the joint hardness are deployment-specific and are committed per-version under Section 7.1.10.

Per-module reactor microstructure (non-limiting). In some preferred embodiments, each module's reactor incorporates microstructure-derived per-device transfer-function vari-

ability of reactor-microstructure-unclonable as disclosed in Section 10.3, contributing an additional per-module hardness component independent of and complementary to the witnessing contribution above. Reactor-microstructure hardness and witnessing-derived hardness are independent contributions to the empirical hardness profile, with the relative weights deployment-specific and committed per-version under Section 7.1.10. An adversary with clones of every module but without scene access at the verifier’s declared meter envelope is rejected to the extent that opened meters depend on residual scene variables not reconstructable within the verifier’s amplitude and phase residual tolerances; an adversary with full scene access but without per-module reactor characterisation is rejected to the extent that opened meters depend on reactor transfer features not reproducible from the adversary’s reactor model under the verifier’s tolerances; an adversary lacking both is rejected to the extent that opened meters depend on either class of variables. The relative weights are deployment-specific, empirically established, and committed per-version.

Effect of dynamic scene reconstruction on vantage-point independence (non-limiting). Modern dynamic scene-reconstruction methodologies, including without limitation neural radiance fields, four-dimensional Gaussian splatting, sparse-view generative completion, and multi-modal scene priors, can reduce the effective vantage-point independence between modules in publicly observable scenes by inferring or hallucinating unseen views from sparse external observations. In deployments where the witness mesh observes publicly observable scenes under known geometry, the per-vantage residual unobserved entropy is correspondingly reduced, and the witnessing contribution to joint hardness is correspondingly reduced. This effect is the structural reason that the witness-mesh primitive’s overall hardness profile is deployment-specific: deployments with sealed, occluded, or otherwise non-externally-reconstructable per-vantage observations retain a large witnessing contribution, while deployments in publicly reconstructable scenes derive most of their joint hardness from other components of the security primitive (latency budget, joint coupled dynamics, verifier coverage, per-module reactor microstructure where applicable).

Residual-entropy evaluation procedure (non-limiting). The per-vantage residual unobserved entropy of Σ_i under the declared adversary surveillance budget is, in some preferred embodiments, estimated by the following procedure for each module M_i at each declared re-evaluation cycle:

1. enumerate the adversary observation channels available under the declared adversary class, including without limitation external surveillance modalities, ambient sensors, scene priors, and any leaked legitimate joint-record windows;
2. define the meter variables of Σ_i that are checked by the verifier through per-module reality binding of Section 7.1.6, including representative non-limiting meter variables such as optical amplitude residuals at the per-module detector, phase residuals at the per-module detector where applicable, local illumination measurements, scene-pose variables, temporal derivatives of the per-module detector output, scene occlusion state,

and where applicable acoustic, RF, thermal, or other side-meter variables associated with the module's vantage of Σ_i ;

3. fit the best declared reconstruction model under the adversary's observation budget that predicts the meter variables from the adversary's available observations, using the strongest adversary-model class committed to the protocol digest;
4. measure the conditional prediction error of the fitted model against the verifier's amplitude and phase residual tolerances ϵ_{amp} and ϵ_{phase} on a held-out evaluation corpus;
5. commit the resulting residual-entropy estimate, with a declared confidence interval and the corpus identifier, as part of the per-version hardness commitment of Section 7.1.10.

The procedure provides an operational method for a person of ordinary skill to evaluate the witnessing contribution to the joint hardness in a specific deployment. Variants of the procedure — for example, using multiple reconstruction-model classes, employing adaptive adversary models, or stratifying across time windows — are non-limiting and may be selected based on the declared adversary class and meter envelope.

7.1.6 Verifier binding

The witness-mesh security primitive presupposes a verifier that checks the joint convolution bundle record at sample-level granularity under the two-seed selective-opening protocol of Section 14.4. The verifier operates under selective opening rather than real-time streaming: each module's full sample-level convolution bundle is committed under a Merkle root $\text{com}_{\text{full}}^{(i)}$ at the close of the evidence window, and the verifier subsequently requests selective opening of randomised atoms or windows under the two-seed protocol; the verifier does not require continuous receipt or processing of the full joint sample stream during the evidence window. The sample-level granularity refers to the resolution at which selective openings, when requested, are evaluated — not to a real-time streaming requirement.

The verifier performs:

1. per-edge directed causal consistency: for each coupling edge $(i, j) \in \mathcal{E}$ and each opened atom or window, verification that the receiving module M_j 's coupling input at sample time t is consistent with the broadcasting module M_i 's fast-loop output at sample time $t - \tau_{ij}$ under the declared latency budget and verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$ of Section 7.1.3;
2. joint-trace consistency: for each opened window, verification that the joint sample sequence $\{(u^{(i)}(t), \mathbf{y}_t^{(i)})\}_{i \in I, t \in [0, T]}$ is consistent with the committed coupling network operating on the recorded peer signals under the declared meter envelope;
3. per-module reality binding: verification that each module's response $\mathbf{y}_t^{(i)}$ is consistent with that module's independently-recorded observation of its physical reality Σ_i at time t , via the meter envelope and declared verisimilitude meters of the security-theory section (Section 12); and
4. full-record commitment: verification of the Merkle commitment $\text{com}_{\text{full}}^{(i)}$ of each module's full sample-level convolution bundle under the protocol of Section 14.4, with selective opening of randomised atoms under the two-seed protocol.

Verifier coverage parameters (declared parameters). The verifier’s coverage of the joint record is characterised by a tuple of declared parameters committed to Π_{dig} at each deployment version:

- $\rho_{\text{open}} \in (0, 1]$, the selective-opening probability or fraction of committed atoms or windows opened by the verifier under the two-seed protocol per evidence window;
- $\kappa_{\text{edge}} \in (0, 1]$, the edge-check coverage or fraction of coupling edges $(i, j) \in \mathcal{E}$ checked for directed causal consistency per opened window, where the coverage is stratified over *directed* edges of $\mathcal{E}$ rather than aggregated over undirected pairs or aggregate edge count, so that for each directed coupling $i \rightarrow j$ the verifier’s per-edge causal-consistency check is a distinct coverage event;
- $\delta_{\text{dec}} \in [1, \infty)$, the decimation factor relating the verifier’s checked sample rate to each module’s fast-loop sample rate f_i , with $\delta_{\text{dec}} = 1$ corresponding to checking every sample;
- ϵ_{amp} , the verifier’s amplitude residual tolerance for joint-trace consistency under the declared meter envelope;
- ϵ_{phase} , the verifier’s phase residual tolerance for joint-trace consistency under the declared meter envelope, where applicable to the inter-module coupling modality.

The witness-mesh security claim of Section 7.1.4 is indexed to the verifier coverage tuple $(\rho_{\text{open}}, \kappa_{\text{edge}}, \delta_{\text{dec}}, \epsilon_{\text{amp}}, \epsilon_{\text{phase}})$ together with $\sigma_{\tau}^{\text{ver}}$ from Section 7.1.3: the joint hardness profile is committed per-version under Section 7.1.10 as a function of these declared parameters under the declared adversary class. Deployments with reduced verifier coverage — for example, lower ρ_{open} , lower κ_{edge} , larger δ_{dec} , or looser ϵ_{amp} or ϵ_{phase} — have correspondingly reduced joint hardness, and the relationship between the coverage tuple and the empirical hardness index is established at each declared re-evaluation cycle.

Detection-probability relation (non-limiting, first-order). For purposes of relating the verifier coverage tuple to the detection of a forged joint record, the verifier’s detection probability against a forgery in which a fraction q_{bad} of *committed* directed edge-atoms would exhibit a residual exceeding the declared tolerances ϵ_{amp} and ϵ_{phase} admits the non-limiting first-order analytical bound

$$P_{\text{detect}} \geq 1 - (1 - q_{\text{bad}} \rho_{\text{open}} \kappa_{\text{edge}})^m,$$

where m is the number of *committed* directed edge-atoms in the evidence window over which independent opening and edge-check trials are applied, ρ_{open} is the per-atom opening probability, and κ_{edge} is the per-opened-window edge-check probability. The bound assumes independence of opened atoms and uniformity of q_{bad} across the opened atoms; both assumptions are imperfect in practice (stratification creates dependencies, and q_{bad} varies across atoms) and the bound is accordingly a first-order analytical anchor rather than an exact relation. Refinements — including stratified-sampling corrections, edge-specific $q_{\text{bad}}^{(i,j)}$ weightings, and decimation-aliased detection terms — are within the scope of this disclosure as non-limiting

embodiments and are committed, where applicable, to the protocol digest as part of the per-version hardness commitment of Section 7.1.10.

Opening-seed mechanics and stratification (non-limiting). In some preferred embodiments, the verifier’s coverage parameters are realised under the following non-limiting mechanics:

1. *Opening seed.* Opened atoms or windows are selected after full-record commitment using a verifier-generated or jointly-generated unpredictable opening seed s_{open} derived from a declared unpredictability source independent of the run seed of the two-seed protocol of Section 14.4. The opening seed is not disclosed to the deployment prior to commitment.
2. *Stratification.* Selection of opened atoms is stratified over (a) time within the evidence window, (b) modules in I , and (c) coupling edges in $\mathcal{E}$, so that no module, edge, or temporal sub-window is systematically excluded or systematically over-represented across selective openings. The stratification scheme is committed to Π_{dig} .
3. *Decimation aliasing constraint.* The decimation factor δ_{dec} is selected so that aliasing against the declared fast-loop sample rates $\{f_i\}$ and any declared modulation frequencies in the inter-module coupling is excluded or bounded by a declared aliasing margin committed to Π_{dig} .
4. *Tolerance calibration.* The amplitude and phase residual tolerances ϵ_{amp} and ϵ_{phase} are calibrated against the declared meter envelope and the empirical noise floor of the measurement chain, with the calibration procedure and measurement chain committed to Π_{dig} .
5. *Failure accounting.* The commitment protocol does not permit selective discard of windows after capture; all captured evidence-window records are committed under $\text{com}_{\text{full}}^{(i)}$, and any declared exclusions (for example for hardware-fault windows under a declared health-monitor protocol) are themselves committed.

Adversarial-discriminator-ensemble check (optional, non-limiting). In some embodiments, the verifier additionally performs an adversarial-discriminator check, in which one or more learned discriminator models trained against declared benchmark forgery model families available at the evaluation date are evaluated against the joint record; the discriminator ensemble’s output is incorporated into the meter envelope as a declared evaluation meter under the meter-partition discipline of the security-theory section. The discriminator ensemble is not a standalone acceptance criterion of the witness-mesh primitive, and any false-acceptance rate is version-specific and committed only for the declared model families, training corpus, verifier tuple, and adversary budget. The discriminator ensemble has no structural security guarantee outside its committed receiver-operating-characteristic evaluation, and is not a substitute for the per-edge directed causal consistency, joint-trace consistency, per-module reality binding, or full-record commitment checks of the verifier specification of this section. In some embodiments, the discriminator ensemble’s training data,

held-out adversary families, false-acceptance threshold, and confidence interval are committed to Π_{dig} as part of the per-version hardness commitment of Section 7.1.10.

Reduced-verifier embodiments and non-conforming verifiers (non-limiting).

Reduction of the verifier to per-module summary statistics, pairwise correlations, low-dimensional aggregates, coarse geometric consistency checks, non-stratified opening, biased opening-seed generation, decimation that aliases against the declared fast-loop rates, or tolerances loose enough to admit generative-surrogate forgeries yields a weaker primitive whose security claim is correspondingly reduced and is not the security claim of this section. A verifier that does not check the per-edge directed causal consistency, joint-trace consistency, per-module reality binding, and full-record commitment of this section under the declared coverage tuple is *non-conforming* with respect to claims of this disclosure that recite the joint-trace verifier with declared coverage tuple. Such a non-conforming verifier may satisfy broader architectural claims of the witness-mesh primitive that do not recite the joint-trace verifier element, but does not satisfy claims that recite the joint-trace verifier limitation. The full-joint-record sample-level verifier under selective opening with the declared coverage tuple is the preferred embodiment for the witness-mesh security claim. In some embodiments, intermediate-fidelity verifiers are employed in resource-constrained deployments, with the meter envelope and declared adversary class adjusted to the verifier’s actual coverage of the joint record, and the resulting hardness profile committed to the protocol digest at each declared re-evaluation cycle.

7.1.7 Distinction from prior art

The witness-mesh security primitive is distinct from the following classes of prior art.

Single-device optical physically unclonable functions. The witness-mesh primitive is distinct from single-device optical physically unclonable function (PUF) constructions, including those cited in the background art (Pappu et al., “Physical One-Way Functions,” *Science*, 2002; Goorden et al., “Quantum-secure authentication of a physical unclonable key,” *Optica* 1, 421–424, 2014; Hui et al., “Non-linear optical scattering PUF,” *Optics Express* 31(24), 40646–40657, 2023; Davis, Letz, Mosk, and Pinkse, EP3252740B1; and related disclosures), in the following respects:

- (a) Optical PUF security rests on the empirical hardness of cloning a single device’s challenge–response mapping. The witness-mesh primitive does not reduce to single-device clonability: the joint reproduction problem of Section 7.1.4 binds at every module’s observation of reality and across all peer couplings, and the joint hardness contribution is not separable into per-module components under the declared meter envelope.
- (b) Optical PUFs are characterised by challenge–response pairs over a fixed challenge space; some optical PUFs are high-dimensional and based on mesoscopic scattering or nonlinear interactions. The witness-mesh primitive is characterised by the joint

convolution bundle record across the network over the evidence window, which is a fundamentally different object from a single-device challenge–response mapping: it requires simultaneous multi-vantage reality witnessing under a coupled live verifier and is not reducible to a finite set of per-device CRPs.

- (c) Published optical-PUF attack methodologies, including transmission-matrix reconstruction, machine-learning surrogate modelling, and neural-network reconstruction of speckle responses, target single-device transfer functions. These methodologies apply to the witness-mesh primitive only insofar as the verifier accepts records reproducible from per-module surrogate models composed under the public coupling network — a regime addressed by the joint-trace verifier of Section 7.1.6 and its declared coverage tuple.
- (d) The witness-mesh primitive incorporates the witnessing requirement of Section 7.1.5, absent from optical PUFs: the security claim binds each module’s response to its observation of physical reality during the evidence window, with the witnessing contribution bounded by the per-vantage residual unobserved entropy under the adversary’s surveillance budget.

In some embodiments, the witness-mesh primitive is combined with optical reactor-microstructure-class single-device hardness, with each module’s reactor incorporating reactor-microstructure unclonability as disclosed in Section 10.3 and related sections. In such combined embodiments, the joint security claim incorporates both the witness-mesh constraint and the per-module clonability constraint, with the contributions independent and complementary.

Integrated photonic interferometric PUFs and on-chip MZI-network PUFs.

The witness-mesh primitive is distinct from integrated photonic interferometric PUFs and on-chip Mach–Zehnder interferometer (MZI) network PUFs, including without limitation Smith and Jacinto, “Reconfigurable Integrated Optical Interferometer Network-Based Physically Unclonable Function,” *J. Lightwave Technol.* 38(17), 4599–4606, 2020 (arXiv:2012.11358); Jacinto and Smith, “Utilizing a Fully Optical and Reconfigurable PUF as a Quantum Authentication Mechanism,” *OSA Continuum* 4(2), 739–756, 2021 (arXiv:2012.10326); Tarik et al., “Realization of Robust Optical Physical Unclonable Function Using a Silicon Photonic Quasicrystal Interferometer”; Mesaritakis et al., “Physical unclonable function based on a multi-mode optical waveguide,” *Sci. Rep.* 8(1), 9653, 2018; and the broader reconfigurable-optical-PUF literature including silicon-photonic moiré-quasicrystal interferometers, electrically- or thermally-reconfigurable scattering PUFs, and phase-change-material- based reconfigurable photonic PUFs. Such constructions describe a single integrated photonic device — in the case of the MZI-network PUFs, a network of Mach–Zehnder interferometers fabricated on a single photonic integrated circuit with computer-controlled phase modulators — used as a reconfigurable challenge–response authentication primitive. The “network” in such constructions is the on-chip optical interconnect within a single device, not a network of physically separate modules. The witness-mesh primitive is distinguished from these constructions by:

- (a) the architectural primitive: the witness mesh comprises a plurality of physically separate Reality Kernel modules, each with its own emitter, reactor, detector, and fast closed loop in causal feedback with its own portion of physical reality, coupled via inter-module signal paths between separately-housed modules; the cited integrated photonic PUFs are single devices with no inter-module signal paths;
- (b) the closed-with-reality property: each witness-mesh module operates a fast closed loop in causal feedback with physical reality Σ_i external to the module during the evidence window, with the reactor in feedback with the loop dynamics; the cited integrated photonic PUFs operate as open-loop challenge-response devices that do not close a feedback loop with external physical reality during authentication;
- (c) the witnessing requirement of Section 7.1.5, which has no counterpart in the cited integrated photonic PUF literature: those constructions authenticate the device or messages from the device, not observations of physical reality external to the device;
- (d) the inter-module continuous-analogue coupling under the trained inter-module coupling network of Section 7.1.2, with declared inter-module latencies τ_{ij} and verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$, which has no counterpart in the cited single-device constructions; and
- (e) the joint convolution bundle record across multiple modules under the verifier of Section 7.1.6 with declared coverage tuple $(\rho_{\text{open}}, \kappa_{\text{edge}}, \delta_{\text{dec}}, \epsilon_{\text{amp}}, \epsilon_{\text{phase}})$, as opposed to challenge-response pairs over a single device.

In some embodiments, an individual witness-mesh module's reactor may be implemented as an integrated photonic device, including a reconfigurable interferometric structure or a quasicrystal interferometer of the kind described in the above-cited literature; the witness-mesh primitive is the multi-module networked construction operating on top of such per-module reactors, not the per-module reactor itself.

Distance-bounding and physical-layer relay-attack literature. The witness-mesh primitive is distinct from distance-bounding protocols and physical-layer relay-attack mitigations, including without limitation Brands and Chaum, "Distance-Bounding Protocols," *Eurocrypt 1993*, LNCS 765, 344–359, 1994; Hancke and Kuhn, "An RFID Distance Bounding Protocol," *SecureComm 2005*, 67–73, 2005; Clulow, Hancke, Kuhn, and Moore, "So Near and Yet So Far: Distance-Bounding Attacks in Wireless Networks," *ESAS 2006*, LNCS 4357, 83–97, 2006; and successor distance-bounding and physical-layer relay-resistance work. Such protocols use round-trip-time measurements between a prover and verifier, with bit-by-bit timed challenges, to upper-bound the prover's physical distance from the verifier and to detect relay attacks; published physical-layer attacks demonstrate that distance-bounding security depends on the specific receiver, demodulation, and channel-design details. The witness-mesh primitive shares the architectural element of latency-bounded physical-layer security with this literature, but is distinguished by:

- (a) the security objective: distance-bounding authenticates the prover's physical distance from a single verifier through a single round-trip-time measurement; the witness-mesh primitive authenticates the joint physical evolution of multiple coupled witnesses bound

- to their individual observations of physical reality, across an entire evidence window;
- (b) the temporal scope: distance-bounding operates over discrete bit-level challenge-response rounds; the witness-mesh primitive operates over a continuous-analogue joint trace of duration T verified at sample-level granularity under the declared coverage tuple of Section 7.1.6;
 - (c) the witnessing requirement of Section 7.1.5, which has no counterpart in distance-bounding: distance-bounding does not bind the prover’s response to any observation of physical reality external to the prover and verifier; the witness-mesh primitive binds each module’s response to that module’s observation of Σ_i ;
 - (d) the multi-witness joint-record structure: distance-bounding typically involves a single prover and single verifier (with multi-prover and multi-verifier extensions described as add-ons); the witness-mesh primitive is intrinsically multi-module with mutually-recursive joint reproduction constraints across the coupling graph $\mathcal{G}$.

The physical-layer relay-attack methodologies analysed in the distance-bounding literature — including without limitation late-commit attacks, overclocking, demodulation-stage attacks, deferred-decision attacks, and analogue waveform manipulation — constitute attack methodologies against which the witness-mesh primitive’s latency-budget contribution must be empirically evaluated. The structural defences against this attack class are the conjunction of the latency budget of Section 7.1.3, the witnessing requirement of Section 7.1.5, and the joint-trace verifier of Section 7.1.6, as disclosed in the latency-budget-alone-is-insufficient paragraph of Section 7.1.4.

Coupled-reservoir-computing and delay-coupled optoelectronic networks; link-inference attacks as a primary attack class.

The witness-mesh primitive is distinct from coupled-reservoir-computing and delay-coupled optoelectronic-network literature, including without limitation Brunner and Larger and successor photonic reservoir-computing systems; Tanaka et al., “Recent advances in physical reservoir computing: A review,” *Neural Networks* 115, 100–123, 2019; Banerjee et al., “Machine Learning Link Inference of Noisy Delay-coupled Networks with Opto-Electronic Experimental Tests,” *Phys. Rev. X* 11, 031014, 2021 (arXiv:2010.15289); Banerjee, Pathak, Roy, Restrepo, and Ott, “Using Machine Learning to Assess Short Term Causal Dependence and Infer Network Links,” *Chaos* 29, 121104, 2019; chaos-synchronisation key-distribution work using coupled nonlinear units; and related disclosures. Such systems describe coupled nonlinear physical dynamical systems with learnable inter-node delays, and demonstrate machine-learning-based reservoir-computing surrogates and topology-and-delay inference from observed nodal time series.

The link-inference and surrogate-emulation methodologies of Banerjee et al. and successor work constitute a primary class of attacks against which the witness-mesh primitive must be empirically evaluated. In particular, an adversary in possession of leaked joint-record windows may, under the public coupling graph $\mathcal{G}$ and the public trained inter-module coupling network, train a reservoir-computing or message-passing surrogate to reproduce the joint nodal-state time series, and may infer the declared inter-module delays $\{\tau_{ij}\}$ from the same

data. The witness-mesh primitive’s structural defences against this attack class are:

- (a) the witnessing requirement of Section 7.1.5: a surrogate that reproduces joint dynamics statistically under the inferred topology must additionally produce, at each module, a response consistent with that module’s actual observation of physical reality during the evidence window;
- (b) the joint-trace verifier of Section 7.1.6: per-edge directed causal consistency at sample-level granularity, evaluated under selective opening with the declared coverage tuple, is evaluated at a fidelity that may exceed, and in committed versions is compared against, the fidelity achieved by declared surrogate-emulation attack classes including the cited literature; and
- (c) the periodic re-evaluation and reconfiguration framework of Section 7.1.10: empirical hardness against the surrogate-emulation attack class is established at each declared re-evaluation cycle and committed to the protocol digest, and coupling-network rotation invalidates accumulated surrogate characterisation at the surface level.

The witness-mesh primitive’s empirical hardness against the link-inference and surrogate-emulation attack class is therefore a declared, time-indexed, attacker-indexed quantity per Section 12 and Section 7.1.10, not a structural guarantee. Architectural elements shared with the cited reservoir-computing literature — delay-coupled inter-module signal exchange, trained coupling parameters, nonlinear nodal dynamics — are repurposed in the witness-mesh primitive for a security objective rather than a computational objective, and are combined with the witnessing requirement, the protocol-digest commitment, the joint-trace verifier with declared coverage tuple, and the per-version hardness commitment, none of which is present in the cited literature.

Continuous-analogue chaos-synchronisation and analogue-coupled secure-communications

literature. The witness-mesh primitive is distinct from continuous-analogue chaos-synchronisation and analogue-coupled secure-communications literature, including without limitation Argyris et al., “Chaos-based communications at high bit rates using commercial fibre-optic links,” *Nature* 438, 343–346, 2005; Soriano, García-Ojalvo, Mirasso, and Fischer, “Complex photonics: Dynamics and applications of delay-coupled semiconductor lasers,” *Reviews of Modern Physics* 85, 421–470, 2013; Stankovski, McClintock, and Stefanovska, “Coupling Functions Enable Secure Communications,” *Phys. Rev. X* 4, 011026, 2014; Nadzinski, Dobrevski, Anderson, McClintock, Stefanovska, Stankovski, and Stankovski, “Experimental Realization of the Coupling Function Secure Communications Protocol and Analysis of Its Noise Robustness,” *IEEE Trans. Inf. Forensics Secur.* 13(10), 2591–2601, 2018; Illing, Panda, and Shreshian, “Isochronal chaos synchronization of delay-coupled optoelectronic oscillators,” *Phys. Rev. E* 84, 016213, 2011; Zhou and Roy, “Isochronal synchrony and bidirectional communication with delay-coupled nonlinear oscillators”; and related disclosures of chaos-based and coupling-function-based secure communications using continuous-analogue inter-device coupling.

Such systems describe two or more nonlinear dynamical systems (typically semiconductor

lasers, optoelectronic oscillators, or related photonic devices) coupled through continuous-analogue signal paths with declared propagation delays, with secure communication of an information signal between the coupled systems by chaos synchronisation, coupling-function-based ciphers, or related dynamical-system primitives. Information security in the cited literature derives from one or more of: the receiver's knowledge of a private coupling function or shared synchronisation key, the difficulty of cloning the nonlinear dynamics from observed time series, or the practical inaccessibility of the inter-device signal path to an eavesdropper.

The witness-mesh primitive shares architectural elements with this literature — including continuous-analogue inter-device coupling, declared propagation delays, and nonlinear dynamics — but is distinguished by:

- (a) the security objective: the cited literature uses coupled analogue dynamics to enable *private communication of an information signal* between two or more parties, with security derived from coupling-function secrecy or synchronisation-key secrecy. The witness-mesh primitive uses coupled analogue dynamics to enable *multi-witness joint attestation of physical-reality observations* to a third-party verifier, with security derived from the joint reproduction problem of Section 7.1.4 under a public coupling commitment;
- (b) public coupling commitment vs. private coupling parameters: the cited literature relies, in various instances, on one or more of private parameters, synchronisation assumptions and pre-shared synchronisation keys, parameter mismatch between transmitter and adversary, chaotic masking, or the practical difficulty of modelling the coupled dynamics from observed signals — with the coupling parameters or synchronisation parameters not disclosed to adversaries. The witness-mesh primitive treats the trained inter-module coupling network and the coupling graph $\mathcal{G}$ as *public commitments* to the protocol digest under the declared threat model, with security derived from joint physical reproduction under public coupling rather than from coupling-parameter secrecy or any of the foregoing;
- (c) multi-witness joint-record structure: the cited literature is typically point-to-point (one transmitter, one receiver) or small symmetric mesh, with information flow from sender to receiver. The witness-mesh primitive is intrinsically multi-witness with the joint convolution bundle record across the network as the security primitive, mutually- recursive across the connected component of $\mathcal{G}$ per Section 7.1.4;
- (d) the witnessing requirement of Section 7.1.5, which has no counterpart in chaos-synchronisation secure communications: the cited literature transmits an information signal generated by the sender, not observations of physical reality external to the coupled-systems network. The witness-mesh primitive binds each module's response to that module's observation of physical reality Σ_i during the evidence window;
- (e) the closed-with-reality property: each witness-mesh module operates a fast closed loop in causal feedback with physical reality external to the module during the evidence window. The cited literature's coupled nonlinear systems close their loops with each

other (and optionally with self-feedback), not with external physical reality;

- (f) the joint-trace verifier with selective opening: the cited literature uses time-evolving Bayesian inference, chaos-synchronisation decoding, or related receiver-side decoding methods to recover the transmitted signal. The witness-mesh primitive uses the verifier of Section 7.1.6 with declared coverage tuple under the two-seed selective-opening protocol of Section 14.4 to evaluate the joint convolution bundle record, which has no counterpart in the cited literature;
- (g) per-version hardness commitment under Section 7.1.10: the cited literature does not commit a per-version empirical hardness estimate against declared adversary classes to a protocol digest as part of the deployment record.

The chaos-synchronisation literature’s attack methodologies — including chaos-synchronisation key-extraction attacks, generalised-synchronisation attacks, parameter-estimation attacks, and time-series-based reconstruction — are methodologies against which the witness-mesh primitive’s empirical hardness must be evaluated where the inter-module coupling exhibits chaotic or near-chaotic dynamics. The witness-mesh primitive’s structural defences against this attack class are the joint-trace verifier with declared coverage tuple of Section 7.1.6, the witnessing requirement of Section 7.1.5, and the per-version hardness commitment of Section 7.1.10. Architectural elements shared with the cited literature — continuous-analogue inter-device coupling, declared propagation delays, nonlinear coupled dynamics — are repurposed in the witness-mesh primitive for multi-witness joint attestation under public coupling commitment, rather than for private communication between parties under coupling secrecy.

Virtual proofs of reality and physical-fact attestation. The witness-mesh primitive is distinct from virtual proofs of reality and related physical-fact attestation primitives, including without limitation Rührmair, Martinez-Hurtado, Xu, Kraeh, Hilgers, Kononchuk, Finley, and Burleson, “Virtual Proofs of Reality and their Physical Implementation,” in *2015 IEEE Symposium on Security and Privacy*, pp. 70–85 (DOI: 10.1109/SP.2015.12), and successor work on virtual proofs of physical statements. The cited literature describes protocols by which a prover establishes a physical statement about its local environment — including without limitation “a certain object in the prover’s system has temperature $X^{\circ}\text{C}$,” “two certain objects in the prover’s system are positioned at distance X ,” or “a certain object in the prover’s system has been irreversibly altered or destroyed” — to a remote verifier over a digital communication channel, with security derived from physically unclonable per-device behaviours of disordered scattering media, temperature-sensitive integrated circuits, quantum systems, or related physical primitives, and without reliance on classical secret keys or tamper-resistant trusted sensor hardware.

The witness-mesh primitive shares with this literature the objective of establishing physical statements to a verifier without relying on a classical secret-key tamper-resistance trust assumption, but is distinguished by:

- (a) single-prover-single-verifier vs. multi-witness joint attestation: the cited literature is

structured as a single-prover-single-verifier protocol in which one physical statement is established by one prover (using one or more sensors or reactor-microstructure-unclonability components co-located on the prover) to one verifier. The witness-mesh primitive establishes a joint physical statement spanning a plurality of physically separate modules whose joint convolution bundle record is the security primitive, with security derived from the joint reproduction problem of Section 7.1.4 across the connected component of $\mathcal{G}$;

- (b) continuous-analogue inter-module coupling: the cited literature establishes individual physical statements through challenge-response interactions between the verifier and individual physical primitives at the prover. The witness-mesh primitive establishes its joint statement through continuous-analogue inter-module coupling among the witness modules during the evidence window, in which each module's fast-loop response is determined in part by peer-module modulation of the inter-module coupling network, with no analogous mutual coupling among physical primitives in the cited single-prover virtual-proofs-of-reality literature;
- (c) the joint-trace verifier with declared coverage tuple of Section 7.1.6 under the two-seed selective-opening protocol of Section 14.4, which has no counterpart in virtual-proofs-of-reality challenge-response protocols; and
- (d) the per-version empirical hardness commitment of Section 7.1.10, including the cross-version transfer-attack evaluation, which is not present in the cited virtual-proofs-of-reality literature.

Sensor physical unclonable functions. The witness-mesh primitive is also distinct from sensor physical unclonable functions (sensor PUFs) and related sensor-binding primitives, including without limitation Rosenfeld, Gavas, and Karri, "Sensor Physical Unclonable Functions," in *2010 IEEE International Symposium on Hardware-Oriented Security and Trust (HOST)*, pp. 112–117, and successor sensor-PUF literature. Sensor PUFs extend the PUF challenge-response paradigm to sensing devices by binding the sensed physical value to a per-device unclonable fingerprint, providing authentication, unclonability, and verification of the sensed value at a single sensor. The witness-mesh primitive's distinguishing structural elements relative to sensor PUFs are:

- the multi-module joint-record structure: sensor PUFs bind a sensed value to a single physical sensor's fingerprint; the witness-mesh primitive binds the joint record of a plurality of mutually-coupled witness modules to the joint physical reproduction problem of Section 7.1.4;
- the inter-module coupling network and joint-trace verifier, neither of which is present in single-sensor microstructure-based authentication; and
- the public coupling commitment, in which the trained inter-module coupling network and the coupling graph $\mathcal{G}$ are public commitments to the protocol digest, vs. the typically-private per-device fingerprint of a sensor PUF.

Where a witness-mesh module's reactor incorporates per-device microstructure variability

of reactor-microstructure-unclonable as disclosed in Section 10.3, the resulting per-module reactor microstructure is one component of the witness-mesh joint hardness profile, complementary to but distinct from the joint-record structure of the witness-mesh primitive disclosed in this section.

Cyber-physical-systems attestation and dynamic watermarking. The witness-mesh primitive is also distinct from cyber- physical-systems attestation and dynamic-watermarking primitives, including without limitation Valente, Barreto, and Cárdenas, “Cyber-Physical Systems Attestation,” (2014), which proposes that a verifier introduce false control signals and compare the resulting plant dynamics against a model to attest the operational correctness of sensors and controllers; Valente and Cárdenas, “Remote Proofs of Video Freshness for Public Spaces,” in *Proceedings of the 3rd ACM Workshop on Cyber-Physical Systems Security & Privacy (CPS-SPC ’17)*, pp. 111–122, which uses physically-manifested visual challenges (such as QR codes or social-media-derived feeds in the camera’s field of view) to attest video-feed freshness for surveillance cameras monitoring public spaces; Hespanhol, Porter, Vasudevan, and Aswani, “Dynamic Watermarking for General LTI Systems,” in *56th IEEE Conference on Decision and Control (CDC)*, December 2017, pp. 1834–1839 (arXiv:1703.07760), which detects malicious sensor attacks on multi-input multi-output linear-time-invariant systems with partial state observations through asymptotic and statistical tests on injected private control signals; Satchidanandan and Kumar, “Dynamic Watermarking: Active Defense of Networked Cyber-Physical Systems,” *Proceedings of the IEEE*, 105(2):219–240, 2017; Porter, Hespanhol, Aswani, and co-authors, “Detecting Generalized Replay Attacks via Time-Varying Dynamic Watermarking” (2020) and successor dynamic-watermarking literature.

The cited literature establishes attestation or attack-detection guarantees for control systems through one or more of: verifier-injected challenge signals into a plant or sensor loop; statistical tests on resulting plant dynamics against a declared plant model; or physically-manifested challenges (such as visual challenges in a camera’s field of view) whose expected response is verifiable from the prover’s recorded output. The witness-mesh primitive shares with this literature the use of declared adversary models, statistical detection tests, and external challenges that manifest physically in the prover’s environment, but is distinguished by:

- (a) single-prover vs. multi-witness structure: the cited literature establishes attestation for a single plant, single sensor stream, or single camera. The witness-mesh primitive establishes joint attestation across a plurality of mutually-coupled witness modules with a joint convolution bundle record as the security primitive;
- (b) inter-module mutual coupling: the cited literature does not couple multiple sensors or plants through a continuous-analogue inter-module coupling network during capture. The witness-mesh primitive’s joint physical evolution under continuous-analogue inter-module coupling is not present in CPS attestation, video-freshness attestation, or dynamic-watermarking primitives;

- (c) public coupling commitment vs. private watermark or challenge: the cited dynamic-watermarking literature relies on a private excitation injected by the verifier that is not disclosed to the adversary, and the cited video-freshness literature relies on visual challenges whose expected manifestation is bound to the prover by visibility in the camera's field of view. The witness-mesh primitive's trained inter-module coupling network and coupling graph are public commitments to the protocol digest, with security derived from joint physical reproduction under public coupling rather than from challenge or watermark secrecy; and
- (d) the joint-trace verifier with declared coverage tuple and the per-version empirical hardness commitment of Section 7.1.10, neither of which is present in the cited literature.

The dynamic-watermarking literature is, in addition, the strongest cited art for attacks against the linear embodiment disclosed in Section 7.1.9 (linear-case reduction to multi-input multi-output system identification) and is referenced in that section accordingly. Where the witness-mesh primitive is deployed with linear per-module dynamics and linear coupling, the joint hardness contribution from coupled dynamics is, as disclosed in Section 7.1.9, evaluated against the cited system-identification and dynamic-watermarking literature; the preferred embodiments are non-affine, history-dependent, or high-dimensional, in which the cited literature's linear-case methods are insufficient to recover the joint dynamics.

Secure localisation, verifiable multilateration, and hidden base-station systems.

The witness-mesh primitive is also distinct from secure localisation and verifiable-multilateration primitives, including without limitation Čapkun and Hubaux, "Secure Positioning of Wireless Devices with Application to Sensor Networks," in *Proceedings of IEEE INFOCOM 2005*, vol. 3, pp. 1917–1928; Čapkun and Hubaux, "Secure Positioning in Wireless Networks," *IEEE Journal on Selected Areas in Communications*, 24(2):221–232, February 2006; Čapkun, Čagalj, and Srivastava, "Securing Localization with Hidden and Mobile Base Stations," in *Proceedings of IEEE INFOCOM 2006*; Čapkun, Rasmussen, Čagalj, and Srivastava, "Secure Location Verification with Hidden and Mobile Base Stations," *IEEE Transactions on Mobile Computing*, 7(4):470–483, 2008; Čapkun, Buttyán, and Hubaux, "SECTOR: Secure Tracking of Node Encounters in Multi-Hop Wireless Networks"; the distance-hijacking and multi-prover relay-attack literature; and successor secure-localisation work. The cited literature establishes secure position verification or secure proof-of-location of one or more provers through verifiable multilateration — in which multiple verifiers measure time-of-flight or signal-strength to a single prover and use distance-bounding to constrain the prover's position — or through hidden or mobile base stations whose positions are not disclosed to adversaries.

The witness-mesh primitive shares with this literature the notion that physical-layer timing constraints among multiple spatially-distributed parties can establish security-relevant properties of a prover, but is distinguished from this literature on grounds that include and extend the v4 distance-bounding distinctions:

- (a) single-prover-multiple-verifier vs. multi-witness joint attestation: the cited literature

establishes the position of a single prover from multiple verifiers' distance-bounding measurements. The witness-mesh primitive establishes a joint physical record across a plurality of mutually-coupled witness modules, in which each module serves both as a prover of its own observation of Σ_i and as a participant in the inter-module coupling that constrains peer modules' fast-loop responses;

- (b) inter-module mutual coupling vs. unidirectional distance-bounding challenges: the cited literature performs challenge-response distance-bounding from each verifier to the prover. The witness-mesh primitive's inter-module coupling is mutual and continuous-analogue across the evidence window, not a discrete challenge-response sequence;
- (c) joint physical evolution: the cited literature does not produce a joint physical record whose joint reproduction is the security primitive; verifier measurements are independent per-verifier distance estimates aggregated to localise the prover. The witness-mesh primitive's joint convolution bundle record under public coupling commitment is the security primitive of Section 7.1.4; and
- (d) the joint-trace verifier and per-version empirical hardness commitment, neither of which is present in the cited secure-localisation literature.

Controlled physical random functions and PUF-modelling-attack literature.

The witness-mesh primitive is also distinct from controlled physical random functions and the PUF-modelling-attack literature, including without limitation Gassend, Clarke, van Dijk, and Devadas, "Controlled Physical Random Functions," in *Proceedings of the 18th Annual Computer Security Applications Conference (ACSAC '02)*, December 2002, pp. 149ff.; Gassend, Clarke, van Dijk, and Devadas, "Silicon Physical Random Functions," in *Proceedings of the 9th ACM Conference on Computer and Communications Security (CCS '02)*, 2002, pp. 148–160; the controlled-PUF and certified-execution literature derived from these works; and the PUF-modelling-attack literature including without limitation Rührmair, Sehnke, Sölter, Dror, Devadas, and Schmidhuber, "Modeling Attacks on Physical Unclonable Functions," in *Proceedings of the 17th ACM Conference on Computer and Communications Security (CCS '10)*, 2010, pp. 237–249. Controlled physical random functions (CPUFs) are PUFs that can only be accessed via an algorithm physically bound to the PUF in an inseparable way, enabling certified execution and challenge-response authentication with the PUF under the bound algorithm. The PUF-modelling-attack literature demonstrates that PUFs can be characterised from collected challenge-response pair sets through machine-learning modelling attacks, motivating mitigations through nonlinearity, controlled access, and reduced challenge-response-pair exposure.

The witness-mesh primitive shares conceptual elements with controlled PUFs — specifically, the binding of an algorithmic transfer function (the inter-module coupling network) to physical primitives (the per-module reactors and detectors) in a manner that constrains the algorithmic function's evaluation by physical hardware — but is distinguished by:

- the multi-module joint-record structure rather than single-PUF challenge-response;
- the continuous-analogue inter-module coupling among physically separate witness

modules, with no analogue in single-device CPUF literature;

- the public coupling commitment, with the trained inter-module coupling network committed to the protocol digest as a public commitment, vs. the typically-private PUF challenge-response model in the cited CPUF literature; and
- the per-version empirical hardness commitment that explicitly evaluates the witness-mesh primitive against PUF-modelling-attack methodologies and successor surrogate-modelling literature, with the resulting empirical hardness estimate committed to the protocol digest.

The PUF-modelling-attack literature is, in addition, one of the declared adversary attack classes against which the witness-mesh primitive’s per-version empirical hardness commitment of Section 7.1.10 is, in some preferred embodiments, evaluated.

Multi-camera attestation and content-provenance systems. The witness-mesh primitive is distinct from multi-camera attestation and content-provenance systems, including the C2PA Content Credentials specification (which defines manifests, assertions, claims, claim signatures, content bindings, and provenance data) and related distributed attestation protocols. Such systems establish multi-witness signature schemes and tamper-evident provenance metadata over independently-captured records, but do not constrain the captured records by physical mutual coupling during capture: the witnesses sign independent records after capture, and the cross-witness consistency check is cryptographic rather than physical. The witness-mesh primitive constrains the joint capture itself through the continuous-analogue inter-module coupling, so that each module’s recorded response is physically determined in part by peer modules’ simultaneous responses, and joint reproduction requires solving the coupled physical-dynamics problem of Section 7.1.4 rather than separately reproducing multiple independent records.

Decentralized proof-of-location and witnessing-zone architectures. The witness-mesh primitive is also distinct from decentralized proof-of-location and witnessing-zone architectures, including without limitation: Brito, Hadachi, Kamm, and Norbistrath, “Decentralized Proof-of-Location systems for trust, scalability, and privacy in digital societies,” *Scientific Reports* 15:19808, 5 June 2025 (DOI: 10.1038/s41598-025-04566-4), which formalises a composable decentralized proof-of-location model based on fault-tolerant witnessing zones, integrating distributed digital signatures, distance-bounding protocols, and consensus mechanisms; and Brito, Castillo, Hadachi, Norbistrath, and Heiss, “Decentralized Proof-of-Location for Content Provenance: Towards Capture-Time Authenticity,” arXiv:2603.27883, 29 March 2026, accepted at the 5th International Workshop on Architecting and Engineering Digital Twins (AEDT 2026) and the companion proceedings of the 23rd IEEE International Conference on Software Architecture (ICSA 2026), which extends the witnessing-zone model with multiple independent observers collectively validating physical events to produce auditable evidence artifacts for capture-time content authenticity in cyber-physical settings.

The cited literature establishes verifiable claims of physical presence, spatio-temporal

synchronisation, collective attestation, and capture-time content authenticity through witnessing-zone architectures in which multiple independent observers (witnesses) attest to a prover's presence or to a physical event, with the witnesses' attestations aggregated through distributed digital signatures, distance-bounding, consensus mechanisms, or related cryptographic primitives. The witness-mesh primitive shares with this literature the multi-witness structure, the spatio-temporal binding of attestations to physical events, and the application to capture-time provenance of physical-world content, but is distinguished by:

- (a) continuous-analogue mutual coupling among witnesses during capture vs. independent observer attestation: the cited literature treats the witnesses as independent attesters whose individual attestations are aggregated cryptographically, with no mutual physical coupling among the witnesses' fast-loop responses during the evidence window. The witness-mesh primitive's modules are mutually coupled through the continuous-analogue, frame-free inter-module signal path of Section 7.1.2, such that no single module's record is independent of the joint record; each module's fast-loop response during the evidence window is physically determined in part by the simultaneous responses of peer modules through the inter-module coupling network. This is a structural property of the joint capture itself, not a post-capture cryptographic aggregation;
- (b) public committed inter-module transfer function vs. cryptographic signature aggregation: the cited literature aggregates per-witness attestations through cryptographic signatures, distance-bounding measurements, or consensus protocols; the trust model is rooted in cryptographic signature verifiability and consensus liveness assumptions among the witness population. The witness-mesh primitive's inter-module coupling network is a public commitment to the protocol digest, with the trust model rooted in the joint physical reproduction problem of Section 7.1.4 under public coupling, evaluated empirically per-version under Section 7.1.10;
- (c) sample-level joint-trace verifier with directed-edge coverage vs. aggregated witness signatures: the cited literature's verification step is signature-aggregation, distance-bound checking, or consensus verification over independent witness attestations. The witness-mesh primitive's verifier of Section 7.1.6 performs sample-level selective opening over committed joint records and checks per-edge directed causal consistency under the declared coverage tuple, with no counterpart in the cited decentralized proof-of-location literature;
- (d) per-version empirical-hardness commitment vs. consensus-based attestation: the cited literature's security derives from cryptographic signature verifiability, distance-bounding correctness, and consensus-mechanism liveness assumptions; security is structural under those assumptions. The witness-mesh primitive's security is deployment-indexed empirical hardness committed per-version against declared adversary classes, including the cross-version transfer-attack evaluation of Section 7.1.10, with no counterpart in the cited literature; and
- (e) closed-with-reality property at each module: the cited literature's witnesses observe the

prover or the physical event as external observers, but do not themselves operate a fast closed loop in causal feedback with the observed physical reality during capture. The witness-mesh primitive’s modules are each closed with their own portion of physical reality Σ_i during the evidence window through the per-module fast loop of Section 7.1.2, and the inter-module coupling is among these closed-loop modules rather than among external attestors.

The cited literature’s distance-bounding, consensus, and signature-aggregation primitives are themselves separately addressed in this disclosure: distance-bounding is distinguished in the distance-bounding subparagraph of this section, and signature-based attestation is distinguished in the multi-camera attestation and content-provenance subparagraph above. The present subparagraph addresses the higher-level architectural combination of these primitives in the witnessing-zone literature.

Distributed and network PUF aggregation. The witness-mesh primitive is distinct from distributed-PUF and network-PUF constructions in which per-device challenge–response pairs are aggregated from multiple devices in a sensor network. Such constructions typically aggregate independent per-device CRPs without mutual physical constraint via fast-loop coupling. The witness-mesh primitive requires fast-loop mutual coupling under the continuous-analogue, frame-free inter-module signal path; aggregation of independently-captured per-device records, even with mutual cryptographic signing, does not implement the witness-mesh primitive.

Sensor-fusion and multi-sensor-calibration systems. The witness-mesh primitive is distinct from sensor-fusion and multi-sensor-calibration systems that establish spatial and temporal consistency across multi-modal sensors. Such systems typically operate on independently-captured records and establish post-capture geometric or temporal consistency. The witness-mesh primitive instead imposes a physical mutual constraint during capture through the continuous-analogue inter-module coupling, and the joint record’s hardness against reproduction is a consequence of the coupled physical dynamics, not of post-capture geometric consistency checks.

7.1.8 Relationship to the two-dimensional hardness index

In the witness-mesh embodiment, the two-dimensional hardness index pair $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ of Section 12 is evaluated against an adversary attempting to forge the joint convolution bundle record across the network under the verifier of Section 7.1.6 with declared coverage tuple $(\rho_{\text{open}}, \kappa_{\text{edge}}, \delta_{\text{dec}}, \epsilon_{\text{amp}}, \epsilon_{\text{phase}})$ and verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$. Specifically:

- k_{dig}^* measures the minimal model capacity required to reproduce the network’s joint dynamics under the committed coupling, given a declared adversary observation budget over the network’s collective inputs and outputs across the coupling graph $\mathcal{G}$.

- m_{ana}^* measures the minimal physical emulation scale required to reproduce the network’s joint analogue response, including the propagation-delay-bounded inter-module signal exchange at the timing fidelity required to satisfy the optical-delay logical clocks of Section 14.2 and the declared latency budget of Section 7.1.3 at verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$.

In some embodiments, the joint hardness index of an N -module witness mesh is empirically estimated by training adversary models against full-network records and measuring the capacity required to reproduce the joint distribution at a declared threshold ε under the verifier of Section 7.1.6, in accordance with the procedure of Section 12. In some embodiments, joint hardness exceeds the sum of per-module hardnesses because the cross-module coupling under the committed inter-module coupling network introduces constraints that are not reducible to per-module characterisation; in some embodiments, joint hardness equals or approaches the sum of per-module hardnesses where the verifier reduces to per-module checks; the specific relationship is empirical and is committed to the protocol digest at each declared re-evaluation cycle.

7.1.9 Preferred non-limiting embodiments

Nonlinear per-module dynamics. In some preferred embodiments, each Reality Kernel module’s fast loop comprises per-module dynamics that are not reducible to a linear or affine transfer under the declared meter envelope, including without limitation the fluorescent and scattering bed of the 1D anchor (Section 7.2), the nonlinear-film, scattering-medium, or phosphor reactor media of the 2D anchor (Section 10.2), the SLM–scattering-medium–camera reactor of Section 10.4, and any of the reactor variants disclosed in Section 10.3 and subsequent reactor sections. The witness-mesh primitive operates with linear per-module dynamics and linear coupling, and the architectural primitive of Section 7.1.2 is not so limited; in linear embodiments, however, the joint dynamics of the coupled network are characterised by a linear time-invariant multi-input multi-output transfer with declared coupling graph and delays, and the joint hardness contribution from coupled dynamics in such embodiments reduces to that of multi-input multi-output system identification under the declared adversary observation budget, including without limitation the methods of Ho & Kalman (1966), Juang & Pappa (Eigensystem Realization Algorithm, 1985), Van Overschee & De Moor (N4SID subspace identification, 1994), Ljung (*System Identification: Theory for the User*, 2nd ed. 1999), and the dynamic-watermarking detection literature of Hespanhol et al. (2017) and successor work. Linear embodiments are accordingly included for architectural breadth and remain within the broader scope of the witness-mesh primitive, but are *not relied upon* for the stronger empirical security effect of the witness-mesh primitive. The stronger empirical security effect is preferentially associated with non-affine, history-dependent, high-dimensional, or heterogeneous embodiments; preferred embodiments are accordingly those in which per-module dynamics, the coupling network, or both are non-affine, history-dependent, and high-dimensional, such that the joint dynamics are not reducible to a linear MIMO transfer under

the declared meter envelope.

Nonlinear coupling network. In some preferred embodiments, the trained inter-module coupling network is itself nonlinear, comprising a frozen small neural network or other declared nonlinear deterministic transfer function from peer fast-loop outputs to per-module modulation input, implemented in analogue hardware where consistent with the continuous-analogue property of the inter-module signal path.

Amplitude modulation as canonical modality. In some preferred embodiments, the inter-module coupling network modulates the receiving module's input by amplitude, consistent with the live amplitude drive $A(t)$ of the 2D anchor of Section 10.2 and the per-sample reactor-detector gain G_{det} of the 1D anchor of Section 7.2 transposed to the inter-module case. Amplitude modulation is a canonical preferred modality but is not the only modality of the witness-mesh primitive; alternative modalities are disclosed in Section 7.1.2.

Phase, frequency, polarization, wavelength, and spatial-mode embodiments. In some preferred embodiments, the inter-module coupling network modulates the receiving module's input by phase, frequency, polarization, wavelength, spatial mode, or a combination thereof, as disclosed in Section 7.1.2. Such embodiments are within the scope of the witness-mesh primitive and may be selected based on the available physical layer, the carrier modality of the inter-module signal path, and the verifier's measurement modality. Phase-modulation embodiments are particularly useful where the inter-module carrier is coherent optical and the verifier has phase-resolving detection; frequency- and wavelength-modulation embodiments are useful for multiplexing multiple peer signals onto a shared inter-module carrier; polarization- and spatial-mode-modulation embodiments are useful for free-space inter-module links with polarization- or mode-resolving detection.

Heterogeneous module construction. In some preferred embodiments, the witness mesh comprises modules of heterogeneous construction with distinct physical reactor classes, distinct nonlinear regimes, and distinct spectral or temporal operating bands, contributing to the joint hardness index by requiring an adversary to model multiple distinct physical reactor classes simultaneously.

Spatial distribution. In some preferred embodiments, the witness mesh is deployed at multiple physical locations with non-overlapping or partially-overlapping observable scenes, and the joint convolution bundle record incorporates the spatial distribution of the network as part of the binding. In some embodiments, the network is collocated within a single enclosure with multiple modules observing different aspects or regions of a shared scene, and the binding incorporates the geometric relationship between modules.

Direct optical inter-module signalling. In some preferred embodiments, the inter-module fast-loop broadcast is implemented as direct optical signalling between modules, with propagation delay determined by free-space or fibre-optic geometry, and with the optical signal carrying the broadcast module’s fast-loop output amplitude, phase, frequency, polarization, or spatial mode (or processed feature thereof) in continuous analogue form, without packet framing or digital re-encoding.

Mixed-modality coupling. In some embodiments, the inter-module broadcast is implemented as a mixed-modality channel combining optical, RF, acoustic, or wired-electrical signalling, with each modality contributing distinct propagation-delay and fidelity characteristics that are committed to the protocol digest.

7.1.10 Lifetime, reconfiguration, and management

The witness-mesh security primitive is, like all empirical hardness claims in this disclosure, a time-indexed and attacker-indexed quantity per Section 12. The witness-mesh hardness at time t is meaningful relative to the adversary class available at time t , and degrades as adversary modelling capability improves and as observation of the network accumulates. Management of the witness-mesh primitive over time follows the framework of Section 12: periodic re-evaluation, network reconfiguration, query throttling, and fleet-level monitoring.

Coupling-network rotation. In some embodiments, network reconfiguration is performed by re-training the trained inter-module coupling network against declared benchmark adversary classes available at the re-evaluation date, freezing the result to a new committed parameter-family version, and committing the new version to the protocol digest. The rotation period is selected based on the empirical characterisation half-life of the prior coupling under the declared adversary observation budget, such that the time required for the declared adversary class to recharacterise the new coupling exceeds the period over which the prior coupling’s empirical hardness exceeds a declared policy threshold. The rotation period is committed to the protocol digest.

Cross-version transfer-attack evaluation (non-limiting). The empirical hardness of a rotated coupling-network version is, in some preferred embodiments, evaluated against an adversary that carries forward, from prior coupling versions, all of the following persistent characterisations to the extent these have been accumulated under the declared adversary observation budget:

- per-module reactor microstructure and transfer function estimates;
- analogue front-end transfer functions and detector noise models;
- module geometry and vantage configuration;
- scene priors and learned reconstruction models for each module’s Σ_i ;
- latency calibration estimates and per-edge propagation characterisations;

- verifier-threshold and tolerance estimates; and
- learned embeddings, surrogate state representations, and intermediate model components trained against the prior coupling that generalise to the rotated coupling.

The cross-version transfer-attack evaluation is committed to the protocol digest as part of the per-version hardness commitment of this section. Coupling-network rotation invalidates accumulated adversary characterisation of the prior coupling at the surface level only, providing a rotation mechanism analogous in part to key rotation in cryptographic systems; rotation is *not* asserted to erase, and does not erase, the persistent characterisations enumerated above, and the empirical hardness of the rotated coupling-network version is established against the cross-version transfer adversary. Persistent characterisation across coupling-network rotations is addressed by the additional management mechanisms of this section, including topology reconfiguration, per-version re-evaluation of the empirical hardness index against updated adversary classes, and physical reactor refresh where applicable.

Topology reconfiguration. In some embodiments, the coupling graph $\mathcal{G}$ is itself reconfigured at declared rotation cycles, with new edges added, removed, or re-weighted, and the new graph committed to the protocol digest. Topology reconfiguration is independent of coupling-network rotation and may be performed at distinct rotation cycles. In some embodiments, topology reconfiguration is implemented at the trained-parameter level (re-weighting existing declared physical paths) without physical re-routing of inter-module signal paths; in some embodiments, topology reconfiguration includes physical re-routing where deployment constraints permit, with the new physical paths' propagation and hardware latencies recharacterised and committed. In further embodiments, topology reconfiguration is implemented as *circuit-switched reconfiguration* of optical and electro-optical interconnects, in which switching elements (including without limitation MEMS mirrors, MEMS lens arrays, micromotor-positioned lens elements, optical-switch fabrics, beam-steering SLMs, and electronically reconfigurable apertures) are commanded to reconnect or redirect signal paths between modules without halting an active scan. In some embodiments, circuit-switched reconfiguration is synchronised to scan phase, so that reconfiguration events occur during declared low-information intervals of the scan trajectory; the duration and placement of such intervals are matched to the response time of the specific switching element, so that for fast-responding elements (including without limitation certain MEMS mirrors and blanking-synchronous apertures) reconfiguration may occur within flyback or blanking intervals, and for slower-responding elements (including without limitation certain spatial light modulators, lens arrays, and optical-switch fabrics) reconfiguration occurs within declared longer reconfiguration windows scheduled outside the active acquisition interval. The reconfiguration schedule, switch states, device-specific response times, and any intermediate transient measurements are committed to the protocol digest. In some embodiments, the circuit-switched topology is varied online across an episode under a declared reconfiguration policy that itself may be trainable or adaptively-controlled.

Module addition, removal, and quarantine. In some embodiments, modules are added to or removed from the witness mesh over the deployment lifetime under the enrolment, revocation, and quarantine framework of the networking section. Modules whose hardness or reliability drops below policy thresholds may be quarantined or removed and the coupling graph updated accordingly.

Per-version hardness commitment. In some embodiments, each committed deployment version — comprising a specific coupling graph $\mathcal{G}$, a specific configured deterministic inter-module coupling transfer function (optionally trained), specific per-module trained parameter families, a specific verifier timing fidelity $\sigma_{\tau}^{\text{ver}}$, and a specific verifier coverage tuple $(\rho_{\text{open}}, \kappa_{\text{edge}}, \delta_{\text{dec}}, \epsilon_{\text{amp}}, \epsilon_{\text{phase}})$ — is associated with a committed empirical hardness estimate $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ established at the declared re-evaluation cycle and committed to the protocol digest, so that downstream verifiers and policy components can act on the version-specific hardness profile. In some embodiments, the per-version hardness commitment additionally records, as committed artifacts in the protocol digest:

- the declared adversary classes against which the empirical hardness was evaluated;
- the declared adversary observation budget;
- the model families employed in the evaluation, including without limitation linear and subspace system-identification baselines, neural state-space and recurrent surrogates, reservoir-computing surrogates, generative-scene-reconstruction adversaries, and analogue-relay adversary models;
- identifiers for the training and evaluation corpora used in the evaluation, in some embodiments including cryptographic hashes of corpus contents committed to Π_{dig} ;
- training/evaluation split seeds, where applicable, so that the corpus partition is reproducible from the committed seeds;
- identifiers for model implementations, including without limitation source-code commit hashes, container image hashes, and software-environment manifests, so that the evaluation pipeline is reproducible;
- calibration transcripts for the verifier-side timing fidelity $\sigma_{\tau}^{\text{ver}}$, the analogue front-end latency, and the residual tolerances ϵ_{amp} and ϵ_{phase} ;
- attack success/failure logs across declared adversary attack classes;
- a replay protocol sufficient to re-run the verifier’s acceptance decision on a sample of opened windows from the evidence corpus;
- the acceptance thresholds applied at the verifier;
- declared confidence intervals on the empirical hardness estimate; and
- negative results material to the estimate, including declared adversary attack classes that succeeded under declared budgets.

The commitment of these artifacts supports auditability of the per-version hardness estimate by a third-party verifier or auditor, without obligating the implementer to a specific numerical hardness multiplier. The empirical hardness estimate is, consistent with Section 12, a time-indexed, attacker-indexed quantity; the per-version commitment records the conditions under

which the estimate was established at the declared re-evaluation cycle.

Control of physical operation through the per-version hardness commitment (non-limiting). In some embodiments, the committed per-version hardness profile is used to control physical operation of the witness mesh, including without limitation:

- selecting or updating the verifier coverage tuple $(\rho_{\text{open}}, \kappa_{\text{edge}}, \delta_{\text{dec}}, \epsilon_{\text{amp}}, \epsilon_{\text{phase}})$ in response to the empirical hardness profile of the deployment version;
- accepting or rejecting evidence-window records at the verifier under acceptance thresholds derived from the per-version hardness profile;
- triggering coupling-network rotation of Section 7.1.10 when the empirical characterisation half-life of the prior coupling falls below a declared policy threshold;
- triggering topology reconfiguration to add, remove, or re-weight coupling edges in $\mathcal{G}$ in response to the per-version hardness profile or to evaluation against cross-version transfer attacks;
- quarantining individual modules whose per-module contributions to the joint hardness fall below a declared threshold under the declared adversary class;
- disabling a deployment version whose committed evaluation no longer satisfies a declared policy threshold, pending re-evaluation against updated adversary classes; and
- gating downstream apparatus that consumes evidence-window records on the per-version hardness profile, such that downstream apparatus operates only on evidence-window records originating from deployment versions whose committed evaluation satisfies a declared policy threshold.

The per-version hardness commitment of this section accordingly operates as a control input to physical apparatus operations, including verifier configuration, coupling-network rotation, topology reconfiguration, module quarantine, deployment-version enablement, and downstream-apparatus gating, rather than as an abstract evaluation or policy artifact.

7.1.11 Cross-references

The witness-mesh primitive cross-references the following sections of this disclosure:

- Section 7.2 (1D anchor): the per-sample fast loop of the 1D anchor is a canonical instantiation of a single witness-mesh module's fast loop closed with reality, and the reactor-detector gain G_{det} is a canonical intra-module instantiation of a trained committed amplitude-modulation transfer function transposed to the inter-module case.
- Section 10.2 (2D anchor): the continuous live amplitude broadcast of the 2D anchor is a canonical instantiation of a single witness-mesh module's fast loop, with the live coupling-path discipline (no per-period wait, no runtime-adaptive element) directly transposed to the inter-module case.
- Section 10.3 (reactor-microstructure-unclonability variants): canonical instantiations of per-module reactor microstructure for the combined witness-mesh-plus-reactor-microstructure embodiments.

- Section 10.5 (CRT/phosphor analogue-memory extension) and Section 10.7 (dual-loop appliance combining a fast loop with a slow analogue-memory loop): optional extensions orthogonal to the witness-mesh primitive.
- Section 10.27 (intra-module multi-reactor coupling): structurally similar to the inter-module witness-mesh coupling but operates within a single Reality Kernel module's extended state X_t ; the witness-mesh primitive operates across modules with shared world state W_t .
- Section 12 (two-dimensional hardness index): provides the framework against which the joint witness-mesh hardness is evaluated.
- Section 14 (Networked Reality Kernels and Transitive Proof-of-Projection): the witness-mesh primitive is the principal refinement of the networked-modules framework.
- Section 14.2 (optical-delay logical clocks): provides the physical timing anchors against which the latency budget of Section 7.1.3 is evaluated.
- Section 14.4 (Merkle commitment and two-seed selective opening): provides the verifier protocol underpinning Section 7.1.6.
- Section 18.1 (multi-reactor network topologies): the coupling graph $\mathcal{G}$ of the witness mesh is a security-purposed instantiation of the multi-reactor topology framework.

7.2 Anchor Embodiment: The Bench-Top 1D Reality Kernel

Before developing the full formalism, a concrete embodiment is described to provide physical intuition. The bench-top 1D Reality Kernel is a minimal but complete implementation that illustrates the two-subsystem topology, the reactor-to-scene amplitude coupling, the trainable parameter families, and the joint-record hardness objective, in a form that can be built from inexpensive commercial components. This embodiment is non-limiting; a 2D generalisation with continuous real-time coupling is described in Section 10.2, and substrate and coupling variants are described in subsequent sections.

Mapping to embodiments (a) and (b). The 1D anchor description that follows in this section, including its per-sample $A(t)$ feedback path and its trainable parameter families, is a concrete realisation of the parallel-subsystem embodiment (a) of Section 4.2. The cascade embodiment (b) of Section 4.2 is illustrated in FIG. 4A by a heavy-stroke forward low-latency coupling arrow running from the scene detector into the UV laser of the reactor subsystem. The forward low-latency coupling, where present, is in addition to the per-sample $A(t)$ feedback described below, and may be implemented in any of the modalities recited in Section 4.2 (electrical direct, mixed-signal, optical gain-pumped, or optical direct relay), with its end-to-end latency declared and committed to the protocol digest. Bench-top instantiations of the 1D anchor described in this section, including the FIG. 4A configuration, typically realise the $A(t)$ feedback path electrically through the conditioning chain described below. The forward low-latency coupling, where engaged, is similarly typically realised elec-

trically in bench-top instantiations; fully-optical and hybrid optical-electrical realisations of either or both coupling paths are within the scope of this disclosure.

Physical layout. In one non-limiting configuration, the bench-top 1D Reality Kernel comprises two physically separate subsystems under a common controller 10:

- 5 1. A *scene subsystem*, open to the world under a declared eye-safety envelope, comprising a scene laser (for example an eye-safe infrared or visible laser), a scene steering mirror mounted on a single-axis micromotor servo, and a scene detector (for example a photodiode).
- 10 2. A *reactor subsystem*, sealed behind a declared safety envelope, comprising: a reactor container located within the sealed reactor envelope, for example an open-topped aluminium cup or can providing an opaque reflective interior, together with a heterogeneous bed within the container (for example mixed fluorescent glass beads or fragments, including without limitation uranium-glass (“vaseline glass”) elements where permitted by applicable safety and regulatory requirements, clear or frosted glass marbles, mirror-finished marbles, and crumpled aluminium foil); a reactor ultraviolet illumination source (for example a near-UV laser at about 405 nm) mounted on the side of the container; a tunable emitter-side focus lens mounted on its own micromotor servo, independent of the scanning mirror; a reactor steering mirror mounted on a separate single-axis micromotor servo; a tunable detector-side focus lens mounted on its own micromotor servo, independent of both the emitter-side focus and the steering mirror; a green-pass or long-pass optical filter; and a reactor detector (for example a photodiode) viewing the container interior through the detector-side focus lens and the filter.

In some embodiments, the reactor detector is mounted near the reactor illumination assembly, approximately co-aligned with the reactor ultraviolet illumination source’s direction of propagation, with precise optical-axis sharing not required. In some embodiments, the reactor detector is mounted at an angle to the specular axis. In some embodiments, these configurations are treated as anchor-equivalent declared detector geometries because the detector response is dominated by wavelength-shifted fluorescence selected by the green-pass filter, which is robust to alignment imprecision. Differences in signal level, glare rejection, or path-length statistics between the two configurations are characterised per embodiment and recorded in the meter envelope. The declared detector-path geometry is committed to the protocol digest.

In some embodiments, the reactor container is wrapped or coated with reflective and opaque layers to suppress stray light and to contain the bed. In some embodiments, shaking or otherwise re-configuring the bed between runs re-randomises internal microstructure, so that a fixed scan program applied to different internal states yields different but statistically related reactor responses. In some embodiments, a deliberate bed reconfiguration event is logged in the protocol digest, and any hardness, calibration, or enrolment claim made after reconfiguration is tied to the resulting reactor state and meter envelope.

Reactor substrate (non-limiting). In the 1D anchor, the reactor substrate is a shallow bed of mixed beads or fragments including fluorescent species. In some embodiments, the bed includes uranium-glass (“vaseline glass”) marbles as the primary fluorescent species, which upon near-UV excitation emit wavelength-shifted green fluorescence with a characteristic persistence time, providing both scattering from the bed’s reflective and refractive elements and wavelength-shifted emission from the fluorescent elements. In some embodiments, other fluorescent or luminous materials — for example europium-doped, terbium-doped, or manganese-doped fluorescent glass beads; stable phosphor beads; other fluorescent glass fragments; or fluorescent polymer beads — may be substituted or combined with uranium-glass, subject to applicable safety and regulatory requirements. The green-pass or long-pass filter in front of the reactor detector selects the wavelength-shifted emission band and rejects the UV excitation, so the reactor detector records a response dominated by fluorescence together with scattered light in the pass band. Implementations of the reactor substrate follow applicable safety, handling, and regulatory requirements.

Spherical, orb, and isotropic reactor geometries (non-limiting). In some embodiments, the reactor is configured as a spherical, ellipsoidal, or otherwise enclosed three-dimensional volume — non-limitingly termed an *orb* — within which the emitter and recorder occupy interior positions and the scene surrounds the module rather than facing it from a preferred direction. The orb interior includes mixed fluorescent, scattering, and absorbing media as in the 1D anchor cup embodiments (Section 7.2), arranged for approximately isotropic coverage of the surrounding scene from the module’s vantage. The emitter is one or more sources arranged for full-solid-angle interrogation (for example a polyhedral arrangement of laser-illuminated fibre couplers, a swept galvo with an enclosing diffuser-reflector boundary, or distributed microemitters embedded in the orb wall); the recorder is correspondingly an arrangement of photodetectors configured for isotropic collection. In PolieBot embodiments (Definition 25) and other mobile or fleet deployments where the scene cannot be confined to a fixed angular sector relative to the module, the orb geometry enables full-solid-angle observation and interrogation without mechanical re-aiming. Convolution-bundle commitment, protocol digests, and meter envelopes apply to the orb embodiment on the same footing as planar, cylindrical, and directional reactor geometries; the geometric variant does not by itself alter the verification or hardness disciplines that apply to other reactor configurations. Device-specific signatures arising from orb fabrication tolerances (sphericity, wall material inhomogeneity, interior medium packing geometry, emitter-and-recorder alignment) contribute to the 2D hardness index in the same way as in non-orb embodiments (Section 12).

7.2.1 Scene subsystem

In the 1D anchor, the scene subsystem comprises an eye-safe scene laser, a scene steering mirror on a single-axis micromotor servo driven by a scene scan profile $\text{scan}_s(t)$, and a scene detector. The scene laser’s instantaneous amplitude is driven by $A(t)$ from the reactor

subsystem (see Section 7.2.3). In some embodiments, the scene detector is a single detector (for example a photodiode) as the anchor default; a camera, a photodiode array, or other spatially resolved detector is a declared extension of the single-detector default. In some embodiments, the scene detector is coaxial with the scene emitter via a polarisation combiner or equivalent.

7.2.2 Reactor subsystem

In the 1D anchor, the reactor subsystem is sealed and comprises, on the excitation path and in optical order: the reactor UV laser, a tunable emitter-side focus lens on its own micromotor servo following a trained profile $f_{\text{emit}}(t)$ periodic in the scan period T , a reactor steering mirror on a separate single-axis micromotor servo driven by a reactor scan profile $\text{scan}_r(t)$, and the sealed container with its heterogeneous fluorescent-and-scattering bed. On the detection path, the reactor subsystem further comprises a tunable detector-side focus lens on its own micromotor servo following a trained profile $f_{\text{det}}(t)$ periodic in T , a green-pass or long-pass filter, and the reactor detector. The reactor UV laser is driven by a trained profile $A_{\text{emit}}(t)$ periodic in T ; $A_{\text{emit}}(t)$ probes the fluorescent and nonlinear response of the bed at each scan phase.

In some embodiments, the detection path and the excitation path are approximately co-aligned, with the reactor detector mounted near the UV laser and looking into the bed along a direction close to that of the UV illumination; in other embodiments, the detection path and the excitation path are angularly separated. In some embodiments, these configurations are treated as anchor-equivalent declared detector geometries because the green-pass filter selects the wavelength-shifted fluorescence response, which is robust to alignment imprecision. The declared path geometry is committed to the protocol digest. In some embodiments, the reactor detector is a single detector as the anchor default.

7.2.3 Per-sample discrete feedback coupling

In the 1D anchor, the scene and reactor subsystems are coupled by a per-sample discrete feedback loop rather than by a continuous broadcast. At each scan step t , the common controller:

1. commands the scene steering mirror to $\text{scan}_s(t)$ and the reactor steering mirror to $\text{scan}_r(t)$;
2. commands the reactor emitter-side focus to $f_{\text{emit}}(t)$ and the reactor detector-side focus to $f_{\text{det}}(t)$;
3. after the steering mirrors and focus actuators have reached their declared settled states, commands the scene laser to emit at amplitude $A(t)$ and the reactor UV laser to emit at $A_{\text{emit}}(t)$;
4. records the scene detector's and reactor detector's responses for step t , yielding $y_s(t)$ and $y_r(t)$;

5. the reactor-detector output $y_r(t)$ is passed through the declared deterministic coupling path, which applies the frozen committed gain to produce the next scene amplitude, $A(t+1) = G_{\text{det}}(y_r(t))$, where G_{det} is the trained reactor detector gain (see Section 7.2.4).

In some embodiments, the common controller sequences this fixed calculation and applies the result to the scene laser drive; the gain law G_{det} is not learned or adapted during the evidence window. In some embodiments, the coupling path — comprising the reactor photodetector, protective conditioning, and the frozen committed G_{det} — holds one sample of buffering inherent to the per-sample discrete feedback loop, and contains no multi-period storage, no runtime-adaptive element, and no runtime-learned element.

In some embodiments, the scene and reactor emit-and-record phases within a step are logically ordered but may overlap in actual timing; in some embodiments, the steering mirrors and focus actuators are required to be at their declared settled states before either detector records, to avoid motion and defocus artefacts. In some embodiments, G_{det} is trained offline and frozen for the committed deployment version. The latency from reactor photocurrent at step t to scene emitter modulation at step $t + 1$ is bounded by a declared settling time and is committed to the protocol digest.

Boot. In some embodiments, the first-sample amplitude $A(0)$ is initialised from one of the following declared sources: (a) a committed public constant, for example a declared offset into the decimal expansion of π ; (b) co-illumination from a distant reactor via the auxiliary noise co-illumination port of Section 7.2.8; or (c) any other declared constant or signal committed to the protocol digest. In embodiments using co-illumination as the boot source, the reactor detector samples the co-illuminated reactor response before the first scene emission, and the frozen committed G_{det} maps that sampled response to $A(0)$ via the same live coupling path used during deployment. In some embodiments, the boot source is recorded in the protocol digest so that a verifier can reproduce the loop from its initial condition.

7.2.4 Trainable parameter families

In the 1D anchor, six trainable parameter families are optimised jointly under hardware-in-the-loop training (see Section 7.2.6) against the hardness objective of Section 7.2.5. The count of six is an addressable-parameter inventory of this anchor embodiment and is not asserted as a guaranteed effective capacity, task-performance bound, or training-scale result; effective capacity, generalisation, and task performance are empirical and embodiment-specific. The trainable parameter families are:

1. The scene scan trajectory $\text{scan}_s(t)$, periodic in T .
2. The reactor scan trajectory $\text{scan}_r(t)$, periodic in T . In some embodiments, $\text{scan}_s(t)$ and $\text{scan}_r(t)$ are initialised identical at training start and are allowed to warp apart under training; the committed trained pair is recorded in the protocol digest.
3. The reactor emitter-side focus lens profile $f_{\text{emit}}(t)$, periodic in T . Controls the focusing of the reactor UV laser onto the fluorescent and scattering bed.

4. The reactor detector-side focus lens profile $f_{\text{det}}(t)$, periodic in T . Controls the region and depth of the bed over which the reactor detector integrates at each scan phase, and may be trained independently from $f_{\text{emit}}(t)$.
5. The reactor UV drive profile $A_{\text{emit}}(t)$, periodic in T , applied upstream of the bed. $A_{\text{emit}}(t)$ probes the substrate's intensity-dependent fluorescent and nonlinear response at each scan phase.
6. The reactor detector gain G_{det} , applied in the live coupling path between the reactor photodetector and the scene emitter drive. G_{det} may be a time profile $G_{\text{det}}(t)$ periodic in T , a nonlinear transfer function $G_{\text{det}}(I)$ acting on instantaneous reactor photocurrent I , or a combined $G_{\text{det}}(t, I)$. In some embodiments, G_{det} is allowed to be nonlinear, for example implementing soft compression at high photocurrents to prevent wipeout, or a dead-zone near zero. Where G_{det} is time-dependent, its phase origin is defined relative to the common scan period T and committed to the protocol digest.

In some embodiments, the trained values defining all six parameter families are committed to the protocol digest as part of the committed parameter-family version.

7.2.5 Hardness objective

In some embodiments, the trainable parameter families of Section 7.2.4 are optimised against a declared adversary family Eve, defined as strong as the available training budget allows; a stronger declared Eve produces a more robust trained configuration. In some embodiments, Eve's attack surface is the full joint record — scene observations, reactor observations, and their mutual consistency under the declared scan and feedback protocol. In some embodiments, Eve may attempt to forge either record or their conditional relationship, and may use white-noise-baseline probing of the public interface to characterise the channel. In some embodiments, the training objective is to maximise the capacity, training-data, and compute budget Eve requires to produce a joint record the legitimate verifier accepts.

In some embodiments, operational hardness claims for the anchor embodiment are quantified by the measured canonical hardness index pair $(k_{\text{dig}}^*(\theta; \varepsilon), m_{\text{ana}}^*(\theta; \varepsilon, r))$ under the declared adversary family and the declared meter envelope, as described in Section 12, evaluated on the reactor subsystem's fluorescent and nonlinear response, the per-sample feedback coupling, and the resulting joint scene-and-reactor record.

7.2.6 Training pathways

In some embodiments, the trainable parameter families are fitted by one or more of the following training pathways, which may be applied sequentially or in combination:

Stage 1 — reactor against white noise (anchor baseline, non-limiting). In some embodiments, the reactor subsystem is characterised in isolation under white-noise inputs, and the reactor-side parameter families — $\text{scan}_r(t)$, $f_{\text{emit}}(t)$, $f_{\text{det}}(t)$, $A_{\text{emit}}(t)$, and G_{det} — are fitted to maximise the capacity budget Eve requires to match the reactor's white-noise-

challenge response. In some embodiments, the scene scan trajectory $\text{scan}_s(t)$ is initialised from the reactor scan $\text{scan}_r(t)$, from a declared calibration scan, or from a terminated or dummy-scene training run, and is subsequently co-trained during scene-coupled or simulated-scene training (Stage 2 or later). In some embodiments, once the reactor-side parameter families and the scene scan are fitted through the relevant stages, the combined parameter-family version is committed and the device is deployed against arbitrary scenes the reactor has never seen, with the reactor serving as a signature attached to the scene illumination via the committed feedback loop. In some embodiments, the white-noise training regime is selected as a broad baseline stressor: Eve’s inability to match the reactor on broadband-noise challenges at the trained capacity budget is treated as evidence of reactor-side empirical hardness under the declared meter envelope. Deployment hardness for scene-driven operation remains evaluated on the committed joint record and reduces to the measured canonical hardness index pair under the declared adversary family.

Stage 2 — digital twin with simulated scenes (declared extension, non-limiting).

In some embodiments, a digital twin of the reactor subsystem is fitted from physical measurements, and the trainable parameter families are further optimised by running the digital twin against a library of simulated scenes. In some embodiments, Stage 2 training targets whole-system properties — task performance, attention allocation, joint-record hardness under scene-driven operation — that are not directly optimised by white-noise training alone. In some embodiments, each Stage 2 acceptance produces a new committed parameter-family version.

Stage 3 — fleet-scale real-time analog reinforcement learning (declared extension, non-limiting).

In some embodiments, once multiple devices are deployed, the trainable parameter families on individual devices or on fleet-coordinated subsets are further updated by real-time analog reinforcement learning using committed physical evidence from deployment. In some embodiments, Stage 3 updates are applied only between declared evidence windows, or within windows expressly declared as training windows. Each accepted update creates a new committed parameter-family version, including a new G_{det} fingerprint where applicable; verification or deployment windows using a given version treat that version as frozen for the duration of the window.

In some embodiments, the training pathway used to fit a given deployed device, the window type under which any update was accepted, and the resulting committed parameter-family version are recorded in the protocol digest.

7.2.7 Yoked operation (orthogonal declared extension)

In some embodiments, Yoked operation is an orthogonal declared extension of the anchor and may be composed with any of the training pathways of Section 7.2.6. In embodiments where direct transfer-entropy and conditional-Lyapunov-exponent estimates are available,

those meters are used directly per the formalism of Section 8.5. In some embodiments, a Yoked-state classifier is trained on the committed joint evidence record as a classifier-mediated declaration embodiment, especially for scene classes where direct two-sided state access is unavailable; the classifier embodiment does not replace the meter-based Yoked formalism of Section 8.5. In some embodiments, the trainable parameter families may be further optimised against scene distributions conditioned on the Yoked declaration signal (meter-based or classifier-based, as declared). In some embodiments, when this extension is engaged, the Yoked declaration method, and the classifier identity and thresholds where applicable, are committed to the protocol digest.

7.2.8 Noise co-illumination (declared extension)

In some embodiments, the sealed reactor envelope further includes an auxiliary noise co-illumination subsystem comprising a dedicated auxiliary light source (for example a low-power laser), an auxiliary scanning mechanism, and an RF input port. In some embodiments, when the noise co-illumination subsystem is engaged, noise light is delivered to the reactor container alongside the trained reactor UV drive $A_{\text{emit}}(t)$; the fluorescent and nonlinear substrate then mixes the trained drive with the co-illuminated noise, and the composite response is delivered through the green-pass filter to the reactor detector and onwards to the live feedback path.

Trainable parameters (local). In some embodiments, the auxiliary scanning mechanism and an auxiliary amplitude gain on the auxiliary light source are locally trainable against standard Eve-simulation training analogous to Section 7.2.5. In some embodiments, these two additional trainable parameter families are committed to the protocol digest when the extension is engaged, distinct from and additional to the anchor’s trainable parameter families.

External signal at the RF port (declared per deployment). In some embodiments, the RF port accepts a signal from one of the following declared sources: (a) a white-noise generator, for example during training-time reactor characterisation; (b) a committed noise profile stored in the protocol digest, for deployments requiring reproducible injected noise; or (c) an external runtime input, for example the output of one or more other devices of the same class or a fleet-level coordination signal. In some embodiments, the external signal itself is not trained by the local device; only the auxiliary scan law and auxiliary amplitude gain, which determine how the external signal is projected into the reactor container, are locally trainable.

Tap point for inter-device sharing (non-limiting). In some embodiments where option (c) is used, the specific tap point on the sharing device from which the external signal is drawn — for example the sharing device’s reactor photocurrent pre- G_{det} , or its coupling-path

output $\text{post-}G_{\text{det}}$ — is committed to the protocol digest on both the sharing and receiving devices.

Relationship to the anchor’s trained regime. In some embodiments, when the noise co-illumination subsystem is engaged during Stage 1 reactor training, the trained reactor is trained in the noise-present regime in which it will be deployed. In some embodiments, the noise co-illumination subsystem also provides a boot source for the first-sample amplitude $A(0)$, as described in the Boot paragraph of Section 7.2.3.

Direct coupling through declared optical feedthroughs (non-limiting). In some embodiments, ambient or external optical signals — including signals from other deployed devices of the same class — may enter the reactor envelope through declared optical feedthroughs, filtered ports, or characterised stray-light paths that remain within the declared safety envelope. Such coupling is separate from the auxiliary noise co-illumination subsystem described above and is addressed by the envelope-management and meter-envelope provisions described in connection with the protocol digest and Section 10.2.

7.2.9 CRT/phosphor analogue-memory extension (declared extension)

In some embodiments, the 1D anchor is further extended by the CRT/phosphor analogue-memory extension of Section 10.5, in which the reactor detector output (or a derived signal) is written to a CRT whose XY deflection is locked to a declared scan coordinate. In the 1D anchor, the default coupling between the reactor detector and the CRT drive is electrical rather than optical, because the green-pass filter architecture enforces wavelength separation between the reactor’s ultraviolet drive and its wavelength-shifted green-fluorescent response, and optical addition of the reactor output onto the CRT would muddy that separation. Optical CRT addition is a declared variant within this extension for embodiments in which the wavelength allocation is compatible.

Mapping to core primitives (1D anchor). In the 1D anchor: the *scene emitter* is the eye-safe scene laser; the *reactor emitter* is the sealed reactor UV laser; each subsystem has its own *scanner* (single-axis micromotor mirror); the *reactor stack* comprises, on the excitation path, the tunable emitter-side focus lens and the sealed container with its fluorescent and scattering bed, and on the detection path, the tunable detector-side focus lens and the green-pass filter; each subsystem has its own *detector*; and the *controller* generates the scan profiles, the reactor drive profile, and the feedback sequencing. The *live coupling path* comprises the reactor detector, protective conditioning, and the frozen committed reactor detector gain G_{det} . The *protocol digest* records the scan profiles, the emitter-side focus profile $f_{\text{emit}}(t)$, the detector-side focus profile $f_{\text{det}}(t)$, the declared detector-path geometry, the reactor drive profile, the reactor detector gain (including phase origin for time-dependent G_{det}), the boot source for $A(0)$, the declared window type, any deliberate bed reconfiguration events, and the

committed parameter-family version. The *meter vector* includes emitter-side and detector-side focus-actuator settling indicators, mirror settling indicators, detector SNR, filter pass-band calibration, geometry-difference characterisation, and boot-state indicators.

7.3 System Architecture

7.3.1 System overview

In the most general terms, a Reality Kernel module includes an emission subsystem, a detection subsystem, one or more physical paths coupling emission to detection (optionally through a reactor medium and/or an external scene), and a controller. The controller executes a control protocol $U_{0:T}$ that specifies, at each time step, the emission configuration (scan coordinate, intensity, wavelength, polarisation, phase, pattern, timing, and gating), the detector configuration (exposure, gain, spectral band, integration window, gating), and any reactor-medium controls (bias, pump power, temperature, mechanical position).

The emission subsystem may include, without limitation, lasers, LEDs, broadband sources, CRTs, projectors, spatial light modulators, and antenna arrays. The detection subsystem may include, without limitation, photodiodes, cameras, photomultipliers, avalanche photodiodes, spectrometers, bolometers, and microphone arrays.

The controller produces a convolution bundle $C_{0:T}$ by pairing each emission event with the corresponding detection result. Each convolution-bundle sample $c_t = (u(t), \mathbf{y}_t)$ records both what was done and what was observed, under a shared timestamp from a common timebase.

In some embodiments, the controller also maintains and updates a reactor latent $Z_\theta(t)$ and logs protocol digests, meter summaries, and optional commitments. The full evidence package for a run includes the convolution bundle, protocol digest, meter envelope, and any commitments or selective-opening artifacts.

7.3.2 Extended state, world kernel, and measurement map

In some embodiments, the system is modelled as a stochastic dynamical system with the extended state X_t (Section 6), evolving under a world transition kernel $\mathbf{P}_{\text{world}}(X_{t+1} \mid X_t, u(t))$ and observed through a measurement map $\mathbf{y}_t \sim p(\cdot \mid X_t, u(t))$. This factorisation is a modelling convenience; the physical system does not necessarily separate into these components, and the factorisation may differ across embodiments. In multi-device settings (Section 14), W_t denotes the shared external world state common to all devices, while X_t denotes the per-device extended state including internal degrees of freedom; W_t is a component of each device's X_t but not identical to it.

7.3.3 Layered optics, memory media, and skip paths

In some embodiments, the optical path between emitter and detector includes multiple layers, each contributing dynamics, memory, or transformation. Non-limiting layers include: scat-

tering plates, nonlinear thin films, phosphor or luminescent persistence layers, fibre-delay loops, liquid-crystal or photorefractive layers, and engineered diffusers.

In some embodiments, skip paths (analogous to residual connections in neural networks) allow a portion of the signal to bypass one or more layers, creating a reference channel that preserves information about the input while the main path undergoes transformation. Skip paths may be implemented optically (beam splitters, fibre couplers) or electronically (signal tapping before and after a processing stage).

In some embodiments, layers can be inserted, removed, or rearranged without changing the Markov-kernel abstraction or the convolution-bundle format, provided the protocol digest and meter envelope are updated to reflect the current configuration.

7.4 Coupling Modes

7.4.1 Direct coupling

In direct-coupling embodiments, the emitter illuminates an external scene and the detector observes the scene’s response directly. The scene and reactor may be in the same optical path (the emitted light passes through the reactor, illuminates the scene, returns through the reactor, and reaches the detector) or in separate paths (scene and reactor are independently illuminated and observed, and the controller correlates their outputs).

7.4.2 Confocal and descanned coupling

In some embodiments, confocal or descanned optical paths provide axial sectioning, spatial filtering, or point-by-point correspondence between emission and detection, improving signal-to-noise ratio and spatial resolution.

7.4.3 Camera-obscura coupling

In some embodiments, an external scene is imaged onto an internal screen or surface (for example via a pinhole, lens, or relay system), and the Reality Kernel module interacts with the projected image rather than the scene directly. This provides environmental isolation while preserving scene-dependent modulation of the reactor’s behaviour.

Reality-side spectral filter (non-limiting). In some embodiments, a spectrally-selective filter is positioned between the external scene and the internal screen, such that the reality-to-screen channel is constrained to a declared spectral band Λ_{ext} . In a canonical embodiment, Λ_{ext} is the visible RGB band. The reality-side filter has two load-bearing functions: (i) it constrains the spectral signature of the projected image to a band whose properties are committed to the protocol digest Π_{dig} , and (ii) it constrains the sensing channel to be unidirectional in the spectral sense, as further described below. The filter parameters (centre wavelength, bandwidth, transmission profile, blocking ratio outside band) are committed to the protocol digest as part of the sensing-layer parameter family.

IR-veracity scan (non-limiting).

In some embodiments, the Reality Kernel module further includes an active interrogation channel that illuminates its own internal screen with a probe beam at a declared wavelength λ_{IR} outside Λ_{ext} , in a canonical embodiment an infrared wavelength selected so that λ_{IR} lies in the blocking band of the reality-side filter. The descanned detector of Section 7.4.2 reads the scattered, reflected, or re-emitted response from the screen at λ_{IR} , producing a *veracity signal* $V_{\text{IR}}(t)$ that recovers geometric, textural, and material properties of what is physically present on the screen, independently of and complementary to the visible-band image $I_{\text{ext}}(t)$ formed by reality. The veracity signal supports detection of tampering, of projection-driver mismatch, and of missing or substituted physical substrate at the screen. The IR-veracity scan parameters — probe wavelength, scan pattern, detector configuration, and the rule by which $V_{\text{IR}}(t)$ is compared against $I_{\text{ext}}(t)$ to produce a verdict — are committed to the protocol digest. The probe wavelength, scan duty cycle, and verdict rule are non-limiting; other interrogation wavelengths and modalities (for example ultraviolet, terahertz, structured polarisation, or acoustic probing where consistent with the screen substrate) may be substituted provided the same load-bearing properties of (i) lying outside Λ_{ext} and (ii) being blocked by the reality-side filter are preserved.

One-way isolation of the sensing channel (non-limiting).

The reality-side filter is positioned such that emissions originating within the Reality Kernel module at wavelengths outside Λ_{ext} — including in particular the IR probe of the IR-veracity scan at λ_{IR} — are blocked from reaching the external scene through the sensing channel by the filter's blocking band, by ordinary spectral selectivity and optical reciprocity. No claim of nonreciprocal optical isolation is asserted; emissions originating within the module at wavelengths inside Λ_{ext} (where present) are reciprocally transmitted by the filter and are addressed by the disclosure-minimisation and emission-budget policies committed to the protocol digest rather than by the spectral filter alone. The reality-to-screen channel is unidirectional in the spectral sense: reality drives the screen within Λ_{ext} , and the module's internal interrogation at λ_{IR} does not exit through the same channel. This one-way property is structural to the filter geometry and is independent of whether any actuator subsystem of the Reality Kernel module is connected to or disconnected from external actuation. The one-way property is committed to the protocol digest as a sensing-layer property and is verifiable by external measurement against the declared filter blocking ratio.

External driveability of the screen and kernel-into-kernel docking (non-limiting).

In some embodiments, the internal screen of the camera-obscura coupling is configured so that, in addition to being driven by an external scene via the reality-side aperture and filter, it may be driven by a separate external apparatus delivering a projection within Λ_{ext} in place of the physics-driven scene image. Where the separate external apparatus is itself a Reality Kernel of the kind disclosed herein, the arrangement implements a *kernel-into-kernel docking* configuration in which a first Reality Kernel module senses through its camera-obscura sensing layer a projection rendered by a second Reality Kernel module. Spectral

matching of the rendered projection to Λ_{ext} ensures that the first module’s visible-band sensing chain cannot distinguish a docked-kernel-driven projection from a physics-driven projection by spectral signature alone. The IR-veracity scan provides the complementary distinguishability instrument: the docking kernel either matches the IR-veracity properties of the physical screen substrate, in which case the docked module’s veracity signal is consistent with physics, or deliberately fails to match them, in which case the docked module’s veracity signal reports the mismatch. Both matched and unmatched IR-veracity regimes are non-limiting embodiments; the choice between them is a configuration parameter of the docking arrangement and is committed to the protocol digest of the docking session. The kernel-into-kernel docking arrangement is disclosed here as an apparatus-level property of the camera-obscura sensing layer and does not depend on any particular use of the docked configuration.

7.4.4 Quantum-sensitive coupling (optional)

In some embodiments, the detection subsystem includes single-photon detectors, photon-number-resolving detectors, or coherent-state receivers, enabling sensitivity to quantum optical properties of the channel (for example photon statistics, quantum correlations, or entanglement witnesses). These capabilities are optional and do not affect the core Markov-kernel abstraction.

7.4.5 Optical Anchor and Continuous Physical Anchor Coupling

In some embodiments, two or more Reality Kernel layers, modules, partitions, or cascade levels are coupled by a continuous physical anchor. The continuous physical anchor provides a bounded-latency physical signal path by which a first layer obtains, challenges, samples, stimulates, witnesses, or constrains a second layer. The anchor is not merely an after-the-fact log comparison; it is a runtime-coupled apparatus pathway whose signal state, timing budget, and anomaly response are declared in, or bound to, the protocol digest.

In a non-limiting optical embodiment, the continuous physical anchor comprises an emitter, an optical path, a modulator or framing element, a target or reactor interface, a detector, a synchronisation source, and a controller. The emitter may emit at any wavelength or band, including ultraviolet, visible, near-infrared, short-wave infrared, mid-infrared, long-wave infrared, terahertz, or a combination thereof. The emitted signal may comprise structured illumination, pulsed illumination, phase modulation, amplitude modulation, polarisation modulation, wavelength modulation, spatial modulation, temporal coding, frequency coding, spread-spectrum coding, challenge-response coding, or any combination thereof. The detector may measure intensity, phase, time-of-flight, spectrum, polarisation, spatial distribution, temporal waveform, fluorescence, phosphorescence, scattering response, speckle response, or another response of the anchored layer.

In some embodiments, an anchor challenge a_t is derived from a protocol state and emitted into the anchored path:

$$a_t = A_\eta(\Pi_t, C_{0:t}, q_t, s_t),$$

where q_t may comprise a challenge, beacon value, timestamp, ledger-derived value, or prior digest, and s_t may comprise a local state of the anchor controller. A measured anchor response b_t is then received:

$$b_t = B_\xi(a_t, X_t, \epsilon_t),$$

where X_t denotes the physical state of the anchored layer or interface and ϵ_t denotes noise, drift, environmental variation, or measurement uncertainty. An anchor meter may compute:

$$m_t^{\text{anchor}} = M_{\text{anchor}}(a_t, b_t, \Pi_t, \Delta t_t),$$

where Δt_t is the measured or bounded age of the anchor response. These equations are illustrative only; any challenge, response, witnessing, synchronisation, or continuous-coupling function may be used.

In some embodiments, the protocol digest commits to an anchor-latency budget. The budget may include one or more of propagation time, emitter rise time, modulator delay, detector integration time, detector electronics delay, analogue-conditioning delay, digital processing delay, queueing delay, decision-window length, synchronisation uncertainty, clock drift, jitter allowance, stale signal age, path-length bounds, or maximum permitted missing-sample duration. The apparatus may reject, flag, quarantine, down-weight, or escalate a signal when an anchor response is absent, stale, delayed, inconsistent, saturated, clipped, spoof-like, out of distribution, or outside a declared meter envelope.

In some embodiments, the continuous physical anchor supports cascade security by making a successful attack on one layer insufficient unless the attacker also satisfies the bounded-latency physical coupling constraints at one or more other layers. The anchor thereby provides a simultaneity-of-attack constraint complementary to bottleneck-minimum per-layer hardness. The local hardness of a layer and the cross-layer anchor condition are separate security properties: the former concerns the difficulty of compromising a layer in isolation, while the latter concerns maintaining physically consistent coupling across layers within the declared timing and meter envelope.

In some embodiments, anomaly handling for the continuous physical anchor is graduated. A first response may increase monitoring cadence, collect additional diagnostic samples, or enter a soft-gate diagnostic mode. A second response may restrict release of governance signals, freeze parameter updates, suspend one or more influence paths, or require selective opening. A third response may disconnect a cascade interface, enter a rescue or containment mode, require a governance authority signature, or trigger mandatory escalation for hard-floor proximity. The response ladder, confidence thresholds, false-positive handling, retry policy, and recovery conditions may be declared in the protocol digest.

Canonical optical-anchor embodiment (non-limiting). In one non-limiting canonical embodiment, the continuous physical anchor comprises a near-infrared LED or laser-diode emitter operating in a declared band, positioned at a declared standoff from a reactor inter-

face or anchored-layer surface; a modulator applying a declared challenge code (for example a spread-spectrum-coded amplitude pattern with a committed code length and chip rate); an optical path of declared geometry and declared maximum path-length variation; a detector comprising a photodiode and transimpedance amplifier with declared bandwidth and noise
5 floor; a synchronisation source comprising a stable oscillator with declared frequency, phase-noise, and clock-drift parameters; and a controller that derives the per-cycle challenge a_t from the protocol state and computes the per-cycle anchor meter m_t^{anchor} . The declared end-to-end anchor latency budget comprises emitter rise time, modulator delay, optical propagation time, detector integration and electronics delay, and decision-window length, all committed
10 to the protocol digest. The graduated anomaly response of this canonical embodiment is: first response, increase sampling cadence and collect additional diagnostic samples; second response, restrict release of governance signals and freeze parameter updates; third response, disconnect the anchored cascade interface and require a governance authority signature for re-engagement. The wavelength, modulation code, latency budget, response ladder, and other
15 parameters of this canonical embodiment are illustrative and non-limiting; other declared parameter choices are within the scope of the continuous physical anchor as generalised in the following paragraph.

Anchor-modality generalisation (non-limiting). Although optical anchor embodiments are described, the continuous physical anchor generalises to any bounded-latency
20 physical coupling having anomaly-as-detection-signal properties. Non-limiting examples include optical coupling at any wavelength, acoustic coupling at any frequency, ultrasonic coupling, electromagnetic coupling, radio frequency coupling, microwave coupling, terahertz coupling, electrostatic coupling, magnetic coupling, particle-beam coupling, electron-beam coupling, photon-counting coupling, thermal coupling, mechanical vibration coupling, flu-
25 idic coupling, and hybrid multi-modal coupling. The anchor may be continuous, sampled, challenge-response, event-triggered, periodic, asynchronous, or multi-rate, provided that the declared coupling and timing constraints are bound to the protocol digest.

7.5 Scanning and Addressing

In some embodiments, one or more scanning elements address spatial or angular coordinates
30 over time. A scan program is represented by scan coordinates $\mathbf{s}(t)$ and associated control signals, and the system logs enough information to map time indices to calibrated scene coordinates and reactor coordinates.

Scan coordinates and calibration (non-limiting). In some embodiments, $\mathbf{s}(t)$ denotes a 2D or 3D scan coordinate, for example $(x(t), y(t))$ on a raster, $(\theta_x(t), \theta_y(t))$ for galvanometer
35 angles, or a higher-dimensional coordinate that includes focus or wavelength selection. A

calibration map M relates commanded coordinates to physical coordinates:

$$\mathbf{s}_{\text{phys}}(t) = M(\mathbf{s}_{\text{cmd}}(t), \mathbf{m}(t)),$$

where $\mathbf{m}(t)$ is a meter vector capturing alignment, gain, drift, and stability indicators.

Helical and volumetric scan laws (non-limiting). In some embodiments, $\mathbf{s}(t)$ includes an internal polar or cylindrical coordinate such as $(\rho(t), \varphi(t))$ together with an axial or focus coordinate. The controller chooses how (ρ, φ, f) evolve over time, and calibrated optics map these emitter-side coordinates to trajectories through a reactor volume. In a non-limiting cylindrical reactor embodiment, planar probes in (ρ, φ) combined with modulation of f yield approximately helical trajectories in reactor coordinates, providing a volumetric sweep that remains characterisable and effectively injective at the sampling timescale. The scan-law angular coordinate $\varphi(t)$ is unrelated to the RT style parameter $\phi \in \Phi$ defined in Section 8.4.4.

Addressing and synchronisation (non-limiting). In some embodiments, addressing links a timebase to scene and reactor coordinates. A controller maintains a common clock that ties: projector emission indices, detector sample indices, scan mirror positions, and optional rolling-shutter timing models. Synchronisation error is captured in meters and used to gate verification, trigger recalibration, or downgrade to a safer regime.

Carrier-modulated and demodulated probing (non-limiting). In some embodiments, control and/or authentication signals are encoded by modulation of a carrier rather than at baseband, including amplitude, frequency, phase, or polarisation modulation of an optical, acoustic, or RF carrier. Demodulation is performed by matched filtering, lock-in detection, or other synchronous detection aligned to a logged carrier schedule. Carrier parameters and demodulation settings are recorded in the protocol digest.

AOD raster and electronic scanning (non-limiting). In some embodiments, scan coordinates are implemented by electronically steered optics, including acousto-optic deflectors (AODs), electro-optic deflectors, MEMS scanners, galvanometers, spatial light modulators, or phased arrays. In a non-limiting AOD embodiment, beam deflection is controlled by radio-frequency drive signals and the scan program is a time series of commanded frequencies and amplitudes.

Mixed and hierarchical scanning schemes (non-limiting). In some embodiments, scanning is mixed or hierarchical, combining coarse trajectories with fine trajectories. Non-limiting examples include coarse rasters that locate regions of interest followed by local high-density scans, or slow patrol scans interleaved with fast verification micro-sweeps. Such schemes support attention-like scan policies in Limager and Reality Transform regimes and adaptive challenge-response sweeps in Truth Beam.

Rolling shutter and temporal mapping (non-limiting). In some embodiments, a detector uses rolling shutter and each row (or block) has an exposure interval. The protocol digest includes the exposure model, and mapping functions associate each observed sample with a scan coordinate and an emission interval. In some embodiments, rolling-shutter readout is configured with row inversion, row/column pairing, and/or windowing so that exposure windows align to scan dwells.

Global shutter and exposure-window integration (non-limiting). In some embodiments, a detector uses global shutter so that pixels integrate over a common exposure interval. In some embodiments, global-shutter acquisition is combined with rolling-shutter acquisition (for example a rolling-shutter sensor for fine temporal attribution and a global-shutter sensor for robust integrated baselines).

Tunable distortion elements and e-beam deflection (non-limiting). In some embodiments, the optical path between a memory surface (including without limitation a CRT phosphor screen, scanned-laser target, or persistence medium) and the observing detector includes a fixed or tunable distortion element, comprising for example a steering mirror, cylindrical lens, tunable lens, electrowetting or liquid-crystal lens, mechanically rotated optic, anamorphic prism pair, or programmable diffractive element. In tunable embodiments, the distortion element is driven by one or more analogue or digital control parameters (for example rotation angle, drive voltage, piezo extension, or applied magnetic field), allowing the geometric relationship between the memory surface coordinate system and the detector coordinate system to be varied continuously or in declared steps over time, with the drive schedule committed to the protocol digest. In further embodiments specific to CRT and related electron-beam reactor substrates, the electron beam itself is subject to one or more tunable distortion controls applied before the beam reaches the phosphor, including without limitation: auxiliary electric deflection plates and electrostatic field configurations applied in addition to the primary deflection yoke; auxiliary magnetic deflection coils, focus coils, astigmatism-correction coils, and convergence coils with independently controllable currents; quadrupole, sextupole, and higher-order multipole electron-optical elements; tunable accelerating-voltage and beam-current modulation; and time-varying yoke drive applied as a trainable kernel parameter rather than only as a static raster control. Each such electric or magnetic e-beam distortion control contributes additional trainable or adaptively-controlled degrees of freedom to the reactor and, together with the optical distortion element described above, provides a tunable spatial-and-angular mapping between displayed or addressed memory coordinates and detected coordinates. The drive schedule, control parameters, and any commanded changes across an episode are committed to the protocol digest and convolution bundle on the same footing as other reactor controls. In some embodiments, small natural variations in deflection-yoke winding, electric-plate spacing, residual magnetisation, and shim placement contribute to the reactor-microstructure unclonability signature (Section 12). In some embodiments, the disclosed electric and magnetic e-beam distortion controls

are implemented in purpose-built electron- optical assemblies rather than as modifications to commodity or aged CRT hardware; in such embodiments, operating envelopes (including without limitation accelerating-voltage range and slew rate, deflection-coil and electrode drive bandwidths, beam-current limits, and safety interlocks) are declared in the protocol digest and constrained by the physical capabilities of the specific implementation. In other
 5 embodiments, a subset of the disclosed controls is implemented as a bounded retrofit of commodity or hobbyist CRT hardware, with the available control envelope correspondingly narrower and likewise declared.

Operating modes A/B/C (non-limiting). In some embodiments, the same hardware

10 supports multiple scanning and tuning modes. Mode A (settled raster): scan coordinates follow a settled $N \times M$ raster with dwell and settle allowances, supporting repeatable calibration. Mode B (sampled continuous): scan coordinates follow a continuous trajectory (for example Lissajous, spiral, or other smooth curves) and the convolution bundle is sampled under logged timing. Mode C (pure analogue tuning): kernel updates are derived from analogue
 15 proxy signals and applied as actuator-safe set-point changes without explicit digital gradients. Mode transitions are explicitly marked in the protocol digest and meter summaries. Modes A/B/C are scan and tuning modalities that may be used within any of the three operating regimes (Truth Beam, Limager, Reality Transform); regime selection determines the objective applied to the resulting convolution bundle, while mode selection determines
 20 how the scan coordinates and tuning parameters evolve during acquisition.

Terminology note: uses of “mode” in this document. This document uses the term “mode” on three independent axes: (i) *scan/tuning modality* (Modes A, B, C above), which governs how scan coordinates evolve; (ii) *coupling mode* (static vs Yoked, Section 8), which governs device–scene feedback depth; and (iii) *training configuration* (adversarial, coopera-
 25 tive, or mixed training stance, and lifecycle state such as Embodied/Docked/Offline, Section 16.9), which governs the training context during calibration. These three axes are orthogonal: any combination of scan modality, coupling mode, and training configuration is in principle valid and is recorded in the protocol digest.

Lissajous galvanometer trajectories and focus modulation (non-limiting). In

30 some embodiments, galvanometer angles follow a Lissajous scan law:

$$\theta_x(t) = r_x \sin(2\pi f_x t + \phi_x), \quad \theta_y(t) = r_y \sin(2\pi f_y t + \phi_y),$$

where (r_x, r_y) are amplitudes, (f_x, f_y) are scan frequencies, and (ϕ_x, ϕ_y) are phases. In some embodiments, focus is treated as a primary trainable driven in synchrony with the scan, for example a focus control $\zeta(t) = \zeta_0 + \kappa_{\text{focus}}(t)$, and the focus schedule is logged so that meter scores condition on the correct focus modulation regime.

Bidirectional addressing surfaces (non-limiting). In some embodiments, addressing surfaces are bidirectional: they can be written and read. Non-limiting examples include CRT or phosphor surfaces (with XY deflection and intensity drives) and other media that preserve a spatiotemporal state.

5 7.6 Multi-Channel Observation and Invariance Profiles

In some embodiments, observations are multi-channel and include intensity, spectral channels, polarisation, time-of-flight or phase, and other modalities. The system may also record reduced-dimensional summaries, field-valued measurements, or feature vectors derived from raw observations and meters.

10 **Field-valued and reduced-dimensional readouts (non-limiting).** In some embodiments, an observation at time t is a field $\mathbf{y}_t(\mathbf{s})$ over scan coordinates, while in other embodiments it is a reduced-dimensional vector $\mathbf{y}_t \in \mathbb{R}^d$ obtained by sampling, pooling, or feature extraction. The choice of readout is logged as part of the protocol digest.

Observation layers (non-limiting). In some embodiments, the system represents ob-
 15 servations using a layered vocabulary: (i) raw stream (sensor outputs as produced by transducers), (ii) conditioned stream (raw outputs aligned to sweep/challenge context with provenance and calibration identifiers), (iii) projection stream (primitive measurement values such as samples, windowed statistics, or demodulated correlator outputs), (iv) feature stream (derived representations for meter evaluation or inference), and (v) signature stream (compressed artefacts intended for storage, comparison, or verification). These layers are concep-
 20 tual rather than mandatory stages.

Sampling, multi-rate acquisition, and alignment (non-limiting). In some embodiments, different modalities operate at different sampling rates and may be asynchronous. The conditioned stream includes timestamps, sweep indices, and alignment metadata, and
 25 the system performs resampling and time-alignment so that cross-channel statistics are computed under a consistent timebase.

Noise, drift, distortion, and injection taxonomy (non-limiting). In some embodiments, meters account for non-idealities including: measurement noise (shot, thermal, amplifier, quantisation), timing and sampling distortion (jitter, clock drift, latency, missing sam-
 30 ples), drift and calibration changes, nonlinearities and saturation, dispersion and memory artefacts, crosstalk and interference, environmental perturbations, and adversarial injection or tampering. Meter vectors include indicators for these categories so that verification and hardness claims are explicitly conditioned on the observed operating envelope.

Detector pooling and compressive detection (non-limiting). In some embodiments, detector-side pooling compresses high-dimensional optical interactions into one or a few channels, enabling compressive operation with single-pixel or near-single-pixel readouts. Non-limiting examples include pooling a compact photodiode array to a single time-continuous stream, aligning rolling-shutter integration windows to scan dwell timing, and using global-shutter frames as integrated baselines.

MIMO transforms (non-limiting). In some embodiments, the module is operated as a multiple-input multiple-output (MIMO) system with multiple emitters, multiple detectors, and multiple observed channels. The system may estimate or learn an effective mixing model that maps emission-control degrees of freedom to observed channels, and use such a model to design protocols that concentrate task-relevant information into a reduced set of channels while still logging the raw multi-channel record for auditability.

Multi-camera observation of a shared reactor (non-limiting). In some embodiments, a single reactor or scene output is observed by multiple detectors in parallel via beam splitters, fibre splitters, or relayed imaging paths. Each detector may use a different shutter mode, spectral bandpass, polarisation analysis, aperture, exposure, or viewpoint, yielding channels with different invariance profiles. In some embodiments, the continuous detector stream produced by a swept scan functions as an analogue “video feed” that can be fanned out to multiple downstream processing or modulation paths.

Invariance profiles (non-limiting). In some embodiments, the system characterises an invariance profile that describes how observations change under controlled transformations, for example changes in scan density, wavelength, polarisation, focus, or viewpoint. A non-limiting goal is to design protocols so that task-relevant quantities are stable to known nuisance transformations while remaining sensitive to authenticity-relevant microstructure.

Fractional Fourier transform as a concrete multi-channel example (non-limiting). In some embodiments, a tunable optical configuration realises a family of transforms parameterised by an order α_{FFT} that interpolates between spatial and Fourier-like representations.

Multi-channel constraints for verification and sensing (non-limiting). In some embodiments, the verifier uses cross-channel consistency constraints, requiring that multi-spectral or multi-polarisation observations agree with a shared underlying scene model and with meter-bounded stability.

7.6.1 Channel separation and side-channel mitigation (non-limiting)

Channel tags (non-limiting). In some embodiments, each separated processing pathway is assigned a channel tag u_{ch} that is bound into the protocol digest and challenge record

and propagated into meter inputs and convolution-bundle metadata. Channel tags enable channel-specific replay detection, audit, and calibration.

Multiplexed channel separation (non-limiting). In some embodiments, multiple logical channels share the same physical medium but are separated by time-division, frequency-
 5 division, and/or code-division multiplexing. Channel assignments are logged in the protocol digest and may be rotated periodically.

Slots and guard intervals (non-limiting). In some embodiments, per-channel emission and scoring are scheduled in fixed-duration slots with guard intervals sized to measured timing jitter. The device pads or equalises processing so that each slot occupies constant
 10 wall-time, mitigating timing side-channels.

Power equalisation (non-limiting). In some embodiments, instantaneous power draw is equalised across slots by adding controlled dither, dummy operations, or compensating illumination, so that power-consumption traces are less informative about channel identity or kernel state.

Shutter-based channel isolation (non-limiting). In some embodiments, electro-optical shutters are placed at strategic points in the optical path, including pre-reactor shutters, inter-reactor shutters between cascaded stages, and pre-detector shutters. Shutter timing protocols enable channel separation for baseline maintenance, style-reset sequences, reference-versus-transform acquisition, and differential measurements.

Dual-camera reference channel (non-limiting). In some embodiments, a beam splitter feeds a primary detector and a secondary detector that serves as a reference channel. The secondary detector may include controllable shutter glasses to acquire only designated time slots or calibration windows.

8 Operating Regimes

A Reality Kernel supports three operating regimes that share the same physical operator but differ in what they optimise. Each regime answers a different question about the scene through the physical channel:

Truth Beam (Illumination)

The device interrogates the scene and asks: *who are you?* In some embodiments, the
 30 optimised quantity is Fisher information about the device’s configuration parameters $I(\theta)$, making forgery detectable under bounded adversary resources.

Limager (Perception)

The device attunes itself to the scene and asks: *what is out there?* In some embodiments,

the optimised quantity is mutual information $I(S; Y)$ between scene variables and the convolution-bundle output.

Reality Transform (Creation)

The device acts on the scene and says: *make it look like this*. In some embodiments,
 5 the optimised quantity is the divergence $d(Y, Y^*)$ between output and target, subject to meter envelopes.

The parenthetical labels (Illumination, Perception, Creation) are non-limiting mnemonic descriptors and are not alternative regime names; the canonical regime names are Truth Beam, Limager, and Reality Transform throughout this specification.

10 These three regimes can operate in two coupling modes: *static* and *Yoked*. In static operation (the default for most embodiments), the scene is treated as having fixed or slowly varying properties, and the device senses, verifies, or renders based on snapshot or time-averaged observations. In Yoked operation, the feedback loop is tuned so that device and scene are driven toward a coupled dynamical state: the convolution bundle encodes not only
 15 what the scene *is* but how it *responds* to being driven, and the resulting joint dynamics are empirically assessed via meters rather than asserted *a priori*. Yoked operation adds a dynamic dimension to any regime: Yoked Limager senses dynamical scene properties such as resonances, damping, and transfer functions; Yoked Truth Beam uses the dynamics of scene-reactor coupling as a security primitive; Yoked Reality Transform renders patterns
 20 that entrain with scene dynamics.

In some embodiments, a single run traverses multiple regimes and coupling modes (for example static verification followed by Yoked sensing), with regime changes and coupling-mode transitions explicitly marked in the protocol digest and meter summaries.

Reactor complexity is application-dependent (non-limiting).

Not all applications
 25 require empirical hardness. The three-regime framework and convolution-bundle data model apply across the full reactor complexity spectrum, from identity (standard camera/projector) through linear media (gain stages, colour filters, LTI systems with memory) to strongly non-linear reactors (saturation, bistability, chaotic mixing, and ultrafast Kerr-effect switching in epsilon-near-zero media as described in Section 10). Empirical hardness—the difficulty of
 30 forging convolution-bundle traces—arises from reactor nonlinearity and manufacturing microstructure, and is relevant primarily to Truth Beam and to applications requiring provenance or attestation. A Limager operating with a linear or identity reactor is a sensor whose convolution-bundle format, meter infrastructure, and protocol-digest logging still apply; the device simply offers no forgery resistance. A Reality Transform with a well-characterised
 35 linear medium can perform physical computation—optical matrix multiplication, analogue inference, or physical gradient propagation—where *fidelity* of the input-output relationship matters more than unclonability. A multi-reactor network may combine linear and nonlinear elements, using linear subsystems for computation and nonlinear subsystems for signing or attesting the result. The degenerate cases in Section 8.9 define baselines for hardness claims;

they are also legitimate operating configurations for applications where hardness is not the objective.

8.1 Operational Stances: Iris, Eve, Demeter, and the Harmonia Coupling Modifier

The three operating regimes are not merely distinct objective functions: each is the stable deployed result of a generative process governed by a characteristic *operational stance*, and each continues to operate in that stance once deployed. Three primary stances are recognised, plus a coupling modifier (Harmonia), named after archetypes that recur throughout this specification.

Definition 30 (Operational stance). *An operational stance is a meta-objective that governs how a Reality Kernel loop selects and weights its primary objective at any given moment. Three primary stances and one coupling modifier are defined:*

Harmonia (coupling stance — Yoked modifier)

The loop is oriented toward joining: it tends the feedback between device and scene toward a coupled state in which neither partner's output can be fully understood without the other. Harmonia does not replace any of the three primary stances; she is a modifier applied across the simplex Δ_2 , orthogonal to the Iris–Eve–Demeter axis. Any point on Δ_2 may be operated in the Harmonia modifier: an Eve-hardened Truth Beam operated under Harmonia becomes a channel whose forgery resistance is amplified by the joint attractor structure; an Iris-expanded Limager operated under Harmonia encodes how the scene responds to being driven, not merely what it is; a Demeter-tended Reality Transform operated under Harmonia entrains scene dynamics toward the target rendering rather than projecting onto a passive surface. Harmonia is the stance of the yoke itself: the bond that transforms two separate systems into a coupled whole whose joint evidence is stronger than the sum of its parts. The corresponding coupling modifier is the yoke depth $\alpha \in [0, 1]$, with $\alpha = 0$ the static (unmodified) case and $\alpha \rightarrow 1$ the maximum commanded coupling depth (Yoked classification is meter-derived from the CLE/TE co-condition, not from α alone). A distinction is drawn between the declared target α_{target} (the coupling strength the controller schedules and commits to in the protocol digest) and the achieved $\hat{\alpha}$ (a meter-derived empirical quantity computed from the convolution bundle). The protocol digest commits to the target range and estimator methodology; the bundle provides the measurements from which $\hat{\alpha}$ is estimated.

Iris (information-maximising stance)

The loop is oriented toward maximum clarity about what is actually present in the scene or channel. No adversarial agenda and no growth agenda are imposed: the objective is to resolve the world as completely as the physical channel permits. Iris does not judge what she sees; she transmits it with full fidelity. The information-maximising stance produces

and characterises the Limager regime. The corresponding regime weight is λ_L on the simplex Δ_2 ; the pure Iris stance is the vertex $(0, 1, 0)$.

Eve (adversarial-hardening stance)

The loop is oriented toward worst-case robustness: it selects challenges, probes, and operating points that maximise distinguishability between genuine physical response and adversarial forgery under a declared attacker family and meter envelope. Eve does not punish; she hardens. A scene-reactor loop hardened by the Eve stance becomes a Truth Beam: a channel whose empirical security is calibrated by systematic adversarial pressure at every layer of the architecture (Section 9). The corresponding regime weight is λ_{TB} ; the pure Eve stance is the vertex $(1, 0, 0)$.

Demeter (managed-development stance)

The loop is oriented toward appropriate development of the scene or agent: it expands capability and rendering when the system is stable and flourishing, and contracts toward stabilisation when risk is elevated. Demeter is bidirectional and care-giving; she tends toward a target rather than interrogating or maximising. The default expression of the Demeter stance applied to a scene is stable stylistic projection mapping: controlled rendering that maintains fidelity to a target style while adapting to scene state. A scene-reactor loop trained by the Demeter stance becomes a Reality Transform. The corresponding regime weight is λ_{RT} ; the pure Demeter stance is the vertex $(0, 0, 1)$.

Stances, regimes, and the simplex (non-limiting). The three primary stances correspond exactly to the three vertices of the regime-weight simplex $\Delta_2 = \{(\lambda_{TB}, \lambda_L, \lambda_{RT}) : \lambda_i \geq 0, \sum_i \lambda_i = 1\}$ already defined in Section 6. A mixed stance — any interior point of Δ_2 — produces a regime that simultaneously exhibits properties of two or three canonical stances: for example a loop with high λ_{TB} and non-zero λ_L interrogates the scene adversarially while also extracting maximal information about scene structure. The Yoked modifier (Section 8.5) applies independently of stance mixture: any point on Δ_2 may be operated in static or Yoked coupling mode. The Harmonia stance names and governs this orthogonal dimension: where Iris, Eve, and Demeter govern *what objective* the loop pursues, Harmonia governs *how deeply the loop is coupled to its scene partner*. The full operational state of a Reality Kernel session is therefore characterised by a point on Δ_2 (the primary stance mixture) together with a yoke depth $\alpha \in [0, 1]$ (the Harmonia dimension).

Stances as generative processes and as ongoing operation (non-limiting). Each stance is simultaneously (i) the *training or configuration process* that produces a deployed regime, and (ii) the *ongoing operational character* of that regime at runtime. A deployed Truth Beam was Eve-hardened during configuration and continues to operate in the Eve stance: its runtime challenge selection, acceptance-boundary calibration, and adversarial re-evaluation all express the same adversarial-hardening principle that shaped its initial configuration. Similarly, a deployed Limager was Iris-expanded and continues to operate in

the Iris stance, maximising information gain at every step. A deployed Reality Transform was Demeter-trained and continues to operate in the Demeter stance, bidirectionally tending scene state toward the target rendering.

Stances at every scale (non-limiting). The same three primary stances (plus the Harmonia coupling modifier) apply at every scale of the Reality Kernel architecture. At the *device level*, they select the operating regime as described above. At the *protocol level*, challenge selection, sweep design, and disclosure policy may be governed by any stance or mixture. At the *fleet and training level*, the same stance vocabulary applies to agent training trajectories: adversarial training configurations (Eve stance), cooperative training configurations (Demeter stance), and undirected exploration (Iris stance) are vertex instantiations of the same simplex. The regime-weight simplex Δ_2 , together with the Harmonia coupling dimension α , thus serves simultaneously as a coordinate system for device operation, protocol design, and training configuration.

8.2 Truth Beam (Verification Regime)

Illumination. Truth Beam is the regime of radical transparency: the device interrogates the scene with the full force of its physical channel, and the question it asks is the simplest and hardest—*was this produced by a genuine physical device under logged conditions?* Like a beam of light that reveals what it touches without being changed by it, Truth Beam maximises the sharpness of the likelihood function around the true physical state. The device emits structured probes, reads the scene’s response, and evaluates whether the response is consistent with a genuine physical interaction or could have been fabricated by a bounded adversary.

Scope of “Truth” (non-limiting). Throughout this document, *Truth Beam* and related terminology (“was this produced by a genuine physical device?,” “verification,” “Semantic Truth Beam”) denote **provenance-consistent physical interaction under declared threat models**, not semantic truth adjudication of depicted content. What the mechanism certifies is a causal provenance chain: that a convolution-bundle trace was produced by a specific physical device, under logged protocols, within declared meter envelopes, and is consistent with genuine physical interaction rather than bounded-adversary fabrication. It does not certify—and should not be read as asserting—that an event depiction is unstaged, unselectively framed, or free of strategic construction by the scene’s participants. “Real-but-misleading” outcomes (genuine physical traces of deliberately arranged scenes) survive Truth Beam verification by design, because the device certifies the physics of the interaction, not the intent behind it. The Semantic Truth Beam extension (Section 8.2.3) evaluates consistency of a semantic claim q with committed evidence, which is a provenance and consistency check on the claim’s evidential support, not an oracle for propositional truth.

In Truth Beam embodiments, the convolution bundle is used for authenticity assessment.

A verifier evaluates whether an observed record is consistent with the physical channel induced by the device, the logged protocol, and the meter envelope, and rejects records that are likely to be forged, replayed, or materially altered.

In some embodiments, Truth Beam uses a verisimilitude or consistency model operating on the convolution bundle or derived features, optionally together with meter vectors and protocol digests, to produce an accept or reject decision, a confidence score, or a continuous hardness score. In some embodiments, this score is used for verification-gated actions, selective opening escalation, or trust-weight updates.

8.2.1 Noise and drift envelope (non-limiting)

In some embodiments, Truth Beam explicitly models an operating envelope that includes shot noise, read noise, and slow drift. A non-limiting measurement model for a scalar sensor channel is

$$y_t = (1 + \delta_t) \mu_t + n_t^{\text{shot}} + n_t^{\text{read}},$$

where μ_t is the expected response under the protocol and scene, δ_t is a slow multiplicative drift term, n_t^{shot} is a shot-noise component, and n_t^{read} is a read-noise component. In some embodiments, shot noise is modelled as a Poisson-like component whose variance scales with signal level, while read noise is modelled as an approximately additive Gaussian component dominated by electronics and digitisation.

In some embodiments, drift is modelled as a slowly varying process, for example

$$\delta_{t+1} = \rho_{\text{AR}} \delta_t + w_t,$$

with ρ_{AR} near one and w_t a small disturbance term. In some embodiments, meters include drift indicators, gain estimates, timing alignment indicators, and stability flags so that verification claims are conditioned on an observed envelope rather than asserted unconditionally.

In some embodiments, an extended noise model includes slow multiplicative drift combined with affine jitter:

$$y_t = a_t(\mu_t + b_t) + n_t,$$

where a_t is a slow gain drift and b_t is a small offset jitter, both estimated or bounded by meters, and n_t is a non-limiting residual noise term.

8.2.2 Hardness and attacker models (non-limiting)

In some embodiments, hardness is reported as an empirical difficulty of producing a forged record that passes the verifier under a bounded attacker model. For example, an attacker may be bounded by query budget, compute budget, model class, or training data access.

In some embodiments, these notions connect to the digital hardness index $k_{\text{dig}}^*(\theta; \varepsilon)$ defined elsewhere in this document, with hardness claims interpreted as conditioned on the protocol digest, meter envelope, and a stated attacker family.

In some embodiments, hardness is strengthened by serial composition of challenges, protocol randomisation, cross-channel constraints, and selective opening checks that force consistency not only in local windows but also across time and across coupled channels.

8.2.3 Semantic Truth Beam (non-limiting)

In some embodiments, Truth Beam is applied to hard inference verification, where the object being verified is a semantic claim or inference output rather than only raw sensor traces. For example, a device or agent may assert a claim q about a scene (a label, a structured description, a detected event, or a property estimate) together with a commitment to the evidence window and protocol digest used to produce that claim.

In some embodiments, a semantic claim is verified by routing challenges to physically grounded evidence that is relevant to the claim. For example, the verifier selects protocols or openings that are conditioned on the claim type, the claimed region of interest, or a semantic routing rule, and then checks consistency of opened windows with the claim, the protocol digest, and the meter envelope. In some embodiments, the same mechanism is used to verify that an inference is not a post hoc fabrication by requiring that the claim be bound to committed evidence prior to the opening seed.

In some embodiments, semantic Truth Beam includes explicit grounding mechanisms that bind semantic outputs to committed evidence and make certain classes of semantic forgery more difficult:

1. *Semantic routing through the physical kernel*: candidate semantic claims (or compact digests thereof) are re-encoded as emission patterns and passed back through a Reality Kernel run, and the resulting response is compared against a reference derived from the original measurement or claimed evidence window.
2. *Inference commitment*: a semantic extraction head commits to intermediate artifacts (for example attention maps, feature activations, derivation traces) using commitment mechanisms described herein.
3. *Spatiotemporal grounding*: each semantic claim is annotated with specific scan coordinates, time indices, and/or disclosed atoms in the convolution bundle that support the claim.
4. *Adversarial semantic hardness*: semantic extraction heads and verifiers are trained adversarially so that producing an alternative plausible-looking semantic description that passes the verifier is empirically difficult for a bounded adversary.

In some embodiments, projection hardness is assessed via a challenge–commit–open flow tied to emission digests: a verifier issues a challenge seed; the device deterministically selects an emission schedule under the seed-driven protocol, commits to an emission root R^{emit} , executes the physical run, commits to an observation root R^{obs} , and later opens selected emission and observation atoms with inclusion proofs for meter-conditioned consistency checks. The commitment binds the full protocol digest (including scan-law parameters and emission schedule), the declared meter envelope $\mathcal{M}_{\text{accept}}$, and the protocol seed(s) used to derive the

emission schedule (following the two-seed structure of Section 14.4 in some embodiments), so that openings can reconstruct and verify the full run configuration.

8.2.4 Graduated verification levels (non-limiting)

In some embodiments, the same device supports multiple verification security levels by controlling which regions of the configuration space Θ the challenge policy probes:

Level 0 (ambient).

Static challenge-response with no feedback. The device behaves as a microstructure challenge-response oracle (in the sense used in the PUF literature). The challenge policy operates in low-curvature regions of the attractor bundle where attractor geometry changes slowly with θ . This level is fast, cheap, and suitable for low-security applications.

Level 1 (trajectory).

A short closed-loop trajectory in a moderate-curvature region. The verifier checks attractor basin identity over a brief feedback window. In some embodiments, spoofing would require matching transient dynamics, not just steady-state responses.

Level 2 (bifurcation).

The challenge policy sweeps near a known bifurcation surface in Θ . Fisher information is concentrated near bifurcations, so spoofing would require tracking the exact topology change. Hardness increases sharply with proximity to the bifurcation set B .

Level 3 (holonomy).

A full cyclic sweep around a bifurcation, with accumulated geometric phase measured from the convolution bundle. If coexisting attractors give rise to non-Abelian holonomy, the accumulated phase is path-dependent and is conjectured to be significantly harder to forge from finite surrogate data, though the precise scaling with path complexity remains an open problem. Holonomy-based hardening is optional and is not required for core Truth Beam, Limager, or Reality Transform operation; Level 1 and Level 2 verification are independently enabled.

In some embodiments, the verification level is selected per session based on the required security level, and the selected level is recorded in the protocol digest. In some embodiments, levels are composed sequentially (a fast Level 0 gate followed by a slower Level 2 deep check if the initial gate passes). The cost of verification scales with the curvature of the interrogation path through Θ , not with device complexity: a single device serves all levels.

8.2.5 Honey-pot and canary verification (non-limiting)

In some embodiments, the verification protocol deliberately operates in a regime where a specific attacker model class appears to succeed, while a secondary check detects the attack:

1. The verifier presents a low-curvature challenge that a linear or low-order surrogate model can match.
2. Simultaneously, the verifier monitors higher-order structure in the convolution bundle (for example transfer entropy spectra, speckle higher-order statistics, or frequency-resolved causal signatures).
3. A genuine device passes both checks. An attacker’s surrogate passes the first but fails the second because the higher-order structure is expected to require, under the declared attacker families, reproducing the full nonlinear device dynamics.
4. The pattern of what the attacker matches and what the attacker fails to match reveals the attacker’s model class and capacity.

In some embodiments, honeypot verification inverts the usual security strategy: instead of making every challenge maximally hard, some channels are deliberately easy in order to create a diagnostic surface that characterises attacker capabilities. The diagnostic output is logged in the meter envelope.

8.2.6 Anti-Truth Beam: provenance negation (non-limiting)

In some embodiments, the system provides a complement to positive verification: the ability to establish, under declared attacker families and calibrated error tolerances, that an observation was *not* produced by any device in a declared family.

In some embodiments, the attractor bundle of a device family defines a region in observable space. Anti-Truth Beam checks whether a submitted observation lies outside this region. The boundary of the region is determined by the bifurcation set B —the same structure that makes positive verification hard also makes negative verification operationally feasible, because the bifurcation set partitions the observable space into topologically distinct basins. As with all empirical hardness claims in this document, negative-provenance decisions are conditioned on the current attacker class and fleet-calibrated acceptance/error rates, and are time-indexed rather than unconditionally durable (see Section 12).

Non-limiting applications include: detecting synthetic or manipulated media that was not produced by a physical Reality Kernel (deepfake detection); determining that a submitted measurement did not originate from an enrolled device (provenance negation); and identifying devices whose physical configuration has changed (for example through tampering or ageing that moves the device’s operating point across a bifurcation boundary).

8.3 Limager (Perception and Sensing Regime)

Perception. Limager is the regime of receptivity: the device attunes itself to the scene and asks *what is out there?* Like a still pool whose surface reflects the sky precisely because it does not impose its own shape upon it, Limager maximises the information flowing from scene to output. The device actively selects scan laws, integration times, coherence regimes, and attention policies to maximise the information flowing from scene to output, much as a listening ear turns toward a sound source.

In Limager embodiments, the Reality Kernel is used for perception and sensing. The system selects protocols and processes the convolution bundle to infer latent scene variables, reconstruct geometry or materials, estimate state, or optimise information acquisition. Limager can operate with or without supervised labels, and supervised labels may be used at any granularity. In some embodiments, Limager is operated as a standalone boundary-state estimation and calibration method (Section 8.3.7), independent of verification or rendering objectives.

8.3.1 Latent scene variables and inference targets (non-limiting)

In some embodiments, the inferred variable is a latent scene description ξ (distinguished from the extended state X_t ; see Section 6), for example geometry, pose, reflectance parameters, spectral parameters, or a learned representation. Given observations $C_{0:T}$ and a protocol $U_{0:T}$, inference is performed by a model that estimates ξ or a distribution over ξ , optionally conditioned on meters and protocol digests.

8.3.2 Information maximisation objective (non-limiting)

In some embodiments, Limager selects or adapts protocols to maximise information gain about ξ under meter and safety constraints. A non-limiting objective is:

$$\max_{P_{U_{0:T}} \in \mathcal{U}} I_{\theta}(\xi; C_{0:T} \mid U_{0:T}),$$

where $\mathcal{U}$ encodes admissible protocols (for example eye-safety bounds, thermal bounds, scan-rate bounds, and stability bounds). In some embodiments, tractable proxies are used, for example mutual-information lower bounds, predictive uncertainty reduction, or reconstruction-error reductions measured on committed windows.

Sparse, compressed-sensing, and plug-and-play reconstruction (non-limiting).

In some embodiments, Limager reconstruction is performed by sparse or compressed-sensing recovery under declared sparsity, incoherence, or measurement-matrix assumptions (Candès–Tao 2006; Donoho 2006), by model-based iterative reconstruction, or by plug-and-play priors (Venkatakrishnan, Bouman, and Wohlberg 2013) in which a denoiser or score-like prior is inserted into the reconstruction loop. The reconstruction algorithm, regularisation weights, denoiser identity, training-corpus digest, stopping rule, and residual diagnostics are committed to the protocol digest, and reconstructions used for verification are gated by the same meter and selective-opening policies as other Limager-derived outputs.

Geometric reconstruction and registration tools (non-limiting). In some embodiments, Limager or digital-twin state estimation composes standard geometric estimation tools, including structure-from-motion, visual odometry, simultaneous localisation and mapping (SLAM), bundle adjustment, iterative closest point (ICP), photometric registration,

optical-flow estimation, or differentiable rendering-based pose refinement. These tools act as optional estimators of scene geometry, pose, correspondence, or motion, while the inventive apparatus remains the committed, meter-bounded Reality Kernel loop. The estimator family, feature representation, correspondence policy, robust-loss function, outlier rejection rule, and validation residuals are recorded with the derived-scene-state identifier.

8.3.3 Joint exposure and flash control for HDR imaging (non-limiting)

In some embodiments, camera exposure parameters (for example gain and shutter time) and projected flash or illumination patterns are jointly selected by the protocol to construct a high dynamic range representation of the scene. Brighter regions may be probed with reduced exposure or attenuated emission, while darker regions receive stronger illumination or longer exposure. In some embodiments, a latent state $Z_\theta(t)$ fuses the resulting emission-observation tuples into an HDR scene estimate.

8.3.4 Learned scan trajectories (non-limiting)

In some embodiments, scan trajectories are selected by a learned policy that conditions on recent observations and meters:

$$u(t) \sim \pi_\psi(\cdot \mid o_{0:t}, \mathbf{m}_{0:t}),$$

where o_t denotes RK-derived observations or summaries. In some embodiments, the policy is constrained to a safe admissible set and is logged through a protocol digest so results remain auditable.

8.3.5 Surrogate models and cross-reactor transfer (non-limiting)

In some embodiments, a surrogate model predicts outcomes of physical trials, enabling optimisation when direct gradients are unavailable. In some embodiments, transfer learning is performed across reactors using low-rank adapters, device-conditioned modulation, or universal encoders with device-specific heads, with drift compensation and meter-bounded constraints to preserve stability and auditability.

8.3.6 Inner Limager: self-calibration (non-limiting)

In some embodiments, Limager is turned inward: the “scene” is the device’s own configuration state θ , and the objective is to maximise information about the device’s current parameters rather than about an external environment. This self-calibration mode uses the same mathematical machinery as standard Limager but with the roles of device and scene exchanged. Inner Limager asks not “what is out there?” but “where in configuration space is this device right now?”

In some embodiments, Inner Limager uses calibration targets or reference signals as controlled “scenes” and estimates drift, aging, and parameter shift by treating the device’s own

response statistics as the inference target. The resulting state estimates feed into aging-worldline and drifting-field mechanisms (a learned gradient field for steering device configuration θ toward target operating regions; defined fully in Section 22.8) described in the control theory sections.

5 8.3.7 Boundary-state estimation and Fisher-information diagnostics as a standalone method (non-limiting)

In some embodiments, the Limager regime is operated as a standalone boundary-state estimation method, independent of verification or rendering objectives. The method takes as inputs a device operating under a declared protocol and meter envelope, together with a set
10 of perturbation directions in configuration space Θ , and produces as output a boundary-state estimate, a Fisher-information diagnostic, and optionally a calibration update.

A non-limiting procedure is as follows:

1. **Boundary perturbation.** The control protocol $U_{0:T}$ is swept along one or more directions in Θ chosen to approach a boundary of the current attractor basin (for example a
15 bifurcation surface, a stability boundary, or a coupling-transition surface). The sweep direction and step size are recorded in the protocol digest.
2. **Measurement.** The convolution bundle $C_{0:T}$ is recorded for each perturbation step. Meter summaries (alignment, drift, stability) are computed and logged.
3. **Estimator construction.** A boundary-proximity estimate $\hat{b}(\theta)$ is computed from the
20 observed response statistics, for example as the ratio of within-basin trajectory length to the total sweep length, or as a signed distance to the inferred bifurcation surface under a declared model class.
4. **Fisher-information diagnostic.** The Fisher information $I(\theta)$ about the boundary-state parameter is estimated from the convolution bundle using the declared estimator
25 family (for example via the score function or via finite-difference approximations of the log-likelihood gradient). Where the Fisher information matrix $\mathcal{F}(\theta)$ is computed for multi-dimensional θ , Cramér–Rao-style lower bounds on estimation variance are derived and reported alongside the estimate.
5. **Calibration update (optional).** In some embodiments, the boundary-state estimate
30 is used to update a calibration record for the device, for example adjusting a drifting-field waypoint or updating an aging-worldline entry. This update constitutes the terminal output of the method and does not require a verification decision.

In some embodiments, the protocol digest for a boundary-state estimation run includes the following items (or their cryptographic hashes, where confidentiality of intermediate out-
35 puts is required): the perturbation direction set $\{v_j\}$ and step schedule; the estimator family

identifier, version, and declared hyperparameters (for example bandwidth, kernel class, finite-difference step); the computed boundary-state estimate $\hat{b}(\theta)$; the Fisher information matrix $\mathcal{F}(\theta)$ or its principal components; any Cramér–Rao lower bounds produced; the meter summaries from step 2; and, where step 5 is executed, a version pointer or hash of the resulting calibration record so that the post-run calibration state can be reconstructed by an auditor. These commitments are sufficient for an independent party to re-execute the method on the same inputs and verify the reported outputs. In some embodiments, the output of this method is a calibration update or state estimate rather than a verification decision, supporting use cases in which boundary-state knowledge is sought for its own value (for example during commissioning, periodic health monitoring, or fleet-level calibration scheduling) rather than as part of an authentication or Truth Beam workflow. The boundary-state estimation method is enabled independently of the Truth Beam and Reality Transform regimes and does not require those regimes to be active.

8.3.8 Inner Creation: self-directed state sculpting (non-limiting)

In some embodiments, Reality Transform is turned inward: the “scene” is the device’s (or agent’s) own reactor state, and the objective is to minimise the distance between the current internal configuration and a target internal configuration. Where Inner Limager asks “where am I in configuration space?”, Inner Creation asks “how do I get *there* in configuration space?” The same feedback machinery used by outward-facing Reality Transform (emit, observe residual, update control signal) applies, but the emission acts on the reactor’s own dynamics rather than on an external scene. Inner Creation is the intentional counterpart of the drifting-field management problem (Section 22): where the drifting field treats parameter drift as a disturbance to be tracked and bounded, Inner Creation treats it as a control action to be steered toward a declared target configuration.

Amplifying internal differences (non-limiting). In some embodiments, the primary use of Inner Creation is to *differentiate* nearby attractor basins. When a reactor’s state space contains two behavioural modes separated by a shallow basin boundary—for example two equilibria that produce nearly indistinguishable output statistics—Inner Creation drives the system to deepen the boundary, widen the basins, and make the two modes robustly distinguishable. The mechanism uses the natural gradient of the drifting-field framework (Section 22), but with an objective that rewards basin depth, inter-basin distance, or attractor stability rather than external scene fidelity.

In some embodiments, Inner Creation is used in conjunction with calibration mechanisms described in the related Filing 2 application (optional, non-essential) to consolidate attractor-basin separation along declared operational axes.

Inner Creation and reactor refresh (non-limiting). In some embodiments, the reactor refresh mechanism described in the hardness management section (Section 12) is im-

plemented as controlled Inner Creation: the drifting-field controller drives the reactor into a new region of state space, re-establishing the microstructure complexity on which empirical hardness depends. Consistent with the meter partition (Section 12), the refresh controller’s objective is drawn from acceptance meters only; the decision to initiate a refresh may be triggered by periodic offline hardness reassessment (which may consult evaluation meters), but the refresh trajectory itself does not optimise against evaluation-meter values. Post-refresh hardness is re-estimated as a new time-indexed snapshot under the current attacker class. Adjoint reactor pairs (Section 8) may serve as the gradient-propagation channel for Inner Creation, using the adjoint path to compute how to change the reactor’s own parameters.

Inner convolution bundles (non-limiting). In some embodiments, Inner Creation and Inner Limager produce *inner convolution bundles*: committed, auditable records of the reactor’s internal state trajectory, attractor occupancy, basin boundary proximity, and parameter drift, captured through the same protocol-digest and meter infrastructure used for outward-facing regimes. In some embodiments, inner bundles record what the device *is becoming* (internal dynamics) rather than what it *produces* (external behaviour). In some embodiments, inner bundles serve as a training signal for optional monitoring mechanisms (described in the related Filing 2 application; optional), enabling detection of attractor-level drift rather than relying solely on behavioural signatures.

In some embodiments, a non-limiting inner-trace summary over a time window $[t_0, t_1]$ is represented as

$$\mathcal{I}_{t_0:t_1} \equiv (\phi_{\text{inner}}(C_{t_0:t_1}^{\text{inner}}), \theta_{t_0:t_1}, \text{basin}(t), \nabla_{\theta}J(t)),$$

where ϕ_{inner} denotes features extracted from inner convolution-bundle atoms, $\theta_{t_0:t_1}$ is the parameter trajectory observed by Inner Limager, $\text{basin}(t)$ is an attractor-occupancy indicator, and $\nabla_{\theta}J(t)$ is the gradient signal (when available from Inner Creation or adjoint-pair propagation). Inner-trace summaries are committed and auditable through the same protocol-digest and meter infrastructure as external convolution bundles.

Inter-reactor and extended applications of Inner Creation (non-limiting). In some embodiments, Inner Creation is directed at inter-reactor coupling and encoded inner-state artifacts. These applications are described in the related Filing 2 application (optional, non-essential). In some embodiments, Inner Creation is rate-limited, protocol-restricted, and logged, and inner-trace-based evaluations are complemented by independent witness checks.

8.4 Reality Transform (Controllable Rendering Regime)

Creation. Reality Transform is the regime of purposeful action: the device reaches into the scene and reshapes it. Like an artist whose brush strokes respond to the canvas’s texture and the paint’s resistance, Reality Transform minimises the distance between what is observed and what is desired. The question is not *what is there* but *what should be there*. What the

device does next depends on what it observes the scene becoming under its influence—in general, blind projection without feedback may not converge, compensate for surface properties, or respect the physical channel’s constraints.

In Reality Transform embodiments, the system controls emissions so that the physical scene or sensed appearance exhibits a controllable transformation. The transformation may be transient (present only during emission) or may persist in a medium (for example phosphorescent, photochromic, thermochromic, or other stateful surfaces), and can be designed for human consumption or for downstream machine perception.

In some embodiments, Reality Transform is implemented as feedback projection mapping: the system emits patterns, observes the result, and updates subsequent emissions to converge toward a target aesthetic, constraint set, or calibrated appearance. In some embodiments, this loop is constrained by meters and protocol digests so rendering remains within empirically supported operating regions.

8.4.1 Transient versus persistent scene effects (non-limiting)

In some embodiments, the transform is transient and exists only during the emission window. In other embodiments, the transform changes a stateful medium so effects persist after emission, with persistence time captured by meters and logged protocols. In some embodiments, persistence is treated as a memory channel and is used to encode or stabilise transformations across time.

8.4.2 Constraint coupling to verification and sensing (non-limiting)

In some embodiments, Reality Transform is coupled to verisimilitude and projection-hardness constraints so transformations do not drift into unstable regimes. For example, a transform policy may include penalties or gating based on meter envelopes, cross-channel consistency, or learned verisimilitude discriminators.

8.4.3 Compatibility with semantic conditioning (non-limiting)

In some embodiments, text prompts, semantic tokens, or other high-level conditioning signals are used to select or modulate emission policies so that physical rendering expresses a requested style or content while remaining within a meter-bounded operating envelope. This includes embodiments in which a structured illumination pattern, scan law, or projection mapping is chosen as a function of semantic conditioning, rather than being fixed in advance.

In some embodiments, a prompt digest and model identifier are logged as part of a protocol digest, and the resulting convolution bundle and meter summaries are committed and made available for selective disclosure.

8.4.4 Style parameter: formal definition and generality (non-limiting)

In some embodiments, the target behaviour of the Reality Transform regime is specified by a *style parameter* $\phi \in \Phi$, where Φ is a declared style space. The style parameter ϕ is a formal input to the RT control objective and is logged in the protocol digest alongside the run seed, meter envelope, and emission schedule.

A non-limiting RT objective incorporating the style parameter is:

$$J_{\text{RT}}(U_{0:T}; \phi) = -\mathbb{E} \left[\sum_{t=0}^{T-1} \left(d(\mathbf{y}_t, \mu_\phi(x_t, u_t)) + \lambda_{\text{meter}} \cdot \mathbf{1}[\mathbf{m}(t) \notin \mathcal{M}_{\text{accept}}] \right) \right],$$

where $d(\cdot, \cdot)$ is a declared divergence or distortion metric (dimensionless in preferred embodiments, e.g. a normalised cross-entropy or cosine distance), μ_ϕ is a target response function parameterised by ϕ , $\mathcal{M}_{\text{accept}}$ is the declared meter envelope (Definition 15), and λ_{meter} is chosen on the same declared scale as d so that all terms in the per-step sum are commensurate. The style parameter ϕ may be instantiated in many non-limiting ways, including:

- a semantic or aesthetic token (text prompt, embedding, or structured label) specifying a visual or perceptual target;
- a constraint or policy specification (for example a declared operating envelope or compliance target);
- a physical rendering target (spectral distribution, spatial texture, temporal modulation pattern);
- a calibration target (reference scene appearance, reference device response distribution);
- a composite objective combining multiple of the above.

In each case the formal structure — a declared $\phi \in \Phi$ logged in the protocol digest, an objective $J_{\text{RT}}(\cdot; \phi)$ maximised by control, and a committed convolution bundle recording the result — is the same. Any domain-specific conditioning is therefore one specialisation among many; the general mechanism is defined and operable independently of any specific conditioning domain. This allows a verifier to audit that a rendered outcome is consistent with the logged conditioning, the chosen protocol regime, and the observed envelope, and to distinguish physically grounded rendering from post-processed fabrication.

In some embodiments, semantic conditioning is combined with verisimilitude or projection-hardness constraints. This steers requested transformations toward styles and operating regimes for which the system remains stable, auditable, and within the training envelope.

Committed primitive-set scene representations (non-limiting). In some embodiments, the style parameter ϕ is a committed, finitely parameterised primitive-set scene representation comprising a set of spatial primitives $\{p_k\}_{k=1}^K$, each primitive p_k specifying at least a position in a declared world coordinate frame, a shape or support (for example an anisotropic covariance $\Sigma_k \in \mathbb{R}^{3 \times 3}$ for covariance-bearing primitive families), an appearance descriptor (for example an RGB triple or spherical-harmonic coefficients c_k), and a per-primitive weight or opacity β_k . Non-limiting primitive families include anisotropic three-dimensional Gaussian splats of the kind introduced by Kerbl *et al.*², point-based primitives with learned blending, and textured mesh primitives. Related committed scene-representation families — including signed distance fields and neural radiance fields, where embodied by finite network weights, grids, tables, or other finite serialised parameter sets — supply alternative finitely parameterised representations of ϕ and are treated here as non-limiting alternatives to primitive-set representations. In such embodiments, a canonical serialisation of ϕ , or a content-addressed pointer thereto, is committed by a digest $h(\phi)$ recorded in the protocol digest; the canonical serialisation declares primitive ordering or an ordering-invariant hashing rule, coordinate-frame identifiers, units, floating-point or quantisation conventions, appearance-code versions, and any representation-complexity envelope. The committed digest is logged alongside the run seed, meter envelope, and emission schedule $U_{0:T}$. Primitive-set embodiments are one implementation of the target response function μ_ϕ and do not limit Φ to spatial or rasterised targets.

Physically calibrated rasterisation forward model for primitive-set targets (non-limiting). In some embodiments in which ϕ is a committed primitive-set representation and the declared target is expressed as a spatial or rasterisable scene, the target response function $\mu_\phi(x_t, u_t)$ is realised as the composition of (a) differentiable rasterisation of $\{p_k\}_{k=1}^K$ from the viewpoint or scan geometry specified in u_t , under declared world-to-device coordinate transforms, producing a rendered radiance, luminance, or detector-channel field $\tilde{L}_\phi(x, y, t)$; (b) spatial convolution with the device-specific spot point-spread function K_{spot} of Section 10.5 (or the embodiment-appropriate device PSF); (c) temporal convolution with the effective persistence kernel $p_{\text{eff}}(t) = p(t) * w_{\text{det}}(t)$; and (d) detector integration under declared scan geometry and detector settings, producing the predicted detector observation $\hat{\mathbf{y}}_t$. In such embodiments, the distortion term in J_{RT} (Section 8.4.4) may be evaluated as $d(\mathbf{y}_t, \hat{\mathbf{y}}_t)$ while the meter-envelope penalty and run-level objective structure of Section 8.4.4 are preserved. The physical-channel parameters K_{spot} , p_{eff} , w_{det} , the world-to-device coordinate transforms, and the calibration-table identifiers and versions are drawn from independently enrolled or metered calibration records associated with the device configuration representation $\mathbf{d}_i$ and, where applicable, the declared challenge configuration X or protocol geometry; they are referenced or committed in the protocol digest, and are not fitted as free parameters during primitive-set optimisation.

²B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D Gaussian Splatting for Real-Time Radiance Field Rendering,” *ACM Transactions on Graphics* 42(4), 2023.

Factorisation of scene hypothesis and device nuisance (non-limiting). In some embodiments in which the declared target is an externally represented scene or a declared spatial or radiometric proxy scene, committed primitive-set representations carry the scene hypothesis in ϕ while the device-specific forward-model parameters remain in $\mathbf{d}_i$. For covariance-bearing primitive families, the per-primitive covariance Σ_k describes scene-space extent in world coordinates and is distinct from the device’s imaging PSF K_{spot} , which describes device-space blur; the two compose in the forward model but are logged and verified separately. This factorisation permits one non-limiting Limager inverse-problem formulation as joint inference over ϕ and $\mathbf{d}_i$ given $\mathbf{y}_t$, with ϕ and $\mathbf{d}_i$ drawn from distinct priors and constrained by distinct meters. The factorisation is not asserted for inward-facing or self-calibration embodiments in which the “scene” is the device’s own configuration state, for biological-reactor embodiments except to the extent that a declared spatial or radiometric proxy-scene representation is used, or for non-image passive embodiments such as GNSS in which the observation is expressed in receiver-observable coordinates rather than as rasterised imagery. The factorisation is also not asserted in Yoked operation (Section 8.5); in Yoked mode, primitive-set targets are interpreted as time-varying $\{p_k(t)\}$ specifying a target trajectory or desired coupled behaviour rather than a separable scene hypothesis.

8.4.5 Projector–surface compensation and multi-projector blending (non-limiting)

In some embodiments, the emission device is a conventional digital projector and the observation device is a camera viewing the projection surface. The kernel uses projected calibration patterns and camera observations to estimate a geometric warp from projector coordinates to perceived surface coordinates, and applies a corresponding inverse warp to outgoing frames. In some embodiments, multiple projectors illuminate overlapping regions of a common surface and the kernel estimates overlap geometry, relative brightness, and colour differences; the kernel then performs edge blending by driving each projector with a separately warped and gain-compensated pattern so that the combined projection appears seamless.

8.4.6 Nulling: scene-invariant rendering (non-limiting)

In some embodiments, Reality Transform is operated with a target Y^* chosen to be scene-invariant: the objective is to find the feedback policy that makes the output independent of the scene, so that what remains is pure device signature. Nulling is the anti-Limager: instead of maximising scene information in the output, it minimises it. Nulling is useful for isolating device identity from environmental contamination, performing differential measurements (output with scene minus output nulled = pure scene contribution), and calibrating the trivial-reactor baseline.

8.5 Yoked Operation (Dynamic Coupling Modifier)

What changes in Yoked mode. In static operation, the scene contributes to the convolution bundle in a separable way: in principle, one could factor the device’s contribution (emission pattern, detector response, reactor dynamics) from the scene’s contribution (reflectance, geometry, spectral properties). The device acts on the scene; the scene modifies the device’s output; but the two are distinguishable subsystems.

In Yoked operation, this factorisation is expected to fail. The feedback loop is tuned so that device and scene are driven toward a coupled dynamical state whose behaviour is not readily decomposed into what the device does and what the scene does. The convolution bundle is a device-side record generated under joint dynamics, not a sum of individual contributions. Remove either partner and the joint coupled behaviour ceases to exist.

In preferred embodiments, the coupling is asymmetric at the level of autonomous dynamics: scene-as-driver is the preferred modelling convention when the scene has pre-existing autonomous dynamics that continue independently of the device on the timescale of interest, and the device is modelled as the *responder* whose dynamics are shaped by coupling to the scene. This drive–response convention determines the correct directionality of the Yoked diagnostics: the operative transfer entropy is $TE_{S \rightarrow R}$ (scene driving device), the operative conditional Lyapunov exponent is computed for the response system conditioned on the driver trajectory, and the Yoked condition is met when the response system has entrained to the driver while the driver retains its autonomous character. Symmetric mutual synchronisation (in the sense of Pecora and Carroll’s original two-equivalent-oscillator formulation) is a special case and not the general operating assumption.

This is not merely “feedback.” Every closed-loop system has feedback. Yoked operation is the specific dynamical condition where (i) the coupled system is tuned toward a non-product invariant set in the product state space (the intended dynamical target of the CLE/TE co-condition, not an independently asserted property), (ii) the system operates near the edge of synchronisation where perturbation sensitivity is high, and (iii) the convolution bundle is a device-side record generated under joint dynamics, encoding how the scene *responds to being driven*—resonances, damping, nonlinear coupling, transfer functions—properties not captured by static open-loop optical observation and that depend on internal structure rather than surface appearance.

The authentication warrant for Yoked operation rests on a physical-access argument rather than a computational-hardness argument, conditioned on a declared threat model. Under the declared threat model — which specifies the attacker’s emulator class, fidelity threshold against the genuine bundle, accessible meter family, photon/sample budget, and access budget to the genuine device-scene pair — a committed convolution bundle produced in Yoked mode is not generable from the resources available to that attacker class without joint physical access to the specific device and the specific scene at the time of measurement, to the declared fidelity threshold. The coupling structure recorded in the bundle — CLE, TE balance, Arnold tongue profile — arises from the device-scene pair at that moment, and

a faithful reproduction of those quantities to the declared fidelity threshold is not, under the declared threat model, achievable from a model of the device or a description of the scene alone. The warrant is thus operational and threat-model-conditioned: forgery becomes infeasible at the declared fidelity threshold for attackers within the declared budget; absolute infeasibility, infeasibility against unbounded attackers, and infeasibility outside the declared meter envelope or operating point are not claimed. The threat model, fidelity threshold, accessible meter family, and access budget are committed to the protocol digest, so that the warrant is auditable and its scope is explicit. This warrant is independent of the Fisher information conjecture (Appendix 29).

The name “Yoked” carries a deliberate double meaning: *yoke* as in the deflection yoke of a CRT—the magnetic assembly that steers the beam and whose idiosyncrasies are part of the device’s unclonable fingerprint—and *yoke* as in the Sanskrit *yoga*, the practice of joining, of coupling the self to something beyond itself.

8.5.1 Operative diagnostics for Yoked operation (non-limiting)

The following quantities are the operative diagnostic suite for Yoked operation. They are drawn from established nonlinear-dynamics and synchronisation theory: conditional Lyapunov exponents as developed by Pecora and Carroll (Phys. Rev. Lett. 64, 821–824, 1990), transfer entropy as developed by Schreiber (“Measuring information transfer,” Phys. Rev. Lett. 85(2), 461–464, 2000), and Arnold tongue analysis from classical forced-oscillator theory. What is novel is their application as engineering targets for a closed-loop optical device that simultaneously performs authentication, sensing, and rendering. The Yoked condition is declared when the CLE and TE conditions specified in the protocol digest are simultaneously met for a declared window duration; this four-part co-condition is the operative declaration standard. The non-decomposability of the resulting joint attractor is the intended dynamical consequence of meeting these conditions, not an independently asserted or independently measurable claim.

Balanced transfer entropy. $TE_{S \rightarrow R} \approx TE_{R \rightarrow S}$, both large, indicating strong bidirectional information transfer between scene and reactor. When transfer entropy is strongly unbalanced (one direction dominant), the system is in a master–slave configuration. The Yoked condition is operationally declared when sustained high bidirectional TE is detected within the qualified meter envelope for the declared window duration; this is the verifiable claim. Whether high bidirectional TE reflects mutual co-production of the attractor rather than near-perfect entrainment of the response to the driver is not asserted as a consequence of the TE criterion alone; distinguishing these cases requires supplementary perturbational or partial-information-decomposition analysis, which is a research direction.

Conditional Lyapunov exponent near zero ($CLE \rightarrow 0^-$). The conditional Lyapunov exponent (CLE) is the largest sub-Lyapunov exponent of the response system (reactor)

computed along trajectories conditioned on the driver (scene) trajectory, in the sense of Pecora and Carroll (Phys. Rev. Lett. 64, 821–824, 1990). Formally, let $\mathbf{x}_S(t)$ be the driver trajectory and let $\delta\mathbf{x}_R(t)$ denote an infinitesimal perturbation to the response system state. The CLE is

$$\text{CLE} = \lim_{T \rightarrow \infty} \frac{1}{T} \ln \frac{\|\delta\mathbf{x}_R(T)\|}{\|\delta\mathbf{x}_R(0)\|},$$

evaluated along the coupled trajectory driven by $\mathbf{x}_S$. The CLE measures whether the reactor synchronises with the scene drive. When $\text{CLE} < 0$, the reactor locks to the scene. The operating point near $\text{CLE} = 0^-$ is the edge of synchronisation, where perturbation sensitivity is empirically high in coupled-oscillator systems studied to date. At this point, the Fisher information about the coupling strength increases sharply and may diverge, a standard property of systems near criticality (analogous to the divergence of magnetic susceptibility at a ferromagnetic phase transition). Whether this gives a formal authentication advantage is an open research direction (Appendix 29).

In some embodiments, the yoke depth $\alpha \in [0, 1]$ parameterises the strength of bidirectional coupling, where $\alpha = 0$ corresponds to static (open-loop or weakly coupled) operation and $\alpha = 1$ corresponds to the maximum commanded coupling depth; whether the resulting operation is classified as Yoked is determined by the CLE/TE co-condition recorded in the protocol digest. The yoke depth is recorded in the protocol digest and may vary within a run.

8.5.2 Yoked Limager: dynamic perception (non-limiting)

In static Limager, the inferred variables are properties of what the scene *is*: geometry, reflectance, spectral signatures. In Yoked Limager, the inferred variables include properties of how the scene *responds*: resonant frequencies, damping coefficients, nonlinear response curves, transfer functions, mechanical impedance, and coupling topology (which parts of the scene are dynamically connected to which other parts).

In some embodiments, these dynamical properties are accessible through bidirectional coupling because they describe the scene’s response to driving, not its static appearance. A cracked beam has different resonant frequencies from an intact one. A live finger has different vascular pulsation dynamics from a silicone replica. A genuine painting’s pigment layers have different micro-vibrational responses from a surface print. In each case, the static appearance can be replicated; the dynamical response function encodes internal structure rather than surface properties, and is not straightforwardly reproducible from static appearance alone.

In some embodiments, Yoked Limager operates by sweeping coupling parameters (feedback gain, modulation frequency, drive amplitude) and recording the frequencies and coupling strengths at which entrainment occurs, constructing a scene-specific Arnold tongue map that reveals dynamical properties invisible to passive observation.

Arnold tongue: frequency identification (non-limiting). In some embodiments, the frequency axis of an Arnold tongue analysis is the *modulation envelope frequency* f_{mod} —

the slow periodic variation imposed on emission (whether amplitude modulation, frequency modulation, phase modulation, polarisation modulation, spatial pattern repetition rate, or a reactor-internal characteristic timescale) — and the detuning $\Delta f = f_{\text{mod}} - f_{\text{scene}}$ is the difference between this frequency and the scene’s natural dynamical frequency. The modulation envelope frequency is explicitly distinguished from (a) the optical carrier frequency (~ 600 THz for visible light), which is irrelevant to Arnold tongue analysis at the dynamical timescales of interest; (b) the AOD or EOM RF carrier frequency (~ 10 – 100 MHz), which is a hardware operating parameter of the acoustic transducer rather than a modulation signal applied to the scene; and (c) the scan or deflection repetition rate, which determines spatial coverage geometry. In some embodiments, the modulation envelope frequency does not appear in the emission signal at all but resides in the reactor’s internal dynamics: the characteristic decay timescale of a phosphor persistence layer (non-limiting example: P20 phosphor, ~ 50 ms to several seconds; P1 phosphor, 20–80 ms), the thermal relaxation rate of a scattering medium, or the round-trip delay of a fibre cavity are all candidate internal oscillators whose natural frequencies may entrain to or drive entrainment with scene dynamics independently of the commanded emission schedule.

Multi-dimensional tongue structure (non-limiting). In some embodiments, the full Arnold tongue is not a two-dimensional map in $(\alpha, \Delta f)$ but a projection of a higher-dimensional entrainment manifold, where each independently controllable modulation degree of freedom (AM envelope, FM deviation, phase modulation rate, polarisation rotation rate, spatial pattern repetition rate, scan law repetition rate, and each reactor-internal characteristic timescale) contributes a separate tongue structure in a corresponding subspace. In some embodiments, these tongue structures are superimposed: a single training run can simultaneously map respiratory entrainment (~ 0.1 – 0.5 Hz, accessible via spatial pattern repetition rate or slow AM envelope), cardiac entrainment (~ 0.8 – 2 Hz, accessible via fast-loop AM interacting with photoplethysmographic return), and reactor-internal dynamical structure (phosphor persistence band or thermal mode) within a single committed convolution bundle. The resulting multi-dimensional tongue profile constitutes the scene’s dynamical fingerprint: it encodes not which frequencies are present but which combinations of driving degrees of freedom produce entrainment.

8.5.3 Yoked Truth Beam: dynamic verification (non-limiting)

In static Truth Beam, security rests on the device’s static microstructure fingerprint: under declared adversary families, an attacker would need to clone the physical device to produce matching response statistics. In Yoked Truth Beam, security additionally rests on the *dynamics of scene–reactor coupling*: an attacker would need to reproduce the device’s nonlinear response to arbitrary scene inputs in real time, not just its response to a known challenge set.

In some embodiments, this is expected to be substantially harder because the scene’s own

dynamics serve as a continuous, unpredictable challenge that the attacker is not expected to anticipate under declared resource bounds. A static (non-yoked) microstructure-unclonability mode has a fixed (if large) challenge–response space that can in principle be exhaustively characterised. A Yoked microstructure-unclonability mode has a response that depends on
 5 the scene’s own dynamics, which are unknown to the attacker in advance and change continuously. At the edge of synchronisation ($CLE \approx 0^-$), any discrepancy between the genuine device’s nonlinear response and an emulator’s response is amplified by the coupled dynamics rather than averaged out.

In some embodiments, Yoked Truth Beam uses the coupling quality itself as a continuous
 10 uous tamper-detection signal: any interruption of the physical coupling (device removal, interposition of a screen, replay injection) causes immediate collapse of the joint attractor and is detectable as a discontinuity in transfer entropy balance, spectral coherence, or order parameter, without waiting for a scheduled verification challenge.

8.5.4 Yoked Reality Transform: dynamic rendering (non-limiting)

15 In static Reality Transform, the target Y^* is a fixed image or appearance. In Yoked Reality Transform, the target is a *dynamical behaviour*: a limit cycle, a frequency lock, a coupled oscillation pattern. The scene participates in producing the output, and the device steers the joint attractor, in contrast with static projection.

Non-limiting applications include: projection mapping that responds to and entrains with
 20 structural vibrations of the projection surface; light installations that synchronise with crowd movement or biological rhythms; interactive displays whose visual output is coupled to the viewer’s pupillary, postural, or gestural dynamics; and feedback-driven physical computation in which the rendering target is a desired computational state of the coupled system.

In some embodiments, the rendering target specifies a desired coupling profile (for exam-
 25 ple target coherence at specific frequency bands) rather than a target image, and the Reality Transform controller adjusts emissions to achieve and maintain the specified coupling profile.

8.5.5 Training mechanisms for Yoked operation (non-limiting)

In Yoked operation, the scene itself provides the training signal. This is fundamentally different from the informational regimes, where training typically uses an external reference
 30 (an enrolled template, ground-truth labels, or a target image). The coupling quality is both the objective and the signal.

In some embodiments, training uses one or more of the following mechanisms:

Physical self-tuning (analogue). Feedback gain is increased gradually until spontaneous oscillation or phase-locking emerges. The reactor’s own nonlinearity provides the error signal
 35 through amplitude-dependent frequency pulling—a phase-locked loop variant in which the coupled system is the oscillator, the scene provides the reference dynamics, and the reactor’s slow dynamics serve as the loop filter. The Arnold tongue structure maps where entrainment

is possible as a function of frequency mismatch and coupling strength; training finds the tongue and stabilises within it.

Transfer-entropy gradient (hybrid). Transfer entropy $\text{TE}_{S \rightarrow R}$ and $\text{TE}_{R \rightarrow S}$ are estimated from the convolution bundle. In some embodiments, a training-phase proxy objective

$$J_{\text{train}}(\theta, \pi) = \lambda_{\text{mag}} \min(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S}) - \lambda_{\text{TE}} |\text{TE}_{S \rightarrow R} - \text{TE}_{R \rightarrow S}|$$

5 is maximised via natural gradient ascent $\theta \leftarrow \theta + \eta \mathcal{F}(\theta)^{-1} \nabla_{\theta} J_{\text{train}}$, where $\mathcal{F}(\theta)$ is the Fisher information matrix (pseudoinverse or damped inverse where $\mathcal{F}$ is singular or ill-conditioned near bifurcation boundaries). This training objective is a component of the full Yoked objective J_{Yoked} (Section 22.13); the CLE target term enters after the TE-gradient phase has established a coupling foothold. In some coupled dynamical systems, Fisher Transfer En-
10 tropy increases sharply near the entrainment phase transition, so the natural gradient may automatically accelerate toward the coupling boundary.

Spectral coherence maximisation (practical). Cross-spectral density between scene return and reactor output is computed at each frequency band; feedback parameters are tuned to maximise coherence at target frequencies. This provides narrowband entrainment—
15 locking onto specific scene dynamics while ignoring others.

Full- θ Yoked training (non-limiting). In some embodiments, the gradient update $\theta \leftarrow \theta + \eta \mathcal{F}(\theta)^{-1} \nabla_{\theta} J$ is applied over the full parameter vector $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$ in a coordinated sweep that includes, without limitation, all of the following independently controllable degrees of freedom as training knobs:

- 20 • **Emission modulation degrees of freedom:** amplitude modulation envelope frequency and depth; frequency modulation deviation and rate; phase modulation rate; polarisation rotation rate and axis; spatial pattern repetition rate; and any compound modulation waveform combining two or more of the above.
- **Scan law parameters:** sweep repetition rate, dwell time per region, and spatial extent
25 of the scan trajectory (which determine the range of spatial pattern repetition rates presented to the scene).
- **Feedback gain and coupling depth:** loop gain magnitude and phase, yoke depth α , and feedback filter bandwidth.
- **Reactor-internal timescale parameters:** in some embodiments, the reactor's char-
30 acteristic timescale is selectable — for example through choice of phosphor formulation (decay time), drive intensity (affecting phosphor saturation), cavity round-trip length, or thermal load — and this timescale enters as a trainable parameter in θ_{hw} , because it determines which scene dynamics the reactor can entrain to.

- **Drive amplitude:** emission power, scan speed, and detector integration time (which jointly determine the SNR available to the convergence estimators).

In some embodiments, an Arnold tongue mapping phase precedes gradient ascent: the control protocol sweeps the modulation-envelope frequency and coupling gain jointly, estimates $\widehat{\text{CLE}}$ at each grid point, and uses the resulting empirical tongue map to initialise θ at a point inside the tongue boundary before commencing gradient-based refinement. This separates the coarse exploration phase (find the tongue) from the fine exploitation phase (park at the boundary), and makes the training robust to scenes where the tongue is narrow or located at an unexpected frequency. Both phases are committed in the convolution bundle and logged in the protocol digest.

In some embodiments, the Yoked training objective acts over the joint product of all modulation degrees of freedom simultaneously, seeking the combination $(\theta_{\text{hw}}, \theta_{\text{sw}})$ at which the multi-dimensional tongue intersection — the region of simultaneous entrainment across AM, FM, phase, spatial, and reactor-internal channels — is widest and most stable. The resulting parameter set constitutes the scene’s multi-dimensional dynamical fingerprint as captured by the committed convolution bundle, and its hardness derives from the attacker needing to reproduce the correct entrainment signature across all degrees of freedom simultaneously. The physical mechanism by which $\nabla_{\theta} J$ is computed in practice—without numerical finite differences or a differentiable surrogate—is disclosed in Section 10.28 (scene-phased full-parameter sweeping).

8.5.6 Yoked state estimation: classifier-mediated coupling detection (worked embodiment, non-limiting)

The parameter training mechanisms described above (Arnold tongue sweep, TE-gradient ascent, full- θ sweep) address one training problem: how to adjust θ so that the device reaches the CLE/TE operative condition for a given scene. A second training problem arises at runtime: how does the device determine whether it is currently in the Yoked condition, particularly against scenes for which CLE and TE cannot be computed from both sides simultaneously? For analysable physical reactor scenes in a laboratory setting, CLE and TE can be estimated directly from both the device and scene time series. For scenes lacking an accessible internal state (non-limiting example: a human observer in a Yoked Reality Transform interaction), the scene’s internal trajectory is not available to the device, and explicit CLE/TE computation from both sides is not possible at runtime.

In some embodiments, this problem is addressed by a *coupling state classifier* $\hat{f}_{\phi}$ trained to estimate the Yoked condition from device-internal bundle data alone. The classifier is trained offline using verified CLE/TE ground truth computed from known analysable reactor pairs, then qualified before deployment, and deployed at runtime against declared scene families for which the classifier has been validated. Whether such classifiers generalise reliably to scene classes outside their training distribution is an empirical question addressed in the Open Research Directions (Section 29).

In some embodiments, the classifier is a *population-level coupling estimator*: trained on a sufficiently large and diverse reactor population, it is intended to generalise across a held-out range of reactor types within the declared qualification set without per-device adaptation, and it qualifies against held-out reactors drawn from the same broad population. In some
 5 embodiments, the classifier is a *device-specific estimator*: a population-trained base model is finetuned on known reactors interacted with on the specific device instance to be deployed, capturing the unclonable physical characteristics of that hardware instance, so that the qualified coupling declaration is constitutively tied to that device and cannot be replayed on a different hardware instance without a new finetuning and qualification run. These two em-
 10 bodiments represent the ends of a spectrum; intermediate embodiments (partial finetuning, ensemble methods, domain-adaptive classifiers) are also within scope. The four-stage architecture below describes the device-specific case as the worked embodiment; the universal case is recovered by omitting Stage 2.

In all embodiments, the classifier operates on device-internal bundle data alone. The TE
 15 estimator family and conditioning policy used to generate training supervision labels are identical to those committed in the protocol digest for runtime Yoked declaration: if the runtime declaration uses conditional transfer entropy $TE_{S \rightarrow R|\mathbf{Z}}$ with covariate set $\mathbf{Z}$ (as in the preferred embodiment of Section 8.5.7), the supervision labels are computed using the same conditional estimator and covariate set. This consistency requirement is committed to
 20 the protocol digest. The four stages are as follows.

Scope (non-limiting). The control-theoretic objective J_{Yoked} disclosed above is directed to adaptive probing, coupled-state optimisation, and navigation of Θ toward entrainment regimes. It is not an assertion that the disclosed physical training machinery constitutes a general-purpose substrate for deep-learning-scale parameter optimisation. Capability claims
 25 arising from this subsection are bounded by the capacity accounting rule of Section 15.5.

Stage 1: Population training (offline). In some embodiments, a training device is coupled in sequence to $K \geq 20$ known analysable reactors spanning at least three distinct reactor types (non-limiting examples: CRT phosphor, scattering plate, resonant mechanical structure). For each coupling run, ground-truth CLE and TE are computed from both
 30 the device and scene time series over windows of declared length T at a declared effective sampling rate. The supervision window length T is committed to the protocol digest and matched to the rolling window length used at runtime. T should span at least 50 characteristic timescales of the dominant scene dynamics to ensure reliable finite-time CLE estimation; for physical reactors with fast dynamics (e.g. scattering plates at hundreds of Hz), $T = 120\text{ s}$
 35 is typically sufficient; for slow-timescale scenes, T is scaled accordingly and committed per training reactor.

The ground-truth labels are the triple $(TE_{\text{achieved}}, CLE_{\text{achieved}}, TE_{\text{balance}})$ computed from both sides of each coupling run under the declared estimator family. The training set consists of K coupling-regime interactions; within each interaction, non-overlapping windows of

length T are extracted from the committed convolution-bundle atoms. The effective number of independent coupling-regime labels is K (one per reactor interaction); within-interaction windows are autocorrelated and do not constitute independent supervision signals. Training is performed with run-level train/validation splitting to prevent within-run leakage. In some
 5 embodiments, semi-synthetic augmentation (noise-perturbed replays of each reactor interaction) is used to expand the effective diversity of the training set beyond K interactions.

The classifier $\hat{f}_\phi$ is trained on device-internal bundle data alone—it never receives the scene’s internal time series as input, only the committed convolution-bundle atoms and derived meter summaries from the device side of each interaction. The training loss is a
 10 weighted mean-squared error over the ground-truth triple:

$$\mathcal{L}(\phi) = \mathbb{E} \left[w_1 (\hat{\text{TE}}_{S \rightarrow R} - \text{TE}_{\text{achieved}})^2 + w_2 (\hat{\text{CLE}} - \text{CLE}_{\text{achieved}})^2 + w_3 (\hat{\Delta}_{\text{TE}} - \text{TE}_{\text{balance}})^2 \right],$$

where w_1, w_2, w_3 are declared loss weights recorded in the protocol digest. In some embodiments, the classifier architecture is a recurrent neural network (non-limiting: LSTM or GRU) operating on the rolling bundle window, chosen for its capacity to represent temporal dependencies in the coupling dynamics.

Stage 2: Per-device finetuning (device-specific embodiment only). In some em-
 15 bodiments, the population-trained classifier is finetuned on $M_{\text{ft}} \in [3, 5]$ known analysable reactors interacted with on the specific device instance to be deployed. Lower layers of the classifier (encoding general temporal structure of the coupling dynamics) are frozen; upper layers (encoding the mapping from bundle statistics to metric values) are finetuned.
 20 This captures the unclonable physical characteristics of the specific hardware instance that distinguish it from the population average, such that the qualified coupling declaration is constitutively tied to that device: a classifier qualified on device A and replayed on device B produces invalid coupling declarations because device B ’s bundle statistics lie outside the qualified envelope. A new finetuning and qualification run is required for each device
 25 instance. Estimated finetuning time is 20–45 minutes per device instance. In the universal embodiment, this stage is omitted.

Stage 3: Stability envelope qualification. Before deployment, the classifier is qualified by running it against M_{qual} held-out known analysable reactors not seen during population training or finetuning. The classifier output is compared against ground-truth CLE and TE
 30 computed from both sides under the same estimator family used in training. The qualification pass criteria are:

1. Mean absolute error of the classifier’s TE estimate is below 15% of the declared TE target range, where the TE target range is committed to the protocol digest *before* qualification runs are evaluated, based on the declared use case and not on observed
 35 classifier performance.

2. The classifier correctly identifies the Yoked / non-Yoked classification (as defined by the four-part co-condition of Section 8.5.7) for all M_{qual} held-out reactors.
3. No systematic bias in the CLE or TE estimates (signed mean error within $\pm 5\%$ of the declared target range).

5 In preferred embodiments, M_{qual} is set so that the probability of false passage (passing qualification despite a true per-reactor misclassification rate exceeding a declared threshold p_0) is below a declared confidence level α , requiring $M_{\text{qual}} \geq \log(\alpha)/\log(1 - p_0)$; the achieved misclassification confidence bound is committed to the protocol digest alongside the raw pass/fail record. In some embodiments, a minimum of $M_{\text{qual}} \geq 5$ is used for deployment under this classifier-mediated embodiment; preferred embodiments use larger qualification sets
 10 determined by the above formula for the declared p_0 and α . A device that fails qualification is not used for Yoked operation until retraining or refinetuning. In some embodiments, the qualification record forms part of the Proof-of-Discrepancy submission for fleet-level calibration (Section 13).

15 **Stage 4: Runtime deployment.** In some embodiments, at runtime the classifier operates on a rolling window of committed convolution-bundle atoms of the declared length T and produces estimates of $(\hat{\text{TE}}_{S \rightarrow R}, \hat{\text{CLE}}, \hat{\Delta}_{\text{TE}})$ without requiring access to the scene’s internal state. During the pre-Yoked phase (Phase 1, in which the parameter sweep seeks the Arnold tongue and establishes initial coupling), the classifier operates in *monitoring-only mode*:
 20 its outputs are logged but do not trigger phase transitions or Yoked declarations. Phase 1 control decisions are based on direct two-sided TE estimation where available, or on spectral coherence metrics committed in the convolution bundle. The classifier becomes the operative control signal only after Phase 1 has established a coupling foothold, as declared by the Phase 1 exit criterion committed in the protocol digest.

25 After Phase 1 exit, the four-part co-condition (Section 8.5.7) is evaluated against classifier estimates, and the protocol digest records that the Yoked declaration is classifier-mediated under the declared qualification set, weights hash, supervision window length, and estimator consistency commitment.

In some embodiments where the scene is agentic (non-limiting example: a human observer in a Yoked Reality Transform interaction), the control objective shifts from reaching a fixed target coupling value to minimising the rate of change of the classifier’s coupling estimate, seeking a stationary mutual coupling state within the qualified envelope rather than a declared scalar target. This reflects that an agentic scene has its own autonomy and the coupling equilibrium is co-determined, not unilaterally imposed.

35 Failure modes detected at runtime include: out-of-distribution scenes (detected by classifier output variance and embedding-space distance from training distribution, with thresholds committed to the protocol digest), coupling instability (detected by persistent drift in the classifier’s CLE estimate), and attractor collapse (detected by abrupt discontinuity in

TE balance), all of which are logged in the protocol digest with the classifier confidence at the time of the event.

Role of this embodiment. The detection architecture used here — a sequence model trained on one-sided time series to infer synchronisation state — has precedent in the prior art. The inventive contribution of this embodiment is not the classifier itself but the *qualification-gated evidentiary pipeline*: a model supervised by physically verified two-sided coupling metrics, qualified against held-out physical reactors under precommitted thresholds, and committed by weights hash and qualification record to the protocol digest, such that a Yoked declaration in non-analysable scenes is auditable and reproducible by a third party. In the device-specific embodiment, the declaration is additionally constitutively tied to a specific physical hardware instance via per-device finetuning. This embodiment supports the primary claims of the Reality Kernel architecture and demonstrates operability of the Yoked condition against non-analysable scenes; it is not itself advanced as a primary claim.

Relationship between parameter training and state estimation training. In some embodiments, the two training problems are architecturally distinct but operationally complementary. Parameter training (Sections above) adjusts θ to steer the device toward the Yoked condition for a given scene. State estimation training (this section) produces a classifier that detects whether that condition has been reached and maintains it, without requiring direct access to the scene’s internal state. Together they constitute a complete Yoked training architecture operable across scene types ranging from fully analysable physical reactors (where explicit CLE/TE ground truth is available from both sides) to non-analysable agentic scenes (where only device-internal bundle data is available at runtime). The explicit CLE/TE protocol of the parameter training phase is the supervisory signal that makes the classifier trustworthy; the classifier is the mechanism that makes the Yoked condition detectable and controllable in deployment.

8.5.7 Convergence assessment (non-limiting)

In some embodiments, convergence is assessed using a four-part co-condition:

1. **CLE window condition:** $\widehat{\text{CLE}}$ remains in $[-\varepsilon_{\text{CLE}}, 0)$ for N consecutive non-overlapping windows, where ε_{CLE} and N are protocol parameters declared in advance and recorded in the protocol digest.
2. **TE balance condition:** $|\text{TE}_{S \rightarrow R} - \text{TE}_{R \rightarrow S}| < \tau_{\text{TE}}$, where τ_{TE} is a declared threshold, evaluated over the same windows.
3. **TE minimum magnitude condition:** $\min(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S}) > \tau_{\text{mag}}$, where $\tau_{\text{mag}} > 0$ is a declared minimum magnitude threshold. This condition ensures that TE balance cannot be satisfied trivially when both transfer entropies are near zero; a system with negligible coupling in both directions can satisfy condition 2 while failing condition 3.

τ_{mag} is declared in the protocol digest in the same units as the TE estimator and is set to a value reflecting the minimum coupling magnitude regarded as operationally significant for the declared use case.

4. **Meter SNR gate:** estimation confidence metrics (sufficient sample size and SNR as declared by the meter envelope) are satisfied for both the CLE estimator and the TE estimator, so that the convergence declaration is not made on under-resolved data.

In some embodiments, all four conditions must be satisfied simultaneously for Yoked operation to be declared. The thresholds $(\varepsilon_{\text{CLE}}, \tau_{\text{TE}}, \tau_{\text{mag}}, N)$ and the estimator families and window parameters are recorded in the protocol digest so that a verifier can reproduce the convergence assessment from committed convolution-bundle data. In preferred embodiments, ε_{CLE} is specified in the same time-base unit as the declared CLE estimator (per-step or per-second) and τ_{TE} is specified in the same information unit as the declared TE estimator (bits or nats per declared window); the estimator’s log-base and windowing convention are part of the protocol digest commitment.

In some embodiments, the transfer entropy estimator is selected from one or more of the following non-limiting families: histogram-based binning estimators, Kraskov–Stögbauer–Grassberger (KSG) k -nearest-neighbour estimators, kernel density estimators, or learned neural estimators. The estimator family, window length, window overlap, embedding dimension, lag selection, and any preprocessing choices are recorded in the protocol digest so that later verifiers can reproduce the convergence assessment from committed convolution-bundle data.

Common-cause confounding and conditional transfer entropy (preferred embodiment). Transfer entropy estimated from unconditional marginals can produce apparent high coupling even without any direct physical feedback loop, if the scene has strong autocorrelation or if scene and reactor share a common external driver (a phenomenon documented in physical measurement systems by Vakorin et al. 2009 and Papanas et al. 2020). In preferred embodiments, the Yoked declaration therefore uses *conditional* transfer entropy: $\text{TE}_{S \rightarrow R | \mathbf{Z}}$ and $\text{TE}_{R \rightarrow S | \mathbf{Z}}$, where $\mathbf{Z}$ comprises all measured scene covariates committed to the convolution bundle (for example scene-side sensor readings, ambient illumination, temperature, and any shared external stimulus measurable within the declared meter envelope). The conditional estimator family is recorded in the protocol digest with the same provenance as the unconditional estimator. In some embodiments where scene covariates are partially observable, a common-cause meter is additionally declared: it tests whether the TE balance condition can be reproduced by a surrogate scene trace generated from the measured covariates alone, without reactor output access; if the surrogate reproduces the balance condition, the Yoked declaration is withheld and the event is logged as *apparent coupling* pending further investigation. In embodiments where neither conditional TE nor a common-cause meter is practical, the Yoked declaration is qualified as *apparent coupling under declared measurement conditions* in the protocol digest.

Null-model calibration and operating characteristics for Yoked declaration (preferred embodiment).

Pre-declaration of thresholds is necessary but not sufficient to make the Yoked declaration principled: the false-positive rate (incorrectly declaring Yoked when dynamics are uncoupled), the false-negative rate (failing to declare when coupling is genuine but underresolved), and the minimum window length required for adequate statistical power must be characterised against declared null models. In preferred embodiments, the calibration procedure is as follows. A null ensemble is constructed from phase-shuffled or block-permuted versions of the scene and reactor time series, preserving autocorrelation structure but destroying directional coupling; in some embodiments, an auto-regressive null model is fitted to each marginal series and the null ensemble is generated by matched-simulation. The four-part co-condition is applied to the null ensemble; the empirical distribution of the co-condition outcome defines the null operating characteristics. In preferred embodiments, τ_{TE} , τ_{mag} , ε_{CLE} , and N are jointly set so that the false-positive rate under the null ensemble is below a declared target δ_{FP} (for example $\delta_{\text{FP}} < 0.01$), and the minimum window length is set so that the false-negative rate against a declared alternative coupling model is below δ_{FN} . The null ensemble, the achieved $(\delta_{\text{FP}}, \delta_{\text{FN}})$ under the anchor embodiment, and the minimum window length are committed to the protocol digest alongside the threshold values. In embodiments where empirical calibration is not performed before deployment, the Yoked declaration is qualified as *uncalibrated* in the protocol digest and carries reduced evidentiary weight.

Statistical uncertainty and bootstrap confidence intervals (non-limiting).

In some embodiments, block-bootstrap confidence intervals are computed for both the TE balance statistic and the CLE estimate, using non-overlapping convolution bundle windows of declared length. The interval halfwidth at a declared coverage level (for example 95%) is committed to the protocol digest alongside the point estimate. The SNR gate (condition 4 above) is necessary but not sufficient for adequate statistical power; in preferred embodiments, the declared thresholds $(\varepsilon_{\text{CLE}}, \tau_{\text{TE}})$ are set at a declared multiple $\kappa \geq 2$ of the empirical bootstrap halfwidth, ensuring that the Yoked declaration is not made unless the point estimates are statistically separated from zero under the declared estimator.

The minimum window length T_{min} is a committed protocol parameter. Near $\widehat{\text{CLE}} = 0^-$, finite-window CLE estimation is least reliable: the $O(1/\sqrt{T})$ standard error of the estimator is of the same order as the quantity being estimated, and the probability of a false sign-flip (estimated CLE crossing zero when true CLE is just below zero) is highest in precisely the operating region the Yoked condition targets. In preferred embodiments, T_{min} is set so that the bootstrap halfwidth of the CLE estimate is below $\varepsilon_{\text{CLE}}/\kappa$; this is the condition under which the multiplier κ in the threshold-setting rule above provides the declared false-positive coverage. The value of T_{min} is scene-class-dependent and must be characterised against the null ensemble for each declared deployment class; it is committed to the protocol digest per scene class alongside the achieved halfwidth.

Lexicographic priority in the Yoked objective (non-limiting). The scalar Yoked objective J_{Yoked} combines CLE proximity, TE magnitude, and TE balance (Section 22.13). In some embodiments, competing objectives are resolved by a declared lexicographic priority committed to the protocol digest: CLE proximity is primary (the system is not declared
 5 Yoked unless $\widehat{\text{CLE}} \in [-\varepsilon_{\text{CLE}}, 0)$); TE balance is secondary; TE magnitude is tertiary. This ordering prevents configurations with high TE but diverging CLE from being declared Yoked. The lexicographic rule is non-limiting and may be replaced by an alternative priority rule declared in the protocol digest.

8.5.8 Limits of Yoked operation (non-limiting)

10 The capabilities of Yoked operation are bounded by three physical constraints:

Bandwidth. The feedback loop has a characteristic timescale set by the round-trip time. Scene dynamics faster than the loop bandwidth are averaged or aliased; dynamics slower than the loop bandwidth can be tracked and coupled to. For embodiments with a two-rate stack such as the CRT/phosphor extension (Section 10.5), the fast optical or modulation loop can
 15 couple to dynamics up to hundreds of MHz, while the slow camera or sensor loop can couple to dynamics from approximately 0.1 Hz upward; the bench-top 1D anchor (Section 7.2) is a single-rate variant whose loop bandwidth is determined by the unified optical-and-detector return path.

Scene cooperativity. In preferred embodiments, the scene responds dynamically to the
 20 device’s output. A rigid, specularly reflecting surface reflects the device’s output without dynamical engagement. Richer Yoked operation is facilitated by a scene with its own dynamics—resonances, damping, nonlinearity—that interact with the device’s output. Living tissue, mechanical structures, reactive media, and other devices are non-limiting examples of cooperative scenes.

25 In some embodiments, a low-cooperativity scene (non-limiting example: a rigid wall or planar diffuser) exhibits few and narrow entrainment tongues, typically at structural resonance modes in the 10–500 Hz range; training is constrained by SNR and photon budget, and the multi-dimensional tongue map is comparatively sparse. In some embodiments, a high-cooperativity scene (non-limiting example: living human tissue) is expected, based on known
 30 physiological coupling characteristics, to exhibit tongue structures across multiple frequency bands simultaneously, including respiratory coupling (~ 0.1 – 0.5 Hz, typically accessible via spatial pattern repetition rate or slow AM envelope), cardiac coupling (~ 0.8 – 2 Hz, accessible via fast-loop AM interacting with photoplethysmographic return signal), skeletal and postural dynamics, and higher harmonics. Whether these tongue structures are broad and
 35 overlapping or narrow and isolated depends on the strength and nonlinearity of the relevant physiological oscillators and has not yet been directly characterised in a projector-camera feedback loop; this is a predicted embodiment pending experimental confirmation (see Ap-

pendix 29). Tongue widths and positions are expected to shift with physiological state, so that mapping benefits from being performed within a single session. The full multi-dimensional tongue map for a human body, committed in a convolution bundle and logged in a protocol digest, would constitute a dynamical identity record encoding the scene’s response to a diversity of driving conditions rather than its appearance under any fixed illumination.

Signal-to-noise ratio. In some embodiments, coupling quality measurement benefits from sufficient photons per measurement window to estimate transfer entropy or spectral coherence reliably. Meters should track estimation confidence and gate Yoked claims on sufficient SNR.

Non-limiting training time estimates. In some embodiments, the time required to complete Yoked training from cold start to convergence declaration depends on scene cooperativity, estimator window length, and convergence threshold parameters. For the bench-top 1D RK anchor embodiment (Section 7.2), non-limiting estimates for the major phases are: Arnold tongue mapping (coarse grid sweep): 5–20 min wall time; initial lock acquisition (finding the tongue and stabilising within it): minutes of wall time for a low-cooperativity scene, seconds to minutes for a high-cooperativity scene once inside the tongue; TE gradient ascent to the coupling boundary: 20–60 min wall time (bottleneck is TE estimator requiring ~ 100 – 300 samples per window at the relevant timescales); convergence declaration ($N = 5$ – 15 windows at ~ 1 min per window): 5–15 min. Total first-run time is typically 1–4 hours for a new scene; re-training to a previously characterised scene is typically 30–60 min. These estimates are non-limiting and will vary with hardware, scene properties, and declared convergence thresholds; the relevant parameters are declared in protocol digests so that convergence claims are reproducible.

8.6 Continuous Interpolation

In some embodiments, the operating regime is set by continuously varying weights on a composite objective rather than selecting a discrete mode. A composite objective combines Truth Beam, Limager, and Reality Transform terms with weights $(\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}})$ on the two-simplex Δ_2 , and a yoke depth $\alpha \in [0, 1]$ that controls the strength of bidirectional dynamical coupling. The full operating point is $(\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}}, \alpha)$, allowing smooth transitions among verification, sensing, and rendering, and between static and Yoked operation, using the same hardware. In preferred embodiments, the weights, yoke depth, and any transition schedule are recorded in the protocol digest so that meters and audits condition on the declared regime mix and coupling mode.

The regime space has the geometry of a triangular prism: the TB–L–RT triangle at each value of α , extruded along the yoke-depth axis. The base face ($\alpha = 0$) is standard static operation. The top face ($\alpha = 1$) is the maximum commanded coupling depth, at which any regime is fully augmented with dynamic coupling; runs at the top face are classified as Yoked

operation only when the CLE/TE co-condition is satisfied as recorded in the protocol digest.
Key practical protocols occupy identified regions:

Static sensing

($\alpha = 0$, high λ_L): standard computational imaging and active perception.

5 Yoked sensing

($\alpha \rightarrow 1$, high λ_L): resonance-based scene characterisation, liveness detection, dynamic material identification.

Static verification

10 ($\alpha = 0$, high λ_{TB}): standard microstructure-based authentication against enrolled fingerprints.

Yoked verification

($\alpha \rightarrow 1$, high λ_{TB}): continuous tamper detection, coupling-based authentication, scene-coupled security.

Static rendering

15 ($\alpha = 0$, high λ_{RT}): standard projection mapping and style transfer.

Yoked rendering

($\alpha \rightarrow 1$, high λ_{RT}): entrainment-driven display, responsive environments, co-creative projection.

Verified Yoked sensing

20 (high α , balanced λ_{TB} and λ_L): dynamical scene sensing with simultaneous authentication of the coupling.

8.7 Information Operations Taxonomy

The three operating regimes and the Yoked modifier correspond to composite objectives built from more primitive information operations. This section enumerates non-limiting primitive
25 operations, each of which defines a distinct trainable axis with its own loss function. In some embodiments, a protocol specification assigns a weight vector across these axes, and the optimisation targets the weighted multi-objective function on the Fisher-Rao geometry of Θ .

8.7.1 Regime-level operations (summary)

30 The three regime-level operations and one coupling-mode operation are described in detail in the preceding subsections:

Read (Limager).

Maximise $I(S; C_{0:T})$: extract information about the scene.

Write (Reality Transform).

Minimise $d(Y, Y^*)$: impose a target appearance on the scene return.

Authenticate (Truth Beam).

Maximise separation between genuine and forged convolution bundles under a bounded
 5 attacker model.

Co-create (Yoked coupling mode).

Maximise the minimum of bidirectional transfer entropy $\min(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S})$: establish a joint attractor that neither scene nor reactor could produce alone. Yoked is a coupling-mode modifier applicable to any of the three regimes, not a fourth regime (Section 8).

10 8.7.2 Primitive information operations (non-limiting)

In some embodiments, the following primitive operations are individually trainable and composable. Each primitive defines a loss function over the convolution bundle and the device’s configuration parameters, and may be combined with any regime-level objective.

Erase (non-limiting). In some embodiments, the system actively destroys targeted information in the convolution bundle. The objective is $\min I(C_{0:T}; S_{\text{target}})$ subject to a declared utility constraint $I(C_{0:T}; S_{\text{utility}}) \geq I_0$ over declared non-target scene features S_{utility} , so as to drive the output toward statistical independence from specific target features while preserving information about non-target features sufficient for the declared utility model. The trade-off form parallels the information-bottleneck objective recited elsewhere in this disclosure and is not asserted to identify a unique optimum. The feedback loop drives the scene-reactor system into a regime where the target features are in the null space of the observation map. This is not passive omission (“do not record”) but active destruction: the coupled dynamics are driven to a region where the target information is thermodynamically dissipated. In some embodiments, the erasure process is designed so that reconstruction of
 20 the target information is physically costly in proportion to the thermodynamic dissipation incurred in driving the coupled dynamics to the null-space region; that dissipation provides a physical lower bound on the resource cost of any hypothetical reconstruction attempt under the declared attacker model and information assumptions. No claim of computational infeasibility in the complexity-theoretic sense is asserted.

30 Non-limiting applications include physical redaction (erasing faces, identifiers, or other personally identifiable information from the convolution bundle at the physical layer, before digital processing, so that the information is rendered statistically non-inferable under the declared reconstruction model class and metered operating envelope), privacy-preserving sensing (reading structural information while erasing identity information, with the information bottleneck enforced by physics), and secure disposal (driving reactor memory media to
 35 saturation to erase accumulated scene history, with thermodynamic verification that erasure occurred).

Encrypt (non-limiting). In some embodiments, the system makes the convolution bundle conditionally accessible. The objective is $\max I(C_{0:T}; S \mid \theta)$ subject to $\min I(C_{0:T}; S \mid -\theta)$: the output is informative about the scene if the device parameters θ are known; uninformative otherwise. The reactor’s nonlinear transfer function acts as a physical cipher. The same
 5 device can invert the transform because θ is known; a different device is unlikely to succeed because the forward map $S \rightarrow C_{0:T}$ given θ passes through bifurcation structure that creates exponentially many branches, making inversion without θ ill-posed.

Non-limiting applications include physical-layer encryption for optical communication (the convolution bundle is the ciphertext; the device is the key), device-locked content
 10 (output is meaningful only through the originating device or a physical clone, which is microstructure-clone-hard to produce), and physical key exchange (two devices that share a coupling history can establish shared secrets from the joint attractor without digital key exchange protocols).

Obscure (non-limiting). In some embodiments, the system irreversibly destroys semantic content in the scene return while preserving signal energy. The scene remains physically
 15 illuminated but its semantic content is scrambled beyond reconstruction. The objective is to minimise parsability by any downstream model. The mechanism drives the reactor into a chaotic regime where scene features are nonlinearly mixed, with the maximum Lyapunov exponent indicating the rate of trajectory divergence in the reactor’s phase space, producing
 20 rapid mixing and loss of parsable scene structure.

Obscure is distinct from Erase in that Erase targets specific features (selective) while Obscure destroys all semantic content (non-selective). Obscure is distinct from Encrypt in that Encrypt is reversible with knowledge of θ while Obscure is irreversible: the information is thermodynamically dissipated.

25 Non-limiting applications include anti-surveillance (making a scene unparsable to hostile optical sensors by driving the reflected light field into a chaotic regime), physical adversarial perturbation (injecting modulations into the physical light field that break a sensor’s model, as distinct from digital adversarial examples), and communication denial (preventing any receiver from extracting useful information from the optical channel).

30 **Compress (non-limiting).** In some embodiments, the system reduces the effective dimensionality of the convolution bundle while preserving task-relevant information. The objective is $\min d_{\text{eff}}$ subject to $I(C_{0:T}; S_{\text{task}}) \geq \tau$: find the smallest attractor that preserves the target information. The attractor itself is the compressed representation, and the attractor dimension d_{eff} is the physical bottleneck width. This implements an information bottleneck
 35 enforced by physics, in contrast with a digital regulariser.

Non-limiting applications include bandwidth-limited transmission (compress scene information into a minimum-dimension attractor before transmission, with a matched reactor at the receiver acting as a decoder) and resource-constrained verification (reduce d_{eff} to the minimum needed for Truth Beam, discarding irrelevant scene complexity).

Amplify (non-limiting). In some embodiments, the system increases differential sensitivity to targeted scene features. The objective is $\max \partial C_{0:T} / \partial S_{\text{target}}$. The mechanism exploits critical amplification near a bifurcation, where the reactor acts as a tunable nonlinear amplifier whose gain is Fisher-curvature-dependent. In some embodiments, near the bifurcation boundary, gain increases sharply for features aligned with the critical eigenvector, subject to hardware saturation limits and noise floors.

Amplify is distinct from Read: Read maximises total scene information; Amplify maximises differential sensitivity to a specific feature, possibly at the cost of saturating others. Non-limiting applications include sub-threshold detection (detecting signals below the noise floor by exploiting critical amplification) and resonant enhancement (amplifying scene features near a specific frequency by tuning the reactor’s natural frequency to match).

Filter (non-limiting). In some embodiments, the system selectively passes information about desired scene features while actively suppressing undesired features. The objective is $\max I(C_{0:T}; S_{\text{pass}})$ subject to $\min I(C_{0:T}; S_{\text{reject}})$. The reactor’s attractor structure defines a physical passband: features that couple strongly to the attractor pass; features orthogonal to the attractor’s stable manifold are rejected. The attractor’s geometry is the filter shape.

Non-limiting applications include spectral selection (passing one frequency band of scene dynamics while rejecting others), semantic filtering (passing structural features while rejecting textural features, or the reverse), and source separation (if the scene contains multiple superimposed sources, training multiple filter configurations, each passing one source, with the reactor’s nonlinear dynamics acting as a physical implementation of independent component analysis).

Predict (non-limiting). In some embodiments, the system extrapolates scene dynamics forward in time. The objective is $\min_{\pi} \mathbb{E} \|S_{t+\tau} - \hat{S}_{t+\tau}(C_{0:t})\|^2$. If the reactor is entrained with the scene (Yoked operation) and the coupled system occupies a known attractor, the attractor’s geometry constrains future states. Reactor memory media store the trajectory needed for prediction. The prediction horizon is set by the conditional Lyapunov exponent: chaotic scenes yield short horizons (exponential divergence); periodic or quasiperiodic scenes yield long or indefinite horizons.

Non-limiting applications include predictive maintenance (entraining with a machine’s vibration signature and predicting future state from attractor dynamics), biological forecasting (entraining with physiological rhythms and predicting upcoming state changes), and predictive sensing (forecasting where the scene will be and pre-pointing the illumination to optimise the next measurement).

Sign and watermark (non-limiting). In some embodiments, the system embeds device identity into the convolution bundle. Two modes are distinguished:

Overt signing: the device’s identity is explicitly present in the convolution bundle. The device’s holonomy—accumulated geometric phase from cyclic parameter sweeps—is a natural

device-specific signature that is verifiable by any party with access to the family’s attractor bundle description.

Covert watermarking: the device’s microstructure-specific response embeds a subtle, broadband signature in the convolution bundle that survives downstream processing (compression, cropping, colour correction) but is detectable only by a matched device or with knowledge of the detection key. In some embodiments, the watermark is trained adversarially to survive a declared suite of transforms while remaining undetectable by a discriminator.

Non-limiting applications include provenance (every convolution-bundle window carries an empirically hard-to-forge physical signature of the originating device), chain of custody (successive devices add their signatures; the output carries a physical audit trail), and anti-deepfake (genuine Reality Kernel output carries a physical watermark that synthetic content is impractical to reproduce without the actual device’s θ).

Forget (non-limiting). In some embodiments, the system selectively erases accumulated information from reactor memory media (for example CRT phosphor state, phase-change material configuration, or other persistent media) while preserving other stored information. The objective is $\min I(\text{Memory}_{\text{new}}; S_{\text{target}})$ subject to preserving $I(\text{Memory}_{\text{new}}; S_{\text{other}})$. Selective beam deflection or localised overwrite targets specific memory locations. Each erasure cycle incurs an irreducible thermodynamic cost.

Forget is distinct from Erase: Erase targets scene information in the *output*; Forget targets accumulated information in the *memory state*. Erase operates on the present; Forget operates on the past.

Non-limiting applications include rolling-window operation (continuously forgetting information older than a declared horizon while retaining recent history, implementing a physical first-in-first-out buffer), privacy compliance (auditable erasure of specific data from device memory under declared hardware and environmental assumptions), and selective pattern retention (forgetting undesired acquired patterns while retaining desired ones, a physical analogue of managing catastrophic forgetting in neural networks).

Create (non-limiting). In some embodiments, the system generates output in the absence of external scene input. The scene loop is either disabled ($S^{\text{scene}} = S_{\text{term}}$, as in Polie-
Puter mode) or presents a featureless input, and the reactor’s own dynamics—driven by internal noise, spontaneous emission, or stochastic fluctuations amplified near a bifurcation—produce structured output.

The mechanism exploits symmetry breaking at bifurcation points: when the reactor’s operating point crosses a bifurcation boundary, infinitesimal noise is amplified into macroscopic pattern selection. The specific pattern selected is determined by the reactor’s microstructure and the instantaneous noise realisation, making each creation event device-specific and non-reproducible. In some embodiments, the creation process is guided by a bias or seed embedded in the control protocol (for example a weak stimulus that biases which branch of the bifurcation is selected) while the detailed structure is filled in by the reactor’s autonomous

dynamics.

Non-limiting applications include physical random number generation (the bifurcation-amplified noise provides entropy that is grounded in physical process rather than algorithmic pseudorandomness), generative art (the reactor produces patterns that are neither predetermined by software nor directly copied from a scene, but emerge from the device's own
 5 nonlinear dynamics), and stochastic search (the reactor's exploration of its own attractor landscape, guided by a weak objective signal, discovers configurations that directed optimisation might miss).

Create is distinct from Write: Write imposes a target pattern Y^* determined externally;
 10 Create produces a pattern from within the device's own dynamics, with no external target. The convolution bundle records the creation event, and protocol digests record the control conditions under which creation occurred.

Invert (non-limiting). In some embodiments, the system or an external party attempts to recover the scene S from the convolution bundle $C_{0:T}$. Inversion difficulty depends on
 15 knowledge of the device parameters θ : with known θ , inversion is a well-posed (if nonlinear) optimisation (this is the legitimate operator's situation); without known θ , inversion requires simultaneously estimating θ and S from $C_{0:T}$, and the bifurcation structure creates exponentially many branches that are unidentifiable from finite data.

The ratio of inversion difficulty (unknown θ divided by known θ) is the security margin.
 20 This ratio is maximised at high Fisher curvature (near bifurcations) where the mapping is maximally θ -sensitive. In some embodiments, the Invert primitive is used as an internal diagnostic: measuring how easily a digital model can invert the device's output provides a lower bound on hardness.

8.7.3 Composite operations (non-limiting)

25 In some embodiments, primitive operations are composed to produce higher-level capabilities. Non-limiting composites include:

Redact (Read + Erase).

Identify specific content in the convolution bundle via a Limager head, then actively destroy it via the Erase primitive, producing a convolution bundle that contains task-
 30 relevant information but physically lacks the redacted content.

Physical steganography (Write + Watermark).

Render target content via Reality Transform while simultaneously embedding a covert device signature via the Watermark primitive.

Predictive sensing (Predict + Read).

35 Forecast scene state from Yoked coupling dynamics, then optimise the next Limager measurement for the predicted state.

Authenticated encryption (Encrypt + Sign).

Produce device-locked content (meaningful only with knowledge of θ) that simultaneously carries an overt provenance signature.

Privacy-preserving authenticated sensing (Read + Erase + Authenticate).

5 Authenticate the scene and device via Truth Beam, extract task-relevant information via Limager, and physically erase personally identifiable information via the Erase primitive, all within a single feedback session. The convolution bundle that exits the session is designed so that the erased content is destroyed in the analogue domain before digital recording, making reconstruction infeasible under the declared threat model.

10 8.7.4 Training vector space and regime geometry (non-limiting)

In some embodiments, a protocol specification is a point in a multi-dimensional objective space, with one axis per primitive operation. A weight vector

$$(w_{\text{read}}, w_{\text{write}}, w_{\text{auth}}, w_{\text{yoke}}, w_{\text{erase}}, w_{\text{encrypt}}, w_{\text{obscure}}, \\ w_{\text{compress}}, w_{\text{amplify}}, w_{\text{filter}}, w_{\text{predict}}, w_{\text{sign}}, w_{\text{forget}}, w_{\text{create}})$$

defines the Pareto trade-off for a specific task. The three operating regimes and the Yoked modifier occupy specific faces of this polytope (for example, pure Limager is the face where w_{read} dominates). Named composites occupy identified edges or interior points. Training is the process of finding the operating point in this polytope that maximises a task-specific utility function, using the natural gradient defined by the Fisher information metric of Θ .

8.8 Cross-Scene Regime Combinations (Non-limiting)

In some embodiments, a device with multiple optical ports (scene-facing channels, reactor-facing channels, or distinct field-of-view sectors) operates different ports in different regimes simultaneously. In some embodiments, two or more devices observe the same scene from different vantage points, each in a potentially different regime.

The resulting combination space has the structure of a product of per-port regime simplices. For a device with two ports A and B, each port occupies a point on the $(\lambda_{\text{TB}}, \lambda_{\text{L}}, \lambda_{\text{RT}}, \alpha)$ prism described in Section 8.6, and the joint operating point lies on the product of two such prisms.

Non-limiting cross-port combinations.

Authenticated rendering (Truth Beam on port A, Reality Transform on port B).

30 Rendering is gated on continuous verification: the Reality Transform output is produced only while the Truth Beam port confirms device and scene authenticity. If the verification score drops below a threshold, rendering is suspended or reverted to a safe default.

Differential sensing (Limager on port A, Limager on port B).

Two scenes or two views of the same scene are compared through the same device. The device's transfer function cancels in the difference, isolating scene-to-scene variation. In some embodiments, one port observes a reference and the other observes a sample,
 5 enabling device-mediated comparison with cancellation of systematic device effects.

Scene translation (Limager on port A, Reality Transform on port B).

Information read from scene A is used to drive the rendering applied to scene B. The device performs a physical scene-to-scene translation: the style, content, or structure of one scene modulates the appearance of another through the reactor's nonlinear dynamics.

10 Multi-surface rendering (Reality Transform on port A, Reality Transform on port B).

Two scene surfaces are rendered simultaneously with shared reactor dynamics. Temporal coordination through the reactor produces coupled visual effects that may be difficult or impractical to achieve with a pair of independent projectors.

15 Yoked cross-scene coupling (Yoked on port A, Yoked on port B).

Two scenes are dynamically coupled through the device. The device mediates a physical interaction between scenes that are not in direct contact. Transfer entropy flows $S_A \rightarrow D \rightarrow S_B$ and $S_B \rightarrow D \rightarrow S_A$ create a device-mediated coupling channel.

20 Multi-device combinations (non-limiting). In some embodiments, multiple devices observe the same scene, each operating in a potentially different regime:

Mutual device authentication.

Two devices both operate in Truth Beam on the same scene. Cross-device consistency of the convolution bundle, conditioned on the shared scene, provides a basis for mutual authentication without a trusted third party.

25 Consensus sensing.

Two or more devices operate in Limager on the same scene. Agreement across independent device observations, each with distinct θ , strengthens scene claims beyond what any single device can provide.

Distributed authenticated rendering.

30 One device operates in Truth Beam to authenticate the scene; a second device operates in Reality Transform, conditioned on the first device's verification output. Verification and rendering are physically separated but cryptographically linked via committed convolution-bundle atoms.

35 Temporal sequencing (non-limiting). In some embodiments, a device transitions between regimes over time within a single session. Non-limiting sequences include: a Truth

Beam phase followed by Limager followed by Reality Transform (authenticate, then sense, then render); interleaved Truth Beam checks during Yoked operation (integrity monitoring during entrainment); and alternating Limager and Reality Transform passes (iterative refinement where each sensing pass informs the next rendering pass). In some embodiments, the protocol digest records the regime sequence, transition times, and any gating conditions.

8.9 Degenerate and Boundary Cases

The following limiting cases define the boundary conditions of the regime framework and the baselines against which capability and hardness aspects of this disclosure are measured.

Trivial reactor. If the reactor transfer function is the identity ($g_{\text{reactor}} = \text{id}$), all regimes collapse to conventional optics: Limager becomes a standard camera, Reality Transform becomes a standard projector, Truth Beam provides no physically unclonable function, and Yoked operation reduces to projector–camera feedback (which has been described in the projection mapping literature). This is the baseline that every hardness and capability claim is measured against.

Fully linear reactor (non-limiting). If the reactor has a linear (but non-identity) transfer function—for example a gain stage, colour filter, or linear time-invariant (LTI) medium with memory—the input–output map is reproducible by standard linear-systems estimation (finite impulse response identification) given a sufficient number of challenge–response pairs. Such a reactor provides limited empirical hardness because emulation is straightforward once the transfer function is characterised. Hardness claims in this document rely on reactor *nonlinearity* (saturation, bistability, chaotic mixing), *bifurcation topology*, and *idiosyncratic microstructure* that make finite-data emulation intractable under the declared attacker families. A fully linear reactor is therefore a second baseline, stronger than the identity case but weaker than any nonlinear embodiment.

Trivial scene. If the scene is absent, featureless, or replaced by a calibration target, Limager returns no useful information about the external world, Reality Transform has no target substrate, and Yoked operation has no dynamic partner. In such embodiments, Inner Limager (self-calibration) and Truth Beam remain operational as the primary meaningful modes. The device becomes a pure microstructure challenge-response oracle. In some embodiments, this is the declared mode for device enrolment, commissioning, and self-calibration.

Broken feedback (open loop). If the feedback path is interrupted, all closed-loop features are lost. Limager reverts to passive sensing. Reality Transform reverts to static projection. Truth Beam loses closed-loop hardness and reverts to static challenge-response (Level 0). Yoked operation is not available in this mode. In some embodiments, the system detects feedback interruption and falls back to a reduced-capability mode comprising at

least static challenge-response verification, and the interruption event is logged in the meter envelope.

Edge of synchronisation. When the conditional Lyapunov exponent is near zero ($CLE \approx 0$), Fisher information peaks, and all regimes are simultaneously at their most powerful and most fragile. This is the operating point of maximum information capacity but also maximum sensitivity to perturbation. In some embodiments, the operating policy targets a neighbourhood of this boundary rather than the boundary itself, trading off peak capacity against robustness, with the distance from the boundary recorded in the meter envelope as a stability margin.

Bifurcation crossing during operation. If the device’s operating point drifts across a bifurcation surface B during a session, the attractor changes topology mid-operation. The regimes respond differently to this event: for Truth Beam, it is a verification failure (the expected attractor has changed); for Limager, it is a recalibration event (switch to a new attractor model); for Reality Transform, it may be a discontinuity in the output; for Yoked operation, it may be a desirable discovery event (the coupled system finding a new dynamical regime) or a catastrophic instability depending on the application. In some embodiments, the protocol digest records whether a bifurcation crossing occurred during the session, and meters report pre- and post-crossing attractor identity.

Identical-twin devices. Manufacturing tolerance may produce devices whose attractor bundles are metrically indistinguishable at a given curvature resolution. Higher verification levels (Level 2 or Level 3) or longer trajectories resolve this, but there exists a minimum interrogation complexity below which two devices are functionally equivalent. This is not a failure of the framework; it bounds the security claim. The Cramér–Rao bound on device-parameter estimation from finite observations sets this resolution floor.

8.10 Human-Coupled Operations (Non-limiting)

In some embodiments, a human observer or participant is included in the feedback loop as an active agent rather than a passive viewer. The human’s sensory and motor channels become part of the coupled system, and the information operations taxonomy (Section 8.7) extends to include human-mediated primitives. Each human-coupled primitive defines a distinct training objective involving the human’s response.

HC-Read (non-limiting). The human reads the Reality Kernel’s output through one or more sensory channels (visual, auditory, haptic). The objective is $\max I(C_{0:T}; R_{\text{human}})$, where R_{human} denotes the human’s measurable response (gaze, reaction time, task performance, physiological signal). The system optimises its output to maximise the human’s information

uptake, adapting illumination, rendering, or sonification to the human’s perceptual capabilities and attentional state. In some embodiments, the human’s response is measured via eye tracking, EEG, galvanic skin response, or task performance metrics, and these measurements are logged in the convolution bundle as auxiliary channels.

5 **HC-Write (non-limiting).** The human provides input that the Reality Kernel incorporates into its feedback loop. Input channels include gaze direction, gesture, voice, touch, physiological signals (heart rate, respiration, muscle tension), and explicit control inputs. The objective treats the human’s input as a scene signal: $\max I(H_{\text{input}}; C_{0:T})$, where H_{input} is the human’s input stream. In some embodiments, multiple human input channels are fused
10 through the reactor’s nonlinear dynamics, producing a physical multimodal integration that is device-specific and logged.

HC-Verify (non-limiting). The human participates in the verification loop by confirming or denying the plausibility of the Reality Kernel’s output. The human acts as an auxiliary verifier whose judgment is logged alongside automated meter scores. In some embodiments,
15 the system presents verification challenges that require human perceptual judgment (for example detecting subtle visual artefacts that automated metrics miss), and the human’s response is bound to the protocol digest. In some embodiments, HC-Verify implements a human-in-the-loop gate where automated operations proceed with human confirmation.

HC-Entrain (non-limiting). The human and the Reality Kernel mutually entrain: the
20 device’s output influences the human’s physiological state, and the human’s physiological response modulates the device’s feedback loop. This is distinct from software-mediated biofeedback: the physical coupling between the device’s optical output and the human’s neural or physiological response IS the intervention, with no digital intermediary in the coupling path. The objective is $\max \min(\text{TE}_{D \rightarrow H}, \text{TE}_{H \rightarrow D})$, where D and H denote device and human
25 respectively. Non-limiting applications include neurofeedback (EEG-optical coupling where brain rhythms and reactor dynamics mutually entrain), rehabilitation (motor-optical coupling for movement training), and biofeedback (entraining physiological rhythms with stable reactor dynamics for regulation).

HC-Create (non-limiting). The human guides the creation process (Section 8.7.2) by
30 providing weak biases that influence which branch of a bifurcation the reactor selects, while the detailed structure is filled in by the reactor’s autonomous dynamics. The human’s creative intent is expressed through continuous low-bandwidth input (for example a joystick, gaze direction, or emotional valence derived from physiological signals) that biases the reactor’s symmetry-breaking, producing output that is neither fully determined by the human (the
35 reactor contributes its own nonlinear dynamics) nor fully autonomous (the human’s intent shapes the large-scale structure). The convolution bundle records both the human’s input and the reactor’s response, making the co-creative process auditable.

8.11 Coupling-history effects and physical key exchange (non-limiting)

In some embodiments, the joint attractor of two coupled systems produces a shared dynamical history that is empirically hard to reconstruct without access to both systems and their coupling record. In some embodiments, coupling-history features are used for trust-topology construction, physical key exchange from correlated measurements, and cost-graded verification where closely coupled devices verify more cheaply than loosely coupled devices. Higher-layer coordination and agent-autonomy applications of coupling-history reification are described in the related Filing 2 application (optional, non-essential).

9 Meters and Training Games

Meters may act directly on the convolution bundle or on derived latents. In some embodiments, meter families include: (i) verisimilitude discriminators that distinguish physical from synthetic traces, (ii) hardness estimators that operationalise empirical difficulty of simulation or forgery under bounded adversaries, (iii) calibration probes that track alignment and drift, and (iv) coupling meters that track Yoked operation quality.

9.1 Yoked-specific meters (non-limiting)

In some embodiments, meters specific to Yoked operation include:

Transfer entropy balance

$TE_{S \rightarrow R}$ and $TE_{R \rightarrow S}$ estimated from the convolution bundle, together with their ratio and absolute values.

Conditional Lyapunov exponent estimate

An estimate of CLE from the time series, indicating proximity to the synchronisation boundary.

Spectral coherence

Cross-spectral density between scene return and reactor output, at each frequency band.

Order parameter

A scalar summary of coupling quality, for example the Kuramoto order parameter $r = |N^{-1} \sum_j e^{i\phi_j}|$ computed over coupled oscillator phases.

Yoke depth

The declared target α_{target} logged in the protocol digest and the meter-derived achieved $\hat{\alpha}$ estimated from the convolution bundle.

Coupling estimation confidence

An estimate of the statistical reliability of coupling quality measurements, gating Yoked claims on sufficient SNR.

Training may include supervised learning, self-supervised learning (including without limitation SimCLR, MoCo, BYOL, SwAV, DINO, masked-autoencoder methods such as MAE, masked language modelling, joint-embedding predictive architectures including JEPA, and joint vision-language methods such as CLIP, ALIGN, and SigLIP), contrastive objectives, generative modelling (including diffusion, score-based, flow-matching, or consistency models), reinforcement learning for protocol selection, and hardware-in-the-loop optimisation such as SPSA, evolution strategies, natural evolution strategies (NES), covariance matrix adaptation evolution strategy (CMA-ES), population-based training (PBT), genetic algorithms, particle swarm optimisation, and rank-structured evolution-strategy methods (including without limitation rank- r perturbation schemes that recover batched-matrix-multiplication arithmetic intensity for hyperscale training) for non-differentiable parameters.

9.2 Roles and multi-agent framing (non-limiting)

In some embodiments, design and operation are framed as a multi-agent interaction. Alice selects configuration and protocols and trains control policies or adaptation procedures. Bob trains or operates verifiers and meters that evaluate the convolution bundle for plausibility, calibration, or hardness. Eve acts as an adversary attempting to spoof, simulate, or perturb the convolution bundle or downstream decisions under a bounded resource model. This framing is used primarily for verification (Truth Beam), but may also be used for Limager and Reality Transform training when adversarial robustness or distribution control is desired.

Eve as a system-wide hardening principle (non-limiting). In some embodiments, Eve is not confined to a single protocol layer but serves as the adversarial-hardening archetype operating at every level of the Reality Kernel architecture. At the physical-channel level, Eve attempts to spoof or forge convolution bundles under bounded resources, driving the empirical hardness that underpins Truth Beam verification. At the information-theoretic level, Eve observes a degraded view of the physical channel (the wiretap lens of Section 21.4), and protocol design aims to widen the Bob–Eve advantage gap. At the meter-training level, Eve’s best-effort attacks under frozen meters define the hardness index that calibrates security claims. At the semantic level, a semantic Eve attempts to produce alternative descriptions, inferences, or behavioural outputs that pass semantic verification (Section 8.2.3); in PoliePuter mode, semantic Eve probes the agent’s reasoning and decision-making under adversarial prompts, perturbations, and distribution shifts, deliberately pushing toward unsafe or deceptive operating regimes so that the system is hardened against regions that natural drift might otherwise reach undetected. At the training level, adversarial training environments (described in the related Filing 2 application; optional) instantiate the same adversarial principle in the agent’s training trajectory, stress-testing behavioural policies under declared floors and meter envelopes. In each case, the adversarial role is constructive: by systematically probing failure modes under bounded attacker families, Eve drives the system toward empirically robust configurations at every layer of the architecture.

Eve injector and jammer embodiment with kappa-only training variant (non-limiting). In some embodiments, Eve operates as an active injector or jammer rather than only as a passive observer: Eve is permitted to inject a bounded signal $j_\psi(c)$ into the optical, electromagnetic, acoustic, or coupled physical path on which the reactor operates, subject to a declared injection power constraint, a declared spectral support, a declared spatial-or-channel mask, and a declared duty-cycle envelope, all of which are committed to the protocol digest. The injected signal is treated as part of the adversarial setting under which Bob's meters and Truth Beam verification are required to retain calibrated performance, and Eve's admissible attack family includes constant, swept-frequency, modulated, and replay-style injections. In further embodiments, a *kappa-only training variant* is supported in which only the physical kernel parameters κ of the reactor are updated during a training round, while ancillary parameters including optical and electrical attenuators, dampers, emitter counts, amplifier biases, and timing-channel keys are held externally fixed at declared settings recorded in the protocol digest. Kappa-only training is used when training-induced changes to ancillary parameters would otherwise confound hardness or robustness measurements, or when a fixed-ancillary baseline is required for cross-device comparability. In some embodiments, the Eve injector embodiment and the kappa-only training variant compose: Eve injects under declared envelopes while only κ is updated, producing an adversarial-robust trained reactor under fixed ancillary settings.

The system-wide counterpart to Eve's adversarial principle is the *Iris* principle: at every layer where Eve probes for failure, Iris expands information capture, ensuring that the full signal content of the physical channel is available for verification, sensing, and calibration. A well-designed Reality Kernel loop alternates between Iris expansion (maximising what the channel can resolve) and Eve hardening (ensuring that what is resolved cannot be forged), with Demeter providing the bidirectional management that maintains stability across the alternation. The three primary stances (and the Harmonia coupling modifier) are defined formally in Section 8.1.

In some embodiments, Reality Kernel design and operation are treated as a two-timescale game. On a slow timescale, Alice commits to a hardware and protocol configuration θ^A (and associated admissible protocol classes), after which Bob and Eve adapt toward best responses within bounded resource classes (meters, models, and query budgets). This can be read as a Stackelberg game with Alice as leader and Bob/Eve as followers, where empirical hardness indices are defined against these best-response followers. On a fast timescale, for fixed physical configurations, Bob and Eve interact with the live device in repeated challenge-response episodes under logged protocol digests.

Programmable-scene jamming game (non-limiting). In some embodiments, the scene includes a programmable surface such as an e-ink display that renders a time-indexed plaintext, watermark, or procedural field whose parameters encode a message. In a non-limiting multi-agent game, Bob is a cooperative decoder trained to recover the plaintext from the convolution bundle under the declared protocol, while Eve attempts to decode the

same plaintext from a degraded view and/or jam the channel by injecting co-illumination or structured interference. Alice selects kernel and protocol families to increase a secrecy-margin objective under meter-envelope constraints:

$$\max_{\kappa, \pi} \left(\mathcal{U}_{\text{Bob}}(\kappa, \pi) - w_{\text{adv}} \mathcal{U}_{\text{Eve}}(\kappa, \pi) \right), \quad \text{subject to meter envelopes and safety bounds.}$$

Multi-beam dampers (non-limiting). A power-allocation vector $a \in \mathbb{R}_+^B$ distributes probe or carrier power across B beams or logical channels distinguished by space, time, frequency, and/or polarisation. A damper vector $\delta \in [0, 1]^B$ attenuates designated beams. Alice optimises (a, δ) jointly with κ and protocol choices to increase Bob’s SNR and reduce Eve’s leverage, subject to $\sum_b a_b \leq P_{\text{max}}$ and per-beam safety bounds.

9.3 Meter families (non-limiting)

In some embodiments, meters operate on convolution-bundle windows and/or derived latents and output scalar scores. Verisimilitude meters score whether a trace is consistent with the empirical distribution of physical runs. Hardness meters estimate difficulty for a specified adversary family to generate traces that pass verification, optionally producing a hardness index. Calibration meters track alignment, drift, timing integrity, and cross-device consistency. Meters may be learned discriminators, energy-based models, statistical tests, or hybrids. In some embodiments, meters are distilled or simplified for embedded operation after training.

Sequential hypothesis-testing meters (non-limiting). In some embodiments, verifier decisions are made by sequential hypothesis-testing procedures rather than by a fixed-length batch score. A sequential probability ratio test, anytime-valid confidence sequence, e-value process, alpha-spending rule, or other declared sequential test updates the accept, reject, or continue state as committed bundle windows arrive. The null and alternative hypotheses, stopping boundaries, maximum sample count, error budgets, multiple-testing correction, and policy for inconclusive runs are declared before the window opens and are included in the verifier’s protocol digest.

9.4 Threat model taxonomy (non-limiting)

Non-limiting attack families include replay of previously recorded traces, model-based simulation using learned emulators, deepfake synthesis, adversarial perturbations of genuine traces, co-illumination interference, hardware tampering, and semantic misattribution. In preferred embodiments, replay is mitigated using freshness challenges and commitments, while simulation and perturbation are addressed using verisimilitude and hardness meters and protocol design.

9.5 Verisimilitude discriminator training (non-limiting)

In one embodiment, a discriminator is trained to distinguish physical convolution-bundle windows from synthetic or attacked windows. Negative examples may include simulator outputs, time-shuffled segments, injected spikes outside hardware envelopes, and quantised or clipped traces. After training, the discriminator may be used as a soft penalty when training controllers and emulators so that rollouts remain within empirically supported regions.

Definition 31 (Verisimilitude functional). *A verisimilitude functional for a given configuration θ is any measurable map*

$$V_\theta : \mathcal{C} \times \mathcal{X}_{\text{aux}} \rightarrow \mathbb{R}$$

from the convolution bundle (and any auxiliary metadata $\mathcal{X}_{\text{aux}}$, such as protocol digests or claimed labels) to a real-valued score, such that higher scores indicate better agreement with the empirical distribution of physical runs under the same or comparable protocol classes. In many embodiments V_θ is realised by a learned discriminator, but hand-crafted statistics are also non-limiting embodiments.

Verisimilitude checker for projector–camera loops (non-limiting). In some embodiments, a projector–camera loop is treated as a special case of a Reality Kernel in which a subset of the control $u(t)$ is an emitted image or pattern and a subset of the observation $\mathbf{y}_t$ is a captured image. A verisimilitude checker is a learned meter that outputs a score indicating whether windows of loop data are consistent with the empirical distribution of genuine, correctly functioning loops under comparable protocol digests. In preferred embodiments, the verisimilitude checker is trained to be sensitive to physical tampering, miscalibration, spoofing, and relay artefacts that preserve superficial pixel statistics but break loop consistency.

Adversarial autoencoder for loop data (non-limiting). In some embodiments, verisimilitude training uses an adversarial-autoencoder pattern: an encoder maps loop windows to latents, a generator/decoder reconstructs one or both sides of the loop, and a discriminator distinguishes genuine loop windows from reconstructed, simulated, or attacked candidates. A reconstruction loss encourages predictive closure of the loop under the logged protocol, while an adversarial loss encourages the latent representation to respect empirical physical constraints rather than only pixel-level similarity.

Multi-pair coordinated training (non-limiting). In some embodiments, a plurality of projector–camera pairs operate in a shared physical scene, and models are trained with cross-pair constraints: candidate reconstructions or simulations are constrained to be mutually consistent across pairs under the same scene coupling and protocol digest.

Human response channels as auxiliary loop evidence (non-limiting). In some embodiments, the loop includes an auxiliary response channel derived from a human observer,

including gaze direction, head pose, task performance, comfort ratings, or physiological signals. The response channel is logged as auxiliary evidence and may condition the verisimilitude checker so that “plausible for a human” and “physically consistent” are jointly enforced under declared policies.

- 5 **External and virtual-view checks (non-limiting).** In some embodiments, given a learned 3D representation or a digital twin, the system predicts what an additional camera should observe under a declared emission schedule, and compares predicted versus measured sequences as an additional consistency term.

Illustrative empirical verifier-stack behaviour. In some embodiments, the verisimilitude meter is evaluated using plural independently trained verifier paths that are not co-trained with one another and that do not share the same optimisation loop. As a non-limiting illustration, independently trained residual, binder, and feature-distance verifier paths may each separate genuine committed-bundle/emission pairings from shuffled-emission pairings and from trained-adversary pairings. Cross-deployment or cross-session evaluation may be
 10 used to test whether the verifier paths are responding to substrate-physics structure rather than to a session-specific artefact. Such empirical observations are reported as part of a declared metric envelope and do not limit the apparatus to any particular architecture, dataset, numerical threshold, or performance value.
 15

Single-meter failure modes as motivation. In some embodiments, optimisation
 20 against a single verifier or single binder-confusion loss can produce Goodhart-style failure modes. Non-limiting examples include anti-binder texture artefacts that confuse a target binder without corresponding to physically valid emissions, and over-smoothed identity collapse in which continued optimisation converges toward a degenerate or near-identity transform that remains visually plausible while failing under an independent verifier path.
 25 The multi-path independence, held-out evaluation, and protocol-digest commitment discipline disclosed herein are configured to make such single-meter pathways detectable. The named failure modes are illustrative only; other failure modes may arise under different adversary architectures, losses, modalities, or deployment conditions.

Load-bearing verifiers and diagnostic instruments. In some embodiments, load-bearing verifier paths are trained or calibrated against committed-bundle ground truth,
 30 curated negative examples, declared distractor sets, adversary-generated examples, or other governance-relevant labels. Non-limiting examples include binder-style emission-recovery models, correct-versus-distractor retrieval models, feature-distance detectors, PatchCore-class detectors, supervised or semi-supervised anomaly detectors, and verisimilitude meters
 35 trained against declared negative example sets. In further embodiments, unsupervised methods including autoencoder-based substrate-signature detection, reconstruction-error scoring,

clustering, density estimation, and latent-space drift analysis are used in parallel as diagnostic instruments for depth-localisation, drift monitoring, calibration, exploratory audit, or spot-check review. Such diagnostic instruments may support the verifier stack without necessarily serving as load-bearing verifiers under the protocol digest.

5 9.6 Domain Discriminators and Domain Generators

In some embodiments, a Reality Kernel apparatus comprises a domain discriminator and a domain generator. A domain discriminator is a model, meter, classifier, verifier, or scoring function configured to estimate whether a bundle, feature, latent state, view, emission, or derived artefact belongs to, or is consistent with, a declared domain. A domain generator
 10 is a model, transformer, simulator, renderer, or adapter configured to produce, translate, normalise, bridge, or augment artefacts across domains while preserving one or more declared constraints.

In some embodiments, domains comprise physical deployments, sessions, rigs, environments, spectral bands, modalities, virtual worlds, simulation environments, reactor states,
 15 governance states, privacy release channels, calibration regimes, observer populations, or adversary curricula. A domain label d may be explicit, inferred, hidden, soft, hierarchical, or represented as a continuous embedding. A discriminator may score:

$$s_d = D_\psi(C_{0:T}, U_{0:T}, Z_{0:T}, O_{0:T}, \Pi; d),$$

where s_d indicates domain membership, consistency, likelihood, distance, confidence, or anomaly score. A generator may produce a domain-transformed artefact:

$$\tilde{C}_{0:T}^{(d')} = G_\phi(C_{0:T}^{(d)}, d, d', \Pi),$$

20 where d and d' denote source and target domains. These equations are non-limiting; the discriminator and generator may operate on raw bundles, reduced features, latent representations, emissions, labels, metadata, views, or meter outputs.

In some embodiments, the discriminator and generator are trained or evaluated jointly. For example, a generator may attempt to translate a bundle from a first deployment domain into a second deployment domain, while a discriminator tests whether the translated
 25 bundle preserves substrate evidence, emission identity, protocol consistency, or governance-relevant signals. In some embodiments, the discriminator is used to reject generator outputs that match superficial domain statistics while failing a binder, held-out verifier, physical re-execution, temporal-coherence meter, or anchor-consistency meter.

30 In some embodiments, the domain-generator primitive supports counterfactual and privacy-preserving governance. For example, a generator may produce a neutralised, redacted, stylised, compressed, or domain-normalised representation for audit, while a domain discriminator or held-out meter verifies that the representation retains the declared governance signal and does not introduce a forbidden transformation. In further embodi-

ments, a domain discriminator detects when a deployment has drifted outside an enrolled domain and triggers calibration, selective opening, increased monitoring, rescue mode, or governance escalation.

Domain primitive generalisation (non-limiting). The domain discriminator may be

any scoring or decision function, including without limitation a classifier, binary discriminator, multi-class discriminator, contrastive model, density estimator, normalizing flow, energy-based model, one-class detector, diffusion residual scorer, retrieval model, conformal predictor, Bayesian model, or ensemble. The domain generator may be any transform or synthesis model, including without limitation an image-to-image translator, video-to-video translator, diffusion model, GAN, flow model, autoregressive model, renderer, simulator, style-transfer model, domain-adaptation network, feature normaliser, calibration transformer, physics model, or optimisation-based procedure. The term domain is not limited to visual domains and may include optical, acoustic, electromagnetic, temporal, semantic, governance, privacy, simulation, or physical deployment domains.

Composition with virtual-camera and virtual-emitter primitives (non-limiting).

In some embodiments, the domain discriminator and domain generator primitives of this subsection are instantiated using the virtual camera and virtual emitter primitives disclosed in Section 17.3, with the virtual camera or virtual emitter acting as a domain generator and an external or held-out meter acting as a domain discriminator. Conversely, the virtual-camera and virtual-emitter primitives of Section 17.3 may use the domain discriminator and domain generator of this subsection as held-out or adversarial verifiers, for example in verisimilitude games and re-capture or re-emission audits.

9.7 Reality Encryption and Neutral Recording Generation

In some embodiments, a Reality Kernel apparatus implements reality encryption by emitting, selecting, modulating, or transforming a physical stimulation pattern such that, under a declared meter envelope and attacker model, a resulting recording is intended or declared to be verifiable, reconstructable, decryptable, classifiable, or semantically useful only to an authorised opener satisfying a declared key, protocol-state, model-state, physical-response-model, authority-token, or selective-opening condition. As with other security and hardness properties disclosed in this specification, the “only to an authorised opener” property is empirical under the declared attacker family and resource budget, is time-indexed rather than unconditionally durable, and does not constitute a formal cryptographic guarantee unless a specific embodiment expressly specifies and parameterises such a guarantee. The encrypted object may be the emitted signal, the captured response, a latent representation, a derived label, a meter output, a reconstruction path, or any combination thereof.

In some embodiments, an emission u_t is derived from a protocol state, key material, and

optional payload:

$$u_t = E_\omega(K_t, \Pi_t, p_t, q_t),$$

where K_t may comprise a cryptographic key, threshold key, PUF-derived value, physical one-way function output, session key, authority key, or derived protocol state; p_t may comprise a payload, challenge, watermark, governance marker, or null payload; and q_t may comprise a
 5 timestamp, randomness beacon, ledger value, prior digest, or external challenge. A recording y_t produced in response to u_t may be bound into a committed bundle:

$$C_{0:T} = \{(u(t), y(t))\}_{t=0}^T.$$

A verifier or authorised opener may recover or test a protected signal:

$$r_t = R_\kappa(C_{0:T}, K_t, \Pi_t, A_t),$$

where A_t may comprise authority records, selective-opening records, attestation records, or proof records. These equations are illustrative only and do not limit reality encryption to
 10 conventional encryption, to digital payloads, or to any particular cryptographic primitive.

In some embodiments, the apparatus comprises a neutral recording generator. A neutral recording generator produces a released recording or representation that preserves one or more declared verification, governance, calibration, or audit signals while suppressing, mask-
 ing, normalising, stylising, encrypting, or removing one or more protected content signals.
 15 The neutral recording may be produced at capture time, after capture, in a governance partition, in a privacy partition, or through a coupled physical-digital process. A neutral recording $\bar{C}$ may be generated as:

$$\bar{C}_{0:T} = N_\phi(C_{0:T}, \Pi, \lambda_{\text{priv}}, \lambda_{\text{verify}}),$$

where λ_{priv} denotes declared privacy constraints and λ_{verify} denotes declared verification constraints. The neutral recording may be accepted when it satisfies:

$$M_{\text{verify}}(\bar{C}_{0:T}, C_{0:T}, \Pi) \in M_{\text{accept}} \quad \text{and} \quad M_{\text{privacy}}(\bar{C}_{0:T}, C_{0:T}, \Pi) \in P_{\text{accept}}.$$

20 The particular form of the privacy and verification meters is non-limiting.

In some embodiments, reality encryption and neutral recording are combined. For example, a device may emit a key-conditioned structured illumination pattern whose captured response supports later verification by an authorised party, while ordinary viewers receive a neutralised recording, a classifier-only result, a perceptual hash, a cohort statistic, or a
 25 proof that a floor condition was not violated. In other embodiments, the emitted pattern is public but the readout model is protected; the emitted pattern is protected but the readout model is public; both are protected; or neither is protected but the selective-opening path is authority-gated.

Reality-encryption generalisation (non-limiting). Reality encryption is not limited to any particular cipher, key size, medium, or payload format. Key material may be pre-shared, derived from correlated measurements, derived from a PUF or physical one-way function, established interactively, generated by a threshold protocol, derived from a ledger
 5 or randomness beacon, or produced by a post-quantum key-establishment mechanism. The protected transformation may use symmetric encryption, asymmetric encryption, hybrid encryption, secret sharing, threshold cryptography, commitment schemes, watermarking, steganography, spread-spectrum modulation, structured illumination, learned codecs, physical unclonability, selective disclosure, zero-knowledge proof, proof of correct computation,
 10 or any combination thereof.

Neutral-recording generalisation (non-limiting). A neutral recording may comprise a redacted video, transformed video, stylised recording, synthetic view, latent representation, embedding, hash, perceptual hash, caption, scene graph, classifier output, audit flag, proof object, accumulator state, compressed representation, or selective-opening handle. Neutrality
 15 may refer to privacy neutrality, semantic neutrality, identity neutrality, policy neutrality, domain-neutrality, calibration neutrality, or any declared suppression or invariance condition. The load-bearing property is not any single neutralisation technique, but the apparatus capability to produce a released representation whose retained and suppressed signal classes are declared, tested, and bound to the protocol digest.

20 **Composition with virtual-camera and virtual-emitter primitives (non-limiting).**

In some embodiments, the neutral-recording generator and the reality-encryption primitives of this subsection are composed with the virtual-camera and virtual-emitter primitives of Section 17.3 to produce a released representation that is generated from, or audited against, a virtual view or a virtual emission rather than from a raw captured recording. In such
 25 embodiments, the released representation inherits the disclosure-regime tags committed to the protocol digest and remains testable by the meters of this section.

9.8 Edge-based structural meters (optional)

In some embodiments, a decoder maps the convolution bundle to an image-like representation and an edge or contour extractor produces a structural map. Losses encourage preservation
 30 of major contours while permitting style variation, enabling Reality Transform constraints that preserve content stability.

9.9 Structural and semantic conditioning extractor family (non-limiting)

In some embodiments, the edge-based structural meter of the preceding subsection is one
 35 member of a broader family of declared structural and semantic conditioning extractors

that consume an image-like or feature representation derived from a committed convolution-bundle window via RTDec and produce a declared conditioning channel or structural representation. Non-limiting members of the family include:

- 5 • **Soft-edge extractors**, including holistically-nested edge detection (HED) and learned soft-edge variants, producing a continuous-valued edge probability map.
- **Hard-edge and contour extractors**, including Canny, Sobel, structured forests, and related operators.
- **Line-art and scribble extractors**, producing sparse or stylised line representations of scene content.
- 10 • **Straight-line and segment extractors**, including M-LSD-style line-segment detection.
- **Monocular depth estimators**, producing per-pixel metric or relative depth maps.
- **Surface-normal estimators**, producing per-pixel surface-orientation vectors, optionally derived by photometric stereo from declared multi-illumination probe sequences.
- 15 • **Semantic, instance, and panoptic segmentation estimators**, producing pixel- or region-wise class, instance, or panoptic labels.
- **Optical-flow estimators**, producing per-pixel motion vectors between declared frame pairs.
- **Disparity and stereo-correspondence estimators**, where geometry permits.
- 20 • **Body-surface UV-coordinate estimators** (such as DensePose-style maps), producing per-pixel correspondences to a declared body-surface chart.
- **The pose-feature extractor** Pose of the pose-conditioned Reality Transform of Section 17.0.1.
- 25 • **Other declared structural or semantic measurement heads** whose identity, weights or version hash, input preprocessing, calibration state, output representation, and any confidence- or visibility-mask rule are committed to the protocol digest.

In some embodiments, the family is operated as a declared set $\text{ControlExt} = \{E_1, E_2, \dots\}$ in which each E_j is an extractor of one of the above types, declared by its identity, version, input preprocessing, output representation, and calibration state, with the set membership
 30 committed to the protocol digest. The set is open: members may be added, deprecated, or version-bumped under the applicable authority-separation policy and the version-history discipline of the protocol digest.

Conditioning role versus meter role (non-limiting). In some embodiments, a member E_j of **ControlExt** participates in the RT compound objective in either or both of two declared roles:

Meter role. A loss term penalises divergence between the extracted control signal $E_j(Y_t^{\text{dec}})$ on the rendered output and a declared target control signal, encouraging the rendered output to preserve a structural or semantic property. The edge-based structural meter of Section 9.8 is one such instantiation; depth, normal, segmentation, flow, and related preservation terms are further instantiations.

Conditioning role. An extracted control signal from a declared reference (a committed reference frame, a committed probe-channel response under the temporal-multiplexed paired-acquisition corpus of Section 17.2.3, or a declared external reference whose provenance type is committed) is supplied as a conditioning input to the controller or to a generator component, steering the rendered output toward a structurally or semantically constrained target.

In some embodiments, the same extractor E_j is operated in both roles within a single run, with the conditioning-role reference and the meter-role target separately declared and committed.

Composition with the RT compound objective. In some embodiments, each member of **ControlExt** contributes a term to **RTTerm** under the RT-against-twin disclosure of Section 17.1.4, with its own declared weight $\lambda_{\text{rt},s}$, declared meter-partition assignment (acceptance, evaluation, or shaping), and declared confidence- or visibility-mask rule. The acceptance / evaluation partition is preserved so that controller optimisation against any single extractor does not expose all evaluative surfaces simultaneously.

Composition with twin-side training. In some embodiments, each member of **ControlExt** inherits the twin-side evaluation of the RT-against-twin disclosure of Section 17.1.4: the extractor is evaluated on $C_{\text{Win}_t}^{\text{Twin}}$ for the digital twin or $C_{\text{Win}_t}^{\text{AnaTwin}}$ for the analogue twin, with the synthetic-window construction and any domain-gap correction (digital twin) or matching-fidelity envelope (analogue twin) committed to the protocol digest. Control signals extracted from twin-produced bundle windows are provenance-typed as twin-predicted or analogue-twin-derived, consistent with the provenance-typing discipline applied to $C_{0:T}^{\text{Twin}}$ and $C_{0:T}^{\text{AnaTwin}}$.

Negative-example construction. In some embodiments, negative examples for control-signal discriminator and meter-residual threshold validation are constructed by applying declared spatiotemporal edits to committed recordings or to committed twin-predicted windows, with each edit family, intensity parameter, and random-seed policy recorded or hash-identified in the protocol digest, following the pattern of the verisimilitude-training disclosure

of Section 9.5 and the pose-conditioned negative-example discipline of the pose-conditioned Reality Transform of Section 17.0.1.

Scope note (non-limiting). For avoidance of doubt, members of `ControlExt` may be drawn from conventional architectures and trained models. The disclosed technical contribu-
 5 tion is the committed-evidence, meter-bounded, and twin-coupled composition of declared structural or semantic conditioning extractors with the Reality Transform compound ob-
 jective, the temporal-multiplexed paired-acquisition training-data pipeline of the temporal-
 multiplexed paired-acquisition corpus of Section 17.2.3, and the provenance-typing discipline
 applicable to extracted control signals across physical, twin-predicted, analogue-twin-derived,
 10 and generator-produced bundle windows, rather than the underlying edge-, depth-, normal-,
 segmentation-, flow-, line-, or related estimator architectures.

9.10 Canonical meter game (non-limiting)

In some embodiments, a canonical evaluation game measures how well a bounded generator
 can imitate physical outputs under a fixed challenge distribution. A hardness index may be
 15 defined as the minimal simulator capacity and training budget required to make simulated
 outputs indistinguishable from physical outputs under a competent discriminator on held-
 out tests. The index is empirical and depends on the chosen simulator class, training budget,
 and meter strength.

9.11 Physical sources of empirical hardness (non-limiting)

20 In some embodiments, empirical hardness is supported by intrinsic noise and chaos, history-
 dependent media, complex configuration manifolds, and multi-channel dependence struc-
 ture across invariance profiles. These mechanisms support persistent empirical difficulty for
 bounded adversaries under the engineering security framing of this disclosure.

Example noise and drift model (non-limiting). For concreteness, one operating model
 25 treats each convolution-bundle sample as a noisy observation of an underlying ideal response
 after passage through optics and media, with a combination of: (i) Poisson shot noise, (ii) ad-
 ditive read noise on detector channels, and (iii) slow multiplicative drift factors. Misalignment
 and vibration may be modelled as small random affine perturbations of projected and imaged
 coordinates. Meters and objectives are trained under these models so that acceptance regions
 30 distinguish “honest noisy channel” from “attacked channel” statistically, without assuming
 noise-free operation.

9.12 Digital twins and surrogate-guided optimisation (non-limiting)

In some embodiments, a digital twin is maintained to approximate the behaviour of a phys-
 ical reactor, scene loop, or coupled system under protocol choices. The twin may be differ-

entiable, partially differentiable, or purely predictive, and may be updated using committed convolution-bundle windows, meter summaries, and protocol digests. The twin is assistive rather than authoritative: the physical Reality Kernel remains the arbiter for verification, hardness, and meter-bounded validity.

5 In some embodiments, the twin supports protocol selection by predicting expected observation summaries and uncertainty under candidate protocols, and choosing protocols that improve information gain, verisimilitude, or projection-hardness subject to meter-bounded constraints. In some embodiments, the twin is used to reduce physical trial count by proposing candidate protocols, after which a subset is validated on hardware.

10 **Physics-based propagation surrogates (non-limiting).** In some embodiments, the digital twin includes a physics-based wave-propagation engine (for example ray tracing, beam-propagation, or wave-optics solvers) together with learned priors for speckle statistics, aberrations, or alignment drift. Such hybrids explore large regions of configuration space in simulation, after which hardware-in-the-loop optimisation fine-tunes around promising
15 configurations.

Neural and point-based twin renderers (non-limiting). In some embodiments, the digital twin’s renderer or scene model is implemented as a neural radiance field (Mildenhall et al. 2020), a hash- or grid-accelerated radiance-field variant such as Instant-NGP (Müller et al. 2022), a signed-distance-field representation, a mesh- or point-based differentiable
20 renderer, or a three-dimensional Gaussian splatting representation (Kerbl et al. 2023). Twin parameters are updated against committed convolution-bundle windows under the declared matching-fidelity envelope, and the renderer family, finite parameter serialisation, training corpus digest, optimiser settings, and validation residuals are recorded with the twin version identifier.

25 9.13 Hardware-in-the-loop optimisation (non-limiting)

In some embodiments, protocol parameters are optimised with hardware in the loop when gradients through the physical channel are unavailable or unreliable. A non-limiting approach uses stochastic approximation (for example SPSA) to estimate gradient directions of a scalar objective $J(u)$ computed from committed windows and meter summaries. In some
30 embodiments, $J(u)$ combines task performance terms with evidence-quality terms, including information-gain proxies, verisimilitude or projection-hardness scores, and penalties for meter violations.

A non-limiting two-sided SPSA-style estimate uses two evaluations per step:

$$\hat{\nabla} J(u) \approx \frac{J(u + c\Delta) - J(u - c\Delta)}{2c} (\Delta^{-1}),$$

where Δ is a random perturbation vector, $c > 0$ is a perturbation scale, and Δ^{-1} denotes

elementwise inversion.

Bayesian experimental design for protocol selection (non-limiting). In some embodiments, protocol selection is performed by Bayesian experimental design (Chaloner–Verdinelli 1995; Foster et al. 2019), with the digital twin or fleet surrogate acting as the predictive model and an expected-information-gain, expected-utility, knowledge-gradient, Thompson-sampling, or related acquisition function selecting the next candidate protocol. The acquisition function is constrained by meter envelopes, optical-radiation safety bounds, thermal limits, query budgets, and disclosure policies, and its prior, posterior-update rule, uncertainty representation, and optimisation budget are recorded alongside the run.

Minimax adversarial training objective (non-limiting). In some embodiments, adversarial hardening is written in minimax form:

$$\min_{\theta \in \Theta} \max_{A \in \mathcal{B}_{\text{surr}}} \mathbb{E} \left[\mathcal{L}_{\text{hard}}(M_{\text{hard}}, C, \tilde{C}_A) \right],$$

where $\mathcal{B}_{\text{surr}}$ is a family of surrogate adversaries, C are genuine committed windows, and $\tilde{C}_A$ are adversary-generated candidates. Optimisation alternates: update A to increase spoof success, update meters to reduce false accepts, and update permissible parameters to increase difficulty subject to meter envelopes.

Projected-gradient surrogate adversaries (non-limiting). In some embodiments, the surrogate adversary family $\mathcal{B}_{\text{surr}}$ includes projected-gradient perturbation adversaries of the kind used in robust-optimisation adversarial training (Madry et al. 2018). The projected-gradient attack is applied to a differentiable surrogate, digital twin, learned feature representation, or differentiable component of the protocol, and its perturbation set, step size, restart count, iteration count, and projection operator are declared in the protocol digest. Candidate adversarial windows produced by the surrogate are supplied as negative examples or hard cases to the meter-update loop, while physical actuation remains bounded by the apparatus safety and meter envelopes.

Curriculum and escalation (non-limiting). In some embodiments, adversarial training follows a curriculum: begin with weak adversaries (replay and low-capacity generators), then escalate to stronger adversaries (higher-capacity models, physics-informed generators, partial side information, and model inversion). Escalation triggers include meter-disagreement spikes, anomalous drift indicators, or sudden decreases in verisimilitude on held-out reference sweeps.

Hybrid digital–analogue co-training (non-limiting). In some embodiments, training updates both digital parameters and analogue/physical degrees of freedom, potentially on different timescales. Non-limiting analogue update mechanisms include: a “training beam”

that writes correction patterns into memory media derived from error signals, pump or gain modulation that adjusts cavity or amplifier operating points as a function of meter objectives, and feedback illumination schedules that drive photorefractive or photochromic media toward configurations that improve declared meter outcomes. These updates are bounded by safety
 5 and stability meters and are logged for audit.

Physical actuation binding (non-limiting). In some embodiments, “updating the kernel” includes applying actuator-safe set-point changes to one or more hardware controls, including without limitation: V_{G1} , V_{G2} , V_{anode} , deflection and auxiliary coil currents, focus and astigmatism controls, mask selection or SLM patterns, MEMS mirror angles, liquid-lens
 10 or MEMS-lens voltages, spectral filter drives, and per-emitter gains in addressable arrays. The controller enforces rate limits and limit projections before committing updates and logs applied set-points and calibration identifiers in the protocol digest.

Common random numbers and robust aggregation (non-limiting). In some embodiments, zeroth-order evaluations reuse the same validation challenges, protocol seeds, and
 15 any internal randomness when evaluating an objective at perturbed parameters $u + c\Delta$ and $u - c\Delta$. This common-random-numbers discipline reduces estimator variance. Each perturbation side may be evaluated using multiple replicates under matched seeds, and replicate scores aggregated using a robust estimator (for example median-of-means).

Run records and audit fields (non-limiting). In some embodiments, training, calibration, and hardware-in-the-loop optimisation consume structured run records as the primary
 20 substrate. Run records extend meter records by binding sweep/control identifiers, challenge identifiers, provenance metadata, calibration state histories, and digital twin metadata to the underlying committed windows and meter outputs. Operation is treated as a two-timescale system, separating slow maintenance (re-enrolment, deep recalibration, twin training, meter
 25 updates, and sweep-library evolution) from fast runtime adaptation (bounded adjustments gated by meters), with both timescales logged.

9.14 Frozen-meter adversarial tuning (non-limiting)

The auxiliary diagnostics in this subsection and in the Scale-to-threshold subsection below (including $H_{\text{err}}(\kappa)$, $H_{\text{steps}}(\kappa; \tau)$, $C^*(\kappa)$, $c^*(\kappa)$, and fidelity-threshold τ framing) are operational training-loop tools, not instances of the canonical two-dimensional hardness index
 30 pair $(k_{\text{dig}}^*(\theta; \varepsilon), m_{\text{ana}}^*(\theta; \varepsilon, r))$ defined in Section 12. They serve as surrogate objectives during meter and device training; the canonical hardness index characterises the resulting device under the declared threat model and evaluation meters.

In some embodiments, an objective term in $J(u)$ is defined using a neural meter trained
 35 in a two-phase procedure. In a first phase, a conditional meter model is pretrained across a range of device/kernel states and protocol classes. The pretrained meter is then frozen and

used as a fixed-capacity adversary or verifier. In a second phase, the physical configuration and/or protocol family is updated to increase the frozen meter’s difficulty.

In discriminator-driven embodiments, a frozen discriminator induces a separability score measuring how well it separates physical convolution-bundle windows from generated or simulated windows under matched challenges. In reconstruction-driven embodiments, a frozen reconstructor induces a modelling-error hardness $H_{\text{err}}(\kappa)$ and an optional compute-hardness $H_{\text{steps}}(\kappa; \tau)$ defined as the minimum number of bounded reconstruction steps required to reach a fidelity threshold τ :

$$H(\kappa) = H_{\text{err}}(\kappa) + \gamma H_{\text{steps}}(\kappa; \tau).$$

In some embodiments, a bank of frozen meters is maintained with varied architectures, seeds, datasets, and sampler configurations, and hardness is computed as a minimum or low-quantile score across the bank to reduce overfitting to a single meter. In some embodiments, a minimum freeze window duration T_{freeze} is declared in the protocol digest and enforced: a meter is not updated or replaced during any active verification or PoD confirmation window, so that an adversary who queries the system during a frozen window cannot steer the meter by slow drift before the window closes. The freeze window start time and duration are committed to the protocol digest before the window opens. A drift meter (a separate, lightweight monitor for meter output stability) is declared alongside each frozen meter; if the drift meter signals anomalous output shift during a freeze window, the freeze is extended and the event is logged. In some embodiments, the hardness index is additionally bounded under an adaptive-query model in which the adversary is permitted Q queries per freeze window: the declared query-rate limit Q/T_{freeze} and the empirically estimated meter-learning rate are recorded in the protocol digest so that the claimed hardness margin under this model is traceable.

Operating-point consistency meter (non-limiting). The slow-drift attack exploits the fact that acceptance meters adapt to gradual operating-point shifts over timescales longer than the rotation period, potentially steering the device toward a region of lower hardness without triggering any single-window alarm. In some embodiments, this attack is bounded by an *operating-point consistency meter* $m_{\text{drift}}(\theta_t, \theta_{\text{ref}})$ that tests whether the device’s operating point θ_t has remained within a declared tolerance $\varepsilon_{\text{drift}}$ of its reference point θ_{ref} during the evaluation window. The reference point is set at the start of each hardness evaluation freeze window. In some embodiments, the maximum tolerable drift over a freeze window of length T_{freeze} is derived from the AR(1) drift model (Section 9.11): $\varepsilon_{\text{drift}} = z_{\alpha} \sigma_{\text{drift}} \sqrt{T_{\text{freeze}}}$, where σ_{drift} is the estimated drift rate and z_{α} is the declared confidence quantile. If m_{drift} exceeds $\varepsilon_{\text{drift}}$, the hardness evaluation is invalidated for that window, the event is logged, and the evaluation is rescheduled after a fresh reference point is established. The operating-point consistency meter and its threshold are committed to the protocol digest and are evaluated concurrently with the frozen evaluation meters. A declared operating envelope in Θ space (separate from meter space) is maintained; operating-point drift is distinguished

from physical device degradation by comparing against the fleet-wide distribution of drift rates.

Freeze window duration bound from AR(1) drift model (non-limiting). Given the AR(1) drift model $\delta_{t+1} = \rho_{\text{AR}}\delta_t + w_t$ (Section 9.11), the expected squared drift over a window of length W starting from a reference point is:

$$\mathbb{E}[|\delta_W - \delta_0|^2] = \sigma_w^2 \frac{1 - \rho_{\text{AR}}^{2W}}{1 - \rho_{\text{AR}}^2},$$

where σ_w^2 is the noise variance of the AR(1) process. In some embodiments, the bias induced on the hardness index estimate by drift during an unfrozen window of length W' is bounded as a function of $\mathbb{E}[|\delta_{W'} - \delta_0|^2]$ times the empirically estimated sensitivity $\partial k_{\text{dig}}^*/\partial \delta$; the maximum unfrozen window duration W'_{max} is then set so that this drift-induced bias satisfies $\text{bias}(W'_{\text{max}}) \leq \eta_{\text{freeze}}$, where η_{freeze} is a policy-declared fraction of the hardness threshold. In some embodiments, W'_{max} is committed to the protocol digest alongside ρ_{AR} , σ_w^2 , and η_{freeze} , making the freeze window duration a derived and auditable quantity rather than an ad hoc engineering choice.

Frozen GAN meter (non-limiting). In some embodiments, a frozen meter is realised by a conditional GAN trained across kernel states. A frozen discriminator defines a separability score $S_D(\kappa)$ for distinguishing physical windows from generator windows. A frozen generator defines a divergence score $S_G(\kappa)$ measuring perceptual discrepancy when attempting to imitate physical responses. Non-limiting GAN objectives include hinge-loss, Wasserstein, and feature-matching variants.

Frozen diffusion meter (non-limiting). In some embodiments, a frozen meter is realised by a diffusion reconstructor. Non-limiting hardness proxies include modelling error and a step-count hardness defined by the minimum number of reverse-diffusion steps required to reach a fidelity threshold under a fixed bounded-step sampler (for example DDIM with $\eta = 0$).

Normalisation and keyed scoring (non-limiting). In some embodiments, meter inputs are standardised by a fixed non-learned normalisation, and normalisation or feature extraction may be keyed or salted so that disclosed scores are harder to replay or overfit without access to the key; key identifiers are logged while the key itself is not disclosed.

9.15 Scale-to-threshold hardness (non-limiting)

In some embodiments, hardness is quantified using a scale-to-threshold protocol: an attacker or meter model class is grown in capacity until it reaches a performance threshold τ on held-out challenge windows. An optional shrink-test estimates the minimal capacity $C^*(\kappa)$ that

still meets τ . The device is updated to maximise $C^*(\kappa)$, thereby forcing attackers to deploy larger models to achieve the same competence. The same protocol family may be used in an assistive direction for cooperative decoders by updating κ to minimise an analogous $C^*(\kappa)$.

Classifier-centred curricula (non-limiting). In some embodiments, scale-to-threshold tuning is organised as a curriculum over a family of decoders or classifiers ordered by capacity c :

$$c^*(\kappa) = \min\{c : \text{Perf}(C_c; \kappa) \geq \tau\}.$$

An assistive phase updates κ to reduce $c_{\text{Bob}}^*(\kappa)$, while an adversarial phase updates κ to increase $c_{\text{Eve}}^*(\kappa)$. A single objective may increase a gap such as $c_{\text{Eve}}^* - w_{\text{adv}} c_{\text{Bob}}^*$ subject to meter envelopes and actuator projections.

Constant-time scoring and DoS guards (non-limiting). In some embodiments, evaluation is padded to a fixed wall-time so that latency does not reveal the number of iterations required. Denial-of-service guards cap per-score wall-time, maximum sampler steps, and query budgets, and a rate limiter constrains the frequency of challenge–response queries.

Randomised-smoothing certificates over meter features (non-limiting). In some embodiments, randomised-smoothing certificates (Cohen, Rosenfeld, and Kolter 2019) are computed over meter features, verifier scores, or latent summaries under a declared perturbation distribution. The certificate supplies an auxiliary certified-robustness radius or bound alongside the empirical scale-to-threshold index $C^*(\kappa)$, and is valid only for the declared feature map, perturbation distribution, base-meter version, confidence level, and sampling budget. The smoothing noise family, variance or scale parameter, number of Monte-Carlo samples, and certified norm are recorded in the protocol digest.

9.16 Continual calibration, drift compensation, and transfer learning (non-limiting)

In some embodiments, calibration is continual: alignment, gain, drift, and timing parameters are updated online using meter feedback and committed windows. Calibration updates are gated by stability meters, and calibration state identifiers and versions are logged. Drift compensation may use replay buffers of previously validated windows with regularisation against a reference parameter state. In some embodiments, catastrophic-forgetting mitigation is implemented using methods from the continual-learning literature, including without limitation elastic weight consolidation (EWC, Kirkpatrick et al. 2017), synaptic intelligence (SI), memory-aware synapses (MAS), online EWC, learning-without-forgetting (LwF), gradient-episodic-memory methods (GEM, A-GEM), experience replay, generative replay, dark experience replay (DER), progressive networks, PackNet, and modular continual-learning methods. Transfer learning across reactors is supported via device-conditioned modulation, low-rank adapters (including without limitation LoRA, QLoRA, DoRA, and LoHa), prefix tun-

ing, prompt tuning, adapter tuning, IA³, BitFit, and other parameter-efficient fine-tuning (PEFT) methods, or universal encoders with device-specific heads. In some embodiments, per-deployment adaptation uses meta-learning methods, including without limitation model-agnostic meta-learning (MAML), Reptile, ANIL, first-order MAML (FOMAML), and meta-
 5 reinforcement-learning methods such as RL² and PEARL.

Probability calibration of meter outputs (non-limiting). In some embodiments, meter outputs that are interpreted as probabilities or confidence scores are calibrated using Platt scaling, temperature scaling (Guo et al. 2017), isotonic regression, beta calibration, or histogram/binning methods. Calibration parameters are versioned with the meter identifier,
 10 evaluated on held-out committed windows stratified by device, protocol class, and operating envelope, and logged with expected calibration error, maximum calibration error, reliability-diagram summaries, and the calibration-data digest. A meter whose calibration error exceeds a declared threshold is downgraded, frozen pending review, or excluded from acceptance decisions under the meter-partition policy.

Conformal and selective-prediction meters (non-limiting). In some embodiments, meter outputs are wrapped by conformal-prediction, selective-classification, or abstention mechanisms that convert raw scores into prediction sets, rejection regions, or audit-escalation triggers under a declared calibration distribution. The conformal score, calibration split, exchangeability or stratification assumptions, coverage level, abstention threshold, and recalibration schedule are recorded with the meter version. Where exchangeability is not plausible
 20 because of drift, device heterogeneity, or protocol adaptation, the conformal guarantee is limited to the declared stratum or replaced by a weighted, covariate-shift, or online calibration variant, and the limitation is recorded in the protocol digest.

Uncertainty-budget and traceability records (non-limiting). In some embodiments, calibration records include an uncertainty budget that decomposes meter uncertainty into
 25 repeatability, detector noise, alignment, wavelength or spectral calibration, timing jitter, temperature drift, model discrepancy, reconstruction error, and cross-device transfer components. The uncertainty budget may follow a GUM-style propagation or Monte-Carlo propagation method, and may include traceability metadata for reference artefacts, calibration
 30 targets, timing references, optical power measurements, and environmental sensors. The budget is recorded as audit metadata and as a gating input for claims that require metrological confidence, without requiring that every embodiment operate as a certified metrology instrument.

9.17 Energy-based loss functions (optional)

35 In some embodiments, training losses are derived from a physically grounded energy functional rather than coordinate-wise penalties. Around each ground-truth configuration S a

Boltzmann-inspired energy landscape at temperature parameter τ motivates the loss:

$$\mathcal{L}_E(\theta) = E(\hat{S}_\theta; S) - E(S; S).$$

If E is invariant under a symmetry group G , then $\mathcal{L}_E$ inherits this invariance. For discrete scenes, a tractable energy is $E_{\text{dist}}(S) = \sum_{i < j} V(\|s_i - s_j\|)$, which is automatically invariant to translations, rotations, and permutations of identical elements.

- 5 **Koopman-style latent dynamics (non-limiting).** In some embodiments, a latent representation is learned in which closed-loop evolution is approximately linear: $z_{t+1} \approx Kz_t$, supporting prediction, control design, and anomaly or tamper detection via deviations in the learned linear evolution or changes in the spectrum of K .

9.18 Distributional alignment and robustness (optional)

- 10 **Optimal-transport divergences (non-limiting).** In some embodiments, discrepancies between response distributions are quantified using optimal-transport losses, including Sinkhorn divergences between batches of simulated and physical convolution-bundle windows. Unbalanced optimal transport supports missing data, and Gromov–Wasserstein variants compare relational structure across modalities.
- 15 **Certified robustness against bounded perturbations (non-limiting).** In some embodiments, a verifier is constructed to admit robustness certificates against bounded perturbations of convolution-bundle features, for example via randomised smoothing or Lipschitz-constrained architectures.

9.19 Self-adapting Reality Kernels (non-limiting)

- 20 In some embodiments, a Reality Kernel is operated as a continuously adapting physical process rather than a system with a strict separation between training and inference. A learned self-edit head proposes adaptation actions:

$$\pi_{\text{edit}} : c_{0:t} \mapsto a_t^{\text{edit}},$$

- where $c_{0:t}$ denotes the emission–observation history and a_t^{edit} denotes a proposed self-edit. Non-limiting self-edits include proposed updates $\Delta\theta_t$ to slow parameters within bounded
 25 ranges, schedules for biasing native analogue relaxation, and bounded write programmes in embodiments that include a fabrication loop. A wrapper map applies edits subject to hard safety and calibration constraints:

$$\theta_{t+1}, M_{t+1} = F_{\text{edit}}(\theta_t, M_t, a_t^{\text{edit}}),$$

where M_t denotes slow media state.

Meta-reward lens (non-limiting). In some embodiments, edits are evaluated over a horizon H using a meta-level return:

$$R^{\text{meta}}(a_t^{\text{edit}}) = \sum_{\tau=t}^{t+H} \left(-\ell_{\text{task}}(\tau) + \lambda_{\text{hard}} H_{\text{dig}}(\theta_{\tau}) - \lambda_{\text{cal}} C_{\text{drift}}(\theta_{\tau}) \right),$$

where ℓ_{task} is a task loss, H_{dig} is an empirical hardness score, and C_{drift} is a calibration-drift penalty. The self-edit policy is trained by reinforcement learning or other optimisation to increase $\mathbb{E}[R^{\text{meta}}]$ while remaining within the admissible edit envelope.

Overnight self-editing pattern (non-limiting). In some embodiments, a practical experimental schedule alternates between supervised “daytime” use and constrained “overnight” self-edit phases, supporting empirical characterisation of stability, overfitting, drift, and benefits of self-adaptation under different meta-reward choices and constraint policies.

9.20 Federated and multi-agent learning (non-limiting)

In some embodiments, learning is federated across a network of devices, with updates weighted by trust and audit outcomes. Devices may accept aggregated updates only when accompanying evidence meets verification requirements, and when meter envelopes indicate stable operation.

Federated averaging, secure aggregation, and privacy accounting (non-limiting).

In some embodiments, the federated update loop is implemented by Federated Averaging (McMahan et al. 2017) over committed-window-derived gradients or parameter deltas, with per-device contributions protected by secure aggregation (Bonawitz et al. 2017) and with the aggregated update accepted only when the disclosed aggregate and accompanying evidence satisfy meter-envelope compliance. In some embodiments, fleet-update releases are governed by differential-privacy accounting, including a moments-accountant or DP-SGD analysis (Abadi et al. 2016), Rényi differential privacy (Mironov 2017), or Gaussian differential privacy (Dong–Roth–Su 2022), with clipping norms, sampling rates, noise scales, composition rule, privacy budget, and unit of protection recorded in the protocol digest. In some embodiments, FLAME-style backdoor filtering, Krum or Multi-Krum, coordinate-wise median, trimmed-mean, Bulyan, clustering-based anomaly filters, or other Byzantine-robust aggregation rules are composed with the meter-gated acceptance step, and the chosen aggregation rule, rejection criteria, and false-rejection tolerance are committed before the round opens.

Privacy accounting and selective-disclosure consistency (non-limiting). In some embodiments, the relationship between a declared differential-privacy budget and the selective-disclosure policy is itself committed in the protocol digest: the selective-disclosure

policy shall not permit openings, witness sets, or audit reconstructions that are inconsistent with the declared DP guarantee or that would, in aggregate, exceed the declared composition budget. In some embodiments, the auditor or relying party is provided with the DP-accounting state alongside any selectively disclosed evidence, so that the privacy-budget posture is itself part of the evidence record.

Fleet-update poisoning diagnostics (non-limiting). In some embodiments, fleet updates are accompanied by poisoning and backdoor diagnostics computed on held-out committed windows, neutral probe corpora, canary challenges, or synthetic-but-committed adversarial cases. Diagnostics include update-norm clipping compliance, cosine- or Mahalanobis-distance outlier scores, influence estimates, gradient-sign agreement, cluster membership, backdoor trigger success on held-out probes, and degradation of calibration or verisimilitude meters. The fleet accepts, rejects, downweights, or quarantines the update according to a predeclared rule, and all rejected or quarantined updates remain available for audit under the selective-disclosure policy.

In some embodiments, the system trains verisimilitude or consistency discriminators using committed windows. A non-limiting training loop alternates: sample committed windows, label them as consistent or inconsistent (for example by controlled perturbations, mismatch injections, or cross-device disagreement), update discriminator parameters to reduce classification error, and periodically validate against fresh physical runs under meter-bounded regimes.

In some embodiments, learned scan policies are trained under constraints: policies propose candidate protocols, meters enforce admissibility, and selective opening is used to audit that learning did not exploit unlogged channels or post hoc fabrication. In some embodiments, training is framed as a multi-agent game between agents proposing protocols or interpretations and verifiers or discriminators enforcing consistency with Reality Kernel evidence, with trust weights and hardness scores shaping the effective payoffs and update weights.

10 Representative Reactor Embodiments (Non-limiting)

This section describes non-limiting physical embodiments that implement the Reality Kernel operator $P_\theta(\cdot \mid S, U_{0:T})$. Unless stated otherwise, each embodiment produces a convolution bundle $C_{0:T}$ as defined in Section 6, where each sample bundles the control input $u(t)$ and an observation vector $\mathbf{y}_t$ that may include multiple detector ports. The bench-top 1D Reality Kernel of Section 7.2 provides a concrete teaching anchor. The 2D generalisation is described in Section 10.2. The subsections below describe additional reactor substrates, memory extensions, multi-rate embodiments, and alternative physical realisations.

10.1 Embodiment overview

In various embodiments, the Reality Kernel module is implemented using one or more of: (i) a scene-facing loop (structured emission toward an external scene and detection of a response), (ii) a reactor loop (an engineered subsystem contributing mixing, memory, nonlinearity, or history dependence), and (iii) optional media layers (slow or persistent memory media) that modify the effective channel over time.

In one practical organisation, some embodiments aim for minimal electronics and treat the dominant computation as optical or physical, with electronics used primarily for timing, gain, digitisation, and control signalling. Other embodiments introduce stronger nonlinearities, longer memory, or higher bandwidth, and may employ additional sensors or actuators. In a non-limiting multi-rate configuration, a fast all-optical loop implements a convolution-like kernel in a feedback cavity (for example a gain-pumped cavity with AOD/EOM providing fast sweep and coupling control), while one or more slower analogue-memory loops—for example CRT/phosphor stacks, laser-addressed luminescent screens, or other persistence media—provide additional state and mixing on longer timescales. In such multi-rate stacks, the fast loop supplies high-throughput convolution-like mixing and scattering, and the slow loop stores or reshapes states between passes, all under the same Reality Kernel control and logging abstraction.

Form factors and deployment scenarios (non-limiting). In some embodiments, Reality Kernels are realised as handheld scanners or probes, desktop or bench-top units, kiosk or public verification stations, integrated mobile devices (for example smartphones and tablets using existing cameras and screens or flashes as emitters), wearables (for example body cameras and smart glasses), fixed infrastructure (for example security cameras and sensor beacons), and airborne or satellite platforms for verified remote sensing. In preferred embodiments, each form factor records protocol digests and meter envelopes so that exported evidence is interpreted under the correct operating regime and sensor geometry.

Carrier and spectral variants (non-limiting). In some embodiments, the carrier includes one or more of: UV (for example fluorescence excitation, surface inspection, and sterilisation verification), visible, NIR, short-wave infrared (SWIR, for example through-fog and moisture-sensitive imaging), mid-infrared (MIR, for example molecular spectroscopy and gas sensing), far-infrared (FIR) and terahertz (THz, for example non-destructive evaluation and screening), RF and microwave (for example channel probing and through-wall sensing), and acoustic or ultrasound. In preferred embodiments, the selected bands, filters, and calibration identifiers are recorded in protocol digests so that verifiers interpret the convolution bundle under the correct bandpass and operating envelope.

Embodiment-family conformance summary (non-limiting). The following non-limiting table maps representative embodiment families to supported primitives, typical

regimes, and key constraints. This table is illustrative and non-exhaustive; additional families appear throughout the specification and combinations are contemplated. *Legend:* Regimes: TB = Truth Beam, L = Limager, RT = Reality Transform. Primitives: R = Read, W = Write, Sign = Sign, Erase = Erase.

Family	Primitives	Regimes	Typical meters	Notes
Hybrid proj.-det.	R/W/Sign/Erase	TB, L, RT	Drift, timing, TE	Full-featured reference
CRT/phosphor	R/W/Sign/Erase	TB, L, RT	Phosphor decay, persistence	Analogue memory; Erase via natural decay
Fibre-delay	R/Sign	TB, L	Speckle, polarisation	Compact; RT limited
Acoustic/cymatic	R/W/Sign	TB, L, RT	Mode spectrum, scattering	Non-optical; SASER variant
PIC/waveguide	R/Sign	TB, L	Ring resonance, loss	On-chip; wafer-scale
Metasurface	R/W/Sign	TB, L, RT	Spectral, polarisation	Flat optics; programmable
Chromic/electro.	R/W	L, RT	Colour change, switching	Slow; persistent state
Sweeping reactor	R/W/Sign	TB, L	Modulation freq., phase	Swept parameter (T, V, path)
GNSS passive	R/Sign	TB, L	Pseudorange, carrier phase	Passive; no emitter control
Extended-scale	R/W/Sign	TB, L, RT	Domain-specific	Geological, atmospheric, bio

- 5 Notes in the table are typical and non-limiting; they do not exclude alternative configurations, faster/slower variants, or additional primitives beyond those listed.

10.2 Generalisation: shared-scan reactor-modulated scene kernel (2D)

The 2D generalisation of the 1D anchor of Section 7.2 extends the bench-top construction
 10 along three axes: the single-axis micromotor-mirror scanners are replaced by galvanometer pairs providing 2D scan coverage, the per-sample discrete feedback loop is replaced by a continuous real-time amplitude drive $A(t)$, and the fluorescence-specific reactor substrate is replaced by a substrate-open reactor stack with a generic nonlinear medium. The 2D anchor embodiment (FIG. 6) comprises two physically separate subsystems driven from a shared
 15 clock: a scene subsystem 210 open to the world under a declared eye-safety envelope, and a reactor subsystem 220 sealed behind a declared safety envelope.

In the parallel-subsystem embodiment (a) of Section 4.2, the architectural coupling between the scene subsystem 210 and the reactor subsystem 220 is exclusively the real-time amplitude drive $A(t)$ produced from reactor detector 44 photocurrent and delivered to scene
 20 emitter 24. The shared scan program produced by common controller 10 and fanned out to

galvanometer pairs 25 and 26 is a synchronisation mechanism, not an architectural coupling within the meaning of this section: the shared scan program causes the two subsystems to traverse their respective scan trajectories in phase, and is structurally distinct from the architectural feedback path $A(t)$. In some embodiments, $A(t)$ is produced by a declared live conditioning chain comprising protective conditioning (for example transimpedance amplification and a safety limiter) followed by a frozen committed reactor detector gain G_{det} , which is trained offline as one of the trainable parameter families for the 2D anchor (see Section 10.2.3) and held fixed for the committed deployment version. In the parallel-subsystem embodiment (a), the live coupling path contains no per-period wait, no forward low-latency coupling from scene detector 43 to reactor source, and no runtime-adaptive or runtime-learned element; the scene emitter's instantaneous amplitude at time t is therefore the reactor subsystem's response at time t as shaped by the declared live conditioning chain, up to a bounded and declared settling time. The cycle is closed in embodiment (a) by the $A(t)$ feedback path alone.

In the cascade embodiment (b) of Section 4.2, the architectural coupling between the scene subsystem 210 and the reactor subsystem 220 additionally includes a forward low-latency coupling from scene detector 43 to the reactor source of subsystem 220, illustrated in FIG. 6 as a heavy-stroke arrow. The forward low-latency coupling is realised in any of the non-limiting implementations recited in Section 4.2 (optical gain-pumped; optical direct relay; electrical; mixed-signal), each with its end-to-end latency declared and committed to the protocol digest. In the cascade embodiment (b), the cycle is closed by both the forward low-latency coupling and the $A(t)$ feedback path operating jointly, with the scene observation entering the reactor stage in real time via the forward coupling and the reactor response being returned to the scene-emitter amplitude in real time via $A(t)$.

In some embodiments, the 2D anchor scene subsystem and reactor subsystem are described more particularly in Sections 10.2.1 and 10.2.2. Unless a declared extension is engaged and committed to the protocol digest, the reactor subsystem and scene subsystem of embodiment (a) are coupled exclusively by the live reactor-detector-to-scene-emitter amplitude drive $A(t)$ (with the shared scan program acting as a synchronisation mechanism rather than as an architectural coupling), and the reactor subsystem and scene subsystem of embodiment (b) are coupled by both the forward low-latency coupling and $A(t)$, in each case rather than by bidirectional cross-conditioning beyond what is declared.

In some embodiments, the shared scan program is a Lissajous trajectory in (x, y) with declared frequencies, phase offsets, and amplitudes, committed to the protocol digest. In some embodiments, the same scan program drives both galvanometer pairs one-to-one, so that the reactor subsystem and the scene subsystem traverse the same trajectory in their respective coordinate systems at the same phase of the shared scan period T . In some embodiments, the scan trajectory shape is one of six trainable anchor parameter families (see Section 10.2.3).

In some embodiments, spatial or temporal warping of the reactor scan relative to the scene scan — translation, rescaling, or phase offset — is a declared extension of the one-

to-one shared-scan default and is committed to the protocol digest when used. In some embodiments, a runtime-adaptive transfer function in the live coupling path beyond the declared live conditioning chain, scene-side amplitude modulation in addition to reactor-driven modulation, and bidirectional cross-conditioning between the subsystems are further
 5 declared extensions, each committed to the protocol digest when used.

Safety envelope. In some embodiments, the scene-facing path is operated under declared eye-safety, thermal, scan-rate, and stability bounds, while higher-power or otherwise restricted reactor illumination is confined to the sealed reactor subsystem. In some embodiments, protocol digests and meter summaries record whether illumination occurred in a
 10 sealed internal path or in a scene-facing path, together with the applicable interlock, safety-envelope, and meter-envelope status.

In some embodiments, a handheld or portable probe embodiment is provided (PolieProboscis), in which a compact projector–detector pair and optional sealed reactor loop are integrated into a single enclosure for close-range scanning, verification, or environmental
 15 sampling. PolieProboscis operation is constrained to an eye-safe scene-facing regime, while any higher-power operation is confined to sealed internal paths under interlocks.

10.2.1 Scene subsystem (2D)

In the 2D anchor, the scene subsystem 210 comprises an illumination source 24 selected to meet a declared eye-safety envelope (for example a lower-power infrared or visible source),
 20 a collimating lens, a galvanometer pair 25 driven by the shared scan program, a scan lens defining a field of view, and a scene detector 43. The scene emitter’s instantaneous amplitude is driven by $A(t)$ from the reactor subsystem 220. The controller drives the galvanometer pair 25 according to the shared scan program (for example a Lissajous trajectory) and records scene responses synchronised to the scan timing. In some embodiments, the scene
 25 detector 43 is a single detector, for example a photodiode, as the anchor default; a camera, a photodiode array, or other spatially resolved detector is a declared extension of the single-detector default. In some embodiments, the scene detector is coaxial with the scene emitter via a polarisation combiner or equivalent.

10.2.2 Reactor subsystem (2D)

30 In the 2D anchor, the reactor subsystem 220 is sealed behind a declared safety envelope and comprises, on the excitation path and in optical order: a reactor illumination source, a galvanometer pair 26 driven by the shared scan program, a tunable emitter-side focus lens, a scattering element, and a nonlinear medium 30. On the detection path, the reactor subsystem further comprises a tunable detector-side focus lens, a detector-side aperture, and
 35 a single reactor photodetector 44, for example a photodiode. Because the reactor subsystem is sealed, the reactor illumination source is selected to drive strong nonlinearity in the reacting medium and is free from the constraints governing the scene illumination; for example, the

reactor illumination source may be a UV source, a high-intensity laser, or any other source compatible with the sealed safety envelope and the medium's excitation requirements. In some embodiments, the reactor illumination source is driven by a trained profile $A_{\text{emit}}(t)$ periodic in the shared scan period T (see Section 10.2.3); $A_{\text{emit}}(t)$ probes the nonlinear medium's intensity-dependent response across different phases of the scan period.

Reactor-side collimation (non-limiting). In some embodiments, the reactor illumination source is a laser or laser-diode module producing a beam of sufficient collimation for galvanometer scanning without a separate collimating lens between the reactor source and galvanometer pair 26. In other embodiments, a collimating element is positioned between the reactor source and galvanometer pair 26; its presence, optical parameters, and calibration state are declared and committed to the protocol digest. The asymmetry between the scene loop's explicit collimating lens (Section 10.2.1) and the reactor loop's optional or absent collimating lens shown in FIG. 6 reflects the different illumination-source choices available in the two loops under the declared safety envelope and is not a feature limiting the apparatus.

In some embodiments, the tunable emitter-side focus lens (for example a liquid lens) follows a trained profile $f_{\text{emit}}(t)$ periodic in the shared scan period T . In some embodiments, the tunable detector-side focus lens follows a trained profile $f_{\text{det}}(t)$ periodic in T and controls the region and depth of the reactor response over which the reactor photodetector integrates at each scan phase. In some embodiments, the detector-side aperture follows a trained profile $a_{\text{det}}(t)$ periodic in T , implemented for example as a fast-actuated iris, an LCD-based aperture, a MEMS-based aperture, or other programmable spatial stop; the aperture provides spatial selectivity of the reactor response analogous to a programmable confocal pinhole in the detection path. In some embodiments using a mechanical iris or other slower actuator not capable of per-scan-period modulation, a_{det} is a committed scalar (for example a committed fixed aperture diameter) rather than a time-varying profile, and is committed to the protocol digest as such.

In some embodiments, the detector-side aperture is positioned before the detector-side focus lens, after the detector-side focus lens, or at an intermediate image plane, provided that the declared configuration determines the spatial selectivity of the reactor response and is committed to the protocol digest.

The detector-side focus lens and detector-side aperture shape the reactor response before conversion to photocurrent; they are not part of the post-photodetector live coupling path from the reactor detector to the scene emitter. The post-photodetector live coupling path begins at the reactor photodetector output and proceeds through protective conditioning and frozen committed G_{det} , as described in Section 10.2.4.

In some embodiments, the scattering element is any strong random optical mixer, for example ground glass, a diffuser, or a heterogeneous scattering bed. In some embodiments, the nonlinear medium 30, additional reactor-medium elements, or combined reactor-medium layer may include, without limitation: (i) a scattering plate such as ground glass or a diffuser; (ii) a thin-film nonlinear element, for example graphene-coated or other intensity-dependent

media; (iii) a temporal or persistence layer such as phosphor or other luminescent media; (iv) a delay or mixing element such as fibre-delay loops; (v) an epsilon-near-zero (ENZ) thin film such as indium tin oxide (ITO), whose dielectric permittivity can be switched on femtosecond timescales via the optical Kerr effect, enabling ultrafast temporal modulation of the reactor's optical properties; (vi) a gas discharge or plasma medium (for example a sealed gas cell, a plasma display panel cell, a dielectric barrier discharge layer, or an arc discharge electrode pair) whose ionisation dynamics, filament formation, and emission spectra provide strongly nonlinear, device-specific, and temporally rich reactor behaviour; (vii) a microwave or radiofrequency cavity with at least one tunable boundary condition (for example a piezoelectrically actuated endwall or an electrically tuned metamaterial surface), whose electromagnetic mode spectrum provides a calculable but device-specific transfer function modulated by the boundary geometry; and/or (viii) any combination of the foregoing. The phosphor option is listed as one non-limiting reactor-medium element among others and is not a privileged default for the 2D anchor.

In some embodiments, the reactor photodetector 44 is a single detector as the anchor default; a photodiode array or other spatially resolved detector is a declared extension. In some embodiments, the 2D anchor is defined by the topology of the reactor stack and the training protocol, not by the specific choice of nonlinear medium, scattering element, or reactor illumination source; the fluorescence-specific substrate of the 1D anchor (Section 7.2.2) is one non-limiting substrate that may be used in the 2D topology.

In the 2D anchor, the reactor emitter-side focus profile $f_{\text{emit}}(t)$ is implemented by a tunable liquid lens driven from a scalar kernel coordinate $\kappa_{f,e}(t)$ synchronised to the shared scan period. Similarly, the reactor detector-side focus profile $f_{\text{det}}(t)$ is implemented by a tunable detector-side focus lens driven from a scalar kernel coordinate $\kappa_{f,d}(t)$ synchronised to the shared scan period. Other controls are held fixed or updated on slower timescales. The trained values defining $f_{\text{emit}}(t)$, $f_{\text{det}}(t)$, and a_{det} (whether a profile or a committed scalar) over the scan period are committed to the protocol digest.

10.2.3 Trainable parameter families (2D)

In the 2D anchor, six trainable parameter families are optimised jointly against the hardness objective of Section 7.2.5 using the training pathways of Section 7.2.6:

1. The shared scan trajectory shape — Lissajous parameters (frequency ratio, phase offsets, amplitudes), or more generally the shared scan path traversed by both galvanometer pairs. Periodic in the shared scan period T .
2. The reactor emitter-side focus lens profile $f_{\text{emit}}(t)$. Periodic in T .
3. The reactor detector-side focus lens profile $f_{\text{det}}(t)$. Periodic in T . Controls the region and depth of the reactor response over which the reactor photodetector integrates at each scan phase, and may be trained independently from $f_{\text{emit}}(t)$.
4. The reactor detector-side aperture profile $a_{\text{det}}(t)$. Periodic in T where the aperture implementation supports fast actuation; alternatively a committed scalar where a fixed

mechanical aperture is used. Provides spatial selectivity of the reactor response analogous to a programmable confocal pinhole. A fixed-aperture embodiment is the scalar degenerate case of the detector-side aperture parameter family; the scalar may be selected or trained during configuration and is then frozen in the committed parameter-family version, preserving the six-family count.

- 5 5. The reactor emitter drive profile $A_{\text{emit}}(t)$, applied upstream of the reactor stack. Periodic in T . $A_{\text{emit}}(t)$ probes the nonlinear medium's intensity-dependent response at each scan phase and is distinct from the reactor detector gain G_{det} because, for a nonlinear reactor, pre-stack scaling and post-stack scaling are not equivalent.
- 10 6. The reactor detector gain G_{det} , applied in the live coupling path between the reactor photodetector and the scene emitter drive. G_{det} may be a time profile $G_{\text{det}}(t)$ periodic in T , a nonlinear transfer function $G_{\text{det}}(I)$, or a combined $G_{\text{det}}(t, I)$, allowed to be nonlinear per Section 7.2.4. Where G_{det} is time-dependent, its phase origin is defined relative to the shared scan period T and committed to the protocol digest.

15 The 2D anchor also has six trainable parameter families, but their composition differs from the 1D anchor. The 1D anchor has separate scene and reactor scan trajectories; the 2D anchor has one shared scan trajectory and adds the detector-side aperture profile $a_{\text{det}}(t)$. Thus the 2D default comprises one shared scan, one reactor emitter-side focus profile, one reactor detector-side focus profile, one detector-side aperture profile, one reactor drive profile, and one frozen committed coupling-path gain. Scan-warping between subsystems, when used, is a declared extension (Section 10.2.10) rather than part of the default trainable set. In some embodiments, the trained values defining all six parameter families are committed to the protocol digest as part of the committed parameter-family version.

10.2.4 Common controller, live coupling path, and time alignment (2D)

25 In the 2D anchor, the common controller 10 generates a single shared scan program and fans it out to the galvanometer pairs in both subsystems. The *live coupling path*, as used in this subsection, refers specifically to the $A(t)$ return path from reactor detector 44 to scene emitter 24 drive. It is a deterministic low-latency path comprising the reactor photodetector, protective conditioning (for example transimpedance amplification and a safety limiter), and the reactor detector gain G_{det} of Section 10.2.3. G_{det} is trained offline and frozen for the committed deployment version. At runtime, the live coupling path contains no per-period wait and no runtime-adaptive or runtime-learned element. In the parallel-subsystem embodiment (a) of Section 4.2, the live coupling path is the sole architectural coupling between the scene and reactor sides, and there is no forward low-latency coupling from scene detector 43 into the reactor stage. In the cascade embodiment (b) of Section 4.2, the live coupling path is supplemented by a separately disclosed *forward low-latency coupling* from scene detector 43 to the reactor source of subsystem 220 (see Section 4.2); the forward low-latency coupling is itself declared and committed to the protocol digest and is not runtime-adaptive. The live coupling path and the forward low-latency coupling are structurally distinct: the live cou-

pling path is the reactor-detector to scene-emitter return signal, and the forward low-latency coupling, where present, is the scene-detector to reactor-emitter forward signal.

In some embodiments, G_{det} is implemented as a fixed analog shaping network whose transfer function matches the trained profile. In other embodiments, G_{det} is implemented
 5 as a deterministic high-rate sampled transfer function or look-up table operating on reactor photocurrent, provided that the sampled implementation introduces no per-scan-period buffering and has a declared bounded latency. In either implementation, the latency from reactor photocurrent to scene emitter modulation is bounded by a declared settling time and is committed to the protocol digest. The scene-detector and reactor-detector measurement
 10 records are separate from the live coupling path and follow the timestamping conventions below.

In some embodiments, the common controller drives the scene subsystem and the reactor subsystem from a shared scan program and timebase. Scene and reactor measurements are timestamped and optionally buffered to compensate for exposure, readout, propagation, or
 15 processing delays. The controller pools the aligned measurements into the observation vector $\mathbf{y}_t$ and logs the convolution bundle $c_t = (u(t), \mathbf{y}_t)$. In embodiments using derived latents, the controller updates $Z_\theta(t)$ from c_t and optionally uses $Z_\theta(t)$ to determine subsequent controls.

10.2.5 Light-field hybrid embodiment (non-limiting)

In some embodiments, the projector-role emission path and detector-role sensing path of the
 20 hybrid embodiment are each realised as four-dimensional light-field subsystems: a light-field projector (Section 10.23) emits a controllable field $E(x, y, u, v)$ into the scene branch, and a plenoptic detector (Section 10.24) records the returning field $L(x', y', u', v')$ in a single exposure. The reactor branch may independently be two-dimensional or four-dimensional. The convolution-bundle sample $c_t = (u(t), \mathbf{y}_t)$ therefore pairs a four-dimensional com-
 25 manded emission with a four-dimensional observed response, and the per-step transfer kernel maps between emitted and observed light-field coordinate spaces rather than between two-dimensional intensity fields.

In some embodiments, the projection-side microlens array and the detection-side microlens array are optically combined through a beam splitter so that the emit and capture
 30 paths share an MLA plane; in other embodiments, the two arrays are independent, with registration established by a calibration sub-protocol logged in the protocol digest.

Where a scene branch includes a camera-obscura screen, relay screen, display surface, or other two-dimensional intermediate surface, the angular-consistency primitive applies to that intermediate physical surface unless the relay optics preserve calibrated angular information
 35 from the external scene. In embodiments claiming external scene parallax through such a relay, the relay geometry, aperture, field stop, and angular-preservation calibration are recorded in the protocol digest and evaluated by declared meters.

Angular-consistency verification primitive (non-limiting). In some embodiments, a Truth Beam sub-regime uses the plenoptic capture as an angular-consistency check: scene emissions that are consistent with a three-dimensional physical scene, as observed over the declared sub-aperture baseline, exhibit measured parallax relationships across sub-apertures (u, v) . Planar presentation spoofs—including printed images, ordinary flat displays lacking calibrated angular multiplexing, planar projection surfaces, and planar retro-reflective replay surfaces—lack the depth-dependent parallax relationships expected under the declared pose, illumination, and calibration state, and therefore produce an angular-consistency residual under the declared meter. In some embodiments, the angular-consistency residual is included as a component of the verisimilitude meter and committed alongside the convolution bundle. This geometric exclusion is limited to planar or effectively planar replay and print-spoof classes; high-fidelity angular replay spoofs, including lenticular, holographic, light-field, angularly multiplexed planar displays, volumetric displays, three-dimensional printed replicas, textured mannequins, and parallax-mimicking surface assemblies, are not excluded by this geometric argument alone and are handled, where relevant, by the existing meter envelope, declared attacker model, and empirical hardness evaluation.

Four-dimensional reciprocity and adjoint-matching (non-limiting). In some embodiments, a calibration and verification sub-protocol emits a declared four-dimensional light field $E(x, y, u, v)$ and verifies that the captured field $L(x', y', u', v')$ is consistent with the adjoint, radiometric-adjoint, or declared transport-adjoint model of the forward reactor or scene transfer kernel under the declared configuration θ . In some embodiments, the reciprocity residual is logged in the meter envelope and treated analogously to the phase-conjugation residual of interferometric regimes, while remaining compatible with incoherent operation where phase-stable interferometric geometry is not available.

Effective-rank headroom (non-limiting). In some embodiments, replacing a two-dimensional emitted or observed intensity field with a calibrated light-field channel increases the raw addressable channel description from spatial coordinates to spatial-angular coordinates. The product of the declared spatial and angular resolutions on the emit and capture sides is treated only as an upper bound on the hypothetical physical rank available under the four-dimensional channel geometry, not as a task-useful rank claim. Declared Tier 2 effective rank is measured under the declared meter set from the committed physical evidence, for example by correlation, singular-spectrum, or independence analyses of the recorded channel. Declared Tier 3 task-useful rank is measured separately for the relevant task and deployment, and is not inferred from the spatial-angular resolution product. The protocol digest records the effective (x, y, u, v) resolution, calibration state, vignetting, cross-talk, and sub-aperture registration on the emit and capture sides so that any downstream rank statement is conditioned on the committed channel geometry and the measured evidence for that deployment.

Compatibility with existing operating regimes (non-limiting). In some embodiments, the light-field hybrid embodiment remains compatible with all three operating regimes (Truth Beam, Limager, Reality Transform) and with Yoked operation: in Limager, four-dimensional capture provides passive depth and refocusing evidence in a single exposure; in Reality Transform, the four-dimensional emitted field supports near-eye and integral-imaging rendering, view-dependent projection mapping, and parallax-consistent immersive rendering; in Yoked operation, the angular response of the scene to controlled angular emission constitutes a new coupling dimension whose stability is gated by the same conditional Lyapunov exponent discipline described elsewhere in this specification.

10.2.6 Sealed reactor pumps and eye-safe external beams

In many embodiments, high-intensity or tightly focused beams are confined to internal optical paths within an enclosed reactor housing, while the scene-facing emission is operated in an eye-safe or otherwise bounded exposure regime appropriate for the deployment context. One or more internal pump sources may drive nonlinear or history-dependent reactor behaviour inside the enclosure. From the channel-model perspective, internal pumps modify the effective configuration θ and thereby modify the induced conditional law $\mathbf{P}_\theta(\cdot \mid S, U_{0:T})$.

10.2.7 High-intensity etching, engraving, and ionising variants

In some variants, one branch of the module is configured for controlled interaction with matter at higher intensities, for example to engrave, etch, locally ablate, expose latent layers, or create persistent fiducials on a target substrate. In such variants, safety and hardware interlocks constrain the permissible working distance, angle, duty cycle, and total exposure. The interaction modifies the scene state over time and therefore changes the realised convolution bundle and, where present, media state M_t .

10.2.8 Temporal modulation and time-frequency reactor operation (non-limiting)

In some embodiments, the reactor medium is modulated in the time domain, not only the spatial domain. Recent experimental work has demonstrated that thin-film epsilon-near-zero (ENZ) materials—such as indium tin oxide (ITO), which is widely used as a transparent conductor in display panels, touchscreens, and LCD screens—can switch optical state on timescales approaching an optical cycle (approximately 10 fs) via the optical Kerr effect, producing temporal apertures whose interaction with probe light yields frequency-domain interference patterns (Tirole et al., “Double-slit time diffraction at optical frequencies,” *Nature Physics* 19, 999–1002, 2023; Galiffi et al., *Adv. Photonics* 4, 014002, 2022). This result is relevant to Reality Kernel embodiments in three respects.

First, temporal modulation extends the reactor’s degrees of freedom: the kernel $\mathbf{P}_\theta$ operates on the spatio-temporal field jointly, not merely the spatial distribution of light. The

emitter can encode structured information in pulse timing as well as spatial pattern, and the recorder can measure spectral interference patterns that encode both scene response and reactor temporal dynamics.

Second, time-varying media can enable nonreciprocal responses under appropriate spatiotemporal modulation conditions: light propagating in one direction through a temporally modulated reactor experiences a different transformation than light in the opposite direction. In some embodiments, the reactor is operated in a temporal modulation regime specifically to exploit this non-reciprocal property as a hardness mechanism for the verification regime. The emitter-to-recorder optical transfer function then differs physically from the recorder-to-emitter transfer function, providing a directional attestation primitive that does not depend on computational complexity assumptions (see also Section 12 for the general reciprocity-breaking hardness discussion).

Third, in some embodiments the reactor’s temporal modulation is periodic with period T , producing discrete frequency sidebands (Floquet modes) separated by $1/T$. In such time-crystal-like regimes, the kernel operates on a discrete spectral lattice, and the periodic modulation provides a natural clock cycle for optical processing. Topologically protected frequency-conversion pathways arising from the Floquet band structure may, in some embodiments, provide signatures that are robust against noise but empirically hard to forge without the physical medium. Time-refraction effects arising from single-cycle refractive-index modulation have been experimentally demonstrated in ENZ media (Lustig et al., “Time-refraction optics with single cycle modulation,” *Nanophotonics* 12(12), 2221–2230, 2023), confirming that sub-cycle temporal interfaces produce measurable frequency shifts suitable for convolution-bundle characterisation.

ITO is highlighted as a non-limiting example because it is already present in the screen-based reactor embodiments described elsewhere in this specification. Other ENZ or time-varying metamaterial substrates may be used. All temporal modulation parameters (pulse timing, modulation depth, repetition rate, pump profile) are logged in the convolution bundle alongside spatial controls. A complementary approach—the sweeping reactor (Section 10.27)—modulates reactor *geometry* rather than material properties, at much lower frequencies (Hz–kHz rather than THz), and uses phase-sensitive (lock-in) demodulation to extract the derivative of the reactor’s physical transform with respect to the swept parameter. The two approaches are orthogonal: temporal modulation of material properties amplifies hardness and can enable nonreciprocal responses, while parametric geometry modulation provides precision differential characterisation of the reactor’s transfer function.

10.2.9 Gas discharge and plasma reactor embodiments (non-limiting)

Gas discharge and plasma media constitute a broad class of nonlinear reactor media whose dynamics are inherently chaotic, device-specific, and temporally rich. Plasma exhibits negative differential resistance, avalanche ionisation, recombination cascades, and self-organised filamentary pattern formation—all strongly nonlinear. Manufacturing variation (electrode

geometry, gas pressure and composition, electrode surface condition, tube or envelope contamination) produces device-specific challenge–response behaviour suitable for reactor-microstructure hardness assessment. Non-limiting plasma reactor embodiments include the following.

- 5 **Plasma display panel (PDP) as micro-reactor array (non-limiting).** In some embodiments, a plasma display panel is used as a combined emitter and reactor array. A PDP comprises a grid of sealed gas discharge cells, each producing ultraviolet emission that excites phosphor coatings to yield visible light. Each cell is an independent micro-reactor with device-specific gas pressure, electrode gap, and phosphor grain distribution. The panel simultaneously functions as a structured emitter (it displays arbitrary patterns) and as a reactor array (each cell’s discharge dynamics are nonlinear and manufacturing-dependent). In some
10 embodiments, the PDP is driven near the cells’ firing threshold, where avalanche ionisation is most sensitive to perturbation and the system operates near a bifurcation surface; Fisher information about cell identity peaks in this regime. A camera observing the PDP output while the controller sweeps driving waveforms across the panel produces convolution bundles
15 encoding the joint emitter–reactor dynamics of thousands of micro-reactors.

- Near-threshold arc discharge (non-limiting).** In some embodiments, a gas discharge source (xenon arc, carbon arc, or other arc lamp) is operated near its breakdown voltage, where the discharge dynamics are maximally sensitive to electrode geometry, gas density,
20 and surface micro-features. Near the ionisation threshold, Townsend avalanche dynamics exponentially amplify small perturbations into macroscopically distinct discharge paths, producing the highest information yield per probe and the hardest conditions for an attacker to reproduce. In carbon-arc embodiments, the electrodes are consumed during operation—carbon evaporates, the crater shape evolves, and carbon nanoparticles participate in the
25 discharge—producing a self-modifying reactor whose challenge–response mapping drifts irreversibly because the physical medium is being destroyed and reformed. This built-in irreversibility provides a naturally time-indexed hardness source: the reactor’s state at any moment reflects its operational history and is not expected to be practically rewound without replacing or materially altering the reactor under the declared maintenance, access, and
30 observation assumptions. Xenon arc sources are commonly used light sources in cinema and high-end DLP projectors; in some embodiments, the arc’s intrinsic dynamics are observed alongside the projected content as additional convolution-bundle channels.

- Laser-induced plasma as transient reactor (non-limiting).** In some embodiments, a focused laser pulse ablates a surface, producing a transient plasma plume whose emission
35 spectrum encodes the elemental composition of the ablated material (the physical basis of laser-induced breakdown spectroscopy, LIBS). In some embodiments, a second structured illumination probe is directed through or onto the expanding plume, so that the plume serves as a transient three-dimensional reactor medium: it scatters, absorbs, and re-emits

the probe light according to its spatially and temporally varying composition and density. A spectrometer and/or camera records both the plume’s self-emission and the transmitted or scattered probe light. This produces a convolution bundle in which the reactor is created by the emission itself, exists for microseconds, carries information about the scene (the ablated surface), and is practically unreproducible because each ablation event removes material. In some embodiments, scanning the ablation point across the surface produces a spatially resolved composition–dynamics map logged as a committed convolution-bundle sequence.

Dielectric barrier discharge and atmospheric plasma (non-limiting). In some embodiments, a dielectric barrier discharge (DBD) panel or atmospheric-pressure plasma source produces self-organised filamentary patterns whose spatial structure depends on gas composition, driving frequency, dielectric surface properties, and electrode geometry. These patterns are chaotic, history-dependent, and responsive to proximity of external objects (a nearby grounded conductor restructures the discharge pattern, analogous to scene-coupled dynamics in Yoked operation). In some embodiments, a plasma ball or similar sealed discharge device is used as a reactor whose visible filament dynamics are observed by a camera; the filaments’ response to nearby objects or touches provides a scene-coupled observation channel.

Fluorescent tube as cascaded frequency-conversion reactor (non-limiting). In some embodiments, a fluorescent tube is used as a cascaded reactor: electrical excitation produces gas discharge plasma, the plasma emits ultraviolet radiation, and phosphor coatings downconvert UV to visible light. The three-stage nonlinear cascade (electrical $\rightarrow$ plasma $\rightarrow$ UV $\rightarrow$ visible) amplifies device-specific manufacturing variation at each stage. Gas pressure, electrode condition, and phosphor grain distribution all contribute to the device-specific spectral output. At mains frequency (50/60 Hz magnetic ballast) or tens of kilohertz (electronic ballast), the discharge is periodically modulated, producing temporal dynamics analogous to the Floquet-regime operation described in the temporal-modulation embodiments above.

In one non-limiting bench-top embodiment, the scan law and a portion of the optical kernel are implemented by a single moving element in a removable shroud. A compact source (for example a laser diode module) illuminates a sealed container holding a heterogeneous scattering medium (for example mixed glass or reflective elements). A micro-servo-mounted focus-and-steering optic provides both focus control and one-dimensional steering by tilting about an axis. A photodetector views the container interior through spectral filtering to emphasise selected bands (for example fluorescence bands or scattered light bands). The servo angle defines a scan law over reactor volume, and the focus setting of the optic defines a kernel coordinate. Shaking or otherwise reconfiguring the internal medium between runs re-randomises microstructure and thereby changes the response distribution while preserving a family resemblance under similar protocols.

Sealed-housing fluorescent-medium variant (non-limiting, by cross-reference).

In one non-limiting variant, the sealed container is a metal housing holding a shallow bed of mixed marbles or fragments. Suitable fluorescent and reflective media, including those discussed in detail at the bench-top 1D anchor (Section 7.2), may be used in the present 2D embodiment subject to the same safety, handling, and regulatory considerations described there. Geometry-specific aspects of the present 2D embodiment include the sealed metal housing; the servo-defined steering schedule across a two-dimensional bed; photodetector observation through a green-pass or long-pass optical filter so that the measured response emphasises wavelength-shifted fluorescence together with scattered light from the non-fluorescent media; and a shake-randomisation procedure between runs that yields different but statistically related photodetector time series under a fixed sweeper program.

10.2.10 Hardness, training, and extensions (by reference)

The hardness objective (Section 7.2.5), the staged training pathways (Section 7.2.6), the Yoked orthogonal extension (Section 7.2.7), the noise co-illumination extension (Section 7.2.8), and the CRT/phosphor analogue-memory extension (Section 10.5) apply to the 2D anchor by direct reference, mutatis mutandis: references in those sections to the 1D per-sample discrete feedback loop are read in the 2D case as references to the continuous live reactor-detector-to-scene-emitter amplitude broadcast of Section 10.2, and the 2D trainable parameter families of Section 10.2.3 substitute for the 1D families. In the 2D anchor, the CRT extension's default coupling between reactor detector output and CRT drive is not constrained by the 1D anchor's green-pass wavelength-separation argument, because the 2D anchor's reactor substrate is substrate-open; both electrical and optical CRT coupling are available as declared configurations.

10.3 Layered reactor stack embodiments

In some embodiments, the reactor loop is implemented as a layered stack designed to amplify memory, mixing, nonlinearity, and empirical hardness. A representative non-limiting configuration may include, in sequence: (i) a scattering plate, (ii) a nonlinear thin-film layer, (iii) a temporal persistence layer, (iv) a delay-network layer (for example fibre-delay loops with selectable couplings), and (v) one or more readout heads operating on the convolution bundle and/or derived latents.

Individual layers may be characterised and tuned in isolation and then assembled. Slow physical parameters may be fine-tuned using hardware-in-the-loop optimisation such as SPSA, while digital meters and decoders are trained on logged traces. Spectral multiplexing may be used to separate measurement bands (for example a band reserved for verification probing and a separate band for presentation or internal pumping), with per-band gain and filtering to maintain detector dynamic range.

10.4 SLM–scattering-medium–camera embodiment (non-limiting)

In some embodiments, the reactor is implemented as a spatial light modulator (SLM) directing structured illumination through a scattering or nonlinear medium, with a camera or detector array capturing the output. The SLM provides programmable, protocol-controlled emission patterns; the scattering medium provides a high-dimensional, physically fixed, device-specific transfer function; and the camera captures the convolution bundle. In such embodiments, the SLM pattern is included in the control protocol $U_{0:T}$ and recorded in the protocol digest, while the medium’s transfer function contributes the analogue complexity and history dependence that generate empirical hardness and computational richness.

This architecture supports all three operating regimes on the same hardware. In Truth Beam mode, the device-specific speckle or scattering signature provides physically unclonable challenge-response behaviour. In Limager mode, the SLM patterns are optimised to maximise information about a scene placed adjacent to or within the scattering medium. In Reality Transform mode, the SLM patterns are optimised to control the output field subject to the medium’s physical constraints. Yoked operation is possible when the medium includes dynamic elements (for example a liquid crystal layer, flowing suspension, or biological specimen whose optical properties evolve in response to the illumination).

In some embodiments, the scattering medium is sealed and characterised at enrolment, providing a stable enrolled microstructure signature. In other embodiments, the medium is exposed to environmental or biological inputs, and the scene-dependent variation in the scattering transfer function becomes the measurement signal. Meters record speckle contrast, mode-mixing statistics, and temporal stability so that verification and sensing claims are conditioned on the observed regime.

10.5 CRT/phosphor analogue-memory extension embodiment

In some embodiments, the anchor embodiment of Section 7.2 or Section 10.2 is extended by adding a CRT/phosphor analogue-memory loop atop the anchor’s live coupling. The CRT/phosphor loop provides a slow, spatially indexed analogue-memory channel, read out by a camera, and does not replace the anchor’s live coupling. In this extension, the reactor photodetector signal (or a derived signal) is written to the CRT via XY deflection locked to a declared scan coordinate: in the 1D anchor, to $\text{scan}_r(t)$, $\text{scan}_s(t)$, or a declared derived coordinate; and in the 2D anchor, to the shared scan program. The phosphor persistence provides a spatial accumulation of reactor response over multiple scan periods that is subsequently read by the camera. In the 1D anchor, the default coupling between reactor detector and CRT drive is electrical, because the green-pass filter architecture enforces wavelength separation. In the 2D anchor, both electrical and optical CRT coupling are available; the declared choice is committed to the protocol digest.

Mapping to FIG. 4 and to embodiments (a) and (b). FIG. 4 illustrates the CRT/phosphor extension as a fast optical loop (laser source 21, AOD/EOM deflector 22,

relay optics 23, reactor medium 30, photodetector 41) cascaded with a slow analogue-memory loop (CRT reactor 31 with electron gun, yoke, and phosphor screen 32, observed by camera 42), both driven from common controller 10. The heavy-stroke arrow shown in FIG. 4 from photodetector 41 to the slow-loop reactor stage is the *forward low-latency*
 5 *coupling* of embodiment (b) of Section 4.2: it carries the fast-loop scene-side observation directly into the analogue-memory stage as the load-bearing forward path of the cascade embodiment. The forward low-latency coupling in FIG. 4 may be implemented in any of the modalities recited in Section 4.2 (electrical direct, mixed-signal, optical gain-pumped, or optical direct relay), and its end-to-end latency is declared and committed to the protocol
 10 digest. The parallel-subsystem embodiment (a) of Section 4.2 is obtained from FIG. 4 by omitting the heavy-stroke forward arrow and relying on the controller’s scan protocol to synchronise fast-loop and slow-loop interrogations, with the slow-loop reactor stage driven by the controller alone rather than by the fast-loop photodetector signal.

Hobbyist and entry-level embodiments (non-limiting). For hobbyist, educational,
 15 or cost-sensitive builds, a cathode-ray tube provides a non-limiting analogue-memory reactor option. Aged and imperfect CRTs provide a rich, high-dimensional optical memory: over their lifetime, phosphors, shadow masks or aperture grilles, glass envelopes, and deflection assemblies accumulate idiosyncratic burn-in patterns, stains, charge traps, and hysteretic behaviours that together define a unique physical fingerprint. Such tubes, including surplus
 20 or second-hand displays, are therefore attractive as low-cost, high-entropy reactor elements even before any deliberate modification.

Augmenting CRT reactors with additional memory layers (non-limiting). In some embodiments, CRT reactors are enhanced with additional optical memory layers that preserve the unique character of each tube while deepening its state space. A thin luminous
 25 or photoreactive coating may be applied over the phosphor or faceplate so that prior illumination leaves slowly decaying traces that modulate subsequent excitation. Dispersions of scatterers, dyes, or textured films may be deposited in front of the screen to add further spatial structure, mixing, and history dependence.

Sources of reactor-microstructure uniqueness in CRT reactors (non-limiting). In some embodiments, reactor-microstructure uniqueness arises from manufacturing tolerances and wear mechanisms including: gun alignment tolerances, phosphor grain structure, shadow-mask or aperture-grille tolerances, magnetic-material idiosyncrasies in yokes and shims, and glass-envelope geometry variations. Together these yield device-specific response diversity in which small control changes induce effectively uncorrelated output changes under meter-
 35 bounded operation.

In a preferred configuration, the CRT is an XY-deflection tube (for example an oscilloscope-style tube) providing separate analogue inputs for horizontal deflection $X_{\text{CRT}}(t)$, vertical deflection $Y_{\text{CRT}}(t)$, and intensity drive $Z_{\text{CRT}}(t)$. In some embodiments, these signals

are produced by a controller and are synchronised with reactor optics (for example galvanometer angles) so that the CRT face coordinates and the reactor scan coordinates share a stable mapping. The intensity drive may be modulated by processed scene signals or by internal controller state to encode additional history into phosphor persistence.

In some embodiments, an optical readout path relays a region of the CRT face into the reactor loop. Relay optics image the CRT face (or a portion of it) into the entrance pupil of reactor optics, after which additional tunable elements and scan mirrors transform the field before it reaches a reactor detector. The reactor detector may be a photodiode with a transimpedance amplifier producing a continuous analogue measurement, or a camera producing sampled frames whose timing is logged.

In some embodiments, the CRT subsystem is used as a bidirectional surface: it can be written by the electron-beam excitation pattern and read optically, and it may be combined with further photoreactive or scattering layers to increase state-space richness. The CRT/phosphor stack is one example of a slower luminescent analogue-memory medium in a multi-rate reactor stack that may be coupled to faster optical reactor loops, alongside alternatives such as laser-addressed luminescent screens or other luminescent persistence media.

10.5.1 Trainable electron-optical controls and calibration (non-limiting)

In some embodiments, the CRT includes an electron gun and electron-optical controls, and the controller treats electron-optical set-points as tunable or trainable kernel parameters. Non-limiting trainable parameters include: control-grid bias $V_{G1}(t)$, screen-grid bias $V_{G2}(t)$, anode (EHT) voltage V_{anode} , deflection coil currents and waveform shaping (including harmonic corrections and lookup tables), focus and astigmatism controls (optionally address-dependent), auxiliary coil currents producing controlled nonuniform fields, per-address dwell and blanking schedules, and per-channel or per-gun spectral-tap gains (for example (w_R, w_G, w_B) in RGB-gun embodiments). In some embodiments, additional tunable degrees of freedom include mechanically positioned ferromagnetic shims or auxiliary magnets near the yoke to introduce controlled geometric warps, and stencil or mask elements (including turrets or programmable masks) to implement piecewise spatial apertures or mask bases.

In some embodiments, an auxiliary diagnostic camera observes the CRT face (or beam) to estimate spot size, convergence, and alignment, enabling closed-loop calibration while the main observation stream remains a single-pixel photodiode/TIA stream and/or a separate scene-facing camera stream.

Nonlinearity and mixing mechanisms (non-limiting). In some embodiments, nonlinearity and mixing arise from controlled geometric warps (shims, auxiliary magnets, auxiliary coils), spatial apertures and mask selection (turrets or programmable masks implementing a mask basis), and bias shaping via $V_{G1}(t)$ schedules that yield nonlinear transfer curves and

saturable regions. Spectral diversity across multiple guns or sources yields multiple natural “taps” with slightly different spot statistics, and these mechanisms interact with phosphor persistence so that nonlinearity is coupled to analogue memory.

Time-token injection (non-limiting). In some embodiments, a time-token $\tau_{\text{tok}}(t)$ is injected into the physical channel, for example as a small-signal modulation onto $V_{G1}(t)$ (or another bias) with a damping coefficient d_τ that bounds its effect on gain and stability, and/or as an auxiliary optical beam summed into the scene or reactor path. The token couples a timing or authenticity signal into the physical response without requiring a separate out-of-band channel, and token schedules, envelope limits, and calibration identifiers form part of θ and are logged in the protocol digest.

Beam PSF and persistence convolution (non-limiting). In some embodiments, instantaneous luminance at a phosphor plane is modelled as

$$L(x, y, t) \propto Z_{\text{CRT}}(t) K_{\text{spot}}(x - X_{\text{CRT}}(t), y - Y_{\text{CRT}}(t)),$$

where $K_{\text{spot}}(\cdot, \cdot)$ is an effective spot point-spread function. Emitted intensity incorporates persistence as a temporal convolution

$$E(x, y, t) = \int_{-\infty}^t L(x, y, \tau) p(t - \tau) d\tau,$$

and the imaged observation is obtained by relay optics and detector integration, with calibration and receptive-field models recorded in the protocol digest.

Phosphor impulse response and detector weighting (non-limiting). In some embodiments, phosphor persistence is modelled by an impulse response

$$p(t) = \sum_i a_i e^{-t/\tau_i}, \quad t \geq 0,$$

and dwell and blanking schedules shape an effective temporal kernel $p_{\text{eff}}(t) = p(t) * w_{\text{det}}(t)$, where $w_{\text{det}}(t)$ denotes detector temporal weighting due to sensor integration, transimpedance amplifier dynamics, filtering, and digitisation aperture. In some embodiments, $w_{\text{det}}(t)$ is estimated by applying a known sub-threshold optical impulse (for example a calibration LED pulse or a low-amplitude electrically driven probe) and fitting a causal parametric IIR, FIR, or spline model; the estimated model is stored as a calibration table and its version is logged.

Receptive-field and spot-PSF calibration (non-limiting). In some embodiments, the detector spatial receptive field $\rho(u, v)$ is estimated before or during operation by spot-scanning the scene with a small bright probe, by micro-pattern projection, or by projecting a

structured calibration grid, and fitting a parametric, spline, or lookup-table model to the resulting response. The estimated receptive field is used to correct signed, differential, bipolar, and pooled measurements, and is committed to the protocol digest together with the estimation procedure and its calibration epoch. In some embodiments, the beam spot point-spread function $K_{\text{spot}}(\cdot, \cdot)$ at the phosphor plane is estimated by knife-edge measurement, slanted-edge modulation-transfer-function methods, micro-grid calibration, or related techniques, and the estimated PSF parameters are stored in lookup tables used by focus, convergence, astigmatism, and drift-compensation control loops. Calibration epochs are versioned and their identifiers are recorded in the protocol digest.

Face-camera diagnostic channel (non-limiting). In some embodiments, a separate diagnostic imager is directed at the CRT phosphor face (or other persistence-medium surface) to provide closed-loop measurement of spot size, beam position, convergence, drift, and uniformity, independently of the primary sensing channel that observes the relayed or scene-projected signal. The face-camera channel is read on a dedicated calibration clock, its samples and exposure schedule are committed to the protocol digest as a separate atom from the primary observation, and its outputs feed the trainable electron-optical and timing controls. In some embodiments, the face camera operates at a lower sample rate than the primary detector and supplies slow-loop calibration corrections without participating in the fast scoring or reconstruction loop.

Safety and burn-in prevention (non-limiting). In some embodiments, blanking interlocks disable beam emission on deflection failure to reduce burn-in risk. EHT discharge procedures are supported by bleeder resistors and explicit discharge workflows, and shielding reduces ancillary emissions. Thermal and ageing compensation uses slow bias-trim loops and periodic recalibration to keep spot geometry and gain within a corridor; drift tracking may use PID- or Kalman-style updates of lookup tables between training rounds.

Sources of CRT and electron-beam unclonability signature (non-limiting). In some embodiments, the per-device unclonability signature that contributes to the reactor-microstructure uniqueness property (Section 12) arises from one or more of the following sources, individually or in combination: electron-gun alignment and assembly tolerances, including cathode-to-grid spacing, accelerating-electrode alignment, and gun-to-screen registration; phosphor microstructure, including grain-size distribution, packing density, layer-thickness variation, and dopant micro-distribution; shadow-mask, aperture-grille, or slot-mask tolerances, including hole or slot pitch variation, edge profile, and registration to the phosphor pattern; deflection-yoke and shim magnetic idiosyncrasies, including winding tolerances, residual magnetisation, and shim placement; focus-coil and convergence-coil winding tolerances; auxiliary electric-deflection-plate spacing and surface finish variations; glass-envelope geometry, including thickness variation, residual strain, and curvature; vacuum-level micro-variation across batches; and component ageing histories that diverge from device

to device under different operating regimes. Each such source is individually sufficient to contribute to per-device challenge-response variation, and in combination they provide the high-dimensional uniqueness signature on which proof-of-projection, proof-of-discrepancy, and related verification protocols rely.

5 **Programmable-mask placement and stencil control (non-limiting).** In some embodiments, a programmable mask is positioned at one or more of: proximate the CRT phosphor screen on the viewing side as a programmable relay stencil; proximate the electron-gun aperture or anode pathway as an electron-compatible beam-stencil reference; in the relay optical path between the phosphor and the detector as an aperture or filter element; or
 10 in the scene-projection path as a structured-illumination element. At the phosphor-screen, relay-path, and scene-projection placements, the programmable mask comprises without limitation a liquid-crystal display panel, liquid-crystal- on-silicon panel, digital micromirror device, or other spatial light modulator. At the gun-aperture or anode-path placement, the electron-compatible beam-stencil reference is implemented without limitation by a config-
 15 ured electrostatic field (for example through patterned biased electrodes or microfabricated electron-optical apertures), by a configured magnetic field (for example through patterned coil or shim arrays), or by a patterned physical aperture mask compatible with the in-vacuum electron-optical environment. Each programmable-mask configuration provides rapid stencil updates under declared control, contributing trainable or adaptively- controlled degrees of
 20 freedom that are committed to the protocol digest together with the mask state and its update schedule.

Enumerated focusing and beam-shaping hardware (non-limiting). In some embodiments, focus, astigmatism, convergence, and beam-shape adjustment is implemented by one or more of: micromotor-driven lens positioning, MEMS-actuated lens elements, elec-
 25 trowetting or liquid-crystal tunable lenses, piezoelectric lens-positioning actuators, voice-coil actuators, varifocal liquid lenses, and deformable-mirror or membrane-mirror elements on the optical side; and on the electron-optical side by trainable focus-coil current, trainable astigmatism-correction multipole currents, trainable convergence-coil currents, and trainable accelerating-voltage modulation. Each focusing or beam-shaping degree of freedom may be
 30 operated as a fixed calibrated setting, a slow drift-compensated setting, a trainable kernel parameter, or an adaptively-controlled runtime parameter, with the operating mode and parameter trajectory committed to the protocol digest.

10.5.2 Signed-weight realisation via differential exposure (non-limiting)

In some embodiments, signed or bipolar weights are realised without sacrificing CRT dynamic
 35 range by driving two complementary exposures E^+ and E^- and performing differential detection:

$$y(t) = \int \left[E^+(u, v, t) - E^-(u, v, t) \right] \rho(u, v) du dv,$$

where $\rho(u, v)$ denotes a detector sensitivity (or pooling) function. In some embodiments, E^+ and E^- correspond to alternating masks, alternating bias offsets, alternating write patterns, or alternating line/field passes, and the differential signal is formed by subtracting consecutive detector samples or by differencing two detector channels.

- 5 **Temporal and channel differencing embodiments (non-limiting).** In some embodiments, signed weights are realised by temporal differencing (positive and negative contributions assigned to distinct time slots) or by channel differencing (positive and negative contributions assigned to distinct channels such as wavelengths, polarisations, or optical paths). A signed measurement is formed by subtracting calibrated readouts under logged
10 timing or channel tags.

Persistence guard (non-limiting). In some embodiments, a minimum inter-polarity delay (a persistence guard) is enforced between E^+ and E^- exposures to bound crosstalk induced by phosphor decay. The guard interval is derived from fitted phosphor time constants and is logged as part of the scan timing model and protocol digest.

- 15 **Multi-CRT summation (non-limiting).** In some embodiments, two or more CRTs are optically combined, with E^+ assigned to one tube and E^- to another, realising signed weights directly in the optical domain. Small misalignments between tubes may be treated as additional mixing degrees of freedom rather than solely as error sources, and are accounted for in calibration and meter summaries. In some embodiments, per-tube misregistration pat-
20 terns define a device-specific convolution kernel that contributes to reactor-microstructure uniqueness of the combined assembly.

Acoustic and RF analogues (non-limiting). In some embodiments, differential or signed-weight readouts are realised in other physical domains by pairing a steered emitter with a persistence element and differential detection. Non-limiting examples include acous-
25 tic paths with decay (reverberant cavities or damped delay lines) and RF cavities whose Q -factor produces multi-timescale memory.

10.5.3 CRT as a two-dimensional analogue memory

In some embodiments, the reactor includes a CRT or similar scanned display device whose phosphor integrates and decays in response to excitation, thereby implementing a two-
30 dimensional analogue memory. The deflection waveforms and beam intensity control are treated as components of the control input $u(t)$, and the phosphor luminance field is treated as part of the reactor or media state.

10.5.4 Scan law parameterisation for CRT addressing

Let the phosphor define a bounded domain $\Omega \subset \mathbb{R}^2$. A scan law may be expressed as a time-parametrised curve

$$\gamma(t; \eta) = (x(t; \eta), y(t; \eta)) \in \Omega,$$

where η parameterises frequencies, phases, amplitudes, envelopes, or basis coefficients. Non-limiting examples include Lissajous trajectories, chirps, piecewise dwell-and-jump schedules, and finite basis expansions. The parameters η may be included in θ and may be held fixed, updated on a slow timescale, or adapted online.

Address-space parameterisation (non-limiting). In some embodiments, addressing is expressed as a composition of maps

$$(\rho, \varphi) = \Gamma_{\text{addr}}(t; \eta), \quad (x, y) = \Psi(\rho, \varphi),$$

where (ρ, φ) denotes a non-limiting internal address coordinate, and Ψ maps addresses to phosphor coordinates. In some embodiments, Γ_{addr} and Ψ are chosen so that the induced address map covers the active phosphor region to within hardware resolution and noise, and is invertible or approximately invertible for writing, reading, and verification under the logged scan law and timing model.

10.5.5 XY-deflection synchronisation with external scanners

In one embodiment, an XY-deflection CRT provides analogue inputs for horizontal and vertical deflection, and the same underlying drive signals are also used, after calibration scaling, to drive galvanometer scanners in another reactor path. This synchronises CRT coordinates and optical scan coordinates so that CRT patterns can act as a spatially indexed memory layer aligned to a scan address space.

10.5.6 Channelised memory using line scans and rolling-shutter readout

In one embodiment, a one-dimensional scanner produces a time-varying intensity profile that is written into one or more CRT lines. A rolling-shutter camera reads the CRT surface with timing chosen relative to the line-write schedule, producing a time-staggered sampling of the phosphor memory. The resulting camera frames and/or derived features are included as channels of $\mathbf{y}_t$. Colour CRTs may be used to store multi-channel features by writing distinct values into different phosphors.

Channelised memory with commodity hardware (non-limiting). In one illustrative embodiment, a commodity barcode scanner is coupled to a line-addressable CRT that is observed by a rolling-shutter camera. Each barcode scan produces a one-dimensional intensity profile; the CRT driver writes this profile into a chosen horizontal line or narrow

band of the screen. The rolling-shutter camera then reads the CRT line by line, converting the two-dimensional pattern on the tube into a time-staggered convolution-bundle stream. In this configuration the CRT functions as a channelised optical memory: the controller can decide where to place the most recent scan, effectively multiplexing many scan histories into a single optical buffer.

Global-shutter and rolling-shutter observation (non-limiting). In some embodiments, the CRT surface is observed by a global-shutter sensor (approximating an instantaneous snapshot) and/or by a rolling-shutter sensor (in which each sensor row integrates the CRT at a slightly different time). In some embodiments, the rolling-scan direction is arranged to be substantially parallel, perpendicular, or oblique to a CRT sweep direction by mechanical mounting and/or by introducing a fixed or tunable optical element (for example a plane mirror, steering mirror, cylindrical lens, or tunable lens) into the optical path. In tunable embodiments, the distortion element is driven by one or more analogue control parameters, allowing the geometric relationship between CRT and sensor to be varied continuously over time; the element's drive schedule is treated as part of θ and recorded in the protocol digest.

10.5.7 Raster CRT variants

In some embodiments, a raster CRT is used instead of an XY-deflection tube. The CRT internal raster is treated as part of the reactor dynamics, and external scanners sample the raster-refreshed phosphor field along slower trajectories, yielding temporally down-sampled but spatially consistent views of the CRT state.

10.6 Optical and Other Physical Long-Term Memory Substrates

In some embodiments, the Reality Kernel apparatus comprises a physical long-term memory substrate configured to retain one or more committed-bundle records, protocol-digest records, meter records, calibration records, attestation records, governance records, or derived evidentiary artefacts. The physical long-term memory substrate may be configured so that a write operation produces a persistent physical state change that is thereafter readable many times and is not rewritable by ordinary runtime operation of the apparatus. Such a substrate is referred to herein as a write-once-read-many, or WORM, substrate. The WORM property is an apparatus property of the storage medium, the write/read path, or both, and is not merely a software retention policy.

In some embodiments, an entry written to the physical long-term memory substrate comprises, or is bound to, a digest of a convolution bundle $C_{0:T}$, a protocol digest Π , a meter vector or meter envelope, a calibration state, a verifier state, a governance-partition state, an authority signature, an external timestamp, a randomness-beacon value, a ledger-derived value, or any combination thereof. In some embodiments, the written entry further comprises a physical-substrate binding value derived from the measured physical response of the memory substrate, so that later retrieval verifies not only the digital content of the

entry but also the association between the entry and the physical memory medium on which it was written.

In a non-limiting embodiment, a record r_k is written to a physical memory substrate by applying a write operation W to a local region or mode of the substrate:

$$\mu_{k+1} = W(\mu_k, r_k, \Pi_k),$$

- 5 where μ_k denotes the physical memory state before writing, and Π_k denotes the protocol digest or digest component associated with the write event. A later read operation R produces a readout $\hat{r}_k$ and a substrate-response value ρ_k :

$$(\hat{r}_k, \rho_k) = R(\mu_{k+1}, \Pi_k).$$

A verification function may then test consistency between the recovered record, the substrate-response value, and one or more committed digests:

$$\text{Verify}_{\text{mem}}(\hat{r}_k, \rho_k, \Pi_k, h(r_k)) \in \{0, 1\}.$$

- 10 The foregoing equations are illustrative only; the storage, readout, and verification functions may be implemented by any physical, digital, optical, magnetic, molecular, silicon, hybrid, or cryptographic mechanism.

In some embodiments, access to the physical long-term memory substrate is physically isolated, logically isolated, cryptographically isolated, temporally isolated, or any combination thereof from an agent behaviour generation pathway. For example, an agent may cause
 15 operational events that are recorded by the apparatus, while lacking direct read, write, erase, or rewrite authority over the memory substrate or its governance-side index. In some embodiments, retrieval requires an authority token, a governance-partition command, a threshold signature, a hardware-attested read path, a selective-opening proof, a proof of authorised
 20 access, or another access-control condition bound to the protocol digest.

The physical long-term memory substrate may be used to store durable evidence while preserving privacy by storing encrypted records, commitments, accumulators, digest paths, selective-opening handles, reconstruction-threshold records, cohort-publication records, redacted records, or derived verification results rather than raw sensor records. In some em-
 25 bodiments, raw records remain unavailable unless a declared authority condition is satisfied, while digest, meter, or accumulator records remain available for ordinary governance audit.

Physical memory substrate generalisation (non-limiting). The long-term memory substrate is not limited to an optical memory. In some embodiments, the substrate comprises any physical storage medium in which read-many semantics, tamper evidence, sub-
 30 strate binding, or write-once behaviour is enforced at least in part by an apparatus property rather than by policy alone. Non-limiting examples include holographic storage, fluorescent storage, photochromic storage, photorefractive storage, phase-change storage, multi-

layer optical-disc storage, perpendicular magnetic recording, magnetic tape, MRAM-WORM storage, DNA storage, polymer storage, anti-fuse memory, one-time-programmable read-only memory, eFuse memory, ceramic or glass-based archival marks, physically unclonable storage media, and hybrid physical-digital storage. Optical, magnetic, molecular, silicon, mechanical,
 5 chemical, and hybrid storage species are examples only.

Representative write/read examples by substrate class (non-limiting). The following non-limiting examples illustrate write, read, and substrate-response operations in five representative substrate classes. In each, the write operation produces a persistent physical state change, the read operation interrogates that state, the WORM property is an apparatus property of the medium and write path, and the substrate-response value ρ_k is measured
 10 at read time and bound into the protocol digest of the corresponding write event.

- *Optical / holographic class.* In some embodiments, the write operation comprises exposing a photosensitive medium (for example a photopolymer, photorefractive crystal, or photochromic film) at a declared write fluence sufficient to produce a refractive-index, absorption, or fluorescent state change; the read operation comprises illuminating the
 15 written region under a declared reference geometry and measuring the diffracted, transmitted, or emitted optical response; ρ_k is a measured optical response value (for example diffraction efficiency, fluorescence intensity, or holographic correlation peak) interpreted under a committed optical-response model.
- *Magnetic / MRAM-WORM class.* In some embodiments, the write operation comprises driving a write current sufficient to set a magnetic-tunnel-junction or domain-wall cell into a declared polarisation state under an apparatus-enforced one-shot write path (for example a one-time-programmable fuse on the write line); the read operation comprises measuring the tunnelling or magnetoresistive response under a declared read bias; ρ_k
 20 is a measured electrical response value (for example a tunnelling-magnetoresistance ratio) interpreted under a committed cell-response model.
- *Molecular / DNA / polymer class.* In some embodiments, the write operation comprises a declared synthesis or polymerisation sequence producing a persistent chemical record (for example a synthesised oligonucleotide strand, a templated polymer sequence, or a covalent reaction product); the read operation comprises a declared sequencing, spectroscopic, or affinity-readout process; ρ_k is a measured readout response (for example a sequencing-quality score, mass-spectrometric peak, or affinity-binding response) interpreted under a committed readout-response model.
 30
- *Silicon / anti-fuse / eFuse / OTP class.* In some embodiments, the write operation comprises applying a one-shot programming voltage sufficient to permanently rupture a dielectric, blow a metal fuse, or set a one-time register; the read operation comprises measuring the post-program resistance, threshold, or logic level under a declared read condition; ρ_k is a measured electrical response value (for example post-program resistance or threshold-voltage shift) interpreted under a committed device-response model.
 35
- *Ceramic / glass / PUF-mark class.* In some embodiments, the write operation com-
 40

prises a declared laser-induced, ion-beam-induced, or mechanically engraved permanent mark in a ceramic, glass, or other archival substrate; the read operation comprises an optical or microscopic measurement of the mark geometry, scattering response, or PUF response under a declared imaging geometry; ρ_k is a measured response (for example a speckle correlation against an enrolled response or a geometric fingerprint of the mark) interpreted under a committed PUF- or mark-response model.

The foregoing classes and operations are illustrative only and do not limit the substrate, write path, read path, or substrate-response mechanism of any disclosed embodiment.

10.7 Bench-top dual-loop evidence appliance

In some embodiments, a worked bench-top embodiment (FIG. 4) comprises a single Reality Kernel module with a fast optical loop and a slow analogue-memory loop under a common controller. The slow loop instantiates the generic CRT and phosphor analogue-memory mechanisms described in Section 10.5; the present subsection specifies how those mechanisms are combined with a fast scattering loop, analogue rate conversion between the two timescales, and a committed dual-loop evidence architecture.

Fast optical loop. In some embodiments, the fast loop comprises a visible-wavelength source 21 (for example a laser diode module), an addressing element 22 (for example a galvanometer mirror or acousto-optic deflector), relay optics 23, a disordered scattering medium 30 (for example a ground-glass plate, semi-opaque diffuser, or sealed container of mixed scattering elements), and a detector 41 (for example a frame camera or photodetector with transimpedance amplifier). In some embodiments, the addressing element defines a scan law (raster, Lissajous, spiral, or pseudo-random) over the reactor volume, and the fast-loop timescale is set by the addressing element's repositioning time (microsecond scale for an AOD, sub-millisecond for a galvanometer).

Analogue-domain computation prior to digitisation (non-limiting). In some embodiments of the fast optical loop, the loop performs convolution-like mixing, scattering, interference, and nonlinear transformation in the analogue optical and electrical domains, with no analogue-to-digital conversion in the signal path between the source and the photodetector output. In some embodiments, the detector produces a continuous analogue electrical signal (for example a current or voltage from a transimpedance amplifier) that encodes the result of the physical computation, and digitisation occurs only after the physical mixing and transformation are complete. In some embodiments, this fully-analogue computational signal path within the fast loop is an architectural distinction from many hybrid photonic computing systems, which digitise optical signals at layer boundaries for digital readout, re-encoding, and re-projection between stages. In some embodiments, the absence of intermediate analogue-to-digital conversion within the fast-loop computational path means that the committed bundle encodes a measurement of the analogue physical output, preserving

temporal and amplitude resolution up to the detector’s analogue bandwidth, and that the physical transformation is not discretised or truncated by converter quantisation noise or sample-rate limitations at intermediate stages. In some embodiments, other parts of the system (for example the routed-ensemble digital re-projection path or the slow-loop CRT write path) may include analogue-to-digital or digital-to-analogue conversion stages; the fully-analogue property described here applies specifically to the fast optical loop’s computational signal path.

Analogue rate conversion between fast and slow loops (non-limiting). In some embodiments, the rate conversion between the fast optical loop (nanosecond to microsecond timescale) and the slow analogue-memory loop (tens of Hz write cadence) is accomplished in two stages, both operating in the analogue domain. In some embodiments, a first stage performs analogue bandwidth reduction on the fast-loop photodetector output, producing a lower-bandwidth analogue signal that represents a declared feature of the fast-loop response. In some embodiments, analogue bandwidth-reduction mechanisms include one or more of: (i) envelope detection, in which a rectifier and low-pass filter extract the amplitude envelope of the fast-loop signal; (ii) sample-and-hold, in which a capacitor captures the fast-loop voltage at a trigger event and holds it for the slow-loop write cycle; (iii) integrate-and-dump, in which the fast-loop signal is integrated over a declared window and the accumulated charge is transferred to the slow-loop input; (iv) analogue peak detection, in which a peak-detector circuit captures the maximum excursion of the fast-loop signal during a declared window; and (v) analogue matched filtering, in which a passive or active filter with a template impulse response correlates the fast-loop signal with a declared feature profile. In some embodiments, a second stage maps the reduced-bandwidth analogue signal to a CRT write signal (for example an XY deflection voltage pair and a Z-axis intensity voltage) that deposits the result into the slow analogue-memory loop at the CRT refresh cadence.

Slow analogue-memory loop. In some embodiments, the slow loop comprises a CRT reactor 31 or related persistence display and a camera 42 that observes the phosphor screen 32. In some embodiments, the CRT phosphor screen 32 provides analogue temporal memory with a persistence time constant that depends on the phosphor species. In some embodiments, the camera captures frames at a rate slower than the fast-loop bandwidth, producing a committed bundle that encodes a time-integrated view of the evolving phosphor state.

Colour-phosphor parallel temporal filter bank (non-limiting). In some embodiments using a colour CRT or colour phosphor screen comprising two or more phosphor species with measurably distinct persistence time constants (for example, in phosphor sets where τ_R , τ_G , τ_B — instances of the general persistence time constants τ_i described above — differ by at least an order of magnitude), each phosphor species implements an independent temporal convolution channel: the same excitation pattern is deposited into all phosphor species simultaneously, and each species integrates and decays at its own characteristic rate,

so that at any readout instant the luminance from each phosphor species represents a differently time-weighted history of the excitation sequence. In some embodiments, a colour camera separates these channels, yielding a set of parallel temporal filter outputs from a single phosphor screen. In some embodiments, the number of independent temporal channels is equal to the number of phosphor species with sufficiently distinct time constants, and whether three species provide measurably independent temporal information is an empirical property of the specific phosphor set that must be validated by measuring inter-channel correlation under representative excitation sequences.

Tuneable readout colour weighting (non-limiting). In some embodiments, the camera used to observe the persistence display is augmented with a tuneable colour filter or colour-channel gain control that allows the relative contribution of each phosphor channel to the readout to be adjusted, thereby selecting among different temporal weighting profiles without modifying the phosphor screen itself.

Channelised optical memory (non-limiting). In some embodiments, the colour-phosphor temporal filter bank is used as a channelised optical memory in which the controller selectively addresses different phosphor channels (for example by varying the electron-beam colour convergence, masking, or excitation spectrum) so that independent information streams are deposited into different temporal-persistence channels and read out through spectral separation.

Signed-weight realisation via differential exposure (non-limiting). In some embodiments, signed effective weights are realised by differential exposure: two phosphor patches, spectral bands, temporal windows, or spatial regions are driven with the same excitation magnitude but opposite intended signs, and a subsequent readout stage computes their difference (for example by subtracting two camera colour channels, two spatial regions, or two temporally interleaved frames). In some embodiments, the resulting signed readout approximates a single bipolar weight without requiring a bipolar physical medium. In some embodiments, whether differential exposure produces signed weights with sufficient precision, dynamic range, and stability for a given task is an empirical property that must be validated per embodiment.

Multi-CRT summation (non-limiting). In some embodiments, multiple CRTs or persistence displays are observed simultaneously by a single camera, and the overlapping phosphor images are summed optically on the detector surface. In some embodiments, this optical summation realises element-wise addition of multiple analogue-memory states without digital intervention.

XY-deflection synchronisation with external scanners (non-limiting). In some embodiments, the CRT XY deflection voltages are synchronised with the fast-loop addressing

element (AOD, galvanometer, or SLM) so that each fast-loop scan point maps to a declared position on the CRT phosphor surface. In some embodiments, this synchronisation enables the CRT to serve as a spatially resolved analogue memory of the fast-loop computation results.

- 5 **Channelised memory with commodity hardware (non-limiting).** In some embodiments, the channelised optical memory and colour-phosphor filter bank described above are implemented using commodity colour CRT monitors and webcams or industrial cameras with Bayer-pattern colour filter arrays. In some embodiments, commodity hardware operates within declared calibration envelopes and its limitations (for example colour cross-
- 10 talk, limited persistence uniformity, and limited spectral separation) are characterised and metered.

Operating modes and commitment. In some embodiments, the dual-loop appliance operates in all three regimes (verification, sensing, transformation) and produces a committed bundle covering both loops. In some embodiments, the protocol digest records the fast-loop

15 scan law, the analogue rate-conversion mechanism, the CRT write parameters, the camera readout cadence, and the phosphor persistence model.

Bring-up and validation (non-limiting). In some embodiments, the bring-up procedure includes: (a) measuring the CRT phosphor persistence time constant per colour channel, (b) measuring the CRT spot size and positional accuracy, (c) measuring the camera-

20 to-CRT geometric alignment and applying a calibration warp, (d) verifying the analogue rate-conversion stage by driving the fast loop with known excitation sequences and comparing CRT-deposited patterns against predicted patterns, and (e) establishing a baseline verisimilitude score using the trained discriminator on committed bundles from the validated configuration.

25 **Servo-swept scanning reactor variant (non-limiting, by cross-reference).** In some embodiments, a compact bench-top variant of the present module follows the bench-top 1D fluorescence-substrate anchor of Section 7.2, used as the reactor for the present multi-rate / dual-loop architecture in place of the CRT and scattering combination. The geometry, source arrangement, lens-and-servo control, sealed container, spectral filtering, scan-law construc-

30 tion, and fluorescent-persistence properties of that anchor apply unchanged in the present variant.

Computational interpretation as gated recurrence (non-limiting). In some embodiments, the bench-top dual-loop evidence appliance is interpreted as a gated recurrent architecture in which: (i) the fast optical loop implements a nonlinear input-to-hidden trans-

35 formation via scattering and analogue mixing, (ii) the analogue rate-conversion stage acts as a gating mechanism that selects which features of the fast-loop state are deposited into

long-term memory, (iii) the CRT phosphor screen implements analogue hidden-state memory with exponential decay, (iv) the camera readout implements a measurement of the hidden state, and (v) the controller’s emission at the next time step may depend on the camera readout, closing the recurrence. In some embodiments, the key distinction from digital gated recurrent architectures is that the gating, state storage, and recurrence operate in the analogue physical domain rather than in digital arithmetic, and the committed bundle provides an auditable record of the recurrent dynamics.

10.8 Fibre-delay reactor embodiments

In some embodiments, the reactor includes one or more fibre-delay networks that implement recurrent temporal kernels in the optical domain. A fibre-delay reactor may include splitters that distribute an optical field into multiple delay loops with different delays $\{\tau_\ell\}$, optional phase shifters and attenuators, and combiners that recombine delayed fields. Parameters may include coupling ratios, phase shifts, selected taps, and, in some variants, selectable loop lengths. These parameters form a subset of θ and may be tuned at design time or as slow controls.

Impulse-response lens (non-limiting). In some embodiments, in a small-signal or narrowband regime, a fibre-delay network is modelled as a linear temporal filter with a sparse impulse response:

$$h_\theta(t) = \sum_\ell a_\ell e^{i\varphi_\ell} \delta(t - \tau_\ell),$$

where a_ℓ and φ_ℓ are non-limiting per-loop amplitude and phase parameters (part of θ and/or slow controls). An input field envelope $E_{\text{in}}(t)$ yields an output

$$E_{\text{out}}(t) = \int h_\theta(\tau) E_{\text{in}}(t - \tau) d\tau,$$

so the reactor implements a physically recurrent temporal kernel whose effective impulse response is determined by the loop delays and couplings.

10.9 Acoustic and cymatic reactor embodiments

In some embodiments, a non-optical or multi-physics reactor is used, including acoustic and cymatic variants. A speaker–microphone pair may drive and sense a membrane, plate, or fluid layer at a carrier frequency with a controlled envelope. The membrane patterns may be observed optically, producing visual features that are included as channels of $\mathbf{y}_t$. Acoustic features (for example envelope or band-energy features over windows) may also be included. Optical and acoustic controls may be coupled through a shared controller and shared latent, and hardness or verisimilitude meters may be trained on the joint multi-channel convolution bundle.

Acoustic control and convolution-bundle notation (non-limiting). In some embodiments, acoustic control at step t is expressed as

$$U_t^{\text{ac}} = (s_{\text{ac}}(t), a_{\text{drv}}(t)),$$

where $s_{\text{ac}}(t)$ denotes a carrier or waveform choice and $a_{\text{drv}}(t)$ denotes a drive envelope or amplitude schedule. Acoustic observations may be represented as an audio channel of the convolution bundle, while camera-observed membrane patterns may be represented as a visual channel. Meters and downstream heads may operate on either channel or on the joint law to exploit cross-domain constraints.

Infrasonic channels and cymatic pattern analysis (non-limiting). In some embodiments, acoustic channels include infrasonic components (below approximately 20 Hz) and mid- to high-frequency modes capable of exciting visible cymatic patterns in fluid, granular, or membrane media. Non-limiting uses include long-range acoustic sensing, detection of architectural or geophysical resonances, and acoustic-to-visual frequency analysis via optical scanning of cymatic patterns. Safety constraints limit infrasonic emission power and duty cycle to levels below those associated with adverse physiological effects.

Ultrasonic phased arrays and haptic fields (non-limiting). In some embodiments, the acoustic channel is ultrasonic and is driven by a phased array that steers and focuses pressure fields in air, water, or tissue. In Reality Transform embodiments, focused ultrasound produces haptic or vibrotactile sensations and the convolution bundle logs the delivered field schedule for audit; in Limager embodiments, the same array implements ultrasound imaging or elastography under logged steering and safety envelopes.

Parametric acoustic emission and self-demodulation in air (non-limiting). In some embodiments, two or more ultrasonic carriers at frequencies $f_1, f_2, \dots$ are emitted along a common axis and undergo nonlinear self-demodulation in air, generating audible difference-frequency components at $|f_i - f_j|$ whose beam directionality is set by the ultrasonic carrier wavelengths rather than by the audible wavelengths. In Reality Transform embodiments, the swept parametric audio beam delivers spatially localised sound to one or more listeners without enclosing loudspeakers, with the delivered field schedule (carrier frequencies, modulation envelope, beam direction, exposure duration) committed to the protocol digest and convolution bundle on the same footing as optical sweeps. In Truth Beam embodiments, the directional beam serves as an acoustic challenge directed at a specific scene region, with the return signal recorded by one or more microphones as an acoustic observation channel and folded into the convolution bundle. In Limager embodiments, the parametric beam supports directional acoustic imaging without the enclosure geometry required by audible-frequency arrays. Device-specific tolerances of the ultrasonic emitter array (element mismatch, phase calibration, near-field beam structure, carrier intermodulation profile) contribute reactor-

microstructure uniqueness in the acoustic channel symmetric to the optical-emitter case (Section 10.19). Safety envelopes limit carrier intensity, duty cycle, and listener exposure to levels below applicable thresholds for ultrasonic and audible exposure.

Phonon lasers and coherent acoustic emission (non-limiting). In some embodi-
 5 ments, the acoustic emitter is a phonon laser (sound-amplification-by-stimulated-emission-of-radiation device, SASER) in which coherent acoustic phonons are generated by stimulated emission in a semiconductor superlattice, optomechanical cavity, trapped-ion chain, or crystalline resonator, rather than by a driven transducer. The resulting emission is a coherent, narrowband acoustic beam whose frequency, linewidth, and threshold behaviour are deter-
 10 mined by the phononic cavity geometry, lattice parameters, and gain-medium properties— analogous to optical laser operation but in the acoustic domain. Device-specific fabrication tolerances in superlattice period, cavity dimensions, defect distributions, and mechanical Q factors provide reactor-microstructure uniqueness in the acoustic channel, symmetric to the optical emitter-as-reactor case (Section 10.19). Near-threshold operation exposes stochastic
 15 phonon emission and mode-competition noise suitable for stochastic sampling in PoliePuter mode. In Limager embodiments, the coherent acoustic beam provides high-resolution probing of material elastic properties, subsurface structure, or biological tissue via phonon scattering and absorption; in Truth Beam embodiments, the device-specific phonon emission finger- print provides verification of the acoustic source. In Yoked embodiments, acoustic backscat-
 20 ter or material response feeds back to the SASER’s gain or detuning, creating a closed-loop phononic–material coupled system. In some embodiments, a SASER is co-integrated with an optical reactor path, providing a joint opto-acoustic channel whose cross-domain correlations supply additional hardness constraints.

10.10 Magnetic and other multi-physics bistatic reactor embodi- 25 ments

In some embodiments, a reactor is implemented using a multi-physics channel rather than a purely optical channel. Non-limiting examples include magnetic, electrochemical, mechanical, photoacoustic, opto-acoustic, and other coupled sensing and excitation channels in which an emission input and a sensing output are linked through a physical medium with hysteresis,
 30 relaxation, or amplitude-dependent nonlinearity. These bistatic embodiments preserve the same high-level structure as optical reactors: a controller selects a protocol $u(t)$, an excitation is applied through a transducer, and a measurement is recorded as an observable stream whose summaries are committed and audited.

10.10.1 Mechanical and inertial microstructure embodiments (non-limiting)

35 In some embodiments, the reactor medium includes embedded inertial microstructures or MEMS-scale masses whose motion is driven by reactor excitation and by external pertur-

bations (for example vibration or acceleration). The resulting mechanical state evolves with slow relaxation, providing a mechanical memory coupled to an optical, magnetic, or acoustic path. One or more inertial sensors (accelerometers, gyros) may be logged as auxiliary channels and used by meters to distinguish external perturbations from internal reactor evolution.

10.10.2 Aqueous, hydrogel, and liquid-crystalline reactor media (non-limiting)

In some embodiments, the reactor medium comprises aqueous solutions, hydrogels, or liquid-crystalline phases whose optical, electrical, or mechanical properties vary with structure, temperature, solute concentration, or preparation history. Such media may exhibit ordering near interfaces, metastable structural states, sensitivity to electromagnetic or acoustic fields, and nonlinear optical responses arising from molecular alignment. Reactor behaviour in these media is characterised empirically. Liquid-crystal phases with well-documented ordering transitions are preferred for reproducible embodiments.

10.10.3 Biological and optogenetic reactor surfaces (non-limiting)

In some embodiments, the reactor medium includes biological surfaces or bio-hybrid layers that can be stimulated and recorded. Non-limiting examples include optogenetic stimulation with optical recording, fluorescence or reflectance readouts, and calcium-indicator measurements, where structured stimulation patterns cause measurable responses whose dynamics provide mixing, persistence, and channel constraints.

Biophoton detection and biological reactor coupling (non-limiting). In some embodiments, the system detects ultra-weak photon emission from biological tissues, arising from oxidative metabolic processes and electronically excited molecular species. Single-photon-sensitive detectors measure emission rates, spectral characteristics, and spatial patterns from living samples, and these readouts are appended to the convolution bundle as additional channels. In some embodiments, structured illumination (including beams carrying orbital angular momentum or other higher-order spatial modes) is directed through biological tissue, and mode mixing introduced by the tissue's refractive-index inhomogeneity is recorded as part of the convolution bundle. In such embodiments, the tissue acts as a scattering medium whose transfer function contributes device-specific and specimen-specific features to the observation, and the preserved or transformed spatial mode structure provides sensitivity to microstructural properties of the biological sample. OAM and higher-order spatial modes require sufficient spatial coherence to propagate recognisably; in thick or highly scattering tissue, mode structure is destroyed over path lengths exceeding a few transport mean free paths. These embodiments are therefore most applicable to thin tissue preparations, surface measurements, or waveguide-coupled samples where the optical path through scattering media is short.

Optically responsive biological reactor media (non-limiting). In some embodiments, the reactor medium comprises a biological preparation in which one or more cellular or subcellular components are both optically addressable and optically emissive, so that the control input $u(t)$ modifies a biological state and the resulting optical emission encodes the consequence of that perturbation as channels of $\mathbf{y}_t$. Non-limiting classes of optically responsive protein that may serve as reactor elements include: naturally occurring emitters such as luciferase variants and fluorescent proteins (GFP and its derivatives) that report gene expression state or protein localisation; genetically encoded state reporters such as GCaMP calcium indicators and genetically encoded voltage indicators (GEVIs) whose emission intensity or spectrum encodes electrophysiological activity; optogenetic actuators such as channelrhodopsin variants (ChR2, step-function opsins, red-shifted opsins) that transduce structured illumination into ionic currents; and subcellular-targeted constructs such as mitochondria-directed opsins engineered for import to the inner mitochondrial membrane using one or more mitochondrial targeting sequences, enabling structured illumination to modulate mitochondrial energetic state, calcium handling, and effective neuronal response gain. In neural tissue and neural organoid embodiments, the preparation may additionally provide endogenous optical signals including NADH and FAD autofluorescence reporting metabolic redox state, membrane-potential-sensitive dyes such as TMRM or JC-1, genetically encoded mitochondrial reporters, and ultra-weak biophoton emission associated with oxidative metabolic processes, all appended as channels of $\mathbf{y}_t$ and summarised in the protocol digest under declared meter envelopes. In some embodiments a two-stage control structure is employed: a first emission pattern sets the gain state of a declared target region (for example by modulating mitochondrial energetic state or by driving an optogenetic actuator to a declared operating point); a second emission pattern then probes the preparation's response under that gain state; and the detection channel records the consequence. The gain-setting and probe patterns, their wavelength bands, power envelopes, and timing are committed in the protocol digest before the run begins, supporting post-hoc audit of illumination conditions independently of the biological readout. In preferred embodiments, safety envelopes include irradiance limits, duty-cycle limits, and phototoxicity-monitoring channels appended to $\mathbf{y}_t$; these embodiments do not assert specific therapeutic or clinical outcomes.

Piezoelectric biological interfaces (non-limiting). In some embodiments, the reactor couples to piezoelectric biological materials (including bone, collagen, tendon, and engineered tissues) that generate electrical signals in response to mechanical stress and conversely deform in response to applied fields.

Chemical and olfactory reactor surfaces (non-limiting). In some embodiments, a proxy scene or reactor surface integrates volatile chemistry, fluid chemistry, or other non-optical fields over time and exposes an evolving state that the Reality Kernel interrogates optically and/or electrically. Non-limiting examples include gas-sensor plates, optoelectronic olfactory receptor arrays, microfluidic channels with dye indicators, and living tissue cultures.

The controller may emit tracer odorant pulses or other chemical stimuli under a logged schedule, and detectors record temporal response traces as additional channels of $\mathbf{y}_t$. In Truth Beam embodiments, the stimulus schedule and response statistics define a challenge–response hardness source; in Limager embodiments, the same channels support inference; and
 5 in Reality Transform embodiments, the controller steers masking or room-profile matching by emitting bounded stimuli under safety and policy constraints.

10.10.4 Bioelectric and brain-scan embodiments (non-limiting)

In some embodiments, the scene and/or a ponderable cell comprises a bioelectric or neuro-physiological field, such as electroencephalography (EEG), magnetoencephalography (MEG),
 10 electrocorticography (ECoG), electromyography (EMG), peripheral nerve recordings, or related modalities. The module may include a sensor cap, electrode array, field probes, or transducer surfaces whose effective spatial sensitivity is time-varying under control. In these embodiments, the control input selects, without limitation, montages or spatial filters, sampling schedules, carrier frequencies, drive amplitudes within safety envelopes, stimulation
 15 sites (where permitted), or structured sweep protocols across the array, and the resulting observations are appended to the convolution-bundle record as channels of $\mathbf{y}_t$.

Frequency-locked EEG–optical coupling (non-limiting). In some embodiments, bioelectric coupling is implemented as a frequency-locked optical channel in which recorded multi-channel neural signals are combined using swept weighting waveforms defining a time-
 20 varying montage or spatial filter. Intermediate combined signals modulate one or more optical carriers that traverse a reactor loop, so that scattering and interference progressively merge multiple channels into a reduced set of optical outputs. The resulting detector readout provides channels included in the convolution bundle and can be trained to track a desired low-dimensional functional of the neural signals.

Neural entrainment and sensory pacing (non-limiting). In some embodiments, the
 25 controller generates sensory stimuli (visual flicker, auditory pulses, tactile signals, or combinations) at frequencies selected to influence oscillatory activity via entrainment, optionally with closed-loop adaptation based on measured EEG or behavioural signals. In preferred embodiments, stimulation is constrained by safety policies (including photosensitive-seizure
 30 risk controls) and is logged with sufficient detail for later audit.

10.10.5 Safety envelope and calibration (non-limiting)

In some embodiments, multi-physics and biological embodiments operate under explicit safety envelopes and calibration policies. Protocols may be constrained by maximum energy, duty cycle, exposure time, thermal bounds, and interlock states, and meters record
 35 compliance with these bounds together with drift indicators and calibration status. Sealed

or physically confined reactor loops are used when appropriate to isolate the reactor medium from the environment.

10.11 Resonant and frequency-domain reactor embodiments

In some embodiments, the reactor is structured as a resonant cavity (optical, acoustic, electromagnetic, or mechanical) with characteristic modes and quality factor. The system may encode information via amplitude, frequency, phase, polarisation, or modulation of carriers. The convolution bundle may be analysed in the time domain and/or frequency domain (for example via Fourier or wavelet transforms), and frequency signatures may serve as features for Limager tasks, calibration, or verification. In Yoked operation, resonant reactors are particularly natural: the resonant modes provide the spectral structure that Arnold-tongue analysis maps, and changes in the scene that alter the cavity’s resonant response are directly encoded as shifts in the entrainment diagram.

In a non-limiting example, the reactor is a Fabry–Pérot optical cavity (two partially reflecting mirrors separated by a controlled gap) containing an absorptive or scattering medium (for example a dye-doped polymer film, nanoparticle suspension, or vapour cell). The emitter is a tuneable laser whose wavelength is swept across one or more cavity resonances; the detector records transmitted intensity as a function of wavelength. The resonant linewidth, free spectral range, and finesse depend on mirror reflectivities, cavity length, and the medium’s absorption and scattering properties—all of which carry device-specific fabrication signatures. In Truth Beam mode, the measured transmission spectrum is compared against a stored device profile to verify device identity. In Limager mode, changes in the cavity’s resonant response (for example shifts caused by a sample placed inside the cavity or by temperature variation) are used to infer scene or sample properties. Protocol digests record laser sweep parameters, mirror alignment indicators, and temperature. Meters record cavity stability, mode-hop indicators, and signal-to-noise ratios.

10.12 Gain-pumped cavity Reality Kernel embodiments

In some embodiments, the reactor is realised as a gain-pumped optical cavity in which state evolves through repeated optical passes prior to detection. A gain medium may be provided by a semiconductor optical amplifier, fibre amplifier, doped waveguide, Raman gain, parametric gain, or other amplifying structure. Feedback elements and output couplers define the cavity dynamics. Control inputs may include pump power, phase shifter settings, modulator settings, tunable filters, and optional weak injection of seed fields. The detector observes one or more outputs of the cavity and records them as channels of the the convolution bundle. Such embodiments may be operated in Truth Beam, Limager, or Reality Transform regimes. Optional quantum-sensitive variants and entangled or squeezed-light variants are described in the quantum coupling section (Section 25).

Threshold, bistability, and mode competition (non-limiting). In some embodiments, gain-pumped cavities exhibit threshold behaviour, bistability, and mode competition that produce rich, history-dependent signatures. In a non-limiting rate-equation lens for mode amplitudes $\{a_k(t)\}$,

$$\dot{a}_k(t) = (g_k(p(t)) - \ell_k) a_k(t) - \sum_j \beta_{kj} |a_j(t)|^2 a_k(t) + w_k(t),$$

- 5 where $p(t)$ is a pump control, ℓ_k are loss terms, β_{kj} model gain saturation and cross-saturation (mode competition), and $w_k(t)$ denotes noise. Meter envelopes record pump and thermal margins, stability indicators, and saturation proxies; protocol digests record pump schedules and tuning settings so that verification and reconstruction are conditioned on the intended cavity regime.

10 10.13 Multi-pass folded-cavity reactor embodiments (infinity-mirror architecture)

Iterative optical feedback loops using incoherent light and commodity components—spatial light modulators of the type used in miniature LCD panels, LED-class light sources, and compact cameras—have been reported to generate nonlinear input–output transformations
15 through cumulative multi-pass interaction, without requiring exotic nonlinear materials or high-power lasers. However, reported systems do not provide convolution-bundle logging, multi-regime objective selection, or committed evidence coupling. In some Reality Kernel embodiments, the principle of building effective nonlinearity from individually linear or weakly nonlinear components is adapted as a reactor architecture with the addition of these disciplines.
20

In some embodiments, a reactor is implemented as a compact folded-cavity or “infinity mirror” arrangement in which light is repeatedly routed through one or more miniature optical elements (for example LCD-type spatial light modulators, diffusers, thin-film layers, and/or polarisation elements) arranged between partially reflecting surfaces. Each pass
25 through the element stack contributes an incremental nonlinear transformation; the cumulative effect of N passes produces an effective nonlinear mapping whose complexity grows with N . A compact camera or photodetector captures the output after a controlled number of passes or at steady state.

In some embodiments, the spatial light modulator in the multi-pass loop is driven by
30 the Reality Kernel controller, so that the per-pass transformation is a trainable or protocol-controlled function. The SLM pattern may be updated between passes (if the SLM refresh rate permits) or held fixed for a burst of passes and then updated, providing two timescales of control analogous to the fast-loop/slow-loop architecture of the bench-top 1D Reality Kernel (Section 7.2).

35 In some embodiments, the multi-pass reactor is coupled to the scene loop so that scene return signals modulate the SLM pattern, the source intensity, or an auxiliary element in

the cavity, creating a scene-coupled feedback reactor from commodity components. In Yoked operation, the scene’s dynamical response feeds back into the cavity parameters, and the multi-pass nonlinearity amplifies coupling signatures that would be invisible in a single-pass system.

5 In some embodiments, reactor-microstructure uniqueness arises from manufacturing tolerances in the SLM pixel structure, partial-reflector coatings, cavity alignment, and any diffuser or scattering elements in the multi-pass path. These tolerances accumulate multiplicatively over N passes, so that small per-element variations produce large and device-specific cumulative effects on the output pattern.

10 In some embodiments, the multi-pass cavity is sealed and miniaturised (for example integrated into a module comparable in size to a small camera), making it suitable for portable or embedded Reality Kernel deployments where a CRT reactor is impractical.

10.14 Phase-change material reactor embodiments

In some embodiments, reactor layers include one or more phase-change materials (PCMs) whose optical properties (refractive index, absorption, or both) switch between two or more stable states under optical or electrical stimulus. Non-limiting PCM materials include antimony selenide (Sb_2Se_3), germanium-antimony-telluride (GST, for example $\text{Ge}_2\text{Sb}_2\text{Te}_5$), antimony sulphide (Sb_2S_3), and vanadium dioxide (VO_2). These materials provide *nonvolatile optical memory at zero static power*: once switched, the optical state is retained without energy input over the device’s declared retention window (subject to material- and device-specific retention, endurance, and resistance-drift characteristics), in contrast to phosphor persistence which decays on millisecond-to-second timescales. In some embodiments, the PCM layer is addressed by a focused laser beam or structured illumination pattern from the Reality Kernel controller, so that spatial patterns of amorphous and crystalline regions are written, read, and overwritten as part of the reactor’s state evolution.

25 In some embodiments, PCM layers are integrated into photonic integrated circuits (for example as cladding overlays on silicon waveguides or microring resonators) to provide programmable, nonvolatile weight elements in an on-chip reactor. In some embodiments, multi-level states (for example six-bit phase levels demonstrated with nitrogen-doped Sb_2Se_3) increase the state-space richness of the reactor medium.

In some embodiments, PCM reactor layers are combined with other reactor elements (for example scattering plates, phosphor layers, or nonlinear thin films) in a layered reactor stack, where the PCM layer provides nonvolatile memory and the other layers provide mixing, persistence, or nonlinearity. In some embodiments, reactor-microstructure uniqueness arises from stochastic crystallisation nucleation, grain boundary formation, and defect distributions in the PCM layer, which vary from device to device even under identical fabrication processes.

10.15 Spiking photonic reactor embodiments

In some embodiments, one or more reactor elements are implemented as spiking photonic neurons—optical devices that produce discrete spike-like output pulses in response to input signals exceeding a threshold, with excitability, refractory periods, and history-dependent
 5 dynamics analogous to biological neurons.

In some embodiments, spiking photonic neurons are implemented using vertical-cavity surface-emitting lasers (VCSELs) biased near threshold, where polarisation switching or injection-locking dynamics produce GHz-rate spike trains whose timing and rate encode input intensity and history. VCSELs are manufactured at wafer scale for data-communication
 10 and smartphone applications, providing a sub-dollar, mass-producible nonlinear reactor element. In some embodiments, reactor-microstructure uniqueness arises from device-specific threshold characteristics, polarisation switching points, and thermal response curves that vary with fabrication tolerances.

In some embodiments, spiking photonic neurons are implemented using microring resonators on silicon-on-insulator platforms, where carrier-induced nonlinear dynamics produce self-pulsation and spike generation at rates of hundreds of MHz. In some embodiments, spiking
 15 neurons are implemented using resonant tunnelling diode (RTD) opto-electronic circuits, providing THz-bandwidth nonlinear excitability.

In some embodiments, arrays of spiking photonic neurons are coupled through optical
 20 waveguides, free-space paths, or fibre links to form spiking neural network reactors in which the collective spiking dynamics provide mixing, temporal memory (through refractory periods and synaptic integration), and nonlinear computation. In some embodiments, the spiking dynamics of the reactor are coupled to the scene loop so that scene return signals modulate the spike rates or timing, providing a natural interface for Yoked operation in which the
 25 coupled scene-reactor spiking dynamics form a joint attractor.

10.16 Perovskite and adaptive-luminescence reactor embodiments

In some embodiments, reactor media include perovskite-based or other adaptive-luminescence materials whose photoluminescent response evolves as a function of excitation history, providing neuromorphic dynamics (for example facilitation, depression, potentiation, and
 30 spike-like threshold behaviour) directly in the optical domain.

In some embodiments, mixed-halide perovskite nanocrystals are embedded in a porous host matrix (for example macroporous rare-earth oxide such as $\text{Y}_2\text{O}_3:\text{Eu}^{3+}$), where interface-mediated halide migration under optical excitation creates progressively evolving energy barriers that modulate photoluminescence intensity and spectrum. Such materials simultaneously
 35 provide: (i) nonlinear optical response (threshold and saturation behaviour), (ii) programmable memory (history-dependent PL evolution), (iii) spectral mixing (host and guest emission bands that shift with state), and (iv) reactor-microstructure uniqueness (stochastic nanocrystal distribution and interface morphology). In some embodiments, these properties

are obtained from a single solution-processable layer, providing a low-cost reactor medium suitable for mass production.

In some embodiments, tin-based perovskite thin films (for example PEA_2SnI_4) provide giant nonlinear refractive indices (on the order of cm^2/GW) on flexible substrates, enabling compact nonlinear reactor layers via solution processing.

In some embodiments, persistent luminescent materials (for example $\text{SrAl}_2\text{O}_4:\text{Eu}^{2+}, \text{Dy}^{3+}$ or $\text{Li}^+/\text{Dy}^{3+}$ co-doped $\text{Sr}_2\text{SiO}_4:\text{Eu}^{2+}$) provide optical memory with controllable decay times from seconds to hours, extending the temporal integration window of phosphor-based reactors beyond that achievable with conventional CRT phosphors.

In some embodiments, mechanoluminescent materials (for example $\text{ZnS}:\text{Mn}^{2+}$ nanocrystals or doped strontium silicate) emit light in response to mechanical stress, providing a transduction pathway from acoustic, vibrational, or tactile scene properties into the optical domain. In some embodiments, mechanoluminescent reactor layers are used in Yoked operation to couple to structural vibrations, acoustic fields, or haptic interactions with the scene.

10.17 Photonic integrated circuit reactor embodiments

In some embodiments, the reactor is implemented as a photonic integrated circuit (PIC) comprising programmable interferometric elements (for example arrays of tuneable Mach-Zehnder interferometers in a mesh topology, microring resonator arrays, or multimode interference couplers) fabricated on a semiconductor substrate (for example silicon on insulator, silicon nitride, or indium phosphide).

In some embodiments, the PIC reactor provides programmable linear transformations (matrix-vector multiplication) combined with intrinsic nonlinearity from detector saturation, carrier effects in resonators, or integrated nonlinear elements. In some embodiments, the PIC reactor provides tuneable feedback paths through on-chip waveguide loops, creating recurrent dynamics suitable for reservoir computing and temporal memory.

In some embodiments, reactor-microstructure uniqueness arises from fabrication-induced random phase variations in the interferometric mesh, resonance offsets in microring arrays, and waveguide dimension tolerances that produce device-specific transfer matrices even for nominally identical designs. In some embodiments, the PIC reactor is coupled to the scene loop through fibre or free-space optics, with the controller programming the PIC configuration as part of the scan protocol and logging the PIC state in the protocol digest.

10.18 Detector-as-reactor: stochastic sampling in the image sensor (non-limiting)

In some embodiments, the detector itself — for example a CMOS image sensor — serves as a stochastic sampling element within the optical path. Every pixel in a CMOS sensor exhibits multiple noise sources whose statistics are device-specific and physically grounded:

- **Shot noise:** Poisson-distributed photon arrival statistics (quantum in origin).
- **Read noise:** thermal and $1/f$ noise in the readout electronics, with per-pixel variance determined by transistor geometry and fabrication tolerances.
- **Dark current:** thermally generated charge carriers whose rate varies per pixel due to crystal defects, implant variations, and localised contamination.
- **Fixed-pattern noise (FPN):** systematic per-pixel offset and gain variations from manufacturing, providing a spatial fingerprint that is stable over time and unique to each sensor.

In some embodiments, the system deliberately operates the sensor in a noise-dominated regime — for example at very low illumination, very short exposure, high gain (ISO), or elevated temperature — so that the readout at each pixel is primarily a sample from the pixel’s noise distribution rather than a deterministic record of incident light. In this regime, the emitter (projector) programs the *energy landscape* of the sampling process: the structured illumination pattern sets the mean photon flux at each pixel, biasing the noise distribution without overwhelming it. The sensor’s physics performs the sampling, and the convolution bundle records both the illumination control parameters and the noisy readout, preserving the standard emission-observation format. This energy-landscape interpretation is strongest when the forward model from projector pixel values through the scene or reactor to sensor photon flux is stable (for example in a sealed reactor or a static calibration scene), so that the dominant source of per-frame variation is sensor noise rather than unmodelled scene dynamics. In embodiments where the scene is dynamic, the interpretation holds per-frame under the assumption that scene changes between consecutive frames are slow relative to the sensor’s integration time.

Energy-based model interpretation (non-limiting). In some embodiments, the per-pixel readout in the noise-dominated regime is modelled as a sample from a conditional distribution

$$y_i \sim p(y_i \mid \lambda_i(\theta, u), \sigma_i^{\text{read}}, d_i, \text{FPN}_i),$$

where λ_i is the expected photon flux at pixel i (set by the projector pattern u and the scene/reactor configuration θ), σ_i^{read} is the pixel-specific read noise, d_i is the pixel-specific dark current rate, and FPN_i is the pixel-specific offset. The projector pattern provides the programmable bias (analogous to the energy-function parameters in an energy-based model), and the pixel noise statistics provide the stochastic dynamics that produce samples. Changing the projector pattern changes the distribution from which the sensor samples, enabling programmable stochastic computation within the existing RK optical path.

reactor-microstructure uniqueness from sensor noise (non-limiting). In some embodiments, the combination of fixed-pattern noise, dark current non-uniformity, and per-pixel read noise statistics constitutes a physically unclonable fingerprint of the specific sensor die. This fingerprint is measurable from the convolution bundle under controlled illumina-

tion (for example a series of flat-field exposures at different levels), and provides the basis for Truth Beam verification: a verifier can distinguish samples produced by a specific sensor from samples produced by a different sensor or by software simulation, because the spatial correlation structure, per-pixel higher moments, and dark-current hot-pixel patterns are device-specific. In some embodiments, because dark current and hot-pixel patterns vary with temperature, exposure time, and sensor aging, enrollment protocols include calibration at multiple operating points and verification uses fuzzy extraction or helper-data algorithms to accommodate drift while preserving uniqueness.

Regime mapping (non-limiting). The three regimes and Yoked modifier apply to the detector-as-reactor mode. Reality Transform in PoliePuter mode corresponds to generating structured random patterns whose statistics are shaped by the projector (the Create primitive, Section 8.7.2): each frame is a device-specific sample. Limager uses the noise-dominated readout for inference: the projector pattern is chosen to maximise information gain about a scene or data variable encoded in the illumination, and the sensor’s stochastic response is the measurement. Truth Beam verifies that the noisy readout was produced by a specific physical sensor under a specific illumination protocol. Yoked operation couples the sensor’s noise-driven output back to the projector, creating a closed-loop stochastic system whose joint dynamics sample from a distribution that depends on both the sensor’s noise profile and the scene’s response to structured illumination.

Denoising chain via temporal iteration (non-limiting). In some embodiments, the projector–sensor loop is iterated: the readout from frame t conditions the projector pattern for frame $t + 1$, implementing a temporal denoising chain within a single projector–camera pair. Each iteration is one sampling step whose conditional distribution depends on the previous sample (through the projector update) and the sensor’s noise (through the readout). This maps to the multi-reactor pipeline (Section 18.1) unfolded in time rather than in space, with each temporal frame as one reactor stage. The convolution bundle logs every frame, providing a complete record of the denoising trajectory.

Scope limitation (non-limiting). The embodiments of this section support declared sensing, attestation, and hardness tasks benchmarked under the filing’s task procedures. They do not support, and shall not be construed to support, capacity or scaling claims of the kinds bounded by Section 15.5, and do not imply any increase in capacity quantities.

10.19 Emitter-as-reactor: stochastic sampling in the source (non-limiting)

In some embodiments, the emitter itself — for example a semiconductor laser, VCSEL, superluminescent diode (SLD), LED, or micro-LED — serves as a stochastic sampling element, symmetric to the detector-as-reactor mode (Section 10.18). Every emitter exhibits

device-specific noise sources whose statistics depend on fabrication:

- **Spontaneous emission noise:** quantum-mechanical spontaneous decay into the lasing or emission mode, setting a fundamental noise floor that is source-specific through cavity geometry, gain-medium composition, and facet reflectivities.
- **Mode-partition noise:** in multi-mode sources, stochastic competition between longitudinal or transverse modes produces intensity fluctuations whose partition statistics depend on the cavity’s mode spectrum and gain curve.
- **Relative intensity noise (RIN):** intensity fluctuations above the shot-noise floor, arising from carrier-photon dynamics in semiconductor sources, with magnitude and spectral shape determined by the specific die’s relaxation oscillation frequency, damping factor, and parasitic elements.
- **Turn-on jitter and threshold fluctuations:** when pulsed or modulated near threshold, the turn-on delay varies stochastically from pulse to pulse, with statistics set by the spontaneous emission factor and carrier lifetime of the specific device.

In some embodiments, the system deliberately operates the emitter in a noise-dominated regime — for example biased just below or at threshold, pulsed with short current pulses near the lasing onset, or operated in a chaotic regime via delayed optical feedback — so that the emitted optical field at each time step is primarily a sample from the source’s noise distribution rather than a deterministic output. The controller programs the bias point, modulation waveform, or feedback strength (the “energy landscape”), and the emitter’s physics performs the sampling. The detector reads out the resulting optical field, and the convolution bundle records both the control parameters and the observed emission, preserving the standard emission-observation format.

reactor-microstructure uniqueness from emitter noise (non-limiting). In some embodiments, the combination of RIN spectral shape, mode-partition statistics, threshold characteristics, and turn-on jitter constitutes a physically unclonable fingerprint of the specific emitter die. This fingerprint is measurable from the convolution bundle under controlled modulation sequences (for example a series of threshold-crossing pulses at different bias levels), and provides the basis for Truth Beam verification: a verifier can distinguish the noise signature of a specific emitter from that of a different emitter or a software simulation. In some embodiments, because emitter noise characteristics (particularly RIN magnitude, threshold current, and mode-partition ratios) vary with temperature, bias current noise from drive electronics, optical feedback from external reflections, and device aging, enrollment protocols include characterisation at multiple operating points and controlled bias conditions, and verification uses robust feature extraction (for example phase-space reconstructions or spectral entropy measures) together with fuzzy extraction or helper-data algorithms to accommodate drift while preserving uniqueness. In some embodiments, the measurement path is specified as the integrated emitter-plus-readout chain (emitter die, coupling optics, and detector bandwidth) to prevent trivial counterarguments that the fingerprint reflects the measurement setup rather than the emitter.

Regime mapping and symmetry with detector-as-reactor (non-limiting). The regime mapping is symmetric to the detector-as-reactor case. In PoliePuter / Create mode, each emission event is a device-specific noise sample shaped by the programmed bias. In Limager mode, the emitter’s noise-driven output illuminates a scene, and the detector reads
 5 back the scene’s modulation of that stochastic illumination, providing information about both the scene and the source. In Truth Beam mode, the emitter’s noise fingerprint is verified. In Yoked operation, the detector’s readout feeds back to the emitter’s bias point, creating a closed-loop stochastic system in which both emitter noise and detector noise contribute device-specific signatures.

Combined emitter-and-detector stochastic mode (non-limiting). In some embodiments, both the emitter and the detector operate simultaneously in their noise-dominated regimes, so that the convolution bundle records the convolution of two independent device-specific noise processes — one at the source and one at the sensor. The combined noise signature is richer than either alone and provides reactor-microstructure diversity from two
 15 independent physical mechanisms within the same optical path. In some embodiments, the two noise contributions are partially separable by modulating the emitter at known frequencies and examining the detector readout at those frequencies versus at other frequencies, enabling independent characterisation of source and detector fingerprints from a single convolution-bundle record.

20 **10.20 All-optical stochastic sampling reactor embodiments (non-limiting)**

In some embodiments, the reactor is implemented as an array of coupled optical elements — for example degenerate optical parametric oscillators (DOPOs), bistable optical cavities with Kerr nonlinearity, coupled semiconductor laser arrays biased near threshold, or arrays
 25 of microring resonators with optical feedback — whose quantum vacuum fluctuations and spontaneous emission noise drive stochastic sampling from a programmable distribution.

Optical parametric oscillator arrays (non-limiting). In some embodiments, an array of DOPOs is implemented in a fibre ring, a free-space cavity, or a photonic integrated circuit. Each DOPO is pumped above threshold so that it selects one of two degenerate phase states
 30 (0 or π relative to the pump), with quantum vacuum fluctuations determining which phase is selected on each pump pulse. Programmable optical or electronic coupling between DOPOs (for example through delay lines, beam splitters, spatial light modulators, or measurement-feedback circuits) implements an interaction graph, so that the collective phase configuration of the array samples from an energy-based model whose energy function is defined by the
 35 coupling weights and local biases (pump detunings). In some embodiments, the sampling distribution corresponds to an Ising model $E(\sigma) = -\sum_{ij} J_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i$ where $\sigma_i \in \{+1, -1\}$ maps to the two DOPO phase states, J_{ij} are the programmed couplings, and h_i

are the programmed biases. In some embodiments, continuous-variable extensions beyond binary phase states are implemented by operating DOPOs in a regime where amplitude fluctuations are significant, or by using non-degenerate parametric oscillators with richer phase-space structure.

- 5 **Bistable optical cavities (non-limiting).** In some embodiments, arrays of Kerr-nonlinear optical cavities exhibit optical bistability: two stable output states for a given input, separated by an unstable branch. Quantum noise (vacuum fluctuations and spontaneous emission) drives stochastic switching between stable states at rates determined by the barrier height (set by cavity detuning, nonlinear coefficient, and input power). Programmable coupling between cavities through waveguides, evanescent fields, or feedback
10 paths implements an interaction graph as in the DOPO case, and the collective switching dynamics sample from the corresponding energy-based model.

- Native quantum noise and security (non-limiting).** In some embodiments, the all-optical stochastic reactor operates in a regime where quantum vacuum fluctuations are a
15 significant or dominant contribution to the noise that drives sampling. In idealised single-mode OPO/DOPO theory, vacuum fluctuations seed the phase selection; in practical implementations, the effective noise budget may also include contributions from pump amplitude and phase noise, technical noise in measurement-feedback electronics (for CIM-style architectures), intracavity loss fluctuations, and thermal phonon coupling to the nonlinear medium.
20 The degree to which quantum noise dominates depends on the operating regime (proximity to threshold, cavity finesse, pump stability, and temperature) and is a design parameter rather than an automatic guarantee. In embodiments where quantum noise is the dominant contribution, the sampling has consequences for both security and sampling quality. For security: the noise statistics depend on cavity geometry, mirror reflectivities, nonlinear
25 coefficients, and other fabrication-dependent parameters, providing reactor-microstructure uniqueness. In the quantum-noise regime, an adversary attempting to characterise these parameters faces a measurement-budget constraint: fully characterising the cavity response at quantum precision requires probe resources that scale unfavourably with the number of modes and the desired fidelity. The security advantage is operational and measurement-
30 budget-limited, strongest when verification protocols exploit quantum correlations (for example photon-number statistics or quadrature correlations) that require many samples to estimate precisely. For sampling: quantum tunnelling through the barrier between bistable states can occur on timescales inaccessible to classical thermal activation, potentially improving mixing times for multi-modal distributions.

- 35 **Convolution bundle and regime mapping (non-limiting).** In some embodiments, the convolution bundle records the phase or intensity state of each optical element at each pump cycle (or measurement window), together with the programmed coupling and bias parameters, pump power, and timing information. The regime mapping follows the detector-

as-reactor pattern (Section 10.18): Reality Transform in PoliePuter mode corresponds to generating samples from a target distribution (the Create primitive, Section 8.7.2); Limager in PoliePuter mode corresponds to inference from data encoded in the coupling parameters; and Truth Beam in PoliePuter mode corresponds to verifying that samples were produced
 5 by a specific physical device whose quantum noise statistics carry device-specific signatures.

Denoising chains and multi-stage architectures (non-limiting). In some embodiments, multiple optical sampling stages are chained so that the output of one stage conditions the input (coupling parameters or bias fields) of the next, implementing a denoising or iterative refinement architecture. This maps to the multi-reactor chain topology (Section 18.1),
 10 with each optical stage as one reactor in the pipeline. In some embodiments, stages operate at different timescales (for example a fast parametric stage followed by a slower cavity-relaxation stage), providing a temporal hierarchy analogous to the fast-loop / slow-loop architecture of the anchor embodiment.

Hybrid optical-cavity and detector-reactor architectures (non-limiting). In some
 15 embodiments, optical parametric oscillators or bistable cavities provide fast, quantum-noise-driven sampling for a subset of variables, while the detector-as-reactor mode (Section 10.18) provides spatially resolved, pixel-level stochastic readout with device-specific noise fingerprints. The optical cavities generate samples at GHz rates; the sensor integrates and reads out at frame rates, providing a natural temporal hierarchy analogous to the fast-loop / slow-
 20 loop architecture of the anchor embodiment. The convolution bundle logs both the optical cavity state trajectories (if accessible through homodyne or photon-counting measurements) and the sensor readout, and the resulting noise statistics combine quantum vacuum fluctuations from the cavities with shot noise and sensor-specific fixed-pattern noise from the detector, providing reactor-microstructure diversity from multiple independent physical sources
 25 within the optical path.

10.21 Reconfigurable metasurface reactor and optic embodiments

In some embodiments, one or more optical elements in the Reality Kernel module are implemented as metasurfaces—planar arrays of sub-wavelength nanostructures fabricated on a substrate by lithographic processes compatible with semiconductor manufacturing. Metasur-
 30 faces may serve as reactor layers (providing scattering, phase shaping, and polarisation transformation with device-specific nanostructure randomness), as emitter optics (beam shaping, steering, and structured illumination generation), or as detector optics (polarimetric filtering, spectral sorting, and computational sensing).

In some embodiments, metasurface elements are dynamically reconfigurable through in-
 35 tegration with phase-change materials, liquid crystals, electrochemical actuation, or MEMS, providing tuneable reactor behaviour under controller command. In some embodiments, metasurface elements are static but device-specific, with reactor-microstructure uniqueness

arising from sub-wavelength fabrication tolerances in nanostructure geometry, placement, and material composition. In a non-limiting example, a silicon-on-insulator metasurface fabricated by electron-beam or deep-UV lithography provides a spatially varying phase and amplitude transfer function at a design wavelength; the fabrication-dependent deviations
 5 from the nominal design (pillar width, height, sidewall angle, and placement jitter) constitute a device-specific fingerprint readable in the far-field diffraction pattern. In Truth Beam mode, the objective is to verify that a measured far-field response matches a stored device profile under a sequence of illumination angles or wavelengths. In Limager mode, the metasurface serves as a computational optic whose known transfer function enables computational
 10 reconstruction of scene properties from compressive measurements. In Reality Transform mode, a reconfigurable metasurface reshapes emitted wavefronts under closed-loop feedback to achieve target scene illumination.

10.22 Chromic and optically writable reactor layer embodiments

In some embodiments, reactor layers include one or more chromic materials whose optical
 15 properties change in response to light (photochromic), electrical stimulus (electrochromic), temperature (thermochromic), or mechanical stress (mechanochromic). Non-limiting examples include spiropyran and diarylethene photochromic films, conductive-polymer electrochromic films, and VO₂ thermochromic layers.

In some embodiments, photochromic reactor layers are directly written by the Reality
 20 Kernel's emission subsystem (for example by UV or visible structured illumination), creating spatial patterns of optical state that persist for controllable durations and that modulate subsequent light passing through the layer. In some embodiments, this provides a projector-writable optical memory that is read by the same or a different wavelength band, with the written pattern decaying or being erased under controlled illumination.

25 In some embodiments, electrochromic reactor layers are electrically addressed by the controller, providing fast (<1 s) and high-contrast (>50% transmittance change) optical switching with stability over thousands of cycles. In some embodiments, electrochromic layers are combined with photovoltaic elements to create self-powered tuneable reactor layers.

10.23 Emitter embodiments beyond lasers and conventional projectors (non-limiting)

30 In some embodiments, the emission subsystem includes one or more of the following emitter types, each providing capabilities distinct from conventional laser diodes, broadband sources, and digital projectors:

Micro-LED arrays. In some embodiments, the emitter is an individually addressable
 35 micro-LED array with pixel pitches of 10 μm or less, providing peak brightness exceeding 10⁶ nits under declared duty-cycle and thermal envelopes, device-level emitter modulation

in the sub-nanosecond to nanosecond range under appropriate drive electronics, and direct electrical control of each emitter element. Micro-LED arrays enable structured illumination patterns to be generated without a separate spatial light modulator. In some embodiments, fast device-level emitter modulation is combined with photodiode-class or transimpedance-
 5 amplifier-class detector loops to support local high-bandwidth feedback, while system-level closed-loop bandwidth, including detector readout, controller update, and committed-bundle logging, operates at a slower rate declared in the protocol digest. In some embodiments, micro-LED arrays include multiple wavelength bands (for example RGB or UV/ visible) on the same substrate, enabling spectral multiplexing of emission patterns. In some em-
 10 bodiments, the plurality of individually addressable micro-LED emitters is partitioned into kernel taps, with per-emitter intensities and switching times defining trainable or adaptively-controlled tap weights distributed across the array, and with per-emitter wavelength assignments fixed by material composition or binning so that the array provides a wavelength-partitioned set of taps. The kernel-tap partition operates either alone or in parallel with
 15 scanning, DMD, SLM, or other emitter modalities, and is configured to interoperate with the channel-separation and side-channel mitigation discipline of the present disclosure.

VCSEL arrays. In some embodiments, the emitter is a two-dimensional array of vertical-cavity surface-emitting lasers (VCSELs), providing coherent, individually addressable, polarisation-controlled emission from a monolithic semiconductor chip at wafer-scale manu-
 20 facturing cost. VCSEL arrays may simultaneously serve as emitters and as spiking reactor elements (see spiking photonic reactor embodiments above), combining emission and non-linear dynamics in a single component.

Microwave-driven continuously tunable mode-locked semiconductor lasers. In some embodiments, the emitter is a monolithic mode-locked semiconductor laser in which
 25 a microwave driving signal applied along the laser cavity induces a spatiotemporal gain modulation, generating intracavity mode-locked pulses with a continuously tunable group velocity, so that the output provides a coherent pulse train with a continuously tunable repetition rate and, equivalently, a frequency comb with a continuously tunable mode spacing, under electrical tuning control and without mechanical cavity adjustment. In some embod-
 30 iments, the laser is a terahertz quantum cascade laser (THz QCL) and the demonstrated tuning range of the repetition rate is approximately 4–16 GHz; in other embodiments, the same spatiotemporal-gain-modulation mechanism may in principle be implemented in other suitable semiconductor laser classes (for example mid-infrared quantum cascade lasers or interband semiconductor lasers with suitable fast-gain components) through modulation-
 35 optimised geometries. In some embodiments, the microwave drive waveform is treated as a component of the emitter control input $u(t)$ and is committed to the protocol digest together with the resulting repetition-rate or comb-mode-spacing schedule, the laser bias and temperature set points, and the relevant device-geometry identifiers. In some embodiments, meters record drive-signal amplitude and phase stability, comb-line linewidth, repetition-

rate lock status, and power stability across a declared tuning range. In some embodiments, continuously tunable comb emission from such an emitter is combined with the wavelength-multiplexing and dispersive primitives of Section 19.7 to support, when paired with a second comb or suitable local-oscillator/reference path, dual-comb or multi-heterodyne spectroscopic interrogation of the reactor medium or of the scene as a declared observation channel. Supporting background includes Senica et al., “Continuously tunable coherent pulse generation in a semiconductor laser,” *Nature* (2026).

Digital micromirror devices (DMDs). In some embodiments, the emitter path includes a DMD providing binary amplitude modulation at pattern rates exceeding 30 kHz, optionally in UV wavelength bands (363–420 nm) suitable for driving fluorescent or photochromic reactor media. In some embodiments, DMD-based structured illumination is combined with camera-based detection in a single compact module.

Spatial light modulators (SLMs) of various types. In some embodiments, the emission or reactor path includes one or more spatial light modulators including, without limitation: liquid-crystal on silicon (LCOS) phase-only modulators, MEMS piston-motion modulators, electro-optic modulators based on lithium niobate or organic electro-optic materials, acousto-optic modulators, and magneto-optic modulators with sub-micron pixel pitch. In some embodiments, the SLM type is selected based on the required modulation speed, pixel count, wavelength range, and whether amplitude, phase, or polarisation modulation is needed.

Light-field, integral-imaging, and holographic-stereogram projectors. In some embodiments, the emission subsystem is a light-field or integral-imaging projector comprising a high-density emitter (for example a DMD, micro-LED array, LCOS panel, or phase-only spatial light modulator) with a microlens array bonded to or positioned immediately in front of the emitter plane, such that each microlens is fed by multiple emitter pixels and the emitted field is controllable in both spatial (x, y) and angular (u, v) coordinates. A single emission configuration therefore specifies a four-dimensional light field $E(x, y, u, v)$, subject to the emitter’s spatial and angular resolution limits and the MLA geometry. In some embodiments, the projector is realised as a stacked or tensor light-field display in which two or more layered spatial modulators jointly encode the emitted angular distribution by multiplicative composition. In some embodiments, the projector is realised as a holographic-stereogram or computer-generated-hologram display in which a phase spatial light modulator reconstructs an approximation of the complex emitted field at the display plane. In some embodiments, the emitted light-field configuration is recorded in the protocol digest together with the MLA pitch, effective angular resolution, and calibration state, and meters record angular uniformity, cross-talk between sub-apertures, and emitter-to-MLA registration. In some embodiments, a light-field projector is paired with a light-field (plenoptic) detector of matched geometry (Section 10.24) to realise a four-dimensional input–output architecture

(Section 10.2.5), in which the reactor or scene sits between two controllable four-dimensional fields and the convolution bundle records both the emitted and received angular distributions at each time step.

Dual-wavelength excitation and readout (non-limiting). In some embodiments, the
 5 emitter operates at a first wavelength band selected to excite fluorescence, phosphorescence, or other wavelength-shifting processes in the reactor medium, and the detector observes the reactor through a bandpass, long-pass, or notch filter that passes the wavelength-shifted emission while rejecting the excitation wavelength. In some embodiments, the excitation band is in the near-ultraviolet (for example approximately 365–420 nm) and the readout
 10 band is in the visible (for example approximately 500–700 nm), or the excitation is in the visible and the readout is in the near-infrared. In some embodiments, the spectral separation between excitation and readout enables the system to observe reactor dynamics (for example phosphorescence decay, fluorescence intensity, spectral shifts, or bleaching kinetics) that would be masked by specularly reflected or scattered excitation light in a single-wavelength
 15 configuration.

In some embodiments, a dual-wavelength embodiment further includes a reference channel in which a second detector or a second spectral band of the same detector observes the excitation wavelength reflected or scattered by the reactor, providing a simultaneous measure of the reactor’s elastic scattering response alongside the inelastic (wavelength-shifted)
 20 response. In some embodiments, the ratio or difference between elastic and inelastic channels is used as a meter input for calibration, drift monitoring, or reactor-state estimation.

In some embodiments, the protocol digest records the excitation wavelength, the emission filter specification, the detector spectral response, and any reference-channel parameters. In some embodiments, dual-wavelength operation is compatible with all three operating
 25 regimes: in verification mode, the wavelength-shifted response provides an additional layer of physical specificity that increases empirical hardness against emulators operating at a single wavelength; in sensing mode, fluorescence or phosphorescence dynamics carry information about reactor state, temperature, chemical environment, or medium composition; and in transformation mode, the excitation beam may be used to write persistent changes
 30 (for example photobleaching, photochromic switching, or phosphorescence loading) into the reactor medium while the readout beam observes the result.

10.24 Detector embodiments beyond conventional cameras (non-limiting)

In some embodiments, the detection subsystem includes one or more of the following detector types, each providing invariance profiles or temporal characteristics distinct from
 35 conventional frame-based cameras:

Event cameras (dynamic vision sensors). In some embodiments, the detector is an event camera in which each pixel independently and asynchronously reports brightness changes that exceed a threshold, producing a stream of timestamped events rather than synchronous frames. Event cameras provide microsecond temporal resolution and dynamic range exceeding 120 dB, and inherently detect changes in the scene’s or reactor’s optical state—precisely the signal of interest in a feedback loop. In some embodiments, event cameras are used as the primary detector in Yoked operation, where the asynchronous event stream directly encodes the temporal structure of bidirectional coupling dynamics. In some embodiments, event cameras are combined with conventional frame-based cameras to provide both high-speed change detection and spatial context.

Single-photon avalanche diode (SPAD) arrays. In some embodiments, the detector is an array of SPADs providing photon-counting sensitivity, picosecond timing resolution, and dynamic range exceeding 150 dB through weighted photon-counting or time-correlated methods. In some embodiments, SPAD arrays with temporal gating (gate widths of nanoseconds or less) are used to separate prompt scatter from delayed luminescence in reactors containing fluorescent or phosphorescent media, or to perform time-of-flight measurements within the reactor path. SPAD arrays extend the quantum-sensitive coupling modes described elsewhere in this specification.

Hyperspectral and multispectral cameras. In some embodiments, the detector is a snapshot hyperspectral camera providing tens to hundreds of spectral bands per pixel at video rates. In some embodiments, hyperspectral detection enables the system to resolve spectral shifts in reactor emission (for example phosphorescence wavelength shifts with temperature or excitation history) and to construct spectrally resolved invariance profiles for cross-channel verification and sensing.

Polarimetric cameras. In some embodiments, the detector includes a polarimetric sensor capable of resolving linear and optionally circular polarisation state of incident light, providing polarisation-resolved invariance profiles. In some embodiments, polarimetric detection is combined with polarisation-controlled emission to create polarisation-multiplexed channels with distinct scene and reactor sensitivities.

Light-field (plenoptic) cameras. In some embodiments, the detector is a light-field or plenoptic camera comprising a microlens array (MLA) bonded to or positioned immediately in front of a two-dimensional image sensor, such that each microlens spans multiple sensor pixels and pixels beneath a given microlens sample distinct angular coordinates or ray directions within that microlens’s sub-aperture set. A single exposure therefore records a sampled four-dimensional light-field representation $L(x, y, u, v)$, where (x, y) denotes the spatial coordinate on the MLA plane and (u, v) denotes the angular coordinate within each microlens’s sub-aperture set. In some embodiments, the plenoptic observation is included as additional

channels of $\mathbf{y}_t$, so that the convolution bundle records not only per-pixel intensity but per-pixel angular distribution within a single exposure. In some embodiments, the plenoptic observation supports depth estimation, refocusing, and sub-aperture view synthesis without active time-of-flight illumination, and provides a measured angular-consistency constraint on scene emissions that planar printed images, ordinary flat displays lacking calibrated angular multiplexing, planar projection surfaces, and other effectively two-dimensional replay surfaces do not reproduce under the declared pose, sub-aperture baseline, and calibration state. In some embodiments, the MLA pitch, microlens focal length, and sensor pixel pitch are recorded in the protocol digest, and meters record sub-aperture alignment, vignetting across the MLA field, and angular-resolution calibration indicators.

Wavefront sensors and digital holographic imaging sensors. In some embodiments, the detector is a wavefront sensor that resolves local phase gradients or complex field structure of the incident beam, including, without limitation, Shack–Hartmann sensors (MLA-over-sensor with per-microlens spot-displacement readout), pyramid wavefront sensors, curvature wavefront sensors, and phase-diversity sensors. In some embodiments, the detector is a digital holographic imaging sensor that records a complex-valued field on a two-dimensional array, for example through off-axis reference-beam interference, phase-shifting interferometry, or in-line holography with computational reconstruction. In some embodiments, the resolved phase-gradient or complex-field observation is included as additional channels of $\mathbf{y}_t$, enabling phase-conjugation and adjoint-matching primitives as verification constraints, and supporting sensing regimes in which intensity-only measurements under-determine the reactor or scene state. In some embodiments, a plenoptic, wavefront, or holographic sensor is combined with conventional intensity, spectral, or polarimetric channels so that the convolution bundle carries simultaneously an intensity image, an angular or phase-resolved representation, and auxiliary channels with distinct invariance profiles.

Computational and in-sensor processing detectors. In some embodiments, the detector performs computation on the optical signal before or during digitisation, for example through analogue photonic matrix operations on the sensor, spectral inner products computed by on-chip filter arrays, or optical reservoir computing performed in the detector’s own optical front-end. In some embodiments, such detectors reduce the data rate and latency of the feedback loop by extracting task-relevant features directly in the optical or analogue electronic domain.

10.25 Printed and fabricated operator embodiments (non-limiting)

In some embodiments, at least a portion of the mapping from emitted fields to measured responses is realised by a designed operator rather than discovered primarily by end-to-end learning. Non-limiting examples include diffractive phase plates, metalens-like stacks, holographic or volume gratings, engineered scattering layers, ultrafast-laser-written photoelastic

micro-vortex arrays in thermoplastic polymer substrates, and hybrid refractive–diffractive assemblies.

In some embodiments, an input field encoded by a projector, microdisplay, spatial light modulator, or patterned emitter is propagated through one or more printed phase layers or diffractive layers and then measured on a detector surface. The overall effect may approximate a known analytic mapping, such as a Hankel-type transform, a diffusion kernel on a disc, or a Laplace–Beltrami propagation on a curved manifold. The configuration parameters θ include geometric and material parameters of the fabricated layers.

Fabricated four-dimensional (light-field) operators (non-limiting). In some embodiments, a fabricated operator is configured to implement a designed four-dimensional transform between an emitted light field $E(x, y, u, v)$ and a captured light field $L(x', y', u', v')$. Non-limiting examples include metasurface stacks configured as angular–spatial couplers, dual-MLA sandwich elements with an engineered intermediate phase layer, holographic volume gratings configured as Bragg-selective angular multiplexers, and diffractive optical neural network layers trained to approximate declared four-dimensional transfer kernels. In some embodiments, the configuration parameters θ include the MLA pitches, intermediate-layer phase profile, volume-grating recording parameters, and inter-layer spacings, all of which are recorded in the protocol digest so that downstream reconstruction and verification are conditioned on the fabricated four-dimensional geometry.

In some embodiments, a fabricated operator is used in a low-recurrence (single-pass) configuration. In other embodiments, the same fabricated operator is placed in a cavity or feedback loop, so that repeated passes yield Markovian dynamics. These embodiments remain instances of the Reality Kernel formalism.

Ultrafast-laser-written photoelastic dispersive operators (non-limiting). In some embodiments, a designed dispersive operator is realised by ultrafast-laser-induced photoelastic micro-vortex structures written into a thermoplastic polymer substrate. In some embodiments the substrate is polycarbonate; in other embodiments other thermoplastic polymers are substituted, subject to applicable photoelastic-response and fabrication requirements. In some embodiments, each such structure produces a rich spectral response over a broad bandwidth (for example approximately 400–1550 nm), occupies a compact footprint (for example approximately $10\ \mu\text{m} \times 10\ \mu\text{m}$), and is substantially independent of viewing angle. In some embodiments, arrays of such structures are registered directly on a CMOS image sensor to form a per-pixel dispersive operator, and the write trajectory, laser fluence schedule, polymer specification, and post-write spectral characterisation are recorded in the protocol digest as part of the operator’s configuration parameters θ . In some embodiments, the device-specific microstructure induced by the ultrafast-laser writing process is treated as a physically unclonable contribution to the operator, such that two nominally identical write programmes on two substrates yield distinguishable per-pixel spectral signatures that can themselves participate in attestation and hardness claims under the rules set out else-

where in this specification. Supporting background includes Zhang et al., “Optical dispersion using micro-vortices in thermoplastic polymers for integrated microspectrometers,” *Nature Electronics* (2026).

Fabrication and use of microstructure-unclonability cassettes and authentication

tokens (non-limiting). In some embodiments, the system includes a fabrication loop in which a beam engraves, modifies, or otherwise writes microstructure into a substrate to form a reactor-microstructure-unclonability cassette (analogous to a physically unclonable function or PUF in the prior-art sense) or authentication token. The engraving branch steers a higher-power beam across a substrate under a programmed trajectory while measuring the evolving response in situ. The resulting emission–observation history forms part of a canonical verification signature. After fabrication, the engraved substrate can be re-inserted into a Reality Kernel module as a microstructure-unclonability cassette, and a lower-power probe protocol is executed and compared against a stored reference.

Additive-manufactured volumetric microstructure-unclonability bodies (non-

limiting). In some embodiments, a microstructure-unclonability cassette is produced by additive manufacturing—for example a 3D-printed translucent body whose internal microstructure is intentionally disordered or process-variant (stochastic infill, controlled bubbles or scatterer distributions, layered exposure variations, or embedded particulate inclusions). When probed by structured illumination, such bodies act as volumetric scattering microstructure-unclonability media: responses are highly instance-specific and difficult to reproduce without the same fabrication microstructure.

Physical signing and watermarking (non-limiting).

In some embodiments, an engraving loop is used to apply physical signatures, serial numbers, or watermarks to objects. The engraved structure is treated as part of the scene during subsequent verification or Lim-
ager passes, providing additional microstructure that can increase empirical distinctiveness.

Training engraving and probing policies (non-limiting).

In some embodiments, the system trains not only a verifier for a given microstructure-unclonability cassette but also an engraving policy and a probing policy intended to increase empirical hardness. Alice selects engraving policies that determine scan laws, dwell patterns, pass schedules, and power envelopes; Bob selects probing patterns and meters to verify the engraved item; and Eve attempts to counterfeit cassettes under a bounded resource model.

Subtractive microfabrication (non-limiting).

In some embodiments, the controlled beam loop is used as a subtractive manufacturing tool to create channels, relief structures, alignment features, or diffractive patterns on a workpiece. The system may alternate lower-power probe exposures and higher-power modification exposures while observing results in

situ. The resulting fabricated element may serve as a static analogue operator in later Reality Kernel runs, or as a functional component in a larger reactor stack.

10.26 Extended-scale and environmental reactor embodiments (non-limiting)

- 5 In some embodiments, the Reality Kernel framework extends beyond laboratory-scale optical systems to encompass larger-scale physical systems whose dynamics satisfy the Markov-kernel abstraction $P_\theta(\cdot \mid S, U_{0:T})$. The following non-limiting embodiments illustrate breadth of applicability.

Geological and geophysical reactors (non-limiting). In some embodiments, a geological
10 ical formation or geophysical system acts as the reactor: seismic wave propagation through rock strata, tectonic strain accumulation and release, magma chamber dynamics, or glacial ice flow. The “emission” is a controlled or instrumented perturbation (for example a seismic source, a borehole pressure pulse, or an acoustic transducer), and the “detection” is a geophone array, strainmeter, GPS, or distributed fibre-optic sensor. The geological medium’s internal
15 structure (fracture networks, layering, mineralogical variation) provides device-specific nonlinear transfer functions suitable for all three operating regimes: Truth Beam for structural monitoring and tamper detection (detecting changes in the medium’s transfer function), Limager for subsurface characterisation (using the medium’s response to active probing), and Reality Transform for controlled modification (for example guided fracturing or injection).
20 Protocol digests and meter envelopes record source parameters and sensor configurations.

Atmospheric and ionospheric reactors (non-limiting). In some embodiments, the atmosphere or ionosphere acts as the reactor. Optical signals (laser, structured light, or natural illumination) propagate through turbulent or scintillating media whose refractive
25 index varies in space and time. The medium’s turbulence profile provides a natural time-varying microstructure signature (device-specific, time-varying, hard to clone exactly). In some embodiments, ionospheric scintillation of satellite signals provides the “scene” and the observable is the convolution bundle of transmitted and received signal structure. Atmospheric embodiments include lidar systems where backscatter from aerosols, clouds, or precipitation acts as the reactor response.

30 **GNSS-based passive Reality Kernel embodiments (non-limiting).** In some embodiments, a Global Navigation Satellite System (GNSS) receiver operates as a passive Reality Kernel in which the satellite constellation provides the emission protocol and the atmosphere, ionosphere, and local multipath environment collectively act as the reactor medium. The “emission” is the known transmitted signal structure (spreading codes, navigation mes-
35 sages, carrier frequencies, and timing) of one or more GNSS constellations, and the “detection” is the receiver’s correlation, tracking, and measurement engine. The convolution bundle

pairs the known transmitted structure with the received observables (pseudoranges, carrier phases, Doppler shifts, and signal-strength indicators) under a shared timebase derived from the GNSS signals themselves.

In some embodiments, the ionospheric and tropospheric propagation channel introduces
 5 signal delays, scintillation, and multipath that are location-specific, time-varying, and difficult to clone exactly, providing a natural physically unclonable channel analogous to device-specific reactor behaviour in active embodiments. Meters evaluate the consistency of received signal characteristics across multiple satellites, frequencies, and time windows; anomalies such as unexpected delay profiles, carrier-phase discontinuities, or signal-strength patterns
 10 inconsistent with the declared antenna environment are flagged for review.

In some embodiments, the system operates in a Truth Beam mode where the objective is to verify that GNSS observables are consistent with a declared location, time, and propagation environment, enabling tamper-evident positioning. In some embodiments, the system operates in a Limager mode where the objective is to characterise the propagation chan-
 15 nel itself (for example ionospheric total electron content, tropospheric wet delay, or local multipath signatures) using the known satellite signals as structured probes. Protocol dictates record receiver configuration, antenna characteristics, constellation selection, and any corrections applied (for example differential corrections or atmospheric models), and meter envelopes record signal-quality and consistency indicators.

20 **Fisher information in passive GNSS Truth Beam (non-limiting).** In the passive GNSS embodiment, the controllable configuration parameters θ include receiver-side variables such as antenna orientation and gain pattern, constellation and signal selection (which satellites and frequencies to track), tracking-loop bandwidth and integration time, elevation mask angle, and multipath mitigation settings. Truth Beam operation in this context selects
 25 θ to maximise the Fisher information of the observable residual distribution with respect to the hypothesis that the receiver is at the declared location and time—equivalently, to maximise the statistical power to distinguish a genuine location from a spoofed one. This is a constrained form of the general Truth Beam objective (maximise $I(\theta)$ about the scene-plus-device configuration) in which the “emitter” is not under the operator’s control but the
 30 receiver’s measurement strategy is.

In some embodiments, a Reality Kernel combines an active optical or RF emission path with a passive GNSS reception path, and the controller fuses evidence from both paths. The GNSS path provides externally anchored position and timing while the active path provides device-controlled scene or reactor probing; cross-consistency between the two paths
 35 strengthens verification by constraining the evidence to be consistent with both the declared physical interaction and the declared spatiotemporal location.

Aquatic, fluidic, and chemical-oscillator reactors (non-limiting). In some embodiments, a body of water, flowing fluid, or microfluidic channel acts as the reactor. The fluid’s turbulence, mixing, particle suspensions, or chemical composition provides the nonlinear

transfer function. Non-limiting examples include: river or ocean current monitoring where structured light interacts with the water column; microfluidic devices where reagent mixing in a channel produces concentration-dependent optical response; and underwater acoustic systems where sound propagation through a water column with varying temperature, salinity, and current provides the channel. In some embodiments, the reactor medium comprises a chemical oscillator such as a Belousov–Zhabotinsky reaction, a chlorite–iodide–malonic acid oscillator, or other reaction-diffusion system whose spatiotemporal dynamics (spiral waves, target patterns, turbulence) provide a high-dimensional, history-dependent, and intrinsically analog nonlinear transfer function. Structured light illumination perturbs the photosensitive variant of the reaction, providing the “emission” channel; optical imaging of the reaction’s colour or fluorescence changes provides the “detection” channel. The reaction’s attractor structure, bifurcation diagram under illumination intensity, and spatial pattern repertoire are medium-specific and difficult to clone or simulate exactly.

Biological ecosystem reactor applications (non-limiting). In some embodiments, a living biological system acts as the reactor: coral reef fluorescence under structured UV illumination, bioluminescent organisms responding to mechanical or chemical stimulation, mycelial networks whose electrical or chemical signalling responds to environmental probes, plant canopy photosynthesis and transpiration dynamics under controlled lighting, or microbial communities whose metabolic activity modulates optical properties of their medium. These systems provide intrinsically nonlinear, history-dependent, and clone-resistant transfer functions suitable for environmental monitoring (Limager), ecosystem health verification (Truth Beam), and bio-responsive installations (Reality Transform). Implementations at larger ecosystem scale may use embodiment-specific instrumentation, calibration, safety, environmental, and regulatory controls appropriate to the selected biological medium and deployment environment. Such implementations are non-limiting applications of the reactor-medium class and do not limit the smaller-scale biological, chemical, optical, or instrumented-reactor embodiments otherwise described in this disclosure. These embodiments differ from organoid computing and biological reservoir computing systems (e.g. DishBrain, FinalSpark neuroplatform) in that the biological medium here functions as a reactor in a cryptographically committed optical feedback loop—its optical response statistics constitute the convolution bundle—rather than as an autonomous computational substrate optimised for task performance.

Social and collective-behaviour reactors (non-limiting, limited scope). In some embodiments, a group of interacting agents (human, robotic, or mixed) acts as the reactor, with group dynamics providing the nonlinear transfer function. The “emission” is a structured stimulus presented to the group (visual display, auditory signal, information feed), and the “detection” is measurement of the group’s collective response (voting patterns, movement trajectories, communication patterns, physiological synchrony). The group’s internal social structure, trust topology, and communication network provide device-specific proper-

ties. Yoked operation is natural: the group’s response modifies the next stimulus, creating a bidirectional coupling between the display system and the social dynamics. **Scope note (instrumented measurement applications; non-limiting):** In preferred embodiments, this embodiment uses physical measurement equipment (sensors, displays, physiological mon-
 5 itors) providing the emission and detection channels; human behaviour itself does not constitute a “reactor” in the patent-law sense absent such instrumentation. Purely abstract social dynamics without physical instrumentation performing nonlinear signal transformation, and systems in which the “reactor” is an economic, political, or informational process without a physical substrate generating measurable nonlinear responses to controlled physical stimuli,
 10 are not required and are not the focus of the embodiments described herein. This paragraph describes applications to human or mixed-agent behaviour via physical instrumentation; deployment scale is selected according to available instrumentation, safety, consent, calibration, and governance requirements.

10.27 Sweeping reactor embodiments (non-limiting)

15 In some embodiments, the reactor is operated in a *sweeping* configuration in which at least one physical parameter λ of the reactor (here λ denotes a generic swept physical parameter such as temperature, bias voltage, or optical path length; it is unrelated to the regime weights $\lambda_{\text{TB/L/RT}}$) is periodically modulated at a frequency f_{mod} while the emitter–recorder subsystem captures the resulting signal modulation:

$$\lambda(t) = \lambda_0 + \delta\lambda \cdot \sin(2\pi f_{\text{mod}} t).$$

20 Phase-sensitive (lock-in) demodulation of the recorder output at f_{mod} extracts the local derivative of the reactor’s physical transform with respect to the swept parameter:

$$S(f_{\text{mod}}) \propto \left. \frac{\partial T}{\partial \lambda} \right|_{\lambda_0} \cdot \delta\lambda,$$

where T denotes the physical transform applied by the reactor. By slowly stepping λ_0 through a range of interest, the full transfer function $\partial T / \partial \lambda (\lambda_0)$ is mapped—providing a differential characterisation of the reactor’s physics that is inaccessible to static measurements.

25 The sweeping reactor principle provides three intrinsic advantages. First, common-mode rejection: any contribution to the measured signal that does not vary at f_{mod} is rejected by the demodulation process, including static offsets, slowly drifting environmental perturbations, and constant systematic biases. Second, access to derivative information: fine structure in the reactor’s transfer function (dispersive slopes, resonance lineshapes, mode crossings)
 30 is resolved directly as a continuous function rather than sampled at isolated points. Third, orthogonality to existing static measurements: a smooth geometry-dependent contribution to the reactor’s output that is degenerate with other systematics in a fixed-geometry configuration is selectively measured by the sweeping reactor while geometry-independent offsets

are rejected. A complementary approach—temporal modulation of reactor material properties at THz frequencies (Section 10.2.8)—addresses a different timescale regime; the two approaches are orthogonal and may be combined.

Null-subtraction and adaptive sweep control (non-limiting). In a preferred implementation, the measurement operates in a null-subtraction mode: a model prediction for the signal at f_{mod} is subtracted from the demodulated output in real time, so that the lock-in output represents the *residual* between the measured transfer function and the model prediction. This concentrates the instrument’s dynamic range on deviations from the model. In some embodiments, the DC operating point λ_0 is stepped under feedback control: if the residual is consistent with noise, the system advances with minimal dwell time; if the residual exhibits statistically significant structure, integration time is increased at that operating point. In some embodiments, the modulation amplitude $\delta\lambda$ is adapted to maintain operation in the perturbative regime near resonances and to maximise sensitivity between resonances. In an advanced implementation, the drive waveform (which may include multiple frequency components, chirps, or shaped pulses) is optimised by a machine learning system to maximise sensitivity to residual structure, constituting a learned emission protocol within the emitter–reactor–recorder architecture (analogous to learned scan laws in other embodiments).

Residuals as device-specific signatures (non-limiting). From the perspective of the Reality Kernel framework, the residual between a model prediction and the measured transfer function is analogous to the device-specific microstructure signature that provides hardness in other reactor embodiments. A reactor whose transfer function is perfectly predicted by a finite-dimensional model has zero residual and provides no empirical hardness beyond the model. A reactor whose transfer function includes contributions not captured by the model—whether from manufacturing imperfections, surface roughness, material inhomogeneity, or any other physical contribution not captured by the finite-dimensional model—produces a nonzero residual that is device-specific, difficult to predict or reproduce without access to the physical reactor, and therefore contributes to the 2D hardness index (Section 12; see also Section 23 for microstructure identification protocols). The sweeping reactor provides a precision tool for measuring these residuals as continuous functions of the swept parameter. In some embodiments, the scalar sweeping principle is extended to simultaneous multi-parameter sweeping with the scene’s return signal as the lock-in reference rather than an internally declared fixed frequency; this generalisation is described in Section 10.28.

Plenoptic readout of sweeping-reactor demodulation (non-limiting). In some embodiments, the recorder of a sweeping-reactor configuration is a light-field (plenoptic) detector (Section 10.24) rather than a single-pixel photodetector or a conventional frame-based camera, and lock-in demodulation at the modulation frequency f_{mod} is performed independently on each sub-aperture channel of the plenoptic observation. The demodulated output

is therefore a four-dimensional derivative field

$$\left. \frac{\partial L(x, y, u, v)}{\partial \lambda} \right|_{\lambda_0}$$

rather than a scalar or two-dimensional derivative, where (x, y) denotes the spatial coordinate on the MLA plane and (u, v) denotes the angular sub-aperture coordinate.

In some embodiments, per-sub-aperture demodulation is used only where the declared photon budget, sampling cadence, and sub-aperture signal-to-noise ratio support the selected lock-in estimator; otherwise sub-apertures are pooled or treated as lower-confidence channels under the meter envelope.

In some embodiments, the plenoptic sweeping-readout configuration provides three capabilities that are not available under single-pixel or conventional two-dimensional readout:

1. *Angular separation estimates for scene and reactor contributions.* Contributions to the demodulated signal whose origin is geometrically coupled to the scene (for example, parallax-dependent return from a three-dimensional scene) and contributions whose origin is intrinsic to the reactor (for example, cavity-mode shifts under boundary modulation, or reactor microstructure response) may exhibit different angular profiles under the declared optical model. In some embodiments, regression of the demodulated field against declared angular basis functions estimates these contributions as additional channels of the meter envelope, and the angular-basis residual records how well the declared separation model fits the committed evidence. This provides a measured separation mechanism orthogonal to the f_{mod} -based rejection described above, rather than an assumption that plenoptic geometry automatically separates the contributions.
2. *Sub-aperture derivative redundancy.* Each sub-aperture yields a separately indexed estimate of the local derivative, or of the view-conditioned component of $\partial T / \partial \lambda$, at a displaced viewing angle. In some embodiments, the cross-sub-aperture consistency of these estimates is computed under declared meters and committed alongside the convolution bundle as a within-device consistency statistic, complementary to fleet-level proof-of-discrepancy confirmation statistics defined elsewhere in this specification. In some embodiments, sub-aperture consistency below a declared threshold is logged as a degraded-confidence indicator rather than treated as operational failure; the consistency statistic is a measured quantity under the declared meter set, not an assumed property of the plenoptic geometry.
3. *Native readout for angular-parameter sweeps.* Where the swept parameter λ itself governs angular structure—for example, MLA translation, MLA tilt, emitted sub-aperture selection, or the phase offset of an angular-multiplex element—a plenoptic recorder reads the derivative $\partial L(x, y, u, v) / \partial \lambda$ directly, whereas a single-pixel or angle-averaging recorder averages over the angular coordinate and loses the signal of interest. In some embodiments, a light-field projector and plenoptic recorder (Section 10.2.5) are used as a matched pair for such angular-parameter sweeps, with the emission-side angular configuration recorded in the protocol digest alongside the demodulated derivative

field.

In some embodiments, the protocol digest records the plenoptic geometry (MLA pitch, effective angular resolution, sub-aperture assignment) alongside the sweep schedule, and meters record sub-aperture registration stability during the sweep, vignetting profile, and angular-basis residual indicators. In some embodiments, the plenoptic sweeping-readout configuration is compatible with the null-subtraction and adaptive-sweep-control strategies described above, with the null model extended to a four-dimensional prediction and the adaptive controller responding to residual structure in any sub-aperture channel.

Cavity-coupled charged-particle embodiment (non-limiting). In a non-limiting embodiment, the reactor is a cylindrical microwave cavity of radius R and variable axial length $L(t) = L_0 + \delta L \cdot \sin(2\pi f_{\text{mod}} t)$, where the movable endwall is displaced by a piezoelectric actuator. The emitter is a single electron (or small ensemble) confined in a Penning trap within a static magnetic field B_0 , undergoing cyclotron motion at frequency $\omega_c = eB_0/m_e$ and radiating microwave emission at ω_c . The cavity's electromagnetic mode spectrum modifies the cyclotron frequency by a calculable shift $\Delta\omega_c(L)$ that depends on the cavity geometry. The recorder is a cryogenic heterodyne receiver that downconverts the cyclotron radiation and applies digital lock-in demodulation at f_{mod} to extract the derivative $d\Delta\omega_c/dL$ at each operating point L_0 .

Key design features of this embodiment include:

- A non-contacting quarter-wave choke groove at the piston-cylinder interface, which presents a low-impedance RF boundary without sliding electrical contact, eliminating particulate generation and time-variable resistive losses that would produce spurious signals at f_{mod} .
- Independent calibration of the piston displacement $\delta L(t)$ via optical interferometry and of the magnetic field at f_{mod} via an independent magnetometer, enabling separation of geometry-dependent and field-dependent contributions.
- Null tests with the charged particle removed, with the particle far from cavity mode resonances, and with different piston materials, to identify and bound spurious signals.

The experimental transfer function $\{L_0, d\Delta f_c/dL(L_0)\}$ is compared to the standard cavity quantum electrodynamics prediction. Residuals are examined for systematic smooth trends (indicating geometry-dependent contributions not captured by the mode sum), mode-correlated deviations (indicating uncharacterised cavity imperfections or other device-specific microstructure), or null residuals consistent with noise (indicating that the reference cavity model accounts for the measured response at the instrument's resolution, and that device-specific signatures must be sought at finer sensitivity or in different measurement channels). From the Reality Kernel perspective, the cavity's manufacturing tolerances, surface finish, and mode-coupling imperfections constitute a device-specific signature measured with high fractional sensitivity (in some embodiments, sub-part-per-billion or finer, depending on cavity quality factor and readout configuration). Implementation of this embodiment uses cryogenic infrastructure (for example a dilution refrigerator or adiabatic demagnetisation re-

frigerator providing base temperatures below 100 mK), ultra-high vacuum (below 10^{-10} Pa), a superconducting solenoid for B_0 , and single-electron detection via image-current amplifiers or quantum-jump spectroscopy. These requirements are standard in precision measurement laboratories and do not require novel infrastructure beyond the Reality Kernel control and logging overlay.

Alternative sweeping reactor implementations (non-limiting). The sweeping reactor principle extends beyond the microwave-cavity embodiment. Non-limiting alternative implementations include:

- *Optomechanically modulated optical cavity*: a Fabry–Pérot cavity with one mirror on a piezoelectric or MEMS actuator, modulated at f_{mod} while an optical source illuminates the cavity and a photodetector records the transmitted or reflected signal. Lock-in demodulation extracts the derivative of the cavity transfer function with respect to length.
- *Acoustically modulated fluid channel*: a microfluidic or mesoscale channel with an acoustically driven wall, probed optically or acoustically. The demodulated signal encodes the derivative of the fluid’s response with respect to channel geometry.
- *Electrically tuned metamaterial boundary*: a metasurface whose electromagnetic response is modulated by applied voltage (for example varactor-loaded elements) at f_{mod} , forming one wall of a resonant structure. The tuning parameter is an applied electric field rather than a physical displacement, enabling modulation at higher frequencies (up to MHz) and eliminating mechanical fatigue.
- *Thermally swept photonic crystal*: a photonic crystal whose band gap is thermally modulated while an optical source probes the transmission. The derivative measurement maps the temperature sensitivity of the photonic band structure.
- *Radiation-pressure gradient measurement in asymmetric cavities*: a microwave cavity of non-uniform cross-section (for example a truncated conical frustum, a stepped-diameter cylinder, or a cavity containing a movable dielectric insert) driven at a resonant mode and mounted on a high-sensitivity force transducer. One cavity dimension is modulated at f_{mod} , and the force transducer output is demodulated to extract the derivative of the net radiation pressure with respect to the modulated parameter. This embodiment does not assert that the apparatus produces anomalous forces; the measured signal is the geometry-dependent radiation-pressure derivative predicted by standard electrodynamics, used as a device-specific signature. The lock-in architecture rejects thermal radiation pressure, outgassing forces, electromagnetic interference, and Lorentz forces from stray fields, providing systematic rejection inaccessible to static force measurements in asymmetric cavities.

Multi-reactor sweeping configurations (non-limiting). In some embodiments, two or more sweeping reactors are operated simultaneously, each modulated at a distinct frequency $f_{\text{mod},k}$ with a controlled phase relationship. Cross-demodulation of reactor j ’s output

at reactor k 's modulation frequency extracts the inter-reactor coupling transfer function: how modulation of reactor k 's geometry affects the physical response observed in reactor j . By sweeping separation, relative orientation, or the DC operating point of either reactor, the full inter-reactor coupling is mapped as a continuous function of geometric parameters.

5 Residuals between measured coupling and the prediction from standard coupled-mode theory constitute a two-reactor analogue of the single-reactor device-specific signature, contributing to the 2D hardness index (Section 12).

Non-limiting spatial configurations include:

- *Collinear*: two or more reactors aligned along a common axis, providing maximum axial coupling sensitivity and a baseline measurement of the coupling's distance dependence.
- *Ring with progressive phase*: N reactors equally spaced around a ring, each modulated with a phase offset of $2\pi/N$ from its neighbour, so that the combined field-energy distribution rotates around the ring. This configuration sources circulating energy currents and separates coupling components with distinct angular symmetry.
- 15 • *Nested concentric (Matryoshka)*: a smaller reactor enclosed within a larger reactor, so that the inner reactor operates within the electromagnetic environment modified by the outer reactor's boundary conditions. The inner reactor's response is sensitive to the outer reactor's geometry through any channel that mediates inter-boundary coupling (Section 10.27).
- 20 • *Incommensurately spaced (Fibonacci)*: N reactors along a line at spacings proportional to successive Fibonacci numbers (or another maximally irrational ratio), providing logarithmically distributed length scales with minimal periodic aliasing. This arrangement optimises extraction of the coupling's distance scaling law (power-law exponent) from a finite number of reactor pairs.

25 In some embodiments, the inter-reactor phase relationships are optimised to maximise the area enclosed by the combined state trajectory in the multi-reactor configuration space, which determines the sensitivity to coupling effects that depend on the cyclic sequence of deformations rather than instantaneous geometry. In some embodiments, sum and difference frequencies ($f_{\text{mod},j} \pm f_{\text{mod},k}$) are also monitored to detect nonlinear inter-reactor coupling.

30 These configurations describe multiple reactors within a single Reality Kernel module or collocated assembly; for multi-device networked operation using protocol-mediated coupling, see Section 14.

Reference experiment: Fisher information at a synchronisation transition (non-limiting). In some embodiments, a reference experiment measures Fisher information as

35 a function of coupling strength across a synchronisation transition between two Reality Kernel modules, or between an Reality Kernel module and a cooperative scene partner. In one non-limiting protocol, two Reality Kernel modules (or one Reality Kernel module and a cooperative scene partner) are configured in Yoked mode with a continuously adjustable coupling-strength parameter $\alpha \in [0, 1]$ of the kind defined in the Yoked-operation discussion

40 of the present disclosure. The parameter α is swept from zero (uncoupled) through a can-

didate synchronisation threshold α_c to full coupling ($\alpha = 1$) and back, in small increments, with the sweep schedule recorded in the protocol digest. At each value of α , a sufficient convolution bundle window is recorded from both modules to estimate the Fisher information matrix $\mathcal{I}(\alpha)$ from the empirical score function under a declared statistical model; transfer entropy estimates in both directions ($\text{TE}_{1 \rightarrow 2}(\alpha)$ and $\text{TE}_{2 \rightarrow 1}(\alpha)$), conditional Lyapunov exponents, and attractor-dimension estimates are computed from the same convolution bundle segment. The trace $\text{tr}[\mathcal{I}(\alpha)]$ is plotted as a function of α and committed to the protocol digest together with all intermediate estimates, the estimator family used for each quantity, and the declared statistical model. In some embodiments, the predicted signature is a peak in Fisher information at or near $\alpha \approx \alpha_c$, accompanied by a conditional Lyapunov exponent approaching zero from negative values on the synchronised side and a sharp increase in transfer entropy; for continuous transitions, the peak width is predicted to scale with a critical exponent related to the universality class of the transition, while for discontinuous transitions, a sharp spike with hysteresis between the sweep-up and sweep-down branches is predicted. The reference experiment provides a non-limiting protocol for characterising critical signatures of synchronisation transitions on Reality Kernel platforms; it connects the precision-metrology framing of the sweeping reactor (Section 10.27) with the Yoked-operation regime and provides a concrete embodiment in which the apparatus's sensitivity to dynamical-coupling phenomena can be characterised against declared theoretical predictions.

Nested reactor (Matryoshka) architecture (non-limiting). In some embodiments, independently of the sweeping reactor configuration, the reactor element comprises two or more concentric or enclosed physical structures, each constituting a reactor in its own right, arranged so that inner reactors operate within the electromagnetic, acoustic, or thermal environment established by outer reactors. This nested (Matryoshka) architecture is distinct from layered reactor stacks (Section 10.3) in which layers are traversed sequentially by the signal: in the Matryoshka configuration, the inner reactor is *immersed in* the modified environment of the outer reactor rather than receiving light that has passed through it.

The physical significance is that the inner reactor's transfer function depends on the boundary conditions imposed by the outer reactor. When the outer reactor's boundaries are modulated (as in the sweeping reactor configuration), the inner reactor's operating environment changes continuously. The inner reactor's response—measured by its own emitter-recorder subsystem—is thus a probe of how the outer reactor's geometry affects the physical environment within it. In the electromagnetic case, this includes the cavity mode spectrum and the local density of states as modified by the outer boundaries.

Non-limiting Matryoshka embodiments include:

- A microwave cavity containing a smaller Penning trap cavity (the inner reactor probes the electromagnetic environment established by the outer cavity via cyclotron frequency shifts).
- A Fabry–Pérot optical cavity enclosing a scattering medium reactor (the inner reactor's microstructure response depends on the intracavity field structure set by the outer

cavity mirrors).

- An acoustic resonator containing a smaller optical reactor (the acoustic mode structure modulates the mechanical environment of the optical components, coupling acoustic and optical degrees of freedom).
- Concentric electromagnetic cavities at different frequency bands (for example an RF cavity containing a microwave cavity, or a microwave cavity containing an optical cavity), where the outer cavity modulates the environment at one frequency scale and the inner cavity probes at another.

In Matryoshka embodiments, each reactor’s convolution bundle is recorded independently, and the cross-coupling between reactors is extracted via the multi-frequency demodulation scheme described above. The nested architecture provides a natural realisation of hierarchical reactor composition within the Reality Kernel algebra (Section 12). In some embodiments, compromise of the outer reactor (characterisation of its full transfer function by an adversary) partially degrades the inner reactor’s effective hardness, because the adversary can predict the inner reactor’s operating environment. However, the inner reactor retains its own device-specific microstructure, which outer-reactor compromise does not directly reveal. Periodic reconfiguration of the outer reactor’s boundary conditions (analogous to key rotation) invalidates the adversary’s accumulated characterisation. In preferred embodiments, inner-reactor hardness is monitored independently so that degradation from outer-reactor compromise is detected through the fleet monitoring mechanisms described in Section 14.

10.28 Scene-phased full-parameter sweeping and physical gradient computation (non-limiting)

10.28.1 Motivation and relationship to scalar sweeping

The sweeping reactor embodiments described in Section 10.27 operate by modulating a single scalar parameter λ of the reactor at a fixed, externally declared frequency f_{mod} , extracting the scalar derivative $\partial T / \partial \lambda$ at each operating point λ_0 via lock-in demodulation. This provides a differential characterisation of the reactor’s transfer function along one dimension of configuration space at a time.

In some embodiments, this principle is extended in two coupled ways: (i) the modulation is applied simultaneously across multiple components of the full configuration parameter vector $\theta = (\theta_{\text{hw}}, \theta_{\text{sw}})$, and (ii) the lock-in reference signal is derived from the scene’s own return dynamics rather than from an externally declared fixed frequency. The resulting operating mode is termed *scene-phased full-parameter sweeping*. It generalises scalar sweeping in the same way that the Jacobian generalises the scalar derivative: where scalar sweeping extracts $\partial T / \partial \lambda$ along one direction in Θ , scene-phased full-parameter sweeping extracts the directional derivative of the coupled transfer function along the direction the scene is actually traversing in configuration space during operation.

10.28.2 Mathematical formulation

Let $\theta(t)$ denote a time-varying modulation of the full parameter vector:

$$\theta(t) = \theta_0 + \delta\theta(t),$$

where θ_0 is the nominal operating point and $\delta\theta(t)$ is a small perturbation applied across all or a selected subset of admissible parameters. In the scalar sweeping embodiment, $\delta\theta(t) =$
 5 $\delta\lambda \cdot \mathbf{e}_\lambda \sin(2\pi f_{\text{mod}} t)$ for a fixed unit vector $\mathbf{e}_\lambda$ and a single declared frequency f_{mod} .

In some embodiments of scene-phased full-parameter sweeping, the modulation direction $\delta\theta(t)$ and the reference signal for lock-in demodulation are both derived from the scene's return signal $r(t)$:

1. A reference signal $\phi_{\text{ref}}(t)$ is extracted from $r(t)$ by bandpass filtering around the scene's
 10 dominant dynamical frequency f_{scene} , where f_{scene} is estimated by a declared spectral-peak or phase-tracking method (for example a spectral coherence maximiser, a PLL, or a power spectral density peak detector) applied to $r(t)$. In some embodiments, the transfer entropy estimators described in Section 8.5 are used as a band-selection gate to identify frequency ranges with strong scene-to-reactor coupling, with spectral-peak
 15 or phase-tracking within that band used to extract the scalar frequency estimate f_{scene} .
2. The modulation $\delta\theta(t)$ is generated with phase locked to $\phi_{\text{ref}}(t)$, so that each component of θ oscillates at f_{scene} with a declared amplitude $\delta\theta_k$ and a declared phase offset ψ_k relative to the scene reference.
3. Phase-sensitive demodulation of the convolution bundle signal $C_{0:T}$ at $\phi_{\text{ref}}(t)$ extracts
 20 the in-phase (real) and quadrature (imaginary) components of the scene-phased response.

The demodulated in-phase output is proportional to the directional derivative of the physical transform T along the modulation direction at the current operating point, to first order in $\delta\theta$:

$$S_{\text{IP}} = G_L \operatorname{Re}[\nabla_{\theta} T|_{\theta_0} \cdot \tilde{\delta\theta}] + O(\|\delta\theta\|^2),$$

25 where G_L is the lock-in gain and $\tilde{\delta\theta}$ is the complex modulation amplitude. This first-order expansion holds under the assumptions that $\|\delta\theta\|$ is small relative to the operating point and that T is sufficiently smooth in a neighbourhood of θ_0 ; higher-order terms appear at harmonic and intermodulation frequencies and can be monitored or suppressed by appropriate modulation depth and bandwidth choice. By cycling $\delta\theta$ through a spanning set of directions
 30 in Θ (or by applying a simultaneous multi-tone modulation with adequate spectral separation, one tone per parameter dimension), the full Jacobian $\nabla_{\theta} T|_{\theta_0}$ is estimated as a function of the coupled operating point.

10.28.3 The lock-in reference is the scene

A defining feature of this embodiment, distinguishing it from scalar sweeping, is that the
 35 demodulation reference is the scene's own dynamical signal rather than a fixed oscillator

internal to the device. This has several non-limiting consequences.

First, common-mode rejection applies not only to signals that do not vary at f_{mod} (as in scalar sweeping) but also to signals that vary at f_{mod} but are not phase-coherent with the scene: spurious device-internal oscillations at the modulation frequency are rejected because they lack the scene’s phase relationship, substantially improving rejection of internal artefacts.

Second, the extracted gradient is conditioned on the scene’s current dynamical state. Two scenes with identical static appearance but different internal dynamics produce different demodulated outputs, because the reference phase $\phi_{\text{ref}}(t)$ encodes the scene’s dynamics. The gradient measurement is scene-specific in the same sense that the Arnold tongue is scene-specific: it reflects the coupled transfer function at the current joint operating point, not the device’s transfer function in isolation.

Third, when the device is operating in Yoked mode (Section 8.5)—specifically when $\widehat{\text{CLE}} \in [-\varepsilon_{\text{CLE}}, 0)$ and transfer entropy balance is confirmed—the reference signal $\phi_{\text{ref}}(t)$ is coherent with the reactor’s own dynamics by construction, because the two are phase-locked to the same joint attractor. In this regime, the modulation $\delta\theta(t)$ is automatically phase-aligned with the joint attractor’s oscillation, and the demodulated gradient reflects the sensitivity of the *coupled* transfer function to parameter variation rather than the device’s transfer function alone.

10.28.4 Connection to Yoked training: physical gradient computation

In some embodiments, scene-phased full-parameter sweeping serves directly as the physical mechanism for computing the gradient $\nabla_{\theta}J$ required by the Yoked training update rule (Section 8.5.5):

$$\theta \leftarrow \theta + \eta \mathcal{F}(\theta)^{-1} \nabla_{\theta}J.$$

In some embodiments, $\nabla_{\theta}J_{\text{train}}$ (where J_{train} is the declared TE-proxy training objective; see Section 8.5.5) is not estimated numerically by finite differences or via a differentiable surrogate model, but is estimated from the lock-in output of the scene-phased sweep. The physical mechanism is: a small perturbation $\delta\theta(t)$ phase-locked to the scene induces a measurable first-order change in the convolution bundle statistics (specifically in the TE estimators computed over declared measurement windows), and the demodulated response provides a directional-derivative estimate of J_{train} along $\delta\theta$, to first order in the modulation depth. The full-gradient estimate for the training update is assembled from these directional-derivative measurements across a spanning set of modulation directions; the outer-loop quantities J_{task} and the CLE penalty are updated on their respective slower timescales as described in Section 8.5.5. Cycling through a spanning set of modulation directions recovers the gradient estimate as a physical measurement, without requiring a differentiable surrogate model of the physical channel.

In some embodiments, this operating mode supplies the measured signal from which the gradient estimate is extracted by the declared demodulation and estimation procedure. The

amplification of sensitivity near the entrainment phase transition—where coupling sensitivity to parameter changes is largest, as described in Section 8.5—means that the gradient signal is strongest precisely at the operating point where the Yoked training update is most informative. For periodically forced limit-cycle oscillators of the class disclosed in this section, and
 5 under the declared phase-dynamics assumptions of Section 8.5, the edge-of-synchronisation condition and the maximum-gradient condition coincide at the Arnold tongue boundary; in some embodiments, this coincidence is empirically observed for the disclosed reactor classes and is not asserted as a universal property of all oscillator or media classes.

In some embodiments, an empirical preconditioner matrix $\hat{\mathcal{F}}(\theta)$ is formed from the covariance of demodulated gradient estimates across multiple scene-phased sweep cycles. Under
 10 a declared likelihood model where the demodulated measurements provide an estimate of the score, this covariance approximates the Fisher information matrix; otherwise it serves as a Gauss–Newton-style preconditioner that adapts the update direction to the observed gradient geometry. In some embodiments, the pseudoinverse or damped inverse of $\hat{\mathcal{F}}(\theta)$ is
 15 computed numerically from these empirical estimates, with damping applied near bifurcation boundaries where $\hat{\mathcal{F}}(\theta)$ is ill-conditioned.

10.28.5 Parameter scope: what lives in θ

In some embodiments, the parameter vector θ subject to scene-phased sweeping includes all or any subset of the following non-limiting parameter classes:

20 Emission amplitude modulation frequency and depth

The frequency and amplitude of the slow intensity envelope imposed on the emission. This is the parameter class addressed by scalar sweeping in its most common form.

Emission frequency modulation (FM)

The centre frequency of the emission carrier, modulated at the scene reference phase.
 25 In some embodiments, this corresponds to the frequency detuning parameter Δf of the Arnold tongue (Section 8.5), making scene-phased FM sweeping a direct instrument for mapping the Arnold tongue boundary from within the coupled system.

Emission phase and polarisation

Phase offset and polarisation state of the emitted field, each swept in phase with the
 30 scene reference. These parameters access degrees of freedom in the coupled transfer function that are orthogonal to amplitude and frequency modulation.

Scan law parameters

The timing, trajectory, and repetition rate of the spatial scan pattern. Sweeping the scan repetition rate in phase with the scene extracts the coupling’s dependence on
 35 spatial sampling rate, complementing the frequency-domain Arnold tongue map with a spatial-domain coupling map.

Feedback gain α (the yoking dial)

The bidirectional coupling strength parameter $\alpha \in [0, 1]$. Sweeping α in phase with the scene reference directly measures the derivative of the coupling objective with respect to coupling depth, which is the primary control parameter for navigating toward and stabilising at the tongue boundary.

Reactor internal timescale parameters

Parameters governing the reactor’s internal dynamics that are under operational control: in phosphor reactors, drive intensity and duty cycle (which affect effective decay time); in cavity reactors, mirror separation or finesse; in delay-line reactors, loop length. Sweeping these in phase with the scene extracts the coupling’s dependence on the reactor’s own dynamical timescale, extending the Arnold tongue map from the (detuning, coupling strength) plane to the full multi-dimensional entrainment manifold.

Learned parameters θ_{sw}

Neural network weights, look-up table entries, or calibration offsets that are differentially connected to the convolution bundle statistics. In some embodiments, scene-phased perturbations of θ_{sw} and physical demodulation of the resulting convolution bundle changes implement a physical analogue of backpropagation, without requiring a differentiable surrogate model of the physical channel.

10.28.6 Multi-frequency parallel gradient extraction (non-limiting)

In some embodiments, multiple parameter components are modulated simultaneously using a bank of modulation tones $\{f_k\}$ referenced to a shared clock derived from the scene reference phase $\phi_{\text{ref}}(t)$. The number of gradient components that can be extracted in a single measurement window is bounded above by the time-bandwidth product of the measurement channel, and in practice is substantially smaller than this bound because of minimum tone separation, per-tone SNR, modulation-depth limits, and calibration stability across the measurement window. Accordingly, this disclosure does not assert that gradient extraction is constant-time in parameter count. The disclosed mechanism supports extraction of selected gradient components, or low-dimensional projections of $\nabla_{\theta} J$, in parallel, subject to the bounds above.

10.28.7 Convergence and meter gating

In some embodiments, scene-phased full-parameter sweeping is gated by the same convergence conditions used to declare Yoked operation (Section 8.5.7): $\widehat{\text{CLE}} \in [-\varepsilon_{\text{CLE}}, 0)$ for N consecutive windows and $|\text{TE}_{S \rightarrow R} - \text{TE}_{R \rightarrow S}| < \tau_{\text{TE}}$. Gradient measurements taken outside this regime are flagged in the protocol digest as pre-convergence sweeps; gradient measurements taken within this regime are flagged as post-convergence and are suitable for use in the Yoked training update rule. In some embodiments, pre-convergence sweeps are used to

navigate toward the Yoked operating regime (coarse Arnold tongue mapping) while post-convergence sweeps are used for fine-grained gradient ascent within the regime.

10.28.8 Relationship to the sweeping reactor

Scene-phased full-parameter sweeping is a strict generalisation of the sweeping reactor (Section 10.27). Where the sweeping reactor fixes: (i) the modulation to a single parameter λ ; (ii) the reference frequency to an externally declared f_{mod} ; and (iii) the extracted quantity to the scalar derivative $\partial T / \partial \lambda|_{\lambda_0}$; scene-phased full-parameter sweeping relaxes all three constraints: (i) modulation is applied across all of θ ; (ii) the reference is the scene’s own return phase $\phi_{\text{ref}}(t)$; and (iii) the extracted quantity is the full Jacobian $\nabla_{\theta} T|_{\theta_0}$ projected along the scene’s dynamical direction.

The two approaches are complementary and may be combined: scalar sweeping with a fixed reference provides device-specific signature characterisation under controlled conditions; scene-phased full-parameter sweeping provides in-situ coupled gradient computation during Yoked operation. In some embodiments, the same hardware supports both modes under controller selection, differing only in the source of the lock-in reference signal and the set of parameters being modulated.

In some embodiments, a plenoptic-readout front-end (Section 10.2.5; see also Section 10.27 for the scalar-sweep case) applies to both scalar and scene-phased operation. In such embodiments, the light-field-valued transfer map may be written as

$$T_{\theta} : E(x, y, u, v) \mapsto L_{\theta}(x', y', u', v'),$$

so the scene-phased measurement estimates the Jacobian of this transfer map in the observed light-field coordinates, equivalently $\nabla_{\theta} L_{\theta}(x', y', u', v')|_{\theta_0}$ for the declared emitted field and operating point, rather than only a scalar derivative. Cross-sub-aperture consistency of the measured Jacobian is recorded as a within-device consistency statistic under the declared meter set and is not assumed from the plenoptic geometry.

10.28.9 Summary

Assertions elsewhere in this disclosure concerning “full- θ ” or “whole-parameter” gradient extraction shall be construed as the extraction of the projection of $\nabla_{\theta} J$ onto the declared tone-bank subspace, under the bounds of Section 10.28.6. No assertion of constant-time gradient-per-parameter extraction is made or implied.

10.29 Scope-preserving notes (non-limiting)

The following non-limiting statements clarify scope and operational interpretation: verification and security-related embodiments rely on empirical hardness metrics defined against declared adversary families and budgets. Hardness indices and meter outputs are interpreted

relative to chosen model classes, divergences, and compute/query budgets, and may be recalibrated as these choices change. Embodiments requiring scan addressing, timing alignment, or proof-of-projection assume maintained calibration and synchronisation within declared envelopes. Quantum-sensitive observables or non-classical effects are optional enhancements
 5 and are not required for core Truth Beam, Limager, or Reality Transform operation. Learned meters may themselves be attacked; robustness and recalibration are ongoing empirical concerns addressed by training curricula, audits, and conservative deployment policies.

10.30 Non-optical architecture-portability demonstrations

The following non-optical examples illustrate how the commitment, calibration, corroboration, and verification architecture disclosed in the preceding sections can be mapped
 10 to non-optical reactor families. These examples identify the component families, commitment plumbing, and architectural correspondences for each non-optical domain. They are architecture-portability demonstrations showing that the same architectural structure can be mapped to non-optical physics domains; they do not constitute full make-and-use bench
 15 recipes at the same level of concrete operating detail as the optical core embodiments above. Concrete operating parameters, calibration procedures, and validation protocols for each non-optical medium are embodiment-specific engineering tasks within the skill of the art once the architectural mapping is established.

Acoustic reactor example (non-limiting). In some embodiments, the reactor is an
 20 acoustic scattering assembly comprising a speaker or ultrasonic phased array as emitter, a disordered solid, fluid, or granular medium as reactor, and one or more microphones or acoustic sensors as detectors. In some embodiments, the controller executes a time-indexed acoustic excitation protocol and produces a committed convolution bundle $C_{0:T}$ binding each emitted acoustic waveform to each measured acoustic response under a shared time
 25 base, with the protocol digest recording excitation frequency, amplitude envelope, phasing schedule, and medium identity.

In some embodiments, the same meter families used for optical embodiments — verisimilitude meters, forgery-difficulty meters, calibration-consistency meters, and timing-integrity meters — are computed from the acoustic bundle using meter heads trained on committed
 30 acoustic evidence rather than optical evidence. In some embodiments, the two-dimensional hardness index $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ is evaluated against a declared acoustic-emulator attacker family. In some embodiments, the acceptance and evaluation meter partition is maintained, and the acoustic module supports Level 0 (static challenge-response) and Level 1 (trajectory) verification under the same graduated verification architecture described in Section 8.2.4.

35 In some embodiments, a fleet of acoustic modules performs proof-of-discrepancy using the same method object, independent physical re-execution, and quorum confirmation workflow described in Section 13.3. In some embodiments, the portable bilateral reproducibility protocol described in Section 16.7 applies between an acoustic module and an optical mod-

ule using the same Event, Envelope, and Portable Record structure, with the cross-physics normalisation procedure — how acoustic and optical observations are transformed into a common confirmation space — being an embodiment-specific engineering parameter.

In some embodiments, these non-optical examples illustrate that the physical channel formalism, commitment infrastructure, meter partition, graduated verification, fleet corroboration, and bilateral portability are architecturally compatible with physical media that contribute memory, nonlinearity, resonant structure, stochasticity, or device-specific microstructure to a committed emission-observation channel. In some embodiments, whether a specific non-optical medium supports the full verification and corroboration stack at the same depth as the optical core is an empirical property of that medium, its associated transducers, and the medium-specific calibration and validation procedures applied.

Magnetic reactor example (non-limiting). In some embodiments, the same architectural mapping extends to the magnetic domain. In some embodiments, the module comprises a magnetic excitation source, a magnetic or ferromagnetic reactor medium whose transfer function depends on composition, geometry, domain structure, and hysteresis state, and one or more magnetic sensors (for example Hall-effect sensors, fluxgate magnetometers, or magneto-optical sensors). In some embodiments, the controller executes a time-indexed magnetic excitation protocol and produces a committed convolution bundle binding each excitation waveform to each measured magnetic response. In some embodiments, hysteresis and Barkhausen noise in the ferromagnetic medium provide device-specific, history-dependent signatures analogous to those provided by scattering and nonlinearity in optical reactors. In some embodiments, the same meter families, graduated verification levels, fleet corroboration, and bilateral portability protocols apply to the magnetic embodiment under the same architectural mapping, with meter heads and calibration procedures adapted to magnetic observables.

Extension to further non-optical reactor families (non-limiting). In some embodiments, the architectural mapping demonstrated for acoustic and magnetic domains extends to any reactor family that contributes memory, nonlinearity, resonant structure, stochasticity, or device-specific microstructure to a committed emission-observation channel, including but not limited to radio-frequency, mechanical, fluidic, chemical, biological, and multi-physics embodiments. In some embodiments, the common requirement is that the medium’s transfer function is sufficiently complex and device-specific to support verisimilitude and hardness metering under a declared attacker model.

11 Cryptographic Overlays and Commitment Plumbing (Optional, Non-limiting)

This section describes optional cryptographic overlays that may be used to bind, commit, timestamp, selectively disclose, aggregate, and govern Reality Kernel evidence. These overlays are not required to implement the Reality Kernel operator and do not replace the physical grounding mechanisms described elsewhere. Instead, they provide standard attestation plumbing on top of the convolution bundle and related summaries.

11.1 Overview and role separation

In various embodiments, cryptographic mechanisms are used for at least one of: (i) integrity of exported logs and summaries, (ii) authenticity of device outputs and identity assertions, (iii) freshness and replay resistance, (iv) selective disclosure and privacy-aware sharing, (v) rate limiting of outputs so that cryptographic material is physically bound to device operation, and (vi) multi-device attestation and quorum mechanisms in distributed deployments. The physical channel remains the source of empirical hardness for forgery, while cryptography provides tamper evidence and protocol structure.

11.2 Hash-chain and one-way state update

In some embodiments, the system maintains a compact cryptographic state that evolves one-way with each emission–observation step, binding the run history into a tamper-evident chain. A chain state χ_t is updated each step using the convolution-bundle sample $c_t = (u(t), \mathbf{y}_t)$ as defined in Section 6:

$$\chi_{t+1} = F(\chi_t, c_t, \text{meta}_t),$$

where F may be instantiated as a cryptographic hash or sponge construction, a MAC or PRF under a device-held key, a verifiable random function (VRF), or a ZK-friendly algebraic hash, and meta_t denotes auxiliary metadata (for example timestamps, sensor identifiers, or calibration state). The terminal value χ_T serves as a compact fingerprint of the run.

In some embodiments, the emission policy is driven directly by the evolving chain state χ_t , forming a closed loop: $u(t+1) = g(\chi_{t+1}, \text{ctx}_{t+1})$, causing future emission patterns to be a one-way function of prior observations, yielding replay resistance.

One-way function family agility (non-limiting). The system may support migration between one-way function families. For example, if a hash family is deprecated, the controller may switch to a new F while preserving the physical layer. In preferred embodiments, the choice of F is versioned and included in committed metadata, so verifiers can interpret tags consistently across device lifecycle.

11.3 Batching, Merkle commitments, and time anchoring

In some embodiments, the run is partitioned into disclosure atoms and each atom is hashed to produce per-atom digests. These are aggregated using a Merkle tree, yielding a root that commits to the entire run while enabling inclusion proofs for any chosen subset of atoms. Time anchoring is applied to commitments to provide externally verifiable freshness via public timestamping services, blockchain inclusion, verifiable delay functions (VDFs), or other time-locked attestations.

In some embodiments, commitment roots are submitted to append-only transparency logs that publish signed tree heads over time, providing publicly auditable evidence that a device published commitments by a given time and did not later equivocate.

11.4 Record formats and ledger integration (non-limiting)

In some embodiments, exported evidence is packaged as structured records whose fields are bound by commitments and signatures. Non-limiting record types include: enrolment records (device identity binding and baseline profiles), run records (protocol digests, calibration identifiers, meter summaries, and commitment references), PoP/tPoP export records (compact exports referencing committed windows plus time anchors and quorum signatures), rotation records (signed policy updates with compatibility flags), and revocation/quarantine records (signed lifecycle decisions with supporting meter evidence).

11.5 Physically rate-limited cryptographic outputs

In some embodiments, the system produces cryptographic outputs whose generation rate is constrained by physical measurement and control, rather than only by computation. Rate-limited outputs are derived from correlated measurements, meter-bounded variability, and commitment schedules. Rate limiting is enforced by requiring that each output be bound to a fresh time anchor, a fresh protocol digest, or a fresh convolution-bundle window, so outputs are not readily generated arbitrarily fast without reproducing the physical interaction. This also limits the rate at which a device can contribute quorum evidence in multi-device attestation deployments.

In some embodiments, the sensor cadence (for example the frame rate of the observation camera and the sweep rate of the scan program) provides an upper bound on the rate at which fresh convolution-bundle windows can be produced. This bound is evaluated under declared attacker families that may know the device architecture, declared profile, calibration summaries, and protocol class, but do not have the ability to reproduce the corresponding physical measurement event under the declared timing, coupling, and meter conditions. Accordingly, the rate limit is not dependent solely on secrecy of the device profile; it is grounded in the physical cost of producing fresh measurement windows at the declared cadence.

11.6 Secure aggregation, threshold attestation, and multi-device computation

In some embodiments, multiple devices participate in aggregation and attestation protocols using threshold signatures (a quorum jointly produces a signature so that no single device can unilaterally authorise), secure multi-party computation (devices contribute private measurements and jointly compute an aggregate without revealing individual inputs), or homomorphic encryption (devices submit encrypted meter scores that are aggregated in encrypted form and decrypted only as a collective result). In some embodiments, outputs of these protocols are treated as conditioning variables for subsequent Reality Kernel operation.

11.7 Abort handling and physical failsafes

In some embodiments, an abort transitions the device into a safe regime by disabling emission, parking scan mirrors, reducing gain, and recording an abort marker. Abort also triggers a bounded flush of buffers and a commitment step for already-recorded windows, so partial runs remain tamper-evident. Sealed reactor loops include interlocks and watchdogs, and scene-facing emissions include eye-safety constraints.

11.8 Selective disclosure and privacy-aware sharing

In some embodiments, the system supports privacy-aware sharing by committing to large records and selectively disclosing only the minimum required windows, features, or summaries for an audit or verification task. Disclosure policy may be role-dependent. Selective disclosure is combined with two-seed opening to reduce anticipatory manipulation, and with escalation policies that request additional openings only when summary-level checks fail.

Post-quantum cryptographic primitives (optional). In some embodiments, cryptographic components are instantiated with post-quantum primitives, for example hash-based signature schemes such as XMSS or SPHINCS+, providing defence in depth for digital bindings while the physical channel provides empirical hardness.

Decentralised identity and registry integration (optional). In some embodiments, device identity is integrated with decentralised identity or registry systems, assigning each device an identifier that resolves to a record containing public keys, an enrolment-profile digest, policy version identifiers, and revocation status.

11.9 Security under profile disclosure: cadence-bounded, scene-coupled, and network-anchored forgery cost

In some embodiments, disclosure of the device profile does not by itself defeat verification, because the relevant security cost is the cost of producing fresh measurement windows under

the declared protocol, timing, and meter conditions. A first component is cadence-bounded freshness; a second component arises in declared Yoked embodiments; and a third component arises in declared network-anchored embodiments. Accordingly, the relevant question is not whether θ is secret, but whether an attacker can reproduce the declared fresh physical interaction under the declared digital, physical, and latency budgets.

12 Security Theory: Hardness Indices and Empirical Security

This section consolidates the mathematical foundations of empirical security for Reality Kernel systems. Security is grounded in physical hardness rather than computational complexity assumptions.

Security-related embodiments in this disclosure are stated as engineering security claims grounded in committed physical evidence, declared attacker classes, meter envelopes, and periodically recalibrated hardness evaluations. Formal proof-system properties, zero-knowledge properties, and reduction-based guarantees are neither required nor implied except where expressly stated for a particular digital primitive.

12.1 Two-dimensional hardness index

In some embodiments, empirical hardness is characterised by a pair $(k_{\text{dig}}^*, m_{\text{ana}}^*)$, where $k_{\text{dig}}^*(\theta; \varepsilon)$ is the minimal model capacity (parameter count, FLOPs, or training budget) required to drive the balanced classification error $\text{err}_k^*(\theta)$ below a threshold ε against frozen evaluation meters on held-out challenge windows, and $m_{\text{ana}}^*(\theta; \varepsilon, r)$ is the minimal physical emulation scale required to reduce the analogue distinguishability advantage $\text{Adv}_{m,r}^{\text{ana}}(\theta)$ below the same threshold ε . Both quantities are formally defined in Section 21 (digital hardness) and the analogue hardness paragraph therein. The pair captures both computational and physical-access dimensions of forgery difficulty, and both are conditioned on a declared threat model and meter envelope rather than asserted unconditionally. In some embodiments, temporal modulation of reactor properties (Section 10) amplifies both dimensions: k_{dig}^* increases because an attacker model would need to capture near-optical-cycle switching dynamics in addition to spatial structure, and m_{ana}^* increases because temporal probes add an independent challenge dimension orthogonal to spatial patterns. Where time-varying media enable nonreciprocal responses, the attacker is additionally denied an equivalent reverse-path measurement under the declared threat model, further increasing the effective m_{ana}^* .

Hardness as a time-indexed, attacker-indexed quantity (non-limiting). Empirical hardness $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ is not a permanent property of a device; it is a snapshot conditioned on the current state of attacker modelling capability and the attacker’s accumulated physical-access budget. As emulator architectures improve and as challenge–response data accumu-

lates through legitimate use or leakage, the effective $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ for a given device and meter family may decrease. The physical device, by contrast, does not improve autonomously. Security is therefore a *race condition*: the device’s hardness at time t is meaningful only relative to the attacker class available at time t . In some embodiments, this is managed by (i) periodic
 5 re-evaluation of fleet-derived hardness thresholds against updated emulator families, (ii) reactor refresh or reconfiguration that changes the challenge–response mapping, (iii) query throttling and selective disclosure to limit the effective m_{ana}^* available to any single observer, and (iv) fleet-level monitoring for population-wide hardness degradation trends. Nothing in this specification should be read as asserting that empirical hardness is time-invariant or
 10 unconditionally durable. A novel contribution is the management framework itself—periodic re-evaluation, reactor refresh, query throttling, and fleet-level monitoring—which provides systematic tools for maintaining useful hardness over time, not an assertion that any particular hardness value is permanent.

Query-complexity bound for the digital hardness index (non-limiting). The query
 15 throttle rate required to maintain the hardness index at a declared target k_{thresh}^* depends on the rate at which adversary model capacity can be effectively increased per additional query to the target device. In some embodiments, this query-complexity curve $k_{\text{dig}}^*(\theta; \varepsilon; Q)$ — the digital hardness index as a function of adversary query budget Q — is estimated empirically for the declared reactor type and adversary model family by training models of
 20 increasing capacity k on CRP datasets of increasing size Q and recording the capacity at which balanced classification error drops below ε . Because k_{dig}^* decreases with Q as additional queries provide larger training sets to the adversary, the slope $\partial k_{\text{dig}}^* / \partial Q$ is non-positive, and the query throttle threshold Q^* is set as the Q at which the magnitude of the slope $|\partial k_{\text{dig}}^* / \partial Q|$ exceeds a policy-declared sensitivity $\Delta_{\text{policy}} > 0$; beyond Q^* , additional queries
 25 meaningfully accelerate an adversary’s model improvement and the throttle engages. In some embodiments, the query-complexity curve and the derived throttle threshold are committed to the protocol digest and re-estimated at each declared re-evaluation cycle.

Average-case summary use of the hardness index pair (non-limiting). The hardness index pair $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ is used as an average-case summary of security under declared
 30 regularity conditions: the challenge protocol draws challenges uniformly from a declared set $\mathcal{X}_{\text{prot}}$, and $\delta_{\text{irr}}(X, i)$ has bounded variance across $\mathcal{X}_{\text{prot}}$. Under these conditions, k_{dig}^* characterises the adversary’s average-case difficulty under the uniform challenge distribution and the pair provides a useful scalar summary for the declared average case. The scalar pair is not a sufficient characterisation for all applications: as detailed in Section 21.2, two devices
 35 can have identical scalar indices while having very different hardness profiles, and full hardness curves are required for cross-device comparison, throttle calibration, adaptive-attack analysis, and applications sensitive to tail risk or non-uniform challenge distributions. When the regularity condition does not hold — that is, when δ_{irr} is concentrated on a small region of the challenge space — the hardness index is supplemented by a concentration diagnostic:

in some embodiments, the spatial variance of δ_{irr} across the security domain $\mathcal{X}_{\text{sec}}$ is committed to the protocol digest, and concentrated distributions trigger a challenge-space diversity protocol.

Race condition formalisation: Nakamoto analogy (non-limiting).

The security condition stated above — hardness holds as long as $k < k_{\text{dig}}^*(\theta; \varepsilon)$ for the declared device and threshold — is analogous in structure to Nakamoto consensus security, which holds as long as the adversary controls fewer than 50 % of total hash power. In both cases, security is a quantity-vs-threshold race condition rather than a reduction-based proof. The disanalogy is also instructive: in Nakamoto consensus, adversary power is measured as a fraction of total network power, and the attack probability decreases geometrically in block depth k at rate $(q/p)^k$. In the Reality Kernel setting, adversary capability is an absolute scale (model capacity, physical emulation scale, query count) rather than a fraction of fleet capacity, and security degrades in discrete steps when a new model class crosses the threshold, rather than continuously. In some embodiments, the management tools for this race condition are: periodic re-evaluation (Nakamoto-analogous to honest-miner majority monitoring), reactor refresh (analogous to changing the hash function), query throttling (analogous to proof-of-work difficulty adjustment), and fleet-level hardness monitoring (analogous to network hash-rate tracking). The PoD architecture is structured so that an adversary who controls only a minority of the participating fleet, under declared assumptions of device-population diversity, independence of manufacturing and calibration provenance, Sybil-resistance of the participation set, and integrity of the verifier-acceptance pipeline, faces a degradation of attack power against committed PoD results that grows with the number of independent re-executions; absolute corruption resistance is not asserted, and the threat model under which this property is claimed is committed to the protocol digest. Failure modes outside the stated threat model — including correlated manufacturing defects, coordinated firmware compromise, model-update poisoning, and verifier capture — are treated as separate risks managed by the fleet-governance and meter-partition mechanisms described elsewhere in this disclosure.

Meter partition for hardness evaluation (non-limiting).

In some embodiments, the meter vector $\mathbf{m}(t)$ is partitioned into *acceptance meters* $\mathbf{m}^{\text{accept}}(t)$ (used for admissibility gating during operation) and *evaluation meters* $\mathbf{m}^{\text{eval}}(t)$ (used for hardness assessment, adversarial testing, and frozen-meter discrimination). In embodiments using the stated partition, hardness indices are computed from evaluation meters; acceptance meters are excluded from the set against which hardness is assessed under that partition. This partition is structured to avoid self-referential hardness definitions in which a meter’s inclusion in the acceptance set simultaneously determines and is determined by the hardness index. In some embodiments, evaluation meters are held out from the optimization loop (including the drifting field), so that the system is not expected to adapt to satisfy the metrics used to judge it under the declared optimisation loop constraints. The partition is declared in protocol digests and bound

into commitments so that auditors can verify which meters served which role.

Residual indirect gaming risk (non-limiting). The meter partition is designed to block the direct Goodhart optimisation pathway: under the stated partition, no gradient flows from evaluation meters into the optimisation loop, and trust-weight updates explicitly
 5 exclude evaluation meters (Section 14). However, acceptance meters and evaluation meters measure the same underlying physical channel, so they are statistically correlated. An agent or adaptation process optimising acceptance meters will exert *indirect* pressure on evaluation meters through those correlations. This is a residual risk inherent in any holdout-evaluation design. In some embodiments, the risk is bounded by (i) maintaining low correlation between
 10 acceptance and evaluation meter sets (achieved by selecting meters that capture complementary aspects of channel behaviour), (ii) periodic rotation of which meters serve which role, so that indirect gaming pressure is reduced on a fixed evaluation set over extended operation, and (iii) fleet-level threshold adaptation, which reactively adjusts evaluation benchmarks if population-level metric drift is detected. Rotation schedules and correlation audits are
 15 logged in protocol digests. Each protocol digest declares the partition in force for its epoch; rotation means the subsequent epoch's digest declares a new assignment, while commitments and audit records made under prior partitions remain valid and interpretable under the partition that was in force when they were created. The partition is a first-order defence; these mechanisms provide second-order adaptivity against learned correlations.

Governance-Partition Isolation (non-limiting). In some embodiments, the Reality
 20 Kernel apparatus comprises a governance partition isolated from one or more agent behaviour generation pathways. The governance partition is configured to receive, derive, store, verify, or emit governance-side signals, including without limitation meter outputs, floor-proximity flags, protocol digest entries, committed-bundle summaries, calibration records, attestation
 25 records, authority decisions, rescue-mode triggers, selective-opening decisions, or audit tokens. The isolation property means that an agent behaviour pathway cannot directly read, write, erase, rewrite, suppress, or directly influence the governance partition except through declared evidence interfaces committed to the protocol digest.

In some embodiments, the governance partition receives declared inputs from one or more
 30 apparatus components while remaining outside the optimisation loop that updates agent policy, behaviour, reward, planning, self-modelling, or actuation parameters. For example, an agent may produce actions that are sensed by the apparatus and therefore affect future meter values, but the agent does not possess direct authority to modify the meter implementation, the governance logs, the authority signatures, the committed thresholds, or the governance-
 35 partition state. This separates an evidentiary causal effect from a direct control path.

In some embodiments, governance-partition isolation is implemented by one or more of: a separate processor, a separate die, a separate package, a separate sensor path, a separate power domain, a hardware security module, a trusted execution environment, a micro-controller, a read-only measurement path, an append-only log, a one-way data diode, an

air-gapped or optically isolated channel, a separate virtual machine, a separate process, a separate key domain, a separate signing authority, a threshold authority, a hardware-rooted attestation path, or a physically isolated memory substrate. The foregoing examples are non-limiting; the load-bearing property is the absence of an undeclared direct agent-side read,
 5 write, or influence path over the governance partition.

In some embodiments, the governance partition stores or emits a governance signal g_t derived from a committed bundle, meter vector, or protocol digest:

$$g_t = G_{\text{gov}}(C_{0:t}, m(t), \Pi_t, A_t),$$

where A_t may include authority records, attestation records, selective-opening authorisations, or other governance-side inputs. The behaviour-generation pathway may receive only
 10 a declared, possibly lossy, delayed, thresholded, or redacted projection $\tilde{g}_t$ of g_t :

$$\tilde{g}_t = R_{\text{decl}}(g_t, \Pi_t),$$

where R_{decl} is a declared release function committed to the protocol digest. These equations are illustrative only and do not limit the governance partition to any particular computational architecture or release function.

In some embodiments, the governance partition is used to support the faithfulness of
 15 privacy-preserving audit. For example, a privacy layer may release only a conclusion, a commitment, a proof, an accumulator state, a selective-opening result, or a lossy channel output, while the governance partition preserves the evidence that the released signal was computed from the declared committed bundle and not fabricated by the agent pathway.

Isolation-mechanism generalisation (non-limiting). Governance-partition isolation
 20 generalises to physical, logical, cryptographic, temporal, procedural, optical, electrical, software, firmware, or hybrid separation mechanisms. Physical examples include separate dies, separate packages, one-way links, sealed modules, and air gaps. Logical examples include separate processes, execution contexts, virtual machines, containers, privilege domains, and enclaves. Cryptographic examples include separate key domains, threshold signing authori-
 25 ties, attested measurement roots, signed release policies, and append-only authenticated logs. Temporal examples include committed past states that cannot be modified from runtime behaviour pathways. No particular isolation technology is essential unless expressly recited in a claim.

12.2 Physical rate-limiting

30 In some embodiments, the sensor cadence provides an empirical upper bound on new-window production rate. Combined with commitment binding, this yields a physically rate-limited token stream: a device can produce at most n independently verified tokens per unit time, where n is determined by sensor cadence and protocol overhead. Under declared attacker

families without the ability to reproduce the corresponding physical measurement event under the declared conditions, exceeding this rate is empirically infeasible.

12.3 Fleet latent space calibration

In some embodiments, a fleet of Reality Kernels jointly calibrates a shared latent space by exchanging committed reference sweeps under standardised protocols. Fleet-calibrated latent spaces enable cross-device comparison of meter outputs, hardness indices, and anomaly thresholds. In some embodiments, fleet calibration uses secure aggregation so that individual device responses are not disclosed to the fleet, only population statistics. In some embodiments, a fleet-derived hardness threshold is computed as a quantile of the cross-device hardness distribution, providing a data-driven acceptance boundary that adapts as the fleet evolves. In some embodiments, the fleet-calibrated latent space provides the device configuration representations used in the proof-of-discrepancy protocol (Section 13), and the irreducible residual defined in Section 13.2 provides a device-specific hardness contribution that complements the population-level hardness index.

12.4 Reality Kernel algebra and modular composition

In some embodiments, Reality Kernel modules are treated as composable operators with explicit input and output interfaces. A module may be specified by (i) an emission or actuation interface, (ii) an observation interface producing the convolution bundle and meter summaries, and (iii) a policy interface defining admissible control protocols.

Modules as kernels with typed interfaces (non-limiting). For descriptive convenience, an RK module K may be modelled as a controlled Markov kernel

$$K_{\theta}(\cdot \mid x, u) \in \mathcal{P}(\mathcal{Y}),$$

mapping an extended state x and a control u (including scan law, projection schedule, gains, alignment, and other commands) to a distribution over observable records or summaries in a space $\mathcal{Y}$. In some embodiments, $\mathcal{Y}$ includes the convolution bundle $C_{0:T}$, meter vectors, timing anchors, and protocol digests. In some embodiments, interface compatibility is enforced by policy-defined port types and meter-envelope constraints, so composition is asserted only within a compatibility class.

Serial and parallel composition (non-limiting). If two modules K_1 and K_2 expose compatible interfaces, their serial composition may be written informally as

$$K_{\text{comp}} = K_2 \circ K_1,$$

where outputs or effective fields of K_1 are treated as part of the inputs available to K_2 (for example through a shared scene, a shared platform state, or direct reactor–reactor ports). In some embodiments, modules are also composed in parallel, written informally as $K_{\text{par}} = K_1 \otimes K_2$, where both modules operate over the same event window and their outputs
 5 are combined by an aggregation rule.

Containment (nesting) composition (non-limiting). In some embodiments, a third composition mode—containment—is distinguished from serial and parallel composition. In containment composition, written informally as $K_{\text{nest}} = K_2 \triangleright K_1$, the outer module K_1 establishes boundary conditions (electromagnetic, acoustic, thermal, or other physical constraints)
 10 that modify the operating environment of the inner module K_2 . Both modules produce independent convolution bundles and meter summaries, but K_2 ’s transfer function depends on K_1 ’s boundary state. Unlike serial composition, the signal does not pass sequentially from K_1 to K_2 ; instead, K_2 is *immersed in* the environment defined by K_1 . Modulating K_1 ’s boundaries (as in sweeping reactor operation, Section 10.27) continuously changes K_2 ’s operating
 15 environment, enabling cross-module coupling measurements. Nested (Matryoshka) reactor embodiments (Section 10.27) instantiate this composition mode. The algebraic properties of $\triangleright$ (for example whether $(K_3 \triangleright K_2) \triangleright K_1$ is equivalent to $K_3 \triangleright (K_2 \triangleright K_1)$, and how $\triangleright$ interacts with serial and parallel composition) are governed by the physical coupling topology of the specific embodiment and are treated as empirical engineering targets rather than formal
 20 identities, consistent with the algebraic-laws discussion below.

Composition without shared secrets (non-limiting). In some embodiments, modules are composed without shared secrets by binding evidence to public commitments and policy-logged protocol digests. Each module contributes (i) commitments to disclosure atoms, (ii) protocol digests $\Pi^{(i)}$, and (iii) meter summaries, and a higher-level composition oper-
 25 ator combines these contributions using quorum aggregation and selective opening. This supports deployments in which modules can be assessed independently and later combined using public logs, inclusion proofs, and policy-defined trust weighting.

Summaries and auditability (non-limiting). In some embodiments, a summary map Σ_{sum} is applied to raw records to reduce bandwidth, for example

$$\Sigma_{\text{sum}} : (C_{0:T}, \Pi, \mathbf{m}) \mapsto \mathbf{s},$$

30 where $\mathbf{s}$ is a compact summary. In some embodiments, commitments and selective disclosure provide an auditable tie between summaries and raw atoms, so that a verifier can request openings when summary-level checks fail or when escalation criteria are met.

Algebraic laws as engineering targets (non-limiting). In some embodiments, desirable algebraic properties are treated as empirical engineering targets rather than exact

identities. For example, associativity and commutativity may hold only approximately under meter-bounded operation, and are evaluated using the stability metrics defined in the following subsection.

12.5 Markov property metrics for engineered stability (optional)

- 5 In some embodiments, Reality Kernel designs and protocol families are engineered to exhibit approximate stability properties that make serial composition and networked operation more predictable. Such properties may be quantified by non-limiting empirical functionals defined over an effective transition operator induced by a protocol.

10 For descriptive convenience, let K_u denote the effective one-step operator under a protocol choice u (which may bundle scan law, projection schedule, gains, alignment, and other controls), acting on an internal extended state X and inducing an observable record (or feature summary) through $C_{0:T}$ and meters. Let $d(\cdot, \cdot)$ denote any non-limiting discrepancy on induced observables or summaries (for example, a distance on meter vectors, a feature-space distance on the convolution bundle, or a divergence between induced distributions).

- 15 **Kernel interpretation for the following functionals (non-limiting).** For the purposes of the following functionals, K_u is interpreted as an effective transition operator on a declared summary or representative state space (for example a sufficient statistic, an embedding into a fixed Hilbert space, or a declared finite-dimensional projection of the observable record), rather than as the underlying law over observable records; the choice of summary
20 or representative is committed to the protocol digest, and the functionals are evaluated on the chosen representative space under a declared metric d .

Approximate idempotence. A non-limiting idempotence functional under a protocol u is

$$\nu_{\text{idemp}}(u) = \mathbb{E}_{X \sim \mu}[d(K_u \circ K_u(X), K_u(X))],$$

- where μ is a chosen reference distribution over operating states (for example, a replay buffer
25 over recent conditions).

Contractivity. A non-limiting contractivity functional may be defined as a Lipschitz-like bound:

$$\nu_{\text{contr}}(u) = \sup_{X \neq X'} \frac{d(K_u(X), K_u(X'))}{d(X, X')}.$$

Mixing. A non-limiting mixing functional over a horizon n is

$$\nu_{\text{mix}}(u; n) = \sup_{X, X'} \text{TV}\left(\mathbf{P}_{\theta, u}^{(n)}(\cdot \mid X), \mathbf{P}_{\theta, u}^{(n)}(\cdot \mid X')\right),$$

- where $\mathbf{P}_{\theta, u}^{(n)}$ denotes the n -step induced law under parameters θ and protocol family u , and
30 TV is total variation distance (or another non-limiting divergence).

Approximate commutativity under serial composition. For two protocol choices u and v , a non-limiting commutativity functional is

$$\nu_{\text{comm}}(u, v) = \mathbb{E}_{X \sim \mu}[d(K_v \circ K_u(X), K_u \circ K_v(X))].$$

12.6 ε -nice kernels and tolerance specification

In some embodiments, a Reality Kernel instance is characterised as ε -nice under a family of protocols when the Markov property functionals defined above are bounded by ε :

$$\mathcal{K}_\varepsilon = \left\{ P_\theta : \nu_{\text{idemp}} \leq \varepsilon, \nu_{\text{contr}} \leq \varepsilon, \nu_{\text{mix}} \leq \varepsilon, \nu_{\text{comm}} \leq \varepsilon \right\}.$$

Instances in $\mathcal{K}_\varepsilon$ exhibit stable behaviour that simplifies composition and network operation. In practice, ε is understood as a vector of per-functional tolerances ($\varepsilon_{\text{idemp}}, \varepsilon_{\text{contr}}, \varepsilon_{\text{mix}}, \varepsilon_{\text{comm}}$), each scaled to the range and units of its respective functional; the single-symbol notation is a shorthand for the requirement that all four tolerances are simultaneously satisfied.

12.7 Self-modification channel limitation (non-limiting)

In some embodiments, the Inner Creation mechanism (Section 8.3.8) permits a reactor to reshape its own attractor landscape. The same physical channel that enables self-calibration could in principle be used to sculpt misleading attractor patterns. Physical constraints (microstructure unclonability, thermal noise, bifurcation topology) raise the cost of such misrepresentation, and in preferred embodiments Inner Creation is rate-limited, protocol-restricted, and logged. All layers of the evidence architecture are ultimately grounded in the same physical channel; this is an inherent limitation of any evidence system built on a shared physical substrate. Extended analysis of this limitation is described in the related Filing 2 application (optional, non-essential).

13 Fleet Model Improvement and Proof-of-Discrepancy Protocol

This section describes methods (FIG. 5) by which a fleet of Reality Kernel devices jointly improves a shared predictive model of device behaviour, using a distributed protocol in which participants discover and submit reproducible experimental methods that expose errors in the predictive model, and in which, in preferred embodiments, verification of submitted methods includes independent physical re-execution on distinct devices.

Terminological note (non-limiting). The term *proof-of-discrepancy* is used throughout this document as a term of art denoting an empirically calibrated fleet-confirmation protocol, not a cryptographic proof system with formal soundness and completeness properties.

What the protocol provides is: (i) committed evidence that a discoverer’s device produced a challenge–response measurement with an asserted residual against the fleet predictive model, (ii) independent physical re-execution on one or more distinct unclonable devices in the verifier role, scored against the fleet residual-margin estimator, and (iii) auditability via the submission record’s commitment binding and the verifier’s confirmation record. It does not provide, and should not be read as asserting, a zero-knowledge or interactive proof in the cryptographic sense. Achieving formal proof-system guarantees would require additional identity, Sybil-resistance, and adversary-model assumptions beyond those specified here. The “discrepancy” in the protocol name refers to the protocol residual that fleet residual-margin calibration commits and that the verifier physically reproduces; it is distinct from the discrepancy term in Bayesian calibration in the manner of Kennedy and O’Hagan, which is treated as a nuisance parameter rather than as a committed protocol resource (see the novelty discussion at Section 12.3).

13.1 Predictive model with device configuration representation

In some embodiments, a predictive model $\mathcal{Q}$ accepts as input a challenge configuration X (comprising at minimum a control protocol or protocol seed specifying emission patterns, timing, gain settings, and other controllable parameters) and a device configuration representation $\mathbf{d}_i$ drawn from a fleet of device representations (a vector or structured object encoding device-specific properties), and produces a predicted response:

$$\hat{Y} = \mathcal{Q}(X, \mathbf{d}_i).$$

The device configuration representation $\mathbf{d}_i$ is a learned or inferred quantity that captures how device i differs from a population-level model of device behaviour. In some embodiments, $\mathbf{d}_i$ is a vector in a continuous latent space; in some embodiments, $\mathbf{d}_i$ is a structured representation comprising sub-vectors corresponding to distinct physical subsystems (for example scattering medium properties, detector characteristics, and optical path parameters).

In some embodiments, the predictive model $\mathcal{Q}$ is implemented as a neural network, a Gaussian process emulator, a conditional generative model, a physics-based simulator with adjustable parameters, or any combination thereof. The specification does not restrict the architecture of $\mathcal{Q}$; what is required is that $\mathcal{Q}$ accepts both a challenge specification and a device-specific representation and produces a predicted response that can be compared with a physical measurement.

Training the predictive model (non-limiting). In some embodiments, the predictive model is trained on challenge–response data collected from a population of devices under standardised protocols. Training data comprises tuples $(X_n, \mathbf{d}_{i_n}, Y_n)$ where Y_n is the measured response of device i_n to challenge X_n . In some embodiments, $\mathbf{d}_i$ and the parameters of $\mathcal{Q}$ are jointly optimised from fleet data. In some embodiments, training uses a hierarchical

structure in which population-level parameters are shared and device-level representations $\mathbf{d}_i$ are inferred per device.

Regularisation of device configuration representations (non-limiting). Without regularisation, $\mathbf{d}_i$ can over- or under-fit the available challenge-response data from device i , inflating or collapsing $\delta_{\text{irr}}(X, i)$ and thereby mis-calibrating the hardness index. In some embodiments, a Gaussian process prior is placed on $\mathbf{d}_i$: the prior mean is the population-level representation $\mathbf{d}_0$ and the prior covariance is a declared kernel (for example a radial basis function with length-scale and output-scale hyperparameters committed to the protocol digest). The posterior over $\mathbf{d}_i$ given the calibration set then provides shrinkage toward the population mean, with the degree of shrinkage decreasing as more calibration data are collected. In some embodiments, an explicit bias-variance diagnostic is computed: the posterior predictive variance on held-out challenges is committed to the protocol digest, and a bias-variance alarm is triggered if the variance exceeds a declared threshold (indicating under-fitting) or if the in-sample residual drops below the noise floor (indicating over-fitting). These diagnostics are included in the hardness index so that the security margin is bounded from below by the degree of regularisation and from above by the noise floor, not by an uncharacterised fit quality.

In some embodiments, the dimensionality of $\mathbf{d}_i$ is explicitly bounded by a declared maximum d_{max} , motivated by the number of resolvable physical degrees of freedom of the reactor (estimated from the scattering mean free path, beam diameter, and camera resolution for optical embodiments). Representations with dimension exceeding d_{max} are not permitted; the bound is committed to the protocol digest and is constant across re-evaluation cycles unless a hardware change is declared. In some embodiments, the regularisation prior also imposes an ℓ_2 norm constraint $\|\mathbf{d}_i - \mathbf{d}_0\|_2 \leq r$ for a population-level radius r estimated empirically from fleet data; representations outside this radius correspond to devices with non-representative manufacturing history that the fleet model is not expected to cover.

In some embodiments, a sample-complexity bound characterises how quickly $|\delta_{\text{irr}}|$ shrinks as the adversary’s calibration set size n_{adv} grows: for a device satisfying the learnability-exclusion condition, the expected reduction in $\|\delta_{\text{irr}}\|^2$ per additional CRP under the declared adversary model is empirically estimated on the anchor embodiment and committed to the protocol digest. The query throttle rate is set so that n_{adv} remains below the declared threshold n_{adv}^* at which the shrinkage rate would reduce the security margin below σ_{sec}^2 , a policy-declared security floor.

Generalisation of $\mathcal{Q}$ to novel challenges (non-limiting). The hardness claims for novel challenges — challenges outside the adversary’s training distribution — require not only that the fleet model $\mathcal{Q}$ generalises, but that the irreducible residual δ_{irr} remains substantial on those challenges. In some embodiments, the generalisation of δ_{irr} to novel challenges is characterised using the residual-margin estimator (Section 12.3), which implements fleet residual-margin calibration on the committed fleet calibration data. In a Gaussian-process

instantiation, the posterior predictive variance at an unseen challenge X_{new} provides a lower bound on the expected squared residual, $\text{Var}[\delta_{\text{irr}}(X_{\text{new}}, i) | \text{data}] \geq \sigma_{\text{post}}^2(X_{\text{new}})$, where σ_{post}^2 is the posterior variance computed from the fleet calibration data. Other estimator classes (neural posteriors, physics-based predictive distributions, or hybrids) may instantiate the residual-margin estimator equivalently; the estimator class is not load-bearing. The estimator output is treated as a committed protocol resource confirmed under the Proof-of-Discrepancy protocol by independent physical re-execution on distinct unclonable devices, not as a nuisance term to be marginalised. This distinguishes fleet residual-margin calibration from Bayesian calibration in the manner of Kennedy and O’Hagan, in which the discrepancy term is typically estimated as a nuisance and integrated out. In a Gaussian-process instantiation, the predictive uncertainty serves as the operationally used security margin: regions of the challenge space where σ_{post}^2 is high under the GP prior (long-range challenge correlations not yet covered by fleet data) are expected to be hard to forge. In some embodiments, the hardness claim on novel challenges is conditioned on challenges at distance $> r_{\text{adv}}$ from the adversary’s training set in the declared challenge-space metric, where r_{adv} is bounded by the adversary’s query budget and the density of the challenge space under the declared protocol distribution; this bound is reported in the protocol digest alongside the hardness index.

Fleet scaling and GP sample complexity (non-limiting). The GP posterior variance converges as fleet calibration data accumulates. For a d -dimensional challenge-device input space $(X, \mathbf{d}_i)$ and a kernel with length scale ℓ , the posterior standard deviation at an unseen point at distance r from the nearest training point decays as $O(r^\nu)$ for a Matérn kernel with smoothness ν . The effective input dimension is bounded by d_{max} , estimated from the number of resolvable spatial modes N_{modes} and the declared device-representation dimension; in practice d_{max} is the dominant sample-complexity driver. No general closed-form convergence theorem for heterogeneous fleets is claimed; the operational approach is to use sparse or inducing-point GP approximations (non-limiting) to maintain tractable inference as fleet size grows, with the number of inducing points and the approximation error committed to the protocol digest. Fleet scalability is therefore an empirical and implementation claim, not a formal convergence claim of this specification.

Novelty of the fleet Proof-of-Discrepancy protocol relative to prior art (non-limiting clarification). The closest statistical prior art for fleet residual-margin calibration is the Bayesian calibration framework of Kennedy and O’Hagan (2001) and the Gaussian-process emulation literature that follows from it. Fleet residual-margin calibration is distinguishable from that framework on three load-bearing points. First, Bayesian calibration in the manner of Kennedy and O’Hagan treats the discrepancy term as a nuisance to be estimated and marginalised, with inferential interest in the calibrated parameters of the computer model; fleet residual-margin calibration treats the protocol residual as the committed resource of interest, with the residual-margin estimator providing the operational security margin that the Proof-of-Discrepancy protocol commits and physically re-executes.

Second, Bayesian calibration in the manner of Kennedy and O’Hagan does not involve physical re-execution on distinct unclonable devices as a confirmation step; fleet residual-margin calibration confirms the committed residual by independent physical re-execution on distinct unclonable devices as its verification primitive. Third, Bayesian calibration in the manner of Kennedy and O’Hagan operates at the level of a single computer model against physical observations; fleet residual-margin calibration operates at fleet level, with the residual decomposition separating shared model discrepancy from device-specific variation and from the irreducible clone-resistant residual maintained as a security primitive. No cited prior calibration system, authentication, or model-improvement system uses independent physical re-execution on distinct unclonable devices as the verification primitive for confirming a model discrepancy. In particular: (i) Optical PUF apparatus including that of Davis et al. (EP3252740B1) operates as a single-device, single-function authentication system and does not describe a fleet-level protocol in which multiple devices jointly improve a shared predictive model or in which the irreducible residual is deliberately maintained as a security margin. (ii) Optical random projection systems (including those of LightOn and related photonic reservoir computing systems of Brunner and Larger) treat the optical layer as a computational substrate for machine-learning inference; the internal state of the optical layer is not committed as a tamper-evident record and those systems do not support multi-regime operation under a shared physical operator. (iii) Chained optical transformation systems such as those of Naughton, Hennelly, and Dowling use fixed optical elements in a CFB-mode encryption scheme; they do not describe a closed feedback loop with an evolving reactor state, a logged control protocol, or a convolution bundle in which each sample pairs emission parameters with measured responses. The combination of (a) physical re-execution on independently manufactured devices as the confirmation primitive, (b) the five-term error decomposition separating shared model discrepancy, device-specific variation, and irreducible clone-resistant residual, and (c) the protocol digest commitment binding the confirmation record to the physical channel event is, to the best of the inventors’ knowledge, absent from the cited prior art and from any combination thereof.

Physical Irreducibility Conjecture (non-limiting, research direction). The residual-margin estimator (Section 12.3) provides an operationally used lower confidence bound on δ_{irr} under the declared estimator class; in a Gaussian-process instantiation, this bound is the GP posterior variance under the declared GP model class. A deeper and as yet unproved question is whether a stronger, model-class-independent lower bound exists. The following is stated as a research conjecture rather than a claim of the present disclosure.

Physical Irreducibility Conjecture. Let a device satisfy the following physical conditions: (i) the scattering medium has a transport mean free path ℓ_s and wavelength λ such that $N_{\text{modes}} \leq c_0 \Delta^2 / \lambda^2$ resolvable spatial modes exist in the measurement aperture of diameter Δ ; (ii) the device dynamics have at least one positive Lyapunov exponent (sensitive dependence on initial conditions); (iii) observations are corrupted by instrument noise $\sigma_{\text{noise}}^2 > 0$. Then

there exists a function $\sigma_{\text{phys}}^2(N_{\text{modes}}, k, T) > 0$ depending only on these physical parameters such that, for any learning procedure using a model class with at most k parameters, the minimax prediction error on novel challenges satisfies

$$\min_{f \in \mathcal{H}_k} \mathbb{E}[\|Y_i(X) - f(X, \mathbf{d}_i)\|^2] \geq \sigma_{\text{phys}}^2(N_{\text{modes}}, k, T) > 0 \quad \text{for all } k < k^*(N_{\text{modes}}, T, \sigma_{\text{noise}}^2),$$

and this lower bound is independent of the adversary model class and holds for any learning procedure given T observations.

Even without a model-class-independent lower bound, the device retains a measurable empirical irreducibility: the irreducible residual δ_{irr} is directly observable as the held-out discrepancy surviving independent physical re-execution under declared model classes, query budgets, and meter envelopes, and is reported as a bound at the declared credibility level.

Absent the conjecture, the bound remains an operational measurement.

If true, this conjecture would constitute a bridge between three bodies of results that are individually well-developed but not yet connected: (i) Cramér–Rao bounds and channel capacity for the optical scattering channel as a function of N_{modes} and T ; (ii) the Kaplan–Yorke conjecture relating Lyapunov spectrum to attractor fractal dimension; and (iii) PAC learning lower bounds converting channel capacity (information about θ extractable in T observations) into a model-complexity threshold k^* . The proof technique most likely to proceed: (a) establish an optical channel converse bounding $I(\theta; C_{0:T})$ by $N_{\text{modes}} \cdot \log(1 + \text{SNR})$; (b) apply Fano’s inequality to convert this to a minimax lower bound on any estimator of θ ; (c) use the resulting bound on estimable degrees of freedom to lower-bound prediction error for any k -parameter model. In some embodiments, this programme is pursued empirically by reporting $\text{err}_k^*(\theta; Q)$ as a function of Q (query count) and k (model capacity) across a well-characterised family of scattering tokens with known grain structure, testing the prediction that error curves plateau at positive values scaling with ℓ_s/λ and N_{modes} . The empirical plateau (if observed) would constitute evidence in favour of the conjecture and would ground the security claims of the present disclosure on a physical law rather than on current model-class limitations alone.

Extension to unseen devices (non-limiting). In some embodiments, a device configuration representation $\mathbf{d}_j$ for a previously unseen device j is inferred from a calibration set of challenge–response pairs $\{(X_n, Y_n)\}_{n=1}^{N_{\text{cal}}}$ obtained by executing a calibration protocol on device j . In some embodiments, inference is performed by optimising $\mathbf{d}_j$ to minimise discrepancy between $\mathcal{Q}(X_n, \mathbf{d}_j)$ and Y_n for the calibration challenges, with the parameters of $\mathcal{Q}$ held fixed. In some embodiments, inference is performed by an amortised encoder—a learned function that maps a set of challenge–response pairs directly to a device configuration representation without iterative optimisation.

In some embodiments, the number of calibration challenges required to infer a device configuration representation to a given accuracy is determined empirically and depends on the dimensionality of $\mathbf{d}$, the information content of each challenge–response pair, and the

complexity of device-to-device variation. The calibration set need not exhaustively cover the challenge space; it need only contain sufficient information to locate the device in the learned representation space.

Fleet geometry in representation space (non-limiting).

In some embodiments, the set of inferred device configuration representations $\{\mathbf{d}_i\}$ for a fleet of devices exhibits structure reflecting manufacturing processes, material properties, and fabrication conditions. Devices from the same manufacturing batch or process may cluster in representation space; devices from distinct processes may occupy distinct regions. In some embodiments, the geometry of the representation space is used for fleet management, including identification of device families, detection of anomalous devices, allocation of computational or sensing tasks based on device characteristics, and assessment of fleet diversity.

13.2 Five-term error decomposition

In some embodiments, a population-level representation $\mathbf{d}_0$ is defined (for example as a learned population mean in the representation space, or as the representation that minimises average prediction error across the fleet). The population-level prediction and the device-specific prediction are then:

$$\hat{Y}_0(X) = \mathcal{Q}(X, \mathbf{d}_0), \quad \hat{Y}_i(X) = \mathcal{Q}(X, \mathbf{d}_i).$$

The per-device residual—the quantity directly observable after inference of $\mathbf{d}_i$ —is:

$$e_i(X) = Y_i(X) - \hat{Y}_i(X).$$

In some embodiments, the measured response $Y_i(X)$ comprises a noise-free physical response $Y_i^{\text{true}}(X)$ and measurement noise $\varepsilon(X, i)$, such that $Y_i(X) = Y_i^{\text{true}}(X) + \varepsilon(X, i)$. In some embodiments, a noise-reduced residual $\bar{e}_i(X)$ is obtained by repeating the measurement under the same conditions and averaging, or by applying a denoising operator consistent with a noise model, so that $\bar{e}_i(X) \approx \mathbb{E}[e_i(X)]$.

In some embodiments, the measured response of device i is decomposed into five terms with three functionally distinct discrepancy components:

$$Y_i(X) = \underbrace{\hat{Y}_0(X)}_{\text{population model}} + \underbrace{\delta_{\text{dev}}(X, i)}_{\text{device signature}} + \underbrace{\delta_{\text{sys}}(X)}_{\text{shared discrepancy}} + \underbrace{\delta_{\text{irr}}(X, i)}_{\text{irreducible residual}} + \varepsilon(X, i),$$

where $\varepsilon(X, i)$ denotes measurement noise and the three discrepancy components are defined on noise-reduced quantities as follows:

- (i) **Device-specific captured variation (device signature)** $\delta_{\text{dev}}(X, i)$: the difference

between the device-specific prediction and the population prediction,

$$\delta_{\text{dev}}(X, i) = \hat{Y}_i(X) - \hat{Y}_0(X) = \mathcal{Q}(X, \mathbf{d}_i) - \mathcal{Q}(X, \mathbf{d}_0).$$

This term captures how device i differs from the fleet average in ways the model can represent. It is a modelled quantity (computed entirely from the predictive model) and provides a computational characterisation of the individual device reflecting its microstructure, fabrication history, and physical properties.

- (ii) **Shared model error (shared discrepancy)** $\delta_{\text{sys}}(X)$: the component of the noise-reduced per-device residual $\bar{e}_i(X)$ that is common across devices. In some embodiments, the shared model error is *defined as* the robust central tendency of the noise-reduced residuals across the fleet under a declared aggregation rule (for example a mean, trimmed mean, or median) committed in the protocol digest:

$$\delta_{\text{sys}}(X) := \text{agg}_i \bar{e}_i(X),$$

where agg denotes the declared and pre-committed aggregation functional (for example robust mean, trimmed mean with declared trim fraction, or fleet median). The aggregation rule, trim parameters, and any subpopulation stratification are recorded in the protocol digest as protocol parameters before the discovery procedure begins. Under this definition, $\delta_{\text{sys}}(X)$ is an exact functional of the observed fleet residuals, not a latent quantity; the equality sign in the decomposition above is exact by construction. Shared model error indicates where the predictive model $\mathcal{Q}$ itself is inadequate, regardless of which device is measured. Regions of the challenge space where $\delta_{\text{sys}}(X)$ is large are the primary targets of the proof-of-discrepancy protocol. In some embodiments, the aggregation is performed within subpopulations of devices sharing similar configuration representations, to capture errors that are common within a manufacturing family but not necessarily fleet-wide; the subpopulation partition is also pre-committed.

- (iii) **Irreducible residual** $\delta_{\text{irr}}(X, i)$: the device-specific component of the noise-reduced residual after removing the shared model error:

$$\delta_{\text{irr}}(X, i) = \bar{e}_i(X) - \delta_{\text{sys}}(X).$$

This residual represents aspects of device i 's physical response that persist even when the best available device configuration representation $\mathbf{d}_i$ is used and the shared model error is accounted for. It arises from the device possessing more internal degrees of freedom than the model can access through the available measurement channels.

After subtracting the baseline prediction $\hat{Y}_0(X)$ and the modelled device signature $\delta_{\text{dev}}(X, i)$ from the raw observation, the remaining per-device residual retains only the three unmodelled

terms. Accordingly, the observed (noisy) per-device residual satisfies:

$$e_i(X) = \delta_{\text{sys}}(X) + \delta_{\text{irr}}(X, i) + \varepsilon(X, i).$$

Identifiability of the additive decomposition (non-limiting). The five-term decomposition above is additive by construction: $\delta_{\text{sys}}(X)$ is defined as the fleet aggregate of residuals, and $\delta_{\text{irr}}(X, i)$ is the per-device residual after removing δ_{sys} . Without additional statistical
 5 structure, the three discrepancy components ($\delta_{\text{dev}}, \delta_{\text{sys}}, \delta_{\text{irr}}$) are not jointly identifiable from finite samples in general: shifting mass between δ_{dev} and δ_{irr} can leave the observed residuals unchanged. In some embodiments, the identifiability gap is addressed by placing independent Gaussian process priors on $\delta_{\text{sys}}(X)$ and the device-specific component of the residual, with
 10 orthogonal covariance kernels (for example distinct length-scale and output-scale hyperparameters committed to the protocol digest); the resulting posterior decomposes the observed residual into components whose credible intervals are disjoint at the declared credibility level. In this Gaussian-process instantiation, the residual decomposition implements fleet residual-margin calibration (Section 12.3) applied to the per-device residual, which is informed by the Bayesian calibration literature in the manner of Kennedy and O’Hagan but extends that
 15 literature in the respects identified in the novelty discussion above: the protocol residual is the committed object, the residual-margin estimator is the inferential apparatus, and the verifier physical re-execution is the confirmation operation. Where independent GP priors are not used, the decomposition is a definitional accounting identity and the components should be interpreted as defined quantities rather than independent physical quantities; all
 20 downstream claims (hardness index, device identity, PoD confirmation) are stated in terms of observable functionals of the residual rather than in terms of the latent decomposition.

Approximate orthogonality diagnostic (non-limiting). For the functional roles of the decomposition to be cleanly separable, the device-signature component $\delta_{\text{dev}}(X, i)$ and the irreducible residual $\delta_{\text{irr}}(X, i)$ should be approximately orthogonal: $\mathbb{E}_X[\delta_{\text{dev}}(X, i) \cdot \delta_{\text{irr}}(X, i)] \approx$
 25 0 over challenges X sampled from the protocol distribution. If this condition fails — for example because $\mathbf{d}_i$ was over-fit to the calibration set, causing representational error to contaminate δ_{irr} — the security primitive is confounded with modelling artefacts. In some embodiments, after model convergence, the empirical correlation $\widehat{\text{Corr}}[\delta_{\text{dev}}(\cdot, i), \bar{e}_i(\cdot) - \delta_{\text{sys}}(\cdot)]$ is computed on a held-out challenge set and committed to the protocol digest; a declared
 30 orthogonality threshold ρ_{orth} triggers a security alert if the empirical correlation exceeds it, indicating that further regularisation of $\mathbf{d}_i$ is needed before the irreducible residual can serve as a security primitive.

Security domain and discovery domain partition (non-limiting). The three functional roles of the discrepancy — security primitive, device identifier, and model improvement
 35 signal — create a structural tension: using challenges for PoD submissions exposes them to model improvement (role iii), which may absorb part of δ_{irr} into a more expressive δ_{dev} under

future model updates, shifting the enrolled authentication signature. In some embodiments, this tension is managed by partitioning the challenge space into two declared sub-domains: (a) a *security domain* $\mathcal{X}_{\text{sec}}$ where challenges are never disclosed in PoD submissions, δ_{irr} is protected from model improvement, and device authentication is performed; and (b) a *discovery* domain $\mathcal{X}_{\text{disc}}$ where challenges are used for PoD submissions and model improvement. The partition $(\mathcal{X}_{\text{sec}}, \mathcal{X}_{\text{disc}})$ is declared and committed in the protocol digest before enrolment. The security domain is designed to be representative of the device’s physical microstructure but non-overlapping with the discovery domain; in some embodiments, the partition is drawn randomly and committed cryptographically at enrolment time. The learnability-exclusion condition must hold independently on $\mathcal{X}_{\text{sec}}$, evaluated against adversary families that have access to PoD submissions from $\mathcal{X}_{\text{disc}}$ but not from $\mathcal{X}_{\text{sec}}$.

Functional roles of the three discrepancy components within the five-term decomposition (non-limiting). In some embodiments, each component of the decomposition serves a distinct functional role within the fleet protocol:

Shared model error drives model improvement. Regions of the challenge space where $\delta_{\text{sys}}(X)$ is large indicate where the predictive model is most inadequate. Discovering and characterising these regions is the primary target of the proof-of-discrepancy protocol described in Section 13.3. When shared model error is identified, confirmed by independent physical reproduction, and incorporated into an updated model, the fleet’s collective predictive capability improves.

Device-specific captured variation provides classification and specialisation. The device signature $\delta_{\text{dev}}(X, i) = \mathcal{Q}(X, \mathbf{d}_i) - \mathcal{Q}(X, \mathbf{d}_0)$ is a modelled quantity that characterises how a device differs from the fleet average. Because δ_{dev} depends on the current model version, it may change when the model is updated even if the physical device has not changed. In some embodiments, δ_{dev} is therefore used primarily for fleet management, device classification, and task allocation (assigning computational or sensing tasks to devices whose specific variation patterns make them well-suited to those tasks), rather than for security-critical authentication. In some embodiments where δ_{dev} is used for device identification, the identification is bound to a specific model version.

Irreducible residual provides an empirically calibrated security margin against cloning under declared attacker families and meter envelopes, and in some embodiments serves as the primary authentication primitive. The irreducible residual $\delta_{\text{irr}}(X, i)$ captures the component of the per-device response that is not well-approximated by the predictive model under declared resource bounds, and is therefore difficult to forge using a digital model alone. It provides a security margin against adversaries who attempt to simulate or clone a device’s responses using a model-based approach. Unlike δ_{dev} , which is explicitly computed from $\mathcal{Q}$, the irreducible residual reflects physical properties of the device that the model does not capture; as a result, its magnitude tends to remain comparatively stable across model updates, though it is not definitionally invariant to model changes and may shift if a substantially more expressive model is deployed. In some embodiments, the magnitude of the irreducible

residual at a given challenge is used as a hardness metric: challenges where the irreducible residual is large are challenges where the device’s response is most difficult to forge under the current model class. In some embodiments, device authentication and provenance attestation are grounded in the irreducible residual rather than in the model-derived device signature.

5 **Relationship to prior calibration formulations (non-limiting).** Fleet residual-margin calibration relates to prior calibration approaches as follows. Bayesian calibration in the manner of Kennedy and O’Hagan provides the closest statistical antecedent and supplies the Gaussian-process emulation tooling that may instantiate the residual-margin estimator; in such frameworks, the discrepancy term is typically treated as a nuisance parameter to be
 10 estimated and marginalised over, with the goal of reducing its impact on parameter inference and prediction. Frequentist calibration approaches and adversarial benchmarking are more distant. Metrological inter-comparison and proficiency testing share the fleet-level scope but do not commit the protocol residual or perform verifier physical re-execution. The five-term decomposition in Section 13.2 extends the Bayesian calibration literature by treating the
 15 protocol residual not as a deficiency to minimise but as a structured resource whose components serve distinct and complementary roles within the fleet protocol; none of the prior approaches treats the protocol residual as a committed resource confirmed by independent physical re-execution on distinct unclonable devices, which is the load-bearing distinction of fleet residual-margin calibration. This functional reframing—from “model error as statistical
 20 nuisance” to “model error as multi-role protocol resource”—is a distinguishing feature of the disclosed system.

In some embodiments, the conditions under which this reframing is valid are stated explicitly as follows. The three functional roles create exactly one structural tension: as the fleet model $\mathcal{Q}$ improves over time (role iii, δ_{sys} decreases), a more expressive model can potentially
 25 absorb part of what was δ_{irr} into a better δ_{dev} (threatening role i, the security primitive). The condition under which roles i and iii remain non-conflicting is that the learnability-exclusion condition holds permanently: the fleet model never achieves sufficient expressiveness to substantially reduce δ_{irr} for the device population under the declared adversary families. This condition is not definitionally guaranteed and must be monitored empirically: in some em-
 30 bodiments, $|\delta_{\text{irr}}|$ is tracked across model update cycles, and a declared learnability-exclusion violation threshold triggers a security alert if $|\delta_{\text{irr}}|$ drops by more than a stated fraction between consecutive model updates. The functional reframing is valid under the learnability-exclusion condition; the monitoring protocol is the empirical check on whether the condition continues to hold.

35 **Model-update stability criterion and subspace separation (non-limiting).** In some embodiments, the following model-update stability criterion is declared in the protocol digest and checked after each fleet model update: (i) on the *improvement subspace* (challenge set $\mathcal{X}_{\text{disc}}$ used for PoD submissions), the shared model error δ_{sys} must decrease or remain stable—a model update that increases δ_{sys} on the improvement subspace is

rejected and triggers diagnostic audit; (ii) on the *verification subspace* (challenge set $\mathcal{X}_{\text{sec}}$ reserved for authentication), the lower confidence bound on $|\delta_{\text{irr}}|$ must not collapse beyond a declared tolerance η_{sec} between model versions — a larger collapse triggers a security alert and halts model deployment pending re-evaluation; (iii) the cross-subspace information leakage (estimated as conditional mutual information between model-improvement updates and verification-subspace residuals) must remain below a declared leakage threshold I_{leak} , committed in the protocol digest. In some embodiments, cross-leakage is bounded by the disclosure policy on improvement-subspace challenge configurations (submitted PoD records do not include raw verification-subspace traces), and the residual leakage is empirically upper-bounded on the anchor embodiment. These three conditions, checked jointly after each model update, constitute the model-update stability criterion; a model version that fails any condition is not deployed until the failure is diagnosed and remediated.

Decomposition is architectural, not post-hoc (non-limiting). In some embodiments, the five-term decomposition arises naturally from the architecture of the predictive model and the structure of fleet data. The device signature δ_{dev} is determined by the difference between device-specific and population-level model predictions. The shared model error δ_{sys} is the fleet-wide central tendency of the per-device residuals. The irreducible residual δ_{irr} is the device-specific component that remains after the shared error is removed. No separate decomposition algorithm beyond standard fleet-level residual analysis is required.

13.3 Proof-of-Discrepancy protocol

In some embodiments, a distributed protocol—referred to herein as a proof-of-discrepancy protocol—rewards participants for discovering reproducible experimental methods that expose errors in the shared predictive model. The protocol operates as follows in non-limiting outline:

Method submission. A participant (the “discoverer”) identifies a challenge configuration X^* and associated operating conditions under which the predictive model’s prediction $\mathcal{Q}(X^*, \mathbf{d}_i)$ diverges from the measured physical response $Y_i(X^*)$ by more than a declared threshold, after accounting for measurement noise and known device-specific variation. The claimed discrepancy is evaluated on the per-device residual $e_i(X^*) = Y_i(X^*) - \mathcal{Q}(X^*, \mathbf{d}_i)$ or a statistic derived therefrom. The discoverer submits X^* and associated operating conditions as a reproducible experimental method—a specification sufficient for another participant to execute the same experiment on a different device. In some embodiments, the submission is committed (for example via a cryptographic commitment scheme) before the verifiers are selected, to reduce the risk of retroactive modification.

Controlled execution conditions (non-limiting). In some embodiments, proof-of-discrepancy methods are executed against a standardised characterisation target, an

internal reactor or controlled scattering medium, or an environment specification sufficiently constrained that different parties can reproduce it (for example a supplied calibration artefact or a specified internal operating mode). The protocol need not require reproduction of arbitrary real-world scenes; the subject of the predictive model is the device’s own physical response under controlled conditions.

Submission commitments (non-limiting). In some embodiments, each submission commits to the challenge configuration X^* , the version identifier of the predictive model against which the discrepancy is claimed, the calibration protocol identifier, and any device firmware or operating mode identifiers relevant to reproducibility. In some embodiments, submissions further specify environment constraints, allowed operating ranges, and data processing procedures sufficient for independent reproduction under comparable conditions. Verification is performed against the committed model version, even if a newer version is available at the time of verification.

Non-limiting submission record format: method-object tuple for discrepancy

submissions. In some embodiments, the submitted method object is represented as a tuple comprising at least a challenge configuration, a device or source-class identifier, a committed protocol digest, a model-version identifier, a measured-response commitment, an asserted residual or scoring statistic, environmental or calibration constraints, and any required reproduction instructions or selective-opening commitments.

In some embodiments, a submission record includes the following fields (serialised in a canonical binary format such as CBOR or a length-prefixed binary encoding):

challenge_config Challenge configuration X^* (operating point coordinates, scan protocol parameters, environmental setpoints).

model_version Cryptographic hash of the predictive model version Q against which the discrepancy is claimed.

device_id Device configuration representation $\mathbf{d}_i$ or anonymised device identifier.

protocol_digest Per-run protocol digest Π_{dig} specifying calibration identifiers, firmware version, and data-processing pipeline.

measured_response Committed hash of the discoverer’s measured response $Y_i(X^*)$.

asserted_residual Summary statistic of $e_i(X^*)$ (for example mean, variance, or a test statistic exceeding the declared threshold).

environment_spec Environmental constraints sufficient for independent reproduction (temperature range, humidity, vibration class, or “internal calibration target” indicator).

timestamp Externally anchored timestamp (for example GNSS time or a trusted timestamping service).

commitment Cryptographic commitment binding all preceding fields.

In some embodiments, a verifier’s confirmation record includes the same fields for the verifier’s device plus a cross-reference to the discoverer’s commitment hash. In some embodiments, the discrepancy is scored by a statistical test (for example a two-sample test com-

paring discoverer and verifier residuals against model predictions, with a pre-registered significance level recorded in the protocol digest).

Verifier selection and physical re-execution. In some embodiments, one or more verifiers are selected from the fleet. Each verifier executes the submitted method X^* on their own device under the specified operating conditions and records the physical response. In preferred embodiments, verification includes physical re-execution of the experiment; computational re-evaluation of the model alone does not constitute verification in these embodiments. In some embodiments, verifiers are selected by a randomised process to reduce or deter collusion between discoverer and verifier.

Discrepancy confirmation. In some embodiments, fleet-level confirmation uses an explicit confirmation statistic $T(X^*)$ defined as a robust aggregate of the per-verifier residuals at the submitted challenge configuration X^* . A non-limiting confirmation statistic is:

$$T(X^*) := \text{agg}_{j \in \mathcal{V}} \bar{e}_j(X^*),$$

where $\mathcal{V}$ is the set of participating verifiers, $\bar{e}_j(X^*)$ is the noise-reduced residual of verifier j at X^* , and agg is the pre-registered aggregation rule (for example fleet median, trimmed mean with declared trim fraction, or a quantile-based statistic). This is the verifier-side instantiation of the shared discrepancy functional $\delta_{\text{sys}}(X)$ (Section 13.2) evaluated at the submitted challenge X^* ; the confirmation statistic is numerically large when the verifiers' residuals are collectively large relative to model predictions, consistent with a genuine shared model discrepancy. The discrepancy is *confirmed* if $T(X^*) > \tau_{\text{confirm}}$, where τ_{confirm} is a pre-registered confirmation threshold.

In some embodiments, two non-limiting verifier-population embodiments are supported.

(i) *Matched-verifier embodiment.* All verifiers are of a declared common device class with a declared common response space, and the residuals $\bar{e}_j(X^*)$ are aggregated directly. (ii) *Heterogeneous-verifier embodiment.* Verifiers may be of different declared device classes; in such embodiments, residuals are placed into a declared common response metric committed in the protocol digest before aggregation. A non-limiting heterogeneous-verifier example: each verifier's $\bar{e}_j(X^*)$ is divided by its declared per-device measurement-noise standard deviation $\hat{\sigma}_j$, producing a dimensionless $z_j(X^*) = \bar{e}_j(X^*)/\hat{\sigma}_j$, and $T(X^*) = \text{agg}_{j \in \mathcal{V}} z_j(X^*)$. Other declared common metrics (response-norm normalisation, declared calibration-uncertainty normalisation, or a committed common-metric tag) are non-limiting alternatives. The choice of embodiment, the declared common metric where applicable, and the per-device normalising quantities are committed to the protocol digest before verification begins.

In some embodiments, a quorum rule additionally requires that at least q out of n verifiers individually exceed a per-device exceedance threshold τ_{device} . The parameters $(n, q, \tau_{\text{confirm}}, \tau_{\text{device}}, \text{agg})$ are pre-committed in the protocol digest before verification begins and constitute part of the PoD method record. In some embodiments, the aggregation

functional is chosen with its breakdown point explicitly stated in the protocol digest. In the convention used here, the finite-sample breakdown point of an aggregator is the largest fraction of contamination below which the aggregator’s output remains bounded under worst-case contamination; for the sample median this convention gives a breakdown point of 0.5, with up to $\lfloor (n-1)/2 \rfloor$ colluding verifiers tolerable when strict tolerance is required, or up to $\lfloor n/2 \rfloor$ when the convention permits boundary contamination. A trimmed mean with trim fraction α has breakdown point α . The required breakdown point β is declared as the maximum tolerated collusion fraction under the adversary model and must satisfy $\beta \leq$ (breakdown point of the declared aggregator). In some embodiments, the recency of verifier device calibration is additionally declared: each verifier’s $\mathbf{d}_j$ must have been re-estimated within a declared calibration window before the PoD submission, to bound the drift bias in $\bar{e}_j(X^*)$.

In some embodiments, the adversary model for the confirmation protocol is stated explicitly as a Byzantine fault model: at most f of the n verifiers in $\mathcal{V}$ are assumed to be colluding adversaries (Byzantine), while the remaining $n-f$ are honest. The quorum parameters (n, q) are chosen to satisfy $n \geq 3f + 1$ and $q \geq 2f + 1$, or an alternative declared tolerance level is committed in the protocol digest with an explicit derivation. In some embodiments, the protocol-level Byzantine tolerance is

$$f_{\text{BFT}} = \min(\lfloor (n-1)/3 \rfloor, \lfloor (q-1)/2 \rfloor).$$

Where a declared robust aggregator with breakdown point β is used, the operational tolerated collusion level is further capped at

$$f_{\text{oper}} = \min(f_{\text{BFT}}, \lfloor n \cdot \beta \rfloor).$$

The declared confirmation rule, quorum parameters (n, q) , the aggregator breakdown point β , and the resulting f_{oper} are committed in the protocol digest. Consistency of $T(X^*)$ across independently operated verifier devices provides evidence that the discrepancy reflects shared model inadequacy rather than a device-specific artefact, measurement error, or fabricated result.

Physical anchoring relative to adversarial benchmarking (non-limiting). The PoD protocol is related to adversarial benchmarking approaches such as Dynabench, in which human contributors submit examples that fool the current model, driving iterative improvement. A key structural difference in the physical setting is that a submitted challenge X^* must reproduce on independently manufactured verifier devices that have not been inspected by the submitter; this physical reproduction requirement anchors the PoD dataset to physical reality in a way digital adversarial datasets are not. A contributor cannot retroactively alter a device’s physical response, so the submitted challenge must generalise to independent physical instantiations, not merely fool the current digital model. This turns the Dynabench

bias — contributors optimise against the current model’s failure modes — into a *coverage bias* in the physical setting: submitters choose physically accessible challenges from the discovery domain, which may underrepresent challenge regions that are physically difficult to reach (extreme operating conditions, rare material states). In some embodiments, a coverage
 5 scoring component in the incentive mechanism rewards submissions from underrepresented regions of the challenge space; the coverage metric and the definition of “underrepresented” are declared in the protocol digest.

Model update. In some embodiments, confirmed discrepancies are incorporated into the predictive model. The model is updated to reduce shared model error in the region of the
 10 challenge space identified by the confirmed discrepancy, while preserving the model’s fidelity elsewhere. In some embodiments, model updates are validated against held-out challenge–response data before deployment, to detect regressions. In some embodiments, model updates are logged with provenance information linking each update to the specific confirmed discrepancy that motivated it.

Model versioning and archival (non-limiting). In some embodiments, model versions
 15 are append-only and archived, so that prior confirmed discrepancies remain verifiable against the model version under which they were submitted. In some embodiments, pending submissions are evaluated against the model version committed at the time of submission, even if a newer version has been deployed, to avoid retroactive invalidation or alteration of in-progress
 20 verification by model updates.

Incentive structure (non-limiting). In some embodiments, the protocol includes an incentive mechanism that rewards discoverers for confirmed discrepancies. In some embodiments, rewards are scaled by one or more of: the magnitude of the confirmed discrepancy, the novelty of the challenge region (whether the discrepancy occurs in a region of the challenge
 25 space not previously known to be problematic), and the informativeness of the discrepancy for model improvement (for example measured by reduction in predictive error after the model update). In some embodiments, verifiers are also rewarded for faithful execution of the verification protocol. In some embodiments, the incentive mechanism uses tokenised rewards; in some embodiments, incentives are reputational, contractual, or institutional. The
 30 specification does not restrict the form of the incentive mechanism.

13.4 Physical reproduction as the verification primitive

A distinguishing feature of the proof-of-discrepancy protocol is that, in preferred embodiments, verification includes independent physical re-execution of the submitted experimental method on devices that are physically distinct from the discoverer’s device. This section elab-
 35 orates on the properties and implications of this verification primitive.

Why physical reproduction is the preferred verification mechanism (non-limiting). Computational re-evaluation—running the predictive model on the submitted challenge and checking whether the model’s own output diverges from its training data—does not constitute verification of a model–reality discrepancy, because the discrepancy is by definition a property of the gap between the model and physical reality, not a property of the model alone. In preferred embodiments, physical re-execution on an independent device provides a particularly strong verification primitive for establishing what physical reality actually produces in response to a given challenge, because it is grounded in measurement rather than model evaluation alone. Independent physical reproduction on a device that is physically distinct from the discoverer’s device further guards against fabrication: if the discoverer has tampered with their device or measurement apparatus, the fabricated result will not reproduce on the verifier’s independently manufactured and operated device.

Role of physical unclonability (non-limiting). In some embodiments, devices in the fleet exhibit physical unclonability arising from manufacturing stochasticity (for example in the microstructure of scattering media, the precise geometry of optical paths, or the fabrication-specific properties of detector elements). Physical unclonability is designed so that each device’s detailed physical response to a given challenge is unique and is computationally infeasible to replicate by constructing a duplicate device under declared adversary families. This property strengthens the verification primitive: an adversary who fabricates a discrepancy claim by manipulating their own device is unable, under the declared resource bounds, to predict or control what an independently manufactured verifier device will produce when executing the same method, because the verifier device’s microstructure differs in unpredictable ways.

In some embodiments, physical unclonability also supports the expectation that the irreducible residual $\delta_{\text{irr}}(X, i)$ (Section 13.2) is substantial across the fleet: under the declared observability and resource limits, no practical model is expected to eliminate the gap between its predictions and the physical response of a device whose internal degrees of freedom exceed what is accessible through the measurement interface.

Distinction from computational, cryptographic, and statistical verification (non-limiting). The physical reproduction verification primitive is distinct from: computational verification (re-executing code or re-evaluating a function), cryptographic verification (checking a digital signature, zero-knowledge proof, or hash), and statistical verification (checking consistency of reported data against distributional assumptions). Physical reproduction involves the verifier physically operating hardware and performing a measurement. In some embodiments, the physical reproduction verification primitive is combined with one or more of these other verification methods—for example, cryptographic commitments may be used to bind the discoverer’s claim before verification, and statistical tests may be used to assess whether the reproduced measurements are consistent with the claimed discrepancy—but in preferred embodiments, physical re-execution serves as the primary verification step.

13.5 Difficulty scaling from physics

In some embodiments, it is expected—and is stated as a research conjecture in the following paragraph—that the difficulty of discovering new proof-of-discrepancy submissions scales with the fidelity of the predictive model, creating a natural progression analogous to difficulty adjustment in computational proof-of-work protocols but grounded in physics rather than arbitrary computation. This expectation motivates the PoD Monotonicity Conjecture; it is not itself an established result.

PoD Monotonicity Conjecture (non-limiting, research direction). The following is stated as a research conjecture. Let $\mathcal{Q}_t$ denote the fleet predictive model at time t , and let D_t denote the expected number of candidate challenges a computationally bounded discoverer must test before finding a challenge X^* satisfying $|\delta_{\text{sys}}(X^*; \mathcal{Q}_t)| > \tau_{\text{confirm}}$ (the confirmation threshold). Let $\|\mathcal{Q}_t\|$ be a measure of model quality (for example, mean squared prediction error on held-out challenges, decreasing as the model improves). *PoD Monotonicity Conjecture.* D_t is monotone non-decreasing as $\|\mathcal{Q}_t\|$ decreases: improving the fleet model reduces the density of discoverable discrepancies, so each successive confirmed discovery requires more search effort. Formally:

$$\frac{dD_t}{d(-\|\mathcal{Q}_t\|)} \geq 0$$

in expectation over the challenge space distribution and fleet device heterogeneity, under regularity conditions on the challenge space and the confirmation criterion.

For a Gaussian process model $\mathcal{Q}$, this conjecture is tractable: as the GP posterior contracts (more fleet data reduces posterior variance uniformly), the expected number of challenges required to find one with posterior variance exceeding τ_{confirm} increases monotonically. Extension to neural network models requires additional conditions: the model must improve uniformly across the challenge space (no pathological residual-concentration), and the confirmation criterion must remain stable as the model updates. In some embodiments, the conjecture is treated as an empirical claim monitored in the protocol digest: the fleet logs an empirical D_t estimate (from discoverer attempt counts) and reports whether D_t is increasing across consecutive model update cycles. A sustained decrease in D_t following a model update is a signal that the update may have introduced model misspecification or that the challenge space is not being covered adequately.

A formal proof of the PoD Monotonicity Conjecture, even in the simplified GP case, would constitute a new distributed systems result: it would establish that the PoD protocol has a *physically-grounded difficulty adjustment* property analogous to—but mechanistically distinct from—proof-of-work difficulty adjustment, grounded in the depletion of modellable discrepancy mass rather than in an adjustable computational parameter.

Even if discrepancy-discovery difficulty is not monotone across model updates, the PoD protocol continues to provide a physically anchored mechanism for discovering, replicating, superseding, and withdrawing discrepancy claims. Model improvement therefore rests on

reproducible correction cycles rather than on any assumed monotone search-difficulty law.

Resolution-dependent difficulty (non-limiting). As the predictive model improves through incorporation of confirmed discrepancies, the remaining shared model error $\delta_{\text{sys}}(X)$ decreases in magnitude and becomes concentrated in regions of the challenge space where the physics is most complex or where measurement resolution has not yet been sufficient to reveal the discrepancy. Discovering the next discrepancy may require finer measurement resolution, more precise control of operating conditions, or exploration of previously untested regions of the challenge space. This creates a difficulty progression driven by the physics of the measurement interface rather than by an adjustable computational parameter.

Depth of physical complexity (non-limiting). In some embodiments, the physical complexity of the devices is such that increasing measurement resolution may reveal additional discrepancies, because the devices possess more internal degrees of freedom than any model can capture through a finite measurement interface. In some embodiments, empirical evidence indicates that discrepancies persist across accessible scales: real scattering media, optical systems, and other high-degree-of-freedom physical substrates exhibit behaviour at finer resolutions that is not fully predicted by models trained at coarser resolutions. In some embodiments, this property is verified empirically by demonstrating that increasing measurement resolution consistently reveals new discrepancies that were not predicted by the current model.

13.6 Learnability-exclusion condition

In some embodiments, the physical devices used in the proof-of-discrepancy protocol satisfy a practical learnability-exclusion condition: no known feasible learning procedure achieves prediction accuracy above the verification threshold when given access to a bounded number of challenge–response observations from the device. Informally, a computationally bounded adversary is not expected to construct a model that reliably predicts the device’s response to unseen challenges with sufficient accuracy to fabricate verification-passing results.

Physical basis (non-limiting). In some embodiments, the practical learnability-exclusion condition is grounded in physical properties of the device: volumetric multiple scattering through disordered media, controlled optical nonlinearities, partially observable internal state, and a number of effective internal degrees of freedom that exceeds the information accessible through the measurement interface. These properties are intended to make the challenge–response mapping difficult to approximate by functions in classes that are efficiently learnable from input–output examples under known learning algorithms; this learnability-exclusion condition is evaluated empirically against declared attacker families and meter envelopes. In some embodiments, this condition is interpreted

through a complexity-theoretic lens as resistance to polynomial-time characterisation, but the operative definition is empirical rather than formal.

Empirical verification (non-limiting). In some embodiments, the practical learnability-exclusion condition is verified empirically rather than proved mathematically. Verification comprises training state-of-the-art machine learning models (for example deep neural networks, kernel methods, compressed sensing algorithms, or other approaches representing the best known learning algorithms) on challenge-response data from a target device and demonstrating that prediction accuracy on held-out challenges plateaus below the verification threshold. In some embodiments, this verification is performed periodically as learning algorithms improve, to confirm that the exclusion condition continues to hold under the current state of the art.

Relationship to irreducible residual (non-limiting). The practical learnability-exclusion condition provides the physical basis for the irreducible residual $\delta_{\text{irr}}(X, i)$ described in Section 13.2. If the condition holds, the irreducible residual remains substantial regardless of the sophistication of the predictive model, because the model is not expected to close the gap without information that is physically inaccessible through the measurement interface.

Operative lower bound on $\|\delta_{\text{irr}}\|$ (non-limiting). In some embodiments, the residual-margin estimator (Section 12.3) yields an explicit posterior credible interval on each component of the five-term decomposition including δ_{irr} . In a Gaussian-process instantiation of the residual-margin estimator (informed by the Bayesian calibration literature in the manner of Kennedy and O’Hagan), after the GP posterior is conditioned on all available challenge-response data from the fleet, the lower tail of the marginal posterior on $\delta_{\text{irr}}(X, i)$ provides an operative lower bound: the irreducible residual is declared to exceed $\delta_{\text{irr}}^{\min}$ at posterior probability $(1 - \alpha)$ for a declared credibility level α (for example $\alpha = 0.05$). Other estimator classes (neural posteriors, physics-based predictive distributions, or hybrids) may instantiate the residual-margin estimator equivalently; the estimator class is not load-bearing, but the commitment of the residual and the verifier physical re-execution are. This bound is a function of the training CRP budget n_{CRP} , the declared attacker model class $\mathcal{H}$, and the estimator hyperparameters, all of which are committed to the protocol digest. In some embodiments, the bound is updated periodically as new fleet data arrive and new attacker model families are evaluated; the hardness index (Section 12) includes $\delta_{\text{irr}}^{\min}$ as a component, so that the stated security margin is directly traceable to the residual-margin estimator output rather than to an unquantified claim. An attacker producing a discrepancy below this bound is within the fleet’s calibrated noise floor and does not constitute a Proof-of-Discrepancy event. The bound shrinks if new learning algorithms reduce the unexplained residual and is updated accordingly; no absolute floor is asserted.

13.7 Trust-bootstrapped reference devices

In some embodiments, the proof-of-discrepancy protocol is anchored by one or more reference devices whose measurements are trusted to higher confidence than measurements from general fleet participants. Reference devices may be operated in controlled environments, characterised under metrological standards (for example consistent with ISO/IEC 17025 requirements for testing and calibration laboratories), or operated by trusted parties. Reference-device measurements provide an external ground truth that helps anchor the fleet model against drift into self-consistent but physically inaccurate states.

In some embodiments, the predictive model is periodically validated against reference device measurements, and discrepancies between model predictions and reference device responses are treated as high-priority model update targets. In some embodiments, new fleet participants are initially calibrated against reference device measurements to establish their device configuration representations with known accuracy before participating in the broader protocol.

13.8 Scoring and method evaluation (non-limiting)

In some embodiments, submitted methods are evaluated using a scoring function that reflects one or more of the following properties:

- (i) **Confirmation:** the degree to which the submitted method’s predicted discrepancy is confirmed by independent physical reproduction.
- (ii) **Discovery:** the degree to which the discrepancy is novel—occurring in a region of the challenge space where the model was previously considered adequate, or exceeding previously known discrepancy magnitudes.
- (iii) **Coverage:** the degree to which the submitted method explores regions of the challenge space that have been under-sampled by previous submissions, contributing to broader characterisation of model adequacy.

In some embodiments, the scoring function is derived from a proper scoring rule to incentivise honest reporting of predicted discrepancy magnitudes. In some embodiments, the scoring function is a weighted combination of the above components, with weights adjusted over time to balance model improvement (favouring discovery and coverage) against verification robustness (favouring confirmation).

13.9 Robustness against adversarial participants (non-limiting)

In some embodiments, the proof-of-discrepancy protocol incorporates mechanisms to mitigate adversarial behaviour by participants, including:

Sybil resistance. In some embodiments, the cost of manufacturing and operating a physical device (including acquisition, calibration, and ongoing maintenance) provides a natural Sybil resistance mechanism: creating fake identities would typically require acquiring and operating physical hardware under the declared identity-anchoring scheme, which is more expensive than creating digital pseudonyms. In some embodiments, device identity is verified via microstructure-based challenge–response authentication (Section 23.4) to confirm that each fleet participant controls a distinct physical device. In some embodiments, as a fallback that does not rely solely on manufacturing cost, device identity is additionally anchored by a cryptographic commitment of the device’s enrolled configuration representation $\mathbf{d}_i$ (or a secure hash thereof) published to a tamper-evident log at enrolment time; a physical one-way function (as described in the one-way state update and hash-chain logging subsection) evaluated on the device’s response to a held-out challenge set, whose output is committed and unsealed at identity-claim time, provides an additional Sybil barrier that is robust to cost collapse of the manufacturing barrier. These cryptographic and physical commitment mechanisms are non-limiting and may be combined in any proportion declared in the identity-anchoring policy. In some embodiments, a minimum manufacturing cost threshold $C_{\min}$ is declared in the identity-anchoring policy and committed to the protocol digest; the Sybil resistance claim based on manufacturing cost is conditioned on this threshold, and the policy requires re-evaluation of the cryptographic-commitment fallback whenever market device cost reports indicate costs approaching $C_{\min}$. A cross-verification protocol — in which each new device is co-located with and cross-verified by at least k_{cross} existing trusted fleet devices before receiving voting weight — provides an additional Sybil barrier independent of manufacturing cost.

Collusion resistance. In some embodiments, verifiers are selected by a randomised process from the fleet, so that discoverers are not expected to predict or choose which devices will verify their submissions. The protocol-level Byzantine tolerance, quorum parameters, declared robust-aggregator breakdown point, and operational tolerated collusion level are computed and committed as described above, including f_{BFT} , β , and f_{oper} . A colluding minority below the declared operational tolerated collusion level may delay, but not corrupt, a PoD confirmation under the declared confirmation rule.

Fabrication resistance. Physical unclonability of devices is designed so that fabricated discrepancy claims—where a discoverer reports a discrepancy that does not correspond to a genuine model–reality gap—are unlikely to reproduce on independently manufactured verifier devices. The physical reproduction verification primitive is the primary fabrication resistance mechanism.

Robust aggregation (non-limiting). In some embodiments, model updates derived from confirmed discrepancies are aggregated using robust statistical methods that down-weight or exclude outlier contributions. In some embodiments, contributions from devices

with unusual verification histories are weighted differently from contributions with consistent track records.

Submission economics (non-limiting). In some embodiments, the protocol includes economic mechanisms to discourage frivolous or adversarial submissions, such as requiring
 5 a bond or deposit that is returned upon confirmation and forfeited upon rejection, or rate-limiting submissions based on device identity, reputation, or staking.

Update containment (non-limiting). In some embodiments, confirmed model updates are evaluated in a shadow or quarantine mode before fleet-wide deployment: the updated model is tested against held-out challenge-response data to detect regressions (cases where
 10 a correction in one region of the challenge space degrades fidelity in another). In some embodiments, the challenges used for verification of submitted discrepancies are drawn from a different distribution than the challenges used for model training, to reduce overfitting risk with respect to the distribution of submitted challenges.

13.10 Meter partition for the proof-of-discrepancy protocol (non-limiting) 15

In some embodiments, the challenge space of each device is partitioned into disjoint subspaces allocated to distinct functional roles:

- (i) **Verification subspace:** challenges reserved for device authentication and identity verification. Responses to challenges in this subspace are used for microstructure-based
 20 identification and are not disclosed to the fleet model.
- (ii) **Sensing and computation subspace:** challenges used for computational tasks, reservoir computing, or sensing applications.
- (iii) **Model improvement subspace:** challenges used for fleet model calibration and proof-of-discrepancy submissions.

25 In some embodiments, the partition is supported by protocol-level constraints intended to keep challenges from one subspace from being reused in another, analogous to the security meter partition described in Section 12. The partition is designed so that information disclosed during model improvement does not compromise device authentication, and that verification challenges are not predictable from publicly shared model-improvement
 30 data. In some embodiments, model-improvement disclosures (including submitted methods, confirmed discrepancies, and model updates) are constrained so that they do not leak verification-subspace challenges or any sufficient statistics from which verification-subspace responses could be predicted. In some embodiments, published proof-of-discrepancy records include the challenge configuration, model version, and a discrepancy statistic with confidence
 35 bounds, while raw device response traces are retained under access control or disclosed

via selective opening of committed records consistent with the disclosure policies described elsewhere in this specification.

13.11 Embodiments and non-limiting examples

The proof-of-discrepancy protocol is described in Section 13 (and, where relevant, Section 13.3) in general terms applicable to any physical device fleet satisfying the stated conditions (a predictive model accepting device configuration representations, physical devices with manufacturing stochasticity, and a measurement interface through which challenges can be executed and responses recorded). The following are non-limiting examples of specific embodiments:

Optical scattering embodiment. In some embodiments, the devices are optical scattering systems comprising a spatial light modulator or structured illumination source, a scattering medium (for example a diffuser, multimode fibre, integrating sphere, or disordered volumetric medium), and a detector array. Challenges comprise illumination patterns projected onto the scattering medium. Responses comprise the detected intensity patterns. The predictive model predicts detected intensity patterns from illumination patterns and device configuration representations. Manufacturing stochasticity in the scattering medium’s microstructure provides physical unclonability.

Projector–camera embodiment. In some embodiments, the devices are Reality Kernel modules as described elsewhere in this specification, operated in a regime where the scene is replaced by a characterisation target or where the reactor’s internal scattering properties are the subject of the predictive model. The convolution bundle provides the challenge–response record. Fleet calibration via the proof-of-discrepancy protocol improves the fidelity of the fleet-shared surrogate model used for simulation, training, and cross-device transfer.

Phonon and acoustic embodiment. In some embodiments, the devices are acoustic or phononic scattering systems in which challenges comprise acoustic excitation patterns and responses comprise measured acoustic fields. In some embodiments, the devices are SASER (sound amplification by stimulated emission of radiation) systems or phonon laser systems, where coherent phonon generation provides structured excitation and scattering through disordered solid-state or fluid media provides the high-DOF challenge–response mapping.

Generalisation condition (non-limiting). The proof-of-discrepancy protocol is applicable to any physical device fleet where: (a) individual devices have more internal degrees of freedom than are accessible through the measurement interface, (b) multiple independently manufactured instances exist, (c) a submitted experimental method can be executed on other instances with comparable results modulo device-specific variation, and (d) device-to-device

variation is large enough for distinct device identities yet structured enough to be separable from shared model error.

Terminology correspondence (non-limiting). In the context of Reality Kernel devices as described elsewhere in this specification, the challenge configuration X corresponds to a control protocol or protocol seed specifying the emission schedule and associated parameters; the measured response $Y_i(X)$ corresponds to the convolution bundle or a derived summary thereof. Specifically, $Y_i(X)$ denotes a summary statistic or derived quantity computed from the observation sequence $\{\mathbf{y}_t\}$ produced by device i under control protocol X . The device configuration representation $\mathbf{d}_i$ corresponds to the device’s position in the fleet-calibrated latent space described in the Fleet Latent Space Calibration subsection. The proof-of-discrepancy protocol operates over the same evidence structures (committed convolution bundles, protocol digests, and meter summaries) used elsewhere in this specification.

13.12 Relationship to adjacent frameworks (non-limiting)

Distinction from physical proof-of-work. Physical proof-of-work protocols use physical processes to replace computational hash puzzles for blockchain consensus. The work product in such protocols is the evaluation of a physical one-way function—a solved puzzle. The proof-of-discrepancy protocol differs in that the work product is a reproducible experimental method demonstrating where a predictive model is wrong. The objective is model improvement through error discovery, not consensus through puzzle-solving.

Distinction from adversarial benchmarking. Adversarial benchmarking platforms reward participants for discovering inputs that cause computational models to produce incorrect outputs. The proof-of-discrepancy protocol differs in that: (a) the proof object is a reproducible physical experimental method, not a digital input example; (b) in preferred embodiments, verification includes independent physical re-execution on distinct devices, not computational re-evaluation or human labelling; and (c) the devices exhibit physical unclonability that raises the empirical cost of result fabrication under the declared attacker model.

Distinction from metrological inter-comparison. Inter-laboratory comparisons and proficiency testing schemes use independent physical measurement as a validation mechanism. The proof-of-discrepancy protocol differs in that: (a) participants are incentivised to actively discover model errors, not merely to demonstrate measurement competence; (b) each device is physically distinct and unclonable, rather than measuring a circulated reference artefact; and (c) the protocol treats model–reality discrepancy as a multi-functional resource rather than as a deficiency to be corrected.

Distinction from microstructure-based authentication protocols. PUF-based blockchain and authentication systems use manufacturing stochasticity for device identity and key generation. The proof-of-discrepancy protocol differs in that: (a) model discrepancy itself is the valued protocol output, not merely a security property; (b) devices participate
 5 in a model improvement loop, not only an authentication loop; and (c) verification is by physical reproduction of a submitted method, not by challenge–response authentication against enrolled templates.

Distinction from calibration compliance systems. Calibration-enforcement blockchain systems (for example metrology-aware consensus protocols) embed calibration compliance
 10 into validation logic. The proof-of-discrepancy protocol differs in that: (a) the protocol rewards discovery of where models fail, not attestation that devices are calibrated; and (b) verification is by physical re-execution, not by cryptographic certificate checking.

Distinction from sensor-fusion and shared-scene witnessing systems. Multi-camera, multi-sensor, and Bayesian-fusion systems (including Kalman-filter, particle-filter,
 15 interacting-multiple-model, and factor-graph fusion frameworks) combine multiple sensor measurements into a single fused estimate of an underlying state. Multi-RK corroboration as described in subsequent sections differs in that: (a) the load-bearing output is a set of independently committed convolution bundles, each retaining its own protocol digest, meter envelope, and microstructure-bound provenance, rather than a fused estimate that
 20 collapses per-sensor provenance into a single state estimate; (b) cross-kernel agreement is evaluated at the meter-score and discriminator level (verisimilitude, hardness, freshness, projection-hardness, microstructure identity) rather than at the raw-observation level, so cross-kernel comparison does not require and does not depend on a shared measurement-noise model; (c) each kernel’s evidence is independently auditable and selectively disclosable
 25 under the selective-disclosure policy, supporting verifier-side reconstruction of the cross-kernel argument without exposing fused intermediate states; and (d) corroboration weights are bound to declared trust and reputation parameters committed in the protocol digest, rather than derived from estimator-internal innovation statistics or shared-state covariance. Where a fused state estimate is required for downstream use, a sensor-fusion estimator may
 30 be composed as a derived output downstream of multi-RK corroboration; the fused estimate is in such embodiments a meter-conditioned derived quantity and is not a substitute for the underlying committed bundles.

Distinction from distance-bounding and continuous authentication protocols. Distance-bounding protocols (in the tradition of Brands and Chaum 1993 and Hancke and
 35 Kuhn 2005) establish that a prover is within a bounded physical distance of a verifier by measuring the round-trip time of a cryptographic challenge–response exchange. Continuous-authentication systems extend this with repeated freshness challenges over a session. Reality Kernel freshness mechanisms differ in that: (a) the load-bearing event is an attested physi-

cal light–matter interaction under a declared protocol, recorded as a committed convolution bundle, rather than a location bound or a session-presence proof; (b) freshness is established by time-anchored protocol-digest commitment chained to external time anchors and to prior committed bundles, rather than by round-trip timing of a cryptographic primitive; and (c) the apparatus and a distance-bounding or continuous- authentication mechanism are not mutually exclusive: in some embodiments, such a mechanism is composed alongside the apparatus and its bounded-distance or session-presence attestation is included in the protocol digest as auxiliary evidence, without becoming the load-bearing primitive.

14 Networked Reality Kernels and Transitive Proof-of-Projection

This section describes embodiments in which multiple Reality Kernel modules operate in a network, exchange evidence derived from the convolution bundle, and perform cross-verification of shared events. These embodiments are optional. A single Reality Kernel module can be implemented and used without any networked operation. In some embodiments, networked modules additionally participate in the fleet model improvement protocol described in Section 13, exchanging evidence of model discrepancies and incorporating confirmed corrections into the shared predictive model.

Scaling regime and limitations (non-limiting). In some embodiments, networked deployments use hierarchical aggregation, committee-based attestation, neighbourhood-local trust updates, secure aggregation of population statistics, and amortised monitoring to bound communication overhead while preserving auditable cross-device consistency. Illustrative deployments may range from local clusters to larger federated fleets, with committee size, aggregation depth, witness cadence, and overlap policy selected per target latency, fault model, and deployment scale. Layered aggregation may be used to compress cross-device comparisons into auditable summaries while preserving selective-opening pathways for deeper review where required.

14.1 Network model and shared channel

Let I index a finite set of Reality Kernel modules. For each $i \in I$, let θ_i denote configuration, $u^{(i)}(t)$ denote per-step control, and $C_{0:T}^{(i)}$ denote the convolution bundle. In some embodiments, devices share a physical channel through a shared environment. A shared world state W_t represents external scene factors accessible to multiple devices:

$$W_{t+1} \sim \mathbf{P}_{\text{world}}^{\text{net}} \left(\cdot \mid W_t, (u^{(i)}(t))_{i \in I}, u_E(t) \right),$$

where $u_E(t)$ denotes adversarial co-illumination when present. The symbol W_t denotes the shared world state and is distinct from the per-device extended state X_t defined in Section 6;

X_t includes device-internal components (reactor state, media state, controller state) that are not shared across devices.

14.2 Distributed time and spatiotemporal consistency

In some embodiments, spatiotemporal constraints on inter-device causal links serve as physical timing anchors that bind a run to measurable timing relationships. These function as optical-delay logical clocks: physically grounded partial orders derived from propagation and hardware-latency bounds. A non-limiting spatiotemporal violation functional is

$$\nu_{\text{st}}^{(ij)} = \left(\frac{d_{ij}^{\text{path}}}{\Delta\tau_{ij}} - c \right)^2 + \lambda_{\text{lat}} (\Delta\tau_{ij} - \tau_{\min})_-^2.$$

where $(x)_- := \min(x, 0)$ denotes the negative-part operator (and $(x)_+ := \max(x, 0)$ denotes the positive-part operator, used later in Yoked objectives).

14.3 Projection hardness and proof-of-projection

For devices $i, j \in I$, pairwise projection hardness is defined as

$$k_{\text{proj}}^*(i \rightarrow j) := \min \left\{ k : D\left(\mathbf{P}_{\theta_j}^{(j)}, \hat{\mathbf{P}}^{(i \rightarrow j, k)}\right) \leq \varepsilon \right\},$$

where $\hat{\mathbf{P}}^{(i \rightarrow j, k)}$ is an emulator class indexed by capacity k . A committee $C(j) \subseteq I$ yields an aggregate projection score $H_{\text{proj}}(j) := \min_{i \in C(j)} k_{\text{proj}}^*(i \rightarrow j)$.

Terminological note. The term *proof-of-projection* is used throughout this document as a term of art denoting an empirically calibrated attestation protocol, not a cryptographic proof system with formal soundness and completeness properties. What the protocol provides is: (i) committed evidence that a physical device produced a convolution-bundle trace under declared conditions, (ii) independent physical realisation on one or more distinct unclonable devices in the verifier role, scored by projection hardness against the fleet projection-hardness statistics, and (iii) auditability via selective disclosure, quorum verification, and the submission record’s commitment binding. It does not provide, and should not be read as asserting, a zero-knowledge or interactive proof in the cryptographic sense. Achieving formal proof-system guarantees would require additional identity, Sybil-resistance, and adversary-model assumptions beyond those specified here. The “transitive” extension (Section 14) inherits this character: transitive proof-of-projection denotes a chain of empirical attestations with auditable trust-weight updates, not a transitively composable formal proof.

14.4 Challenge–commit–open protocols for selective disclosure

In some embodiments, devices commit to the convolution bundle and later selectively open subsets for audit. Per-atom digests are computed and aggregated into a Merkle root $R^{(i)}$,

binding emission and observation windows:

$$\text{com}_{\text{full}}^{(i)} = H(\text{rid} \parallel i \parallel \Pi_{\text{dig}} \parallel \mathbf{m}^{(i)} \parallel R^{(i),\text{obs}} \parallel R^{(i),\text{emit}}).$$

Canonical two-seed selective-opening protocol (non-limiting). In some embodiments, a two-seed structure separates the run commitment from the opening decision so that the challenge is structured to reduce the opportunity for anticipatory tailoring to the specific atoms that will later be opened. A non-limiting protocol proceeds as follows:

1. **Run phase.** The device executes a protocol run under run seed s_{run} , derived from the protocol digest and a nonce committed before the run begins. The run seed determines the emission schedule and scan law; it is logged in the protocol digest and not disclosed until step 4 (selective disclosure).
2. **Commitment.** Upon run completion, the device computes and publishes the Merkle commitment $\text{com}_{\text{full}}^{(i)}$ above. The run seed s_{run} remains withheld at this stage.
3. **Challenge (opening seed).** A verifier (or a randomised beacon in some embodiments) supplies an opening seed s_{open} , derived independently of the run seed. s_{open} determines which atoms or windows are to be opened.
4. **Selective disclosure.** The device reveals the requested atoms together with their Merkle inclusion proofs and the run seed s_{run} . The verifier checks that (a) the opened atoms are consistent with the committed Merkle root, (b) the protocol digest reconstructed from s_{run} matches the committed digest, and (c) meter summaries lie within declared envelopes.
5. **Audit escalation (optional).** Under policy-triggered escalation, additional atoms may be opened using further opening seeds supplied by an auditor.

The two-seed structure is referenced throughout this specification wherever “two-seed opening”, “run seed”, or “opening seed” appear; those usages refer to this protocol. In embodiments using this structure, the run commitment is bound to the committed evidence before the opening seed is known to the device, reducing the opportunity for anticipatory tailoring of the run to match future opening challenges.

A non-limiting protocol flow summary: run phase (run seed determines emission schedule and scan law), commitment (Merkle roots and signatures published), challenge (opening seed supplied by verifier or beacon), selective disclosure (requested atoms with Merkle inclusion proofs and run seed), and optional audit escalation.

14.5 Transitive proof-of-projection and reactor pairing

In some embodiments, devices perform continuous or periodic pairing procedures. A non-limiting continuous pairing latent $\sigma_{ij}(t)$ is updated as

$$\dot{\sigma}_{ij}(t) = -\kappa_{\sigma} \sigma_{ij}(t) + F_{\text{pair}}(\mathbf{y}^{(i)}(t), \mathbf{y}^{(j)}(t), u^{(i)}(t), u^{(j)}(t)).$$

A tPoP record for an event includes run identifiers, protocol digests, commitment references
 5 for observation and emission windows, quorum-contributed meter summaries, and pairing latents.

Layered RK networks (non-limiting). In some embodiments, a network is organised into layers with a rolling amortised monitor (where η in this equation is a layer-level smoothing coefficient, not the scan-law parameter or learning rate):

$$\overline{H}_{\ell}(t) = (1 - \eta) \overline{H}_{\ell}(t - 1) + \eta \text{Agg}(\{H_{\text{proj}}(j; t)\}_{j \in \mathcal{L}_{\ell}}).$$

10 **Direct optical coupling between reactors (non-limiting).** In some embodiments, neighbouring devices are coupled optically at the reactor level through free-space apertures, fibre links, or a shared lightguide backplane.

14.6 Trust weights and reputation updates

14.6.1 Cross-prediction and compatibility

15 In an overlap window in which devices i and j observe the same event under related protocols, device i may attempt to predict a view of j 's record (or summary) from its own record and the logged protocols. For example, device i may form a cross-prediction

$$\widehat{C}_{t_0:t_1}^{(j)} = \mathcal{E}_{i \rightarrow j}(C_{t_0:t_1}^{(i)}, U_{t_0:t_1}^{(i)}, U_{t_0:t_1}^{(j)}),$$

where $\mathcal{E}_{i \rightarrow j}$ is a learned emulator, calibrated geometric transfer, or other cross-device predictor, and $t_0:t_1$ denotes the overlap window.

20 A non-limiting per-event loss for the directed edge ($i \rightarrow j$) is

$$\ell_{i \rightarrow j}(e) = \text{clip}_{[0,1]} \left(\frac{D(\widehat{C}_{t_0:t_1}^{(j)}, C_{t_0:t_1}^{(j)})}{\tau} \right),$$

where $D(\cdot, \cdot)$ is any non-limiting discrepancy (for example, a meter-weighted feature distance, a negative log-likelihood under a fitted noise model, or a learned perceptual distance), and $\tau > 0$ is a scale parameter that controls sensitivity. To preserve the meter partition (Section 12), the discrepancy D is computed using acceptance meters $\mathbf{m}^{\text{accept}}$ in embodiments
 25 using the stated meter partition; evaluation meters are excluded from trust-weight compu-

tation because trust weights feed into quorum selection and thereby into the optimisation loop.

14.6.2 Exponential-weights web-of-trust update (non-limiting)

In some embodiments, device i maintains nonnegative trust weights $\{W_{i \rightarrow k}(t)\}_{k \in \mathcal{N}_i}$ over a neighbourhood $\mathcal{N}_i$ of peers. After an event e (or batch of events) the weights may be updated by an exponential-weights rule:

$$\widetilde{W}_{i \rightarrow j}(t+1) = W_{i \rightarrow j}(t) \exp(-b \ell_{i \rightarrow j}(e)), \quad W_{i \rightarrow j}(t+1) = \frac{\widetilde{W}_{i \rightarrow j}(t+1)}{\sum_{k \in \mathcal{N}_i} \widetilde{W}_{i \rightarrow k}(t+1)},$$

where $b > 0$ controls update aggressiveness.

In some embodiments, stabilisers are applied, for example: (i) flooring and renormalisation $W_{i \rightarrow j} \leftarrow \max(W_{i \rightarrow j}, w_{\min})$, (ii) mixing with a uniform prior $W \leftarrow (1 - \mu_{\text{mix}})W + \mu_{\text{mix}}/|\mathcal{N}_i|$ for $\mu_{\text{mix}} \in (0, 1)$, or (iii) clipping to prevent single-event collapse.

The resulting weights define a continuous web-of-trust used to select quorums, weight cross-signatures, schedule higher-cost challenges, or gate access to high-value transitive proof-of-projection records.

14.6.3 Bayesian trust update (non-limiting)

In some embodiments, trust and reputation are updated using Bayesian-style posteriors over a peer's reliability, optionally conditioned on operating regime and meter envelope.

Binary audit outcomes (Beta-Bernoulli, non-limiting). In some embodiments, agreement for a peer j on an event e is reduced to a binary variable $a_{j,e} \in \{0, 1\}$. With a Beta prior $\pi_j \sim \text{Beta}(a_0, b_0)$, the posterior after observing outcomes $a_{j,1:n}$ is

$$\pi_j \mid a_{j,1:n} \sim \text{Beta}\left(a_0 + \sum_{e=1}^n a_{j,e}, b_0 + \sum_{e=1}^n (1 - a_{j,e})\right).$$

A posterior mean, quantile, or lower confidence bound may be used as a reputation score.

Forgetting and regime conditioning (non-limiting). In some embodiments, older evidence is downweighted using exponential forgetting:

$$S_j(t+1) = \gamma S_j(t) + a_{j,t}, \quad F_j(t+1) = \gamma F_j(t) + (1 - a_{j,t}),$$

with $\gamma \in (0, 1)$ (a forgetting factor; not the scan-law function $\gamma(t; \eta)$), yielding a posterior of the same Beta form. In some embodiments, separate reliabilities are maintained per regime, per meter envelope, or per role, and combined by a policy-defined aggregation.

Continuous compatibility scores (non-limiting). In some embodiments, continuous losses or residual scores are used instead of binary variables. The loss $\ell_{i \rightarrow j}(e)$ may be mapped

into a pseudo-likelihood, or discretised into bins (pass, marginal, fail), to update a posterior over reliability. In some embodiments, pairing latents $\sigma_{ij}(t)$ and layer-level rolling monitors modulate how strongly evidence updates trust.

14.7 Additional network mechanisms (non-limiting)

5 **Optical delays as physical logical clocks (non-limiting).** In some embodiments, controlled delay elements impose causal ordering constraints that function as physical logical clocks. A challenge emitted by one device may be routed through a known delay element (for example a fibre delay line) before reaching another device, and devices log timestamps and convolution-bundle evidence for the corresponding response. The known delay bounds
10 constrain allowable histories and help detect fabricated ordering, supplementing or replacing reliance on external global time services.

Phase-locked and frequency-synchronised networks (non-limiting). In some embodiments, multiple devices synchronise to a common frequency reference using phase-locked loops, injection locking, or consensus timing. Synchronisation may enable coherent mea-
15 surements across devices, distributed interferometry, coordinated emissions with controlled interference, and tighter spatiotemporal consistency checks without external clocks.

Near-field resonant coupling between reactors (non-limiting). In some embodiments, devices exchange energy or information through near-field coupling without direct wired or free-space optical links, for example through evanescent coupling, shared resonant
20 cavities, or magnetic coupling between microstructured media.

Entrainment and collective dynamics (non-limiting). In some embodiments, networks exhibit emergent collective dynamics such as synchronisation, correlated oscillations, or collective stability modes. These dynamics may be logged and used as anomaly indicators, additional hardness constraints, or distributed timing signals.

25 **Cross-attestation and overlap-window protocols (non-limiting).** In some embodiments, network integrity is strengthened by cross-attestation: multiple kernels observe a shared scene interval and produce mutually constrained evidence. A non-limiting procedure comprises: (i) each node records a committed observation window under a logged protocol digest; (ii) nodes exchange compact digests (window roots, timing-anchor summaries, meter
30 vectors) over authenticated channels; (iii) a verifier checks overlap-window consistency and accepts when the declared quorum condition (at least k of n witnesses) is met under the stated policy. In some embodiments, cross-attestation is strengthened by structured emissions that are jointly scheduled or beacon-seeded.

Enrolment, identity binding, and certificates (non-limiting). In some embodiments, a device participates in a network only after enrolment. Enrolment binds an identity to a physical device profile and an initial policy regime. A non-limiting procedure comprises identity assignment, profiling (calibration challenges and baseline recording), certification by
 5 an anchor node, and anchoring to a transparency log.

Revocation, quarantine, and trust lifecycle (non-limiting). In some embodiments, network trust is managed over a device lifecycle that includes drift, re-enrolment, and revocation. Devices whose hardness or reliability drops below policy thresholds may be quarantined (reduced weight, restricted participation) pending re-profiling.

10 **Privacy and unlinkability for network witnesses (optional, non-limiting).** In some embodiments, network participation supports privacy by limiting disclosure and reducing linkability. Non-limiting mechanisms include pseudonymous identities, group signatures or anonymous credentials for membership proofs, and secure aggregation of meter outputs so that only quorum aggregates are disclosed. In some embodiments, overlap-window corroboration is performed on committed digests rather than raw the convolution bundle, with
 15 selective opening triggered under audit-escalation policy.

Policy versioning and audit escalation (non-limiting). In some embodiments, network attestation policy is expressed as signed, versioned policy objects that specify meter thresholds, admissible protocol families, quorum requirements, disclosure rules, and revocation procedures. Policy identifiers are bound into protocol digests and signatures so that
 20 downgrades or equivocation are detectable. Audit escalation policies include requesting additional openings, requesting independent witnesses, switching to higher-cost protocols, and temporarily disabling high-impact actions until discrepancies are resolved.

Non-limiting deployment topologies. In some embodiments, networked kernels are
 25 deployed as mobile fleets, fixed installations, environmental beacon networks, tiled immersive chambers, or hybrid topologies. Topology and protocol selection may adapt to deployment context under policy constraints.

Approximate physical common knowledge (non-limiting). In some embodiments, the committed and multiply-witnessed convolution bundles implement a form of approx-
 30 imate common knowledge: multiple bounded-perspective agents can verify not only that an event occurred but that they share the same optically grounded evidence of it, and that each knows the others share it, up to the temporal and perceptual granularity of the commitment protocol. Strict common knowledge in the Aumann sense is unattainable in any practical system with communication latency or perceptual uncertainty; however, the
 35 physically committed evidence provides a stronger epistemic foundation than purely digital broadcast, because each agent's committed convolution bundle is bound to a device-specific

signature that other agents can verify, and the temporal overlap and spatial co-observation constraints make it empirically difficult for any subset of agents to fabricate a consistent multi-perspective record. In some embodiments, the level of approximate common knowledge is parameterised by the commitment protocol’s temporal resolution, the number of independent witness devices, and the spatial co-observation constraints, and is reported as a meter-conditioned confidence level rather than as a binary common-knowledge predicate. In some embodiments, this approximate common knowledge enables agreement protocols without a trusted third party: agents who share physically committed evidence of the same event can reach agreement by exchanging posterior beliefs, because the common evidence base is physically assured to within the protocol’s temporal and spatial granularity rather than assumed.

14.8 Byzantine adversaries and robustness (optional)

In some embodiments, under resource bounds on the adversary and a sufficient fraction of honest, high-hardness witnesses with diverse viewpoints, coordinated forgery becomes difficult because fabricated records would need to remain consistent with many independent, high-weight committed logs.

Non-limiting adversary modes. In some embodiments, adversaries act not only by forging traces or tampering with a single device, but by exploiting network structure, quorum selection, selective disclosure, and cross-device trust updates. Non-limiting adversary modes include:

1. Collusion and Sybil participation to bias quorums or trust-weight updates.
2. Eclipse or routing attacks that isolate a device or force it to interact with a compromised neighbourhood.
3. Replay and timing manipulation, including attempts to re-use old commitments or to desynchronise protocol digests.
4. Selective opening manipulation, including attempts to anticipate which atoms will be audited, or to craft atoms that pass shallow checks while failing globally.
5. Audit flooding, in which an adversary forces repeated escalations to exhaust bandwidth or disclosure budgets.
6. Cross-prediction poisoning, in which an adversary crafts emissions or summaries to distort cross-device residual estimates.
7. Layer targeting, in which higher-layer aggregators are biased even if local witnesses are honest.
8. Policy manipulation, including attempts to change thresholds, decay rates, or quorum eligibility rules.

Non-limiting defences. Defences include:

1. Diversity requirements for quorums (for example spatial diversity, spectral diversity, hardware lineage diversity, or operator independence).
2. Trust-weighted quorum selection with rate limits and minimum diversity constraints.
- 5 3. Delayed opening via a two-seed structure (run seed versus opening seed) to reduce precommitment attacks.
4. Anti-replay binding to run identifiers, protocol digests, timing anchors, and freshness policies.
5. Multi-layer monitoring with amortised background scoring and escalation only on deviations beyond policy thresholds.
- 10 6. Audit budgets and escalation policies that bound disclosure and computational cost.
7. Cross-checks that tie reported meter outputs to peer evaluations on the same shared event.
8. Periodic re-seeding and neighbourhood reshuffling to reduce long-lived eclipse conditions.
- 15

In some embodiments, audit outcomes and detected inconsistencies feed back into trust weights and reputations, reducing future influence of suspected peers and triggering higher-cost challenges only when policy thresholds are exceeded.

15 Routed Physical Ensembles and Optical Mixture-of-Experts

15.1 Expert subunits, structural diversity, and route binding

In some embodiments, a routed physical ensemble comprises multiple physically distinct expert subunits. In some embodiments, each subunit includes a scattering medium or related reactor, associated input and output optics, and a readout configured to extract task-relevant structure from a high-dimensional physical transform. In some embodiments, each expert is itself a parameterised Markov kernel within the broader Reality Kernel framework.

In some embodiments, each expert exhibits structural diversity due to microstructure-based uniqueness of its microstructure: no two expert subunits compute the same transformation. In some embodiments, structural diversity arises from manufacturing variability but task complementarity is an empirical property that must be validated by comparing multi-expert routed performance against single-expert and random-routing baselines on specific benchmarks.

In some embodiments, expert subunits are not limited to disordered scattering media. In some embodiments, expert subunits include phase-change reactors whose transfer function is switched between amorphous and crystalline states, gain-pumped optical cavities whose mode structure provides a distinct transform per cavity geometry, photonic integrated circuits whose fabrication variation produces device-specific transfer matrices, fibre-delay networks whose delay and coupling configuration defines a recurrent temporal kernel, and acoustic or multi-physics reactors as described in Section 10. In some embodiments, a heterogeneous ensemble mixes expert types — for example scattering media, phase-change elements, and PIC subunits — to provide diversity along different physical axes. In some embodiments, the commitment, provenance-binding, and calibration-state attestation layer described in this section applies uniformly to all expert types, because each expert is a parameterised Markov kernel regardless of its physical substrate. In some embodiments, the routing interface — including coupling optics, mode matching, and per-expert calibration — varies by expert type and is determined per embodiment.

In some embodiments, routing identities, expert microstructure identities, calibration-state hashes, and readout or weights version identifiers are committed to the protocol digest and bound into provenance records, so that every computation is traceable to the specific physical path through which it was executed.

Distinguishing feature (non-limiting). In some embodiments, the routed physical ensemble is distinguished from prior optical reservoir computing, photonic neural network, and conditional computation architectures by the integration of committed physical evidence into the routing and computation path. In some embodiments, the selected physical path through the ensemble, the microstructure identity of each participating expert, the calibration state of each expert, and the execution transcript are committed to the protocol digest and bound into provenance records, so that every routed computation is traceable, auditable, freshness-anchored, and integrated into the proof-of-discrepancy, bilateral portability, and fleet corroboration infrastructure described elsewhere in this specification. In some embodiments, this commitment and auditability layer is the architectural distinction, not the optical routing mechanism itself.

15.2 Routing, addressing, and gating

In some embodiments, routing is performed by an addressing subsystem that may include acousto-optic deflectors, electro-optic devices, MEMS scanners, galvanometers, spatial light modulators, camera or reprojection systems, or hybrid optical-digital routing mechanisms.

In some embodiments, a conservative optical embodiment uses a TeO_2 acousto-optic deflector with a beam diameter of approximately 5 mm, acoustic velocity in the range of approximately 600–4200 m/s depending on interaction geometry, access time in the range of approximately 1–25 μs depending on beam diameter and acoustic velocity, RF bandwidth of approximately 50–200 MHz, approximately 50–200 resolvable steering positions (sufficient for

16–64 expert subunits with margin depending on configuration), single-axis diffraction efficiency of approximately 60–85%, and effective random-access routing rate of approximately 40 kHz–1 MHz depending on access time. In some embodiments, two cascaded deflectors provide two-axis steering with approximately 36–64% throughput.

5 In some embodiments, a relay lens between the AOD output and the expert subunit array maps AOD angles to positions on the expert inputs. In some embodiments, coupling optics between each expert and the shared output path collect scattered or transmitted light and direct it to a shared detector or detector array.

10 In some embodiments, routing decisions are made by a gating network that may be fully digital (a learned gating head), fully optical (routing based on optical preprocessing), or hybrid (optical features consumed by a digital gating head). In some embodiments, routing identities and schedules are logged in the protocol digest and bound into provenance and audit records.

Architectural constraint (non-limiting). In some embodiments, the system logs sufficient information to reconstruct which expert subunit(s) were selected, the routing parameters, and the calibration-state hash of each participating expert, so that downstream relying parties can verify or audit the physical computation path. This architectural constraint distinguishes the routed ensemble from an opaque black-box accelerator: every computation is traceable to a specific physical path under a committed protocol.

20 **Candidate relay layout (non-limiting).** In some embodiments, the relay layout includes two cylindrical relay lenses (one per axis), a field lens at an intermediate plane, and a micro-relay array aligned with the expert input facets. In some embodiments, practical alignment tolerance requires active calibration using the same calibration-loop infrastructure disclosed in this specification.

25 15.3 Gain-assisted short-chain coupling, failover, and inter-expert relay

In some embodiments, the per-stage optical loss budget for a routed ensemble limits the number of concatenated expert stages in a single forward pass, because loss accumulates multiplicatively across stages, lens surfaces, deflector passes, and coupling into and out of each expert. In some embodiments, a loss budget analysis includes at least deflector efficiency per pass, relay-lens reflection losses, coupling-into-expert loss, scattering transmission per expert, coupling-out-of-expert loss, and detector quantum efficiency.

35 In some embodiments, a semiconductor optical amplifier (SOA) or other gain medium is inserted between expert stages to partially compensate inter-stage loss and extend the usable chain length beyond what passive optics alone would permit. In some embodiments, the SOA contributes amplified spontaneous emission (ASE) noise that reduces SNR, and the useful operating point is determined by the SOA gain, noise figure, and accumulated ASE

across the chain length, so that there exists a practical maximum chain length beyond which additional stages degrade rather than improve task performance.

In some embodiments, SOA gain is included in the per-stage loss budget. In some embodiments, a two-stage budget template is used:

Table 5: Per-stage budget template for gain-assisted inter-expert coupling (representative values; actual values are embodiment-specific).

Stage element	Typical range	Notes
AOD diffraction (per pass)	60–80%	TeO ₂ , single axis
Relay lens (per surface)	97–99%	AR-coated
Expert coupling (in + out)	10–50%	geometry-dependent
Expert scattering transmission	1–30%	medium-dependent
SOA net gain (if present)	10–20 dB	minus NF penalty
Detector QE	40–90%	wavelength-dependent

In some embodiments, whether SOA-assisted chaining produces measurably better task performance than single-expert operation is an empirical question that must be established per configuration and per task by comparison against single-expert and random-routing baselines.

State augmentation for gain-coupled operation (non-limiting). In some embodiments, when SOA gain is used between expert stages, the extended state tracked by the controller is augmented to include SOA bias current, estimated ASE power, SOA temperature, and gain saturation state. In some embodiments, these parameters are logged in the protocol digest and metered within declared envelopes. In some embodiments, the SOA gain profile drifts with temperature and aging, requiring periodic recalibration using the same drift-monitoring infrastructure described elsewhere in this specification. In some embodiments, thermal management and power-supply stability for the SOA are engineering requirements comparable in kind (though not in scale) to those for the primary emitter and are addressed by standard optoelectronic packaging practice.

In some embodiments, failover modes include rerouting to alternative expert subunits, falling back to single-expert operation, or reverting to digital-only processing when optical chain integrity degrades below a declared meter threshold. In some embodiments, failover decisions are logged and auditable.

15.4 Worked dimensionality-tier routed-ensemble embodiment

This subsection describes one conservative routed-ensemble embodiment analysed at three dimensionality tiers, showing how routing policy changes at each tier.

Tier 1 — raw sensed and addressed channels. In some embodiments, the conservative AOD embodiment described above addresses 64 expert subunits. In some embodiments, each subunit receives input from a spatial light modulator with approximately 10^6 addressable

pixels and produces output captured by a camera with approximately 8×10^6 pixels. In some embodiments, Tier 1 dimensionality is therefore bounded by the hardware channel count: approximately 10^6 input dimensions per expert and approximately 8×10^6 output dimensions per expert, with 64 experts selectable at approximately 800 kHz random-access rate. In some
 5 embodiments, at Tier 1 the routing policy selects among experts based on input modality, task identity, or scheduling constraints, without requiring detailed knowledge of each expert’s computational properties.

Tier 2 — approximately independent channels and effective rank. In some embodiments, the number of channels that carry approximately independent information is
 10 substantially smaller than the raw hardware count, because of étendue or space-bandwidth product constraints, numerical aperture limits, oversampling, control modality (phase-only, amplitude-only, or complex), and inter-pixel correlations. In some embodiments, for typical multiply-scattering systems studied in transmission-matrix experiments, Tier 2 dimensionality is approximately 10^4 to 10^6 per expert, depending on geometry and coherence. In some
 15 embodiments, at Tier 2 the routing policy incorporates measured effective rank per expert and selects among experts based on which experts offer the highest effective rank for the current input domain, accounting for correlations that reduce independent information content below the raw pixel count. In some embodiments, Tier 2 dimensionality is measured experimentally using singular-value decomposition of sampled transmission-matrix blocks or
 20 related effective-rank estimation procedures, and the measured rank is logged as a calibration parameter.

Tier 3 — task-useful effective rank. In some embodiments, the number of dimensions that actually contribute to performance on a specific task, after readout training, is smaller than or equal to the Tier 2 effective rank and is task-dependent. In some embodiments, at
 25 Tier 3 the routing policy selects among experts based on measured task-relevant structure: which experts contribute the most to task performance after readout training, as measured by held-out validation on committed physical evidence. In some embodiments, capability claims for the routed ensemble are tied to measured Tier 3 structure rather than to bulk Tier 1 channel counts or Tier 2 effective-rank estimates. In some embodiments, Tier 2 di-
 30 mensionality is a necessary but not sufficient condition for capability: it bounds the space of possible computations but does not by itself demonstrate that any specific task benefits from the available dimensions. In some embodiments, Tier 3 task-useful rank must be established experimentally on declared benchmarks before capability is asserted, and any claimed routing advantage is validated against single-expert and random-routing baselines on specific
 35 benchmarks using committed physical evidence.

Routing policy and tier interaction. In some embodiments, the routing policy determines which expert subunits are selected for a given input, and the dimensionality tier determines how the resulting output is characterised. In some embodiments, at Tier 1 rout-

ing is coarse: expert selection is based on availability, scheduling, or input modality. In some embodiments, at Tier 2 routing becomes correlation-aware: experts are selected based on measured effective rank and domain coverage, and redundant experts may be bypassed. In some embodiments, at Tier 3 routing is task-specialised: experts are selected and weighted
 5 based on their demonstrated contribution to specific task performance, and the routing policy itself is trained using committed physical evidence from task evaluation. In some embodiments, any claimed routing advantage is validated against single-expert and random-routing baselines on specific benchmarks using committed physical evidence. In some embodiments, whether routing at higher tiers produces measurably better task performance is an empirical
 10 property established per embodiment.

15.5 Capacity accounting rule (non-limiting)

For avoidance of ambiguity across this disclosure, six categories of quantity are distinguished, and no claim herein shall conflate them: (a) raw hardware channel count; (b) effective rank; (c) task-useful rank; (d) control waveform coefficients; (e) simultaneously-measurable gra-
 15 dient components; and (f) stable trained weights. No capability claim in this disclosure shall assert a value in category (f) derived from counting in categories (a), (b), (d), or (e). Any such claim shall be supported by direct task-benchmark measurement of category (c) on committed physical evidence and, where stable trained weights are recited, by separate empirical stability evidence under the declared drift and freeze-window criteria. Migration
 20 between categories requires explicit experimental demonstration.

15.6 Optional semantic extension branch and attested semantic computation (validation-dependent)

In some embodiments, the routed physical ensemble further includes a late, optional, validation-dependent semantic extension branch in which text or other symbolic content
 25 is rendered as spatial optical patterns, processed by selected expert subunits through the same committed physical channel used for nonsemantic routed computation, and decoded by trained readout heads, with digital fusion remaining the default multimodal baseline.

In some embodiments, semantic capability in such an extension resides in the encoder, readout, and any bias-mask parameters rather than in the scattering medium alone. In some
 30 embodiments, the scattering physics is identical to that used for nonsemantic computation; the semantic branch is a domain-specific application of the same core reservoir architecture to rendered textual or symbolic inputs. In some embodiments, the default multimodal embodiment combines perceptual features and semantic features in a digital fusion layer, and that digital fusion path serves as the reference baseline against which any more physical
 35 comparison mechanism is evaluated.

Attested semantic computation record (non-limiting). In some embodiments, the system produces an attested semantic computation record comprising: a committed Truth Beam bundle identifier, a selection-rule digest, a protocol digest, a routing log identifying which expert subunits were traversed and in what order, microstructure identities of participating experts, a calibration-state hash, a readout or weights version identifier, a time anchor, the semantic output, and any confidence, discriminant, or quality statistics associated with the output. In some embodiments, hash-chain linkage and external time anchoring make the semantic run order-sensitive and freshness-sensitive, so that a semantic conclusion cannot be fabricated and back-filled without breaking the chain. In some embodiments, the physical channel is rate-limited, so that the attestation object resists digital-speed brute-force generation or transcript farming.

Worked semantic example (non-limiting). In some embodiments, a non-limiting worked example illustrates the semantic extension branch as follows. A text prompt is rendered as a spatial optical pattern by an encoder. The pattern is projected onto a selected expert subunit via the AOD routing subsystem. The expert subunit scatters and transforms the pattern. A camera captures the output. A trained readout head extracts a semantic feature vector. The entire chain — prompt rendering parameters, routing identity, expert microstructure identity, calibration state, readout weights version, and output — is committed to the protocol digest. In some embodiments, this chain produces an attested semantic computation record as described above. In some embodiments, the same chain may be replayed under declared calibration tolerances for dispute resolution.

Industrial applicability scope (non-limiting). The semantic extension branch disclosed in this section is directed to attested visual input/output around a digital semantic model, bounded-vocabulary semantic grounding, narrow-task continuous streaming semantic operation, and semantic Truth Beam attestation of claims over committed physical evidence windows. This disclosure does not assert, and shall not be construed to assert, that the physical scattering substrate constitutes an open-ended language model, a general-purpose semantic engine, or a substrate for training-scale deep neural networks over natural language.

15.7 Mapping to reactor-network primitives

In some embodiments, the routed physical ensemble instantiates route, compile, parallelise, reduce, and abstract primitives of a broader reactor network algebra. In some embodiments, route corresponds to expert selection, compile to sequential chaining, parallelise to simultaneous activation, reduce to output aggregation, and abstract to trained readout or head extraction. In some embodiments, fleet-level calibration and provenance monitoring extend across the ensemble so that expert identity, routing history, and quality-state transitions are auditable.

15.7.1 Heterogeneous expert-type example (non-scattering, non-limiting)

The following example illustrates how the routed-ensemble architecture can be mapped to expert subunits beyond disordered scattering media. As with the non-optical reactor examples above, this is an architectural mapping showing that the routing, provenance-binding, and commitment infrastructure can apply across expert types; whether a specific heterogeneous combination achieves measurable task-relevant complementarity is an empirical question requiring experimental validation per configuration.

In some embodiments, a heterogeneous routed ensemble includes at least two expert types selected from: (a) disordered scattering media providing high-dimensional random-feature-map transformations, (b) phase-change elements (for example chalcogenide or GST-based reactors) whose transfer function is switchable between amorphous and crystalline states and whose switched configuration is metered and logged, (c) gain-pumped optical cavities whose mode structure provides a transform that depends on cavity geometry, mirror reflectivities, and pump conditions, (d) photonic integrated circuit subunits whose fabrication-variant waveguide couplings provide device-specific transfer matrices, and (e) fibre-delay networks whose delay and coupling configuration define a recurrent temporal kernel.

In some embodiments, the routing interface — including coupling optics, mode matching, alignment calibration, and per-expert metering — varies by expert type and is determined per embodiment. In some embodiments, the commitment and provenance infrastructure (protocol digest, routing log, microstructure identity, calibration-state hash, readout weights version, time anchor) applies uniformly regardless of expert type, because each expert is a parameterised Markov kernel within the common framework.

In some embodiments, the central empirical question for any heterogeneous combination is whether the ensemble achieves measurable task-relevant complementarity — that is, whether routing across expert types produces statistically significant improvement on declared benchmarks compared with single-expert and random-routing baselines using committed physical evidence. In some embodiments, a heterogeneous ensemble that lacks validated complementarity defaults to single-expert or homogeneous-ensemble operation and is logged accordingly.

15.8 Conservative routed optical ensemble

In some embodiments, a conservative routed optical ensemble (non-limiting) comprises a coherent source (for example a 532 nm or 785 nm diode laser), a beam expander, a spatial light modulator with approximately 10^6 pixels, a TeO_2 acousto-optic deflector providing approximately 120 resolvable positions at approximately 800 kHz random-access rate, relay optics (two cylindrical lenses, a field lens, and a micro-relay array), 64 scattering expert subunits arranged on a planar grid, collection optics per expert, a shared CMOS camera with approximately 8×10^6 pixels, and a digital controller connected to the SLM driver, AOD RF driver, and camera via synchronised interfaces.

In some embodiments, the controller implements the following cycle: (a) select an expert subunit by driving the AOD to the corresponding angle; (b) display an input pattern on the SLM; (c) capture the output on the camera; (d) compute routing features from a preliminary digital analysis of the captured output or from a separately trained feature extractor; (e) log the routing decision, expert identity, SLM pattern identifier, AOD drive parameters, and camera frame in the protocol digest; and (f) optionally chain to a second expert stage via gain-assisted relay as described in Section 15.3, with per-stage loss budget as described in Table 5.

In some embodiments, the scan cycle time is dominated by the AOD access time (approximately $1.2\ \mu\text{s}$) plus the camera integration time (approximately $100\ \mu\text{s}$ to $10\ \text{ms}$ depending on exposure requirements), yielding effective scan rates of approximately $100\ \text{Hz}$ to $10\ \text{kHz}$ for the full route-project-capture cycle. In some embodiments, the SLM refresh rate (approximately $60\ \text{Hz}$ for a liquid-crystal SLM or approximately $10\ \text{kHz}$ for a digital micromirror device) may be the bottleneck for input pattern changes; in other embodiments, the same SLM pattern is used across multiple expert selections within a single SLM frame, amortising the SLM refresh latency across expert queries.

In some embodiments, initial bring-up and calibration include: (a) per-expert transmission-matrix sampling using random input patterns and camera capture, yielding a sampled transmission-matrix block per expert from which Tier 2 effective rank is estimated via singular-value decomposition; (b) AOD angle-to-position calibration using a known target in each expert position; (c) relay alignment verification using a collimated beam and position-sensitive detector; (d) SOA gain and noise-figure measurement (if gain-assisted chaining is used); and (e) baseline verisimilitude scoring using the trained discriminator on committed bundles from the calibrated configuration.

Relay-geometry design parameters (non-limiting). In some embodiments, the relay layout is characterised by: working distance from the AOD output aperture to the expert input plane (approximately $100\text{--}300\ \text{mm}$), magnification selected to match the AOD output beam diameter to the expert input aperture, field curvature and distortion across the expert grid (corrected by the field lens and micro-relay array), and vignetting at the grid edges (metered during calibration and compensated by per-position gain adjustment). In some embodiments, the relay is designed for telecentric operation at the expert input plane to minimise angle-dependent coupling variation across the grid.

In some embodiments, the micro-relay array comprises one microlens per expert input facet, with focal length and numerical aperture matched to the scattering medium's input mode profile. In some embodiments, the micro-relay array is fabricated as a single moulded or lithographically defined element and is aligned as a unit during assembly.

Experimental validation requirements (non-limiting). In some embodiments, the routed optical ensemble is validated by: (a) measuring single-expert task performance on declared benchmarks; (b) measuring multi-expert routed performance on the same bench-

marks under the trained routing policy; (c) measuring random-routing baseline performance; (d) reporting whether multi-expert routing produces statistically significant improvement over single-expert and random-routing baselines; (e) reporting Tier 2 effective rank per expert and Tier 3 task-useful rank for the ensemble; and (f) reporting per-stage loss budget and, if applicable, whether gain-assisted chaining improves or degrades net task performance compared with single-stage operation. In some embodiments, all validation uses committed physical evidence and is reproducible under the declared calibration and protocol-digest conditions.

16 Agent Integration (PolieBots) and Verification-Gated Actions

This section describes embodiments in which an autonomous or semi-autonomous agent uses one or more Reality Kernel modules as its primary observation, verification, and action-conditioning channel. References to “PolieBots” are non-limiting examples of mobile platforms carrying one or more Reality Kernel modules. The verification-gated action mechanism and the RK-centred agent loop are core operational mechanisms of this disclosure; optional higher-layer coordination and training architectures are described in the related Filing 2 application (optional, non-essential).

Physical and functional composition (non-limiting). In one non-limiting configuration, a PolieBot apparatus comprises:

1. A *mobile or deployable platform* (for example a wheeled, tracked, legged, aerial, or aquatic chassis; a fixed-installation enclosure; or a wearable or handheld carrier) providing structural support and power for the remaining subsystems.
2. One or more *Reality Kernel modules*, each comprising at least an emitter, a detector, one or more physical paths coupling emission to measurement, and the meter, commitment, and protocol-digest infrastructure described elsewhere in this disclosure (for example the bench-top 1D anchor of Section 7.2, the 2D anchor of Section 10.2, or any of the reactor variants described in subsequent sections).
3. A *locomotion or actuation subsystem* (for example wheels with motors and encoders, propellers, articulated limbs, manipulator arms, or steering surfaces) configured to translate platform motion commands a_t^{platform} into physical movement of the platform or of one or more Reality Kernel modules relative to the scene; in some embodiments the motion commands are derived directly from the analogue output of one or more Reality Kernel modules of Item 2 via declared transducers, without separate digital command synthesis.
4. A *communication subsystem* (for example a wireless radio, an optical link, a wired interface, or a beacon receiver) configured to receive challenges, beacons, or coordination signals from external verifiers or peer agents, and to publish committed bundle

digests, evidence summaries, or coordination payloads; in standalone embodiments the communication subsystem is absent and the apparatus operates without external coordination.

- 5 5. A *decision-making subsystem* that selects per-step actions $a_t = (u(t), a_t^{\text{platform}}, a_t^{\text{tool}})$ based on observation history $o_{0:t}$ and meter envelopes $\mathbf{m}_{0:t}$, comprising one or more of:
 - (i) one or more Reality Kernel modules of Item 2 operating as the primary decision-making mechanism, in which the analogue feedback dynamics of the reactor-scene loop, taken together with declared decoders or transducers from reactor-side observables to platform-side commands, directly determine per-step actions without requiring separate digital policy evaluation (for example a fully optical feedback loop between a reactor and a scene in which emission schedules and platform motion commands are produced by the loop's own dynamics);
 - 10 (ii) one or more processors executing a learned or programmed policy $\pi_\psi(\cdot \mid o_{0:t}, \mathbf{m}_{0:t})$ over observation history and meter envelopes; or
 - 15 (iii) a hybrid configuration in which the Reality Kernel analogue dynamics of (i) and the digital policy of (ii) are composed (for example an analogue inner loop conditioned by a digital outer policy, or a digital policy whose output is filtered or shaped through an analogue reactor stage before actuation);
 together with a verification-gating mechanism that partitions the action space according to evidence sufficiency (unconstrained, RK-verified, hardness-gated, and multi-RK-verified classes as described below in this section), and an audit-policy interface that constrains action eligibility to declared envelopes. The verification-gating mechanism and audit-policy interface are functional requirements and may be implemented digitally, in the analogue feedback dynamics of one or more Reality Kernel modules, or in any combination.
- 20 6. Optionally, an *external-tool interface* configured to invoke tool actions a_t^{tool} (for example calls to remote computation, retrieval, or human-in-the-loop confirmation) under the same verification, hardness, and audit constraints applicable to the platform's other actions.

30 In some embodiments, the platform carries multiple Reality Kernel modules in distinct geometric or spectral configurations, supporting multi-RK-verified actions of the type described below in this section. In some embodiments, the locomotion and Reality Kernel subsystems are mechanically coupled such that platform motion participates in the scan law $u(t)$ (for example a body-mounted sensor whose pose changes as the platform moves); in other embodiments the Reality Kernel modules are independently steered relative to the platform and the scan law $u(t)$ is constructed from module-internal degrees of freedom only.

Fully analogue configuration (non-limiting). In a non-limiting fully analogue configuration, the apparatus reduces to a mobile or deployable platform of Item 1; one or more Reality Kernel modules of Item 2 acting in role (i) of Item 5; a locomotion subsystem of Item 3 driven by analogue transducer output from those Reality Kernel modules; and an

analogue verification-gating subsystem and analogue audit-policy interface of the form described below, implemented within or alongside the analogue feedback dynamics of the same modules. The digital processor of role (ii) of Item 5, the external-tool interface of Item 6, and the communication subsystem of Item 4 may all be absent in such embodiments.

5 In one non-limiting analogue verification-gating subsystem, the gate apparatus comprises: (a) one or more analogue meter transducers configured to convert declared committed-evidence observables (for example photodetector amplitudes, coupling-quality envelope read-outs, or hardness proxies derived from the reactor-scene loop) into continuous voltage or current signals; (b) one or more threshold comparators or window discriminators config-
10 ured to compare each meter signal to a declared fixed threshold or window, the threshold values being set as hardware-declared component values (for example resistor-divider references, voltage-reference diodes, or precision-set potentiometer trims) that are physically inspectable as device-state evidence; (c) one or more analogue latch or sample-and-hold elements configured to record the comparator outcomes for the duration of an action-class
15 evaluation window, providing memory of recent gate state across the loop's response time; (d) one or more analogue limiters or rate-limited switch elements configured to clamp, attenuate, or interrupt the locomotion or emission control path when any of the gated comparator outcomes indicates an out-of-envelope condition; and (e) a policy-selecting switch network configured to route the locomotion or emission control between an unconstrained path, an
20 RK-verified path, a hardness-gated path, and a multi-RK-verified path according to the latched comparator outcomes, the routing implementing the partition of the action space described elsewhere in this section.

In one non-limiting analogue audit-policy interface, gate state is recorded as physical evidence of device behaviour rather than as a digital audit log. Non-limiting recording mechanisms include: a servo-driven mechanical position, ink-on-paper trace, exposed photographic
25 medium, or set-once analogue memory element whose state is determined by the latched comparator outcomes and inspectable post-run; a co-emitted committed companion analogue signal recorded by an external observation channel; or incorporation of the gate state into the optical or electrical signature of the controlled emission such that a verifier with access to the committed output and the declared hardware declaration can recover the gate's
30 effective decision history. The hardware declaration — comprising at least the threshold component values, comparator and latch identities, switch-network topology, and recording-medium identity — is committed under the same protocol-digest discipline applied elsewhere in this disclosure to physical configuration declarations, so that an audit recovers what the
35 analogue gate could and could not have done given its physically declared parameters.

The analogue verification-gating and audit-policy machinery described in this paragraph is non-limiting; embodiments may combine analogue gate elements with digital, optical, or hybrid recording mechanisms. Where verifier-facing audit beyond the committed hardware declaration and committed output channel is required (for example for selectively-openable
40 atom-level evidence, cryptographic commitment plumbing, or remote audit), a digital or

hybrid implementation as described in role (ii) or role (iii) of Item 5 is included rather than the fully analogue configuration of this paragraph.

16.1 RK-centred agent loop

In some embodiments, at each step t , the agent selects an action $a_t = (u(t), a_t^{\text{platform}}, a_t^{\text{tool}})$, where $u(t)$ is applied to RK modules, a_t^{platform} denotes platform motion, and a_t^{tool} denotes calls to external tools. A non-limiting per-step reward is

$$r_t = r_t^{\text{task}} - \lambda_{\text{cost}} \mathcal{C}(u(t)) + \lambda_{\text{info}} \hat{I}_t + \lambda_{\text{hard}} H_t.$$

In some embodiments, the RK control is selected by a learned policy $u(t) \sim \pi_\psi(\cdot \mid o_{0:t}, \mathbf{m}_{0:t})$, so that scan paths and emission schedules adapt to scene context and meter envelopes while remaining compatible with audit policies.

16.2 Verification-gated actions

In some embodiments, the agent partitions its action space into classes with different verification requirements: unconstrained actions (internal computation, reversible actions), RK-verified actions (conditioned on recent verisimilitude checks), hardness-gated actions (conditioned on empirical hardness thresholds), and multi-RK-verified actions (requiring corroboration from multiple independent modules). Failures trigger bounded escalation.

16.3 Continual learning via physical kernel interactions

In some embodiments, the agent logs experience tuples $\tau_t = (o_t, u(t), a_t^{\text{platform}}, r_t, \mathbf{m}(t), \Pi_t)$ and uses these logs for continual learning with bounded drift and audit compatibility. Hardness-regularised objectives include penalties for meter violations and rewards for maintaining or improving empirical hardness during adaptation.

Automated reactor design and meta-learning (non-limiting). In some embodiments, the design of reactor configurations (media selection, geometry, scan protocols, and associated digital heads) is optimised via meta-learning or automated search. A meta-learner proposes reactor configurations; each configuration is instantiated physically and/or in simulation and evaluated on a distribution of tasks; and fitness signals guide search toward configurations that generalise well under declared cost and safety constraints.

Uncertainty quantification and calibration (non-limiting). In some embodiments, the system produces calibrated uncertainty estimates alongside predictions. Non-limiting mechanisms include ensemble queries across multiple reactor passes or heads, stochastic decoding that samples from output distributions, explicit uncertainty heads trained with calibration losses, and conformal-prediction-style wrappers that provide coverage guarantees

under declared assumptions. In some embodiments, intrinsic stochasticity of the physical channel provides a natural source of uncertainty signals.

Streaming and online inference (non-limiting). In some embodiments, the Reality Kernel processes continuous input streams online, producing outputs with bounded latency after each input chunk rather than waiting for complete sequences. Reactor memory maintains running context; an encoder maps arriving chunks to emission patterns; and a decoder produces partial outputs that are refined as more of the convolution bundle arrives.

Self-modelling and introspection (non-limiting). In some embodiments, the system maintains a self-model of its capabilities, limitations, and current calibration state. This supports confidence calibration, knowing when to defer, predicting latency/energy/error characteristics, and detecting drift or degradation from evidence streams and meter outputs.

16.4 Interpretability and internal model probing via RK externalisation

In some embodiments, the agent uses a Reality Kernel as a physically grounded interpretability and debugging mechanism. The agent externalises internal hypotheses, predicted states, or latent variables as emission patterns and compares observed responses to predicted responses. A non-limiting procedure includes:

1. Encode an internal hypothesis or predicted scene property into a structured emission protocol.
2. Execute the protocol on the Reality Kernel and record the convolution bundle.
3. Compare the measured convolution bundle to the agent's predicted convolution bundle under the same protocol.
4. Update belief state or flag mismatch when discrepancy exceeds a threshold.

This provides a model-reality consistency check that is difficult to satisfy by purely semantic manipulation because it is anchored to the physical channel.

Optional extensions (described in the related Filing 2 application; optional, non-essential). In some embodiments, the agent system uses optional training and operational extensions, together with sim-to-real world-model refinement. These embodiments are described in the related Filing 2 application (optional, non-essential). The sim-to-real transfer mechanism, bifurcation-targeted probing, and fleet-driven simulation improvement described there use the same convolution-bundle, meter, and commitment machinery as the core embodiments and are enabled by them, but are not required for core Reality Kernel operation.

Perception hierarchy (non-limiting). In some embodiments, agent perception is organised as a hierarchy of increasing abstraction: raw photon statistics (photon counts, arrival times, spectral distribution); spatial structure (edges, textures, depth, spatial frequency content); temporal dynamics (motion, flicker, rhythmic patterns, transient events); semantic scene interpretation (object identity, scene layout, activity recognition); causal modelling (physical cause-and-effect relationships, intervention predictions); and relational reasoning (relationships between entities, roles, social structure). Each level of the hierarchy corresponds to a different decoder head applied to the convolution bundle, and the agent’s developmental curriculum can be structured to introduce higher levels progressively as lower levels are mastered.

16.5 Offline simulation and physical validation

In some embodiments, offline or partially offline experience is generated in at least two ways. In a first class of embodiments, a physical Reality Kernel module operates on synthetic or modified scenes (for example projected patterns, printed targets, controlled test fixtures, or augmented views), so that the optical and media stack remains physical while the scene content is virtualised. In a second class of embodiments, the Reality Kernel is replaced by a learned emulator or digital twin calibrated to the physical module, and the agent interacts with the emulator as a virtual reactor. A non-limiting method comprises:

1. Train or update an emulator on the logged convolution bundle and control histories, and optionally define families of synthetic scenes for use with a physical Reality Kernel module.
2. Generate simulated rollouts under candidate policies, using at least one of: (i) a virtual Reality Kernel emulator, and (ii) a physical Reality Kernel operating on synthetic or modified scenes.
3. Select validation probes, execute them on the physical Reality Kernel, and compare the observed convolution bundle to emulator predictions.
4. Update policy and world model using both simulated and physical experience, with weighting dependent on validation outcomes and confidence.

Parameter-sandboxed offline sessions (non-limiting). In some embodiments, the physical kernel operates on synthetic or modified scenes during an offline session in which kernel parameters may be updated freely. All parameter updates made during the offline session are reverted upon return to live operation, so that the session constitutes a sandboxed exploration and no parameter excursion persists into the live operating configuration. In some embodiments, classifiers or other derived models trained on the convolution-bundle record produced during the sandboxed session are retained and available for use in subsequent oper-

ation, even though the kernel parameters themselves are reverted. Higher-layer deployment of such classifiers is described in the related Filing 2 application (optional, non-essential).

Calibration and self-model construction during offline periods (non-limiting).

In some embodiments, an offline calibration period is used for physical parameter update (drifting-field compensation and re-commissioning against declared meter envelopes) and for fitting a generative self-model to the device's own logged convolution-bundle history. Unlike sandboxed sessions, parameter updates made during calibration periods are retained. In some embodiments, the resulting smoothed self-model serves as a synthetic scene source for subsequent sandboxed sessions.

Inward-facing gain modulation (non-limiting). In some embodiments, the Reality Transform regime is applied with a local gain-modulation scalar (denoted g_{in} and distinct from the trust score $\sigma_{ij}(t)$, the canonical participant ordering π_{ord} , the tool-call specification tc , and any conventional statistics-notation σ elsewhere in this disclosure) targeting the device's own response gain rather than an external scene's appearance. In some embodiments, this may be applied inwardly (modulating perceptual reactivity), outwardly (attenuating emitted signal), or simultaneously in both directions. In some embodiments, the device remains live during inward-facing operation and may continue to interact with its environment; the distinguishing feature is that the fast feedback loop is deliberately damped, for example via a smoothed self-model constructed during a prior calibration period. In some embodiments, this is a degenerate case of Reality Transform rather than a new regime.

16.6 Multi-RK corroboration for agents

In some embodiments, an agent has access to multiple Reality Kernel modules with distinct configurations, invariance profiles, or vantage points. The agent may request corroboration for a claim by querying multiple kernels and comparing the resulting evidence under spatiotemporal consistency constraints. In some embodiments, the agent uses a committee rule such that high-stakes actions require corroboration from at least k of n kernels, optionally with diversity constraints so that not all corroborators share the same failure mode.

16.7 Bilateral Proof-of-Discrepancy (B-PoD) protocol (non-limiting)

Purpose

Bilateral PoD (B-PoD) exists to let independent devices share and verify physical discoveries even when they are not members of the same fleet. B-PoD makes a jointly encountered physical correction event portable, authentic, and open to later third-party reproduction, so that cross-device trust can be grounded in reproducible physical evidence rather than in shared fleet membership alone.

The primary security function of B-PoD is different from that of fleet-quorum PoD. Fleet quorum PoD provides immediate Byzantine-fault-tolerant consensus by requiring physical reproduction on $n \geq 3f + 1$ independent verifier devices. B-PoD provides deferred public falsifiability: any third party can attempt to reproduce the claimed discrepancy; a fake bond
 5 that does not reflect a genuine physical event is falsifiable by any party who tries to replicate the experiment and fails. These are complementary security functions, not competing alternatives.

Physical Authentication Principle (reminder). See Remark 3. The primary object of B-PoD is the physical discrepancy event. Cryptographic identifiers, signatures, and related
 10 proofs authenticate the event record; they do not by themselves establish the physical truth of the recorded claim.

Native object model

B-PoD uses three distinct objects. Standards-based infrastructure (DID-style identifiers, verifiable credential proof formats, remote-attestation evidence structures) are interoperability
 15 mappings over these objects, not substitutes for them.

B-PoD Event. The primary object. A record of a physical event with authentication attached. In some embodiments, a valid B-PoD Event binds all of the following:

- (i) **Originator identifiers.** Controller-bound identifiers for both participating agents.
- (ii) **Experiment Definition.** Content-addressed specification of the experiment: declared
 20 circuit configuration identifier, envelope version, scan law, protocol parameters, and procedure sufficient for independent replication.
- (iii) **Discrepancy domain and type.** Structured label indicating the relevant physical subdomain (e.g. scattering, nonlinear dynamics, coupling) and the category of discrepancy (e.g. systematic, noise-floor, regime-boundary).
- 25 (iv) **Pre-run commitments.** Each originator's pre-run predictions or model expectations for the declared experiment, committed cryptographically before execution.
- (v) **Independence declaration.** A signed statement by each originator that, during the run generating the claimed discrepancy, its execution environment was not controlled by, inspectable by, or execution-steered by the counterparty except through the agreed
 30 Experiment Definition and permitted protocol messages.
- (vi) **Evidence commitments.** Content-addressed commitments to raw sensor logs, calibration records, processing steps, and meter summaries sufficient to support independent replication.

(vii) **Reproduction criteria.** Declared conditions under which a later party’s run constitutes a same-configuration replication (see Replication Outcome types below).

(viii) **Observed discrepancy statement.** The measured discrepancy relative to declared pre-run predictions, stated in the same units as the corresponding fleet discrepancy metric.

(ix) **Model update claim.** A declaration (not a model disclosure) that at least one originator updated a model or prediction in response to the discrepancy. Model update policy is local to each agent; this field records that an update occurred, not its content.

(x) **Both originator signatures.** Authenticating the event record. The signatures are not the event; they are the authenticity layer on the event.

B-PoD Envelope. The cryptographic carrier for a B-PoD Event. Contains controller-bound identifiers, signing keys, timestamps, nonces, and optional hardware attestation material. DID-style identifiers and remote-attestation-style evidence operate at the envelope layer. The envelope transports the event; it is not the event.

Portable Discrepancy Record. An append-only collection of B-PoD Events plus all subsequent appraisals. The following outcome types are first-class record elements:

- **Same-configuration replication.** An independent party ran the same Experiment Definition (same circuit configuration identifier, envelope version, and procedure as declared in the root event) and either confirmed or disconfirmed the observed discrepancy. This is the only outcome type that is directly confirmatory or disconfirmatory for the root claim.
- **Related-configuration replication.** A run using a related but distinct circuit configuration, envelope, or procedure. Informative but not directly confirmatory. Must be typed distinctly; blending this type with same-configuration replication smears evidence across experiments that are nearby in domain but not the same claim.
- **Methodological challenge.** An argument that the root run or any subsequent replication was not a valid test of the declared claim. May target any event in the record, not only the root.
- **Supersession.** Replaces the operational interpretation of an earlier event with a better-characterised successor while preserving the superseded event in history. Supersession is science working correctly, not a retraction.
- **Withdrawal.** Originator-level reduction of confidence in an earlier claim, without deleting the record. A withdrawal is a signal worth examining; it does not erase the prior event.

Living Scientific Object. A Portable Discrepancy Record is a living scientific object rather than a static credential. Confidence in a B-PoD claim is cumulative and asymmetric: successful replications increase confidence without proving the claim correct; failed replications decrease confidence without conclusively refuting it (failure can arise from methodological differences, uncontrolled variables, or other causes); and the full history of both is part of the object itself.

Restricted-domain commensuration

B-PoD enables cross-device comparison only within the restricted domain defined by a declared experiment, circuit configuration, operating envelope, and procedure. Two devices with otherwise divergent or independently trained predictive models may still jointly determine whether the declared physical system under declared conditions produced the declared outcome. Physical reality does not depend on which model either device holds. B-PoD does not require or imply convergence of broader models or calibration histories; it enables measurement-level comparison within a declared envelope.

This is the architectural mechanism by which devices with different predictive models can share physically grounded corrections: reproducibility under a declared protocol is the common basis. Two devices with genuinely different surrogate models can both execute the same physical experiment and jointly observe the same discrepancy. What bilateral PoD records is that joint observation.

Extended bilateral applications (non-limiting). In some embodiments, B-PoD supports operation by independent or fleet-divergent devices, fleet reintegration through reproducible claims, and relying-party appraisal of Portable Discrepancy Records. These extended applications are described in the related Filing 2 application (optional, non-essential).

16.8 Security considerations for RK-agent systems

In some embodiments, attacks include scene spoofing, co-illumination interference, hardware tampering, communications interference, and adversarial examples targeting meters. Mitigations include multi-channel invariance profiles, multi-RK corroboration, spatiotemporal checks, and gating of actions on validated RK evidence. In multi-agent deployments, some nodes may be compromised or Byzantine. Trust-weighted aggregation, diversity constraints, and periodic audits of calibration and meter agreement may be used to reduce the influence of compromised nodes.

16.9 Agent lifecycle management and trajectory records (non-limiting)

Lifecycle states (non-limiting). In some embodiments, a device or agent transitions between lifecycle states: **Embodied** (external actuators connected and real scene loop active),

Docked (actuators disconnected or logically disabled while RK evidence plumbing remains active), **Offline** (operating on synthetic or modified scenes, with parameter-sandboxed or calibration modes as described in the offline simulation subsection above), and **Re-embodied** (actuators reconnected after compatibility checks). State transitions are logged in protocol digests and meter summaries. The full lifecycle state machine is described in the related Filing 2 application (optional, non-essential).

17 Digital Twin and Reality Merge Pipeline (Optional, Non-limiting)

In some embodiments, the Reality Transform regime and its associated training, deployment, and verification machinery are extended via a digital-twin and reality-merge pipeline. This section describes non-limiting embodiments of multi-factor pose-conditioned Reality Transform, payload-embedding Reality Transform, co-emission across the separation channel, twin training and analogue-twin deployment, style corpus acquisition, generator-and-inverse pairs, scene-state-contrast training-data construction, and associated label-integrity and privacy-discipline mechanisms. These embodiments are optional extensions: core Reality Kernel operation, including the three operating regimes of Section 8, does not require any mechanism described in this section.

17.0.1 Multi-factor pose-conditioned Reality Transform (non-limiting)

In some embodiments, the Reality Transform regime is specialised to articulated-content production by adding declared pose-feature consistency terms to the RT control objective of Section 8.4.4. The pose-feature terms are computed from committed evidence windows of the convolution bundle under a declared extraction rule, and are used by the controller to select or update emission controls subject to the protocol digest and meter envelope. The pose-feature terms do not require any novel pose-estimation architecture: the pose-feature extractor may be any declared measurement head whose identity, weights or version, input preprocessing, confidence thresholds, calibration state, and output coordinate convention are committed to the protocol digest.

In some embodiments, a committed Reality-Transform decoder or extraction map RTDec , whose identity, version, parameterisation, output codomain, and any fixed comparison map are committed to the protocol digest, maps a committed evidence window of the convolution bundle to an image-like or feature representation

$$Y_t^{\text{dec}} = \text{RTDec}(C_{\text{Win}_t}, \Pi),$$

where C_{Win_t} denotes the committed convolution-bundle atoms selected by the declared evidence-windowing rule Win_t , and Π is the protocol digest. The declared output codomain of RTDec , or the declared fixed comparison map applied to it, is chosen so that $d(Y_t^{\text{dec}}, \mu_\phi(x_t, u_t))$

is evaluated in a common representation space. A declared pose-feature extractor **Pose** — whose architecture, weights, input preprocessing, confidence threshold, calibration state, and output coordinate convention are likewise committed — then produces a pose-feature vector, joint-angle vector, keypoint set, parametric-body-model state, hand-pose state, or
 5 related pose representation

$$p_t = \text{Pose}(Y_t^{\text{dec}}),$$

optionally together with a confidence or visibility mask Mask_t . The RTDec identity, pose-feature extractor identity, pose-coordinate convention, time base Δt , output-comparison space, and any confidence-mask rule are committed to the protocol digest.

Compound pose-control penalty. In some embodiments, the controller optimises a
 10 non-limiting pose-conditioned RT objective

$$J_{\text{RT-pose}}(U_{0:T}; \phi) = -\mathbb{E} \left[\sum_{t=0}^{T-1} \left(d(Y_t^{\text{dec}}, \mu_\phi(x_t, u_t)) + \lambda_{\text{dat}} \ell_t^{\text{dat}} + \lambda_{\text{tmp}} \ell_t^{\text{tmp}} + \lambda_{\text{meter}} \mathbf{1}[\mathbf{m}(t) \notin \mathcal{M}_{\text{accept}}] \right) \right].$$

Here $\phi \in \Phi$ is the declared style parameter of Section 8.4.4, μ_ϕ is the target response function defined therein, $d(\cdot, \cdot)$ is the declared divergence or distortion metric, and $\mathbf{m}(t)$ and $\mathcal{M}_{\text{accept}}$ are the meter vector and meter envelope of Definition 15. The sign convention matches J_{RT} of Section 8.4.4: the objective is *maximised*, which corresponds to *minimising* the per-
 15 step weighted-loss sum. The additional per-step pose-consistency contributions ℓ_t^{dat} and ℓ_t^{tmp} are defined in the following paragraphs. During operation, the values of ℓ_t^{dat} and ℓ_t^{tmp} are computed from committed convolution-bundle windows and applied by the controller to select or update the next physical emission or control action of the Reality Kernel module, so that the pose terms are operative control constraints on a physical channel rather than
 20 merely post-hoc mathematical scores. For online control, any pose-consistency term whose declared window requires atoms later than the current control decision is evaluated as a delayed meter, a retrospective update term, or a declared predictive surrogate, and no future atom is used to select a control action before that atom has been committed. The weights λ_{dat} , λ_{tmp} , and λ_{meter} are chosen on the same declared scale as $d(\cdot, \cdot)$ so that all terms in the
 25 per-step sum are commensurate, and are committed to the protocol digest together with the identities of RTDec, **Pose**, the pose-reference set $\mathcal{D}_{\text{pose}}$, and the windowing and normalisation rules used to compute the pose terms.

Dataset-consistency contribution (non-limiting). In some embodiments, the dataset-consistency contribution ℓ_t^{dat} is derived from a functional L_{dat} that compares pose features
 30 extracted from the RT evidence window with pose features extracted from a declared pose-reference set $\mathcal{D}_{\text{pose}}$ under the same declared pose-feature extractor **Pose** and coordinate convention. Non-limiting realisations include:

1. a learned pose-distribution discriminator A_{dat} , whose architecture, weights, training window length, positive set, negative set, held-out validation policy, and version hash

are committed to the protocol digest, and whose score on $p_{0:T}$ derived from committed RT outputs versus **Pose**-features derived from $\mathcal{D}_{\text{pose}}$ defines L_{dat} ; in some embodiments, A_{dat} is trained under a declared binary or multiclass classification loss distinguishing pose windows drawn from $\mathcal{D}_{\text{pose}}$ from pose windows drawn from generated, transformed, or edited RT pose windows, with the loss family and threshold-calibration policy committed to the protocol digest;

2. a declared kernel two-sample statistic

$$\text{MMD}_k^2(P, Q) = \mathbb{E}_{p, p' \sim P} k(p, p') + \mathbb{E}_{q, q' \sim Q} k(q, q') - 2 \mathbb{E}_{p \sim P, q \sim Q} k(p, q),$$

computed over pose-feature vectors or pose-feature windows, with kernel family, bandwidth, batching rule, and confidence-mask policy declared in the protocol digest; and

3. a pose-class or action-class consistency loss, in embodiments where the style parameter ϕ includes a declared pose-class, action-class, or kinematic-intent component, with the classifier identity, label ontology, calibration set, and confidence threshold committed to the protocol digest; in such embodiments, the protocol digest further declares the mapping from ϕ to the target pose-class, action-class, or kinematic-intent distribution, and declares whether the per-step contribution is a cross-entropy, calibrated margin, or other classifier-score loss.

In some embodiments, ℓ_t^{dat} is obtained from L_{dat} by a declared per-step decomposition (for example a windowed discriminator score, a windowed MMD estimate, or a windowed classifier margin), with the decomposition rule committed.

Between-frame consistency contribution (non-limiting). In some embodiments, the temporal-consistency contribution ℓ_t^{tmp} is derived from a functional L_{tmp} that measures inter-frame pose-dynamics plausibility using the committed time base of the convolution bundle. Non-limiting realisations include:

1. a derivative-residual penalty

$$\ell_t^{\text{tmp,deriv}} = \left\| \text{Mask}_t \odot \frac{p_{t+1} - p_t}{\Delta t} \right\|_{\Sigma_v^{-1}}^2 + \left\| \text{Mask}_t \odot \frac{p_{t+1} - 2p_t + p_{t-1}}{\Delta t^2} \right\|_{\Sigma_a^{-1}}^2,$$

where Σ_v and Σ_a are declared velocity and acceleration envelopes or covariance estimates committed to the protocol digest. For boundary indices at which a finite-difference stencil is not fully available, the protocol digest declares whether a one-sided difference, padding convention, or boundary-index exclusion is used; the digest also declares how confidence masks are composed across the velocity or acceleration stencil and whether Σ_v^{-1} and Σ_a^{-1} denote ordinary inverses, regularised inverses, pseudoinverses, or diagonal reciprocal envelopes;

2. a learned pose-dynamics residual

$$\ell_t^{\text{tmp,dyn}} = \|\text{Mask}_{t+1} \odot (p_{t+1} - g_{\text{dyn}}(p_t, u_t, \Delta t))\|^2,$$

where g_{dyn} is a declared pose-dynamics model whose identity, input fields, training corpus, validation corpus, and version hash are committed; in some embodiments, g_{dyn} is trained by a declared one-step or multi-step pose-prediction loss on committed positive pose-sequence windows, with the prediction horizon and teacher-forcing or rollout policy committed to the protocol digest; and

3. a kinematic-plausibility classifier trained against committed physical pose-sequence windows and declared negative examples, including time reordering, frame-interpolation artefacts, pose or keypoint substitutions between protocol-compatible sequences, and local blending of motion-warped frames; in some embodiments, the kinematic-plausibility classifier is calibrated on held-out committed physical pose-sequence windows and held-out edited-negative windows, with the acceptance threshold and false-positive target committed to the protocol digest.

In some embodiments, missing or occluded pose features are handled by the confidence or visibility mask Mask_t rather than by treating all keypoints as equally observed.

Composition with verisimilitude and structural meters. In some embodiments, $J_{\text{RT-pose}}$ is evaluated together with the verisimilitude functional of Definition 31, the edge-based structural meters of Section 9.8, timing-integrity meters, calibration-consistency meters, and spatiotemporal-consistency meters. The pose contributions may be declared as acceptance meters, evaluation meters, or objective terms according to the meter-partition policy for the deployment. The acceptance/evaluation partition is preserved so that controller optimisation against pose consistency does not expose all evaluative surfaces simultaneously.

Training data and negative-example construction (non-limiting). In preferred embodiments, $\mathcal{D}_{\text{pose}}$ comprises committed physical recordings, committed RT runs, or committed offline-session records whose protocol digests are available for audit. In some embodiments, an externally curated pose-reference set may be used as an operator-declared reference envelope, but such a set does not by itself supply physical provenance or empirical-hardness evidence unless its acquisition and validation are separately committed. The protocol digest shall provenance-type $\mathcal{D}_{\text{pose}}$ as committed physical evidence, committed offline-session evidence, or external-reference-only, and verifier-facing attestations shall not treat an external-reference-only set as physical-provenance evidence or as support for empirical-hardness claims. Negative examples for discriminator and kinematic-classifier training, and for validating pose-dynamics residual thresholds, may be constructed by applying declared spatiotemporal edits to committed recordings, with each edit family, intensity parameter,

and random-seed policy recorded or hash-identified in the protocol digest, following the pattern of Section 9.5.

Sandboxed and live operation (non-limiting). In some embodiments, pose-conditioned RT training is performed during a parameter-sandboxed offline session under the offline simulation and physical-validation mechanism of Section 16.5, in which the physical kernel operates on synthetic or modified articulated scenes, pose-reference sequences, or rendered targets. Parameter updates made during the sandboxed session are reverted on return to live operation unless a separate calibration or deployment authority authorises retention under the applicable protocol. Pose-feature extractors, pose-distribution discriminators, kinematic classifiers, and validation summaries trained from the sandboxed convolution bundle record may be retained as derived models where permitted by the protocol digest and meter policy.

Relation to the Limager regime (non-limiting). The compound-objective structure is formally analogous to the Limager regime’s decomposed information-gain objective in that both regimes factorise a scalar control objective into declared terms targeting distinct evidence dimensions, each subject to protocol-digest commitments and meter-envelope discipline. The primitive question is different: Limager selects probes to increase information about scene variables, while pose-conditioned RT selects emissions to realise a declared style target while maintaining pose-distribution, pose-dynamics, structural, and physical-consistency constraints.

Scope note (non-limiting). For avoidance of doubt, the pose-feature extractor **Pose** may be drawn from conventional keypoint, body-model, hand-pose, or related pose-estimation architectures. In such embodiments, the disclosed technical contribution is the use of the declared pose-feature trace as a committed, meter-bounded component of a Reality Transform control objective over a physical evidence substrate, rather than the pose-estimation architecture itself.

17.0.2 Payload-embedding Reality Transform and prior-frame-hash chaining (non-limiting)

In some embodiments, the Reality Transform regime is specialised to carry a declared bit-payload per frame, such that the payload is extractable from the committed convolution-bundle evidence by a declared extraction head while the style fidelity, verisimilitude, and meter-envelope discipline of the surrounding RT objective are preserved. The mechanism is a specialisation of the RT objective of Section 8.4.4: the payload becomes an additional declared evidence dimension on which the compound objective places a meter-bounded term. The payload-embedding Reality Transform is compatible with, and independent of, the run-level hash-chain state χ_t of Section 11.2.

Declared per-frame payload and extractor. In some embodiments, a declared embedder **Embed**, committed to the protocol digest by identity, version, weights, input preprocessing, and output bit-length, produces an emission update that causes frame t of the decoded evidence window to carry a payload

$$\text{Payload}_t \in \{0, 1\}^{N_b},$$

5 where N_b is a declared payload bit-length (for example $N_b \in \{32, 64, 128, 256, 512\}$), committed to the protocol digest. A declared extractor **Extract**, likewise committed by identity, version, weights, and bit-error-rate (BER) threshold, recovers an estimate $\widehat{\text{Payload}}_t$ from the RTDec-decoded frame $Y_t^{\text{dec}} = \text{RTDec}(C_{\text{Win}_t}, \Pi)$:

$$\widehat{\text{Payload}}_t = \text{Extract}(Y_t^{\text{dec}}, \Pi).$$

The BER threshold and any confidence threshold are calibrated on held-out committed
10 physical, parameter-sandboxed, or twin-predicted windows under declared distortion, edit, and compression families, with the calibration corpus and acceptance rule committed to the protocol digest. In some embodiments, **Embed** and **Extract** are jointly trained under the compound RT objective defined below, on committed parameter-sandboxed offline-session evidence, on probe-corpus evidence (Section 17.1.1), or on live RT recordings committed
15 during a declared training phase.

Prior-frame-hash payload structure. In some embodiments, the payload carried by frame t encodes one or more hashes of prior frames, not necessarily the immediate predecessor. Let

$$\text{FrameHash}_t = H(\text{Canon}(Y_t^{\text{dec}}, \text{meta}_t, \Pi), k_{\text{hash}}),$$

where H is a declared cryptographic or keyed hash (for example HMAC, SHA-family, sponge,
20 or a ZK-friendly algebraic hash), meta_t denotes declared per-frame metadata (for example time base, protocol-digest epoch, device identifier), and k_{hash} is a declared key or public parameter committed or referenced in the protocol digest. Here **Canon** is a declared canonicalisation map, committed by identity, version, quantisation rule, colour or feature-space convention, metadata fields, and rounding or tolerance policy, so that frame hashes are re-
25 computable from selectively opened physical evidence windows rather than from uncontrolled raw pixel values.

A declared *lag schedule* $\text{Lag}_t : \{0, \dots, T-1\} \rightarrow \{1, \dots, L\}^{N_{\text{slot}}}$ assigns, for each frame t and each payload slot $j \in \{1, \dots, N_{\text{slot}}\}$, a positive lag value $\text{Lag}_{t,j} \in \{1, \dots, L\}$ indicating which prior frame's hash occupies slot j at frame t :

$$\text{Payload}_t = \text{Pack}\left(\text{FrameHash}_{t-\text{Lag}_{t,1}}, \text{FrameHash}_{t-\text{Lag}_{t,2}}, \dots, \text{FrameHash}_{t-\text{Lag}_{t,N_{\text{slot}}}}; \Pi\right),$$

30 where the packing, slot order, truncation, error-correcting-code, and per-slot weighting rules

of **Pack** are committed to the protocol digest. For prefix frames in which $t - \text{Lag}_{t,j} < 0$, the protocol digest declares whether the corresponding slot is encoded by **Pack** as an omitted-slot sentinel, filled with a committed initialisation hash, or filled under another declared bootstrap convention, with **Pack** producing a fixed N_b -bit payload in all cases.

5 In some embodiments, Lag_t is (i) a fixed constant schedule (for example always $\text{Lag}_{t,j} = 1$), (ii) a declared per-segment schedule committed to the protocol digest, or (iii) a pseudorandom-function output keyed by the protocol digest and the run seed, so that Lag_t is reproducible by a verifier holding the key or public parameters but is not predictable to a party without them. Where a lag schedule or embedding rule depends on a secret or
 10 delayed parameter, the protocol digest commits the key identifier, public parameter, delayed-reveal policy, or verification-access rule under which a verifier may reproduce the schedule for audit. Multi-slot schedules with $N_{\text{slot}} \geq 2$ provide redundancy against short-range edit attacks that attempt to resynchronise to an adjacent frame.

Compound RT objective with payload extractability. In some embodiments, the
 15 controller optimises a non-limiting payload-conditioned RT objective, written in the negative-expected-loss convention of J_{RT} :

$$J_{\text{RT-pay}}(U_{0:T}; \phi) = -\mathbb{E} \left[\sum_{t=0}^{T-1} \left(d(Y_t^{\text{dec}}, \mu_\phi(x_t, u_t)) + \lambda_{\text{pay}} \ell_t^{\text{pay}} + \lambda_{\text{ver}} \ell_t^{\text{ver}} + \lambda_{\text{meter}} \mathbf{1}[\mathbf{m}(t) \notin \mathcal{M}_{\text{accept}}] \right) \right],$$

where:

- the style-target term $d(Y_t^{\text{dec}}, \mu_\phi(x_t, u_t))$ is the RT fidelity term of Section 8.4.4;
- the per-step payload-extractability contribution ℓ_t^{pay} is a declared bit-error or cross-
 20 entropy loss between $\widehat{\text{Payload}}_t$ and the target Payload_t specified by Lag_t and **Pack**, with the loss family, per-slot weighting, and error-correcting-code policy committed to the protocol digest;
- ℓ_t^{ver} is a declared verisimilitude term derived from the functional of Definition 31, penalising embedding artefacts that depart from the empirical distribution of genuine
 25 unembedded RT outputs under the committed verisimilitude discriminator; and
- the meter-violation term is the standard envelope term from Section 8.4.4.

The weights $\lambda_{\text{pay}}, \lambda_{\text{ver}}, \lambda_{\text{meter}}$ are on the same declared scale as $d(\cdot, \cdot)$ and are committed to the protocol digest. Perceptual transparency is promoted and meter-bounded by composition: the λ_{ver} term penalises embedding artefacts that depart from the empirical distribution of
 30 genuine unembedded RT outputs as measured by the committed verisimilitude discriminator, and any acceptance threshold or transparency envelope is declared in the protocol digest.

Composition with pose-conditioned RT. In some embodiments, the payload-extractability and verisimilitude terms are composed with the pose-conditioned RT terms

of Section 17.0.1, yielding a compound objective

$$J_{\text{RT-pose+pay}} = J_{\text{RT-pose}} - \mathbb{E} \left[\sum_t (\lambda_{\text{pay}} \ell_t^{\text{pay}} + \lambda_{\text{ver}} \ell_t^{\text{ver}}) \right].$$

Equivalently, the terms may be written as a single negative-expected-loss objective under the same sign convention as J_{RT} and $J_{\text{RT-pose}}$. The meter-partition policy determines which terms are declared as acceptance meters, evaluation meters, or objective terms; the accep-
 5 tance/evaluation partition is preserved as in Section 17.0.1.

Tamper-evidence properties (non-limiting). In some embodiments, the prior-frame-hash payload chain provides localised tamper evidence: if a frame t^* is post-hoc edited, the FrameHash_{t^*} values committed in the payloads of all frames t satisfying $t - \text{Lag}_{t,j} = t^*$ under the declared **Lag** schedule fail validation against the recomputed hash of the edited
 10 frame, after canonicalisation by **Canon** and subject to the declared extraction confidence, BER threshold, and error-correcting-code policy, thereby localising the tamper event. This localised evidence is orthogonal to, and composable with, the global tamper-evidence provided by the run-level hash chain χ_t of Section 11.2: the per-frame payload chain identifies *which* frame was edited, while χ_T provides the compact run fingerprint. In some embod-
 15 iments, PRF-indexed Lag_t schedules strengthen tamper evidence against anticipatory or segment-resynchronisation attacks; construction of an internally consistent edited segment further requires re-embedding corrected payloads under the declared **Embed** authority, key or parameter policy, BER threshold, and verisimilitude and meter-envelope constraints.

Relation to hash-chain and watermark mechanisms (non-limiting). The payload-
 20 embedding Reality Transform is distinct from, and compatible with: (i) the run-level hash-chain χ_t of Section 11.2, which binds $(u(t), \mathbf{y}_t)$ -level state in a one-way chain external to the rendered content; (ii) the covert device-watermark primitive in the information-operations taxonomy, which embeds a continuous device-identity signature rather than a discrete per-frame payload; and (iii) the per-atom Merkle digests of Section 11.3, which commit to atoms
 25 after recording. In the embodiments described in this paragraph, payload-embedding RT is distinguished by embedding a declared *discrete extractable bit-payload* in each frame of the rendered content itself, while the other listed mechanisms bind provenance, device identity, or post-recording commitments through different surfaces.

Scope note (non-limiting). For avoidance of doubt, **Embed** and **Extract** may be drawn
 30 from conventional deep-steganography or digital-watermarking architectures. In such embodiments, the disclosed technical contribution is the joint specification of a declared prior-frame-hash payload structure, committed lag schedule, canonical frame-hash and packing rules, and meter-bounded RT training or evaluation over committed physical evidence windows or provenance-typed twin-predicted windows, rather than the embedder or extractor

architecture itself.

Positive-residual additive-overlay RT objective (non-limiting). In some embodiments suitable for bright-ambient or minimum-perturbance operation, the Reality Transform objective is restricted to additive overlays whose contribution to the photodetector or imager response, evaluated against a declared ambient or neutral baseline Baseline_t , is non-negative on the aggregate observed signal. The baseline Baseline_t is computed by the controller from declared baseline-acquisition windows committed to the protocol digest — non-limiting choices include short pre-emission ambient samples, neighbouring-pixel neutral references, or temporally adjacent neutral frames — with the baseline-acquisition rule, baseline-window length, and any spectral or spatial pre-filtering parameters all committed to the protocol digest. The compliance functional is the aggregate observed positive excess $\max(\widehat{R}_t - \text{Baseline}_t, 0)$, evaluated over the declared evaluation window and (where applicable) summed across K_{co} co-emitted streams; the controller selects emission schedules under the constraint that this aggregate-excess functional remains within the declared additive-overlay envelope. In some embodiments, the additive-overlay objective composes with the meter envelope of Definition 15 as an additional declared meter term, and any per-stream perturbation budgets are recorded as auxiliary constraints valid only within the declared separation, linearity, and crosstalk envelope. The aggregate formulation prevents a companion stream from complying per-stream while the combined emitted light violates the bright-ambient perturbation objective.

Perception-collapse semantic fallback (non-limiting). In some embodiments, the Reality Transform regime declares a *semantic fallback* regime selected at deployment time when a perception-collapse condition is detected on the verisimilitude or dual-direction-consistency meter envelope. The perception-collapse condition is defined by a declared meter-threshold rule committed to the protocol digest, comprising at least: a verisimilitude floor, a dual-direction-consistency floor, a sustained-window length over which the floors must be violated, and a hysteresis margin governing return to the primary regime. While the fallback is active, the controller restricts emission selection to a declared safe subset of the operating envelope — non-limiting choices include reversion to a low-amplitude additive overlay under the positive-residual objective of Section 17.0.2, suspension of payload embedding, or rotation to a previously verified protocol-digest version — with the fallback entry timestamp, the meter values at entry, the active fallback policy identifier, and the fallback exit timestamp committed to the protocol digest. In some embodiments, the semantic fallback is integrated with the verification-gating mechanism described in Section 16, such that actions classified as RK-verified, hardness-gated, or multi-RK-verified are denied or escalated while the fallback is active. The fallback regime does not replace any audit, commitment, or verisimilitude evidence already accrued before the perception-collapse condition; it restricts the policy of subsequent emissions until the meter envelope returns to compliance.

Intervention-gated recurrent-state hygiene (non-limiting). In some embodiments, recurrent or stateful components of the controller — including learned policies with hidden state, adaptive scan-law generators, and recursive estimators — maintain a declared *state hygiene* discipline under which recurrent state is explicitly resettable, snapshot-able, and auditable. The hygiene discipline comprises at least: a declared snapshot-cadence rule committing recurrent-state digests to the protocol digest at declared intervals; an intervention-gating rule specifying which controller events — non-limiting examples include protocol-digest version rotation, semantic-fallback entry or exit, manual operator intervention, B-PoD discrepancy escalation, or any reset operation declared by the audit policy interface — trigger committed snapshot, partial reset, or full reset of declared recurrent-state components; and a recurrent-state provenance rule under which any synthetic-controller output produced from recurrent state is provenance-typed identically to other declared synthetic artefacts in this disclosure (evidence of the committed input, the declared inference procedure, and the recurrent-state digest at output time, not direct physical- provenance evidence). In some embodiments, the snapshot, reset, and intervention events are themselves committed events of the protocol digest under the same commitment plumbing of Section 11.2, so that recurrent-state hygiene is auditable from the same selectively-openable record as the rest of the run.

17.0.3 Co-emission across the separation channel (non-limiting)

In some embodiments, the Reality Transform regime shares the separation channel **Sep** (Section 17.2.3) with one or more additional declared emission streams operated concurrently with the styled RT emission. The resulting *co-emission* architecture supports RT-loop stabilisation via companion-channel promotion, continuous live paired-evidence acquisition during deployment, tamper-evidence via committed companion streams, and multi-regime concurrent operation. The primary operational use of co-emission is stabilisation of an RT loop exhibiting degradation, rather than continuous multi-stream emission.

Base primitive (non-limiting). In some embodiments, a co-emission configuration comprises $K_{\text{co}} \geq 2$ independent emission streams carried concurrently across **Sep**. For avoidance of doubt, the symbol K is used in the remainder of the filing with other meanings (reactor index, coupling-regime label, dynamics coefficient), and K_{co} is used specifically for the number of co-emitted streams throughout this subsection and the corresponding claims. Each stream is drawn from a declared source-class, drawn from a non-limiting list comprising:

- the Reality Transform styled emission of Section 8.4.4;
- a Truth Beam probe emission (Truth Beam regime of the present filing);
- a Limager probe emission (Limager regime of the present filing);
- a declared neutral probe emission drawn from the probe family **Probe** of Section 17.1.1;

- a committed cryptographic or pseudorandom generator stream (for example a keyed extendable-output function such as BLAKE3 XOF across declared octaves);
- a dedicated human-viewing emission (for example committed low-intensity white or pink illumination) whose role is naked-eye viewer illumination only, used in sub-embodiments in which the Reality Transform is not the human-visible channel.

Each stream is committed to the protocol digest with a declared stream identifier, source-class, amplitude profile, per-stream recovery rule, detector or demodulation assignment, residual-crosstalk envelope, and role assignment specifying what downstream processes (human perception, twin training, verification, tamper-evidence audit, stabilisation, or combinations thereof) the stream feeds. A verifier holding the committed stream declarations and a recording from the declared detector or detector set assigned to the relevant stream can determine whether each declared stream was selectively recoverable under the committed Sep modality and its residual-crosstalk envelope.

RT as human channel (primary configuration, non-limiting). In preferred embodiments, the Reality Transform styled emission serves as the human-visible channel, and one or more companion streams serve computational roles that do not require naked-eye viewing. Companion streams are declared at amplitudes either below a declared perceptual-visibility threshold (in the case of imperceptible companions) or at declared intensities recoverable by appropriate filter hardware (in the case of filtered-viewer configurations). The dedicated human-viewing source-class is used only in sub-embodiments where the Reality Transform is not currently active; in the default case, the Reality Transform is itself the human channel.

Amplitude regimes (non-limiting). In some embodiments, co-emission operates under one of two declared amplitude regimes, each committed to the protocol digest along with corresponding perceptual thresholds and recording recoverability thresholds:

1. **Dark-ambient / photometric-match regime.** The Reality Transform styled emission fills the scene's illumination budget under a declared photometric-match objective; companion streams are additive on top of the style and must remain at declared low amplitudes to preserve the photometric match, typically below the declared perceptual-visibility threshold while remaining above the declared recording-recoverability threshold.
2. **Bright-ambient / minimum-perturbance regime.** The scene is already illuminated by ambient light, the Reality Transform modifies appearance additively, and co-emitted companion streams must operate under the positive-residual additive-overlay objective of Section 17.0.2. The additive-overlay compliance is evaluated on the aggregate observed positive excess $\max(\hat{R}_t - \text{Baseline}_t, 0)$ across all K_{co} co-emitted streams, where Baseline_t is the declared ambient or neutral baseline of Section 17.0.2, not on any single named stream, with per-stream perturbation budgets logged as auxiliary

constraints valid only within the declared separation, linearity, and crosstalk envelope. The aggregate formulation ensures that a companion stream cannot comply per-stream while the combined light violates the bright- ambient perturbation objective.

Stabilisation via companion-channel promotion (primary use-case, non-limiting).

In preferred embodiments, co-emission is operated in a state-transitioned manner: the Reality Transform runs in its ordinary single-stream configuration during steady-state operation, and a companion stream is *promoted* (brought into concurrent co-emitted use) when a committed instability meter M_{instab} indicates loop degradation. The companion is *demoted* (returned to inactive status) when a committed recovery meter indicates loop recovery. The promotion and demotion policies, instability meter, recovery meter, and all associated thresholds are committed to the protocol digest.

In some embodiments, M_{instab} comprises one or more of the following declared statistics, alone or in combination:

1. dual-direction consistency drift from Section 17.2.7: δ_{fwd} , δ_{inv} , or δ_{sym} exceeding declared thresholds;
2. verisimilitude-discriminator pass-rate on a rolling window dropping below a declared threshold, evaluated under the verisimilitude functional of Definition 31 and the discriminator-training composition of Section 9.5;
3. pose-extractor confidence statistics derived from Mask_t of Section 17.0.1 dropping below declared thresholds on a declared evaluation window;
4. twin residual r_t or r_t^{AnaTwin} moving outside the declared approximation envelope or matching envelope of Section 17.1.5.

Companion-class stabilisation roles (non-limiting). In some embodiments, the stabilising effect of companion-channel promotion depends on the companion source-class:

- A neutral-probe companion (drawn from the probe family of Section 17.1.1) provides an independent clean scene response at each frame, from which pose features, scene state, and twin residuals become estimable even when the styled channel produces confusing captures.
- A Truth Beam-probe companion provides continuous substrate physical-verification evidence; drift outside $\mathcal{M}_{\text{accept}}$ on the probe channel is direct evidence that the Reality Transform’s assumed substrate model is in error, distinguishing substrate drift from loop drift.
- A Limager-probe companion provides continuous information-gain measurement about scene state variables, resolving observability gaps in the Reality Transform’s pose-tracking without degrading the styled emission.

- A committed cryptographic or pseudorandom stream companion provides measurement-side audit only: its recovered statistics can be compared against the committed profile to detect measurement-chain or sensor drift, but by itself the cryptographic companion does not recover pose, scene state, or twin residual. In embodiments in which the instability meter indicates scene-state or perception-side degradation rather than pure measurement-side drift, a cryptographic companion alone is insufficient, and the protocol digest declares whether the stabilisation procedure instead promotes an evidence-bearing companion (neutral, Truth Beam, or Limager) or escalates to output-side recovery via the semantic-fallback or recurrent-state-hygiene mechanisms of Section 17.0.1.

Promotion/demotion hygiene and state boundaries. In some embodiments, each promotion event and each demotion event is treated as a recurrent-state hygiene boundary for the Reality Transform controller: the controller’s recurrent state is reset, quarantined, or reweighted under a declared rule committed to the protocol digest, so that intervention-contaminated hidden state does not persist across the boundary. This composes with the intervention-gated recurrent-state hygiene mechanism of Section 17.0.1. In some embodiments, demotion occurs automatically when the recovery meter crosses a committed threshold (auto-demotion variant); in other embodiments, demotion requires external authority approval after the recovery meter crosses its threshold (authority-gated variant). Both variants are committed alternatives.

Dual-side recovery composition. In some embodiments, the co-emission stabilisation mechanism is explicitly composed with the output-side recovery mechanisms of the pose-RT addendum: companion-channel promotion handles the input side (recovering clean evidence when observation signals degrade), while semantic-fallback and recurrent-state-hygiene handle the output side (what to emit when recovery is still in progress). The two sides are independently committed and trigger under independent or shared meters as declared in the protocol digest.

Continuous-operation uses (non-limiting). In some embodiments, co-emission is operated continuously rather than only during stabilisation. Non-limiting continuous uses include:

1. **Live $\mathcal{D}_{\text{pair}}$ acquisition during deployment.** A neutral-probe companion concurrent with a live Reality Transform emission produces paired probe/style atoms under any declared **Sep** modality. Live $\mathcal{D}_{\text{pair}}$ atoms acquired during deployment are committed as deployment-origin paired evidence with run-time role tags, **Sep** modality declaration, recovery envelope, and pre-derivative timestamps recorded before any curated, distilled, synthetic, or training-corpus derivative is generated from them, so that a verifier can distinguish deployment-origin evidence from pre-training evidence and from any subsequent derivative.

2. **Tamper-evidence via committed-statistics companion with externally-anchored commitment.** A committed cryptographic or pseudorandom companion stream provides tamper evidence when the seed commitment is bound to an external anchor **Anchor** — for example a pre-run hash-chain anchored by a verifier-supplied
 5 nonce, a public transparency log entry, a trusted timestamp, or an equivalent external commitment witnessed before the capture window begins. The seed is revealed only after the committed window closes. Local operator-side commitment without **Anchor** is disclosed only as a reduced-assurance sub-embodiment; the primary embodiment requires external anchoring to prevent operator backdating, suppression, or replacement
 10 of the commitment.
3. **Multi-style per-viewer.** Two or more Reality Transform streams with different style parameters $\phi^{(i)}$ are carried on different subchannels, slots, codes, phases, bands, or filter subscriptions under the declared **Sep** modality, such that goggle-equipped viewers
 15 subscribed to different streams experience different styled content of the same underlying scene. Per-stream style-parameter identifiers, goggle filter specifications, and multi-stream amplitude balance are committed to the protocol digest.
4. **Secure recording with triple consistency.** A committed cryptographic companion stream is co-emitted with a Reality Transform styled emission; the recording contains the superposed signal. A verifier holding the recording and the revealed seed recovers
 20 the cryptographic channel via demodulation under the committed **Sep** recovery rule and additionally applies G_{destyle} of Section 17.2.6 to the recovered styled channel (obtained via **Sep** demodulation or a declared prefiltered capture after crypto-channel removal, validated under the committed residual-crosstalk envelope) to produce a synthetic neutral view, yielding three independent consistency sources (the committed
 25 styled atoms, the recovered cryptographic channel, and the G_{destyle} -generated synthetic neutral view). The provenance asymmetry disclosed in Section 17.2.6 is preserved: the synthetic neutral view remains a generator-produced corroborand of declared provenance type, not a substitute for opened physical companion-stream evidence.

Cross-regime co-emission (non-limiting). In some embodiments, co-emission is oper-
 30 ated in cross-regime configurations that may omit a human-visible channel or may combine a computational probe stream with a Reality Transform stream. Non-limiting cross-regime configurations include:

- Truth Beam probe concurrent with Limager probe, providing dual-regime verification on a shared substrate with no human-visible channel;
- 35 • Truth Beam probe concurrent with Reality Transform styled emission, providing continuous physical-verification evidence during styled operation;
- Limager probe concurrent with Reality Transform styled emission, providing concurrent information-gain measurement during styled operation.

Each cross-regime configuration inherits the commitment and recovery discipline of the base primitive.

Goggle-based access control (non-limiting). In some embodiments, co-emission is combined with viewer filter hardware (goggles, polariser eyewear, IR-blocking eyewear, etc.) to produce declared access-control behaviours. Goggle specifications (filter type, passband, shutter timing, polariser orientation, spectral cutoff) are committed to the protocol digest alongside the **Sep** modality. Multi-goggle-set configurations permit different viewers to subscribe to different subsets of the co-emitted streams. In embodiments where a naked-eye viewer (without goggles) is expected to be present, the aggregate amplitude profile across all co-emitted streams is constrained to keep the unfiltered naked-eye experience within a declared safety and navigability envelope, committed to the protocol digest.

Declared bleed-through envelope (non-limiting). For avoidance of doubt, goggle-based perceptual filtering is never absolute: an unfiltered viewer receives residual photons from filter-intended channels through ambient scatter, imperfect filtering, viewing-angle dependence, or scene-induced modality changes (depolarisation, spectral shift). The protocol digest commits a declared bleed-through envelope describing the expected residual amplitudes perceived by an unfiltered viewer; claims of imperceptibility are always relative to this declared envelope rather than absolute.

Independence discipline for composed tamper-evidence (non-limiting). In some embodiments, co-emission composes with the payload-embedding Reality Transform of Section 17.0.2 to produce three orthogonal tamper-evidence mechanisms operating simultaneously: the prior-frame-hash payload chain on the styled channel, the run-level hash-chain state χ_t of Section 11.2, and the independently recoverable cryptographic companion stream. In such embodiments, the three mechanisms are declared under independent seeds, domain separators, and key-derivation roots committed to the protocol digest, so that compromise of one mechanism's keying material does not compromise the others.

Reality-merge composition (non-limiting). In some embodiments, role-authorised companion-stream residuals are logged as reality-merge evidence under Section 17.1.5, and may drive twin refinement, analogue-twin re-matching under the matching-fidelity envelope of Section 17.1.3, or successor protocol-digest version updates, according to the companion stream's declared role assignment in the protocol digest.

Scope note (non-limiting). For avoidance of doubt, the co-emission primitive uses the same **Sep** infrastructure as the paired-capture training-data acquisition of Section 17.2.3. The disclosed technical contribution in this subsection is the joint specification of declared source-class composition, committed per-stream roles, amplitude regimes composing with the additive-overlay objective, committed instability meter and promotion/demotion policy

for RT-loop stabilisation, external commitment anchoring for cryptographic companions, and composition with payload-embedding RT, G_{destyle} , semantic fallback, and recurrent-state hygiene, rather than the underlying separation-channel hardware itself.

In some embodiments, where the Reality Transform subsystem produces articulated content under a declared deployment scope, the trajectory-domain classifier family further includes pose-conditioned classifiers whose inputs comprise committed pose-feature traces, pose-distribution residuals, and pose-dynamics residuals derived from the pose-conditioned Reality Transform embodiments of Section 17.0.1, with the pose-feature extractor, pose-reference-set provenance type, negative-edit families, and residual definitions committed to the session protocol digest. Companion governance embodiments described in the related Filing 2 application may consume these classifier outputs and committed evidence types under their own state and audit-policy machinery; such consumption does not require any state, mode, or governance taxonomy from the related Filing 2 application to be present in the apparatus disclosed here.

In some embodiments, the style parameter ϕ may include an articulated-content component, such as a pose class, action class, kinematic intent, or reference-motion digest, in which case pose-conditioned Reality Transform embodiments may evaluate additional pose-distribution and pose-dynamics terms as described in Section 17.0.1. In some embodiments, the declared Reality Transform objective may additionally carry a declared per-frame payload whose extractability is enforced as part of the compound objective, as described in Section 17.0.2. In some embodiments, training of any such specialised Reality Transform objective is conducted against a declared digital twin under the reality-merge pipeline of Section 17.1.

17.1 Digital twin and reality-merge pipeline (non-limiting)

In some embodiments, the physical Reality Kernel and its derived models are coupled through a pipeline that proceeds, end-to-end, through four stages: (i) acquisition of a committed probe corpus under declared spectrally neutral or spectrally shaped illumination with declared temporal-correlation structure (Section 17.1.1); (ii) training or matching of a twin from the probe corpus, where the twin may be a digital twin (Section 17.1.2), an analogue twin comprising a second physical Reality Kernel module operated in twin-role (Section 17.1.3), or a hybrid digital–analogue twin; (iii) training of a Reality Transform controller against the twin under a compound objective that may include pose-conditioned terms (Section 17.0.1), payload-extractability terms (Section 17.0.2), verisimilitude (Definition 31), edge-structural terms (Section 9.8), and meter-envelope penalties (Definition 15); and (iv) reality merge, in which the trained controller is deployed to the physical substrate and the residuals between twin predictions and physical observations feed back into verisimilitude discriminator training (subject to declared anomaly classification), twin refinement, and successor protocol-digest version updates (Section 17.1.5).

The twin is assistive rather than authoritative: the physical Reality Kernel remains the

arbitrator for verification, hardness, and meter-bounded validity. The pipeline’s correctness discipline is that every stage emits committed, auditable evidence under a versioned protocol digest, so that a verifier can recompute the twin’s identity, the probe-corpus contents, the RT training schedule, and the reality-merge residuals from selectively opened convolution-bundle windows.

17.1.1 Probe-corpus acquisition (non-limiting)

In some embodiments, a *probe corpus* $\mathcal{D}_{\text{probe}}$ is a committed set of input-output pairs $(u_{0:T}^{\text{Probe}}, \mathbf{y}_{0:T})$ in which the emission sequence $u_{0:T}^{\text{Probe}}$ is drawn from a declared probe family **Probe** rather than from a semantically structured RT target, and in which the corresponding reactor or scene response is recorded through the ordinary convolution-bundle pipeline. The probe corpus provides the twin with training evidence that samples declared regions of the reactor’s transfer-function support under a committed coverage rule, without imposing semantic bias on the twin.

Probe families (non-limiting). In some embodiments, the probe family **Probe** is specified along two independent axes:

1. **Spectral / spatial shape.** The illumination may be (a) spectrally neutral, meaning equalised within declared calibration tolerances across the reactor’s sensitive spectral band and declared spatial channels, so that no wavelength or channel is preferentially excited beyond the committed tolerance envelope; (b) pink-noise-shaped, with $1/f^{\text{Exp}_{\text{spec}}}$ power spectral density for a declared exponent Exp_{spec} (for example $\text{Exp}_{\text{spec}} = 1$ for canonical pink noise, $\text{Exp}_{\text{spec}} = 2$ for red noise, and $\text{Exp}_{\text{spec}} = 0$ for white); (c) custom-shaped under a declared filter or code family; or (d) a declared mixture of the above.
2. **Temporal-correlation structure.** The emission sequence may be (a) temporally white (independent samples); (b) temporally pink, with $1/f^{\text{Exp}_{\text{temp}}}$ temporal power spectral density for declared Exp_{temp} , yielding long-range frame-to-frame correlations; (c) driven by a declared autoregressive or state-space generator; or (d) constructed from concatenated segments with declared segment-boundary statistics.

The spectral shape parameters, temporal shape parameters, sample rate, frame rate, total duration, random-seed policy, and any calibration pre-sequence are committed to the protocol digest as part of the probe-corpus identity. In some embodiments, temporally pink correlation is preferred over white because it delivers equal power per octave, exercising slow reactor modes that white noise undersamples in any fixed total observation budget.

Neutrality and semantic non-bias (non-limiting). In preferred embodiments, the probe corpus is drawn from a probe family whose spectral, spatial, and temporal statistics are declared neutral with respect to any downstream semantic category (pose class, action

class, style class, or payload pattern), so that a twin trained on $\mathcal{D}_{\text{probe}}$ does not encode semantic priors that would bias downstream RT training. In some embodiments, semantic non-bias is verified by training declared probe classifiers on $\mathcal{D}_{\text{probe}}$ and confirming that their discrimination accuracy for the target semantic categories remains at or near chance, with
 5 the verification protocol and threshold committed to the protocol digest.

Commitment and provenance (non-limiting). In some embodiments, $\mathcal{D}_{\text{probe}}$ is committed as a Merkle-rooted set of disclosure atoms (Section 11.3), with per-atom digests bound to the probe-family declaration and to the run-level hash-chain state χ_t (Section 11.2) at the acquisition time. The protocol digest provenance-types $\mathcal{D}_{\text{probe}}$ as committed physical ev-
 10 idence, and verifier-facing attestations may rely on $\mathcal{D}_{\text{probe}}$ as a physical-provenance reference for subsequent twin identity, RT-training audits, and empirical-hardness arguments.

17.1.2 Twin training from the probe corpus (non-limiting)

In some embodiments, a digital twin **Twin** is trained to predict the reactor or scene response to declared emissions at the same step:

$$\hat{\mathbf{y}}_t = \text{Twin}(u_{0:t}, \text{meta}_t, \Pi), \quad C_{0:T}^{\text{Twin}} = \{(u(t), \hat{\mathbf{y}}_t)\}_{t=0}^T.$$

15 $C_{0:T}^{\text{Twin}}$ is the *twin-predicted convolution bundle* (synthetic bundle), constructed step-wise from declared emissions and twin-predicted observations so that downstream components such as RTDec, Pose, Extract, the verisimilitude functional, and the edge-based structural meter may be evaluated on twin-predicted windows $C_{\text{Win}_t}^{\text{Twin}}$ during twin-side training. The protocol digest provenance-types $C_{0:T}^{\text{Twin}}$ as twin-predicted or synthetic, and verifier-facing attestations shall
 20 not treat $C_{0:T}^{\text{Twin}}$ as physical-provenance evidence except as a recomputable surrogate derived from a committed twin identity and declared input schedule.

Twin may be (a) a learned autoencoder whose encoder maps emissions to a latent dynamics state and whose decoder predicts the response; (b) a transfer-function model whose parameters are fitted to the probe corpus by system-identification techniques (for example
 25 subspace identification, prediction-error methods, or Gaussian-process state-space models); (c) a physics-based propagation surrogate as described below; or (d) a declared hybrid. In preferred embodiments, the twin is differentiable so that gradients of a downstream RT objective with respect to emissions may be propagated through the twin during RT training.

Training objective on the probe corpus (non-limiting). In some embodiments, **Twin**
 30 is trained by minimising a declared same-step prediction loss on $\mathcal{D}_{\text{probe}}$:

$$\min_{\text{Twin}} \mathbb{E}_{(u_{0:T}^{\text{Probe}}, \mathbf{y}_{0:T}) \sim \mathcal{D}_{\text{probe}}} \left[\sum_{t=0}^T \ell_{\text{twin}}(\text{Twin}(u_{0:t}^{\text{Probe}}, \text{meta}_t, \Pi), \mathbf{y}_t) \right],$$

where ℓ_{twin} is a declared prediction loss (for example squared error, negative log-likelihood under a declared observation model, or a learned perceptual loss). The twin’s identity, architecture, weights or version hash, input fields, training set reference, validation set reference, and approximation-quality metrics on held-out probe-corpus windows are committed to the
 5 protocol digest.

Physics-based propagation surrogates (non-limiting). In some embodiments, the digital twin includes a physics-based wave-propagation engine (for example ray tracing, beam-propagation, or wave-optics solvers) together with learned priors for speckle statistics, aberrations, or alignment drift. Such hybrids explore large regions of configuration space
 10 in simulation, after which hardware-in-the-loop optimisation fine-tunes around promising configurations.

Approximation bounds and assistive status (non-limiting). In some embodiments, the twin’s approximation quality is characterised by a declared envelope on held-out probe-corpus windows and, where applicable, on held-out live recordings. The twin is used as a
 15 differentiable or predictive surrogate during training and protocol selection, but downstream validity claims rest on physical Reality Kernel evaluation and meter-envelope discipline, not on the twin alone.

17.1.3 Analogue twin: a second physical kernel in twin-role (non-limiting)

In some embodiments, the twin is realised as an *analogue twin* **AnaTwin** comprising a second
 20 physical Reality Kernel module whose reactor configuration, scan law, operating point, and invariance profile are committed and matched to the primary kernel through a shared probe corpus. Unlike the digital twin, the analogue twin produces a genuine convolution bundle

$$C_{0:T}^{\text{AnaTwin}} = \{(u(t), \mathbf{y}_t^{\text{AnaTwin}})\}_{t=0}^T$$

with physical provenance: each atom binds a declared emission to a physically observed response through the analogue-twin module’s own emitter, reactor, and detector. Consequently, $C_{0:T}^{\text{AnaTwin}}$ is provenance-typed by the protocol digest as committed physical evidence
 25 of the analogue-twin module’s own emitted-and-observed run, and is not subject to the synthetic-surrogate restriction applicable to $C_{0:T}^{\text{Twin}}$ of Section 17.1.2. For verifier-facing purposes, $C_{0:T}^{\text{AnaTwin}}$ is physical-provenance evidence only of the analogue-twin module’s own run; it is evidence of the primary kernel’s expected response only as a matched cross-kernel
 30 surrogate within the committed matching-fidelity envelope $\mathcal{M}_{\text{match}}$, and it is not physical-provenance evidence or empirical-hardness support for the primary kernel unless independently corroborated by primary-kernel committed evidence.

Matching procedure (non-limiting). In some embodiments, the analogue twin is matched to the primary kernel by executing a shared declared probe family **Probe** (Sec-

tion 17.1.1) on both kernels and fitting operating-point parameters to minimise a declared cross-kernel prediction loss on held-out probe windows:

$$\min_{\theta^{\text{AnaTwin}}} \mathbb{E}[\ell_{\text{match}}(\mathbf{y}_t^{\text{AnaTwin}}, \mathbf{y}_t)] ,$$

where θ^{AnaTwin} comprises the analogue-twin kernel’s declared adjustable parameters (for example gain, bias, alignment, reactor operating point, or scan-law parameters) and ℓ_{match} is a declared prediction or distributional distance. The matching procedure, probe corpus, fitted parameter values, and residual-error statistics on held-out windows are committed to the protocol digest.

Matching-fidelity envelope (non-limiting). In some embodiments, the analogue twin’s fitness for twin-role use is characterised not by a digital approximation envelope but by a *cross-device matching-fidelity envelope* $\mathcal{M}_{\text{match}}$, a declared admissibility bound on residuals computed from paired committed probe-corpus bundles. $\mathcal{M}_{\text{match}}$ is a cross-device admissibility envelope, distinct from the single-kernel meter envelope $\mathcal{M}_{\text{accept}}$ of Definition 15, but it is declared in the protocol digest and bound into commitments using the same declaration and commitment mechanics as that definition. An analogue twin is in-envelope when cross-kernel residuals remain within $\mathcal{M}_{\text{match}}$ across a declared evaluation window.

Reality Transform training against an analogue twin. In some embodiments, the Reality Transform controller of Section 17.1.4 is trained against $C_{\text{Win}_t}^{\text{AnaTwin}}$ by emitting declared controls $u(t)$ on the analogue-twin module and computing the compound objective’s evidence-dimension terms from the resulting genuine bundle. Because C^{AnaTwin} is physical, downstream RTDec, Pose, Extract, and verisimilitude evaluation on $C_{\text{Win}_t}^{\text{AnaTwin}}$ do not require the synthetic-window construction rule of Section 17.1.4 and are not subject to the twin-predicted-bundle verifier-facing restriction. Gradients through the analogue twin are, in general, not analytically available; hardware-in-the-loop optimisation (Section 9.13) is used instead of backpropagation through the twin.

Reality merge with an analogue twin. In some embodiments, when the trained controller is deployed to the primary kernel, the reality-merge residual acquires a different interpretation than in the digital-twin case. For an analogue twin,

$$r_t^{\text{AnaTwin}} = \mathbf{y}_t - \mathbf{y}_t^{\text{AnaTwin}}$$

measures drift, wear, or tamper between matched physical substrates rather than model-approximation error. Systematic r_t^{AnaTwin} outside $\mathcal{M}_{\text{match}}$ triggers re-matching of the analogue twin, replacement or recalibration of the analogue-twin module, or a successor protocol-digest version recording the change, under the applicable authority-separation policy. In some embodiments, r_t^{AnaTwin} is additionally used as an input to the verisimilitude-discriminator train-

ing of Section 9.5, under the same declared anomaly-classification procedure that governs digital-twin residuals.

Cross-attestation composition (non-limiting). In some embodiments, the analogue twin simultaneously serves as a cross-attestation witness under the multi-RK cross-attestation protocols of Section 14 and, where applicable, as a B-PoD witness under the
 5 bilateral Proof-of-Discrepancy protocol of Section 16.7. The analogue-twin role and the cross-attestation witness role are recorded in the protocol digest as separate capability declarations so that a single module may participate in multiple roles without role conflation in verifier-facing attestations.

Hybrid digital–analogue twin (non-limiting). In some embodiments, the digital twin
 10 of Section 17.1.2 and the analogue twin of this section are operated concurrently, with the controller trained against both under declared weightings. This composition is a specialisation of the hybrid digital–analogue co-training discussed in connection with hardware-in-the-loop optimisation (Section 9.13): digital-twin gradients support fast iteration; analogue-twin
 15 evidence supports physically grounded validation; the pair together reduces the reliance of any single validity claim on either digital approximation quality or single-kernel physical provenance. The weightings between digital-twin and analogue-twin terms, and the triggering rules for escalation from digital-twin-only to combined operation, are committed to the protocol digest.

Scope note (non-limiting). For avoidance of doubt, the analogue twin of this section
 20 is a Reality Kernel module as already disclosed; its novelty as claimed in the claim set of the related Filing 2 application lies in the joint specification of role (twin-role rather than primary-role), matching procedure, matching-fidelity envelope, analogue-twin-specific reality-merge interpretation, and composition with digital-twin and cross-attestation ma-
 25 chinery, rather than in any new physical kernel architecture.

17.1.4 Reality Transform training against the twin (non-limiting)

In some embodiments, a Reality Transform controller is trained by optimising a compound objective in which the physical observation $\mathbf{y}_t$ is replaced, during training, by a twin-produced observation: $\hat{\mathbf{y}}_t$ from a digital twin (Section 17.1.2), $\mathbf{y}_t^{\text{AnaTwin}}$ from an analogue twin (Sec-
 30 tion 17.1.3), or a declared weighted combination of the two. In the digital-twin case, gradients of the compound objective flow through **Twin** to the controller; in the analogue-twin case, gradients are estimated by hardware-in-the-loop optimisation (Section 9.13) unless a differentiable analogue-twin model is additionally maintained. During twin-side RT training, any decoder, pose-feature extractor, payload extractor, verisimilitude functional, or structural
 35 meter that would ordinarily consume a physical convolution-bundle window is evaluated on the corresponding twin-produced bundle window – $C_{\text{Win}_t}^{\text{Twin}}$ for the digital twin or $C_{\text{Win}_t}^{\text{AnaTwin}}$ for

the analogue twin – with the synthetic-window construction and any domain-gap correction (for the digital twin) or the matching procedure and matching-fidelity envelope (for the analogue twin) committed to the protocol digest.

The compound objective aggregates declared terms across evidence dimensions:

$$\mathcal{J}_{\text{compound}}(U_{0:T}; \phi) = \sum_{s \in \text{RTTerm}} \lambda_{\text{rt},s} \mathcal{L}_s + \lambda_{\text{meter}} \text{MeterPen}(\mathbf{m}_{0:T}; \mathcal{M}_{\text{accept}}),$$

5 where **RTTerm** is a declared candidate set of evidence-dimension terms, $\mathcal{J}_{\text{compound}}$ is minimised unless the protocol digest declares an equivalent negative-reward convention, and **MeterPen** is a declared meter-envelope-violation aggregate (such as a per-step indicator sum, hinge loss, or smooth barrier penalty). Terms in **RTTerm** are drawn from: (i) the style-target term of Section 8.4.4; (ii) the pose-distribution and pose-dynamics terms of Section 17.0.1; 10 (iii) the payload-extractability term of Section 17.0.2; (iv) the verisimilitude term of Definition 31; (v) the edge-based structural term of Section 9.8; and (vi) any further declared meter term. The weights $\{\lambda_{\text{rt},s}\}$ and the meter-partition policy are committed to the protocol digest. In some embodiments, the protocol digest identifies active terms by assigning nonzero $\lambda_{\text{rt},s}$ values to selected members of **RTTerm** and zero values to inactive members; 15 pose-conditioned RT and payload-embedding RT are two non-limiting such specialisations.

Twin-to-physical transfer (non-limiting). In some embodiments, a controller trained against **Twin** is subsequently deployed on the physical substrate, and a declared fine-tuning phase updates the controller against physical observations under the same compound objective. The fine-tuning schedule, permissible update magnitude, and twin-to-physical transfer 20 meter are committed to the protocol digest.

17.1.5 Reality merge: deployment to physical and residual feedback (non-limiting)

In some embodiments, once the Reality Transform controller is deployed on the physical substrate, the twin is retained as a residual reference. At each step, the residual

$$r_t = \mathbf{y}_t - \hat{\mathbf{y}}_t = \mathbf{y}_t - \text{Twin}(u_{0:t}, \text{meta}_t, \Pi)$$

25 is computed and logged. The residual feeds three downstream processes:

Verisimilitude discriminator training. In some embodiments, residuals outside the declared twin approximation envelope are logged as candidate anomalies or out-of-model physical examples. Such residuals are used as negative examples for the verisimilitude-discriminator training of Section 9.5 only if a declared post-hoc classification procedure 30 assigns them to a forged, edited, simulated, or otherwise non-genuine class; otherwise they are used for twin refinement or meter-envelope revision.

Twin refinement. In some embodiments, committed residuals are used to update the twin by declared continual-learning or re-fitting procedures, with each twin update producing a new versioned twin identity committed to the protocol digest as a successor protocol-digest version. The prior twin is retained as part of the committed version history so that verifiers
 5 may re-evaluate earlier runs under the twin active at the time.

Successor protocol-digest updates. In some embodiments, systematic residuals trigger successor protocol-digest version updates (for example new probe-corpus acquisition, new meter envelope, or new RT fine-tuning schedule) under the applicable authority-separation policy; the prior protocol digest remains the operative digest for runs it already bound.

10 **Relation to hardware-in-the-loop optimisation.** The reality-merge step composes with the hardware-in-the-loop optimisation of Section 9.13: twin-trained controllers provide initial emission policies; hardware-in-the-loop optimisation refines them against physical evidence; residual feedback refines the twin; the cycle iterates under declared schedules.

17.1.6 Sandboxed and live operation (non-limiting)

15 In some embodiments, probe-corpus acquisition, twin training, and Reality Transform training against the twin are performed under the parameter-sandboxed offline-session mechanism of Section 16.5. Parameter updates made to the physical kernel during the sandboxed session are reverted on return to live operation; **Twin**, **Embed**, **Extract**, pose-feature extractors, pose-distribution discriminators, kinematic classifiers, and verisimilitude discriminators trained
 20 from sandboxed evidence may be retained as derived models where permitted by the protocol digest and meter policy.

17.2 Styled training-data generation methods (non-limiting)

In some embodiments, training data for downstream components (digital twin, analogue-twin matching, verisimilitude discriminator, pose-distribution discriminator A_{dat} , pose-dynamics
 25 model g_{dyn} , kinematic-plausibility classifier, and the Reality Transform controller itself) is generated not only from neutral probe-corpus acquisition (Section 17.1.1) but also from one or more *styled training corpora* in which the emitted content carries declared style, aesthetic, or target-distribution structure. The following subsections describe seven non-limiting styled-corpus generation mechanisms, each independently claimable, together with
 30 combined configurations. Unlike the probe corpus of Section 17.1.1, styled training corpora are semantic-content-by-design and their non-neutrality is declared and accepted in the protocol digest rather than verified against; semantic-non-bias testing does not apply to styled corpora.

17.2.1 Externally-sourced curated styled corpus (non-limiting)

In some embodiments, a declared styled corpus $\mathcal{D}_{\text{style}}$ comprises externally-sourced materials (images, video sequences, art assets, authored aesthetic references) selected as representative of the target RT style family. The corpus is identified by content hash, source attribution, declared style-family identifier, and any declared temporal-correlation structure of source sequences, all committed to the protocol digest. An externally-sourced curated corpus is operator-declared and does not by itself supply physical-provenance or empirical-hardness evidence; verifier-facing attestations shall not treat it as physical-provenance evidence unless the acquisition and validation of the source material are separately committed.

17.2.2 Unpaired iterated acquisition with committed reset (non-limiting)

In some embodiments, a styled corpus is generated by an unpaired iterated procedure: an initial noisy transfer function or roughly-trained styling component is applied to declared seed content to produce candidate styled emissions; those emissions are projected through the physical substrate and captured, producing committed (emission, observation) atoms; the committed atoms feed supervised training of the next cycle's styling component. Because the only training signal in each cycle derives from the prior cycle's own captures, the procedure is subject to a compounding-drift failure mode: mode collapse, distribution drift, or verisimilitude-pass-rate decline accumulates across cycles.

Committed degradation detector and reset protocol. In some embodiments, a declared degradation detector monitors committed per-cycle statistics (for example style-distribution distance from a declared reference, verisimilitude-pass rate on held-out evaluation windows, mode-collapse proxy such as effective latent-space coverage) and triggers a reset when one or more statistics cross committed thresholds. On reset, the procedure re-seeds from a fresh neutral probe corpus or from a declared bootstrap corpus, producing a new training trajectory independent of the degraded one. The detection statistics, thresholds, reset policy, and per-cycle protocol digests (chained through χ_t of Section 11.2) are committed.

Relationship to reinforcement-learning-based iterative improvement (non-limiting). In some embodiments, convergent iterative improvement of a styling component under a reward signal is conducted as a reinforcement-learning procedure with declared committed trajectories, rewards, and value estimates; such procedures are distinct from the supervised one-shot-and-reset mechanism described in this subsection and are not required by it.

17.2.3 Separation-channel paired probe/style acquisition (non-limiting)

In some embodiments, a paired corpus $\mathcal{D}_{\text{pair}}$ is acquired by projecting a declared neutral probe emission u^{probe} and a declared styled emission u^{style} simultaneously onto the same

scene through a declared separation channel **Sep** that preserves scene, geometry, lighting, and sensor state across the probe and style captures within a declared invariance tolerance. A single detector (or a detector-pair matched by the separation channel) captures both channels, and the separation property permits recovery of the two captures as paired
 5 $(\mathbf{y}_t^{\text{probe}}, \mathbf{y}_t^{\text{style}})$ atoms bound to the same scene instant.

Separation modalities (non-limiting). In some embodiments, **Sep** is realised under a declared separation modality. The modalities are partitioned into preferred embodiments and alternative embodiments, where the alternatives carry declared bleed or confound envelopes reflecting physical limitations. Preferred realisations of **Sep** include:

- 10 1. **Temporal-interleave (preferred).** Probe and style emitted in interleaved frame slots at rates above a declared perceptual envelope, with phase-locked frame-synchronous capture recovering the two sequences. Robust against bleed when detector-shutter synchronisation is preserved within a declared synchronisation tolerance.
- 15 2. **IR/RGB (preferred).** Companion channel in infrared or near-infrared wavelengths outside a declared human visual response range, captured by IR-sensitive or NIR-sensitive detection, with RT styled emission in the visible range. Non-limiting hardware sub-embodiments include: (a) two separate emitters, one visible-only and one IR-only, sharing scene illumination; (b) a single emitter with visible and IR output separated by sensor spectral response; (c) a wavelength-tagged single emission with dichroic capture
 20 recovering the two wavelength ranges.

Alternative realisations of **Sep**, each subject to a declared bleed, confound, or geometric-restriction envelope committed to the protocol digest, include:

- 25 3. **Spatial (geometrically-disjoint coverage).** Two or more projection sources addressing geometrically disjoint regions of the scene, such that each scene point is addressed by exactly one projector under a declared coverage rule enforced by a committed scene-region mask, occlusion model, or surface-partition map, together with a rigid-body or known-geometry scene assumption. Non-limiting example: rear-projection onto a robot or mannequin surface for calibration or probe acquisition while front-projection delivers styled content; the scene-region mask or surface partition over the rigid-body
 30 geometry ensures disjoint coverage. The geometric-restriction envelope covering scene-geometry assumptions, mask/occlusion/partition identity, and any boundary-overlap tolerance is committed to the protocol digest. This modality does not cover per-eye stereo goggles, which are disclosed below as a deployment pattern across temporal, polarisation, and chromatic modalities rather than as a spatial separation.
- 35 4. **Code-division on a shared beam path.** Probe and style carried by orthogonal modulation codes on a single beam path, with demodulated capture recovering the two

code-separated channels. Requires declared modulation-bandwidth and demodulation-sophistication envelopes; shares the beam path entirely and eliminates geometric parallax between the captures.

5 **5. Polarization (alternative; declared bleed).** Orthogonally-polarized probe and style emissions with polarization-resolved detection. Scene-induced depolarisation, imperfect polariser stack-ups, and viewing-angle-dependent bleed require a declared residual-crosstalk envelope. Not suitable for co-emission configurations in which the companion must be imperceptible to naked-eye viewers without declared tolerance for visible bleed.

10 **6. Chromatic / spectral-within-visible (alternative; declared confound).** Probe and style emitted in non-overlapping bands within the human visual response range, with spectrally-resolved detection. Human visual spectral sensitivity overlap with any "invisible" band strategy, and scene-reflectance-induced spectral shifts, require a declared confound envelope committed to the protocol digest.

15 The separation modality, invariance tolerance, calibration procedure for the separation channel, and any residual-crosstalk, bleed, confound, or geometric-restriction envelope are committed to the protocol digest.

Stereo / per-eye goggles as deployment pattern (non-limiting). In some embodiments, per-eye stereo viewing for viewers equipped with goggle hardware is achieved as
20 a deployment pattern over the temporal-interleave, polarization, or chromatic modalities above, rather than as a standalone separation modality. The goggle filter specification (polariser orientation, shutter timing, spectral passband) is committed to the protocol digest alongside the underlying Sep modality.

Stable multiplexed-loop acquisition (non-limiting). In some embodiments, paired
25 acquisition is iterated over a committed cycle upper bound N under a committed per-cycle evaluation meter, N and the meter committed to the protocol digest. Each cycle reacquires independent probe-channel ground truth through the separation channel Sep; the compounding-drift mechanism of the unpaired iterated method (Section 17.2.2) does not apply, because the separation-invariant property supplies an independent probe anchor at
30 each cycle rather than requiring each cycle to learn from the prior cycle's own drifted output. Empirical result quality per cycle is evaluated under the committed per-cycle meter and is not asserted as monotone or unconditionally convergent.

Commitment and provenance. In some embodiments, $\mathcal{D}_{\text{pair}}$ is committed as a Merkle-rooted set of paired disclosure atoms, with each paired atom binding a matched ($\mathbf{y}_t^{\text{probe}}, \mathbf{y}_t^{\text{style}}$)
35 observation to a common scene instant and to the run-level χ_t at acquisition time. The protocol digest provenance-types $\mathcal{D}_{\text{pair}}$ as committed physical evidence, subject to the declared separation-invariance tolerance.

17.2.4 Slow-teacher / fast-student capability transfer via physically-captured evidence (non-limiting)

In some embodiments, an expensive styling component **Teacher** (whose per-frame inference may require minutes or longer under declared compute constraints) generates styled emissions offline; those emissions are projected onto a physical scene held stable under a declared scene-stability envelope **SceneStab** (for example a mannequin, static scene, or slow-moving subject) and captured through the substrate, producing committed (emission, captured-physical-response) atoms. A fast styling component **Student**, deployable at runtime rates, is then trained to reproduce the captured physical response from the input seed, rather than to reproduce the teacher’s raw output tensor. The student therefore inherits physical-channel transfer-function effects as part of its training signal. For avoidance of doubt, **Teacher** denotes the slow styling component of this subsection and is unrelated to the teacher-forcing policy referenced in the pose-dynamics-model training subsection of Section 17.0.1.

Scene-stability envelope (non-limiting). In some embodiments, **SceneStab** is a declared admissibility bound on scene drift, subject motion, lighting variation, and inter-frame differences across the teacher’s per-frame inference duration. The envelope may be declared quantitatively (for example, bounded subject displacement and bounded inter-frame pixel-difference statistics) or via a committed subjective stability assessment procedure. The envelope, the declared teacher inference-cost class, and the acquisition procedure are committed to the protocol digest.

Student deployment-independence (non-limiting). In preferred embodiments, **Student** is fully independent of **Teacher** at deployment: once trained, the student does not invoke the teacher at runtime. In other embodiments, hybrid deployment may invoke the teacher as a fallback when a committed student-confidence statistic falls below a committed threshold, with the fallback policy committed to the protocol digest.

Committed teacher-student chain. In some embodiments, the teacher identity, version hash, training-seed content, per-frame captured atoms, student identity, training schedule, and held-out validation results are committed as a chained record under the run-level χ_t , so that a verifier can walk from the committed teacher output, through the committed projection-and-capture atom, to the committed student training step, and check consistency.

Composition with pose-conditioned acquisition (non-limiting). In some embodiments, when the stable scene is a slow-moving subject, the acquisition composes with the pose-conditioned RT embodiments of Section 17.0.1: pose-feature traces p_t extracted during acquisition are committed alongside the teacher-student chain, and pose-distribution and pose-dynamics terms may be evaluated on the resulting student training corpus.

Scope note (non-limiting). For avoidance of doubt, Teacher and Student may be drawn from conventional styling-component architectures (including but not limited to conditional GANs, diffusion models, flow-matching generators, or transformer-based generators). The disclosed technical contribution is the committed teacher-student chain through physically-captured evidence, the declared scene-stability envelope, and the resulting physical-provenance training signal, rather than any particular generator architecture.

Knowledge-distillation and model-compression methods (non-limiting). In some embodiments, the teacher-student chain is implemented using knowledge-distillation methods, including without limitation logit distillation (Hinton, Vinyals, Dean 2015), feature distillation, intermediate-representation distillation (FitNets), attention distillation, relational knowledge distillation, contrastive representation distillation (CRD), self-distillation, online distillation, and ensemble distillation. In some embodiments, the student is additionally trained or post-processed under model-compression methods including quantization-aware training (QAT), post-training quantisation (PTQ), magnitude pruning, structured pruning, lottery-ticket selection, weight clustering, low-bit-width weight quantisation (including without limitation 8-bit, 4-bit, and 1-bit and 2-bit weight quantisation methods), mixed-precision training, and sparse-attention pruning. These methods are non-limiting; the apparatus is compatible with future distillation and compression methods applied to the committed teacher-student chain.

17.2.5 Forward counterfactual generator G_{alt} (non-limiting)

In some embodiments, a scene-conditioned generator G_{alt} is trained on $\mathcal{D}_{\text{pair}}$ to predict the styled response that would have occurred at a given scene instant under a candidate styled emission $u_{\text{cand}}^{\text{style}}$, given the committed paired neutral probe $(u^{\text{probe}}, \mathbf{y}_t^{\text{probe}})$ of that instant:

$$\hat{\mathbf{y}}_t^{\text{style}} = G_{\text{alt}}(u^{\text{probe}}, \mathbf{y}_t^{\text{probe}}, u_{\text{cand}}^{\text{style}}, \Pi).$$

The generator is scene-conditioned: the committed probe capture acts as a physically-attested scene embedding, distinguishing G_{alt} from unconditional style generators that marginalise over scenes.

Training-corpus conditioning modes (non-limiting). In some embodiments, G_{alt} conditions on the committed probe capture directly; in other embodiments, it conditions on a declared feature representation derived via RTDec, Pose, or another committed feature extractor applied to the probe capture. The conditioning mode, feature extractor identity, and feature dimensionality are committed to the protocol digest.

Use as scene-conditioned twin. In some embodiments, G_{alt} serves as a scene-conditioned digital twin distinct from the unconditional Twin of Section 17.1.2: its prediction is over styled responses rather than arbitrary physical responses, and its conditioning includes committed

scene evidence. The output $\hat{\mathbf{y}}_t^{\text{style}}$ is provenance-typed as synthetic under the same discipline applied to $C_{0:T}^{\text{Twin}}$ in Section 17.1.2.

Use as verifier-side forward audit. In some embodiments, a verifier holding committed $(u^{\text{probe}}, \mathbf{y}_t^{\text{probe}}, u_t^{\text{style}}, \mathbf{y}_t^{\text{style}})$ feeds the first three to $\mathbf{G}_{\text{alt}}$ and checks whether the predicted $\hat{\mathbf{y}}_t^{\text{style}}$ matches the captured $\mathbf{y}_t^{\text{style}}$ within a declared tolerance. Mismatch is evidence of scene change between probe and style captures (violating the separation-invariance property) or of tampering.

Use for training-data amplification. In some embodiments, $\mathbf{G}_{\text{alt}}$ is applied to unpaired probe captures together with a library of candidate styled emissions to synthesise plausible styled counterfactuals, which are provenance-typed as $\mathbf{G}_{\text{alt}}$ -generated and used for training downstream components.

Scope note (non-limiting). $\mathbf{G}_{\text{alt}}$ may be drawn from conventional conditional-generation architectures; the disclosed technical contribution is the training-data acquisition via separation-paired capture, the scene-conditioning via committed physically-attested probe evidence, and the provenance-typing of outputs, rather than any particular generator architecture.

17.2.6 Inverse destyling generator $\mathbf{G}_{\text{destyle}}$ (non-limiting)

In some embodiments, a scene-conditioned generator $\mathbf{G}_{\text{destyle}}$ is trained on $\mathcal{D}_{\text{pair}}$ in the inverse direction: given a committed styled capture $(u^{\text{style}}, \mathbf{y}_t^{\text{style}})$, it predicts the response that the probe channel would have captured of the same scene at the same instant:

$$\hat{\mathbf{y}}_t^{\text{probe}} = \mathbf{G}_{\text{destyle}}(u^{\text{style}}, \mathbf{y}_t^{\text{style}}, u_{\text{query}}^{\text{probe}}, \Pi),$$

where $u_{\text{query}}^{\text{probe}}$ is a declared reference probe emission (typically a canonical neutral probe). The output is the *synthetic neutral view* of the scene under active styling.

Use for perception-module input enhancement. In some embodiments, a downstream perception module (for example Pose extractor, verisimilitude discriminator, or kinematic-plausibility classifier) that would ordinarily operate on raw styled captures instead operates on $\mathbf{G}_{\text{destyle}}$ -produced synthetic neutral views, recovering near-unstyled perception quality under active RT projection.

Use for continuous baseline estimation. In some embodiments, $\mathbf{G}_{\text{destyle}}$ provides a continuous estimate of the baseline scene appearance Baseline_t used in the positive-residual additive-overlay objective of the pose-RT companion addendum, replacing the interleaved baseline-block acquisition of that addendum between scheduled baseline blocks. The baseline-continuous-estimation policy is committed to the protocol digest.

Use for synthetic-neutral-view secure recording. In some embodiments, a recording captured under active RT projection is accompanied by a synthetic neutral-view version produced by G_{destyle} , and the synthetic version is committed with a declared provenance-type identifying it as G_{destyle} -generated rather than directly captured. The synthetic neutral view
 5 may be disclosed for audit, training, or sharing as a provenance-limited generator estimate of underlying neutral scene content conditioned on committed styled-capture evidence, while any committed styled atoms, if separately retained or selectively opened, remain the physical-provenance record of the styled run.

Downstream evidentiary scope of synthetic neutral views (non-limiting). For
 10 verifier-facing purposes, a G_{destyle} - generated synthetic neutral view is evidence only of the committed styled-capture input, the declared G_{destyle} identity and version, and the declared inference procedure; it shall not be treated as direct physical-provenance evidence of the scene’s neutral appearance, nor as evidence of scene facts not independently supported by selectively opened styled atoms or other committed physical evidence. Opening committed
 15 styled atoms under a declared audit authority does not change the synthetic neutral view’s status: the styled atoms remain physical- provenance evidence of the run, while the synthetic neutral view remains generator-produced output conditioned on them, merely corroborated, contradicted, or qualified by comparison against the opened styled atoms.

Redaction-at-source and selective-disclosure variants (non-limiting). In some em-
 20 bodiments, G_{destyle} is applied at recording time so that the device commits the styled-capture bundle (for RT operation) and the synthetic neutral view (for downstream disclosure) but not a separately-redactable styled original; the configuration is style-private by design. In other embodiments, the styled capture atoms and the synthetic neutral view are separately selectively-openable under distinct audit authorities, so that a verifier with one authority
 25 sees only the synthetic neutral view while a verifier with a higher authority may additionally open the styled atoms.

Use for governance audit under partial disclosure. In some embodiments, a governance-side verifier given only committed styled-capture evidence recovers an estimated neutral scene appearance via G_{destyle} and runs downstream integrity checks on the
 30 estimated neutral view, under declared partial-disclosure rules committed to the protocol digest.

Use for training-data amplification, inverse direction. In some embodiments, unpaired styled captures are destyled to synthesise plausible probe-captures, which are provenance-typed as G_{destyle} -generated and used for training components that expect
 35 neutral-scene input.

Scope note (non-limiting). G_{destyle} may be drawn from conventional conditional-generation architectures; the disclosed technical contribution is the separation-paired training-data acquisition, the scene-conditioning via committed styled evidence, the provenance-typing of outputs, and the declared operational uses (perception enhance-
 5 ment, continuous baseline estimation, synthetic-neutral-view secure recording, governance audit under partial disclosure), rather than any particular generator architecture.

17.2.7 Dual-direction consistency meter (non-limiting)

In some embodiments, both G_{alt} and G_{destyle} are trained on the same $\mathcal{D}_{\text{pair}}$ and deployed together as a *dual-direction consistency meter*. For a committed frame $(u^{\text{probe}}, \mathbf{y}_t^{\text{probe}}, u^{\text{style}}, \mathbf{y}_t^{\text{style}})$,
 10 the meter produces three consistency statistics:

1. $\delta_{\text{fwd}} = \|G_{\text{alt}}(u^{\text{probe}}, \mathbf{y}_t^{\text{probe}}, u^{\text{style}}, \Pi) - \mathbf{y}_t^{\text{style}}\|$ (forward consistency);
2. $\delta_{\text{inv}} = \|G_{\text{destyle}}(u^{\text{style}}, \mathbf{y}_t^{\text{style}}, u^{\text{probe}}, \Pi) - \mathbf{y}_t^{\text{probe}}\|$ (inverse consistency); and
3. δ_{sym} , a declared pair-symmetry statistic evaluating whether the forward and inverse predictions form a consistent round-trip when composed.

15 Tampering, scene change between probe and style captures, or separation-channel degradation tends to break one, two, or all three statistics; the meter partition records which statistic(s) failed, providing a per-frame physical-consistency signal strictly stronger than forward or inverse consistency alone. The meter’s statistics, thresholds, and partition rule are committed to the protocol digest.

20 17.2.8 Composed styled-training-data configurations (non-limiting)

In some embodiments, two or more of the mechanisms of Sections 17.2.1–17.2.7 are composed in a single deployment. Non-limiting combined configurations include:

1. paired acquisition (Section 17.2.3) feeding both G_{alt} and G_{destyle} training, with the dual-direction consistency meter of Section 17.2.7 active at runtime;
- 25 2. paired acquisition feeding slow-teacher / fast-student distillation (Section 17.2.4), with the teacher’s styled output serving as the style channel of each paired capture;
3. unpaired iterated acquisition with committed reset (Section 17.2.2) producing initial training data, which is subsequently augmented by paired-acquisition cycles for stability once a seed styling component has been established;
- 30 4. curated styled corpus (Section 17.2.1) serving as declared reference for style-distribution loss terms while one or more of the above mechanisms supply deployment-matched training evidence.

The active combination, mixing policy, and per-component weightings are committed to the protocol digest.

17.3 Virtual Views and Virtual Emissions (non-limiting)

In some embodiments, the verisimilitude and twin machinery of Sections 9.5, 17.1.2, and 17.2.5 is extended to predicted observations under detector and emitter configurations not physically instantiated during the run. Two dual primitives are disclosed: a *virtual camera*,
 5 being a predicted observation under a detector pose, spectral band, or aperture configuration not physically present; and a *virtual emitter*, being a predicted observation under an emission u^{cand} that was not physically applied at the declared time index. Each primitive admits three operating modes: predictive use, loss-function use, and adversarial or competitive use. The forward counterfactual generator G_{alt} of Section 17.2.5 and the external and virtual-view
 10 checks paragraph of Section 9.5 are special cases of this unified treatment.

17.3.1 Definitions (non-limiting)

Definition 32 (Virtual camera). A virtual camera for a configuration θ is a measurable map

$$\hat{\mathbf{y}}_t^{(\text{cam})} = G_{\text{vcam}}(C_{0:t}, \xi^{\text{cam}}, u(t), \Pi)$$

from a committed convolution-bundle history $C_{0:t}$, a declared virtual-camera configuration
 15 ξ^{cam} (specifying pose, intrinsics, spectral band, gating, polarisation, or sub-aperture set), the emission $u(t)$, and a protocol digest Π , to a predicted observation that an additional camera with configuration ξ^{cam} would have produced at time index t . G_{vcam} may be realised by a digital twin (Section 17.1.2), an analogue twin (Section 17.1.3), a learned 3D representation including but not limited to NeRF-like radiance fields, point-based representations,
 20 Gaussian primitives, mesh-based representations, or hybrid implicit–explicit forms, or by a scene-conditioned generator analogous to G_{alt} extended in the detector axis. The configuration ξ^{cam} , model identity, version hash, and prediction tolerance are committed to the protocol digest.

Definition 33 (Virtual emitter). A virtual emitter for a configuration θ is a measurable
 25 map

$$\hat{\mathbf{y}}_t^{(\text{emit})} = G_{\text{vemit}}(C_{0:t}, u^{\text{cand}}(t), \xi^{\text{emit}}, \Pi)$$

from a committed convolution-bundle history, a candidate emission $u^{\text{cand}}(t)$ that was not physically applied at time index t , a declared virtual-emitter configuration ξ^{emit} (specifying source pose, beam profile, spectral band, polarisation, modulation schedule, or aperture), and the protocol digest Π , to a predicted observation on one or more physical detectors that
 30 the additional emission would have produced. G_{vemit} generalises the forward counterfactual generator G_{alt} of Section 17.2.5 from substitution of a single emission to addition of one or more candidate emissions on top of $K_{\text{co}} \geq 1$ committed concurrent emission streams under the co-emission architecture of Section 17.0.3. The configuration ξ^{emit} , candidate emission specification, model identity, version hash, and prediction tolerance are committed to the
 35 protocol digest.

Provenance typing. Outputs of G_{vcam} and G_{vemit} are provenance-typed as synthetic under the same discipline applied to digital-twin outputs in Section 17.1.2 and to G_{alt} -generated outputs in Section 17.2.5, and shall not be treated as direct physical-provenance evidence of scene facts not independently supported by physically committed atoms.

5 17.3.2 Predictive use (non-limiting)

Virtual-camera prediction. In some embodiments, given a learned 3D representation, digital twin, or analogue twin, G_{vcam} predicts the observation that an additional camera at ξ^{cam} would record under the committed emission schedule, generalising the external and virtual-view checks paragraph of Section 9.5.

10 **Virtual-emitter prediction.** In some embodiments, G_{vemit} predicts the observation that one or more physical detectors would record under an additional candidate emission u^{cand} at ξ^{emit} , generalising G_{alt} from substitutive to additive operation.

Linearity-regime decomposition (non-limiting). In some embodiments, the detection pathway operates in an additive, calibrated, unsaturated regime in which the response to
 15 a multi-emission configuration $\{u^{(k)}\}_{k=1}^{K_{\text{co}}}$ decomposes by superposition into per-stream contributions $\{\mathbf{y}_t^{(k)}\}$ recoverable under the declared separation modality **Sep** of Section 17.0.3. In such embodiments, G_{vemit} is realised by predicting the per-stream contribution $\hat{\mathbf{y}}_t^{(\text{cand})}$ of a candidate additional emission and forming the predicted aggregate observation by linear
 20 superposition over the recovered physical contributions and the predicted candidate contribution. The linearity envelope, including declared upper bounds on saturation, threshold-crossing, dead-time, and crosstalk regimes within which superposition is asserted, is committed to the protocol digest, and a meter monitors linearity-envelope occupancy throughout the prediction window.

Non-additive regime (non-limiting). In some embodiments operating outside the additive regime of Section 17.3.2, G_{vemit} is realised as a learned scene-conditioned generator
 25 trained on co-emission paired captures in which the same scene is recorded under varying multi-emission configurations, and the model learns the non-linear forward map directly. Training-corpus identity, the held-out emission configurations used for validation, and the held-out scene classes are committed to the protocol digest.

30 **Combined virtual-view and virtual-emission prediction (non-limiting).** In some embodiments, G_{vcam} and G_{vemit} are composed to predict the observation of a virtual camera at ξ^{cam} under a virtual emitter at ξ^{emit} , neither of which was physically present during the run, with the composition rule and joint prediction tolerance committed to the protocol digest. As prediction error compounds under composition of two synthetic primitives, the joint
 35 prediction tolerance committed to the protocol digest accounts explicitly for the compounded uncertainty.

Light-field hybrid speciality (non-limiting). In some embodiments deploying the light-field hybrid embodiment of Section 10.2.5, virtual cameras correspond to sub-aperture views (u^*, v^*) not physically read out and virtual emitters correspond to angular-sub-aperture emission configurations $E(x, y, u^*, v^*)$ not physically commanded; predicted virtual
 5 observations are evaluated against measured sub-apertures or measured emission configurations under the declared microlens-array pitch, calibration state, and angular-resolution envelope.

17.3.3 Loss-function use (non-limiting)

Held-out virtual-camera supervision. In some embodiments, an agent, controller, decoder, or world model is trained on input derived from a first physically captured camera
 10 cam_1 and supervised by a loss

$$L_t^{\text{vcam}} = d(\hat{\mathbf{y}}_t^{(\text{cam}_2)}, \mathbf{y}_t^{(\text{cam}_2)})$$

on the predicted virtual-camera output $\hat{\mathbf{y}}_t^{(\text{cam}_2)}$ versus a measurement $\mathbf{y}_t^{(\text{cam}_2)}$ from a second physically captured camera cam_2 held out from the agent's input, where $d(\cdot, \cdot)$ is a declared distance functional with declared scale committed to the protocol digest. In some
 15 embodiments, the second camera is operated in a governance-only role, with its observations excluded from the agent's memory-formation and action-selection pathways and routed exclusively to training supervision and audit logging.

Held-out virtual-emitter supervision. In some embodiments, the dual scheme applies on the emitter axis: a controller is trained to act under a first committed emission stream
 20 and supervised by a loss

$$L_t^{\text{vemit}} = d(\hat{\mathbf{y}}_t^{(\text{emit}_2)}, \mathbf{y}_t^{(\text{emit}_2)})$$

on the predicted virtual-emitter response $\hat{\mathbf{y}}_t^{(\text{emit}_2)}$ versus the physically measured response $\mathbf{y}_t^{(\text{emit}_2)}$ to a second committed emission stream held out from the controller's reactive loop. In some embodiments, the emission held out for supervision is selected from a held-out emission family rotated periodically and not used during prior training rounds.

Asymmetric controller training. In some embodiments, the loss-function uses of this
 25 section compose with the Reality Transform training against the twin of Section 17.1.4: the controller's input is a twin-produced or virtual-camera-produced observation, and the controller's training loss is evaluated against a held-out physical camera or held-out physical emitter. Gradients flow through the twin or through $\mathbf{G}_{\text{vcam}}$ or $\mathbf{G}_{\text{vemit}}$ to the controller, while
 30 the supervisory signal remains physically grounded.

Cross-virtual-real consistency loss. In some embodiments, a consistency loss

$$L_t^{\text{cons}} = d(\mathbf{G}_{\text{vcam}}(C_{0:t}, \xi^{\text{cam}}, u(t), \Pi), \mathbf{y}_t^{(\text{cam})}) + d(\mathbf{G}_{\text{vemit}}(C_{0:t}, u(t), \xi^{\text{emit}}, \Pi), \mathbf{y}_t^{(\text{emit})})$$

is applied to virtual-view and virtual-emission predictions when the corresponding physical configurations are subsequently realised on the device, providing an audit term that physically closes the loop on the virtual primitives.

Multi-pair extension. In some embodiments, the loss-function uses of this section generalise the multi-pair coordinated training paragraph of Section 9.5 from cross-pair consistency between physical pairs to cross-pair consistency between physical pairs and virtual pairs, with the virtual pair configurations committed to the protocol digest.

17.3.4 Adversarial and competitive use (non-limiting)

Virtual-view verisimilitude discriminator. In some embodiments, a verisimilitude functional is extended to V_{θ}^{vcam} , scoring whether windows of predicted virtual-camera observations are consistent with the empirical distribution of physical camera observations of the same scene at the same configuration. Negative training examples include synthetic outputs drawn from competing 3D representations, view-synthesis baselines, and twin variants of differing fidelity, in the manner of the adversarial-autoencoder pattern of Section 9.5.

Virtual-emitter verisimilitude discriminator. In some embodiments, the dual functional $V_{\theta}^{\text{vemit}}$ scores whether predicted virtual-emitter responses are consistent with the empirical distribution of physical responses to the candidate emission, with negative examples drawn from substitutive G_{alt} variants, additive-superposition predictions outside their declared linearity envelope, and twin predictions of differing fidelity.

Per-axis generator-verifier games on virtual primitives. In some embodiments, the minimax structure of the GAN-like adversarial training of Section 9.5 is instantiated on each axis independently, recognising that the camera and emitter discriminators are independent verifiers. A camera-axis game:

$$\min_{G_{\text{vcam}}} \max_{V_{\text{vcam}}} \mathbb{E}[\log V^{\text{vcam}}(\text{real})] + \mathbb{E}[\log(1 - V^{\text{vcam}}(G_{\text{vcam}}))],$$

and an independent emitter-axis game:

$$\min_{G_{\text{vemit}}} \max_{V_{\text{vemit}}} \mathbb{E}[\log V^{\text{vemit}}(\text{real})] + \mathbb{E}[\log(1 - V^{\text{vemit}}(G_{\text{vemit}}))].$$

In further embodiments, the two games are trained jointly with a declared weighting term between the two objectives committed to the protocol digest. The decomposition into per-axis games is non-limiting.

Re-emission and re-capture audit. In some embodiments, virtual primitives are physically validated by the system subsequently emitting u^{cand} at ξ^{emit} or capturing from ξ^{cam} on the physical device, and comparing the previously committed predicted observation to the

freshly captured physical observation. The declared comparison tolerance and the time interval between prediction commitment and physical re-capture are recorded in the protocol digest. In some embodiments, this re-capture is selected adversarially: the system identifies, by gradient ascent on V^{vcam} or V^{vemit} over admissible $(\xi^{\text{cam}}, \xi^{\text{emit}}, u^{\text{cand}})$ within meter envelopes, those virtual configurations on which the generator is least confident, and physically realises those configurations to maximise audit information per re-capture.

Adversarial robustness training (non-limiting). In some embodiments, the agent or controller is trained against adversarial virtual primitives: G_{vcam} and G_{vemit} are perturbed within declared envelopes to produce worst-case predicted observations, and the agent’s training loss is evaluated under these perturbations, yielding controllers that are robust to virtual-primitive hallucination within the declared envelope.

Capacity-accounting scope clause (non-limiting). The training, loss, and adversarial uses of this subsection are directed to physically grounded model improvement, controller robustness, and verisimilitude auditing under the declared meter and protocol-digest discipline. They do not constitute, and shall not be construed as, capacity or scaling claims of the kinds bounded by the capacity-accounting rule of the present specification, and do not imply any increase in capacity quantities.

Virtual-interface generalisation (non-limiting). Virtual cameras and virtual emitters generalise to any declared computational sampling, rendering, stimulation, counterfactual, or projection interface over a physical or simulated apparatus-scene model. Non-limiting examples include pinhole cameras, rolling-shutter cameras, event cameras, spectral cameras, light-field cameras, plenoptic cameras, depth cameras, LiDAR-like samplers, ultrasound samplers, RF samplers, projector models, structured-light emitters, laser scanners, holographic emitters, acoustic emitters, microwave emitters, and hybrid virtual instruments. Scene models may include NeRF-style radiance fields, occupancy fields, signed-distance fields, Gaussian splats, mesh models, point clouds, light fields, plenoxels, hash-grid models, learned latent scene representations, physics simulators, differentiable renderers, and non-neural parametric or non-parametric models.

17.4 Scene-state-contrast training data (non-limiting)

In some embodiments, training data for downstream components (classifiers, segmenters, state-conditional generators, digital twin conditioning, perception-monitor admissibility rules, and governance-state gates) is generated not only by style-variable contrast of the kind disclosed in Section 17.2, but also by *scene-state contrast*: two or more committed corpora acquired through the Reality Kernel substrate under a matched emission protocol or declared matching envelope, where the corpora differ in a declared physical scene-state

variable. The corpora are committed with per-atom state labels and downstream models are trained, validated, or audited on the state contrast.

Scene-state-contrast is anchored to the substrate: each committed atom binds an actual emission to an actual physical observation under a shared time base and protocol digest, with Merkle commitment and hash-chain binding in the usual manner. The novelty is not the concept of contrast-based training (which is generic in supervised and contrastive ML) but the committed physical-evidence acquisition discipline around scene-state variables, the provenance-typing of state-conditional model outputs, the composition with Reality Kernel machinery (twin, analogue twin, perception monitor, co-emission, additive-overlay), and the privacy discipline around sensitive state corpora.

The following subsections describe the primary sub-embodiment (subject-presence contrast), five adjacent sub-embodiments that generalise it, and two state-specific discipline subsections (label-integrity and privacy-discipline) that apply across all sub-embodiments.

17.4.1 Subject-presence contrast (non-limiting, lead embodiment)

In some embodiments, two committed corpora are acquired differing in whether a subject is present in the scene:

- An *empty-scene corpus* $\mathcal{D}_{\text{empty}}$ is acquired by operating the Reality Kernel under a declared emission protocol on a declared scene or scene-class without a subject present; committed atoms record the scene’s response to emission in the subject’s absence.
- An *occupied-scene corpus* $\mathcal{D}_{\text{occupied}}$ is acquired by operating the Reality Kernel under the same or declared-matched emission protocol on the scene with a subject present; committed atoms record the combined scene-plus-subject response.

In preferred embodiments reflecting deployed practice, the two corpora are acquired in *separate sessions* rather than in a matched entry/exit capture session, to preserve acquisition cleanliness and prevent cross-contamination between state classes. In alternative embodiments, matched same-scene acquisition uses declared entry/exit-event markers and a committed geometric-stability envelope, producing tightly-paired per-frame-level state contrast at the cost of more demanding scene management. The acquisition regime (separate-session or matched-capture), emission-protocol identity, scene-class declaration, entry/exit-event markers where applicable, and per-atom occupancy labels are committed to the protocol digest.

Downstream uses (non-limiting). In some embodiments, the contrasted corpora feed one or more of:

- a subject-presence binary classifier producing per-frame or per-window subject/no-subject decisions;

- a subject-segmentation model producing pixel-wise, region-wise, or feature-wise subject/background partitions;
- a subject-conditional generative model producing empty- scene reconstructions from occupied captures (subject removal), or the inverse (scene-plus-subject synthesis), in the manner of the inverse generator G_{destyle} of Section 17.2.6 but conditioned on occupancy rather than style;
- a subject-presence feature for the digital twin of Section 17.1.2, supplying occupancy as a declared conditioning variable so that twin predictions correctly handle occlusion, reflection, and subject-scene interaction patterns;
- validation of upstream components (pose extractor, verisimilitude discriminator, kinematic-plausibility classifier) via physically-grounded occupancy ground truth.

17.4.2 Non-human scene-state contrast (non-limiting)

In some embodiments, the contrasted scene-state variable does not involve a human subject. Non-limiting variables include: object presence/absence; object arrangement class; door/window state; equipment position or operating mode; calibration-target state; display, screen, or signage state; occluder state; surface cover state; wet/dry state; contamination or clean state; lighting configuration or preset; environmental conditions such as smoke, dust, or aerosol density (claimed as declared binned or ranged variants per the continuous-variable discipline below); and safety-zone occupancy by equipment. Each non-human state variable is declared by its identity, value domain, acquisition regime, and matching envelope across corpora, all committed to the protocol digest.

Continuous and multi-state variables. In some embodiments, the contrasted variable is continuous (for example illuminance, spectrum, sun angle, time of day, temperature, humidity, subject distance, object pose) rather than discrete. Such variables are treated as declared binned, ranged, or regression-conditioned variants with a committed calibration envelope distinguishing them from the binary state-class case of subject presence. In other embodiments, the contrasted variable is multi-state but discrete (subject count; activity class; equipment operating mode); multi-state variables require more than two corpora under a matched protocol.

17.4.3 State-transition contrast (non-limiting)

In some embodiments, the contrasted corpora are acquired on the same scene in temporal before/after order across a declared transition event, producing a *state-transition corpus* structurally distinct from separate-session state-class corpora. Non-limiting transitions include subject entry/exit, object placement or removal, lighting or environmental change, equipment movement, calibration update, cleaning, wear accumulation, tamper event, and

weather-exposure transitions. State-transition corpora support transition prediction, change detection, tamper detection, and causal-residual training that separate-session corpora cannot support at the per-instant level.

The transition-event identity, pre-transition and post-transition atom boundaries, transition-time commitment, and any declared scene-stability envelope for the pre- and post-transition intervals are committed to the protocol digest. State-transition corpora may be composed with separate-session corpora to provide both transition-grounded and class-level training evidence.

17.4.4 Device-state contrast (non-limiting)

In some embodiments, the contrasted corpora differ in a declared *device-state variable* rather than a scene-state variable: same scene, same occupancy, but different projector calibration state, focus state, detector gain/exposure, reactor operating point, scan-law parameterisation, emitter aging state, aperture or filter state, actuator drift state, safety-envelope state, or analogue-twin matching state of Section 17.1.3. Device-state-contrast corpora support training of device-state-conditioned twins, drift detectors, tamper classifiers, and calibration-update policies.

The device-state variable identity, pre-contrast and post-contrast device configurations, declared measurement procedure for the state variable, and any matching envelope across contrasting corpora are committed to the protocol digest. This sub-embodiment composes with the trainable parameter-family discipline of the Reality Kernel architecture: device-state variables can be drawn from the committed parameter families.

17.4.5 Differential and residual training data (non-limiting)

In some embodiments, training data is committed not as paired raw atoms but as committed differential or residual atoms derived from two contrasting states: for example per-pixel, per-feature, or per-latent residuals $\Delta \mathbf{y} = \mathbf{y}^{(A)} - \mathbf{y}^{(B)}$ computed under a declared residual-extraction rule. Differential atoms provide a compact training signal and are distinct from paired raw-atom corpora in that downstream models train directly on the residual rather than learning to extract it. Per-pixel or per-frame residuals require matched same-scene or matched-transition acquisition (for example the matched-capture regime of Section 17.4.1); separate-session class-level corpora support distributional residuals, contrastive losses, or classifier contrasts, not exact ground-truth residuals.

The residual-extraction rule, declared feature or latent space, per-atom canonicalisation, and declared usage policy are committed to the protocol digest. Residual atoms inherit provenance-typing as derivatives of the two contrasting corpora and are not independently physical-provenance evidence of either state.

17.4.6 Multi-factor factorial contrast (non-limiting, optional)

In some embodiments, more than one declared state variable is varied across corpora under a declared factorial or blocked acquisition design (for example occupancy $\times$ lighting, subject-identity $\times$ pose, object-arrangement $\times$ weather, device-calibration $\times$ scene-state). The active design, factor levels, confound envelopes across uncontrolled factors, and block or replication structure are committed to the protocol digest. This sub-embodiment is optional and may be deferred where claim bandwidth or acquisition cost does not support full factorial coverage.

17.4.7 State-label integrity (non-limiting)

Per the commitment discipline of the Reality Kernel, per-atom state labels are themselves committed evidence. In some embodiments, the label source is one or more of:

- **Operator-declared labels:** the state is asserted by the operator at acquisition time, with a committed operator identifier and timestamp.
- **Sensor-confirmed labels:** an independent sensor (for example a separate occupancy detector, a calibration witness, or an environmental sensor) confirms the state, with the sensor identity, version, and confidence committed.
- **Externally-witnessed labels:** an independent authority witnesses the state declaration, with the witness identifier and witnessing protocol committed.
- **Bonded-identity-confirmed labels:** where subject identity is relevant, a committed bonded-identity record from the Reality Kernel's bonded human identity extension may confirm presence, consent, or other identity-linked state (subject to the identity-firewall discipline of Section 17.4.8).
- **Classifier-derived labels:** a prior committed classifier produces the state label, with the classifier identity, version, training-corpus digest, and confidence committed.

Label sources of different strength support different audit claims. In preferred embodiments, the label source, authority, and audit method are committed per atom or per commitment epoch, and higher-stakes downstream uses (safety gating, privacy restrictions, provenance attestations) require stronger label sources or multiple independent confirmations.

Anti-poisoning and label-audit discipline (non-limiting). In some embodiments, a declared label-audit procedure is committed, under which a sample of committed atoms is randomly or adversarially selected and the state label is independently re-confirmed. Label-audit failure rates are committed as meter-envelope statistics. A false "empty" label is both a privacy failure and a safety failure, and the protocol digest declares the consequences of label-audit failure (corpus quarantine, retraining, authority downgrade, successor protocol-digest version).

17.4.8 Privacy-discipline and three-object sensitivity envelope (non-limiting)

Subject-presence contrast and related scene-state-contrast training data raise sensitivities beyond those addressed by the generic provenance-typing discipline of the filing. Three sensitive objects require independent handling:

5 **$\mathcal{D}_{\text{empty}}$ sensitivity (non-limiting).** The empty-scene corpus is not necessarily innocuous. It may reveal private scene layout, equipment placement, security posture, absence-of-people patterns, or unobserved-scene conditions that the scene owner has interest in keeping private. In some embodiments, $\mathcal{D}_{\text{empty}}$ is committed under a declared sensitivity envelope distinct from $\mathcal{D}_{\text{occupied}}$, with separate audit-authority levels, retention rules, selective-opening rules,
10 and disclosure granularity. The two corpora are separately Merkle-addressable and separately redaction-capable, so that an audit of one does not automatically expose the other.

$\mathcal{D}_{\text{occupied}}$ sensitivity (non-limiting). The occupied-scene corpus aggregates personally identifying information. In some embodiments, per-subject consent records, bonded-identity commitments where applicable, and subject-consent-based retention rules are committed
15 alongside the corpus, with disclosure restricted to authorities authorised to inspect subject-bearing atoms.

Classifier-output sensitivity (non-limiting). Subject-presence classifier outputs reveal when a subject was present and constitute a surveillance-grade inference in aggregate. In some embodiments, classifier outputs are provenance-typed as model inferences over a de-
20 clared committed evidence window, with classifier identity, version, training-corpus digest, threshold/calibration version, confidence score, time window, and disclosure authority committed. Classifier outputs shall not be treated as direct physical-provenance evidence of occupancy or non-occupancy unless supported by selectively opened atoms or an evidence-bearing companion stream. This is especially important for "no subject present" determina-
25 tions, which are safety- and privacy-sensitive.

Subject-removal reconstruction provenance (non-limiting). In some embodiments, a generator produces a synthetic empty-scene estimate from an occupied capture (subject removal), trained on the contrast of $\mathcal{D}_{\text{empty}}$ and $\mathcal{D}_{\text{occupied}}$ under the same discipline as $\mathbf{G}_{\text{destyle}}$ of Section 17.2.6. Such generator output is evidence only of the committed occupied-capture
30 input, the declared generator identity and version, and the declared inference procedure; it shall not be treated as direct physical- provenance evidence that the empty scene actually had the reconstructed appearance. Selective disclosure may permit the synthetic empty estimate to be shown without opening occupied atoms, while preserving an escalated audit path to the underlying occupied evidence for authorised verifiers.

35 **Crypto-companion mode-commitment proof (non-limiting).** In some embodiments, a co-emitted committed cryptographic companion stream (Section 17.0.3) with

external anchoring proves that the device committed to a declared state-classifier version, training-corpus digest, empty-only mode, occupied-data-disabled mode, or disclosure policy during a capture window. The cryptographic companion proves *mode commitment*; it does not by itself prove the factual absence of a subject during the capture window. Factual ab-

5 sence requires classifier evidence, opened physical atoms, or an evidence-bearing companion (neutral probe, Truth Beam probe, or Limager probe). The protocol digest declares the scope of what the cryptographic commitment proves and the limits of that proof.

Deployment-scope restrictions (non-limiting). In some embodiments, subject-presence classifier outputs are used only within a declared deployment scope committed to the protocol digest: safety gating (emission downgrade, hazardous- capability disablement),

10 data-collection disablement (retention minimisation when subjects present), selective disclosure (redact subject-bearing atoms from routine audits), or policy-triggered fallback to semantic-fallback output. The deployment-scope restriction is itself a committed claim of the device: the classifier output is a governance input for the declared restrictions and is not

15 released as a standalone surveillance feed outside the declared scope.

Identity firewall (non-limiting). Presence, subject-count, subject-segmentation, subject-identity, and bonded-subject-identity are separate inferences with separately escalating sensitivity. In some embodiments, the protocol digest commits independent authority, consent-record, selective-opening, and provenance-typing discipline for each layer. A subject-presence

20 inference does not authorise identity or bonded-identity inference; identity compositions require separately declared authority, separate bonding records, and separate audit trails, and in preferred embodiments are placed under the Reality Kernel’s bonded human identity extension rather than the scene-state-contrast subsection.

18 Application Domains (Non-limiting)

25 This section describes non-limiting application domains. Any use in which a controllable emission and sensing system records the convolution bundle and uses the resulting evidence for verification, perception, and/or controllable rendering is within scope.

18.1 Generative media and immersive content

In some embodiments, Reality Kernels support generation, transformation, and presenta-

30 tion of media grounded in physical interaction. Subsystems include controllable rendering and projection mapping (Reality Transform head selecting emission patterns), hybrid diffusion/flow/consistency models (physical stochasticity as forward corruption with learned reverse process), and semantic-conditioned rendering (text prompts modulating emission policies). In Yoked variants, the scene’s response to rendering informs the next rendering

35 step, creating responsive immersive content.

3D generative models and view consistency (non-limiting). In some embodiments, the system supports 3D reconstruction and rendering, including NeRF-like radiance field models, point-based models, and scene graphs. Multi-view constraints may be enforced by combining channels with distinct invariance profiles and by training decoders to produce view-consistent outputs under changing control protocols. In some embodiments, a light-field hybrid embodiment (Section 10.2.5) provides a concrete hardware instantiation of sampled light-field evidence for radiance-field model fitting and parallax-consistent rendering, in which a plenoptic detector acquires multi-view evidence in a single exposure and a light-field projector synthesises parallax-consistent output, with the convolution bundle committing the sampled four-dimensional input–output correspondence as the evidentiary basis for view consistency.

Audio and multimodal generation (non-limiting). In some embodiments, the system generates or transforms audio, speech, or music using reactor temporal dynamics and audio convolution-bundle channels, optionally coupled to optical channels (for example cymatic media, meaning media whose surface patterns respond to sound pressure, observed by a camera). Text or semantic embeddings may condition emission policies so that physically grounded signals drive audio synthesis or transformation.

PoliePuter mode: reactor as compute substrate (non-limiting). In some embodiments consistent with the open-loop sub-case of Definition 24, the scene loop is disabled ($S^{\text{scene}} = S_{\text{term}}$) and the reactor subsystem is used as a physical compute substrate. A learned encoder maps an input x to a control protocol $U_{0:T}(x)$, the reactor produces the convolution bundle $C_{0:T} \sim P_{\theta}(\cdot \mid S_{\text{term}}, U_{0:T}(x))$, and a learned decoder maps $C_{0:T}$ to a reconstruction $\hat{x}$. The coupled sub-case of Definition 24, in which a continuous abstract or computational scene partner closes a bidirectional loop with the reactor, is treated separately and is governed by the Yoked-mode machinery of Section 8.5; the present paragraph concerns only the open-loop sub-case.

The mathematical structure of the attractor bundle, Fisher–Rao connection, curvature, holonomy, and characteristic classes is preserved in the open-loop sub-case to the extent that those constructions do not depend on scene-coupling dynamics: regime-level results that are stated for general reactor dynamics under $S^{\text{scene}} = S_{\text{term}}$ apply to the open-loop PoliePuter substrate, while results whose enabling derivation invokes scene coupling (in particular Yoked-mode synchronisation diagnostics, CLE and transfer-entropy conditions, the Yoked authentication warrant of Section 8.5, and any property whose proof requires a coupled scene partner) do not automatically carry over and are not asserted for the open-loop sub-case. The following remapping is used where it applies: scene maps to input data (digital-to-analogue converted or natively analogue); scene dynamics map to input signal statistics; convolution bundle maps to computed output; reactor memory maps to working memory or context buffer; protocol digest maps to execution log; and holonomy maps to computation audit trail.

Latency and reconfiguration depth (non-limiting). This disclosure distinguishes propagation depth from reconfiguration depth. Propagation depth contributes latency on the scale of cumulative optical path length. Reconfiguration depth contributes latency on the scale of the slowest active element per stage. Claims of low-latency operation herein
 5 refer to propagation depth and do not assert that reconfigured routed ensembles execute in propagation-scale latency.

Multi-reactor architectures (non-limiting). In some embodiments, N reactors are coupled in a network. The configuration space of the network is $\Theta_{\text{net}} = \Theta_1 \times \cdots \times \Theta_N \times \mathcal{G}$, where $\mathcal{G}$ is the space of coupling graphs specifying which reactor feeds which, with what
 10 strength and delay. Setting a point $(\theta_1, \dots, \theta_N, G) \in \Theta_{\text{net}}$ defines the computation. Non-limiting topologies include:

Chain / pipeline.

Reactors in sequence, $R_1 \rightarrow R_2 \rightarrow \cdots \rightarrow R_N$. Each reactor is one processing stage. The chain depth determines a temporal hierarchy: deeper layers integrate over longer
 15 timescales if each reactor has a distinct time constant. This implements deep reservoir computing with physical layers.

Parallel / SIMD.

Same input routed to N reactors with different θ_i , outputs aggregated. The collection of outputs is a nonlinear feature expansion. Effective reservoir dimension equals $\sum_i d_{\text{eff},i}$
 20 plus coupling-induced dimensions if aggregation is nonlinear. In some embodiments, wavelength-division multiplexing routes different spectral components to different reactors simultaneously; in some embodiments, time-division multiplexing injects input segments sequentially to the same reactor with memory providing context across segments.

Graph / mesh.

Arbitrary coupling topology where a coupling matrix G_{ij} specifies strength and delay from R_i to R_j . Any cycle in G creates recurrence. This topology is isomorphic to the networked kernel framework (Section 14) with data as input instead of scenes; all network
 25 dynamics apply.

Hierarchical / layered.

Graph with strict layer ordering and optional skip connections. Each layer aggregates information from the previous layer and passes to the next, creating a hierarchy of abstraction levels.
 30

Reservoir-of-reservoirs.

Each reactor is itself a network of simpler elements. Two-level hierarchy: micro-dynamics within each reactor, macro-dynamics across the network. Coupling between levels can
 35 introduce additional independent state variables not present in either level in isolation;

the effective computational capacity of the coupled system, measured as the number of accessible linearly independent state variables under the relevant fading-memory framework and readout regime, may correspondingly exceed a simple additive baseline computed from uncoupled isolated readouts for suitable readout and coupling regimes. No
 5 claim of super-additivity in all configurations is asserted; the actual capacity gain is regime-specific and is characterised empirically per deployment.

Recurrent.

Directed cycles in the coupling graph create recurrence at the network level. Unlike single-reactor recurrence (which arises from the feedback loop’s own temporal dynam-
 10 ics), network-level recurrence routes information through physically distinct media with different nonlinearities, time constants, and noise characteristics. In some embodiments, recurrent reactor networks implement physical analogues of gated recurrent units: one reactor modulates the coupling strength of another, providing content-dependent gating of memory and information flow. The recurrence depth is limited by signal-to-noise
 15 degradation per cycle rather than by discrete layer count.

Multi-reactor primitives (non-limiting). In some embodiments, the following operations exist only with $N > 1$ coupled reactors and have no single-reactor analogue:

Route: dynamically modulate coupling strengths G_{ij} to direct information flow through specific reactor sub-networks based on input content, implementing a content-dependent
 20 switch fabric analogous to mixture-of-experts routing. *Compile*: find the network configuration $(\theta_1, \dots, \theta_N, G)$ that minimises a target task loss, using natural gradient descent on the product Fisher metric of the network’s configuration space, where the coupling Fisher information captures how task loss depends on topology. *Parallelise*: partition the input (spatially, spectrally, or temporally) and route partitions to separate reactors for simultaneous process-
 25 ing, with throughput scaling with the number of physical channels. *Reduce*: combine outputs from multiple reactors via voting (fault-tolerant), averaging (noise-reducing), or nonlinear combination (where the aggregator is itself a reactor providing further computation). *Error-correct*: replicate computation across K reactors with nominally identical but physically
 30 distinct parameters (the reactor-microstructure unclonability property), with agreement indicating correctness and disagreement indicating fault or attack; in some embodiments, the attractor manifold itself serves as a codebook for error detection, where off-manifold outputs indicate computational errors. *Abstract*: chain reactor stages with different time constants to create representations at multiple abstraction levels, where early stages capture fast local features and deep stages capture slow global features.

Proof-of-physics (non-limiting). In some embodiments, when a computation is per-
 35 formed physically by a specific device or device network, the output inherently carries three properties: the device’s microstructure-specific attractor signature (the Sign primitive applied to computational output); a thermodynamic dissipation cost arising from the physical

processes of the reactor (with lower bounds under standard physical assumptions; a digital simulation may avoid these specific dissipation costs but is not expected to reproduce the reactor’s microstructure-dependent thermal trace); and a holonomy encoding of the computational path taken through Θ_{net} (what was computed, not merely the result). This triad
 5 provides a basis for attestation that a specific physical computation was performed by a specific physical device. Verification checks whether the output lies on the claimed device’s attractor manifold (Truth Beam on the output) and whether the output’s trajectory holonomy is consistent with the claimed computation’s path through Θ_{net} .

Non-limiting applications include verifiable inference (attesting that a machine-learning
 10 workload was processed by specific physical hardware), regulatory compliance (attesting that a risk model was evaluated on certified hardware), and physical provenance for computed artefacts (the computation’s result is empirically attested, under declared attacker families and meter envelopes, as having been produced by a specific physical device at a specific time).

Data types as scenes (non-limiting). When the input is data rather than a physical
 15 scene, different data types map to different input modalities. Time series map naturally to temporal reactor dynamics via direct analogue injection. Images are scan-converted to temporal signals where the scan pattern (raster, spiral, Hilbert curve, random) determines how spatial information maps to reactor dynamics; different scan patterns activate different
 20 attractor features. Graphs and networks are encoded as coupling topology: data graph nodes map to reactors, data graph edges map to couplings, and graph-level properties (centrality, clustering, community structure) emerge as attractor features of the reactor network’s dynamics. Tabular data modulates different reactor parameters or input channels, with the reactor’s nonlinear mixing providing automatic feature interaction. Text and sequences are
 25 tokenised and injected sequentially, with reactor memory providing a physical context window whose length equals memory persistence time divided by token injection rate. Audio feeds directly as an analogue signal. Multi-modal inputs route different modalities to different reactor ports or wavelengths, with the reactor fusing modalities through nonlinear mixing.

Computational tasks with physical advantage (non-limiting). In some embodi-
 30 ments, physical computation offers structural advantages beyond raw speed. Non-limiting examples include: Boltzmann sampling, where the reactor’s thermal noise is the sampling process itself and the attractor’s basin structure defines the distribution; optimisation via entrainment, where a cost function is mapped to coupling strengths and the reactor net-
 35 work settles to its ground-state attractor; differential equation solving, where the reactor physically integrates its own dynamics at propagation speed rather than numerical integration speed; analogue-to-insights processing, where analogue sensor data is processed in the analogue domain without digital conversion, with latency equal to propagation time and potentially much lower power consumption; and privacy-preserving computation, where data

need not be digitally represented and intermediate states may exist only as transient physical dynamics that are thermodynamically erased after use, with proof of deletion via the Erase primitive (Section 8.7). Cryptographic primitives are a further non-limiting example: bifurcation structure provides natural one-way functions (the forward map $x \rightarrow C_{0:T}$ given θ is

5 computable, but inversion without θ requires disambiguating exponentially many branches), the reactor’s thermal noise provides a physical random number generator at GHz rates, and device-specific attractor signatures provide key material that is grounded in physics rather than computational hardness assumptions.

Adjoint reactor pairs and physical gradient propagation (non-limiting). In some

10 embodiments, two reactor subsystems are configured as a *forward-adjoint pair*: reactor A implements a forward operator f_A (for example a linear optical transform such as a matrix multiplication, a convolution, or a Fourier-domain filter), and reactor B is configured to physically implement the adjoint (transpose, conjugate transpose, or generalised inverse) $f_A^\dagger$ of that operator. Error signals originating from a loss computed on the output of reactor

15 A are propagated backward through reactor B , producing physical gradient estimates without digital Jacobian computation. In these embodiments, reactor A may be linear or weakly nonlinear—linearity is an advantage, not a limitation, because it ensures the adjoint relationship $\langle f_A(x), e \rangle = \langle x, f_A^\dagger(e) \rangle$ holds exactly for linear embodiments, and holds to first order under local linearisation for weakly nonlinear embodiments, yielding unbiased

20 (or approximately unbiased) gradient propagation. Non-limiting physical realisations of the adjoint include: phase-conjugate mirrors that time-reverse an optical wavefront; transposed free-space optical interconnects (reversing source and detector planes); reciprocal fibre-optic couplers traversed in the reverse direction; and spatial light modulators programmed with the transpose of the forward modulation pattern.

25 In some embodiments, a chain of L forward-adjoint pairs implements layered physical backpropagation: the forward pass propagates data $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_L$ through reactors $A_1, \dots, A_L$, a loss $\mathcal{L}(x_L)$ is computed (digitally or by a comparator circuit), and the error signal $\nabla_{x_L} \mathcal{L}$ is propagated backward through adjoint reactors $B_L, \dots, B_1$ to produce per-layer gradient estimates. Parameter updates to each forward reactor are then derived from

30 the conjunction of its stored forward activation and the backward-propagated error at that layer. In some embodiments, the forward and adjoint reactors share a physical medium traversed in opposite directions (exploiting optical reciprocity), reducing the hardware to a single medium per layer.

In some embodiments, the adjoint reactor pair is combined with a nonlinear reactor

35 that provides attestation: the linear pair performs the computation, and a separate nonlinear reactor signs the result via Truth Beam, so that the output carries both a computed result and a provenance certificate. The convolution bundle logs both the forward and adjoint passes, and protocol digests record the adjoint-pair configuration so that the physical gradient computation is auditable. This embodiment is an instance of a broader pattern

40 in which linear and nonlinear reactors play complementary roles within a single module or

network: linear subsystems for high-fidelity computation, nonlinear subsystems for authentication and hardness. Adjoint-reactor pairs are a non-limiting instance of the multi-reactor architectures described above, using the Route, Merge, and Fork coupling primitives to direct forward and adjoint signal flows through the appropriate reactor subsystems. In some
 5 embodiments, forward–adjoint reactor architectures are integrated with attestation reactors so that forward computation, backward gradient propagation, and provenance certification are co-logged within a unified convolution bundle. This arrangement allows physically executed computation and physically grounded attestation to be combined within the same module or network.

10 **Training dimension count (non-limiting).** In some embodiments, the total number of trainable axes per single reactor is the number of information-operation primitives (Section 8.7) plus the number of multi-reactor primitives defined above. For a network of N reactors, the total training dimensionality is $N \times P + |\mathcal{G}|$, where P is the number of per-reactor primitive axes and $|\mathcal{G}|$ is the dimensionality of the coupling graph parameterisation.
 15 In some embodiments, the coupling graph is parameterised by $O(N^2)$ strength-delay pairs; in sparse topologies, the effective dimensionality scales as $O(N)$. This count is an addressable-parameter inventory of the disclosed apparatus and is not asserted as a guaranteed effective capacity, task-performance bound, or training-scale result. Effective capacity, generalisation, and task performance are empirical and embodiment-specific, and any capacity claim else-
 20 where in this disclosure is governed by the capacity-accounting discipline of that subsection rather than by the present axis count. Under the declared per-reactor conditional independence given coupling parameters, the network Fisher metric admits the block-diagonal-plus-coupling decomposition $\mathcal{F}_{\text{net}} = \text{blockdiag}(\mathcal{F}_1, \dots, \mathcal{F}_N) + \mathcal{F}_{\text{coupling}}$, where $\mathcal{F}_{\text{coupling}}$ captures how task loss depends on topology and inter-reactor coupling parameters and may be esti-
 25 mated empirically from cross-reactor score covariances; in some embodiments, the natural gradient defined by this product Fisher metric is used to navigate the parameter space, with the empirical efficacy of such navigation being embodiment-specific and not asserted as a convergence guarantee.

Regime reinterpretation in computational mode (non-limiting). In computational
 30 mode, the three regimes and Yoked modifier reinterpret: Truth Beam becomes verified computation (is this output consistent with computation by a known physical device?); Limager becomes inference and feature extraction (what does this input data contain?); Reality Transform becomes generation and transformation (produce output matching a target distribution); and Yoked operation becomes unsupervised learning (lock the reactor network’s
 35 dynamics to the input’s statistical structure, discovering natural modes and clusters by physical entrainment). Non-limiting interpolations include: Limager plus Reality Transform as a physical autoencoder where the attractor dimension d_{eff} is the bottleneck; Reality Transform plus Yoked as generative entrainment (generating samples entrained with the data distribution); and Truth Beam plus Reality Transform as signed generation where every out-

put carries proof of which device created it. Additional non-limiting interpolations include: Limager plus Yoked as self-supervised learning (the reactor discovers the input’s statistical structure without labels by physical entrainment, then uses that structure for feature extraction); Truth Beam plus Limager as verified inference (extract features from data with
 5 simultaneous attestation that the inference was performed by a specific physical device); and all three regimes with Yoked coupling engaged, operating simultaneously as a physical agent that perceives, generates, verifies, and entrains in a continuous loop.

Scope of computational embodiments (non-limiting). In the disclosed computational embodiments, computation is performed by a physical medium—linear, nonlinear,
 10 or a combination—whose dynamics carry out the computation. Nonlinear media provide empirical hardness and device-specific signatures (the reactor-microstructure unclonability property); linear media provide high-fidelity operators suitable for adjoint-pair gradient propagation and matrix computation. Both are constrained by measured noise, drift, bandwidth, and dissipation inherent to the physical substrate. Systems in which computation is per-
 15 formed by software simulation of dynamics, or in which the “reactor” is a purely social, economic, or abstract process without a physical substrate performing transformation of physical signals, are not required and are not the focus of the embodiments described herein.

Haptic and tactile rendering (non-limiting). In some embodiments, Reality Transform includes non-visual outputs such as haptic fields and tactile patterns that are generated
 20 and audited using the same convolution-bundle machinery. Non-limiting embodiments include ultrasonic phased arrays that generate mid-air haptic sensations, vibrotactile actuators embedded in wearables or furnishings, electrostatic touch surfaces, and pneumatic or mechanical actuator arrays. The control protocol specifies spatial and temporal patterns (including per-element phase and amplitude schedules for phased arrays), and detectors include mi-
 25 crophones, inertial sensors, contact sensors, cameras, and/or physiological channels (where available and permitted) to measure delivered fields and responses. In preferred embodiments, the protocol digest records safety envelopes and exposure limits, and the convolution bundle binds what was delivered to what was observed for later audit or dispute resolution.

Immersive chamber and tiled environments (non-limiting). In some embodiments,
 30 arrays of Reality Kernel modules are arranged in a PolieDeck: a tiled environment (for example walls, ceiling, and floor) instrumented with multiple projector–detector pairs and optional reactor loops, so that the room itself functions as a coupled scene for Limager, Truth Beam, and Reality Transform regimes. In some embodiments, PolieDeck deployments use trust-weighted witnesses and transitive proof-of-projection records to corroborate that
 35 immersive content was physically executed in the chamber, with selective opening used to audit disputed segments with bounded disclosure.

18.2 World models, planning, and control

In some embodiments, Reality Kernels provide a physically grounded interface for planning and control, including learned world models, model-predictive control, and reinforcement learning with physically-derived rewards.

18.3 Materials characterisation and autonomous experimentation (non-limiting)

In some embodiments, a Reality Kernel operating in Limager mode is integrated into an automated experimental workflow in which the system designs illumination protocols, directs structured light at a sample or material under test, captures the convolution bundle, updates an internal model of the sample's optical and physical properties, and selects the next protocol—forming a closed loop that iteratively refines knowledge of the sample without human intervention. In such embodiments, the scan law, emission schedule, and decoder heads are jointly optimised so that each measurement maximises expected information gain (or reduces expected loss toward a declared target) conditioned on all previously committed convolution-bundle windows.

Optical metamaterial and nanophotonic characterisation (non-limiting). In some embodiments, the sample under test is an optical metamaterial, metasurface, photonic crystal, or nanophotonic structure whose far-field emission or transmission profile depends on fabrication parameters. The Reality Kernel operates in Limager mode to map the relationship between fabrication parameters and optical response, using structured illumination to probe the structure and committed convolution-bundle data to build a predictive model. In some embodiments, the feedback loop drives an optimisation cycle in which fabrication parameters are adjusted between measurement rounds, creating a closed-loop discovery process in which the system identifies governing relationships between structure and optical function from empirical data. In some embodiments, the sample under test is a time-varying metamaterial or epsilon-near-zero thin film (Section 10), and the structured illumination includes temporally modulated probes whose spectral interference patterns characterise the material's switching dynamics, ENZ crossing frequency, and nonlinear Kerr response.

Verification of experimental provenance (non-limiting). In some embodiments, Truth Beam mode provides tamper-evident records of the measurement conditions, sample identity, and instrument state for each experimental run. Protocol digests and meter envelopes committed during the measurement campaign form an auditable evidence chain, so that results reported from the closed-loop process can be verified by third parties via selective opening.

18.4 Security, privacy, provenance, and compliance

In some embodiments, Reality Kernels provide physically grounded evidence for authenticated capture and tamper evidence, liveness checks and proof-of-projection, privacy-preserving operation (selective disclosure, secure aggregation, differential privacy), regulatory and audit use cases, and insurance documentation. Yoked variants add continuous tamper detection where ongoing scene-reactor coupling provides a persistent liveness signal that an adversary would need to maintain in real time.

Journalism and public-interest provenance (non-limiting). In some embodiments, Reality Kernels are used in journalism and investigations to produce tamper-evident capture records for images, video, and measurements. The convolution bundle, protocol digests, and meter envelopes are bound into committed records that can be selectively opened under dispute, making it more difficult to substitute or fabricate media after capture.

Remote notarisation and contract execution (non-limiting). In some embodiments, a Reality Kernel supports remote notarisation or contract execution by attesting that a declared person, object, or document was physically present in a declared session under a declared policy. Liveness checks and proof-of-projection protocols reduce replay and relay attacks, and multi-witness cross-attestation is used for higher-stakes sessions.

Collateral and inventory monitoring (non-limiting). In some embodiments, Reality Kernels verify and monitor collateral or inventory (for example pledged goods, equipment, or sealed containers) by executing periodic scan protocols and committing the resulting evidence. Auditors can request selective openings of committed windows when anomalies are detected.

Privacy-preserving operation (non-limiting). In some embodiments, privacy is supported by one or more of: local processing, selective disclosure of committed atoms rather than full traces, secure aggregation of meter outputs, differential privacy mechanisms in the physical or digital layers, and trusted execution environments for digital post-processing.

18.5 Data infrastructure, retrieval, and embedding stores

In some embodiments, the convolution bundle and derived features are stored as an evidence archive and may be indexed for retrieval. Retrieval supports similarity search, anomaly detection, and forensics, with selective opening of committed atoms for audit. In some embodiments, the system uses optical correlation, memory media, or reactor dynamics to implement fast approximate similarity operations on embeddings or signatures derived from the convolution bundle.

18.6 Human interaction and personalisation

On-device adaptation (non-limiting). In some embodiments, a personal device includes a compact Reality Kernel module and adapts models locally using physically grounded signals, while exporting only commitments, aggregates, or privacy-preserving updates.

5 **Assistive interfaces and AR/VR (non-limiting).** In some embodiments, head-mounted displays provide camera feeds and pose telemetry to a Reality Kernel and receive environment maps, occlusion masks, calibration parameters, or stylised overlays. Convolution-bundle logs support audit of what was shown and what was measured.

18.7 Narrative-charged tokens and economic value discovery (non-limiting, contextual application)

10 **Industrial-applicability scope of this subsection (non-limiting).** The technical contribution disclosed in this subsection is physical-provenance measurement and verification, including anti-counterfeiting, authenticity attestation, tamper- evidence over committed physical evidence windows, and cryptographically-bound binding of physical tokens to device-specific optical microstructure signatures. Economic and collective-decision uses described below are presented as non-limiting downstream applications of the physical-provenance machinery; they are not intended as the inventive contribution of this subsection, and no claim of this disclosure depends on those downstream uses for its inventive support. The independent claim hooks of this disclosure rest on the physical-provenance, authenticity-verification, and tamper-evidence functions disclosed elsewhere in the present specification; the present subsection provides contextual deployment description only.

In some embodiments, a narrative-charged microstructure-signed token participates in Reality Kernel verification protocols by binding a physical token's unclonable optical microstructure signature to declared physically anchored provenance data through committed evidence windows. The technical operations performed are: (i) committed binding of the token's microstructure signature to time-stamped scans at a designated site; (ii) committed binding of attestations from designated guardians or co-participants to the token's microstructure signature; (iii) committed binding of shared material provenance to the token's microstructure signature through optical microstructure cross-correlation; and (iv) selective opening of any of the foregoing committed bindings under dispute or audit. Each of these operations is a physical-provenance measurement function; none of them is, by itself, an economic or collective-decision function.

35 **Downstream economic and collective-decision applications (non-limiting, contextual).** In some embodiments, the foregoing physical-provenance machinery may be used as input to downstream economic or collective-decision processes external to the Reality Kernel itself, for example revealed-preference aggregation (auction clearing prices, bid trajectories,

holding times, transfer patterns), measurement-side attention proxies (query frequency, protocol complexity and duration, multi-party attestation rates, information-theoretic features), or governance rules using provenance depth and verification history as inputs to vote-weight assignment or membership pricing in declared interaction sets. These downstream applica-
 5 tions are described as contextual deployment scenarios; they are not claimed as inventive contributions of this disclosure, and the technical anchor for any such downstream application is the physical-provenance and authenticity-verification machinery described above and elsewhere in this specification.

Definition 34 (Narrative-charged microstructure-unclonability artefact (non-limiting)). A
 10 narrative-charged microstructure-unclonability artefact *is a physical token whose unclonable optical signature is committed in a Reality Kernel evidence record together with declared physically anchored provenance data binding the token to a declared event, location, artefact, relationship, or interaction history. In some embodiments, such artefacts may be referred to as “Narravite” crystals or similar, as a non-limiting naming convention.*

Interaction-set membership as a verifiable provenance attestation (non-limiting).

In some embodiments, the system supports verifiable attestations that a token (and by extension its holder) participated in a defined class of events, locations, attestation graphs, or material lineages. Non-limiting attestation forms include co-location with other tokens in a scene scan, temporal priority over a state change established by committed timestamps,
 20 verification by a quorum of designated Reality Kernels, presence in a named ritual or ecological survey under a declared protocol, or shared fabrication substrate detectable through optical microstructure cross-correlation. The technical contribution is the verifiable attestation; downstream pricing, gating, or governance uses of such attestations are external to the Reality Kernel and are not claimed as inventive contributions of this disclosure.

Learned anomaly detectors over committed provenance records (non-limiting, contextual).

In some embodiments, learned predictors operating over committed provenance records may flag anomalous patterns where attention or external usage patterns deviate sharply from the committed provenance profile, supporting fraud detection and authenticity audit in systems that combine physical microstructure-unclonability components and
 30 commitments. The technical contribution disclosed in this paragraph is provenance-anomaly detection over committed physical evidence; any market-surveillance use of such anomaly signals is a downstream contextual application external to the Reality Kernel and is not claimed as an inventive contribution of this disclosure.

18.8 Scientific, environmental, and industrial sensing

35 In some embodiments, Reality Kernels serve in remote sensing and environmental monitoring (airborne/satellite platforms, buoys, underwater vehicles), agriculture (crop health, pest detection, yield estimation), infrastructure inspection, materials characterisation, geophysical

and geological sensing, water-quality monitoring, and scientific sample tracking. RF, radar, and telescope array embodiments are non-limiting extensions.

Wildlife camera traps and conservation sensing (non-limiting). In some embodiments, a Reality Kernel is integrated with a camera trap or other conservation sensor to produce verifiable imagery of wildlife and habitats. Commitments and device identity bindings support audit that disclosed images were captured by an enrolled device under a declared protocol and were not substituted or materially altered after capture.

Pollution source monitoring and environmental compliance (non-limiting). In some embodiments, Reality Kernels provide tamper-evident monitoring of pollution sources and environmental quality, including industrial outfalls and distributed air or water sensors. Protocol digests and meter envelopes document calibration state and sampling conditions, and multi-device corroboration detects missing data, selective reporting, or inconsistent histories.

Acoustic biodiversity monitoring (non-limiting). In some embodiments, the same evidence and commitment pipeline is applied to acoustic monitoring (for example microphones or hydrophones) to produce authenticated recordings of animal vocalisations or anthropogenic noise, enabling auditable biodiversity surveys and anti-poaching monitoring.

Environmental DNA (eDNA) chain-of-custody (non-limiting). In some embodiments, a Reality Kernel is coupled to an eDNA sampling or analysis workflow (for example a cartridge-based sampler, filtration cassette, or lab bench instrument), and the sampling act is logged with protocol digests, device identity, and time/location metadata so that downstream genetic measurements are bound to a verifiable chain of custody.

Carbon and ecosystem accounting (non-limiting). In some embodiments, verifiable environmental observations support carbon and ecosystem accounting, including forest-cover and biomass monitoring, verification of agricultural practices associated with carbon sequestration (for example cover cropping, no-till, or rotational grazing), and monitoring of blue-carbon ecosystems (for example seagrass, mangroves, or kelp) using underwater or aerial deployments.

Citizen science and community networks (non-limiting). In some embodiments, low-cost or smartphone-integrated Reality Kernels enable citizens and community groups to contribute verifiable local observations (air and water quality, habitat changes) while preserving privacy via selective disclosure and secure aggregation; cross-device corroboration reduces the impact of compromised or misconfigured nodes.

Adaptive sampling, local intervention, and pest control (non-limiting). In some embodiments, a controller adapts sampling schedules and illumination patterns based on detected changes, and may trigger bounded interventions (for example alarms, valve actuation, aeration, or non-lethal deterrence) under safety constraints, while logging emissions and responses for audit. In some embodiments, Reality Kernels are integrated with pest or vector-control devices; meters classify organisms transiting a sensing volume and local interventions are gated by logged meter output and policy constraints.

18.9 Medical and biomedical embodiments (non-limiting)

In some embodiments, Reality Kernel architectures are applied to medical or biomedical sensing and intervention systems. The present disclosure concerns device architecture, control, logging, and verification mechanisms; clinical deployment of any embodiment would proceed under the regulatory and validation requirements applicable to the relevant indication. Subsystems include imaging and guided scanning, telemedicine and longitudinal monitoring, closed-loop actuation under safety envelopes, and ultrasonic actuation with biofeedback and audit.

18.10 Scientific computing and industrial systems

In some embodiments, Reality Kernels serve as physical substrates for fast transforms, filtering, or optimisation, including reservoir-style computation.

18.11 Sector-specific deployments

In some embodiments, Reality Kernels are deployed in agriculture, energy, logistics, retail, education, gaming, insurance, telecommunications, and real estate. Yoked operation adds resonance-based scene characterisation (probing how a building, crop, or object responds to structured stimulation), liveness detection (continuous rather than snapshot-based), and distributed coupling proof for sensor networks.

18.12 Tooling, ecosystem, and graceful degradation

In some embodiments, deployments include supporting tools such as convolution-bundle inspection utilities, calibration profilers, SDKs, simulation toolkits, and certification frameworks. The system is designed for graceful degradation when reactor hardware drifts or fails.

Cross-substrate equivalence summary (non-limiting). The following five sections map core Reality Kernel components to five interpretive substrates. The non-limiting table below summarises the correspondences; all entries represent non-limiting analogy-level mappings rather than strict equivalence.

RK component	Optical	Neural	Info-theoretic	Control	Crypto
Reactor Z_θ	Scattering medium	Latent $\mathbf{z}$	Channel state	Plant state	microstructure state
Emission $\mathbf{e}(t)$	Illumination pattern	Input stimulus	Transmitted signal	Reference input	Challenge
Observation $\mathbf{y}_t$	Detected image	Network output	Received signal	Measured output	Response
Config θ	Actuator settings	Weights (analog of θ)	Codebook	Controller gains	Public params / key material
Protocol $U_{0:T}$	Scan schedule	Training protocol	Coding scheme	Excitation signal	Challenge sequence
Meter $\mathbf{m}(t)$	Calibration checks	Loss / diagnostics	Rate / divergence	Error signals	Verification score
Convolution bundle $C_{0:T}$	Committed disclosure atoms	Activations log	Codeword sequence	I/O trajectory	Transcript

19 Optical Primitives Substrate

This section catalogues non-limiting optical primitives that may be realised within a Reality Kernel module and used as building blocks for verification, perception, and controllable rendering. In the embodiments described in this section, each primitive is accessed as part of a sweepable, parameterised kernel: the primitive is implemented as at least one of (i) a *sweeping kernel* that modulates an emitted field prior to scanning or addressing, (ii) a *stationary kernel* or reactor element that is swept by a scanner or addressed illumination after scanning, or (iii) a multi-stage combination of sweeping and stationary kernels. This discussion is not intended to limit the scope to any single optical effect; it shows that a single closed-loop controllable Reality Kernel can traverse a wide range of optical regimes under software control while preserving convolution-bundle logging and the Markov-kernel framing.

19.1 Coherence regimes

In some embodiments, the system operates under a chosen coherence regime that affects measurement sensitivity, stability requirements, and the structure of the convolution bundle. Coherence may be temporal, spatial, or both, and the chosen regime is recorded in protocol digests and meters.

Coherent operation (non-limiting). In some embodiments, coherent illumination enables interference, phase-sensitive signatures, and high sensitivity to small path-length and

alignment changes. This can strengthen hardness by exposing microstructure and coupling that is difficult to emulate, and it can also increase sensitivity to vibration, thermal drift, and alignment.

Partially coherent operation (non-limiting). In some embodiments, partially coherent operation reduces sensitivity to nuisance fluctuations while retaining useful structured signatures. This supports regimes where stability meters are bounded but tight phase lock is not required.

Incoherent operation (non-limiting). In some embodiments, incoherent illumination supports intensity-based measurements with reduced phase sensitivity. This can improve robustness and simplify modelling, while still supporting multi-channel constraints (for example spectral channels, polarisation, and scan-dependent signatures).

Metering and logging (non-limiting). In some embodiments, meters record coherence-related indicators such as source bandwidth estimates, coherence-length proxies, speckle statistics, fringe contrast (when applicable), temperature, vibration, and optical power stability. Protocol digests record the chosen coherence regime and relevant source configuration.

Speckle-statistics calibration (non-limiting). In some embodiments, a scattering-medium reactor is characterised using speckle-statistics measurements of the kind treated in Goodman’s *Speckle Phenomena in Optics* (2007), including first- and second-order intensity statistics, speckle contrast, spatial and temporal correlation length, polarisation-dependent speckle structure, and memory-effect angular range where applicable. The speckle-statistics summary is included in the calibration record and is stratified by wavelength, coherence regime, detector exposure, polarisation state, scan geometry, reactor temperature, and medium history.

19.2 Linear canonical transforms and convolution primitives

In some embodiments, optical propagation and engineered transfer functions implement linear canonical transforms (LCTs) and convolution-like operators that shape how stimuli map into observations.

LCT family (non-limiting). In some embodiments, optical propagation is described by an LCT family with a kernel parameterisation. Fourier, Fresnel, and fractional Fourier transforms are non-limiting special cases. In some embodiments, a tunable parameter acts as an invariance knob, where stable structure persists over a parameter range while mismatched emulators exhibit characteristic deviations.

Kernel form (non-limiting). In some embodiments, a linear canonical transform is written using an integral kernel:

$$(\mathcal{T}_\theta S)(x) = \int K_\theta(x, x') S(x') dx'.$$

In some embodiments, tunable optical elements implement a programmable LCT engine by sweeping θ (for example focal length, propagation distance, aperture, or phase-mask settings), and protocol digests record the sweep schedule so that verifiers interpret observations under the correct operator family.

Convolution and point-spread functions (non-limiting). In some embodiments, the system is approximately shift-invariant over a region of interest, yielding a convolutional mapping between a stimulus and an effective point-spread function (PSF). In other embodiments, the mapping is shift-variant, and the protocol digest records the scan law and optical configuration so that verifiers interpret observations under the correct operator family.

Scan-controlled kernels (non-limiting). In some embodiments, scanning and addressing cause the effective operator to vary over time, producing a controlled family of kernels indexed by scan coordinate, wavelength, focus, or polarisation. Meters capture alignment and stability so that the selected transform regime is reproducible and auditable.

19.3 Correlation and matched filtering

In some embodiments, correlation operations are used as primitives for alignment, verification, detection, and reconstruction, and are implemented optically, digitally, or in hybrid form.

Cross-correlation (non-limiting). In some embodiments, cross-correlation between channels or between predicted and observed windows is used to estimate timing offsets, scan alignment, or scene motion. Correlation peaks, sidelobe ratios, and consistency across channels are recorded as meter-like summaries. In some embodiments, a non-limiting cross-correlation form is

$$\text{corr}_T(S)(\Delta x) = \int S(x) \overline{T(x - \Delta x)} dx,$$

where T is a template (for example an expected response under a logged protocol or a disclosed probe window).

Matched filtering (non-limiting). In some embodiments, a known emitted pattern (or a digest-derived template family) is matched-filtered against observations to detect whether expected structure is present. This supports verisimilitude discrimination, projection-hardness

checks, and protocol binding, particularly when selective openings disclose only sparse windows or derived features. In some embodiments, matched filtering is implemented optically using a correlator architecture (for example a VanderLugt correlator in a 4f optical system) as a non-limiting implementation detail.

- 5 **Audit compatibility — correlation and matched filtering (non-limiting).** In some embodiments, correlation summaries are derived from committed atoms and protocol digests so that a verifier can request openings if correlation-level checks fail or if higher-confidence auditing is required.

19.4 Interferometry and phase-sensitive primitives

- 10 In some embodiments, phase-sensitive measurement primitives are used to increase sensitivity and to impose constraints that are difficult for emulators to reproduce under meter-bounded operation.

Phase sensitivity (non-limiting). In some embodiments, phase $\phi(t)$ depends on optical path length and refractive index variations. Small mechanical or thermal changes can produce
15 measurable fringe shifts. This can strengthen authenticity checks by coupling measurements to fine-grained physical behaviour.

Interferometric configurations (non-limiting). Non-limiting embodiments include two-beam interference, common-path interferometry, heterodyne-style phase measurement, or phase retrieval from controlled phase shifts. Protocol digests record the wavelength, path
20 configuration, modulation schedule, and sampling timing used for phase-sensitive operation.

Homodyne, heterodyne, and balanced detection (non-limiting). In some embodiments, phase-sensitive measurement is implemented using a local oscillator (LO) combined with a signal field at a beamsplitter and measured using balanced photodetection. In these
25 embodiments, the LO phase $\phi_{\text{LO}}(t)$ and amplitude may be swept as part of the control protocol $U_{0:T}$ (analogous to a phase scan law), producing quadrature-like observables that are included as channels of $\mathbf{y}_t$. In some embodiments, heterodyne operation is achieved by offsetting the LO frequency so the beat note encodes phase and amplitude over a logged window.

Meters and stability (non-limiting). In some embodiments, meters record fringe vis-
30 ibility, vibration proxies, temperature, coherence indicators, and phase-lock status (when applicable). Verification is conditioned on these meters, and phase-sensitive regimes may be gated or downgraded when stability is insufficient.

19.5 Polarisation primitives

In some embodiments, polarisation provides additional channels and constraints for both sensing and verification.

Polarisation channels (non-limiting). In some embodiments, the system measures or controls Stokes parameters, polarisation-resolved intensities, or polarisation-dependent responses. Polarisation channels can increase distinguishability between genuine and emulated channels, and can add invariance profiles across controlled polariser angles.

Polarisation optics and logging (non-limiting). In some embodiments, polarisation is controlled by polarisers, waveplates, liquid-crystal modulators, or birefringent elements. Protocol digests record polarisation settings, modulation schedules, and calibration versions. Meters record extinction ratios and polarisation stability indicators.

19.6 Holographic and volumetric storage primitives

In some embodiments, the system uses memory media that store optical states or mappings in a persistent or semi-persistent manner, supporting both reconstruction and hardness.

Volumetric or holographic storage (non-limiting). In some embodiments, volume holograms or interference-based media store patterns or transfer functions. Retrieval can behave as content-addressable memory, where the output depends sensitively on alignment, wavelength, and incident field structure.

Hardness and auditability (non-limiting). In some embodiments, stored media increase analogue hardness because reproducing the same retrieval behaviour, under the stated attacker model and meter tolerances, typically requires matching physical microstructure and alignment. Write and read protocols are logged via protocol digests and meters, and the resulting records are committed so that later audits can verify that a claimed memory state was actually exercised.

State persistence (non-limiting). In some embodiments, persistence times and decay profiles are metered and treated as part of the operating envelope, especially when memory effects influence subsequent emissions or observations.

19.7 Spectral and wavelength-selective primitives

In some embodiments, spectral selectivity provides multiplexing and additional constraints.

Wavelength multiplexing (non-limiting). In some embodiments, the system emits and observes at multiple wavelengths, either sequentially or with multiplexing. Spectral signatures can constrain emulation, improve reconstruction, and help verify that an observation is consistent with a logged protocol regime. In some embodiments, wavelength multiplexing is implemented as a spectral observation primitive using a continuously tunable frequency-comb emitter of the class described in Section 10.23, supporting, when paired with a second comb or suitable local-oscillator/reference path, dual-comb or multi-heterodyne interrogation of the reactor or scene spectral response as a declared observation channel, with the comb-mode-spacing schedule, heterodyne offset, and detection bandwidth recorded in the protocol digest; any verification or hardness use of such records remains conditioned on declared meter envelopes and empirically measured hardness indices, rather than on comb tuning range or waveform-space size alone.

Dispersive and filtering elements (non-limiting). In some embodiments, dispersive optics, tunable filters, diffraction gratings, or spectrally selective materials shape the channel. In some embodiments, compact photoelastic dispersive structures written into thermoplastic polymer substrates (for example polycarbonate) by ultrafast laser pulses provide broadband, view-angle-independent spectral dispersion in footprints of the order of $10\ \mu\text{m} \times 10\ \mu\text{m}$ and integrate directly with a CMOS image sensor for per-pixel spectral analysis, as described in Zhang et al., “Optical dispersion using micro-vortices in thermoplastic polymers for integrated microspectrometers,” *Nature Electronics* (2026). Protocol digests record wavelength schedules, bandwidth, and filter states. Meters record wavelength lock indicators, power stability, and spectral calibration status.

Audit compatibility — spectral and wavelength-selective (non-limiting). In some embodiments, spectral summaries and wavelength schedules are committed and selectively disclosed, enabling verification that disclosed evidence is consistent with the claimed spectral regime.

19.8 Random scattering and random projection primitives

In some embodiments, random scattering media and disordered optical paths implement high-dimensional mixing that behaves like a random projection. This can amplify microstructure sensitivity and increase analogue hardness, because small physical differences can cause measurable changes in the convolution bundle distribution under the same protocol digest.

Random projection viewpoint (non-limiting). In some embodiments, an input field or pattern is mapped through a random medium to a mixed observation that behaves approximately like a random feature map. The resulting readout can be treated as a high-dimensional embedding whose statistics are difficult to emulate without matching the physical medium and alignment. In some embodiments, this is interpreted as a Johnson–Lindenstrauss-style

random projection lens (after suitable normalisation), where distances or inner products are approximately preserved in the embedding, enabling discrimination and learning with small downstream heads. In some embodiments, a local linearised scattering model is used as an interpretation lens:

$$\mathbf{c} = A_{\text{scatter}}(\theta) \mathbf{s} + \mathbf{w},$$

- 5 where $\mathbf{s}$ denotes an input pattern or field parameterisation, $\mathbf{c}$ denotes a corresponding observation vector, $A_{\text{scatter}}(\theta)$ denotes an effective mixing operator induced by microstructure and configuration, and $\mathbf{w}$ denotes noise.

Speckle and stability (non-limiting). In some embodiments, speckle statistics provide useful signatures, and meters record speckle contrast, stability over time, and sensitivity to
10 scan coordinate. Protocol digests record illumination conditions and scan settings so that verifiers interpret speckle and correlation summaries under the correct regime.

Audit compatibility — random scattering (non-limiting). In some embodiments, random-scattering outputs are committed as atoms and opened selectively. In some embodiments, verifiers request openings that test consistency across multiple scan points, wave-
15 lengths, or polarisation settings to reduce the probability of emulator overfitting to a single view.

19.9 Nonlinear optics and saturating media

In some embodiments, nonlinear or saturating media provide amplitude-dependent transformations that increase distinguishability and make simple linear emulators ineffective. Non-
20 limiting examples include saturable absorbers, gain media near saturation, Kerr-like effects, photorefractive effects, and other media with intensity-dependent response.

Nonlinear response as a signature (non-limiting). In some embodiments, the effective mapping from emitted patterns to observed summaries depends on intensity history and local field distribution. This can create protocol-dependent signatures that are hard to
25 match with an emulator that lacks the same nonlinearity and internal state.

Metering and safe operation (non-limiting). In some embodiments, meters record saturation indicators, thermal indicators, drift, and stability bounds. Protocol digests record drive intensity schedules and duty cycles. In some embodiments, nonlinear regimes are gated by safety envelopes and downgraded when meters indicate instability.

30 **Interaction with verification and rendering (non-limiting).** In some embodiments, nonlinear media strengthen Truth Beam by amplifying microstructure sensitivity under

bounded protocols. In some embodiments, nonlinear regimes are also used in Reality Transform, where the nonlinearity shapes the controllable rendering dynamics while remaining meter-bounded and auditable.

19.10 Resonance, frequency-domain and cavity primitives

5 In some embodiments, resonant structures provide frequency-domain selectivity and memory. Non-limiting examples include optical cavities, resonant filters, ring resonators, and other frequency-selective elements whose response depends on alignment, temperature, and internal state.

Frequency response and tuning (non-limiting). In some embodiments, the system
10 probes a resonance by sweeping frequency, wavelength, or modulation rate and recording a response curve. The protocol digest records sweep schedules and tuning parameters. Meters record resonance stability, drift, and lock indicators.

Cavity memory and dynamics (non-limiting). In some embodiments, cavity dynamics introduce temporal memory and mixing. A short burst can influence later response,
15 creating a channel with history dependence. This can increase analogue hardness because matching the same transient dynamics, under the stated attacker model, materially raises the cost of matching both the structure and the operating regime.

Audit and selective opening (non-limiting). In some embodiments, the verifier challenges frequency-domain consistency by requesting openings across multiple probe points,
20 rather than a single point on a response curve. This reduces the probability that an emulator can fabricate a consistent frequency signature without matching the underlying physical response.

19.11 Time-of-flight, phase, and timing-anchor primitives

In some embodiments, timing provides a robust primitive for both sensing and verification.
25 Non-limiting examples include time-of-flight, phase-delay measurements, modulated illumination, and timing anchors derived from scan schedules and detector timing models.

Timing as a constraint (non-limiting). In some embodiments, the convolution bundle includes timing-dependent features that are required to be consistent with the logged protocol and the metered synchronisation state. This can strengthen verification by forcing
30 consistency across emission timing, scan coordinates, and detector readout timing.

Time-of-flight and phase delay (non-limiting). In some embodiments, depth or delay is inferred from phase shifts or time-of-flight signatures. Protocol digests record modulation

frequency, exposure models, and sampling schedules. Meters record timing jitter, clock stability, and alignment indicators.

In some embodiments, time-of-flight is obtained using pulsed illumination with gated detection or time tagging (TCSPC as a non-limiting example), where $g_{\text{det}}(t)$ encodes gate placement and width and the observation includes a per-step histogram of arrival times. In other embodiments, time-of-flight is obtained using continuous-wave modulation and phase delay, including frequency-chirped or multi-tone modulation (FMCW-style as a non-limiting example) in which a beat frequency or demodulated I/Q pair is recorded as part of $\mathbf{y}_t$ under a logged chirp schedule.

Anchoring and freshness (non-limiting). In some embodiments, timing anchors bind evidence windows to externally verifiable time. This interacts with batching and Merkle commitments: the system commits to windows and anchors commitments so that later openings have a clear freshness interpretation.

GNSS positioning and timing as auxiliary channels (non-limiting). In some embodiments, a GNSS receiver (for example GPS, Galileo, GLONASS, BeiDou, or multi-constellation receivers) is coupled to the Reality Kernel and provides position fixes, velocity estimates, and precision timing references as components of the auxiliary observation vector $y_{\text{ext}}(t)$. Position and timing metadata are recorded in the protocol digest alongside device identity and calibration identifiers, binding convolution-bundle evidence to a declared geospatial location and an externally traceable time source.

In some embodiments, the GNSS-derived timing reference serves as a timing anchor that supplements or replaces local oscillator references, and meters monitor the consistency between GNSS-derived time, local clock drift estimates, and scan-schedule timestamps. Detected discrepancies (for example GNSS signal loss, spoofing indicators, or timing jumps) are flagged in the meter envelope and may trigger downgrade to a degraded-confidence operating mode or increased reliance on alternative timing sources (for example a local atomic clock, network time protocol, or cross-device timing overlap).

In some embodiments, GNSS-derived position is used in multi-device corroboration to verify geometric consistency: the claimed positions of two or more Reality Kernels are checked against cross-attestation timing, overlap evidence, and signal-propagation constraints. In some embodiments, differential GNSS or real-time kinematic (RTK) corrections are logged as part of the protocol digest so that verifiers can assess the declared positioning accuracy.

In some embodiments, GNSS pseudorange and carrier-phase observables are logged as raw auxiliary channels (rather than only processed position fixes), enabling retrospective reprocessing, integrity analysis, and detection of multipath or interference signatures. In such embodiments, the raw GNSS observables are treated as additional components of the conditioned observation stream and may be committed alongside the optical convolution bundle.

19.12 Structured illumination families and multiplexing

In some embodiments, structured illumination patterns are selected from families that provide controllable information content, verification signatures, and rendering effects. Non-limiting families include pseudo-random patterns, coded apertures, Hadamard-like patterns, sinusoidal phase shifts, and multi-scale or multi-frequency patterns.

Pattern families and protocol digests (non-limiting). In some embodiments, pattern selection is driven by a seed or by a logged pattern schedule. Protocol digests record seeds, family identifiers, and schedule parameters so that opened atoms can be checked against the claimed emission regime.

Multiplexing (non-limiting). In some embodiments, multiplexing is performed across time, wavelength, polarisation, or spatial tiles. Multiplexing can increase robustness and reduce acquisition cost, while also increasing the constraint set available for verification. Meters record cross-channel alignment and stability so multiplexed observations remain comparable.

Orbital angular momentum and higher-order spatial modes (non-limiting). In some embodiments, structured illumination patterns include beams carrying orbital angular momentum (OAM), Laguerre–Gaussian modes, Hermite–Gaussian modes, or other higher-order spatial mode families. OAM beams carry helical phase fronts characterised by an integer topological charge and provide a theoretically unbounded set of orthogonal states, each of which can serve as an independent computational or encoding degree of freedom alongside time, wavelength, and polarisation. In some embodiments, OAM mode indices are included in the control protocol $U_{0:T}$ and recorded in protocol digests, so that verification and reconstruction can condition on the spatial mode structure of the emission. In some embodiments, mode mixing introduced by the physical channel (scattering, turbulence, refractive-index inhomogeneity) is treated as a device-specific and scene-specific transformation that contributes to the convolution bundle’s distinguishability and hardness properties rather than as noise to be corrected.

Illumination as a trainable computational resource (non-limiting). In some embodiments, the structured illumination pattern itself—not only the optical elements in the feedback path—participates in training or optimisation. The emission schedule $u(t)$ and any spatial, spectral, or polarisation structure of the emitted field are treated as learnable parameters that are jointly optimised with downstream heads (decoders, classifiers, verifiers) via backpropagation through a differentiable model of the physical channel or via policy-gradient methods. In such embodiments, the illumination pattern is not a fixed probe but a co-adapted component of the computational pipeline, and the protocol digest records the pattern parameters so that the computational contribution of the illumination is auditable.

Physical-channel distortion as computation (non-limiting). In some embodiments, the mode mixing, scattering, dispersion, and nonlinear transformation introduced by the physical medium between emitter and detector are treated not as degradation to be compensated but as computation performed by the medium itself. When structured light (including higher-order spatial modes) propagates through a complex medium, the medium's transfer function acts as a high-dimensional, physically fixed transformation analogous to the weights of a neural network layer. In such embodiments, only the readout (decoder or classifier applied to the convolution bundle) is trained, and the physical medium provides the rich feature space. This interpretation unifies the roles of the reactor, the scene, and any intervening medium: all contribute computational transformations to the convolution bundle, and the distinction between “device computation” and “channel distortion” is a modelling choice rather than a physical boundary.

Hybrid verification and sensing (non-limiting). In some embodiments, the same pattern family supports both Limager information acquisition and Truth Beam verification. For example, patterns are chosen to maximise information gain while remaining within regimes where verisimilitude and hardness remain high and auditable.

19.13 Gain media and optical amplification

In some embodiments, the module includes optical gain or amplification, for example an active medium that increases signal energy under pump control. Gain primitives can introduce sensitivity, nonlinearity, and history dependence that strengthen distinguishability and increase analogue hardness, while requiring explicit metering and regime logging.

Gain control and operating regimes (non-limiting). In some embodiments, the gain element is controlled by a pump schedule, bias, or drive waveform recorded in the protocol digest. In some embodiments, the gain is operated in a small-signal regime, a saturated regime, or a near-threshold regime, and the selected regime is treated as part of the protocol and meter envelope. In some embodiments, gain settings, pump power, wavelength schedules, and polarisation settings are recorded as sweepable parameters included in θ and/or $U_{0:T}$.

Amplification noise and saturation (non-limiting). In some embodiments, amplification introduces additional noise contributions, for example amplified spontaneous emission (ASE) and saturation effects. A non-limiting intensity-level description is

$$I_{\text{out}}(t) = G(t) I_{\text{in}}(t) + n_{\text{ASE}}(t),$$

where $G(t)$ is an effective gain (possibly history dependent) and $n_{\text{ASE}}(t)$ is a non-limiting noise term whose statistics depend on operating regime. In some embodiments, saturation is treated as a controlled nonlinearity, and meters record saturation indicators and stability bounds so verification and reconstruction remain conditioned on the observed regime.

Metering, stability, and logging — gain media (non-limiting). In some embodiments, meters record gain-related indicators including output power, pump power, gain estimates, saturation proxies, spectral broadening, temperature, and drift. In some embodiments, these meters gate whether a gain regime is used for verification, sensing, or rendering,
 5 and regime downgrades are logged when stability is insufficient.

Audit compatibility — gain media (non-limiting). In some embodiments, gain-regime runs are committed as atoms, and selective opening can reveal probe windows that test consistency of gain behaviour under the logged protocol. In some embodiments, audits compare expected versus observed amplification statistics (for example correlation and
 10 matched filtering under gain-dependent templates) and verify that disclosed windows are consistent with the protocol digest and meter envelope. In some embodiments, gain primitives increase analogue hardness because emulators would need to reproduce both the amplification behaviour and the associated noise and saturation statistics under the same logged regime.

15 19.14 Cavity and eigenmode primitives

In some embodiments, the module is operated as a cavity or ring-resonator system whose response is governed by resonant modes and eigenstructure. Cavity and eigenmode primitives provide frequency selectivity, temporal memory, and mode-dependent signatures that can strengthen verification and increase analogue hardness, while requiring explicit stability
 20 metering.

Eigenmodes and resonance conditions (non-limiting). In some embodiments, the system supports one or more resonant modes whose frequencies depend on geometry, refractive index, coupling, and boundary conditions. A non-limiting eigenmode form is

$$\mathcal{L}_\theta \psi_k = \lambda_k \psi_k,$$

where $\mathcal{L}_\theta$ denotes a round-trip propagation operator (or other effective cavity operator) under
 25 configuration θ . A non-limiting resonance condition is

$$m\lambda \approx 2nL,$$

where m is an integer mode index, λ denotes wavelength (standard optical notation; not the regime-weight symbol), n is an effective index, and L is an effective path length (for example a cavity length). In some embodiments, the measured response depends on coupling coefficients, loss, and mode competition, and these parameters are treated as sweepable or
 30 calibratable components of θ and/or $U_{0:T}$.

Frequency probing, ring-down, and memory (non-limiting). In some embodiments, the cavity is probed by sweeping wavelength, frequency, or modulation rate and recording a response curve or transient. In some embodiments, ring-down or transient response introduces temporal memory, and a non-limiting time constant is

$$\tau \approx \frac{Q}{\omega_0},$$

5 where Q is a quality factor and ω_0 is a resonant angular frequency. In some embodiments, this history dependence increases analogue hardness because an emulator would need to reproduce both steady-state and transient dynamics under the same protocol and meter envelope.

Metering, stability, and logging — cavity and eigenmode (non-limiting). In some
10 embodiments, meters record resonance stability, temperature, vibration proxies, coupling or tuning indicators, lock status (when applicable), and mode-hop indicators. Protocol digests record sweep schedules, tuning schedules, coupling settings, and calibration versions so that verification and reconstruction are conditioned on the correct cavity regime.

Audit compatibility — cavity and eigenmode (non-limiting). In some embodi-
15 ments, cavity probes are committed as atoms and selectively opened for audit. In some embodiments, audits request openings across multiple probe points or time windows so a verifier can check internal consistency of the claimed resonance response rather than a single point. In networked settings, multiple witnesses can corroborate cavity signatures under shared protocols, and trust-weighted aggregation can reduce the probability of fabricated
20 frequency-domain evidence.

19.15 Summary (optical substrate)

This section enumerates non-limiting optical primitives that can be combined with scan protocols, meter-bounded operation, and commitment and selective disclosure mechanisms.

In some embodiments, the primitive set includes coherence regime selection (coherent,
25 partially coherent, incoherent); linear canonical transform and convolutional regimes; correlation and matched filtering; interferometric and phase-sensitive regimes; polarisation-resolved channels; holographic or volumetric storage; wavelength-selective and dispersive channels; random scattering and random projection mixing; nonlinear and saturating media; resonant and cavity frequency-domain regimes; timing and time-of-flight constraints; and structured
30 illumination families with multiplexing.

In some embodiments, optical gain and amplification primitives and cavity or eigenmode primitives provide additional regimes in which microstructure sensitivity, stability, and memory can be tuned and logged. In some embodiments, photo-reactive and persistent surfaces provide history-dependent state channels whose write/read dynamics are metered and au-

ditable. In some embodiments, explicit modelling of sensor non-idealities and calibration state versioning supports conditioning verification and reconstruction on the appropriate measurement model. In some embodiments, cross-domain coupling primitives provide additional constraints by linking optical behaviour to thermal, mechanical, electrical, or acoustic dynamics within a metered operating envelope.

These primitives are non-limiting and may be combined or gated by policy. In some embodiments, protocol digests record the selected primitive family and configuration, meters record stability and envelope indicators, and commitments enable selective opening to audit that disclosed evidence is consistent with the claimed regime.

Non-limiting summary table.

Primitive	RK realisation	Use / constraint
Coherence regime selection	Choose coherent / partially coherent / incoherent sources; log stability meters	Trade sensitivity vs robustness; phase-sensitive signatures vs intensity-only stability
LCT / convolution operators	Propagation plus engineered transfer functions; sweepable θ implements an operator family	Programmable optical operators; impose structure and invariances; compare emulator mismatch across θ
Correlation / matched filtering	Cross-correlation and template matching on disclosed windows or predicted responses	Timing alignment, scan consistency, and verification checks under selective opening
Interferometry / phase-sensitive probes	Interferometers (Mach–Zehnder, Michelson, Sagnac as non-limiting examples) with logged basis and stability	Fine-grained path-length sensitivity; hard-to-forge phase constraints under meter envelopes
Polarisation-resolved channels	Polarisation optics and multiple detector channels with logged settings	Additional constraints and invariance knobs; multi-channel cross-checks
Spectral / dispersive channels	Tunable filters, gratings, dispersive media; wavelength schedules in protocol digests	Spectral signatures and multiplexing; separate bands for probing, pumping, and presentation
Random scattering / random projections	Disordered media; speckle and scattering-matrix viewpoint; meter stability tracking	High-dimensional mixing and microstructure sensitivity that amplifies analogue hardness
Nonlinear / saturating media	Intensity-dependent elements (saturable absorption, Kerr-like effects, photorefractive media)	Nonlinear signatures and history dependence; amplifies differences between genuine and emulated regimes

Primitive	RK realisation	Use / constraint
Gain / amplification primitives	Gain media near saturation with metered stability; logged duty cycle and thermal margins	Amplify signatures and noise statistics; typically uses explicit stability gating
Cavity / eigenmodes and resonance	Ring/cavity configurations with swept tuning and ring-down probing	Frequency selectivity and temporal memory; mode-dependent signatures for verification
Timing / time-of-flight anchors	Modulation schedules, timing models, and clock stability meters	Freshness and causal ordering constraints; detect replay and enforce protocol alignment

20 Neural-Architecture Substrate Mapping

This section describes non-limiting correspondences between the Reality Kernel formalism and common machine learning architectures. This mapping is not intended to imply that the Reality Kernel must be implemented as a particular digital architecture; it shows that a Reality Kernel can physically instantiate the computational structure of these architectures as special cases. These correspondences are explanatory embodiments and do not limit the core definition of a Reality Kernel.

20.1 Recurrent architectures (RNN, LSTM, GRU)

In some embodiments, the Reality Kernel loop (emit, interact with scene and/or reactor, detect, control, emit) can be treated as recurrent. Multi-timescale media and controller dynamics may serve roles analogous to gated recurrence. Slow media may behave like long-lived state, while fast dynamics behave like short-lived state. Detector gating and emission envelopes may act as write and read controls.

Gating as multi-timescale media (non-limiting). In some embodiments, gated recurrence corresponds to explicit multi-timescale media and explicit control schedules. A non-limiting mapping is:

LSTM component	RK physical analogue
Cell state c_t	Slow-decay media (phosphors, thermal mass, persistent optical states)
Hidden state h_t	Fast-decay media (fluorescence, transient scattering)
Forget gate f_t	Dissipation schedule $\gamma_{\text{slow}}(\theta)$
Input gate i_t	Illumination envelope $\mathbf{e}(t)$ modulating write strength
Output gate o_t	Detector gating $g_{\text{det}}(t)$

In these embodiments, “gates” are enacted by physical dynamics and by the protocol and metered operating envelope, rather than by a purely digital hidden-state update.

20.2 Autoencoders and variational inference (AE, VAE, denoising)

In Limager operation, the physical channel acts as an encoder from scenes to the convolution bundle. Digital heads act as decoders from the convolution bundle (or derived latents) to inferred scene variables. The effective bottleneck is constrained by physical channel capacity, noise, bandwidth, and detector resolution. In stochastic regimes, the channel may function as a variational encoder, and denoising behaviour may arise from physical averaging and controlled probing.

20.3 Adversarial structure (GAN-like)

In Truth Beam operation, a verifier meter may play a discriminator-like role distinguishing genuine physical traces from forged or simulated traces. An adversary that trains a simulator or generator to imitate the convolution bundle plays a generator role. The physical Reality Kernel provides genuine samples, yielding an adversarial training and evaluation structure in which physics constrains the feasible set of outputs.

Role mapping (non-limiting).

GAN component	Truth Beam analogue
Generator G	Reality Kernel $\mathcal{R}_\theta$ producing physical traces
Discriminator D	Verifier / meter head V_ψ
Adversary	Eve's spoofing policy π_{Eve}

Minimax objective lens (non-limiting). In some embodiments, the evaluation game is written in a minimax form:

$$\min_{\pi_{\text{Eve}}} \max_{V_\psi} \mathbb{E}[\log V_\psi(\text{real})] + \mathbb{E}[\log(1 - V_\psi(\text{Eve}))].$$

This is an interpretation lens: it highlights that “the generator” is the physical process and that the adversary’s job is to produce synthetic traces that pass a competent meter under a logged protocol and meter envelope.

20.4 Diffusion, score-based, and flow models

In some embodiments, repeated application of a configured Reality Kernel serves as a forward corruption or mixing process under controlled schedules, for example by increasing scattering, noise, or dissipation. A learned digital head may implement a reverse process or score-like correction operating on the convolution bundle or a latent representation. Flow matching and consistency models may be implemented by learning transport fields between distributions of convolution-bundle traces.

Markov chain form (non-limiting). In a non-limiting notation, repeated application can be written as a Markov chain:

$$x_{k+1} \sim P_{\theta}(\cdot \mid x_k, U_{0:T}^{(k)}),$$

where x_k is an intermediate state or representation (for example a latent, field state, or scene proxy) and $U_{0:T}^{(k)}$ denotes a step-indexed protocol schedule. Here P_{θ} is used as shorthand for the induced transition kernel on representations; x_k plays the role of the scene input at each stage, so the typing is consistent with the core definition $P_{\theta}(\cdot \mid S, U_{0:T})$ when S is instantiated by the intermediate state. In some embodiments, a distinct symbol (for example $T_{\theta}^{(k)}$) is used when the stage-indexed transition differs materially from the single-pass kernel.

20.5 Reservoir computing and random features

In some embodiments, the reactor acts as a physical reservoir whose internal dynamics induce a high-dimensional, task-adaptive feature map of the stimulus and control protocol. For example, the convolution bundle and meter summaries may be treated as a feature vector $\phi_{\theta}(S, U)$ that depends on stimulus S , protocol U , and physical parameters θ .

Physical random features and induced kernels (non-limiting). In some embodiments, the reactor implements physical random features in the sense that inner products in feature space behave as a kernel on inputs:

$$k_{\theta}((S, U), (S', U')) = \langle \phi_{\theta}(S, U), \phi_{\theta}(S', U') \rangle,$$

where $\langle \cdot, \cdot \rangle$ is any non-limiting inner product or similarity measure on extracted features. Such kernels may be used for classification, regression, nearest-neighbour matching, or as components of a verisimilitude discriminator.

Linear readouts and small heads (non-limiting). In some embodiments, a linear readout or small learned head operates on $\phi_{\theta}(S, U)$ to produce decisions or predictions. This matches reservoir computing in which a complex dynamical system provides rich features and only the readout is trained or adapted.

Dual verification and computation on a single physical substrate (non-limiting).

In some embodiments, the same physical medium that provides reservoir-like computational features also provides physically unclonable verification properties. The microstructure that makes a scattering medium, nonlinear cavity, or multi-mode waveguide useful as a computational resource (high-dimensional, device-specific transfer function) is the same microstructure that makes it useful as a microstructure challenge-response medium (device-specific, hard-to-clone challenge-response behaviour). In such embodiments, the distinction between

Truth Beam (verification) and Limager or Reality Transform (computation) is not a hardware distinction but an objective-function distinction: the same convolution bundle produced by the same physical channel is evaluated for authenticity (Truth Beam), for scene information (Limager), or for rendering quality (Reality Transform). This dual use is a direct
 5 consequence of the Markov-kernel abstraction, which treats the physical channel as a single parameterised object $P_\theta(\cdot \mid S, U_{0:T})$ whose output supports multiple downstream objectives without hardware modification.

Audit compatibility — reservoir computing (non-limiting). In some embodiments, the feature map ϕ_θ is computed from committed atoms and meter summaries so that downstream
 10 stream decisions can be audited by selective opening when required.

20.6 Residual and skip connections (ResNet-like)

In some embodiments, optical or physical skip paths split a stimulus into a direct branch and a processed branch and then recombine them, yielding residual-like behaviour. This can be implemented by beam splitters, waveguides, multiplexed paths, or mixed digital and physical
 15 routing.

Residual mixing with controllable bypass (non-limiting). A non-limiting mixing form is

$$E_{\text{out}}^{(k)} = b_k E_{\text{in}}^{(k)} + (1 - b_k) \mathcal{R}^{(k)}(E_{\text{in}}^{(k)}),$$

where $E_{\text{in}}^{(k)}$ and $E_{\text{out}}^{(k)}$ denote the effective field or representation at stage k , $\mathcal{R}^{(k)}$ is a non-limiting transformation implemented physically or digitally, and $b_k \in [0, 1]$ controls the
 20 bypass fraction.

Stability and expressivity (non-limiting). In some embodiments, bypass fractions b_k are used to control stability and mixing, aligning with contractivity and commutativity targets in the Markov-metric view. In some embodiments, bypass paths also carry timing anchors or calibration signals to improve alignment and robustness.

25 20.7 Convolutional and equivariant mappings (CNN-like)

In some embodiments, when an optical or physical transfer function is approximately shift-invariant over a region of interest, the resulting input–output relationship can be modelled as a convolution-like operator. This connects to convolutional neural networks (CNNs), equivariant mappings, and designed filter banks.

Convolutional form (non-limiting). A non-limiting continuous-space convolution form
 30 is

$$(f * g)(x) = \int f(\xi) g(x - \xi) d\xi,$$

with corresponding discrete forms for sampled fields. In some embodiments, convolution-like behaviour arises from Fourier-optic propagation, pupil filtering, engineered point-spread functions, or structured illumination combined with fixed readout optics.

Equivariance as an engineering target (non-limiting). In some embodiments, the mapping T from stimulus to observation is designed to be equivariant to a transformation group G (for example translations, rotations over a limited range, or planar rigid motions), meaning

$$T(\rho(g)f) \approx \rho'(g)T(f),$$

where ρ and ρ' are representations of G acting on inputs and outputs. Approximate equivariance may be assessed empirically under meter-bounded operation and used to select optical configurations, scan laws, and aggregation rules.

Group convolution viewpoint (non-limiting). In some embodiments, convolution is generalised to group convolutions:

$$(f *_G \psi)(g) = \int_{h \in G} f(h) \psi(h^{-1}g) dh,$$

which supports non-limiting embodiments where optical routing and scan protocols implement filtering over transformation parameters (for example angle or pose), rather than only over spatial translations.

Designed operators and controlled scan protocols (non-limiting). In some embodiments, operators are designed by choosing illumination patterns and readout optics so that the induced mapping approximates a desired filter family (for example wavelets, steerable filters, or multi-scale banks). In some embodiments, controlled scan protocols provide the equivalent of receptive-field traversal and multi-scale sampling while preserving convolution-bundle logging and audit compatibility.

20.8 Energy-based and associative memory (Hopfield and Boltzmann-like)

In embodiments where reactor media evolve toward stable attractors, the physical dynamics can be interpreted as minimising an energy-like functional. This connects to energy-based models, Hopfield-like associative memories, and Boltzmann-like sampling interpretations.

Energy and relaxational dynamics (non-limiting). A non-limiting abstract form is a state x evolving under a potential $E(x)$:

$$\dot{x}(t) = -\nabla E(x(t)) + w(t),$$

where $w(t)$ is a non-limiting noise term (including thermal noise, shot noise, or injected structured noise). In some embodiments, stable attractors correspond to stored patterns, calibrated configurations, or preferred operating regimes.

Associative retrieval (non-limiting). In some embodiments, presenting a partial cue via structured illumination causes dynamics to settle into a compatible attractor, yielding associative recall. The resulting convolution-bundle summaries act as readouts of the retrieved state and may be compared to predicted responses for probing or verification.

Training alignment (non-limiting). In some embodiments, energy-based losses or contrastive objectives are used so that digital training aligns with the physical relaxational dynamics observed in the reactor under meter-bounded operation.

20.9 Attention and transformer-like interpretations

In some embodiments, associative retrieval in a reactor serves a role analogous to attention, where the system forms content-dependent weights over stored or reachable patterns and then produces a weighted retrieval.

Attention-style weighting (non-limiting). A non-limiting attention form may be written as

$$a_{ij} = \text{softmax}_j \left(\frac{\langle q_i, k_j \rangle}{\sqrt{d}} \right), \quad z_i = \sum_j a_{ij} v_j,$$

where q_i is a query-like representation, k_j are key-like representations, v_j are value-like representations, and d is a scaling dimension. In some embodiments, q_i, k_j, v_j are derived from convolution-bundle summaries, meter-conditioned features, or reactor-internal states.

Physical correlators and holographic storage (non-limiting). In some embodiments, optical correlation, interference, or convolution implements parts of the dot-product or similarity computations that underlie attention-like weighting. In some embodiments, holographic or interference-based media provide associative storage such that retrieval behaves as content-addressable memory, which can be interpreted as a continuous Hopfield-like mechanism.

Audit and policy constraints (non-limiting). In some embodiments, tokenisation, conditioning, and normalisation functions are constrained by meter envelopes and commitment policies, so that transformer-like interpretations remain compatible with selective opening and trust-weight updates.

20.10 State-space models and continuous-time dynamics

In some embodiments, the Reality Kernel admits a controlled state-space description, with observations arising from the measurement map and the convolution-bundle channel. In the discrete-time formulations below, $u_t := u(t)$ denotes the per-step control of Section 6, and y_t denotes a scalar or vector observation consistent with $\mathbf{y}_t$.

Discrete-time state-space form (non-limiting). A non-limiting form is

$$x_{t+1} = f_\theta(x_t, u_t) + w_t, \quad y_t = g_\theta(x_t, u_t) + v_t,$$

where x_t is an extended state, u_t is a control input (including scan law, projection schedule, gains, and alignment commands), y_t is an observation or summary (including the convolution bundle features and meters), and w_t, v_t are non-limiting process and observation noise terms.

Continuous-time form and Neural ODE interpretation (non-limiting). In some embodiments, continuous-time dynamics are used:

$$\dot{x}(t) = f_\theta(x(t), u(t), t), \quad y(t) = g_\theta(x(t), u(t), t).$$

When f_θ is parameterised by a learned function class, this aligns with Neural ODE or continuous-time state-space modelling viewpoints, with meter summaries providing constraints and validation signals.

Control and estimation (non-limiting). In some embodiments, standard estimation and control tools (filtering, smoothing, observability and controllability analyses, robust control) are applied, subject to the meter-bounded operating envelope and verification-gated action policies described elsewhere in this document.

20.11 World models

In some embodiments, the system is described using a world-model decomposition in which an encoder produces a latent state, a dynamics model predicts latent evolution under actions, and a decoder (or predictor) produces expected observations. This framing supports planning, control, and verification when the primary observation stream is derived from the convolution bundle and meter summaries.

Latent-state decomposition (non-limiting). A non-limiting form is

$$z_t = \text{Enc}_\psi(o_t), \quad z_{t+1} = F_\psi(z_t, a_t) + w_t, \quad \hat{o}_{t+1} = \text{Dec}_\psi(z_{t+1}),$$

where o_t is an RK-derived observation or summary (for example features extracted from the convolution bundle together with meter vectors and protocol digests), a_t is an action

(including RK control components), and w_t is a non-limiting disturbance term.

Meter-bounded and commitment-aware world models (non-limiting). In some embodiments, world-model updates and planning are constrained by meter envelopes and by audit policies. For example, training or adaptation may be conditioned on meter vectors and protocol digests, and may be regularised to preserve verisimilitude and projection-hardness scores. In some embodiments, when commitments exist without openings, planning operates in a provisional mode and upgrades to a verified mode after selective opening confirms consistency.

Planning and control (non-limiting). In some embodiments, planning uses the latent dynamics model for model-predictive control, policy optimisation, or search over action sequences, with costs that include both task costs and evidence-quality terms. This is compatible with the agent integration section and with verification-gated action policies described elsewhere in this document.

20.12 Neural radiance fields and differentiable rendering (NeRF-like)

In some embodiments, Limager and Reality Transform operation are interpreted using scene-representation lenses similar to neural radiance fields (NeRFs), differentiable rendering, and light-field models, with the key distinction that the evidence is produced by a physical swept kernel rather than by purely digital rendering.

NeRF-like representation lens (non-limiting). NeRF-style models represent a scene as a continuous field, for example

$$F_\psi : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma),$$

mapping spatial position $\mathbf{x}$ and viewing direction $\mathbf{d}$ to colour $\mathbf{c}$ and density σ (or other radiometric quantities). In some embodiments, a Reality Kernel operating in Limager mode learns a representation that is analogous in purpose: it predicts or reconstructs scene variables from the convolution bundle under a family of logged control protocols, and it is validated by executing additional physical probes and comparing predicted versus observed convolution-bundle windows under the same protocol digests.

Differentiable rendering lens (non-limiting). In some embodiments, Reality Transform operation is treated as a physically instantiated inverse-rendering or projection-mapping loop: the controller selects emissions, observes responses, and updates projection parameters to reduce a target appearance loss while remaining within meter envelopes. Gradients may be obtained by differentiable surrogates, finite differences, or policy-gradient methods, and

the resulting actuation remains auditable because protocol digests, meter summaries, and committed windows record what was emitted and what was observed.

Explicit scene primitives (non-limiting). In some embodiments, scene representations are explicit (for example point clouds, surfels, Gaussian primitives, meshes, or hybrid implicit–explicit forms) and probe protocols are designed to efficiently constrain those primitives (pose, depth, reflectance, and dynamics) under logged scan schedules.

Light-field and ray-space interpretations (non-limiting). In some embodiments, a Reality Kernel is interpreted as sampling a ray-space or light-field parameterisation, where different control paths correspond to different ray bundles (space, angle, wavelength, polarisation, time-of-flight bins), and the learned model predicts or constrains the induced convolution-bundle distribution under those ray samples.

20.13 Backpropagation through physics and surrogate gradients (non-limiting)

In some embodiments, training and tuning use gradient-like updates even when physical dynamics are not directly differentiable.

Surrogate backpropagation (non-limiting). In some embodiments, a differentiable surrogate $\hat{P}_\theta$ is trained on logged input–output pairs and is used to compute gradients for losses defined on physical artifacts (for example hardness meters, verisimilitude meters, calibration penalties, or reconstruction error). Updates are then applied to admissible parameters (digital and/or analogue) subject to actuator projections and safety envelopes.

Perturbation gradients (non-limiting). In some embodiments, gradients are approximated by finite differences or stochastic perturbations (for example SPSA), where the controller perturbs control paths or parameters, measures the change in a scalar objective, and applies projected updates under variance-control practices and meter gating.

Adjoint and time-reversal analogues (optional, non-limiting). In some embodiments, continuous-time models admit adjoint-style gradient calculations, and reversible or approximately reversible optical subsystems admit time-reversal analogues (for example phase conjugation) that support efficient sensitivity estimation. These are non-limiting implementation patterns and may be combined with digital surrogates and recorded forward trajectories.

Reinforcement learning as a fallback (non-limiting). In some embodiments, reinforcement learning treats the physical device as a black-box environment: rewards are de-

rived from meters and task losses, and a policy is updated to select scan protocols, emissions, and tuning actions that improve objectives under logged regimes.

Modern reinforcement-learning algorithm families (non-limiting). In some embodiments, the policy-update step uses any of the modern reinforcement-learning algorithm families, including without limitation proximal policy optimisation (PPO, Schulman et al. 2017), trust-region policy optimisation (TRPO), advantage actor-critic methods (A2C, A3C, and synchronous and asynchronous variants), soft actor-critic (SAC) and other off-policy entropy-regularised methods, deep deterministic policy gradients (DDPG), twin-delayed DDPG (TD3), Q-learning and its deep variants (DQN, Double DQN, Dueling DQN, Rainbow, prioritised experience replay), distributional reinforcement learning methods (C51, QR-DQN, IQN), IMPALA and other distributed actor-learner architectures (Ape-X, R2D2, SEED-RL), model-based reinforcement learning methods including MuZero-style learned-model-with-search and Dreamer-style latent-dynamics-model RL (Hafner et al.), offline reinforcement learning methods (CQL, IQL, BCQ, AWAC, decision transformer), group-relative policy optimisation (GRPO) and related verifier-based post-training methods used for reasoning agents, and goal-conditioned and hierarchical reinforcement learning methods (HER, options framework, FeUdal Networks, HIRO). The apparatus is compatible with any such method applied to scan-policy training, controller training, or governed-agent training, including future methods within the same algorithmic envelope.

20.14 Summary (neural-architecture substrate)

This section provides a non-limiting mapping between Reality Kernel physical mechanisms and common neural-architecture abstractions. The purpose is explanatory and enablement-oriented: it shows how a single physical substrate (configured by scan protocols, structured illumination, and meter-bounded operation) can instantiate behaviours analogous to widely used computational motifs.

Architecture	RK instantiation	Key physical mechanism
RNN / LSTM / GRU	Core feedback loop with multi-timescale media and gating schedules	Multi-timescale dissipation and metered write/read controls
AE / VAE / denoising	Limager regime with a physical encoder and digital decoder	Channel capacity as bottleneck; noise and controlled probing
GAN-like	Truth Beam meter as discriminator; adversary as generator	Hard-to-forge physical signatures under bounded protocols
Diffusion / score / flow	Physical forward mixing plus learned reverse/score head	Shot noise, scattering, and scheduled dissipation as corruption
Reservoir / random features	Reactor dynamics plus small readout head	High-dimensional mixing; implicit kernel on (S, U) features

Architecture	RK instantiation	Key physical mechanism
ResNet / skip connections	Split/recombine paths with controllable bypass	Beam splitters, multiplexed paths, reference arms
CNN / equivariant	Shift-invariant or group-structured optical operators	PSFs, diffraction, and structured scan-addressed transforms
Energy-based / Hopfield	Relaxation to attractors under structured cues	Physical energy landscape and content-addressable retrieval
Transformers / attention-like	Correlation and associative retrieval over stored patterns	Similarity weighting via correlation/interference and multiplexed channels
State-space / Neural ODE	Controlled continuous-time dynamics with observation map	Physical dynamics with filtering/estimation on the convolution bundle
NeRF / differentiable rendering	Limager/Reality Transform with protocol-conditioned scene models and physical validation	Active probing and meter-bounded physical constraints on predictions
Backprop / surrogate gradients	Surrogate-guided tuning and perturbation-based optimisation	Differentiable twins, finite differences, and metered update envelopes

In some embodiments: (i) reservoir and random-feature viewpoints arise when convolution-bundle and meter-conditioned summaries define a high-dimensional feature map and induced kernel; (ii) residual and skip-path viewpoints arise when bypass fractions mix direct and processed paths in a controllable manner; (iii) convolutional and equivariant viewpoints arise
5 when optical transfer functions approximate shift-invariant or group-structured operators under controlled scans; (iv) energy-based and associative-memory viewpoints arise when reactor media relax toward attractors under structured cues, optionally with noise and injected perturbations; (v) attention-like viewpoints arise when correlation and associative retrieval implement content-dependent weighting over stored or reachable patterns; and (vi)
10 state-space and continuous-time viewpoints arise when the system is modelled as controlled dynamics with observations derived from the convolution bundle and meter summaries.

These mappings do not assert that the physical system is identical to any specific neural architecture. Rather, they provide non-limiting interpretations and design analogies that can guide implementation choices, training objectives, and verification policies while pre-
15 serving convolution-bundle logging, selective disclosure, and the Markov-kernel framing used throughout this document.

21 Information-Theoretic Primitives

This section describes information-theoretic viewpoints and quantities that may be used to characterise Reality Kernel systems. These quantities are non-limiting and may be used as

design and evaluation tools. They do not replace physical grounding or meter-based evaluation. They provide vocabulary for limits and trade-offs across Truth Beam, Limager, and Reality Transform regimes. This section establishes formal relationships with information theory as interpretive vocabulary for limits and trade-offs across regimes; it does not constrain the core definitions or embodiments.

Remark 5 (Notation). *Throughout this section, the symbol Ξ denotes a task-relevant random variable (stimulus, secret, or label) drawn from a design-time distribution. This is distinct from the scene tuple $S = (S^{\text{scene}}, S_{\theta}^{\text{reactor}})$ of Section 6: the scene tuple is a physical configuration; the task variable Ξ is a random variable over which information-theoretic quantities are optimised. In many applications the task variable is derived from or indexed by the scene (for example a label identifying which scene is present), but the two objects are formally distinct.*

21.1 Reality Kernel as a noisy channel

A Reality Kernel (RK) module can be modelled as inducing a conditional distribution over the recorded convolution bundle given an input stimulus (or secret) and a control protocol. Let Ξ denote a stimulus, secret, or task-relevant label drawn from a design-time distribution $\mathbf{P}_{\Xi}$. Let $U_{0:T}$ denote a control protocol (projection sequence, scan law, actuator schedule, gains, alignment parameters, and any other commanded inputs), and let $C_{0:T}$ denote the resulting convolution-bundle record.

The RK induces a channel family parameterised by physical and control parameters θ :

$$C_{0:T} \sim \mathbf{P}_{\theta}(\cdot \mid \Xi, U_{0:T}).$$

When $U_{0:T}$ is treated as given (known to the verifier, logged, or fixed by policy), the induced information transfer from Ξ into $C_{0:T}$ can be quantified by the conditional mutual information

$$I_{\theta}(\Xi; C_{0:T} \mid U_{0:T}).$$

In some embodiments, the protocol itself is randomised (for example, via structured noise or scan randomisation). In such cases $U_{0:T}$ is a random variable, and the same quantity is interpreted as mutual information conditioned on the realised protocol and meter state.

Operational role of $\mathbf{P}_{\theta}$ (non-limiting). The kernel $\mathbf{P}_{\theta} : \mathcal{S} \times \mathcal{U}^{T+1} \rightarrow \mathcal{P}(\mathcal{C})$ serves as a conceptual unification device: it provides a common probabilistic vocabulary for the three operating regimes and grounds the information-theoretic quantities that follow. All operational computation uses specific statistics derived from the convolution bundle under declared estimators — the meter vector, TE estimates, hardness indices, GP posterior quantities — rather than the full distribution $\mathbf{P}_{\theta}$ itself, which is not directly estimable from finite data as an element of the infinite-dimensional measure space $\mathcal{P}(\mathcal{C})$. In some embodiments, $\mathcal{P}(\mathcal{C})$ is equipped with the weak topology induced by bounded continuous test functionals of

the trajectory (for example, finite-lag moment functionals of the convolution bundle), and identifiability of θ from the induced distribution is understood as a meter-conditioned empirical target: under protocol family $\mathcal{U}$ and meter envelope $\mathcal{M}_{\text{accept}}$, two configurations $\theta \neq \theta'$ are distinguishable with probability at least $1 - \delta$ given $n(\delta, \varepsilon)$ trajectories, where $n(\delta, \varepsilon)$ is
 5 estimated empirically for the declared estimator family. The ε -nice kernel characterisation (Section 6) is the approximation regime within which this holds.

Data processing inequality (post-processing cannot add information). If a downstream algorithm (including compression, feature extraction, hashing, learned encoders, or any deterministic or stochastic post-processing) produces $Z = g(C_{0:T}, U_{0:T})$, then a standard
 10 consequence of the data processing inequality is

$$I_{\theta}(\Xi; Z \mid U_{0:T}) \leq I_{\theta}(\Xi; C_{0:T} \mid U_{0:T}).$$

This formalises the design principle that any verification, reconstruction, or attestation pipeline is ultimately limited by the information carried in the recorded convolution bundle under the chosen physical regime and protocol.

Memory, channel dependence, and the DPI (non-limiting). The standard data
 15 processing inequality above assumes a Markov chain $\Xi \rightarrow C_{0:T} \rightarrow Z$. When the convolution bundle exhibits temporal dependence (for example due to reactor persistence within the declared memory depth τ_{mem} ; see the Markov kernel remark in Section 6), the inequality continues to hold in the form stated, since $Z = g(C_{0:T}, U_{0:T})$ is still a deterministic function of the channel output. However, the mutual information $I_{\theta}(\Xi; C_{0:T} \mid U_{0:T})$ itself may be
 20 inflated by temporal dependencies if Ξ and the channel memory are not independent under the protocol; in preferred embodiments, the bundle length T is chosen large enough to average over within-memory-depth correlations, and the effective information $I_{\theta}(\Xi; C_{0:T} \mid U_{0:T})$ is computed with an estimator that accounts for the declared autocorrelation structure (for example a block-structured mutual-information estimator with block length τ_{mem}). The
 25 resulting bound is referred to as the *memory-adjusted DPI bound* and is the operative form used in verification and reconstruction performance statements.

Multi-query DPI composition for adaptive adversaries (non-limiting). The DPI bound stated above applies within a single fixed-horizon run (non-adaptive setting), where $\hat{\Xi}$ is computed from a single bundle $C_{0:T}$. For an adaptive adversary who submits Q queries
 30 sequentially, using each response to choose the next challenge, the aggregate information gain over Q runs is bounded by a composition of the per-run DPI bounds. In some embodiments, this composition is stated as follows: under the declared query throttle rate and selective disclosure policy, and under the assumption that the run-to-run mutual information $I(C_{0:T}^{(j)}; C_{0:T}^{(j+1)} \mid U_{0:T}^{(j)}, U_{0:T}^{(j+1)}) \leq \rho_{\text{run}}$ is bounded by a declared inter-run correlation ρ_{run}
 35 (estimable from the AR(1) drift model and the protocol randomisation structure), the aggregate

gate information gain satisfies $I(\Xi; \{C_{0:T}^{(j)}\}_{j=1}^Q \mid \{U_{0:T}^{(j)}\}) \leq Q \cdot I_\theta(\Xi; C_{0:T} \mid U_{0:T}) + \Delta_{Q,\text{run}}(Q)$, where $\Delta_{Q,\text{run}}(Q)$ is a nonnegative validation-envelope term separately upper-bounded under the declared validation regime, with empirical estimates used to support the declared bound in the operative protocol regime. In the operative protocol regime, the validation target is that both pairwise run-to-run dependence and higher-order transcript leakage across the full collection of runs are empirically driven toward zero; the operative protocol declares and verifies these conditions empirically rather than deriving $\Delta_{Q,\text{run}}$ from a closed-form expression in ρ_{run} . Pairwise run-to-run correlation alone does not in general control higher-order synergistic information across the full collection of runs. The assumption of approximate independence across runs — $\rho_{\text{run}} \approx 0$ — is a declared protocol requirement, verified by checking that the run-to-run correlation in convolution bundle statistics does not exceed ρ_{run} under the declared protocol randomisation; this requirement and its empirical verification are recorded in the protocol digest.

Transfer entropy: coupling diagnostic vs. hardness bound (non-limiting). Transfer entropy $\text{TE}_{S \rightarrow R}$ and $\text{TE}_{R \rightarrow S}$ are used in two distinct roles in this specification, and these roles should not be conflated. As a *coupling diagnostic* (Yoked operation; Section 8.5.7), TE measures the degree to which the scene and reactor exchange information in both directions; the relevant property is balance and magnitude, and the primary concern is statistical power and common-cause confounding. As a *component of Yoked-regime security* (Yoked Truth Beam), high TE indicates that the scene’s response is informationally entangled with the reactor’s history, making replay or surrogate attacks harder; the relevant security parameter is the operative mutual information $I_\theta(S_{0:T}; R_{0:T})$ between scene and reactor under the disclosed coupling and input distribution; the memory-adjusted DPI bound provides an upper bound on the information that any downstream estimator can recover about $S_{0:T}$ from $R_{0:T}$. No converse-style channel-capacity claim (supremum over input distributions) is asserted in this disclosure. In some embodiments, the hardness contributions from the Yoked coupling are quantified separately from the hardness contributions from physical microstructure (Section 12), and the two estimates are combined with declared weights in the hardness index; this separation ensures that a degradation in coupling strength (for example from a scene change) does not silently collapse the hardness claim without triggering a meter alarm.

Because TE is used for both the coupling diagnostic and the security argument, two distinct estimator families with separately declared calibration parameters are required: one optimised for the *difference* $\text{TE}_{S \rightarrow R} - \text{TE}_{R \rightarrow S}$ (bias-cancellation property of shared embedding is the relevant criterion), and one optimised for the *absolute* entropy rate and coupling capacity (bias in the absolute value is the relevant criterion). Both estimator families and their protocol parameters (embedding dimension, lag, window length, k -NN count for KSG estimators) are declared separately in the protocol digest; using a single estimator family for both purposes is a non-conforming embodiment.

Yoked convergence criterion: necessary vs. sufficient (non-limiting). The Yoked convergence criterion (CLE, TE balance, and estimator SNR gate co-conditions) is a necessary condition for Yoked operation but is not alone sufficient for the security properties claimed for Yoked Truth Beam. The TE criterion can in principle be satisfied by a signal
 5 whose TE value matches the genuine bundle but whose higher-order statistical structure differs, in a manner analogous to a hash pre-image attack. In some embodiments, a higher-order structure check (for example, the honeypot verification embodiment described in Section 8.2.5) is integrated as a mandatory co-condition of the Yoked convergence criterion: the joint distribution of consecutive bundle segments must exceed a declared divergence thresh-
 10 old from the null model of uncoupled processes, computed using a declared higher-order estimator (e.g., an energy statistic or a neural mutual-information estimate) committed to the protocol digest. In embodiments where the honeypot check is not integrated, the security argument for Yoked Truth Beam explicitly depends on the higher-order structure checks described in the honeypot embodiment as a parallel monitoring process.

15 21.2 Mutual information and capacity (design-time bounds)

For a fixed horizon T and RK parameterisation θ , a non-limiting notion of design-time capacity is

$$C_T(\theta) = \sup_{\mathbf{P}_{\Xi}} I_{\theta}(\Xi; C_{0:T} \mid U_{0:T}),$$

where the supremum ranges over admissible stimulus or secret distributions $\mathbf{P}_{\Xi}$ (for example, bounded-energy stimuli, bounded alphabet, or application-specific label priors). In some
 20 embodiments, the admissible set also constrains the control protocol distribution (for example, $\mathbf{P}_{U_{0:T}}$ restricted by scan speed, eye-safety, thermal limits, or meter stability), yielding a constrained capacity

$$C_T(\theta; \mathcal{U}) = \sup_{\mathbf{P}_{\Xi}, \mathbf{P}_{U_{0:T}} \in \mathcal{U}} I_{\theta}(\Xi; C_{0:T} \mid U_{0:T}).$$

Cross-regime constraint conjecture: hardness and sensing capacity as complementary quantities (non-limiting, research direction). The single-kernel formulation

25 $\mathbf{P}_{\theta}$ unifies Truth Beam (authentication), Limager (sensing), and Reality Transform (rendering) over the same convolution bundle. A deep implication of this unification, not yet proved, is that the digital hardness index $k_{\text{dig}}^*(\theta; \varepsilon)$ in Truth Beam mode and the sensing capacity $C_T(\theta; \mathcal{U})$ in Limager mode are not independent quantities: both are bounded above by the same physical resource, namely the optical information capacity of the channel, which scales
 30 as $N_{\text{modes}} \cdot \log(1 + \text{SNR})$. The following is stated as a research conjecture.

Cross-Regime Constraint Conjecture. There exists a decreasing function $h(\cdot)$ such that, for a device with digital hardness index $k_{\text{dig}}^*(\theta; \varepsilon)$ and sensing capacity $C_T(\theta; \mathcal{U})$ operating under the same physical channel $\mathbf{P}_{\theta}$,

$$C_T(\theta; \mathcal{U}) \leq h(k_{\text{dig}}^*(\theta; \varepsilon)),$$

where h is a decreasing function of the hardness index.

Informally: a device that is highly secure in Truth Beam mode (high k_{dig}^* , hard to predict from any model) is a constrained sensing device (bounded C_T , because the same physical complexity that prevents modelling also limits extractable scene information). The argument sketch: (i) k_{dig}^* grows with the channel’s resistance to statistical characterisation, which is related to $I_\theta(\theta; C_{0:T})$ via the provisional entropy-rate bound; (ii) $C_T = \sup I_\theta(\Xi; C_{0:T})$ is also bounded by $N_{\text{modes}} \cdot \log(1 + \text{SNR})$ from optical information theory; (iii) if both are bounded by the same physical quantity, an implicit trade-off exists. The precise form of the trade-off depends on the relationship between k_{dig}^* and $I_\theta(\theta; C_{0:T})$, which is established only provisionally (Section 21). In some embodiments, this trade-off is treated as an empirical hypothesis: devices are characterised on both axes (k_{dig}^*, C_T) across a range of protocol configurations, and the resulting cloud of operating points is examined for the predicted negative correlation. A confirmed negative correlation, stable across device families and protocol variations, would constitute empirical evidence for the conjecture and would establish a new design principle: *hardness and sensing capacity are complementary resources of the same physical channel, subject to a conservation-like constraint.*

Limager objective (information maximisation, non-limiting). In Limager-style operation, the task variable may be a latent scene description ξ (geometry, materials, spectral parameters, pose, or a learned representation), where ξ is distinguished from the extended state X_t of Section 6 (which includes scene, reactor, media, and controller state; ξ is typically a component or projection of X_t). A non-limiting objective is to choose protocol parameters to maximise information gained about ξ :

$$\max_{\mathbf{P}_{U_{0:T}} \in \mathcal{U}} I_\theta(\xi; C_{0:T} \mid U_{0:T}),$$

subject to cost, safety, and stability constraints captured by $\mathcal{U}$ and by meter thresholds. Other objectives (for example, reconstruction accuracy, classification accuracy, or downstream task performance) can be used in addition to or instead of information-theoretic criteria.

Variational mutual-information estimation and information bottlenecks (non-limiting). In some embodiments, mutual-information quantities such as $I_\theta(\Xi; C_{0:T} \mid U_{0:T})$ or leakage terms $I(K; C_{0:T} \mid U_{0:T})$ (for a secret K) are estimated or bounded using variational objectives. A learned critic may be trained with contrastive losses (InfoNCE-style as a non-limiting example) or mutual-information neural estimation (MINE-style as a non-limiting example) on pairs of stimuli and convolution-bundle traces, yielding lower bounds on retained information and upper bounds on unwanted leakage under declared protocol and meter regimes. These bounds may be used at design time to characterise a device or at run time to adapt control policies. In some embodiments, an explicit information-bottleneck objective is imposed on intermediate latents: a digital encoder is trained to minimise $I(Z; C_{0:T})$ subject

to preserving a task or verification objective, thereby trading off utility against invariance and hardness.

Digital hardness index for inference from the convolution bundle (non-limiting).

To make empirical hardness statements operational, consider a family of attacker models (or hypothesis classes) $\{\mathcal{H}_k\}_{k \geq 1}$ indexed by a resource scale k (for example, parameter count, available FLOPs, wall-clock time, query budget, or training data budget). Given a predictor $f \in \mathcal{H}_k$ that observes $(C_{0:T}, U_{0:T})$ and outputs $\hat{\Xi} = f(C_{0:T}, U_{0:T})$, define a balanced classification error for binary $\Xi \in \{0, 1\}$ as

$$\text{err}_{\text{bal}}(f; \theta) = \frac{1}{2} \Pr_{\theta}[f(C_{0:T}, U_{0:T}) = 0 \mid \Xi = 1] + \frac{1}{2} \Pr_{\theta}[f(C_{0:T}, U_{0:T}) = 1 \mid \Xi = 0].$$

For multiclass or structured Ξ , this can generalise to an average per-class error or an application-specific loss (for example, label accuracy, key-bit prediction error, or another task-specific misclassification loss).

Define the best achievable error at scale k as

$$\text{err}_k^*(\theta) = \inf_{f \in \mathcal{H}_k} \text{err}_{\text{bal}}(f; \theta),$$

and define the digital hardness index as the minimal scale required to drive error below a target threshold ε :

$$k_{\text{dig}}^*(\theta; \varepsilon) = \min \{k : \text{err}_k^*(\theta) \leq \varepsilon\}.$$

In some embodiments, $k_{\text{dig}}^*(\theta; \varepsilon)$ is reported alongside meter summaries (noise, drift, alignment, and stability) so that hardness claims are explicitly conditioned on an observed operating envelope and on the chosen attacker model family. The meter partition (Section 12) restricts hardness computation to evaluation meters $\mathbf{m}^{\text{eval}}$; acceptance meters are excluded from the attacker’s discriminative toolkit and from the fidelity criterion against which hardness is assessed.

Analogue hardness and cross-reactor emulation (non-limiting). In some embodiments, an attacker attempts to emulate the physical channel using other physical systems, for example by using a different reactor, a coupled reactor network, or a hybrid physical and digital surrogate that maps protocol inputs to output records. This motivates an analogue hardness notion that complements digital hardness.

Let R_{θ} denote a target reactor (or device) operating under a protocol distribution and meter-bounded envelope, and let $\tilde{R}$ denote an emulator constructed from physical resources (for example another reactor $R_{\theta'}$, transducers, calibration procedures, and optional learned pre- and post-processing). Let $\mathcal{T}$ denote a transcript for a run, for example

$$\mathcal{T} = (\text{rid}, \Pi, \mathbf{m}, \text{com}, \text{opened atoms and proofs as applicable}),$$

where Π is a protocol digest, $\mathbf{m}$ are meter summaries, and com denotes commitment references.

A non-limiting analogue distinguishability advantage is parameterised over two separate resource scales: the emulator/attacker construction scale m , governing the physical resources available to the emulator $\tilde{R}$, and the verifier/distinguisher scale r , governing the resources available to the distinguishing or verification procedure $\mathcal{A}$:

$$\text{Adv}_{m,r}^{\text{ana}}(\theta) = \inf_{\tilde{R} \in \mathcal{E}_m} \sup_{\mathcal{A} \in \mathcal{D}_r} \left| \Pr[\mathcal{A}(\mathcal{T}) = 1 \mid \mathcal{T} \sim R_\theta] - \Pr[\mathcal{A}(\mathcal{T}) = 1 \mid \mathcal{T} \sim \tilde{R}_m] \right|,$$

where $\mathcal{E}_m$ is a non-limiting class of emulators constructed at scale m , and $\mathcal{D}_r$ is a non-limiting class of distinguishers or verification procedures available at scale r (for example bounded witnesses, bounded openings, bounded cross-prediction checks, and bounded compute). The infimum-supremum structure makes explicit that the emulator chooses its construction to minimise distinguishability against the worst-case verifier in the declared verifier class. The two-scale form is monotone under declared regularity: increasing m at fixed r does not increase $\text{Adv}_{m,r}^{\text{ana}}$, and increasing r at fixed m does not decrease it.

The analogue hardness index is defined for a declared verifier resource budget r as the minimal emulator scale required to reduce distinguishability below a threshold ε :

$$m_{\text{ana}}^*(\theta; \varepsilon, r) = \min\{m : \text{Adv}_{m,r}^{\text{ana}}(\theta) \leq \varepsilon\}.$$

In some embodiments, the verifier scale r is committed to the protocol digest alongside the threshold ε , and the analogue hardness index is reported as the function $m_{\text{ana}}^*(\theta; \varepsilon, r)$ rather than as a single scalar; the two-dimensional surface $(m, r) \mapsto \text{Adv}_{m,r}^{\text{ana}}(\theta)$ may also be reported for cross-device comparison and for adaptive-attack analysis.

In some embodiments, m is interpreted as a budget over physical degrees of freedom (for example number of coupled subsystems, stability and calibration effort, bandwidth of transducers, achievable timing alignment, or access to reactor–reactor coupling), and $\tilde{R}_m$ includes emulation via another reactor $R_{\theta'}$ together with protocol-dependent pre-distortion and post-processing. In some embodiments, analogue hardness is strengthened by protocol randomisation, serial composition of challenges, and two-seed selective opening that limits anticipatory tailoring. In some embodiments, network witnesses and reactor–reactor coupling edges are used to increase the difficulty of emulation, since the emulator would need to reproduce not only local records but also cross-device compatibility under shared events and coupling.

In some embodiments, hardness scales with serial composition: multiple modules are physically concatenated, each contributing independent microstructure and noise, increasing analogue hardness without shared secrets or joint training, as the joint channel statistics become increasingly harder to emulate, often faster than linearly in practice.

Security oracle game and meter-extraction audit (non-limiting). To make the digital hardness index a falsifiable and attack-stable security parameter, the attacker access model must be stated explicitly. In some embodiments, the following oracle model is committed to the protocol digest as part of the hardness evaluation: the adversary is given

- (i) white-box access to the predictive model $\mathcal{Q}$ and the declared device representation $\mathbf{d}_i$;
- (ii) black-box query access to the physical device, where each query submits a challenge X and receives the response $Y_i(X)$ (no acceptance or rejection decision is returned by the device oracle in the default model); and (iii) a query budget Q declared in the protocol digest. The adversary wins if they produce a bundle accepted by the verification procedure that was not generated by the genuine device. In some embodiments, a stricter oracle variant is also evaluated: the adversary additionally receives acceptance/rejection feedback from the acceptance meter after each submitted bundle (the “acceptance-oracle” variant). Because acceptance meters and evaluation meters are statistically correlated (Section 12), the acceptance-oracle variant provides indirect information about the evaluation meter boundary; the residual attack advantage from this indirect channel is reported as part of the hardness evaluation for the declared acceptance-evaluation meter correlation $\rho_{\text{acc-eval}}$.

In some embodiments, a meter-extraction red-team evaluation is conducted as part of Eve-mode benchmarking: an attacker family is given Q adaptive acceptance-oracle queries and attempts to learn the acceptance meter boundary well enough to construct bundles that pass verification with probability exceeding a declared threshold. Success probability as a function of Q is reported in the protocol digest alongside k_{dig}^* ; the query throttle rate is set so that Q remains below the Q^* at which the success probability crosses a policy-declared tolerance.

Full hardness curves in protocol digest (non-limiting). The scalar index $k_{\text{dig}}^*(\theta; \varepsilon)$ and the scalar analogue index $m_{\text{ana}}^*(\theta; \varepsilon, r)$ are summary statistics of the full hardness curves $\text{err}_k^*(\theta)$ (error vs. model capacity) and $\text{Adv}_{m,r}^{\text{ana}}(\theta)$ (distinguishing advantage vs. physical scale). Two devices can have identical scalar indices while having very different hardness profiles (concentrated vs. diffuse over challenge space, steep vs. gradual capacity curves), so the scalar thresholds alone are not a sufficient characterisation for all applications. In preferred embodiments, the protocol digest includes the full hardness curves (as tabulated or parametric fits) alongside the scalar thresholds, together with confidence bands estimated from repeated evaluation windows. The full curves are required for calibrating the query throttle threshold and for comparing devices across model-update cycles.

Worked example hardness instantiation (non-limiting). In some embodiments, the digital hardness index $k_{\text{dig}}^*(\theta; \varepsilon)$ is instantiated with a concrete attacker family to make the framework operational without overclaiming security. A non-limiting attacker family $\{\mathcal{H}_k\}_{k \geq 1}$ may be defined as neural predictors drawn from a template family (for example U-Net-like conditional predictors for inference from the convolution bundle, or GAN-style or diffusion-style predictors for synthesis), with the scale k measuring one or more of: parameter

count, FLOPs, wall-clock budget, training data budget, or query budget to the device.

In some embodiments, attacker evaluation and reporting are conditioned on a protocol digest and a meter envelope, rather than asserted unconditionally. For example, an implementation may report that under protocol class Π and within a meter envelope $\mathbf{m}$ observed
 5 for the run (for example drift indicators, timing alignment indicators, and stability flags), the best achievable error $\text{err}_k^*(\theta)$ over $f \in \mathcal{H}_k$ remains above a policy threshold unless k exceeds a reported scale. In some embodiments, the same approach is used for semantic Truth Beam, where the prediction target is a claim q or a label rather than raw sensor traces, and audits open claim-relevant atoms and emission digests to rule out post hoc fabrication.

10 In some embodiments, hardness reporting is paired with an analogue hardness notion $m_{\text{ana}}^*(\theta; \varepsilon, r)$, reflecting the physical scale required for cross-reactor emulation under the same protocol and meter envelope. Together, these provide complementary bounds: computational forgery difficulty and physical emulation difficulty.

21.3 Rate-distortion viewpoint for Limager

15 In some embodiments, Limager operation is viewed as reconstruction of scene variables from a physically constrained encoding provided by the Reality Kernel channel. A rate-distortion viewpoint expresses trade-offs between reconstruction error and information retained by the channel. In practice, scan protocols and kernel settings may be chosen to move to an operating point that improves task-relevant reconstruction or classification subject to cost
 20 and stability constraints. In some embodiments, estimation-focused quantities such as Fisher information and Cramér–Rao-style bounds are used as non-limiting tools to characterise how accurately particular scene parameters can be estimated from the convolution bundle under a given protocol and noise model.

Rate-distortion function (non-limiting). In some embodiments, the classical rate-
 25 distortion function for a source Ξ with distortion measure $d(\Xi, \hat{\Xi})$ is

$$R(D) = \inf_{\substack{p(\hat{\Xi}|\Xi) \\ \mathbb{E}[d(\Xi, \hat{\Xi})] \leq D}} I(\Xi; \hat{\Xi}). \quad (1)$$

Physical distortion bound (non-limiting). In the RK setting, the physical channel and any decoder together induce an effective conditional law $p(\hat{\Xi} | \Xi)$, and achievable distortion is bounded below by a distortion–rate function:

$$\mathbb{E}[d(\Xi, \hat{\Xi})] \geq D^*(I_\theta(\Xi; C_{0:T} | U_{0:T})), \quad (2)$$

where $D^*(R)$ denotes the minimal achievable distortion at information rate R (a non-limiting
 30 distortion–rate function). Since $\Xi \rightarrow C_{0:T} \rightarrow \hat{\Xi}$ forms a Markov chain, data processing implies $I(\Xi; \hat{\Xi}) \leq I_\theta(\Xi; C_{0:T} | U_{0:T})$, and the bound above may be read as a physically grounded floor on reconstruction performance under the chosen protocol and meter-bounded regime.

21.4 Wiretap and advantage gap viewpoint for Truth Beam

In some embodiments, Truth Beam verification is viewed through a wiretap-style lens in which a verifier (Bob) has access to a better-conditioned view of the physical channel than an adversary (Eve). This subsection provides a non-limiting interpretation lens; it does not
 5 assert unconditional information-theoretic security.

Let Ξ denote a stimulus or secret, let Y denote a verifier-visible observation (for example the convolution bundle features and meter summaries under a logged protocol), and let Z denote an adversary-visible observation (for example a degraded view due to limited access to the reactor, reduced witness diversity, or restricted openings). A non-limiting design goal
 10 is to increase an advantage gap so that the verifier can reliably discriminate authentic records while an adversary is not expected to efficiently forge records that pass verification under the same protocol digest and meter envelope.

Secrecy-capacity lens (non-limiting). In some embodiments, a secrecy-capacity style expression is used as an interpretation lens:

$$C_s = [I(\Xi; Y) - I(\Xi; Z)]^+,$$

15 where $[x]^+ = \max(x, 0)$. In this interpretation, increasing $I(\Xi; Y)$ corresponds to improving verifier information (for example via additional witnesses, better meter-bounded stability, or richer multi-channel protocols), while reducing $I(\Xi; Z)$ corresponds to limiting adversary information (for example via two-seed opening, bounded disclosures, protocol randomisation, and rate-limited physical outputs).

20 In some embodiments, this lens guides protocol design and policy selection by favouring regimes that widen the verifier–adversary gap, while remaining consistent with meter objectives and audit procedures described elsewhere in the specification.

Wiretap-channel coding embodiments (non-limiting). In some embodiments, the Truth Beam advantage gap is instantiated using wiretap-channel coding constructions of the
 25 kind associated with Wyner (1975) and Csiszár–Körner (1978), with the verifier-accessible observation and the bounded-adversary observation modelled respectively as the main and wiretap channels. In such embodiments, the secrecy-margin objective used by the meter and training games is a secrecy-capacity-style design target under declared protocol randomisation, disclosure policy, witness geometry, and query-throttle constraints; it is recorded
 30 as an operational meter-conditioned margin rather than as an assertion of unconditional information-theoretic security.

Reciprocity breaking as a physical origin of Eve degradation (non-limiting). In some embodiments, the physical channel deliberately introduces nonreciprocal behaviour so that an adversary is denied an equivalent symmetric view of the verifier-visible interaction
 35 in embodiments using the declared nonreciprocal configuration. Non-limiting mechanisms

include optical isolators, Faraday rotators, moving or time-varying media (including epsilon-near-zero thin films under ultrafast temporal modulation as described in Section 10), diffusers with effectively irreversible scattering, and decoherence-inducing components. In these embodiments, reciprocity breaking provides a physical origin of Eve-channel degradation in the wiretap lens, conditioned on meter-bounded stability and logged protocol regimes.

21.5 Entropy rate of the convolution bundle

In some embodiments, when a closed-loop process is stationary under a fixed policy, the convolution bundle admits an entropy-rate interpretation. Higher entropy rate, subject to calibration constraints, may correlate with higher empirical hardness because a simulator would need to reproduce richer stochastic structure to avoid detection. This is not a cryptographic guarantee, but it may be used as a calibration statistic.

Entropy rate as calibration statistic (non-limiting). In some embodiments, the entropy rate h of the convolution bundle $C_{0:T}$ under a declared stationary and ergodic process model and policy π is reported as a calibration statistic alongside the empirical hardness index k_{dig}^* and the meter summaries. The entropy rate is used as a regime and drift indicator, not as a security warrant: high entropy rate is consistent with — but does not entail — empirical hardness, and a forger can in principle match the entropy rate of the genuine source while differing in higher-order statistical structure. In some embodiments, sustained departures of the empirical entropy rate from the declared process model trigger meter-gated re-characterisation or fallback to a more conservative operating regime. Estimation of the entropy rate uses declared symbolisation, per-device and per-envelope stratification, and lower confidence bounds reported alongside the point estimate. No formal lower bound on k_{dig}^* as a function of h is asserted in this disclosure; entropy is a calibration quantity, not a discriminator capacity bound.

Conditional min-entropy, extraction, and pseudorandom generation (non-limiting). In some embodiments, the security-relevant entropy quantity is the smooth conditional min-entropy of the convolution bundle on the security domain, conditioned on all adversary side information E : $H_{\infty}^{\varepsilon}(C_{0:T} \mid E)$. Where the adversary may hold quantum side information (for example an entangled probe whose measurement is deferred), E is treated as a quantum register and H_{∞}^{ε} is the smooth conditional min-entropy against quantum side information in the sense of Tomamichel, Schaffner, Smith, and Renner (2010). Min-entropy supports operationally correct guessing and extraction statements: an adversary guessing $C_{0:T}$ from E in a single attempt succeeds with probability at most $2^{-H_{\infty}(C_{0:T}|E)}$, and a verifier’s acceptance set of size M admits genuine-source mass at most approximately $M \cdot 2^{-H_{\infty}(C_{0:T}|E)}$ under the relevant side-information conditioning. Min-entropy does *not* by itself lower-bound the state size or capacity of a generic discriminator, and no such discriminator-state lower bound is asserted in this disclosure.

In some embodiments, the connection between physical-process min-entropy and derived cryptographic objects is structured in two information-theoretically separated steps. First, by the leftover hash lemma (Impagliazzo, Levin, and Luby 1989; conditional and smooth-min-entropy formulations following Bennett, Brassard, Crépeau, and Maurer 1995 and the generalisations surveyed in Vadhan, *Pseudorandomness*, 2012), a source with smooth conditional min-entropy $H_\infty^\varepsilon(C_{0:T} \mid E) \geq m$ can be *extracted* to a string that is ε -close to uniform from E 's view, using a universal hash function whose seed is independent of the source and may be public; the extracted string has length approximately $m - 2\log(1/\varepsilon)$ bits and requires no computational hardness assumption. Where the adversary may hold quantum side information, the corresponding quantum-proof formulation (Tomamichel, Schaffner, Smith, and Renner 2010) is used, with extraction parameters governed by the same $m - 2\log(1/\varepsilon)$ relation up to the standard quantum-LHL constants. Correlations across $C_{0:T}$ are absorbed into the block min-entropy term and are not assumed to be independent; estimating that block min-entropy from data is a separate empirical problem, conducted under declared symbolisation, per-device and per-envelope stratification, and lower confidence bounds. Second, the extracted near-uniform string may serve as the seed for a computationally secure pseudorandom generator (PRG); the PRG's stretch and security rest on a computational hardness assumption separate from and additional to the extraction step, for example the existence of one-way functions in the sense of Håstad, Impagliazzo, Levin, and Luby 1999. The leftover hash lemma provides the information-theoretic foundation for the extraction; it does not by itself provide PRG stretch.

21.6 Error exponents and confidence

In some embodiments, beyond capacity-style bounds, reliability language characterises how error probabilities decay as observation length increases. In practice, longer windows and stronger protocols may yield improved confidence for verifiers and for decoders, subject to drift and nonstationarity. These concepts may be used to select protocol window lengths, validation budgets, and acceptance thresholds.

Error exponent lens (non-limiting). In some embodiments, an error exponent (reliability function) is written as

$$E(R, \theta) = \lim_{n \rightarrow \infty} -\frac{1}{n} \log P_e^{(n)}(R), \quad (3)$$

where $P_e^{(n)}(R)$ is a minimum achievable error probability at rate R over n channel uses (non-limitingly, n emission–observation bundles or windows of the convolution bundle). In some embodiments, the RK's temporal structure (multi-step protocols, adaptive control, and reactor memory) is used as a feedback-like resource that improves error decay relative to memoryless use of the channel; classical feedback results (for example Schalkwijk–Kailath for Gaussian channels) provide a non-limiting interpretation lens even when they do not

directly increase Shannon capacity.

21.7 Autopoietic interpretation (optional, non-limiting)

In some embodiments, the Reality Kernel is interpreted through the lens of autopoietic (self-producing) systems. The following mapping is non-limiting and does not assert that the Reality Kernel is alive or conscious; it identifies structural correspondences that may guide design and analysis.

The eight necessary structures of an autopoietic system correspond to Reality Kernel components as follows: a *boundary* (the device’s physical housing and optical apertures define a topological boundary separating internal reactor dynamics from external scene); *constitutive processes* (the reactor’s nonlinear dynamics continuously regenerate the attractor structure that defines the device’s identity); *components* (reactor media, emitters, detectors, and optical elements); a *network of processes* (the feedback loop coupling emitter, reactor, scene, and detector); *energy throughput* (optical power, electrical power, and thermal dissipation); *subordination of component production to the network* (reactor media properties are shaped by the operating regime, not independently); *spatial extension* (the device occupies a specific region of physical space with identifiable boundaries); and *organizational closure* (the feedback loop is closed — the device’s output affects its own future input through the scene, and the device’s internal state determines its output, completing the circular causality).

In some embodiments, organizational closure is the structure most directly relevant to the specification: it is the formal property that distinguishes the Reality Kernel’s closed-loop operation from open-loop sensing or projection, and it is the property that generates the attractor structure, Fisher curvature, and holonomy on which all regime-level claims depend.

21.8 Connections and non-limiting notes

In some embodiments, the information-theoretic viewpoint connects to other parts of the specification. Cryptographic overlays provide commitments and selective disclosure for evidence streams. Control-theoretic methods select protocols to improve information gain and robustness. Optical primitives influence channel bandwidth, invariance profiles, and noise structure. The information-theoretic viewpoint is used as a design and analysis lens and is not a claim of reduction-based security.

Connections to other primitive classes. In some embodiments, the information-theoretic view provides a shared vocabulary that connects other parts of this disclosure:

- **Cryptographic primitives:** reactor-microstructure uniqueness and unpredictability map to entropy (including min-entropy and Rényi-entropy lenses) of channel outputs. Commitment-style protocols can be interpreted by requiring binding to correspond to

non-zero information about a committed value for an honest verifier, while hiding corresponds to a degraded-view adversary obtaining negligible information under declared disclosure policies and meter envelopes.

- **Neural architecture primitives:** an information-bottleneck lens, $\min I(X; Z)$ subject to $I(Z; Y) \geq I_0$, provides a non-limiting interpretation of autoencoder-style Limager operation, where Z is a learned representation derived from the convolution bundle and Y is a downstream task variable; reservoir-style computation can be interpreted as exploiting high $I(\Xi; C_{0:T})$ with low-dimensional sufficient statistics.
- **Optical primitives:** channel capacity and rate–distortion trade-offs depend on optical degrees of freedom (spatial, spectral, polarisation, temporal, coherence regime) and on transfer operators (for example LCT/convolutional regimes, scattering matrices, and resonant/eigenmode structure).
- **Control-theoretic primitives:** adaptive control protocols of the form $u(t) = \pi(u_{0:t-1}, C_{0:t-1})$ act as feedback and can improve reliability and error decay and can increase usable information rate under memory and constraint conditions, even when Shannon-style capacity is not increased.
- **Thermodynamic primitives:** Landauer-style principles link entropy production to information erasure; in some embodiments, dissipation schedules and irreversibility provide a physical origin for one-wayness and for costs associated with high-capacity operation.

21.9 Summary table (information-theoretic substrate, non-limiting)

Quantity / lens	RK interpretation	Design use
Conditional mutual information $I_\theta(\Xi; C_{0:T} \mid U_{0:T})$	Information retained about a task variable or secret Ξ by the convolution bundle under a logged protocol and meter envelope	Choose protocols and operating regimes that increase inference power for Limager or increase verifier advantage in Truth Beam
Constrained capacity $C_T(\theta; \mathcal{U})$	Best achievable information transfer given admissible protocol families and safety/stability constraints	Compare hardware and protocol design choices; budget horizons and channel allocations across modalities
Rate–distortion $R(D)$ and distortion–rate $D^*(R)$	Reconstruction trade-off between information rate and achievable distortion under the physical channel and decoder	Select protocol length, bandwidth, and reconstruction targets; interpret “no decoder can beat the channel” limits

Quantity / lens		RK interpretation	Design use
Secrecy-capacity $[I(\Xi; Y) - I(\Xi; Z)]^+$	lens	Verifier–adversary advantage gap, with Bob observing Y and Eve observing a degraded view Z	Guide Truth Beam design toward regimes that widen Bob–Eve gaps using access asymmetries, bounded disclosures, and reciprocity breaking
Entropy rate of $C_{0:T}$		Stochastic richness of logged traces under a stationary or quasi-stationary policy	Calibration proxy for empirical hardness and for detecting drift or regime changes
Error exponent $E(R, \theta)$		How quickly error probabilities decay with observation length under a given rate and regime	Select window lengths, repetition budgets, and acceptance thresholds for desired confidence under drift constraints

21.10 Complexity-theoretic interpretation lenses (optional, non-limiting)

Remark 6 (Oracle complexity classes (non-limiting)). *In some embodiments, the RK channel $P_\theta(\cdot \mid S, U_{0:T})$ is treated as a physical oracle (where S here denotes the scene tuple of Section 6, not the task variable Ξ of the preceding information-theoretic subsections), defining an oracle-relative complexity class such as BPP^{RK} : problems solvable in probabilistic polynomial time with oracle access to a physical Reality Kernel instance under declared protocol classes and meter envelopes. This is an interpretation lens only and does not assert a formal separation of standard complexity classes. It provides vocabulary for reasoning about tasks that are easy with physical access (querying the device) but conjecturally hard to simulate with bounded digital resources under the same disclosure policies.*

Remark 7 (Physical assumptions versus computational conjectures (non-limiting)). *In some embodiments, security and function are grounded in physical assumptions rather than in purely computational conjectures. Non-limiting examples include: one-wayness as a consequence of dissipative irreversibility; true randomness sourced from physical noise (shot noise, thermal noise, or other high-entropy observables) plus extraction; and unclonability sourced from microscopic disorder in reactor media. In some embodiments, analogue relaxation dynamics provide practical heuristic solvers for some optimisation problems without implying that formal complexity-theoretic open problems are resolved. These lenses are intended to clarify that the system’s guarantees (where any are asserted) are operational and meter-conditioned rather than reduction-based theorems.*

Remark 8 (Fundamental problems and physical sidesteps (non-limiting)). *In some embodiments, the Reality Kernel framework replaces open computational conjectures with physical assumptions, while leaving purely mathematical conjectures unchanged. The following cata-*

logue is provided as a non-limiting interpretation lens and does not assert that any formal complexity-theoretic or number-theoretic open problem is solved.

Problems sidestepped (non-limiting).

One-way functions. Computational cryptography assumes one-way functions exist but cannot prove it (a proof that one-way functions exist would imply $P \neq NP$). In some embodiments, dissipative RK channels provide physical one-wayness via thermodynamic irreversibility: information is scattered, absorbed, and thermalised, not merely hidden.

True randomness. Whether cryptographically strong pseudorandom generators exist is unknown. In some embodiments, quantum or other physical noise in RK detectors (for example shot noise, dark counts, thermal noise, or other quantum-limited observables) supplies actual entropy, with extraction and health testing providing usable random bits.

Unclonability. No computational proof establishes unclonable tokens in general. In some embodiments, microstructure-based reactor media provide physical unclonability from microscopic disorder, where the “secret” is the medium itself and its drift-bounded configuration.

Halting problem. Turing’s theorem shows no algorithm decides whether an arbitrary program halts. In some embodiments, physical RK systems terminate under normal operation: bounded energy, finite detector integration time, and dissipation are designed so that each protocol run completes within declared operating constraints, so the halting question does not arise operationally for device queries.

Kolmogorov complexity. The shortest program producing a string is uncomputable. In some embodiments, the RK’s dissipative channel provides physical compression: the map $\Xi \mapsto C_{0,T}$ destroys information measurably under logged protocols, yielding finite “physical description length” even though algorithmic Kolmogorov complexity remains uncomputable.

Gödel incompleteness. Sufficiently powerful formal systems contain true but unprovable statements. In some embodiments, physical systems are evaluated by observable evidence streams rather than self-referential proofs; incompleteness does not constrain what the device can measure or verify about its channel behaviour under declared protocols.

Rice’s theorem. Non-trivial semantic properties of programs are undecidable. In some embodiments, RK observables are physical (detector amplitudes, photon counts, entropy rates, meter summaries), not semantic properties of arbitrary programs.

NP-hard optimisation. *Finding global optima of combinatorial problems is believed hard. In some embodiments, analogue physical systems (for example energy-minimising or relaxation dynamics in reactors) provide native heuristic solvers by physical relaxation without resolving P versus NP.*

- 5 **Busy Beaver.** *BB(n) is uncomputable and grows faster than any computable function. In some embodiments, physical systems have bounded state spaces, bounded energy, and finite time horizons; the unbounded growth regimes assumed by Busy Beaver constructions are not expected to manifest under the declared bounded state, energy, and time constraints of device operation.*
- 10 **Problems not sidestepped (non-limiting).** *Number-theoretic conjectures such as the Riemann Hypothesis remain mathematical statements about integers. In some embodiments, RK modules may exhibit random-matrix-like spectra (for example GOE/GUE-like statistics in strongly scattering regimes), but this does not resolve classical conjectures about $\zeta(s)$ or prime distribution.*
- 15 **The pattern (non-limiting).** *In some embodiments, a physical assumption (thermodynamics, quantum noise, bounded resources, microscopic disorder, drift envelopes) replaces a computational conjecture. The RK does not prove $P \neq NP$; it grounds operational security and function in physics and in meter-conditioned empirical hardness, providing physical analogues of hard problems and oracle-relative separations rather than solutions to open prob-*
- 20 *lems.*

21.11 Transfer entropy and bidirectional information flow (non-limiting)

In some embodiments, transfer entropy is used as a first-class information-theoretic primitive for quantifying directed information flow in closed-loop Reality Kernel operation, rather than
 25 as a post-hoc diagnostic. Let S_t^{scene} denote the scene state at time t (the scene component of the extended state X_t of Section 6; note that S without superscript is reserved for the task-relevant random variable per the Notation remark above). The transfer entropy from scene to reactor R is

$$\text{TE}_{S^{\text{scene}} \rightarrow R}(t) = H(R_{t+1} \mid R_{t-k:t}) - H(R_{t+1} \mid R_{t-k:t}, S_{t-k:t}^{\text{scene}}),$$

and symmetrically for $\text{TE}_{R \rightarrow S^{\text{scene}}}$.

- 30 In Yoked operation (§8.5), the balance $\text{TE}_{S^{\text{scene}} \rightarrow R} / \text{TE}_{R \rightarrow S^{\text{scene}}}$ serves as a primary coupling meter. Balanced transfer entropy (ratio near unity) indicates bidirectional dynamical coupling. In some embodiments, the ratio is logged continuously and gating decisions for Yoked claims are conditioned on it.

Fisher Transfer Entropy (non-limiting). In some embodiments, the Fisher information of the transfer entropy with respect to coupling parameters is tracked. In some coupled dynamical systems, this quantity (here termed Fisher Transfer Entropy for convenience) increases sharply near entrainment transitions (phase-locking onsets and bifurcations in the coupled system), providing a candidate detector for regime boundaries in Yoked operation. In some embodiments, spikes in Fisher Transfer Entropy trigger recalibration, regime-transition logging, or escalation in the protocol digest.

22 Control-Theoretic Primitives

This section describes control-theoretic viewpoints and primitives that may be used to design, analyse, and operate Reality Kernel systems. These tools are non-limiting and do not change the core definition of a Reality Kernel. They provide methods and vocabulary for selecting protocols, ensuring stability, estimating state, and enforcing safety, including under disturbances and adversarial interference. The viewpoints in this section establish formal relationships with control theory as interpretive vocabulary; they do not constrain the core definitions or embodiments, with the exception of the drifting field and few-shot commissioning subsection, which describes an operational mechanism for device configuration and is relied upon for enablement of configuration-related embodiments.

22.1 State-space formulation

In some embodiments, a Reality Kernel is modelled as a controlled dynamical system in which an extended state evolves under controls and produces observations recorded as the convolution bundle. A non-limiting extended state includes scene state, reactor state, media state, and fast internal controller state. Controls include scan coordinates, emission settings, and detector settings. Observations include one or more detector channels. This framing supports reasoning about stability, observability, controllability, and protocol selection under chosen families of admissible controls.

22.2 Controllability and reachability

In some embodiments, controllability is used to characterise what internal states can be driven by admissible control sequences under physical constraints. Fast components (for example scan mirrors and modulators) may be highly controllable, while slow media (for example persistence layers or thermal states) may be only partially controllable over short horizons. In verification embodiments, limited controllability of slow degrees of freedom may increase empirical hardness because an adversary is not expected to quickly steer the physical state to match a target response distribution within the declared control-rate and access budget.

22.3 Observability and identifiability

In some embodiments, observability is used to characterise what aspects of the extended state are inferable from the convolution bundle given known controls. Limager regimes may select protocols to increase observability of task-relevant scene variables. Truth Beam regimes may intentionally reduce invertibility while preserving distinctive cross-channel constraints. In some embodiments, protocol design is treated as improving an observability metric over a chosen state representation, including by selecting scan patterns, spectral settings, or detector gating schedules.

Linearised observability Gramian (non-limiting). In some embodiments, a local linearisation is used as an interpretation lens. Let x_t denote a perturbation about a nominal operating point $\bar{X}$ of the extended state X_t (Section 6), or equivalently a reduced task-relevant projection of X_t within a meter-bounded regime. A non-limiting linearised form is:

$$x_{t+1} = Ax_t + Bu_t, \quad y_t = Cx_t + Du_t,$$

for matrices A, B, C, D determined by an operating point and meter-bounded regime. In these embodiments, a finite-horizon observability Gramian is

$$W_o = \sum_{t=0}^{T-1} (CA^t)^\top (CA^t).$$

In some embodiments, scan protocols and sensing schedules are selected to increase an observability metric such as $\det(W_o)$ or $\text{tr}(W_o)$, subject to admissible control sets and safety/stability meters.

22.4 Stability, dissipation, and boundedness

In some embodiments, stability refers to bounded operation under bounded inputs. Physical boundedness is supported by dissipation and by safety envelopes on permissible power, duty cycle, scan speed, and parameter ranges. In some embodiments, a Lyapunov-like monitor is used, where the monitored quantity is a physical energy, a surrogate energy, a temperature proxy, or a stability score derived from the convolution bundle. When instability risk is detected, the controller may fall back to conservative protocols, reduced power, neutral patterns, or shutdown.

Control-barrier and reachability safety gates (non-limiting). In some embodiments, actuator commands are filtered through control-barrier functions, barrier certificates, reachability analysis, or model-predictive safety filters before being applied to emitters, scanners, reactor drives, or memory-writing media. The safety filter defines a forward-invariant safe set over optical power, temperature, scan speed, mechanical position, stored energy, and meter-state variables, and projects a proposed command into the admissible set when

necessary. The barrier function, safe-set parameters, solver tolerances, override logic, and fallback action are recorded in the protocol digest, and a command that cannot be projected without violating a hard safety constraint causes abort, reduced-power continuation, or a declared degraded mode.

22.5 Filtering and state estimation

In some embodiments, filtering estimates latent state from the noisy convolution bundle. Non-limiting approaches include extended Kalman filtering, unscented Kalman filtering, particle filtering, and learned filters operating on convolution-bundle windows. In some embodiments, the reactor latent $Z_\theta(t)$ is implemented as a learned state estimate updated per emission-observation bundle and used for prediction, control, and meter evaluation.

22.6 Optimal control, MPC, and information gain

In some embodiments, protocol selection is treated as optimal control. Stage costs may include reconstruction error, verification error, energy use, drift penalties, and safety penalties. In some embodiments, model-predictive control rolls out candidate protocols using a learned emulator and executes a selected prefix, updating as new convolution-bundle data arrives. In Limager regimes, controls may also be chosen to maximise information gain about scene variables, reduce uncertainty, or increase robustness to nuisance variation.

Optimal control objective (non-limiting). In some embodiments, protocol selection is written as minimising expected cumulative cost:

$$\pi^* = \arg \min_{\pi} \mathbb{E} \left[\sum_{t=0}^{T-1} c(x_t, u_t) + c_T(x_T) \right], \quad (4)$$

where $c(\cdot)$ is a stage cost and $c_T(\cdot)$ is a terminal cost. Different choices of c recover the three operating regimes as special cases, for example:

- **Limager:** c penalises reconstruction error and control effort (and optionally rewards information gain proxies; information-gain terms enter as negative costs under the minimisation convention).
- **Truth Beam:** c penalises low verifier–adversary distinguishability subject to meter envelopes and safety constraints (equivalently, rewards are expressed as negative stage costs).
- **Reality Transform:** c penalises deviation from a target rendering or style objective while penalising meter violations and instability.

The Yoked objective J_{Yoked} is stated in maximisation form (Section 22.13); this is equivalent to the cost-minimisation formulation above under $J_{\text{task}} = -\sum c(x_t, u_t)$.

22.7 Adaptive control and drift compensation

In some embodiments, the system performs periodic calibration probes and adapts lookup tables, gain settings, and other calibration parameters. Where permitted, limited parameter subsets may be updated under rate limits and re-characterisation schedules. In some
 5 embodiments, updates to slow physical parameters are gated by approval rules or change-control procedures, and re-characterisation of meter behaviour is performed after material configuration changes.

22.8 Drifting field and few-shot commissioning

In some embodiments, the configuration space Θ is equipped with a learned vector field that
 10 guides navigation toward operating points with desired properties. The drifting field $v(\theta)$ assigns to each point $\theta \in \Theta$ a direction of steepest improvement for a declared objective drawn from acceptance meters (for example information gain, coupling strength, or a composite multi-objective; per the meter partition of Section 12, evaluation meters are held out and do not appear in the optimised objective). The field is learned from fleet experience: a population
 15 of devices characterised under diverse conditions provides training data from which the field's dependence on θ is inferred.

In some embodiments, the drifting field enables few-shot commissioning of new devices. A new device undergoes a small number of characterisation runs (for example fewer than ten) that locate its approximate position in the pre-learned bundle geometry. Once positioned,
 20 the drifting field provides a trajectory from the current position to any desired operating regime (for example from an initial low-curvature commissioning point to a high-curvature verification point near a bifurcation surface). The number of commissioning runs required scales with the dimensionality of the local fibre, not with the full complexity of Θ , because the bundle structure constrains where the device can be.

In some embodiments, the drifting field is implemented as the natural gradient of a
 25 declared objective on the Fisher–Rao metric: $v(\theta) = \mathcal{F}(\theta)^{-1} \nabla_{\theta} J$, where J is the objective and $\mathcal{F}(\theta)$ is the Fisher information matrix (or its regularised pseudoinverse). This aligns the navigation with the intrinsic geometry of the configuration space. Because the physical channel $\mathbf{P}_{\theta}$ is stochastic and θ_{hw} may include hardware parameters (lens positions, gain
 30 settings, reactor drive levels) that are not directly differentiable, $\nabla_{\theta} J$ is in practice estimated from finite samples—for example via score-function estimators, surrogate models trained on fleet data, or finite-difference probes applied to acceptance-meter readings under small parameter perturbations (see Section 9.13 for detailed SPSA and zeroth-order estimation methods). In some embodiments, the drifting field is pre-computed for a family of objectives
 35 and stored as a lookup table indexed by approximate device position and target regime. In some embodiments, the drifting field is updated online as the device acquires more data, refining the trajectory estimate.

Self-calibration stability under drift (non-limiting). A device whose sensor model P_θ is calibrated using meter outputs derived from P_θ itself faces a bootstrapping challenge: the instrument is calibrated by its own readings. The specification addresses this through four complementary mechanisms that, taken together, provide bounded tracking error under physical drift:

(i) *Inner Limager* (Section 8.3.6) uses known calibration targets or reference signals as controlled scenes, providing external grounding that breaks the self-referential loop. The reference response anchors the meter outputs to a physical standard rather than to the device’s own prior calibration.

(ii) *Fleet cross-calibration* exchanges committed reference sweeps under standardised protocols across a population of devices, so that no single device’s drift can propagate uncorrected. The fleet provides the external reference that a single device cannot provide for itself.

(iii) *Iterated self-modelling convergence (non-limiting theoretical correspondence)*: in selected embodiments, a depth curriculum is used to drive self-modelling toward a stable fixed point; convergence properties under this curriculum are discussed in the categorical unification section as a conceptual or diagnostic correspondence and are not required by the operational mechanisms otherwise disclosed.

(iv) *The meter partition* (Section 12) is arranged so that evaluation meters used to assess calibration quality are held out from the optimisation loop, thereby reducing the risk of the drifting field chasing metrics that drift with the device.

Under active physical drift, the fixed point itself moves, and convergence is replaced by a tracking problem: the calibration must track the moving target closely enough that meter outputs remain meaningful. In some embodiments, tracking error is bounded by requiring that the drift rate $\|\dot{\theta}\|$ (measured in Fisher–Rao geometry) remain below the contraction rate of the self-calibration loop, so that calibration converges faster than the device drifts. When drift rate exceeds this bound, meters flag an out-of-envelope condition and the device enters a re-characterisation mode (Inner Limager with reference targets) until tracking is re-established. The drift rate, contraction rate estimate, and re-characterisation transitions are logged in protocol digests.

22.9 Adversarial control and robust design

In some embodiments, Truth Beam operation is treated as a game in which the controller selects challenges and an adversary selects disturbances or forgery attempts under bounded resources. Robust control ideas may be applied by treating the adversary as a bounded disturbance and selecting protocols that maximise distinguishability under worst-case interference. Non-limiting tools include adversarial training over protocol families, conservative acceptance rules when uncertainty rises, and protocol randomisation within safety envelopes.

Stackelberg equilibrium lens (non-limiting). In some embodiments, the verification protocol is treated as a Stackelberg game in which a verifier commits to a protocol u (short-

hand for $U_{0:T}$) and an adversary best-responds with a forgery pipeline under bounded resources. Let $\mathbf{P}_\theta^{\text{Bob}}(\cdot | u)$ denote a convolution-bundle distribution induced by a genuine RK under protocol u , and let $\mathbf{P}^{\text{Eve}}(\cdot | u)$ denote a corresponding distribution induced by an adversary's forgery pipeline for the same declared protocol. In these embodiments, an equilibrium
 5 protocol is expressed as

$$u^* = \arg \max_u \min_{\text{Eve}} D_{\text{KL}}\left(\mathbf{P}_\theta^{\text{Bob}}(\cdot | u) \parallel \mathbf{P}^{\text{Eve}}(\cdot | u)\right), \quad (5)$$

choosing protocols that are robust against an adversary's best response within the assumed attacker family.

H_∞ robust control lens (non-limiting). In some embodiments, adversarial interference is modelled as a bounded-energy disturbance acting on sensors, actuators, or the scene
 10 channel. In these embodiments, H_∞ robust control provides a non-limiting interpretation lens: the verifier selects protocols that limit worst-case induced gain from disturbances to verification-relevant residuals, and meters gate operation when disturbances exceed envelope assumptions.

22.10 Safety envelopes and interlocks

15 In some embodiments, a safety envelope bounds at least one of emission power, exposure duration, scan speed, flicker, thermal load, and other hazard-relevant constraints. In some embodiments, the envelope is enforced by a separate interlock layer so that even faulty or adversarial policies are blocked from commanding emissions outside permitted bounds in embodiments using the stated interlock configuration. This is compatible with verification
 20 gating and with cryptographic overlays, but does not require them.

22.11 Non-limiting notes (control-theoretic substrate)

Control-theoretic primitives support all three regimes. Limager uses them for estimation and information gain. Reality Transform uses them for stability and constrained stylisation. Truth Beam uses them to design adversary-robust challenges and safe operation under bounded
 25 interference.

22.12 Summary table (control-theoretic substrate, non-limiting)

Control primitive	RK realisation	Regime / application
Controllability / reachability	Admissible scan laws, emission settings, and slow configuration knobs drive the internal media and observation distribution	Limager: explore measurement space; Truth Beam: limited slow controllability can strengthen hardness

Control primitive	RK realisation	Regime / application
Observability / identifiability	State components inferable from the convolution bundle under known controls; linearised Gramian W_o as an interpretation lens	Limager: choose protocols to increase observability; Truth Beam: preserve distinctive constraints while limiting invertibility
Stability / dissipation	Physical boundedness plus meters and interlocks; Lyapunov-like monitors from energy or stability proxies	All regimes: safe operation, regime downgrades, and consistent calibration under drift
Filtering / state estimation	EKF/UKF/particle filters or learned filters updating a latent $Z_\theta(t)$ from bundles c_t	Limager: reconstruct scene variables; Truth Beam: widen Bob-Eve estimation gap via private ports and meters
Optimal control / MPC	Protocol optimisation under costs combining task objectives and meter penalties; receding-horizon selection with emulators	Limager: information gain and reconstruction; Reality Transform: constrained stylisation; Truth Beam: adversary-robust challenges
Robust / adversarial design	Worst-case or game-theoretic selection of protocols; minimax lenses such as Stackelberg KL gaps; H_∞ approximations	Truth Beam: protocol families resistant to bounded interference and forgery strategies
Safety envelopes and interlocks	Independent enforcement of power, exposure, scan speed, thermal, flicker and other limits	All regimes: blocks or screens unsafe commands and constrains the admissible protocol class $\mathcal{U}$ in embodiments using the interlock layer

22.13 Yoked operation: coupling quality as control objective (non-limiting)

In Yoked operation, the control objective includes a coupling-quality term that drives the system toward bidirectional dynamical coupling between device and scene:

$$\begin{aligned}
 J_{\text{Yoked}}(U_{0:T}) = & J_{\text{task}}(U_{0:T}) \\
 & + \lambda_{\text{TE}} \text{Balance}(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S}) \\
 & + \lambda_{\text{mag}} \min(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S}) \\
 & - \lambda_{\text{CLE}} |\text{CLE} - \text{CLE}_{\text{target}}|.
 \end{aligned}$$

where $\text{Balance}(\cdot)$ penalises asymmetric information flow, $\min(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S})$ rewards large mutual transfer entropy in both directions (the “both large” Yoked signature), and the CLE term penalises deviation from a declared target $\text{CLE}_{\text{target}} < 0$ chosen close to zero so as to maintain operation at the edge of synchronisation. The weights $\lambda_{\text{TE}}, \lambda_{\text{mag}}, \lambda_{\text{CLE}} \geq 0$ are non-limiting hyperparameters that trade off symmetry, magnitude, and convergence depth; in some embodiments they are scheduled or adapted during a coupling curriculum. In preferred embodiments, all TE quantities are reported in a declared information unit (bits or nats) under the declared estimator family and windowing, CLE quantities are reported in the declared time base (per-step or per-second), and J_{task} is normalised to the same declared scale; the λ weights include the necessary unit conversions so that all four terms are commensurate. The normalisation choices are declared in the protocol digest. In preferred embodiments, the objective J_{Yoked} is maximised.

CLE convergence criterion (non-limiting). In some embodiments, Yoked operation is declared converged when the conditional Lyapunov exponent CLE approaches zero from below, per Hart’s criterion. The sign and magnitude of CLE are logged as meter outputs and used to gate Yoked claims.

Non-limiting Balance functional example. In some embodiments, a concrete instantiation of $\text{Balance}(\text{TE}_{S \rightarrow R}, \text{TE}_{R \rightarrow S})$ is the negative absolute log-ratio: $\text{Balance}(a, b) = -|\log((a + \epsilon)/(b + \epsilon))|$, where $\epsilon > 0$ is a regularisation constant, and the objective is maximised (penalising asymmetric information flow). Transfer entropy is estimated using one of the non-limiting estimator families described in Section 8 (for example Kraskov–Stögbauer–Grassberger).

Arnold tongue analysis (non-limiting). In some embodiments, controllability in Yoked operation is characterised by mapping entrainment regions (Arnold tongues) in the space of coupling strength versus detuning (frequency ratio between scene dynamics and reactor dynamics). Protocol design targets operating points within entrainment tongues where coupling is robust, and meters flag excursions toward tongue boundaries where coupling becomes fragile. In some embodiments, tongue boundaries are estimated empirically from sweep data and used to define the admissible Yoked operating envelope.

Active-inference interpretation (non-limiting). In some embodiments, the Reality Kernel loop may be interpreted through the lens of active inference: the convolution bundle $C_{0:T}$ supplies sensory evidence, the control protocol $U_{0:T}$ realises epistemic actions (active sampling to reduce uncertainty) and pragmatic actions (interventions to achieve goals), and the reactor dynamics implement a physically grounded generative model whose predictions are compared against observations via meter scores. Under this interpretation, Limager operation corresponds to epistemic foraging (actions chosen to maximise expected information

gain), Truth Beam corresponds to model-evidence comparison (evaluating whether observations are consistent with the enrolled generative model), and Reality Transform corresponds to pragmatic action (driving the scene toward a target state). Yoked operation corresponds to coupled active inference between two generative models (device and scene) whose joint
 5 free-energy minimum corresponds to the targeted coupled attractor. This interpretation is offered as a non-limiting explanatory lens that may aid design intuition and connect the Reality Kernel formalism to an active research literature; it does not alter the definitions, claims, or formal results elsewhere in this specification, and the Reality Kernel formalism is not derived from or dependent on the free-energy principle. A Markov blanket, in this
 10 context, is the minimal set of variables (active controls and sensory observations) that separates an agent’s internal state from the external world in a probabilistic graphical model. The Markov blanket interpretation (described in the related Filing 2 application; optional, non-essential) provides additional detail on the boundary-probe implications of this lens.

23 Cryptographic-Primitive Substrate

15 This section describes non-limiting cryptographic-primitive constructions that may be implemented using Reality Kernel systems. These constructions derive practical value from a combination of (i) standard cryptographic primitives for commitments, signatures, and transport, and (ii) empirical properties of a physical Reality Kernel channel, including device-specific microstructure, noise, and history dependence. No reduction-based cryptographic guarantee
 20 is required by these embodiments. The constructions are optional overlays and do not alter the core definition of the convolution bundle or the Reality Kernel channel.

23.1 Mapping meter games to analogue cryptographic structures

In some embodiments, multi-agent meter games are mapped directly to physical configurations of mobile or fixed Reality Kernel platforms, including PolieBot configurations, to im-
 25 plement analogue versions of cryptographic structures that exploit real-world asymmetries. In such embodiments, roles analogous to prover, verifier, and adversary are instantiated by physical devices with different access, vantage points, and constraints.

Non-limiting real-world asymmetries that may be exploited include:

- **Access asymmetry:** a verifier device has direct physical access to a reactor, a scene,
 30 or a private detector port that an adversary is assumed not to have access to absent physical compromise of the verifier device.
- **Geometry and viewpoint asymmetry:** devices observe the same event from different vantage points, so a forged history would need to satisfy multi-view spatiotemporal constraints.
- **Timing and causality asymmetry:** observable causal effects and bounded propa-
 35 gation delays constrain permissible transcripts without requiring a shared clock.

- **Interference and power asymmetry:** an adversary may be bounded in co-illumination power, duty cycle, or spectral overlap relative to a verifier's sensing configuration.
- **Microstructure asymmetry:** device-specific optical or multi-physics microstructure yields microstructure-unclonable behaviour that is empirically difficult to emulate without comparable hardware.

In these embodiments, meter outputs (verisimilitude, hardness, calibration, and spatiotemporal consistency) provide the decision statistics for analogue commitment-open protocols, proof-of-projection protocols, device-bound rate limiting, and web-of-trust updates. The result is a family of physically anchored protocol patterns that behave like cryptographic constructions in operation while being grounded in physical constraints and empirically calibrated hardness.

23.2 Non-limiting hardness assumptions and attacker model

In some embodiments, security arguments rely on empirical hardness assumptions concerning physical modelling and forgery rather than on number-theoretic hardness alone. Non-limiting assumptions include:

- **RK microstructure unpredictability:** given access to many observed transcripts from a device, an adversary without the ability to reproduce the corresponding physical measurement event under the declared conditions has difficulty predicting responses to fresh challenges within an acceptance region.
- **RK-distinguishability:** given mixed samples, a competent meter distinguishes physical traces produced by a device family from simulated traces produced by a bounded emulator family, for a chosen operating point and protocol distribution.
- **RK one-wayness by dissipation:** in sufficiently dissipative configurations, reproducing the full joint statistics of the convolution bundle, or inverting a run summary to recover a scene or protocol that matches the same distribution, is empirically difficult without the ability to reproduce the corresponding physical measurement event under the declared conditions.

Security parameter (non-limiting). In some embodiments, an effective security parameter n_{sec} scales with total transcript information content. A non-limiting discrete proxy is

$$n_{\text{sec}} \approx \min(\log_2 |\mathcal{X}|, nT),$$

where $|\mathcal{X}|$ is a count of admissible challenges or protocol seeds (for example distinct scan protocols), n is a non-limiting count of reliably extractable bits per response after noise and quantisation, and T is a non-limiting count of independent challenge-response bundles. In

some embodiments, n_{sec} is tied to an estimated entropy of the joint transcript rather than to any single syntactic parameter.

Negligible-bound lens and design-target framing (non-limiting). In some embodiments, the standard cryptographic notation $\text{negl}(n_{\text{sec}})$ is used as an interpretation lens to express empirical design targets under stated attacker families and meter envelopes. These bounds are not asserted as theorems; they are targets evaluated and re-evaluated empirically as modelling and attack methods improve.

RK microstructure-hardness lens (non-limiting). In some embodiments, for a response space $\mathcal{Y}$ and any efficient adversary $\mathcal{A}$ without the ability to reproduce the corresponding physical measurement event under the declared conditions, a design target is

$$\Pr[\mathcal{A} \text{ predicts correctly}] \leq \frac{1}{|\mathcal{Y}|} + \text{negl}(n_{\text{sec}}),$$

where the probability ranges over fresh challenges and device noise realisations under a declared protocol class and meter envelope.

RK-distinguishability lens (non-limiting). In some embodiments, a design target is that no efficient adversary distinguishes genuine device-produced transcripts from efficient simulations with more than negligible advantage:

$$|\Pr[\mathcal{A}(\text{real}) = 1] - \Pr[\mathcal{A}(\text{simulated}) = 1]| \leq \text{negl}(n_{\text{sec}}).$$

Unclonability via total-variation gap (non-limiting). In some embodiments, a device is treated as implementing a noisy microstructure-unclonable response distribution $f(x)$ on challenges $x \in \mathcal{X}$, and unclonability is expressed by requiring that for any candidate clone $\tilde{f}$, fresh-challenge response distributions are separated in total variation distance on a non-negligible fraction of challenges:

$$\Pr_{x \leftarrow \mathcal{X}}[d_{\text{TV}}(f(x), \tilde{f}(x)) > \varepsilon] \geq 1 - \text{negl}(n_{\text{sec}}).$$

ML-based modelling as first-class attack strategy (non-limiting). In some embodiments, the adversary model explicitly includes machine-learning-based surrogate modelling (emulators trained from transcripts), and empirical hardness indices are reported relative to bounded families of such emulators and their best-achieved performance under the stated protocol distribution and meter envelope. These are empirical assumptions calibrated to realistic noise models, budgets, and adversary capabilities, and may be re-evaluated over time as modelling methods improve.

23.3 Commitments to the convolution bundle and selective disclosure

In some embodiments, the system produces a commitment to the convolution bundle (or to windowed segments) so later parties can verify integrity without receiving the entire trace.

5 The device logs the convolution bundle and produces one or more commitments, including without limitation: a hash commitment to a trace, Merkle roots for windows or atoms, or commitments to summaries together with commitments to underlying atoms.

In some embodiments, selective disclosure is supported by opening requested atoms or windows together with authentication paths, enabling verification against a prior commitment. This supports auditability under bandwidth and privacy constraints.

Hiding and binding lenses (non-limiting). In some embodiments, hiding and binding are expressed as design targets for the commitment-and-opening protocol rather than asserted as unconditional guarantees. Let M denote a committed message variable (or committed statement), and let τ denote a transcript or commitment view available to an adversary. A non-limiting hiding target is

$$I(M; \tau) \leq \varepsilon,$$

for small ε under the stated disclosure policy and adversary view. A non-limiting binding target is that it is infeasible to produce two distinct valid openings consistent with the same commitment:

$$\Pr[\mathcal{A} \text{ outputs valid } (m_0, r_0), (m_1, r_1) : m_0 \neq m_1] \leq \text{negl}(n_{\text{sec}}),$$

where r denotes non-limiting opening randomness and validity is checked by a verifier using committed atoms, protocol digests, and any disclosed windows. Note that the hiding target above is information-theoretic (bounded mutual information), while the binding target uses the computational convention $\text{negl}(n_{\text{sec}})$. This hybrid formulation is intentional: hiding is assessed by information-leakage budgets consistent with the spec’s empirical-hardness posture, while binding relies on the digital commitment layer’s computational properties. The “selective disclosure” described here refers to Merkle-style audit openings and does not presume any particular formal selective-opening security notion from the cryptographic literature unless explicitly claimed and parameterised for a specific embodiment.

23.4 microstructure-based identification and authentication

In some embodiments, a Reality Kernel module functions as a microstructure-unclonable challenge-response oracle. A challenge comprises a control protocol or seed that generates a protocol. A response comprises the measured convolution bundle or a derived signature. Verification is performed by testing membership in an acceptance region, optionally implemented

by a learned discriminator or calibrated statistical test.

In some embodiments, device enrolment produces reference data, a learned verifier, or helper data supporting robust matching under noise. In some embodiments, helper data includes secure sketches and/or fuzzy extractor constructions that derive stable keys or identifiers from noisy physical measurements while preserving unclonability and unpredictability in practice. In some embodiments, microstructure-based authentication is combined with the device configuration representation (Section 13.1), enabling fleet-level identity management alongside device-level verification; the irreducible residual (Section 13.2) provides the security margin that raises the difficulty of model-based forgery of challenge-response pairs.

23.5 microstructure-backed signatures and attestations

In some embodiments, signatures and attestations are implemented using a microstructure-backed process. To attest to a message, the system derives a message-dependent challenge protocol, executes it through the physical kernel, and returns at least one of: a committed transcript, a committed summary, and openings sufficient for verification.

In some embodiments, the secret key is physical possession of a device’s microstructure and calibration state, and public verification material comprises a verifier model, reference commitments, or enrolment summaries. In some embodiments, microstructure-backed attestations are combined with conventional digital signatures for compatibility with standard public-key infrastructure.

23.6 One-way state updates and hash-chain logging

In some embodiments, the controller maintains a state that is updated step-by-step as a one-way function of prior state and logged measurements. A terminal value serves as a compact fingerprint of a run or session. In some embodiments, the emission policy is driven by the evolving state so that challenges are chained to past observations, improving replay resistance.

The one-way update family may include, without limitation, hash functions, sponge constructions, MACs, PRFs, VRFs, and ZK-friendly algebraic hashes, selected based on implementation constraints and, where used, proof-system cost.

One-wayness and preimage lens (non-limiting). In some embodiments, the resulting terminal digest is treated as a hash-like compression of a run transcript. A non-limiting design target is that given a released terminal digest χ , it is infeasible to construct an alternative transcript (or alternative stimulus/protocol pair) that yields the same digest while falling outside a disclosed equivalence relation:

$$\Pr[\mathcal{A}(\chi) = S' : H_\theta(S') = \chi, S' \not\sim S] \leq \text{negl}(n_{\text{sec}}),$$

where H_θ denotes a non-limiting digest function derived from the one-way update rule, and $\sim$ denotes a non-limiting equivalence relation (for example scenes differing only within an accepted tolerance band).

23.7 Physically rate-limited outputs

- 5 In some embodiments, cryptographic outputs are rate-limited by requiring completion of a physical run prior to release. For example, to produce a signature-like output on a message, the device executes a run and releases an output after a successful meter acceptance check, in embodiments using that release policy. This ties output rate to physical time and device availability and reduces offline bulk precomputation.

10 23.8 Key agreement from correlated measurements

In some embodiments, two parties derive shared secrets from correlated measurements. Two devices (or two parties interacting with a shared channel) execute correlated protocols and obtain correlated convolution-bundle-derived bits. They perform reconciliation over a public channel and privacy amplification to derive a shared key. Security depends on an adversary
15 observing a degraded view relative to legitimate parties. This resembles key agreement from correlated randomness rather than conventional public-key exchange.

Key length lens (non-limiting). In some embodiments, letting X and Y denote legitimate party views and Z denote an adversary view of the same interaction under authenticated public discussion and a declared source model, a non-limiting key-length design lens for
20 secret-key agreement from correlated randomness in the wiretap / degraded-source special case is

$$|K| \approx I(X; Y) - I(X; Z) - O(\log(1/\varepsilon)),$$

where ε is a non-limiting security tolerance parameter and $O(\cdot)$ suppresses constant factors. The general secret-key agreement rate is model-dependent (source model, public-discussion model, authentication assumptions, reconciliation leakage, adversary class); this lens is used
25 as a design guide for the special-case regime and does not assert a generic capacity expression. In some embodiments, this lens is used as a design guide for increasing legitimate correlation (for example via private detector ports or richer protocols) while reducing adversary correlation (for example via bounded disclosures and access asymmetries).

23.9 Secure aggregation, threshold schemes, and multi-device at- 30 testation

In some embodiments, multiple devices produce joint outputs using threshold and aggregation mechanisms. Non-limiting examples include threshold signatures (a quorum jointly produces a signature), secure aggregation of meter outputs (individual device scores are

combined without revealing per-device values), and homomorphic aggregation of encrypted scores. These mechanisms support collective attestation and fault tolerance in multi-device deployments.

23.10 Lifecycle: drift, re-enrolment, and revocation

- 5 In some embodiments, device drift and lifecycle are managed explicitly. Registration or enrolment characterises a device and sets verifier material. Operation proceeds under noise and drift constraints. Re-enrolment updates acceptance regions, verifier models, or helper data. Revocation procedures address compromised or retired devices. In some embodiments, physical destruction or irreversible media reset provides strong decommissioning.

10 23.11 Non-limiting notes (cryptographic substrate)

These constructions are intended as optional overlays. They can be mixed and matched and may use conventional cryptographic primitives for transport and standardisation while relying on empirical physical hardness for additional constraint. They do not claim mathematical proof of security. They provide an engineering framework for tamper-evidence, auditability,
15 privacy-aware disclosure, and device-bound scarcity grounded in physical measurement logs.

Witness-indistinguishable proof lens for scene properties (optional, non-limiting).

In some embodiments, the system supports protocols in which a prover demonstrates a scene property without disclosing a full scene description. This is a non-limiting interactive lens and does not assert full zero-knowledge; formal simulation-based definitions and proofs are
20 not required by these embodiments. In some embodiments, formal proof systems (for example zk-SNARKs, zk-STARKs, or Bulletproofs) are used to prove predicates over committed summaries or reactor signatures under a declared disclosure policy. Let $P(S) \in \{0, 1\}$ denote a predicate on scenes (for example “the inspected seal is intact” or “the part includes a required fiducial pattern”). A non-limiting protocol sketch is:

- 25 1. A verifier selects a fresh challenge (for example a protocol seed or scan schedule) and commits to it when required by policy.
2. A prover with physical access executes the corresponding protocol $U_{0:T}$ on a scene S and returns a committed transcript and/or disclosed window summaries sufficient for the verifier to evaluate P .
- 30 3. The verifier checks protocol binding, meter envelope compliance, and that disclosed atoms are consistent with the committed transcript, then accepts if the evidence is consistent with $P(S) = 1$ under the declared regime.

In some embodiments, soundness is grounded in RK hardness lenses: without access to a valid witness scene and a compatible physical channel, producing accepted evidence under fresh

challenges is empirically difficult for bounded adversaries. In some embodiments, witness indistinguishability is supported by the existence of multiple scenes (or multiple micro-variants) satisfying P that yield statistically similar accepted transcripts under bounded disclosure, so the verifier learns the property outcome but not which witness was used beyond
 5 what is inevitably leaked by the disclosed windows.

Family membership proofs via bundle topology (optional, non-limiting). In some embodiments, topological invariants of the attractor bundle provide a basis for proving family membership without revealing device identity. Topological invariants (for example characteristic classes or Euler class of the bundle) are determined by the manufacturing process and
 10 are shared across all devices from the same production family, while individual holonomy depends on each device’s specific microstructure. In some embodiments, a protocol proves “this device belongs to family F ” (the topological invariant matches the family’s declared invariant) without revealing “this is device serial number N ” (the specific holonomy is not disclosed). In some embodiments, the converse is also supported: proving specific device iden-
 15 tity without revealing which family the device belongs to. This separation between topology (family) and geometry (individual) is a natural property of the bundle structure rather than a construction requiring additional cryptographic machinery, though in some embodiments it is combined with conventional zero-knowledge proof systems for formal guarantees.

True random number generation (TRNG) and min-entropy (non-limiting). In
 20 some embodiments, a Reality Kernel is configured to expose a high-entropy physical noise source (for example shot noise, thermal noise, or quantum-limited observables) and produce raw samples Z_t . In some embodiments, an entropy model assumes a conditional min-entropy bound

$$H_\infty(Z_t \mid \mathcal{E}) \geq k,$$

where $\mathcal{E}$ denotes side information available to an adversary. In some embodiments, an ex-
 25 tractor Ext is applied to a window $Z_{1:n}$ to produce output bits that are ε -close to uniform:

$$d_{\text{TV}}(\text{Ext}(Z_{1:n}), U_m) \leq \varepsilon.$$

In some embodiments, health tests are run continuously on the raw stream to detect degradation or manipulation and to trigger fallback or re-characterisation.

Summary table (non-limiting).

Primitive	RK instantiation	Key mechanism (lens)
Commitments and selective opening	Commit atoms/windows of the convolution bundle; open only challenged windows with authentication paths	Tamper-evident provenance; hiding/binding targets under bounded disclosure and meter envelopes

Primitive	RK instantiation	Key mechanism (lens)
microstructure identifica- tion	Reactor microstructure as challenge–response oracle; accep- tance region or learned verifier	Unpredictability and unclonabil- ity from device-specific disorder and noise
microstructure-backed at- testations	Message-derived protocol seed; run produces committed tran- script or summary plus openings for verification	Unforgeability tied to physical ac- cess and meter-validated regime
Hash-chain logging	One-way chain state χ_t updated from bundles c_t ; terminal digest χ_T anchors session history	Replay resistance and tamper ev- idence; preimage-style targets for alternative transcript search
Rate-limited outputs	Release tags/keys only after a physical run passes meters	Physical time and device avail- ability bound output rate; re- duces offline bulk forgery
Key agreement	Correlated RK measurements + reconciliation + privacy amplifi- cation	Secret-key agreement from corre- lated randomness with degraded adversary view
WI / ZK-style proof lens	Interactive challenge–commit– open protocols proving a property $P(S)$ with bounded disclosure	Soundness from empirical hard- ness under fresh challenges; lim- ited witness leakage via noise and disclosure policy
TRNG	High-noise RK regime + extrac- tion + health tests	Min-entropy bounds and extrac- tor closeness as design lenses
Multi-witness attestation	Threshold signatures / aggrega- tion over commitments and meter summaries	Fault tolerance under partial compromise

Committed scene representations and forgeability asymmetry (non-limiting). In some embodiments in which the declared target ϕ is a committed primitive-set representation (Section 8.4.4), a deception attempt against the declared target may require producing (a) a primitive set $\{p_k\}_{k=1}^K$ consistent with a claimed scene and (b) a rendering pathway consistent with the specific enrolled device’s independently calibrated forward-model parameters, including K_{spot} , p_{eff} , w_{det} , and associated world-to-device coordinate transforms drawn from, or referenced by, the enrolled $\mathbf{d}_i$ and the declared challenge configuration X or protocol geometry. The first may, in some embodiments, be performed using rasterisation-based optimisation without reproducing the enrolled physical forward model, whereas the second is evaluated under the declared threat model and meter envelope by the canonical hardness index pair $(k_{\text{dig}}^*(\theta; \varepsilon), m_{\text{ana}}^*(\theta; \varepsilon, r))$ of Section 12, and requires that the primitive-set prediction match the physical measurement not only photometrically for a single view but also consistently across held-out scan geometries, meter windows, detector settings, and temporal persistence behaviour specified by $U_{0:T}$ and the protocol digest. In such embodiments, the following

separation rules apply: primitive parameters are optimised under declared scene-space priors and meter envelopes; device kernels K_{spot} , p_{eff} , and w_{det} are held fixed at their independently calibrated values during verification rather than fitted as free parameters; and primitive covariance, density, opacity, appearance-bandwidth, and cardinality bounds are expressed as a
5 declared representation-complexity envelope, with spatial support, covariance, and density components expressed in declared world-coordinate units, including at least a maximum K , covariance eigenvalue or support-radius bounds, primitive-density or support-overlap limits, appearance-bandwidth bounds, and per-primitive or aggregate opacity/radiance-energy bounds, distinct from the device’s device-space PSF. Where these separation rules are sat-
10 isfied, the deception-symmetry and unified-trust-root limitations of Sections 12.7 and 12.7 specialise to a concrete evidentiary structure: the committed primitive set is the claimed-scene commitment; the enrolled physical channel, its calibration record, and the committed device configuration representation together supply the verifier’s forward model; and verification requires held-out consistency between rasterised prediction and physical measurement
15 under the shared $U_{0:T}$.

24 Categorical Unification of Substrate Mappings (Optional, Non-limiting Theoretical Framework)

Note for prosecution. This section provides a non-limiting mathematical framework for understanding relationships between the substrate mappings described in the preceding sec-
20 tions. It does not define additional claim subject matter and does not constitute a separate embodiment; it is provided as theoretical background for practitioners familiar with category theory and should be read as an interpretive tool rather than a technical teaching. No claim of the present disclosure depends on any result or conjecture stated in this section.

The preceding five sections (Sections 19–23) each map the Reality Kernel formalism
25 onto a different established theoretical framework: optical physics, neural-network architectures, information theory, control theory, and cryptographic primitives. Each mapping is constructed independently, and each reveals structure in the Reality Kernel that the others do not. This section presents a non-limiting theoretical framework, stated as conjectures rather than proven results, that unifies these five mappings as instances of a single categori-
30 cal construction. The framework provides precise language for transferability, compositional structure, and self-modelling, and identifies specific mathematical claims that are testable and falsifiable.

Containment statement. These categorical constructions are an optional descriptive lens and do not introduce additional hardware requirements, operational steps, or claim
35 limitations beyond those already described in the preceding sections. No embodiment requires the categorical framework for implementation. The framework may be removed entirely without affecting the enablement or scope of any disclosed embodiment. Conjectures 2 and 3

apply specifically to embodiments whose reactor dynamics satisfy a Turing-completeness condition; this is an additional design target not achievable with all reactor types, and the categorical framework introduces no such requirement for existing embodiments.

Non-load-bearing status of all conjectures. No core claim in this specification

5 **depends on any of Conjectures 1–5.** Each conjecture is a research-programme item whose truth or falsity does not affect the enablement, operation, or security properties of any described embodiment. The operational work that Conjecture 4 (Mod as contraction) was partly intended to formalise is instead performed by the PoD architecture itself: the PoD protocol physically anchors complexity growth through the confirmation requirement; 10 the security/discovery domain partition prevents self-modelling from eroding verification stability; and the quorum structure bounds individual divergence. The practical theorem—that the PoD architecture keeps the fleet’s collective self-modelling in a bounded regime by design—is independently grounded in the protocol specification and does not require the categorical apparatus. The categorical framework is retained as a research programme; its 15 conjectures are open problems, not established results.

24.1 The weak category of dissipative Hamiltonian systems

Objects. An object of the category **DissHam** is a tuple $(M, \omega, H, \Gamma, \Theta)$ where M is a finite-dimensional smooth manifold (the phase space), ω is a symplectic or pre-symplectic form, $H: M \times \Theta \rightarrow \mathbb{R}$ is a family of Hamiltonians parameterised by a configuration space Θ , and 20 Γ is a dissipation structure (a Rayleigh dissipation function, a Lindblad generator, or more generally any structure that breaks time-reversal symmetry and drives trajectories toward attractors). In some embodiments, stochastic forcing (thermal noise, shot noise, quantum fluctuations) is included as part of Γ . Every Reality Kernel module, every reactor, and every scene whose dynamics are modelled as a controlled dissipative system is an object of the 25 weak category **DissHam**.

Morphisms. A morphism $\kappa: A \rightarrow B$ is a physical coupling: a specification of how system A ’s state influences system B ’s dynamics, parameterised by coupling type (optical, mechanical, electrical, informational), coupling strength $g \in [0, \infty)$, coupling topology (which degrees of freedom of A couple to which degrees of freedom of B), and coupling timescale. 30 The identity morphism is the zero coupling ($g = 0$). Composition of morphisms $\kappa_1: A \rightarrow B$ and $\kappa_2: B \rightarrow C$ is defined by adiabatic elimination of the mediating system B when its dynamics are fast relative to A and C , yielding an effective coupling $\kappa_2 \circ \kappa_1: A \rightarrow C$.

Self-coupling as endomorphism. The Reality Kernel feedback loop — wherein a system observes itself through a measurement function and feeds the observation back into its own 35 dynamics — is a nontrivial endomorphism $\kappa_{\text{self}}: A \rightarrow A$, distinct from the identity. The system with feedback is a different dynamical system from the system without feedback,

even though the underlying phase space is the same. This endomorphism is the formal counterpart of the projector–camera loop in the anchor embodiment.

Weak associativity. Composition is well-defined physically but may not satisfy strict associativity in the categorical sense, because the adiabatic elimination operation depends
 5 on timescale ordering: coupling A to C through mediator B differs from coupling A to C through mediator D even if the effective coupling strengths are the same, because B and D have different dynamics that leave different imprints on the effective coupling. This means **DissHam** may be a weak category (or bicategory) rather than a strict category. The enriched framework described below is tolerant of this: the structure may be more precisely
 10 described as an enriched bicategory or tolerance-enriched category (in the sense of Lawvere’s generalised metric spaces), where hom-objects carry metric structure that quantifies the degree of associativity failure. Formalising this rigorously is an open problem identified in this section.

Timescale separation condition for composition (non-limiting). The adiabatic
 15 elimination that defines morphism composition is valid when the mediating system B relaxes on a timescale τ_B that is much shorter than the timescales τ_A and τ_C of the coupled systems: the operative condition is $\tau_B / \min(\tau_A, \tau_C) < \eta_{\text{sep}}$ for a declared timescale separation ratio η_{sep} (a non-limiting heuristic value is $\eta_{\text{sep}} = 0.1$). When this condition holds, the composition is well-defined and the tolerance-enriched distance between the two compositional orderings
 20 $((\kappa_3 \triangleright \kappa_2) \triangleright \kappa_1)$ and $(\kappa_3 \triangleright (\kappa_2 \triangleright \kappa_1))$ is bounded in the enriched metric. When the timescale condition does not hold — that is, when $\tau_B \sim \tau_A$ or $\tau_B \sim \tau_C$ — adiabatic elimination is not valid, the composition must be replaced by full coupled simulation, and the categorical structure does not apply; this is the honest domain of validity of the **DissHam** framework. In some embodiments, the timescale separation ratio is estimated from the AR(1) drift model
 25 parameters and committed to the protocol digest.

24.2 Fisher–Rao enrichment

In some embodiments, **DissHam** is enriched over the category of Riemannian manifolds (with Cartesian product as monoidal structure): the hom-set between two objects A and B is not merely a set of couplings but a Riemannian manifold $\text{Hom}(A, B)$ of all possible couplings,
 30 equipped with a metric derived from how sensitively the coupled dynamics depend on coupling parameters. This metric is the Fisher–Rao metric on the coupling parameter space. Composition of morphisms is a smooth map between the relevant Riemannian manifolds, and the enrichment coherence conditions (associativity and unitality of composition up to the manifold structure) are satisfied in the weak-coupling regime where joint parameter
 35 spaces factor as products.

The Fisher–Rao metric therefore appears at every level of the framework:

1. On Θ (the configuration space of a single system), as defined in Section 6.

2. On the space of attractors within a single system's bundle.
 3. On $\text{Hom}(A, B)$ (the space of couplings between systems).
 4. On the space of transformations of couplings (meta-couplings, as in a higher-order controller's modulation of system–environment interactions).
- 5 This ubiquity is the signature of a natural structure: the Fisher–Rao metric is not imposed externally but arises from the statistical properties of the physical dynamics at each level. The enriched category is denoted **DissHam**_{FR}.

24.3 The attractor bundle functor

Definition (attractor bundle assignment; lax monoidal functor). The *attractor bundle assignment* $\text{Att}: \mathbf{DissHam}_{\text{FR}} \rightarrow \mathbf{FibBund}_{\text{FR}}$ maps each dissipative Hamiltonian system to its attractor bundle and maps each coupling to the induced map on attractor bundles. Here **FibBund**_{FR} is the category of fibre bundles over parameter spaces equipped with the Fisher–Rao connection.

On objects:

$$\text{Att}(M, \omega, H, \Gamma, \Theta) = (\Theta, \{A_\theta\}_{\theta \in \Theta}, g_{\text{FR}}, \nabla),$$

15 where A_θ is the set of attractors at parameter value θ , g_{FR} is the Fisher–Rao metric on Θ , and ∇ is the connection on the bundle derived from g_{FR} (the e -connection or m -connection of information geometry).

On morphisms: if $\kappa: A \rightarrow B$ is a coupling, the induced map on attractor bundles is $\text{Att}(\kappa): \text{Att}(A) \times \text{Att}(B) \rightarrow \text{Att}(A \otimes_\kappa B)$, mapping the individual attractor bundles to the joint attractor bundle of the coupled system, where $A \otimes_\kappa B$ denotes the coupled system under coupling κ . Note that this is not the standard type signature for a functor's action on morphisms ($F(f): F(A) \rightarrow F(B)$); the product domain and coupled-composite codomain reflect the fact that coupling creates a *joint* system whose attractor structure depends on both partners. The map is more precisely a component of the monoidal product structure (Section 24.3, monoidal product paragraph below) rather than a strict functorial action on hom-sets. In some embodiments, Att is treated as a lax monoidal functor whose coherence conditions are satisfied up to the Fisher–Rao distortion bound of Conjecture 1.

Status and open mathematical questions. The functor is well-defined on objects (existence of attractors for dissipative systems and the Fisher–Rao metric on parameter spaces are both well-established). Its action on morphisms is well-defined in the weak-coupling limit, where joint attractors are perturbations of individual attractors. For strong coupling, the joint attractor structure can be qualitatively different from either individual's (new attractors appear, old ones vanish via bifurcation), and the functorial property — that composition of couplings maps to composition of attractor-bundle maps — is physically plausible (attractors of the coupled system do not depend on the order in which couplings were activated, in the long-time limit) but has not been proven in full generality. A known mathematical obstruction is that attractors change discontinuously at bifurcation points — they appear,

disappear, change dimension, or undergo explosive transitions — so the attractor bundle has singular fibres precisely where the curvature of the Fisher–Rao connection diverges. This is consistent with the physical picture (Fisher information does diverge at phase transitions) but means the functor Att maps into a category of singular bundles rather than smooth
 5 bundles. Possible resolutions include restricting Att to structurally stable regions and treating bifurcation loci as boundary strata; extending via the Conley index (which provides continuation through bifurcations but yields algebraic invariants rather than bundle fibres); or working with the space of invariant measures (which varies continuously even through bifurcations, at the cost of losing geometric structure). The correct resolution is an open
 10 mathematical question whose answer likely depends on whether the intended application is local (near a fixed configuration) or global (across the full bifurcation diagram).

24.4 The five substrate mappings as functors

Each of the five preceding substrate sections defines, in categorical language, a functor from $\mathbf{DissHam}_{\text{FR}}$ (or a subcategory thereof) to a target category specific to that domain:

Section	Functor	Target category
Optical Primitives	F_{opt}	Category of parameterised optical channels
Neural-Architecture Mapping	F_{nn}	Category of recurrent network architectures
15 Information-Theoretic Primitives	F_{info}	Category of noisy channels with Fisher metric
Control-Theoretic Primitives	F_{ctrl}	Category of controlled dynamical systems
Cryptographic-Primitive Substrate	F_{crypto}	Category of interactive proof systems

The repeated appearance of the same formal structures (attractors, bifurcations, Fisher information, holonomy) across all five substrate mappings is consistent with this framing: under the proposed categorical encoding, these structures arise as properties of the source functor Att , and if the relevant preservation properties of the specific functors $F_{\text{opt}}, \dots, F_{\text{crypto}}$
 20 are established, they would be inherited by all five images rather than coincidentally shared. Whether these five mappings are indeed faithful functors and whether they actually preserve the relevant properties of Att in the senses needed are open questions (see the status paragraph above); the framing is offered as an organising hypothesis, not a proven theorem.

24.5 Monoidal structure and composition

25 In some embodiments, $\mathbf{DissHam}_{\text{FR}}$ has additional algebraic structure relevant to Reality Kernel composition (Section 14):

Monoidal product ($\otimes$).

Parallel composition of systems: given A and B , the product $A \otimes B$ has phase space $M_A \times M_B$, parameter space $\Theta_A \times \Theta_B$, and independent dynamics (no coupling). This
 30 is the categorical counterpart of the parallel multi-reactor topology.

Traced monoidal structure.

Feedback loops are formalised by the trace operation: given a morphism $f: A \otimes C \rightarrow B \otimes C$, the trace $\text{Tr}^C(f): A \rightarrow B$ feeds C 's output back as input, yielding the closed-loop system. The Reality Kernel feedback loop (projector $\rightarrow$ scene/reactor $\rightarrow$ detector $\rightarrow$ controller $\rightarrow$ projector) is exactly the trace over the optical channel.

Braiding.

In some embodiments, the monoidal product is symmetric (the order of parallel composition does not matter up to canonical isomorphism), consistent with the physical symmetry that two non-interacting systems can be labelled in either order.

The multi-reactor architectures described in the PoliePuter section — chain, parallel, graph, hierarchical, recurrent, and reservoir-of-reservoirs — correspond to different diagrams in the traced monoidal category, and the multi-reactor primitives (Route, Compile, Parallelise, Reduce, Error-correct, Abstract) are natural transformations between these diagrams.

24.6 The self-modelling endofunctor and fixed-point conjectures

Definition (self-modelling endofunctor). The *self-modelling endofunctor* $\text{Mod}: \mathbf{DissHam}_{\text{FR}} \rightarrow \mathbf{DissHam}_{\text{FR}}$ maps each system A to its best internal model of itself: the subsystem of A whose dynamics track A 's own attractor structure through the feedback loop (the self-coupling endomorphism κ_{self}). This is the formal counterpart of the depth curriculum described in the related Filing 2 application (optional, non-essential), where a PolieBot iteratively refines its model of its own coupling boundaries.

On morphisms: given a coupling $\kappa: A \rightarrow B$, the induced morphism $\text{Mod}(\kappa): \text{Mod}(A) \rightarrow \text{Mod}(B)$ is the coupling between the respective self-model subsystems induced by κ —that is, the restriction of κ to the self-modelling degrees of freedom of A and B . This is a non-limiting definition; the precise form depends on how the self-model subsystem is demarcated within each system's phase space.

Conjectures (open problems, non-limiting). The following conjectures are stated as a theoretical framework. They are specific enough to be tested, proven, or refuted, and they identify precise mathematical claims that would, if established, unify the transferability and self-modelling aspects of the Reality Kernel formalism. Proving them would require rigorous construction of $\mathbf{DissHam}_{\text{FR}}$ as an enriched category (including the weak-associativity issue), proof that Att preserves enriched structure, and analysis of the fixed-point structure — work that is beyond the scope of this disclosure but is identified here as a research programme.

Conjecture 1 (Transferability).

Att is an enriched functor: it preserves the Fisher–Rao metric structure on hom-objects, up to bounded distortion. The distortion bound depends on the topological distance between the attractor bundles of the source and target systems (specifically, on the

difference in their bifurcation-set topology). If Att preserves enriched structure, then: nearby couplings produce nearby attractor-bundle maps (continuity of transfer); the metric distortion under Att is bounded (transfer quality is predictable); and the kernel of Att identifies substrate-independent equivalence classes (couplings that produce the same attractor-bundle map regardless of substrate).

Conjecture 2 (Self-consistent self-model, existence).

For any object A in $\mathbf{DissHam}_{\text{FR}}$ satisfying (a) A 's phase space M_A has sufficient dimensionality to embed a model of its own attractor bundle, and (b) A 's dynamics are capable of universal computation (the system is dynamically Turing-complete), the endofunctor Mod has a fixed point: there exists $A^* \cong \text{Mod}(A^*)$.

Conjecture 3 (Self-consistent self-model, structure).

The fixed point A^* necessarily has an attractor bundle $\text{Att}(A^*)$ containing structures isomorphic to: non-trivial attractors; a bifurcation set of positive codimension; positive entropy production (irreversibility); noise-driven selection at bifurcations; self-referential feedback modifying the attractor structure; and coupling-dependent boundary dissolution. The self-model at the fixed point contains representations of all six structures (the model correctly describes its own attractors, bifurcations, irreversibility, noise, self-reference, and boundary-dependence).

Conjecture 4 (Convergence via depth curriculum).

Under a suitable training curriculum (the depth curriculum of the related Filing 2 application; optional, non-essential), Mod is a contraction mapping on $\mathbf{DissHam}_{\text{FR}}$, and iterated application of Mod converges to the fixed point A^* at a rate determined by the contractivity constant. In the enriched setting, this is a Banach fixed-point theorem: each training iteration brings the self-model closer to self-consistency (the Fisher–Rao distance between the system and its self-model decreases), and the convergence rate is the contraction ratio of Mod.

Note on viability. Conjecture 4 as stated is contested and is retained as an open research problem rather than an operational claim. Three structural objections have been identified. (1) **Capability expansion.** A better self-model reveals previously unmodelled capability dimensions, requiring a strictly more expressive model to represent them. Mod is expansive by construction in parameter space; self-modelling discovers new content rather than converging to a pre-existing fixed point. (2) **Two-speed dynamics.** Self-modelling simultaneously compresses stale structure and expands into newly discovered structure. The net effect on Fisher–Rao distance to any putative fixed point is indeterminate; a clean contraction is not assured. (3) **Co-evolutionary dynamics** (most decisive). Any realistic operating system models interactions with other self-modelling systems that are themselves running Mod and becoming more complex. There is no single fixed point in a fleet of co-evolving agents; the joint fixed point of simultaneous fleet-wide self-modelling is a far stronger requirement than stated and is unlikely to

exist (Red Queen dynamics). The restricted version—locally contractive for an isolated system in a fixed environment near a structurally stable fixed point—avoids these objections but is too weak to be interesting and too far from operating conditions to be useful. **No core claim in this specification depends on Conjecture 4.** The work it was intended to formalise is performed instead by the PoD architecture: physical anchoring bounds complexity growth; the security/discovery domain partition prevents self-modelling from eroding verification stability; quorum structure bounds individual divergence. The practical proposition—that the PoD architecture keeps fleet-level self-modelling in a bounded regime by design—is independently grounded and requires no categorical machinery.

Conjecture 5 (Universality of natural transformations).

The normalised transfer-entropy matrix, synchronisation order parameter, and Fisher information profile are enriched natural transformations of Att. They are substrate-independent in the precise sense that they commute with all morphisms in **DissHam_{FR}**, up to the distortion bound from Conjecture 1.

Even if one or more categorical correspondences hold only approximately or fail for a given embodiment, the disclosed apparatus, calibration loops, drift navigation, discrepancy protocol, and network corroboration remain supported as independently described engineering mechanisms.

Relationship to Lawvere’s fixed-point theorem. Conjecture 2 is related to, but distinct from, Lawvere’s categorical fixed-point theorem, which establishes that in a cartesian-closed category with a point-surjective morphism from an object to its exponential, every endomorphism has a fixed point. **DissHam_{FR}** is unlikely to be cartesian-closed in the strict sense (the exponential object B^A — the “space of all ways A can influence B ” — may not exist as an object in the category). The enriched formulation via Banach contraction would avoid this difficulty: it requires only that the hom-objects are complete metric spaces and that Mod is a contraction, which *if it holds* gives constructive convergence rather than mere existence. However, as noted in the viability note to Conjecture 4, the contraction of Mod is contested on structural grounds (capability expansion, two-speed dynamics, and co-evolutionary Red Queen dynamics). If those objections apply, the Banach-contraction route to Conjecture 2 is also unavailable; existence of a fixed point $A^* \cong \text{Mod}(A^*)$ would then require a different proof strategy—for example, topological fixed-point methods (Schauder, Brouwer, or Kakutani) that require compactness and continuity but not contraction, and which do not provide constructive convergence. Whether such a strategy is available for **DissHam_{FR}** is an open question. The connection to Lawvere remains of theoretical interest because it links self-reference in physical systems to the same diagonal-argument structure underlying Gödel’s incompleteness theorems, the halting problem, and Cantor’s theorem.

Practical implications for Reality Kernel design. If Conjectures 1, 2, 3, and 5 are approximately correct, they have direct engineering consequences: the drifting field (Section 22.8) navigates the Fisher–Rao geometry of Θ using the structure that Att preserves; few-shot commissioning works because Att’s continuity (Conjecture 1) implies that nearby configurations produce nearby attractor bundles; and the five substrate mappings give consistent results because they are all faithful functors out of the same source category. These conjectures provide a theoretical explanation for design choices that are already empirically motivated elsewhere in this specification.

Conjecture 4 (depth curriculum convergence via contraction of Mod) is *not* included in the above list. As noted in the viability note to Conjecture 4, the contraction claim is contested on three structural grounds (capability expansion, two-speed dynamics, and co-evolutionary Red Queen dynamics) and is recorded as an open and probably hard research problem. The empirical observation that depth curricula converge in practice does not depend on Conjecture 4; it is an operational mechanism (independently monitored by dedicated metres, Section 22.8) and requires no categorical machinery. Conjecture 4 offers a *possible* theoretical explanation if it is eventually proved in some restricted form; it is not a precondition for the observation.

This paragraph is conditional throughout on the remaining conjectures. Conjecture 2 applies only to embodiments whose reactor dynamics satisfy a Turing-completeness condition, which is an additional design target not achievable with all reactor types; verifying this condition for specific embodiments is a research target and is not required for the operational mechanisms claimed in this disclosure.

25 Quantum and Non-Classical Coupling (Optional, Non-limiting)

In quantum-sensitive embodiments, the disclosed architecture extends the same committed-bundle, protocol-digest, and multi-regime operator framework to photon-counting, interferometric, and related non-classical channels, thereby allowing quantum-sensitive observables to participate in the same verification, sensing, and controlled-transformation workflows as the classical embodiments.

In such embodiments, quantum Fisher information provides an additional metric on the device configuration manifold that can be used to select protocols enlarging the gap between device-achievable and attacker-achievable estimation precision under declared budgets.

This section describes non-limiting embodiments that incorporate quantum-sensitive or non-classical effects while preserving the core Reality Kernel structure: a swept, controlled operator with configuration parameters θ , a control protocol $U_{0:T}$, and a convolution-bundle record $C_{0:T}$ comprising emission-observation bundles. These embodiments are optional enhancements. Core Truth Beam, Limager, and Reality Transform operation may be implemented in purely classical regimes.

25.1 Overview and quantum classes

In some embodiments, quantum-sensitive channels are included as additional components of the observation vector $\mathbf{y}_t$. These channels include, without limitation, photon counts, arrival-time distributions, coincidence statistics, homodyne or heterodyne quadrature measurements, interference fringe statistics, and other observables derived from quantum-sensitive detection.

In some embodiments, coincidence statistics include time-tagged correlations and second-order intensity correlations derived from click streams. A non-limiting example is a normalised correlation function $g^{(2)}(\tau)$ computed from time tags or binned counts, which can be treated as a derived channel and logged or committed as part of convolution-bundle summaries.

For descriptive convenience, non-limiting quantum-sensitive embodiments are grouped into quantum classes:

- **Q0:** single-photon or few-photon operation using photon counting without requiring entanglement.
- **Q1:** coherent interferometric operation with phase-sensitive observables.

Quantum-flavoured hardness metrics (optional, non-limiting). In some embodiments, hardness is reported under explicit photon-budget or coherence-budget constraints. Non-limiting auxiliary metrics (distinct from the canonical $(k_{\text{dig}}^*, m_{\text{ana}}^*)$ hardness index) include: a minimum photon budget (or minimum integration time) required for a bounded attacker model to meet a fidelity threshold τ under a declared protocol; and residual mismatch in interferometric statistics (for example visibility or phase-estimation error) under a declared phase sweep and calibration regime. These metrics are empirical, conditioned on meter envelopes and protocol digests, and are optional enhancements rather than requirements for core operation.

25.2 Quantum class Q0: Single-photon and photon-counting operation

In Q0 embodiments, a Reality Kernel module operates in a photon-starved regime and one or more detectors record photon-count statistics. Sources include, without limitation, attenuated coherent sources and heralded single-photon sources. Detectors include, without limitation, single-photon avalanche diodes and other photon-counting detectors. The system treats resulting click streams and derived statistics as additional channels in the convolution bundle.

Photon-count observation model (non-limiting). In some embodiments, a detector channel returns a photon count per step (or per window) that is modelled as a conditional

count process. A non-limiting model treats the count N_t (or binned count vector $\mathbf{N}_t$) as

$$N_t \sim \text{Poisson}(\lambda_\theta(S, u(t))),$$

where λ_θ depends on the scene, reactor state, and control settings including, without limitation, wavelength, intensity envelope, scan coordinate, and detector gate $g_{\text{det}}(t)$ (integration time and/or time-bin settings). These counts (or derived statistics such as mean/variance
 5 over repeated micro-sweeps) are included as components of $\mathbf{y}_t$ and therefore of the convolution bundle.

Time-tagging readout (non-limiting). In some embodiments, the photon-counting channel is time-resolved by recording time tags for click events relative to an emitted pulse or modulation cycle (for example time-correlated single-photon counting (TCSPC)). In these
 10 embodiments, the per-step observable may be a list of time tags or a time-bin histogram $\mathbf{h}_t$. In some embodiments, the controller sweeps gate widths, bin edges, pulse repetition rates, or modulation frequencies as part of $U_{0:T}$. Time-resolved channels can strengthen both Limager (depth/delay inference) and Truth Beam (fine-grained timing signatures conditioned on metered synchronisation and drift).

Temporal modulation bridge (non-limiting). In some embodiments, the classical-
 15 regime temporal modulation of epsilon-near-zero reactor media (Section 10) provides a natural bridge to quantum-regime temporal operation. At classical intensities, periodic temporal modulation produces Floquet sidebands; in the photon-starved regime, these correspond to discrete photon-energy channels whose statistics can be resolved by TCSPC or spectral detec-
 20 tion. The same physical mechanism therefore supports both a classical temporal-hardness channel and a quantum-regime timing-statistics channel, logged in the same convolution-bundle format.

In some embodiments, Q0 operation provides hardness amplification because matching
 25 full photon-count statistics across a disordered reactor may be more demanding than matching mean intensities. In some embodiments, shot noise and arrival processes provide entropy sources for protocols. In some embodiments, Q0 supports low-light Limager operation for photosensitive scenes or covert sensing.

25.3 Quantum class Q1: Coherent interferometric operation

In Q1 embodiments, the system includes phase-stable interferometric structure such as
 30 Mach-Zehnder or Sagnac geometries. Phase modulators and optional delay elements are treated as components of the control protocol $U_{0:T}$. In some embodiments, the scan protocol sweeps phase settings jointly with spatial scan coordinates, producing a phase-resolved convolution bundle.

Outputs such as visibility, phase estimates, and fringe statistics are included as channels of
 35 $\mathbf{y}_t$. In some embodiments, calibration sub-protocols maintain phase stability using reference

tones or interleaved calibration segments, and the resulting calibration data is logged and optionally committed together with the convolution bundle.

In some embodiments, non-classical sources or stronger quantum constraints (for example squeezed-light enhancements) are used as optional sensitivity amplifiers; such variants are not required for core operation and are not further detailed in this section.

25.4 Hybrid classical–quantum training and optimisation

In quantum-sensitive embodiments, the device interface remains classical: detectors output classical data (counts, time tags, quadratures). Meters and decoders are trained on these channels as they would be on any other channel.

Optimisation of quantum-control settings and certain reactor parameters may use gradient-free methods, including without limitation SPSA, evolution strategies, and bandit methods, and may also use score-function estimators. In embodiments with small parametrised interferometers, analytic gradient estimators may be used for those limited parameters while treating the rest of the Reality Kernel as an environment.

25.5 Quantum-regime detector-as-reactor operation (non-limiting)

In some embodiments, the detector-as-reactor mode described in Section 10.18 operates in a regime where quantum noise sources — shot noise from photon arrival statistics and, at very low photon counts, vacuum fluctuations — are the dominant contribution to the per-pixel readout variance. In this regime, each pixel’s output is a quantum measurement outcome whose statistics are set by the photon number distribution of the incident field, the detector’s quantum efficiency (which varies per pixel due to fabrication), and the electronic noise floor.

In some embodiments, this quantum-noise regime has consequences for both security and sampling quality. For security: in the photon-starved regime, and under a declared adversary access model committed to the protocol digest — specifying limits on the adversary’s verification photons, calibration photons, observation time, illumination access, detector access, and observability of calibration ground truth — an adversary attempting to characterise the detector’s per-pixel response faces a measurement-budget constraint. Fully characterising the quantum efficiency, dark current, and timing jitter of each pixel to a target precision δ in the shot-noise-limited regime requires a photon budget per pixel scaling as $\Omega(1/\delta^2)$ under independent classical or coherent probing; stronger quantum-metrologically enhanced scaling toward Heisenberg-type $1/N$ -precision is possible in principle for some parameters but requires nonclassical probe states and compatible measurements and is not assumed here. Each characterisation measurement consumes part of the adversary’s finite photon and time budget, so probe photons expended on early pixels are unavailable for later ones; the readout record is altered in the count-statistics sense (additional samples recorded) rather than in the wavefunction-collapse sense, and the security advantage is operational (measurement-budget-limited) rather than information-theoretic (no-cloning). The advantage is strongest

when the verification protocol uses fewer photons per pixel than would be needed for full per-pixel characterisation by the adversary under the declared access model. Adversaries operating outside the declared access model — for example with unlimited time, unlimited illumination access, or full observability of calibration ground truth — are not within the scope of this warrant. For sampling: at very low photon counts, the readout statistics are sub-Poissonian or super-Poissonian depending on the source characteristics, providing additional distributional structure that a classical noise model does not capture.

In some embodiments, the natural metric on the parameter space is the quantum Fisher information $\mathcal{F}_Q(\theta)$. The quantum Fisher information $\mathcal{F}_Q$ defines a Riemannian metric on the manifold of quantum states (the Bures metric, up to a multiplicative constant; see Provost and Vallée 1980, Braunstein and Caves 1994), and for a scalar parameterisation the inequality $\mathcal{F}_Q \geq I(\theta)$ (where $I(\theta)$ is the classical Fisher information obtained from a particular measurement) holds under standard regularity conditions on the parameterisation and the measurement family, with equality attained for a measurement saturating the symmetric logarithmic derivative (SLD) bound; the classical limit, in which the relevant family of states commutes and a common-eigenbasis measurement reduces the problem to classical Fisher information, is one special case of equality. For multi-parameter θ , the corresponding matrix inequality $\mathcal{F}_Q(\theta) \succeq \mathcal{F}(\theta)$ holds for fixed POVMs, but simultaneous attainment of the multiparameter quantum Cramér–Rao bound is not automatic and is obstructed in general by non-commuting SLDs and incompatible optimal measurements (Ragy, Jarzyna, and Demkowicz-Dobrzański 2016, and references therein); the matrix inequality is used herein as design guidance rather than as an operative attainable bound. In the quantum-noise regime, the scalar gap $\mathcal{F}_Q(\theta) - I(\theta)$ (or, in the multi-parameter case, a scalar functional such as $\text{tr}[\mathcal{F}_Q - \mathcal{F}]$) is informative when attainable: protocols supplying the necessary non-classical resources and measurements may achieve estimation precision closer to the quantum Cramér–Rao bound than classical protocols, subject to loss, detector noise, calibration uncertainty, and measurement compatibility. The attractor bundle formalism extends to this regime through the connection *induced by* the QFI/Bures metric (in particular, the Levi-Civita connection of the pullback of the QFI/Bures metric to the device configuration manifold) in place of the connection induced by the classical Fisher–Rao metric, and the curvature of the resulting connection is conjectured to diverge at bifurcation singularities in the same manner as in the classical case (this is a quantum-regime extension of the curvature-divergence property described in Section 24.3; its validity is an open question related to Conjecture 1). The novelty here is not the quantum Fisher information itself (which is a standard tool in quantum estimation theory) and not the QFI/Bures geometry on quantum state space (which is established; see Provost and Vallée 1980 and the geometric-phase literature, including Berry 1984 and Simon 1983, for adjacent fibre-bundle constructions), but the use of the QFI-induced connection as the connection on an attractor bundle over a physical device’s configuration space, used as a design metric for selecting protocols that maximise the gap between the device’s achievable precision and an attacker’s achievable precision in

both Limager and Truth Beam operation.

25.6 Optional enhancements and non-limiting notes

Quantum embodiments are optional and may be mixed with classical channels. In some embodiments, a device supports both classical and photon-counting operation by switching sources and detectors while preserving convolution-bundle logging conventions. The Q0/Q1 classification above is used for description only and does not limit combinations or hybrid operation.

Educational-kit and practical-lab quantum embodiments (non-limiting). In some embodiments, quantum-sensitive operation is implemented at educational, hobbyist, or practical-lab scale rather than at full-laboratory scale. Such embodiments comprise without limitation: attenuated coherent probes derived from continuous-wave or pulsed laser sources at declared mean-photon-per-pulse ranges (for example 10^{-2} to 1 photon per pulse, with attenuation calibrated by committed neutral-density filters or active attenuators); heralded single-photon sources implemented by spontaneous parametric down-conversion crystals (for example beta-barium-borate, periodically poled lithium niobate, or periodically poled potassium titanyl phosphate) pumped by visible or near-infrared lasers, with herald events provided by coincidence detection on the idler arm; avalanche photodiode and single-photon-avalanche-diode detectors operated in Geiger mode with declared dark-count rates, dead times, and timing jitter; time-correlated single-photon counting acquisition with declared picosecond-to-nanosecond timing resolution; spatial- superposition branches separated by wave plates, polarising beam splitters, dichroic mirrors, or Mach-Zehnder or Sagnac interferometric geometries with declared phase-stability budgets; multi-photon extensions including GHZ-state, cluster-state, and W-state preparations by successive entangling operations; and optional stabilisation comprising pilot-tone phase reference, dispersion compensation in optical-fibre interconnects, and active thermal or mechanical feedback. Each such practical-lab embodiment operates under the same committed-evidence, protocol-digest, verification, and hardness discipline as full-laboratory variants of the present section, with per-embodiment parameter ranges, calibration tables, and stabilisation procedures recorded in the protocol digest.

Scope limitation (non-limiting). The embodiments of this section support declared sensing, attestation, and hardness tasks benchmarked under the filing's task procedures. They do not support, and shall not be construed to support, capacity or scaling claims of the kinds bounded by Section 15.5, and do not imply any increase in capacity quantities.

26 Methods of Operation (Non-limiting)

26.1 Claim-relevant agent-integration embodiments (non-limiting)

The following embodiments are non-limiting examples relevant to system and method claims:

1. A method for controlling an agent using a Reality Kernel as a primary observation
channel, including updating belief state from the convolution bundle and selecting
actions that include Reality Kernel control signals and platform actuation.
2. A method for verification-gated action execution in which an action class is permitted
when configured policy thresholds on Reality Kernel evidence are met, said thresholds
comprising one or more of verisimilitude, hardness, freshness, or multi-RK corrobora-
tion.
3. A method for offline agent learning using at least one of (i) an RK-calibrated emulator
and (ii) a physical Reality Kernel operating on synthetic or modified scenes, including
periodic physical validation probes and confidence-weighted updates from simulated
experience.
4. A method for probing an agent internal representation by externalising a hypothesis
as a physical emission protocol and comparing the measured convolution bundle to the
predicted convolution bundle.

27 Additional Human, Deniability, Physical-Layer, and Agent-Layer Extensions (Non-limiting)

Positioning relative to proof-of-personhood systems (non-limiting). In some em-
bodiments, the human-layer records described below are used in application spaces adja-
cent to proof-of-personhood. The embodiments below share a high-level goal with generic
proof-of-personhood systems: Sybil-resistant human-anchored attestation as a substrate for
downstream voting, governance, reputation, and resource-allocation applications. They are
distinguished from such systems in at least six architectural respects. First, the load-bearing
evidence object is a committed human–fleet coupled-trajectory history, not a one-time asser-
tion that a person is globally unique. Second, the record is derived from HC-Read, HC-Write,
and HC-Verify channels bound into convolution-bundle evidence, rather than from an un-
qualified biometric, document, account, or credential alone. Third, recognition and quota
consequences are scoped to a declared fleet or subnet policy catalogue, rather than to a
universal personhood registry. Fourth, token emission and quota accounting are rate-limited
by declared cadence, freshness, meter envelopes, and interaction windows, rather than by
an assumed one-human-one-account property. Fifth, portability is evidentiary rather than
dispositive: a receiving fleet or subnet applies its own declared thresholds to an existing

portable record and any fresh session it elects to run. Sixth, adversarial incentives are directed at detector hardening through declared Eve-style attempts against subnet detectors and interaction surfaces, not at probing applicants or extracting border-admission evidence. Accordingly, the embodiments below may be combined with personhood-adjacent use cases, but their load-bearing claims remain committed physical evidence, scoped policy, physically rate-limited outputs, and declared attacker-model accounting.

27.1 Portable Human Bonding Records (Non-limiting)

Definition 35 (Portable human bonding record (non-limiting)). *A portable human bonding record is a peer-portability record whose authenticated principal is not a device alone, and not a human identity assertion alone, but a committed human–fleet coupled-trajectory history. In some embodiments, the record is built from a cumulative sequence of HC-Read, HC-Write, and HC-Verify sessions, with auxiliary human-response channels logged as convolution-bundle evidence under the protocol digest and commitment infrastructure described in Section 14.4.*

In some embodiments, enrolment comprises N committed human-coupled sessions executed under declared protocols and meter envelopes. The resulting human-response signature is derived from convolution-bundle evidence, including committed auxiliary response channels. Non-limiting response channels include gaze-pattern trajectories, reaction-time distributions, task-performance fingerprints, physiological-channel coherence, haptic timing, voice-timing features, and cross-channel consistency among such signals. In some embodiments, the enrolment record stores commitments to the session atoms, protocol digests, meter summaries, time anchors, and any verifier-selected openings rather than a raw biometric archive.

In some embodiments, verification is performed by any fleet member that runs a fresh HC session under a declared protocol and compares the fresh convolution-bundle response evidence against the committed human-response signature within a declared tolerance envelope. In some embodiments, a passing comparison refreshes the bond, opens the declared quota window for that fleet or subnet, and appends the fresh session to the portable record. In some embodiments, a failing comparison is logged as advisory evidence, causes a downgrade to a lower bonding level, or requires re-enrolment under the receiving fleet’s declared policy.

In some embodiments, bonding level is graduated by analogy to the verification levels of Section 8.2.4. Level 0 is single-session recognition under a low quota and short freshness window. Level 1 is sustained closed-loop HC history across multiple sessions under stable meter envelopes. Level 2 is deeper enrolment including Yoked human-coupled trajectories and cross-channel coherence checks. Level 3 is a declared full-quota bonding level requiring the fleet’s deepest accepted HC trajectory class, optional selective openings, and the cadence specified by that fleet or subnet. Thresholds, session counts, tolerances, and quota functions are declared per fleet or subnet; no universal human-identity threshold is asserted.

In some embodiments, a portable human bonding record includes a staleness window. In some embodiments, a bond not refreshed within the declared period is treated as stale and falls back to Level 0, advisory status, or re-enrolment, depending on the declared policy. In some embodiments, freshness is indexed by the last accepted HC session, the time anchor
 5 of the most recent commitment, and the meter-envelope status of the session that refreshed the bond.

Scope of human-response signatures (non-limiting). Human-response-signature distinctiveness is weaker than optical PUF uniqueness and is not treated as device-grade unclonability. The relevant threat family includes twin-class biometric overlap, relay attacks in
 10 which one human rotates across multiple bonded PolieBots, coerced bonding, coached or rehearsed responses, sensor spoofing, and synthetic-response attacks generated from observed auxiliary channels. Longer HC trajectories, Yoked response windows, and richer auxiliary channels may separate some members of this family under declared attacker bounds, but there exists a minimum interrogation complexity below which two humans or two response
 15 histories are functionally equivalent. This is not a failure of the framework; it bounds the claim. The resolution floor is the cross-fleet tolerance envelope under the declared attacker family, cadence, and meter state, and the bond is time-indexed rather than unconditionally durable.

Plenoptic-parallax liveness primitive (non-limiting). In some embodiments, an HC
 20 session uses a plenoptic (light-field) detector to capture the human participant's face or interaction surface in a single exposure as an auxiliary liveness meter, not as a stored human-response signature and not as a device-grade identity primitive. The session admits the interaction only if the captured four-dimensional light field exhibits angular-parallax relationships consistent with a physically present three-dimensional subject or interaction
 25 surface under the declared pose, distance, and illumination conditions. In some embodiments, the angular-parallax residual is committed as a liveness component of the session meter envelope, and sessions that fall below a declared parallax-consistency threshold are downgraded or refused before any bond-updating step. In some embodiments, the plenoptic liveness check is composed with the other meters described in this subsection rather than
 30 substituted for them; it geometrically bounds specific planar spoofing modes (ordinary flat-display replay lacking calibrated angular multiplexing, printed portrait, planar projection, and planar retro-reflective replay) and does not replace coached-response, relay-attack, or synthetic-response mitigations addressed elsewhere. The parallax residual is not asserted to provide optical reactor-microstructure-grade unclonability, biometric uniqueness, or human-
 35 signature distinctiveness; any human-response signature remains derived from the committed convolution-bundle evidence under the weaker human-layer scoping stated in this subsection.

Cross-fleet promiscuity (non-limiting). In some embodiments, human bonding is promiscuous by default: a human may hold parallel bonded records with multiple fleets, and

each fleet tracks only its own scoped HC history, quota cadence, and tolerance envelope. In some embodiments, the portable record may be offered to a receiving fleet as evidence, but the receiving fleet applies its own declared thresholds to the existing committed record and any fresh HC session it elects to run. Subnet-level exclusivity, disclosure obligations, and cross-subnet accounting are handled by the subnet policy catalogue described in Section 27.2.

27.2 Subnet-Scoped Identity and Attested-Interaction Token Economies (Non-limiting)

Definition 36 (Subnet identity scope (non-limiting)). *A subnet is a declared identity scope over a fleet, contract set, or interaction set, associated with a committed policy catalogue. In some embodiments, the catalogue includes at least: (i) an identity-exclusivity policy; (ii) a quota function mapping attested interaction time or accepted evidence windows to emission rate; (iii) a source/sink contract catalogue with per-contract pricing; and (iv) a cross-subnet policy specifying whether proof-of-projection or tPoP records from another subnet count as evidence, and whether B-PoD handoffs under Section 16.7 are permitted.*

In some embodiments, the identity-exclusivity policy is selected from a declared finite catalogue. Non-limiting examples include: exclusive within one subnet; mutually exclusive across a declared family of subnets; disclosure-required across subnets; and isolated, in which cross-subnet correlation is forbidden except under an expressly committed audit or dispute policy. In some embodiments, the policy is committed in the protocol digest or in an append-only policy catalogue referenced by the subnet’s token records.

In some embodiments, token emission is a subnet-scoped, physically rate-limited PoP or tPoP export denominated in the subnet’s own unit. In some embodiments, the token accrues to a bonded human principal across the subnet’s fleet rather than to a single device. In some embodiments, “PoliePunts” or similar non-limiting labels are used for such subnet-denominated units, in the same sense that “Narravite” is a non-limiting label for narrative-charged microstructure-unclonability artefacts. The label is not load-bearing; the committed source, sink, quota, and cross-subnet policies are load-bearing.

In some embodiments, a token source contract specifies which committed evidence windows produce subnet-denominated credit and at what rate. Non-limiting source examples include attested optical-signature scans of new artefacts, books, samples, or locations; attested presence at declared sites or events; attested sensor contributions including camera-trap, environmental-monitoring, scientific-sample-tracking, or industrial-process windows; B-PoD discrepancy discovery contributions; attested HC-session contribution under declared protocols; and attested participation in interaction-set membership events of the type described for narrative-charged tokens.

In some embodiments, a token sink contract specifies how subnet tokens are spent, burned, escrowed, or locked. Non-limiting sink examples include weighted voting on committed fleet-calibrated latent-space region handles under Section 12.3; priority-queue access for

scarce compute, detector time, or physical sampling resources; curation or moderation weight; access gates for restricted subnet actions; prediction-market settlement under declared-oracle contracts; quadratic voting, conviction voting, or positive-sum voting mechanisms; and rate-limited messaging boards over committed latent-space volumes. In some embodiments, a latent-space region handle is committed as a hash of a centroid and radius, a cluster identifier, or another declared region descriptor in the fleet-calibrated embedding, so that the voting target is cryptographically unambiguous.

In some embodiments, vote weights or access weights are computed from a declared function of provenance depth, verification history, bond freshness, interaction-set membership, and token balance. In some embodiments, the function is committed per subnet and may differ across subnets. Mechanism names such as positive-sum game, auction, quadratic vote, conviction vote, prediction market, decentralised curation rule, or latent-volume message board are descriptive labels for declared contract mechanisms, not load-bearing theoretical commitments. Choice of mechanism is a per-subnet product and policy decision.

Sybil accounting and cross-subnet amplification (non-limiting). In some embodiments, the subnet’s declared exclusivity policy supplies the accounting rule for per-human per-window quota within that subnet, conditioned on the declared attacker family, cadence bound, and tolerance envelope of the underlying bonding records. Under an exclusive policy, accepted records are treated as non-amplifying only inside the declared subnet family and only while freshness and tolerance conditions hold. Under a disclosure-required policy, parallel participation is auditable but may still multiply influence according to the declared accounting rule. Under a promiscuous or isolated policy, the relying party bears explicit responsibility for whether and how third-party subnet records are counted.

Cross-subnet Sybil strength is bounded by the weakest-exclusivity subnet in a human’s portfolio and by the least restrictive cross-subnet policy on which the relying subnet depends. Subnets relying on third-party subnet records inherit those subnets’ attacker-model declarations, refresh cadences, tolerance envelopes, and disclosure policies. Non-limiting threat families include cross-subnet collusion, stale-bond replay, relay use of one human across multiple bonded PolieBots, coerced participation, weak-disclosure laundering, and mutually inconsistent exclusivity declarations. The resolution floor is the declared cross-subnet accounting envelope, not a universal claim of one-human-one-vote identity.

27.3 Red-Team Bounty Primitive for Subnet Detector Hardening (Non-limiting)

In some embodiments, the Eve hardening principle described for physical-channel, information-theoretic, meter-training, and semantic layers is extended to subnet identity and subnet detector operation. The subnet is the attack target. The red-teamer is the attacker. The bounty aligns incentives for discovering detector holes in the subnet’s actual declared interaction surfaces rather than in a separate toy environment.

Definition 37 (Subnet red-team bounty attempt (non-limiting)). *A subnet red-team bounty attempt is a committed adversarial attempt against a subnet detector, executed through ordinary subnet interaction surfaces, and later opened against a committed post-attack subnet state under a declared success/failure oracle.*

In some embodiments, before the attack window opens, the red-teamer commits to a structured target claim that the subnet is being tricked into accepting, where the claim is machine-checkable against committed subnet state. In some embodiments, the pre-attack commitment also binds the attack class from the subnet’s declared attack taxonomy, the declared subnet-interaction budget for the attempt, and a nonce. The commitment is time-
 anchored using the same run-commitment and transparency-log mechanisms used for committed evidence records, and in some embodiments follows the two-seed pattern of Section 14.4: the attack is committed before the post-attack opening condition is known.

In some embodiments, attack execution occurs only through ordinary subnet interaction surfaces. Non-limiting surfaces include physically rate-limited token sources, HC sessions, declared token sinks, proof-of-projection submissions, tPoP exports, B-PoD handoff interfaces, latent-space voting targets, and selective-opening dispute flows. In some embodiments, honeypot verification channels of the type described in Section 8.2.5 are included in the surface so that attacker effort leaves diagnostically useful traces.

In some embodiments, after the attack window closes, the subnet publishes a committed post-attack state hash. The red-teamer then opens the pre-attack commitment and the relevant evidence atoms. The success/failure oracle is declared in the protocol digest. Non-limiting oracle options include automated detector output, a declared reviewer group, an independent verifier quorum, or a hybrid rule that combines detector output with selectively opened evidence windows. In some embodiments, the oracle checks whether the target claim was accepted, whether the acceptance relied on the committed attack class, and whether the attack stayed within the declared interaction budget.

In some embodiments, the subnet commits an attack-class taxonomy in its protocol digest or policy catalogue. Non-limiting attack classes include identity confusion; synthetic provenance, including fabricated attested scans of nonexistent artefacts or locations; relay attack in which one human operates multiple bonded PolieBots in rotation; quota-laundering through cross-subnet amplification contrary to a declared exclusivity policy; semantic injection inducing the subnet to upvote a latent-space region under false pretences; collusion by a coordinated multi-principal group; and coercion-simulation, in which a detector is tested for recognition of a bonded human operating under duress.

In some embodiments, bounty payment is implemented through the same subnet token rails as an additional declared sink. In some embodiments, the bounty-signing key family or signing role is distinct from the key family used for ordinary source, sink, and detector-operation records. This is a structural separation of signing roles within the commitment plumbing of this disclosure; it does not import any companion-filing governance machinery. In some embodiments, payment is conditioned on successful opening, oracle outcome, budget

compliance, and absence of collusion flags.

In some embodiments, paid attempts and defeated attempts both enter the subnet’s detector-training evidence set with their committed evidence chain preserved. The resulting subnet detector-hardness index is conditioned on the observed red-team population, attack-
 5 class diversity, bounty budget, detector version, and interaction budget rather than treated as unconditional detector coverage. The resolution floor is the declared red-team coverage envelope under the attack taxonomy, detector version, interaction budget, and bounty bud-
 get, time-indexed rather than unconditionally durable. Relying parties may discount subnet trust when the red-team population is small, the attack taxonomy is narrow, or the bounty
 10 budget is insufficient for the declared attacker family.

Failure modes and mitigations (non-limiting). Non-limiting failure modes include red-teamer front-running by opening the commitment before the subnet state is fixed, mit-
 igated by requiring opening only after a committed post-attack state hash is published; subnet-operator patching of the detector during the attack window, mitigated by freezing
 15 detector version to the version committed at attack start; and collusion between red-teamers and subnet operators staging fake attacks, mitigated by separated bounty-signing keys, se-
 lective opening of evidence atoms, and statistical audit of red-team success rates against a declared reference distribution.

Out-of-scope note (structural, not policy default). This primitive tests subnet detec-
 20 tors. As a structural property of the primitive, a subnet red-team bounty attempt is addressed to a subnet detector, committed subnet state, declared token surface, quota rule, source/sink contract, or other subnet interaction surface, and is never applied to an applicant under any subnet policy. Applicant-directed evaluation is outside the primitive’s scope by construction. A workflow that presents a bonded human, prospective member, or cross-subnet applicant
 25 as the attack target is not an instance of this primitive. The primitive is not an admission mechanism and not a screening mechanism for bonded humans. Cross-subnet admission, where present, is handled by the receiving subnet applying its own declared thresholds to the existing portable record and any admissible committed evidence; the red-team bounty primitive does not probe applicants.

30 27.4 Physical-layer advantage channels: keyed, trained, and scene-geometric embodiments

The following embodiments, also referred to in some embodiments as “reality encryption,” concern physical-layer advantage channels in which a cooperative observer or decoder (Bob) is made more capable than a bounded adversarial observer or decoder (Eve) under de-
 35 clared resource, detector-geometry, and meter-envelope constraints. These embodiments extend the Alice/Bob/Eve multi-agent framing of Section 9, the wiretap advantage-gap lens of Section 21.4, and the scale-to-threshold Bob/Eve curricula of the preceding subsection,

and they specialise to three progressively less pre-coordinated embodiment families: (i) keyed deterministic-noise obscuration at a declared cadence, (ii) trained neural-network obscuration generators optimised against an explicit Bob/Eve secrecy objective, and (iii) meter-bounded discovery and validation of scene-geometric advantage channels without any pre-shared key or separately declared digital message channel. In each case the disclosed inventive matter is an apparatus-and-procedure configuration, evaluated through committed convolution-bundle records and declared meter envelopes; no claim is made that any particular learned policy will emerge for every scene or every resource bound.

Stage 1: keyed deterministic-noise advantage at declared cadence (non-limiting).

In some embodiments, an auxiliary emission subsystem injects a deterministic obscuration pattern into the scene, the reactor path, or the detector-facing observation path. The obscuration pattern is driven by a committed seed or pre-shared cryptographic key available to the cooperative party but not to the bounded adversary. The cooperative decoder cancels, subtracts, or compensates for the obscuration pattern in its observation pipeline within a declared cancellation-latency budget, while the bounded adversary, lacking the seed or key, is required to invert the obscuration from observations alone within a declared inversion-time budget. In some embodiments, the following quantities are declared as design variables and are logged in the protocol digest: the obscuration update cadence, the detector integration time, the programmable-scene refresh rate (for example an e-ink, LCD, OLED, or micro-LED surface rendering a time-indexed plaintext or watermark as in the programmable-scene jamming game of Section 9), the cooperative-party cancellation-latency budget, and the adversary's inversion-time budget under a declared bounded resource class. A non-limiting design posture is to choose the obscuration cadence and the cooperative-party cancellation latency so that the implied adversary inversion-time budget, under the declared bounded resource class, is empirically insufficient to recover the plaintext within the cadence window. The obscuration generator in this stage is deterministic given its seed; it may be a linear-feedback shift register, a keyed pseudorandom function, a noise profile whose commitment, encrypted form, identifier, or selective-opening policy is stored in the protocol digest, an external runtime input drawn from a sibling device of the same class under a committed tap point, or a cryptographically specified stream generator, without limitation. Where an anchor noise co-illumination embodiment is present, this stage composes with that embodiment; in embodiments not including such an anchor extension, the same keyed-obscuration logic applies to any auxiliary emission path, reactor-facing path, or detector-facing path whose control schedule or profile commitment is bound into the protocol digest under the commitment plumbing of Section 11. Physical rate-limiting (Section 11.5) provides a cadence floor that offline precomputation alone does not remove under a declared attacker model lacking the cooperative key, timely access to the obscuration realisation, or equivalent physical access to the emitting path; this makes cadence a meaningful physical-layer security parameter rather than a purely protocol-level assumption, and composes with the correlated-measurement key-agreement lens of Section 23.8.

Stage 2: trained neural-network obscuration generator under an explicit Bob/Eve secrecy objective (non-limiting). In some embodiments, the obscuration generator is a

trainable neural-network or parameterised functional whose output is optimised end-to-end against an explicit Bob/Eve secrecy objective rather than being fixed by a deterministic seed.

- 5 In a non-limiting form, the obscuration generator G_ϕ produces an obscuration waveform or emission pattern $n_t = G_\phi(k, \xi_t)$ from a cooperative-party key, declared private input, or cooperative-only observation k and an internal state ξ_t ; a cooperative decoder $D_{\psi_B}^{\text{Bob}}$ recovers a plaintext or target variable from its observation given the key; and a bounded adversary decoder $D_{\psi_E}^{\text{Eve}}$ attempts the same recovery from its declared degraded view without the key.
- 10 A non-limiting generator-side objective, evaluated against a best-response Eve decoder, is

$$\mathcal{L}(\phi, \psi_B, \psi_E) = -\mathbb{E}\left[\log P_{D_{\psi_B}^{\text{Bob}}}(\Xi | Y_B, k)\right] + w_{\text{adv}} \mathbb{E}\left[\log P_{D_{\psi_E}^{\text{Eve}}}(\Xi | Y_E)\right],$$

where Ξ is the plaintext or target variable, Y_B and Y_E are the cooperative and adversarial observation views, respectively, and $w_{\text{adv}} \geq 0$ is a non-limiting weighting between Bob recovery and Eve suppression. In some embodiments, the optimisation is minimax, alternating a minimisation over (ϕ, ψ_B) with a best-response maximisation over ψ_E within a de-

- 15 clared bounded resource class for Eve. In some embodiments, physical emission parameters of the auxiliary subsystem (scan law, amplitude gain, spectral profile, polarisation, timing alignment) are co-trained with ϕ using hardware-in-the-loop procedures (Section 9.13) or a differentiable surrogate, so that training pressure acts on physically realisable obscuration waveforms rather than on waveforms that cannot be emitted. In some embodiments, the
- 20 Stage 2 training game composes directly with the Bob/Eve scale-to-threshold curricula of the preceding subsection: adversarial capacity c for Eve is grown toward a declared threshold while cooperative capacity for Bob is reduced, with the gap objective $c_{\text{Eve}}^* - w_{\text{adv}} c_{\text{Bob}}^*$ providing an auditable, capacity-calibrated version of the secrecy objective above. Meter envelopes, eye-safety bounds, and power ceilings constrain the admissible obscuration policies during
- 25 and after training.

Stage 3: meter-bounded discovery and validation of scene-geometric advantage channels without pre-shared keys (non-limiting). In some embodiments, the Reality

Kernel scene is configured as a physical search space for candidate advantage-channel policies among role-assigned agents, without requiring a pre-shared seed, a pre-committed noise

- 30 profile, or a separately declared digital message channel between the cooperative parties; the candidate channel, if any, is the meter-bounded physical scene interaction itself. A first policy, assigned the Alice role, controls one or more emission, projection, scan, wavelength, polarisation, timing, amplitude, or scene-actuation variables; a second policy or decoder, assigned the Bob role, receives observations through a declared detector geometry or observation port;
- 35 and a third policy or decoder, assigned the Eve role, receives observations through a different declared detector geometry, observation port, or resource-bounded view. The scene may include, without limitation, specular reflectors, diffuse reflectors, occluders, shadowing geom-

etry, saturating or non-linear surfaces, wavelength-selective elements, polarisation-selective elements, retroreflectors, dynamic or addressable surfaces, and volumetric scattering media, with their placements, orientations, and spectral properties committed as part of the scene configuration metadata. A non-limiting advantage objective combines cooperative-party recovery performance, bounded-adversary recovery performance, meter-envelope compliance, and physical-safety compliance, and may be optimised through reinforcement learning, evolutionary search, gradient-based search over a differentiable scene surrogate, or any combination of these, subject to declared actuator and emission bounds. No claim is made that any particular policy will reliably emerge for every scene or every resource bound; the inventive matter is the apparatus and procedure for searching, validating, and committing candidate policies, not a guaranteed learning outcome.

Discovery, validation, and commitment procedure (non-limiting). In some embodiments, a Stage 3 embodiment executes the following procedure, producing a committed, auditable evidence record at each step:

1. **Scene configuration.** Configure the physical scene with a declared catalogue of optical affordances (reflectors, occluders, saturating surfaces, wavelength- or polarisation-selective elements, dynamic surfaces) and commit the scene configuration metadata, including affordance placements, orientations, and declared spectral and polarisation properties.
2. **Role-assigned search.** Execute role-assigned search or training over candidate Alice, Bob, and Eve policies under the declared advantage objective, bounded actuator envelopes, and meter envelopes, with each executed episode producing an emission-observation record.
3. **Evidence commitment.** Commit for each candidate policy the convolution bundle $C_{0:T}$, the protocol digest, the Bob-side and Eve-side detector geometries, the meter outputs, and any policy-identifying parameters or parameter digests, using the commitment plumbing of Section 11.
4. **Advantage measurement.** From the committed record, measure cooperative-party recovery performance, bounded-adversary recovery performance, and the advantage gap between them, using a pre-declared evaluation distribution and a pre-declared bounded-adversary resource class.
5. **Affordance ablation.** Run targeted ablation episodes in which selected scene affordances are removed, moved, rotated, spectrally altered, polarisation-altered, or replaced with a null-affordance control, and measure the advantage gap again on each ablated scene.
6. **Acceptance criterion.** Accept a candidate advantage channel only when (a) the advantage gap on the unablated scene satisfies the declared threshold, (b) the advantage

gap changes in the pre-declared direction and by at least a declared diagnostic magnitude, or falls below a declared floor, under a pre-declared subset of ablations identified as diagnostic of the hypothesised physical mechanism, and (c) meter-envelope and safety bounds hold throughout. The ablation-sensitivity requirement functions as a physical-mechanism falsification test: an accepted channel must depend on the identified scene affordance in the committed evidence, rather than on an unlogged digital side-channel or a spurious training artefact.

Viewpoint-specific decoding as the physical origin of the advantage (non-limiting).

In some embodiments, the physical origin of the advantage in Stage 3 is that a single physical emission produces different information channels at different detector geometries, wavelengths, polarisations, or temporal-integration windows. Specular reflection from a bounded region of the scene, for example, can produce a high-intensity return only along a narrow range of detector viewpoints; occluders can deny a detector at one declared geometry access to a region that is fully visible at another declared geometry; saturating surfaces can exceed the linear range of a detector at one declared geometry while remaining within range at another; and wavelength- or polarisation-selective elements can transmit to one declared channel while attenuating another. This viewpoint-dependence connects Stage 3 to the cross-scene regime combinations of Section 8.8, and provides a physical account of why a Bob-side detector geometry can be structurally favoured over an Eve-side detector geometry under a committed emission sequence, without any pre-shared key between the cooperative parties.

Relationship to adjacent technical families (non-limiting).

Without admitting that any particular reference or family constitutes prior art against any claim, the embodiments of this subsection may be understood relative to several adjacent technical families as follows. Stigmergy and related indirect-coordination mechanisms in swarm robotics use persistent low-dimensional environmental markers as coordination memory; the present embodiments instead use meter-bounded, high-dimensional, viewpoint-dependent optical or multi-physics scene affordances recorded in committed convolution-bundle. Emergent-communication methods in multi-agent reinforcement learning, including reinforced or differentiable inter-agent learning over declared discrete or differentiable message channels, optimise messages over an explicit communication action space; the present embodiments have no declared communication action space or shared seed and instead search physical scene affordances, validating candidate channels through committed evidence and scene-affordance ablation. Physical adversarial-patch and projector-based adversarial-attack methods optimise a physical perturbation to induce classifier misbehaviour in a target vision system; the present embodiments optimise and validate a cooperative/adversarial recovery gap under declared detector geometries and committed convolution-bundle, rather than targeting a specific classifier failure mode. Free-space-optical and visible-light-communication physical-layer-security methods design and analyse a predesigned optical link under a channel model

and an eavesdropper model, often with beamforming or precoding at a known transmitter geometry; the present embodiments treat the scene affordances themselves—reflectors, occluders, saturation, polarisation, wavelength selectivity, detector viewpoint—as the discovered channel, validated through the commitment and ablation procedure described above rather than through a communications-system channel model. These observations are non-limiting and are provided to clarify inventive scope rather than to narrow it.

Composition with existing embodiments (non-limiting). The three stages above are not mutually exclusive. In some embodiments, a Stage 1 keyed-noise layer, a Stage 2 trained obscuration generator, and a Stage 3 scene-geometric search proceed concurrently, with the three contributions composing into a combined advantage objective subject to a shared meter envelope and safety bound. In some embodiments, the Stage 2 trained generator is initialised from a Stage 1 deterministic seed and then unfrozen for end-to-end training; in some embodiments, a Stage 3 scene configuration that produces a sufficient Bob/Eve advantage gap is used as a warm start for a Stage 2 training run in which the Stage 2 generator is tuned to widen the gap further. In all cases the inventive matter is the apparatus-and-procedure configuration and the committed evidence record, not a promise of any particular learned outcome.

Positioning relative to existing commitment infrastructure (non-limiting). In some embodiments, the primitives described below extend the commitment and selective-disclosure machinery of Section 14.4 and the hiding and binding lenses of the same section to support use cases in which the evidentiary weight of a genuine live session is deliberately scoped at the time of capture, scoped to a participating party set, or rescaled after capture under a declared trapdoor authority. The underlying designated-verifier, commit–reveal randomness, and chameleon-hash cryptographic primitives are not asserted here as novel in isolation. The contribution described in this section is their composition with convolution-bundle commitments, the two-seed selective-opening protocol, disclosure-regime tags in the protocol digest, proof-of-projection and proof-of-discrepancy audit workflows, and authority-action discipline. Adjacent anonymity-family primitives, including anonymous-set attestations, ring signatures, selective-disclosure credentials, zero-knowledge set-membership proofs, and plausibly deniable encryption, may be composed with the records disclosed here as external credential or confidentiality layers, but are not required for the designated-verifier, mutual N -party liveness, or chameleon-opening embodiments disclosed in this section. The primitives are composable with the canonical two-seed selective-opening protocol (Section 14.4), with convolution-bundle commitments, with proof-of-projection records, and with the bonded-human-identity records of Section 27.1. They do not weaken and are not a substitute for: positive verification under the Truth Beam regime (Section 8.2); provenance negation under the Anti-Truth Beam lens (Section 8.2.5 and the surrounding provenance-negation material); proof-of-discrepancy quorum confirmation under the PoD verifier-selection protocol (Section 13.3); or the human-coupled-trajectory evidence chain of the bonded-human-identity

extension. Each of those workflows operates on committed evidence under disclosure policies explicitly declared in the protocol digest; the deniability primitives below apply only where the protocol digest declares the relevant disclosure regime. The primitives inherit the terminological note of the proof-of-projection section: the deniability, simulatability, and trapdoor properties asserted below are empirical under declared attacker families and resource budgets, are time-indexed rather than unconditionally durable, and do not constitute formal cryptographic proof-system guarantees unless a specific embodiment explicitly specifies and parameterises such a guarantee.

27.5 Designated-verifier liveness via per-block salts (non-limiting)

Definition 38 (Designated-verifier liveness record (non-limiting)). *A designated-verifier liveness record is a convolution-bundle commitment whose liveness-attestation property is interpretable only by a declared verifier V possessing a pre-session per-block salt s_V shared with the submitting device A , and whose published transcript is, under the declared attacker family, adversary view, and resource budget, not distinguishable by any third party with advantage above a declared bound from a transcript producible by V alone using the same salt.*

In some embodiments, before the run commitment is published, device A and verifier V establish a fresh per-block salt s_V over a declared side channel. Non-limiting side channels include a short pre-run handshake session, a pre-registered session key, a key-derivation from a longer-lived shared secret bound to an earlier committed joint session, and a hash chain rooted at a previously attested coupling record. The side channel itself is governed by a declared attacker model, freshness discipline, and compromise assumptions committed to the protocol digest. The salt is bound into the run commitment as an additional field:

$$\text{com}_{\text{DV}}^{(i)} = H(\text{rid} \parallel i \parallel \Pi_{\text{dig}} \parallel \mathbf{m}^{(i)} \parallel R^{(i),\text{obs}} \parallel R^{(i),\text{emit}} \parallel s_V),$$

and is not disclosed to any party outside $\{A, V\}$ under the declared disclosure regime.

In some embodiments, the declared verifier identity, verifier-set digest, or an unlinkable verifier handle h_V is committed in the protocol digest and, where the disclosure regime permits, bound into the designated-verifier commitment, so that the same salt material is not ambiguously reusable across verifier identities, verifier sets, or parallel sessions. Where public disclosure of verifier identity would damage the simulatability property above, h_V is auditor-openable rather than publicly visible, under the declared disclosure regime and adversary view.

In some embodiments, the liveness property claimed by the record is stated as two operational components:

1. *Verifiability by the designated verifier.* V reconstructs $\text{com}_{\text{DV}}^{(i)}$ from the opened bundle atoms, the committed protocol digest, and s_V , and checks agreement with the published commitment under the hiding and binding lenses of Section 14.4.
2. *Simulatability against third parties.* Under the declared attacker family, adversary view,

and resource budget, a third party $T \notin \{A, V\}$ observing any subset of the published transcript, the run seed, and the opened atoms cannot, with advantage above a declared bound, distinguish a record produced by a genuine A – V session from a record V produced unilaterally using an alternative bundle and the same s_V . The declared
 5 advantage bound, the adversary view, and the attacker resource budget are committed to the protocol digest.

In some embodiments, per-block salts are rotated per run, per segment, or per atom. Rotation policy, salt-derivation KDF chain, and salt-anchor identifiers are committed to the protocol digest. In some embodiments, the salt-anchor identifier references an earlier
 10 committed joint session between A and V so that the salt chain is itself auditable under selective opening when the disclosure regime permits. Salt-anchor handles may be unlinkable tokens rather than long-lived identifiers where cross-session linkability is itself part of the declared threat model.

Outer-authentication simulation policy (non-limiting). In some embodiments, any
 15 outer transport signature, device identifier, registry entry, certificate chain, or publication channel used to carry a designated-verifier liveness record is itself marked as designated-verifier scoped, omitted from the third-party comparison view declared in the protocol digest, or simulable by V under a declared transcript-simulation policy; otherwise the simulatability property above is not asserted to extend to the outer authentication layer. The transcript-
 20 simulation policy, the set of outer layers it covers, and any omitted layers are committed to the protocol digest.

Composition with the two-seed protocol (non-limiting). In some embodiments, the designated-verifier salt is a disclosure binding separate from the run seed s_{run} and opening seed s_{open} of Section 14.4. The two-seed machinery continues to govern which atoms are
 25 opened and when; the designated-verifier salt governs *to whom* the opened record is meaningful as a liveness attestation. In some embodiments, a record may be opened to V under the designated-verifier regime and independently re-opened under a public-verifier regime only if the protocol digest declares an explicit re-anchoring protocol at commit time.

Failure modes and mitigations (non-limiting). Non-limiting failure modes include
 30 salt leakage to third parties outside $\{A, V\}$, mitigated by rotation, committed anchoring side-channel identifiers, and declared salt-validity windows; compromise of the salt-establishment side channel itself, including man-in-the-middle capture or replay of handshake material, mitigated by declared side-channel attacker models, freshness counters, and externally anchored salt epochs; replay of a prior $\{A, V\}$ salt or salt-anchor session, mitigated by monotonic coun-
 35 ters, freshness anchors, and committed rotation policies; linkability of multiple designated-verifier records across salt-anchor identifiers or rotation chains, mitigated by unlinkable salt-anchor handles and disclosure-minimised salt-anchor referencing; re-identification of A from outer signatures, run identifiers, protocol-digest metadata, meter envelopes, or tim-

ing metadata notwithstanding the hiding of s_V , mitigated by disclosure-minimised meta-data profiles and by the outer-authentication simulation policy above; collusion between A and a third party to fabricate an A -looking transcript without a live session, which is not a designated-verifier failure and is addressed by the independent proof-of-projection and
 5 quorum-confirmation machinery of the present filing; and unilateral fabrication by V of a record that V then presents to a third party as A 's attestation, which is a feature rather than a bug of the primitive and is addressed at the application layer by the relying party declining to accept designated-verifier records from V as A -originating attestations.

27.6 Mutual N -party liveness via commit–reveal (non-limiting)

10 **Definition 39** (Mutual N -party liveness record (non-limiting)). *A mutual N -party liveness record is a convolution-bundle commitment whose run seed is derived from a committed N -party nonce protocol such that each participating party independently witnesses session freshness from its own unpredictable contribution, while a non-participant observer, under the declared attacker family, adversary view, and resource budget, cannot with advantage
 15 above a declared bound distinguish a genuine N -party session from a record fabricated by a coalition with equivalent knowledge of the combined nonce.*

In some embodiments, the protocol proceeds as follows:

1. **Commit phase.** Each party $k \in \{1, \dots, N\}$ independently generates a nonce n_k from a declared unpredictability source, computes a hiding commitment

$$c_k = H(\text{tag}_{\text{NPL}} \parallel \text{rid} \parallel k \parallel \Pi_{\text{dig}} \parallel L_{\text{part}} \parallel n_k \parallel r_k),$$

20 with fresh randomness r_k , where tag_{NPL} is a declared domain-separation tag, Π_{dig} is the protocol digest, L_{part} is a committed canonical participant-list digest, and k is interpreted under the declared canonical participant ordering. For this commit phase, Π_{dig} denotes the pre-run protocol-template digest and excludes nonce-derived realised fields except where such fields are represented by declared templates or placeholders.
 25 Each party publishes c_k to the joint commit set before a declared commit deadline. The unpredictability source, canonical participant ordering, participant-list digest, and commit deadline are committed to the protocol digest.

30 **2. Reveal phase.** After the commit deadline has passed, each party publishes (n_k, r_k) . A late reveal, a malformed reveal, or a non-reveal is processed according to a declared non-reveal policy.

3. **Combining.** The combined nonce n_{comb} is computed from the revealed $\{n_k\}$ under a declared combining rule. Non-limiting combining rules include domain-separated hash-concatenation

$$n_{\text{comb}} = H(\text{tag}_{\text{NPL-comb}} \parallel \text{rid} \parallel \Pi_{\text{dig}} \parallel L_{\text{part}} \parallel n_{\pi_{\text{ord}}(1)} \parallel \dots \parallel n_{\pi_{\text{ord}}(N)}),$$

where π_{ord} is the committed canonical participant ordering (a permutation of the participant indices; distinct from trust score $\sigma_{ij}(t)$ and from any other body-text use of σ) and $\text{tag}_{\text{NPL-comb}}$ is a declared combining-rule domain-separation tag; bitwise XOR; and threshold rules in which any t -of- N revealed nonces determine n_{comb} under a declared threshold scheme.

4. **Run seed derivation.** The run seed s_{run} of the two-seed protocol of Section 14.4 is derived from n_{comb} and the committed protocol digest.
5. **Run phase and commitment.** The device executes the run under s_{run} and publishes the run commitment. The subsequent opening phase proceeds under the two-seed protocol.

Public-beacon variant and run-seed confidentiality (non-limiting). In the variant in which n_{comb} is publicly revealed before the run, the combined nonce supplies post-commit freshness for the run seed; the opening seed s_{open} remains independently supplied after the run commitment under the two-seed protocol. In some embodiments, where run-seed confidentiality until opening is required for its anti-tailoring role, n_{comb} is combined with a device-side committed nonce or verifiable-random-function output that remains undisclosed until selective opening, so that s_{run} is pre-bound to the joint freshness but is not fully determined by the public beacon alone. The chosen variant (public beacon or device-side combined) is committed to the protocol digest at commit time.

In some embodiments, the liveness property claimed by the record is: for each party k whose nonce n_k was unpredictable to the other $N - 1$ parties at the commit deadline under the declared unpredictability-source assumptions, the run seed is not fixed before k 's commitment by any party or coalition lacking n_k , except with advantage bounded under the declared attacker family, nonce-source quality, reveal policy, and resource budget; therefore the session is alive from k 's perspective under those declared bounds. A non-participant observer sees only the combined nonce and the public transcript, and cannot, with advantage above the declared bound and under the committed adversary view, distinguish a genuine N -party session from one in which a coalition reconstructed transcript fields $\{c_k, n_k, r_k\}$ consistent with the combining rule.

In some embodiments, the primitive is composable with designated-verifier liveness records by designating one or more of the N parties as designated verifiers under the machinery of Section 27.5, with per-block salts bound into the joint commit set. In some embodiments, the primitive is composable with proof-of-projection quorum protocols by using the N parties as an initial freshness-attestation quorum separate from the verification quorum of the PoD protocol (Section 13.3).

Scope of the liveness claim (non-limiting). The primitive binds the run's freshness to N independently committed unpredictability contributions. It does not on its own establish that the N parties correspond to N distinct identities; Sybil resistance, where relevant, is handled by the bonded-human-identity mechanisms of Section 27.1, by device-identity

machinery, or by external-identity anchors declared in the protocol digest.

Failure modes and mitigations (non-limiting). Non-limiting failure modes include last-revealer advantage, in which a party withholds its reveal until after observing other reveals and then chooses whether to reveal so as to bias n_{comb} , mitigated by the commit phase preceding the reveal phase, by non-reveal penalty policies committed in the protocol digest, and by threshold combining rules that remain well-defined when a bounded number of parties fail to reveal; non-uniform or correlated nonce sources that amplify combining-rule bias, including shared random-number-generator infrastructure or shared hardware entropy sources, mitigated by extractor-based nonce conditioning and by committed declaration of independent unpredictability-source identifiers per participant; abort-and-retry grinding across repeated sessions, and threshold-subset grinding in which multiple valid reveal subsets produce different n_{comb} values, mitigated by domain-separated combining hashes, deterministic non-reveal defaults, abort-rate meters committed in the protocol digest, and committed tie-breaking rules over the reveal-subset lattice; rushing during the reveal phase notwithstanding the commit deadline, mitigated by simultaneous or deadline-enforced reveal publication, committed reveal-order rules, and externally anchored reveal timestamps; coordinated pre-agreement in which all N parties jointly choose a target n_{comb} before the commit phase, which defeats the freshness claim against non-participants and is addressed at the application layer by requiring that at least one party be independently selected by a non-colluding anchor (for example a verifier-selected participant drawn from a fleet under the randomised verifier-selection rule); nonce-source compromise, mitigated by declared unpredictability sources with attacker-resource budgets committed in the protocol digest; and anti-tailoring degradation in the public-beacon variant, addressed by the device-side committed nonce or VRF composition described above where anti-tailoring is required.

27.7 Chameleon-hash retroactive selective evidentiary weight (non-limiting)

Definition 40 (Chameleon-hash evidentiary record (non-limiting)). *A chameleon-hash evidentiary record is a convolution-bundle commitment in which one or more binding fields are instantiated using a chameleon-hash function $\text{CH}(\cdot, \cdot; \text{pk})$ under a designated trapdoor authority's public key pk , such that the record is treated as binding, under the hiding and binding lenses of Section 14.4, against parties without knowledge of the trapdoor tk , under the declared attacker family, adversary view, parameterisation, and resource budget, and such that the trapdoor authority may, under a declared authority-action policy, produce an alternative valid opening consistent with the published commitment.*

In some embodiments, the run commitment is computed as

$$\text{com}_{\text{CH}}^{(i)} = \text{CH}(\mathbf{z}^{(i)}, r^{(i)}; \text{pk}_{\mathcal{T}}),$$

where $\mathbf{z}^{(i)}$ denotes the same binding-field tuple used in $\text{com}_{\text{full}}^{(i)}$ of Section 14.4, optionally with layer identifiers and disclosure-regime tags added under the declared layer-composition policy; $r^{(i)}$ is opening randomness; and $\text{pk}_{\mathcal{T}}$ is the public key of a declared trapdoor authority $\mathcal{T}$. The corresponding trapdoor $\text{tk}_{\mathcal{T}}$ is held by the authority under the declared
 5 authority-custody policy.

In some embodiments, the claimed binding-and-plausible-deniability properties are:

1. *Third-party binding.* Under the declared attacker family and resource budget, a party without $\text{tk}_{\mathcal{T}}$ cannot, with advantage above a declared bound, produce two distinct valid openings $(\mathbf{z}, r) \neq (\mathbf{z}', r')$ consistent with the same $\text{com}_{\text{CH}}^{(i)}$. The bound may be expressed
 10 as $\text{negl}(n_{\text{sec}})$ for a conventional digital chameleon-hash instantiation and as an empirical acceptance-probability bound for any physical-evidence component of the claim; the bound, adversary view, parameterisation, and resource budget are committed to the protocol digest.
2. *Authority-side alternative opening.* For a selected chameleon-hash instantiation satisfying the declared trapdoor-collision algorithm, the trapdoor authority $\mathcal{T}$, holding
 15 $\text{tk}_{\mathcal{T}}$, can compute r' such that $\text{CH}(\mathbf{z}', r'; \text{pk}_{\mathcal{T}}) = \text{com}_{\text{CH}}^{(i)}$ for an alternative $\mathbf{z}'$ drawn from a declared alternative-opening policy space. The choice of chameleon-hash instantiation, parameter size, hash domain separation, trapdoor-collision algorithm, and post-quantum or classical security posture is committed to the protocol digest at capture
 20 time.

In some embodiments, the trapdoor authority is non-limitingly the submitting fleet, a designated policy authority separate from the submitter, the submitting party itself (self-trapdoor), a committee holding shares of $\text{tk}_{\mathcal{T}}$ under a threshold secret-sharing or threshold trapdoor-collision- generation scheme, or a committee using threshold signatures as an autho-
 25 risation layer controlling a separately declared trapdoor- custody mechanism, with a declared quorum of t -of- n authority members required before an alternative opening is treated as conforming. The selected authority mode, custody mechanism, and quorum rule are committed to the protocol digest and referenced in each authority-action record.

Scope of the deniability claim (non-limiting). The primitive is evidentiary, not defi-
 30 nitional. It supports retroactive scoping of the weight assigned to a specific opening of a commitment; it does not erase that a commitment was published at time t , that a convolution-bundle capture occurred in the committed operating envelope, or that cross-device corroboration occurred under independent channels. In particular, independent transitive proof-of-projection records, peer-verifier quorum confirmations under Section 13.3, and externally
 35 anchored transparency-log entries binding the commitment to a time anchor remain operative against any subsequent alternative-opening action by the trapdoor authority, and continue to contribute to provenance determinations under the Anti-Truth Beam lens.

Mixed-layer composition (non-limiting). In some embodiments, a chameleon hash is used for the full run commitment, for selected atoms under Merkle inclusion, or for a sum-

mary digest while leaving lower-level atoms under standard commitments. The layers at which chameleon and standard commitments are applied, the declared alternative-opening policy space at each layer, and the trapdoor-authority identity at each layer are committed to the protocol digest. In some embodiments, records using a chameleon hash at any layer carry an explicit disclosure-regime tag in the protocol digest so that a verifier can identify which binding regime applies at which level of the record. Where deniability requires that chameleon-hash records be indistinguishable from standard-hash records in the public commitment stream, the regime tag is auditor-openable rather than publicly visible, and the chameleon instantiation is selected so that $\text{com}_{\text{CH}}^{(i)}$ values are indistinguishable from standard-hash values at the commitment-envelope layer under the declared adversary view. Where the regime tag is auditor-openable rather than publicly visible, a relying party whose acceptance policy requires non-trapdoor binding either requires opening of the regime tag under its acceptance policy or treats the unopened record as potentially trapdoor-bound and assigns it the corresponding lower evidentiary weight.

Authority-action audit discipline (non-limiting). In some embodiments, any alternative opening produced by the trapdoor authority is itself committed in a transparency log anchored externally, under the same external-anchoring convention used elsewhere in this specification for freshness anchors. In some embodiments, authority-action frequency is rate-limited under a declared policy digest and the count of alternative openings produced against a given commitment is bounded and committed. In some embodiments, authority-action openings are themselves subject to selective opening under an auditor-supplied seed analogous to the two-seed protocol, so that conforming relying-party policies can reject or downgrade any alternative opening that lacks the corresponding authority-action audit record, and an off-ledger alternative opening is treated as a policy violation rather than as a conforming evidentiary action. Threshold participation is not treated as evidence of authority honesty. In embodiments using threshold secret-sharing or threshold signatures, each share-use, partial-opening contribution, quorum transcript, signer identity, policy version, and authority-action reason code is committed as an authority-action record; records lacking such audit material are treated as lower-weight or inadmissible under the relying party's declared acceptance policy.

Failure modes and mitigations (non-limiting). Non-limiting failure modes include trapdoor compromise, mitigated by threshold secret-sharing of $\text{tk}_{\mathcal{T}}$, periodic key refresh under a committed rotation policy, and authority-custody procedures committed in the protocol digest; long-term trapdoor exposure and retroactive compromise of historical records, mitigated by epoch-scoped trapdoor keys, committed key-rotation and destruction attestations, and relying-party acceptance policies that discount records whose governing epoch is no longer under declared custody; inadequate forward security across key epochs, mitigated by declared forward-security discipline over epoch transitions and externally anchored destruction attestations; public-stream distinguishability between chameleon-hash and standard-

hash records where deniability requires indistinguishable commitment formats, mitigated by instantiation-choice discipline and by auditor-openable rather than publicly visible regime tags; quantum-adversary degradation of classical chameleon-hash assumptions, mitigated by declared post-quantum instantiations where required and by epoch-scoped migration policies; post-capture expansion of the alternative- opening policy space by the trapdoor authority, mitigated by capture-time commitment of the permitted alternative-opening domain and by authority-action audit records that expose any deviation; misuse of the alternative-opening capability by the trapdoor authority to silently substitute a fabricated opening for a legitimate one, mitigated by the authority-action audit discipline above and by relying-party policies that downgrade the evidentiary weight of chameleon-hash records in contexts where authority actions are not externally auditable; and confusion between binding and non-binding regimes by relying parties, mitigated by the explicit disclosure-regime tagging of the protocol digest and by relying-party acceptance policies that specify which binding regime is required for which downstream decision.

Cross-cutting disclosure-regime discipline (non-limiting). In some embodiments, a session does not freely switch between designated-verifier, N -party, and chameleon-hash regimes after the run commitment is published without a declared re-anchoring protocol. The disclosure regime in force for each layer of a record is committed to the protocol digest at commit time. A relying party accepting a record for a downstream decision consults the disclosure-regime tags and applies its own declared acceptance policy for the regime at each layer. In particular:

1. Designated-verifier records are not acceptable proof-of-discrepancy submissions under the PoD quorum protocol of Section 13.3 unless the designated verifier is itself part of the declared verifier quorum and opens the record to the quorum under the quorum's disclosure policy. Admission of such a record does not alter the pre-committed PoD parameters $(n, q, \tau_{\text{confirm}}, \tau_{\text{device}}, \text{agg})$, the Byzantine-fault-tolerance assumptions (including, where the declared PoD policy is Byzantine-fault-tolerant, the constraints $n \geq 3f + 1$ and $q \geq 2f + 1$), or the acceptance and evaluation meter partition.
2. Chameleon-hash records are not acceptable as positive attestations under the Truth Beam regime unless the relying party's acceptance policy explicitly admits records whose binding fields are under a trapdoor authority, in which case the policy declares how authority-action records and alternative-opening frequencies feed into the acceptance decision. Even where such admission is declared, Truth Beam positive verification remains based on the underlying physical evidence, meter envelope, and independent corroboration; the chameleon layer affects disclosure regime and evidentiary weighting only, and does not alter the underlying physical-verification status of the record.
3. Designated-verifier and chameleon-hash records do not alter the operational semantics of provenance negation under the Anti-Truth Beam lens: a record's content remains subject to cross-device corroboration under independent channels regardless of which deniability regime governs disclosure of that record to third parties.

Interaction with bonded-human-identity records (non-limiting). In embodiments in which records from Section 27.1 are present, designated-verifier liveness may be used for HC-Read, HC-Write, or HC-Verify channels where a human participant wishes to scope attestation of a specific coupled-trajectory event to a specific fleet-side counterparty. In some
5 embodiments, chameleon-hash binding may be used for auxiliary human-response channels (non-limitingly: physiological coherence summaries, reaction-time distributions) where an applicable data-protection policy requires the human participant or a designated authority to retain retroactive capability to produce alternative openings of specific response summaries. In both cases, the cumulative human–fleet coupled-trajectory evidence chain that
10 underpins the bonded-human-identity record retains its commitment structure, and the relying subnet’s acceptance policy declares which disclosure regimes are admissible for which HC-channel classes. The human-layer scope exclusions stated in the bonded-human-identity extension remain in force; the deniability primitives do not introduce admissibility paths for any of the channels excluded there. These uses do not add any new HC-channel class,
15 participant-probing authority, dream-state pathway, Mandela-probe pathway, or applicant-directed verification path beyond those expressly admitted in the bonded-human-identity extension.

Out-of-scope note (structural, not policy default). The primitives of this section are commitment-infrastructure extensions to the present filing. As a structural property: (i) they
20 do not replace, reduce, or substitute for proof-of-projection, transitive proof-of-projection, proof-of-discrepancy quorum confirmation (Section 13.3), or provenance negation under the Anti-Truth Beam lens; (ii) they do not grant the trapdoor authority of Section 27.7, or the designated verifier of Section 27.5, any positive-verification role under the Truth Beam regime; (iii) they govern disclosure regimes for committed evidence and not training objec-
25 tives, governance decisions, admission standards, or behavioural evaluation of agents, each of which is addressed in the related Filing 2 application on empirical ethics training architecture; (iv) all security, binding, hiding, and deniability claims stated in this section remain empirical under declared attacker families, adversary views, and resource budgets, time-indexed rather than unconditionally durable, and do not constitute unconditional cryp-
30 tographic guarantees unless a specific embodiment explicitly specifies and parameterises such a guarantee.

HC-Actuate (non-limiting). In some embodiments, the Reality Kernel issues a prompt, request, or operator-facing instruction to a human participant through a committed emission channel and captures the participant’s response through the same detector chain used for
35 scene observation or through a declared auxiliary tamper-evident channel committed under the same protocol digest, timebase, and meter policy. In some embodiments, the request is rendered through a declared emission channel, including any sensory channel already available for HC-Read or HC-Entrain operation; this paragraph concerns the commitment of the outbound request, rendering parameters, and response-capture window rather than redefin-

ing those human-coupled channels. The request-generating process may be the RK controller or a logged computational partner, including without limitation a PolieBot or PoliePuter-coupled process as described in Definition 25, Definition 24, and Section 16; higher-layer multi-agent coordination remains optional and non-essential to the present paragraph.

5 In some embodiments, a non-limiting worked flow comprises: (i) the request-generating process selects a request payload conditioned on committed observation $o_{0:t}$, meter vector $\mathbf{m}_{0:t}$, and, where applicable, a tool-mediated derivation trace; (ii) the payload is committed to the protocol digest prior to rendering, together with the selected rendering-channel parameters and the expected-response window; (iii) the payload is rendered through the
10 committed emission path, and where the rendering uses audio synthesis, the synthesis voice identity, prosody parameters, and synthesis model fingerprint, provider/model identifier, or version hash where available are committed alongside the payload digest; (iv) the response may enter through any HC-Write input channel, including voice, gesture, locomotion, explicit control input, tool-mediated input, or physiological measurement, and is logged under
15 the channel identity and detector configuration used for that response; (v) request digest, rendering parameters, response atoms, inter-event timing, and any coupled meter readings are bound into a single $C_{0:T}$ window under the protocol digest.

In some embodiments, the request–response binding provides a closed-loop audit object in which the request, rendering parameters, response atoms, channel tags, and inter-event
20 timing are jointly checked against the protocol digest and disclosed atoms, so that post-hoc substitution of a request, response, rendering parameter, or timing value is detectable as a commitment, channel-consistency, or meter-consistency failure.

In some embodiments, HC-Actuate composes with channel-separation primitives of Section 7.6.1, with the request assigned a declared channel tag u_{ch} and, where applicable,
25 with the cross-scene regime combinations of Section 8.8 in which a concurrent Truth Beam, Limager, Reality Transform, or cryptographic companion stream corroborates the request–response event.

In some embodiments, failure to render the request within the declared emission envelope, failure to capture a response within the expected response window, detector saturation,
30 channel occlusion, model-fingerprint mismatch, or response-channel dropout is logged as an HC-Actuate anomaly and causes the request–response episode to be treated as incomplete rather than silently accepted.

In some embodiments, where the response is captured through an auxiliary channel rather than the primary scene-observation detector chain, the episode is marked as a reduced-
35 assurance HC-Actuate sub-embodiment unless the auxiliary channel has an independently declared calibration state, channel tag, timing model, and meter envelope.

In some embodiments, HC-Actuate episodes are subject to declared participant-safety, exposure, privacy, and abort envelopes, and an abort event disables or attenuates the emission path, commits already-recorded atoms, and marks the episode as aborted in the protocol
40 digest.

In some embodiments, meters for HC-Actuate are partitioned into acceptance meters that gate whether the request–response episode may be relied upon, evaluation meters that are held out for later audit, and optional shaping meters used only to adapt future request rendering and not to validate the present response.

5 **Multi-contributor freshness composition (non-limiting).** As a non-limiting extension of the two-seed structure of Section 14.4 and Paragraph 14.4, in some embodiments a freshness value used as, or used to derive, s_{run} or s_{open} of Definition 26 is composed from two or more declared contributors via a commit–reveal protocol. In some embodiments, each contributor publishes a commitment $K_i = \text{Commit}(r_i; \rho_i)$ before the capture window opens,
 10 the commitments are aggregated into the protocol digest alongside declared contributor identities and role tags (for example, operator, verifier, participating agent, or provider class), and the capture window executes.

Where the composed value contributes to s_{run} , contributor reveal to the seed-computing controller, quorum, or escrowed derivation process occurs after all commitments are bound
 15 into the protocol digest but before the first consumption of the run seed by the scan law or emission schedule; disclosure of the revealed values to other verifying parties may remain withheld until selective disclosure under the declared opening policy. Where the composed value contributes to s_{open} , contributor reveal occurs after publication of the run commitment or capture-window close anchor, under the declared opening schedule.

20 In some embodiments, the composed freshness value is obtained via a declared composition rule—for example $s = H(r_1 \| r_2 \| \dots \| r_k)$ or a bitwise XOR over fixed-width representations—with the composition rule itself committed into the protocol digest.

In some embodiments, at least one contributor is a computational process or agentive partner operating under PolieBot or PoliePuter coupling; for purposes of this paragraph, the
 25 disclosed role of that partner is limited to contributing or committing a freshness input, not to broader multi-agent coordination.

In some embodiments, verifier treatment of the composed freshness value is conditioned on the following declared checks: (a) all declared contributor commitments appear in the protocol digest before the capture-window open anchor; (b) freshness assurance under this
 30 sub-embodiment is conditioned on a declared entropy and independence model, and is reduced or invalidated when the disclosed record shows that the composed value was determined entirely by one contributor’s pre-capture state, by a single common-control source, or by any source excluded by the declared freshness policy; and (c) the composition rule is recomputed from disclosed atoms during verification.

35 In some embodiments, if a declared contributor fails to reveal within the committed reveal schedule, reveals a value inconsistent with its commitment, or reveals through a channel failing the timing-anchor policy, the composed value is marked invalid or downgraded according to a predeclared missing-contributor rule. In some embodiments, the missing-contributor rule is selected from abort, reduced-assurance continuation, threshold composition from a declared subset of contributors, or recomputation using a predeclared fallback freshness source,
 40

with the selected rule committed before the capture window opens.

In some embodiments, contributor identities, role tags, commitment values, reveal schedule, composition rule, canonical serialisation format, and any truncation or domain-separation tags are committed in the protocol digest before the corresponding seed is
5 consumed.

In some embodiments, local operator-only contribution, single-contributor contribution, contribution by a single computational provider acting as a declared contributor, or contribution by entities under common control is logged as a reduced-assurance freshness sub-embodiment rather than as independent multi-contributor freshness.

10 In some embodiments, freshness-related meters are partitioned so that minimum freshness checks gate acceptance, while contributor-diversity, common-control, and entropy-budget indicators may be retained as evaluation meters or assurance annotations rather than used by the operating controller as shaping signals.

27.7.1 Logged derivation of session parameters

15 In some embodiments, session parameters consumed by the controller of a physical Reality Kernel run—including scan-law randomisation inputs, challenge-selection indices, emission-schedule seeds, and human-readable presentations of derived seed values, including wordlist encodings—are computed by one or more participating agents or controllers through logged tool calls. For the subset of agent or controller tool calls that derive session parameters for
20 a physical RK run, the request, response, derivation inputs, and post-processing record are logged as a declared auxiliary channel of the convolution bundle rather than treated as an opaque external input.

In some embodiments, the derivation is consumed by, constraining, or verifying a physical RK run or evidence bundle, with the parameter-consumption event committed in the pro-
25 tocol digest Π_{dig} before the capture window opens; the auxiliary channel binds the tool-call record to a specific physical RK run through the run-identifier rid and the timestamp of the parameter-consumption event.

In a non-limiting worked derivation example, the canonicalisation rule normalises the tool response to a UTF-8-encoded JSON string with sorted keys, and the derivation rule is
30 HKDF over the canonical response with Π_{dig} as salt and a declared domain-separation tag as info, with the HKDF output parsed as a 32-bit unsigned integer scan-seed, an 8-tuple of exposure-schedule parameters in declared units, and a 16-bit verifier-cadence value; other canonicalisation and derivation rules are non-limiting alternatives.

In some embodiments, each such tool call is specified in advance by a precommitted tool-
35 call specification $\text{tc} = (\text{name}, \text{args-schema}, \text{provider-policy}, \text{canonicalisation-rule}, \text{derivation-}$ whose hash is bound into the protocol digest before the capture window opens, such that substitution of a different tool call after the fact is detectable on selective opening. In some embodiments, the provider policy declares a provider identity, provider class, permitted provider set, local deterministic tool, or corroboration policy, and the canonicalisation rule

declares how tool responses are normalised before derivation.

In some embodiments, the derivation input to the tool call comprises one or more of: the protocol digest of a parent run, a concurrent run whose referenced windows have closed before the parameter-consumption event, a hash-chain anchor exposed by a previously committed window, an external blockhash or beacon value, or a declared composition of the above with multi-contributor commit–reveal outputs per Paragraph 27.7. In some embodiments, the derivation trace—inputs, tool identifier, model version or provider fingerprint, the response or a canonical response representation sufficient under the declared derivation rule, together with any response hash, selective opening policy, and post-processing steps applied before the derived value enters the protocol—is committed as an auxiliary channel of the convolution bundle such that a verifier can recompute the derivation from disclosed atoms.

In some embodiments, the same-session case comprises a configuration in which an agent participating in a capture session derives a later session parameter from the chain state of one or more already-closed prefix windows of that session, and the derivation trace is committed before the derived parameter is consumed. In some embodiments, this configuration produces a bounded same-session consistency loop wherein (a) session parameters are verified against the chain state, (b) chain state is verified against the committed atoms, and (c) committed atoms are verified against the declared session parameters, such that the three consistency checks are evaluated against the same protocol-digest, hash-chain, and selective-opening commitment family rather than against an uncommitted off-chain parameter source.

In some embodiments, same-session derivation is constrained by a declared causality envelope including both a time-order constraint and a consumption constraint: a chain-state value may be read only from a window whose close anchor precedes the first controller event that consumes the derived parameter, and this constraint is enforced by the verifier during selective opening. In some embodiments, violation of the causality envelope—for example, a tool-call record whose committed input references a window timestamped after its consumption event—is surfaced as a protocol-digest anomaly regardless of the correctness of the derived value.

Where the derived value is obtained through a semantic or routed ensemble branch, the semantic-computation attestation record of Section 15.6 may be used as the tool-output attestation object; this subsection addresses derivation of session parameters rather than semantic correctness.

In some embodiments, the tool-call specification further declares a canonical serialisation format for inputs and outputs, a domain-separation tag for each derived parameter type, and a deterministic post-processing rule, so that independent verifiers recompute the same derived value from disclosed atoms.

In some embodiments, if the tool provider returns an error, timeout, noncanonical response, response outside the declared schema, or response whose provider fingerprint does not match the committed tool-call specification, the derived parameter is rejected or the run is marked as reduced-assurance under a predeclared fallback rule.

In some embodiments, nondeterministic tool calls are admitted only when their nondeterminism is itself declared and committed, for example by committing provider sampling parameters, provider-side randomness commitments, or a canonical response-selection rule; otherwise the nondeterministic output is treated as non-recomputable and cannot serve as
 5 an acceptance-critical derivation input.

In some embodiments, derivation checks that gate acceptance of the run are partitioned from evaluation checks that assess provider reliability, tool-call latency, model drift, or common-mode failures, so that the participating agent cannot optimise directly against every derivation-quality surface.

10 **Heterogeneous-provider corroboration of canonical derived values (non-limiting).**

In addition to multi-RK corroboration of physical evidence, in some embodiments corroboration of canonical derived values produced through logged tool-call specifications—including the seed derivations of Section 27.7.1, tool-call outputs contributing to protocol randomisation, and commitment values advanced as freshness contributions—is obtained by executing
 15 the same committed tool-call specification on two or more provider instances having declared diversity or non-common-control properties, such as distinct organisational operators, distinct model families, distinct serving clusters, or separately attested execution environments, whose identities, model versions, and serving infrastructure fingerprints are committed to the protocol digest alongside each response.

20 Provider corroboration verifies agreement on a canonical derivation under a committed tool-call specification; it does not by itself add fresh entropy unless a provider is separately declared as a contributor under Paragraph 27.7.

In some embodiments, agreement among provider responses on the canonical derived value is a non-entropy-bearing derivation-corroboration meter or freshness-assurance sub-
 25 meter, while entropy contribution and contributor diversity are accounted for separately under the multi-contributor freshness composition rule.

In some embodiments, disagreement is logged as a protocol-digest anomaly and may be surfaced through a discrepancy-reporting or audit-escalation channel; the disagreement is not treated as a proof-of-discrepancy unless the applicable PoD requirements of Section 13.3,
 30 including physical re-execution where required, are separately satisfied.

In some embodiments, provider diversity and contributor diversity are recorded as distinct fields: provider diversity corroborates execution of a committed derivation rule, while contributor diversity is evaluated under the freshness-composition rule of Paragraph 27.7.

In some embodiments, the committed tool-call specification declares canonicalisation
 35 rules for provider responses, including whitespace, encoding, numerical precision, truncation, domain-separation tags, and schema validation, so that provider agreement is assessed on the same canonical value.

In some embodiments, if provider responses disagree, fail to arrive within the declared response window, or fail schema validation, the derived value is not silently majority-resolved;
 40 instead the run follows a predeclared rule selected from abort, reduced-assurance continua-

tion, manual audit, deterministic local recomputation, or re-execution under a fresh provider set.

In some embodiments, provider-diversity claims are downgraded when providers share a common model lineage, serving infrastructure, operator, training corpus, tool backend, or
 5 upstream deterministic implementation likely to create a common-mode failure.

In some embodiments, provider-corroboration meters are partitioned from acceptance meters so that a provider-comparison mechanism can be used for held-out audit or reliability scoring without becoming an optimisation target for the participating agent.

In some embodiments, a single-provider execution of the committed tool-call specification
 10 remains a reduced-assurance sub-embodiment where the tool trace, canonical response, and post-processing rule are still committed and recomputable.

28 Industrial Applicability

The disclosed systems and methods are applicable to at least the following fields: physical evidence and provenance verification in media production, journalism, and legal proceed-
 15 ings; active sensing and computational imaging for metrology, inspection, and autonomous navigation; projection mapping and immersive display for entertainment, architecture, and education; hardware security and device authentication using physically unclonable properties; continuous tamper detection for critical infrastructure monitoring, evidence custody, and nuclear safeguards; liveness detection and anti-spoofing for biometric systems; networked
 20 sensing and distributed verification for environmental monitoring, smart cities, and defence; physical computing and reservoir computing using optical or hybrid media; and distributed model improvement and fleet-calibrated device management through physically verified discrepancy discovery.

Non-limiting Statement

25 The embodiments and examples described herein—including all main-body sections, appendices, and optional modules—are non-limiting. The scope of protection, once claims are filed, is defined solely by the claims. No feature, embodiment, example, numerical value, operating range, material choice, protocol, algorithm, or theoretical correspondence described in this specification is intended to limit the scope of the claims unless expressly recited therein.
 30 Where the specification describes a feature as present “in some embodiments,” the absence of that feature in other embodiments is equally contemplated. The terms “preferred,” “advantageous,” and “particularly suitable” indicate non-limiting design preferences and do not exclude alternative implementations. Dimensions, quantities, frequencies, and other numerical parameters are illustrative and may be varied without departing from the principles
 35 described. Theoretical frameworks, substrate mappings, categorical constructions presented herein are interpretive tools and do not constrain hardware or software implementation.

29 Declared Open Problems

The following are open empirical questions, explicitly not asserted as established results: survival of simultaneous tone multiplexing; the effective-rank to task-useful-rank mapping; drift-bias tolerance of physical gradient methods; and controllability-scaling calibration. These open questions are not required for the Tier A or Tier B claims of Section 3 and bear only on Tier C capacity claims within the bounds of Section 15.5.

This appendix collects research directions and unproven conjectures that are intellectually connected to the Reality Kernel architecture but are not claims of this disclosure. No embodiment outside this appendix depends on any result stated here.

29.1 Yoked channel authentication capacity (conjecture)

The following is an unproven conjecture stated as a research direction. Let α denote the coupling strength between scene and reactor, and let $\mathcal{I}_F(\alpha)$ denote the Fisher information about α in a bundle $C_{0:T}$:

$$\mathcal{I}_F(\alpha) = \mathbb{E} \left[\left(\frac{\partial}{\partial \alpha} \log p(C_{0:T} \mid \alpha) \right)^2 \right].$$

The operating hypothesis is that $\mathcal{I}_F(\alpha) \rightarrow \infty$ as $\text{CLE}(\alpha) \rightarrow 0^-$, by analogy with susceptibility divergence at a phase transition (fluctuation–response theorem of statistical mechanics). If this divergence holds, then any emulator attempting to reproduce the coupled dynamics with coupling parameter $\tilde{\alpha} \neq \alpha$ incurs an expected log-likelihood gap that grows without bound as the genuine system approaches the synchronisation boundary.

This motivates the *Yoked Channel Authentication Capacity Conjecture*: a Yoked channel operated at $\text{CLE} \in [-\varepsilon_{\text{CLE}}, 0)$ achieves an authentication capacity $C_{\text{Yoked}}(\alpha)$ (in the sense of bits of identity information attributable to the joint attractor rather than to any product-state emulator of equal physical scale) that is strictly positive and increases as $\text{CLE} \rightarrow 0^-$. Whether C_{Yoked} can be computed in closed form for a representative class of Yoked systems, and whether it can be lower-bounded via the mutual information $I(S_{0:T}; R_{0:T})$ in Yoked mode relative to the maximum mutual information achievable by a decomposable system of equal physical scale, are open questions. A positive result would establish Yoked coupling as a channel primitive in the information-theoretic sense.

All operational claims in this specification regarding Yoked operation are conditioned on the empirically measured TE and CLE proxy suite under declared meter envelopes and are independent of the truth of this conjecture.

Even if no closed-form Yoked authentication capacity is established, Yoked operation remains supported by the declared transfer-entropy, conditional Lyapunov, coupling-quality, and tamper-detection criteria recorded in the protocol digest.

29.2 Classifier-mediated Yoked state estimation: cross-scene generalisation

The worked embodiment of Section 8.5.6 trains a coupling state classifier on known analysable reactor pairs and deploys it against non-analysable scenes. Several open ques-
 5 tions must be resolved by experiment before this architecture can be relied upon for deployment against novel scene classes:

Whether device-side bundle data identifies two-sided CLE and TE across unseen scene classes; how generalisation error scales with reactor diversity rather than raw reactor count K ; which reactor families are needed to cover human-relevant dynamics; whether the declared
 10 supervision window length T is sufficient for CLE convergence in each declared scene class; what the true run-level effective sample size is after within-interaction autocorrelation is accounted for; what end-to-end latency is achievable under on-device compute constraints; and whether a concrete agentive-scene control law converges stably or requires empirical bounding.

The minimal adequate experiment for the most important of these — cross-scene classi-
 15 fier generalisation — is as follows. Train the classifier on $K \geq 50$ known analysable reactors spanning at least five distinct scene classes (non-limiting examples: phosphor screen, scattering plate, resonant mechanical structure, living plant tissue, artificial vascular phantom). For each reactor, compute ground-truth CLE and TE from both device-side and scene-side
 20 signals simultaneously; use device-side bundle data alone as classifier input. Evaluate on $M \geq 20$ held-out reactors from at least two scene classes entirely absent from training. A positive result requires: area under ROC > 0.85 for binary Yoked / not-Yoked classifica-
 25 tion on held-out scene classes; Pearson $r > 0.7$ between predicted and ground-truth CLE on a continuous scale; and latency within the declared rolling window period. A negative result requiring restriction of the embodiment to declared scene families: AUC below 0.6
 on held-out classes while remaining high within training classes (indicating the classifier is recognising scene-class signatures rather than coupling state), or systematic CLE sign-flip error rate above 0.2 near the Yoked boundary.

These are experimental questions whose resolution would strengthen the embodiment
 30 but whose absence does not affect any other claim in this disclosure.